Buzzing culture: drone technology to empower underserved audiences to connect with heritage and culture

Isilda Almeida

A thesis submitted in partial fulfilment of the requirements of the University of Brighton for the degree of Doctor of Philosophy

With Heritage partners: National Trust, South Downs National Park Authority

July 2025

Abstract

Drone technology is progressively more embedded in civil society, including heritage and cultural organisations. As these institutions engage with drones to complete functional tasks associated with the preservation and documentation of material culture, they are also working to decentre their systems and ways of working. New practices are focused on addressing inequalities in the access to and participation in heritage and culture, by underserved communities, in ways that foster capacity building and ownership in these groups.

This thesis*, Buzzing culture: drone technology to empower underserved audiences to connect with heritage and culture*, investigates the ways in which UAVs can facilitate agency in underserved communities, in order for them to shape their heritage and cultural experiences, as part of answering the principal research question "How can drone technology empower underserved audiences to connect with heritage and culture".

The methodological approach in this project was guided by theories including collaborative design and iterative research, which determined the qualitative methods used. The application of these theories was supported by the values of empowerment, agency, and inclusion. Semi-structured interviews are used to capture drone users' and heritage experts' perceptions of UAVs opportunities, barriers, and risks in engaging underserved communities with heritage and culture. The views of heritage experts are further explored through a UK-wide online survey which makes it possible to test the generalisability of the interview data.

As underserved communities are at the centre of this project, the study involves co-design research with people living with dementia, culminating in a drone livestreaming activity, in which participants instruct the drone pilot on the direction of the drone. The perspective of Global Majority women is investigated through an autoethnographic study. Both include the use of creative techniques and visual content analysis.

Key findings suggest concerns for loss of institutional power and control are at the centre of heritage organisations' reluctance to adopt drone technology, as a new approach to empower

underserved audiences to connect with heritage. Drones’ technical capabilities of drone mindedness, drone sensing, spatial mobility and verticality, control, and automation lead to opportunities for the technology to facilitate a deeper connection with landscape and place and engage with heritage.

Nevertheless, to empower communities, these capabilities must be used with agency by underserved communities. When communities are agents/producers instead of spectators /consumers, their drone video content creates a distinct visual language that emerges as a counternarrative to the spectacle created by the auratic views in institutional drone content. The latter project heritage as perfect, unique, and universally relevant.

Through the co-design research livestream, the production of a framework for the analysis of visual language and power in drone video content, and the identification of specific characteristics in drone video content generated by underserved audiences, this thesis contributes new empirical and conceptual knowledge. These empirical and conceptual theorisations address a gap in digital mobilities, visual studies and arts and media scholarship.

Table of Contents

Lists of Tables and Figures

Tables

Figures

Acronyms and definitions

CAA – Civil Aviation Authority

DCMS – Department for Digital, Culture, Media and Sport

NT – National Trust

SDNP – South Downs National Park

SDNPA – South Downs National Park Authority

STEM – Science Technology Engineering and Maths

UAV – Unstaffed Aerial Vehicle

Acknowledgements

I would like to thank:

My supervisors for their guidance, committed support and feedback: Dr Lara Perry, Dean, School of Humanities and Social Science, University of Brighton; Dr Frauke Behrendt, Associate Professor in Transitions to Sustainable Mobility, Eindhoven University of Technology; Prof Olu Jenzen, Professor of Media and Digital Culture, University of Southampton; and Dr Patricia Prieto-Blanco, Lecturer in Digital Media Practice, Lancaster University.

Heritage Partners and supervisors, for their commitment to the research, being open, resourceful, and generous: Anooshka Rawden, Cultural Strategy Manager, South Downs National Park, and Keith Challis, Remote Sensing Coordinator, National Trust.

Dr Anne Galliot, Writing Coach, for their support, encouragement, and hard work which got me here.

Alison Milton, my mentor, for listening with enthusiasm and compassion, really seeing me, and contributing insightful and relevant wisdom into my research.

Anne-Marie Lovejoy, Director, Lovejoy Centre, and the Centre's staff and volunteers, Chantelle, Shirley, and Sarah, for all the support, laughter, and enthusiasm.

The Lovejoy Centre clients and their families, for being open, adventurous, and curious. I would not have done the PhD I dreamed of, without you: Alfie, David, Delia, Eileen, Maggie, Carmen, Sheila, Thay, and Viv.

All the SDNPA and NT staff who participated in interviews and supported the research activities and access to their resources, all heritage practitioners who took part in the online survey for this research, and all the drone experts who agreed to be interviewed, for their generosity, time, and openness.

Alison Pritchard, Site Manager, Monks House, National Trust, for taking risks and being curious, and allowing me to include their property in my research.

Dr Anna Jackman, Lecturer in Human Geography, University of Reading, for the important reflections on diversity in academic research which led me to think more creatively about participant recruitment and capturing richer data.

Professor Julia Hildebrand, Assistant Professor of Communication at Eckerd College in St. Petersburg, Florida, for their generosity, openness and encouragement.

Sam Moore, Drone Pilot, Visual Air, for their expertise, creativity and for sharing my vision and excitement about the research.

Shellie Baxter, Director, The Drone Project, for the shared vision, determination, and commitment to changing the lives of Global Majority girls and women through Science and Engineering.

Dr Elisa Serafinelli and Dr Lauren O'Hagan for their time and generosity in discussing their research.

James Sainsbury, Curator of Archaeology, Worthing Theatres and Museum for the inspiring Highdown Hill walking tour, the imparted knowledge, access to their learning collection and helping make the experience richer.

The Engineering and Physical Sciences Research Council for funding the Centre for Doctoral Training in Science and Engineering in Arts, Heritage, and Archaeology (SEAHA) which my Doctoral Project was a part of.

The former Centre for Digital Cultures and Humanities, University of Brighton for funding the autotechnographic films.

Writing Our Legacy for funding the Research and Development of creative work for my research as part of their *Shifting The Gaze* programme.

The University of Brighton Doctoral College and SEAHA staff for their continued and faultless support.

Dr Maya Angelou's estate and family.

Prof David Nash, Co-Director, EPSRC Centre for Doctoral Training in Science and Engineering in Arts, Heritage, and Archaeology for their support, reassurance, and compassion.

Dave Woods for all the support, patience, and resourcefulness

Fiona Sutton, former Deputy Head, Doctoral College, University of Brighton, for listening, being kind, supportive and generous.

Emma Newell, Doctoral College Admissions Officer, University of Brighton, for their efficiency, friendship, compassionate listening, and kindness.

Ronnie Williams, Library Assistant, University of Brighton, for the welcome, the care, the shared stories and most of all, the conversations.

Rachel Long, Poet, for giving me inspiring feedback on the writing for my autotechnographic study.

Bryony Benge-Abbot, interdisciplinary artist and researcher, for supporting my research and sharing their art, research, and expertise.

Associação da Calçada Portuguesa, @VioletVik, Seb the Nomad and Caroline Ribeiro for their permission to use their drone footage.

My husband, Ozzie and Clara who flew drones with me, comforted me, made me laugh and kept me grounded.

To my mums whom I will always love and remember:

Penny Walker, as the kindest person I have ever known, for unconditional love and unwavering support and wishing racism did not exist; and Maria Carolina R. for passing on excellent values of integrity, honesty, the drive to do better and achieve more, and for always speaking up against injustice.

Author's Declaration

I declare that the research contained in this thesis, unless otherwise formally indicated within the text, is the original work of the author. The thesis has not been previously submitted to this or any other university for a degree, and does not incorporate any material already submitted for a degree.

Student Name: Isilda Almeida

Signed: [redacted]

28th November 2025

Chapter One: Introduction

Chapter introduction

In our society, drone technology is increasingly used by organisations, businesses, communities, and individuals. The technology's capabilities are rapidly developing, presenting risks resulting from misuse whilst offering innovative ways of preserving, documenting, and experiencing place. This research investigates the use of UAVs in empowering underserved groups to connect with heritage. It builds on current academic work examining what the naturalisation of drones into non-militarised contexts means in terms of regulatory frameworks, and for the different stakeholders of aerial space.

This investigation took place in the aftermath of COVID-19 and the murder of George Floyd. Both events highlighted the experiences of inequality, injustice, and exclusion some communities in the UK face. This is particularly true of disabled, working class and people of the Global Majority. The latter are defined as "all ethnic groups except white British and other white groups, including white minorities" (NCVO, 2024). Heritage and cultural sectors have been proactively exploring ways to collaborate with underserved communities through co-production approaches to exploring and representing their narratives. This is a corrective measure to address how organisational systems and processes have amplified exclusion.

Drones in society, and heritage and cultural participation are interrelated through the shared context of devolution of institutional power. The growing use of Unstaffed Aerial Vehicles (UAVs) has prompted agential negotiations over aerial space, between governmental agencies, civic institutions, commercial organisations, communities, and individuals. The contexts above present a unique research opportunity to develop scholarship where these fields mutually inform each other's practices.

1.1 Important concepts and ideas in this research

This is an interdisciplinary doctoral project grounded on Museum and Heritage scholarship, Culture, Media, and Digital Mobilities' Studies. It is driven by two contexts: first, the inequalities in the access to and participation in heritage; second, the ubiquity of drones in

civil society (including use in Heritage practices) and the inherent power and agential shifts in the hierarchies of aerial space. This research is centred on interrogating how drone technology can be used to empower underserved communities to connect with heritage and culture.
To explore this, it is important to define key concepts and ideas: drone technology, empowerment, underserved communities, and heritage. Drone technology refers to battery operated, camera equipped (Mademlis *et al*., 2019, para. 1) autonomous and remotely operated aerial vehicles (D'Andrea, 2014, para. 1) used commercially and by hobbyists (Hildebrand, 2021).

In relation to the concept of empowerment, when considering the ability of drones to facilitate community empowerment, conceptualisation of the latter by Zimmerman (1995) as a multilayered process is crucial. Empowerment involves a perception of collective control, proactivity, awareness of community resources, collaboration and development of skills towards achieving social change. Control, in the sense of collective decision making on issues that are relevant to groups, is particularly in deficit in underserved communities. These are defined in the thesis as groups with a shared experience of being left out, also described as marginalised or excluded (Humm and Schrögel, 2020) by institutional systems and processes.

Finally, in this study, Heritage is considered to be, as defined by the United Nations Educational, Scientific and Cultural Organisation (UNESCO) "[…] artefacts, monuments, a group of buildings and sites, museums […] of […] symbolic, historic, artistic, aesthetic, ethnological or anthropological, scientific and social significance." (UNESCO, 2009). This includes tangible and intangible heritage artefacts, sites, or collections. This study considers both heritage and culture (festivals, celebrations).

1.2 Research questions

This study is guided by a principal research question: "How can drone technology be used to empower underserved audiences to connect with heritage and culture?" This question brings together three separate elements: the capabilities of drone technology, heritage and cultural organisations' perceptions of the technology, and underserved audiences in both their relationship with heritage and with UAVs. The aforementioned aspects were articulated as three further sub questions investigating: What do heritage organisations perceive to be the barriers, risks, and unique opportunities of drone technology to engage underserved, non-visiting audiences? What unique opportunities does drone technology offer the interpretation of and engagement with heritage? And, finally, what is the role of drone technology in the empowerment and engagement of underserved communities to access heritage?

1.3 Methodological approach

This research was designed as an iterative study using qualitative methods, including semi-structured interviews, online surveys, and participatory approaches integrating creative methods, co-design research and autoethnography. Each research activity was informed by the previous stages of data collection and shaped the design of the next stage.
The methodological approach was also guided by the research questions in this study. Semi-structured interviews and online surveys with heritage experts were used to capture data to answer the question: “What do heritage organisations perceive to be the barriers, risks, and unique opportunities of drone technology to engage underserved, non-visiting audiences?”

Interviews captured the views of research partner organisations- the National Trust and the South Downs National Park Authority. A section of the partners’ interview data was integrated with the results from semi-structured interviews with drone users. This was as part of exploring research question: “What unique opportunities does drone technology offer the interpretation of and engagement with heritage?” Participants in drone interviews included heritage practitioners operating drones as a significant part of their role, commercial and hobby users.

Investigating the question of the role of drone technology in the empowerment and engagement of underserved communities to access heritage required the deployment of an approach that was itself empowering. This was undertaken through participatory techniques - co-design research with people living with dementia - integrated with creative methods. A researcher-led autoethnographic study informed by this thesis author’s Global Majority identity and gender, supported the exploration of the question on the use of drone technology, in the empowerment of underserved communities to access heritage.

The research methods applied in this research produced qualitative data (interviews and online surveys), and empirical data (observations, reflections, creative visual media content). As part of the iterative research process, the data was interpreted using both inductive and deductive methods. Semi-structured interviews were analysed through the application of thematic analysis and coding. Online surveys were considered through descriptive statistical analysis and thematic coding of qualitative responses, and the co-design research and livestream were analysed through the thematic analysis and coding of participants' discussions. Visual methods were employed to analyse visual language in pre-existing, publicly available drone visual content, and drone videos instigated by (Pauwels, 2010) and produced within the research project.

The research approaches were supported by collaborative work with project partners. NT and SDNPA (research partners) supported the research by providing access to their strategic documents and policies, advising on the dissemination of interview recruitment materials, supporting the delivery of co-research activities, and facilitating access to the most suitable drone flight sites in their care.

The research with the Lovejoy Centre was designed with the Centre staff and shaped by a period of observation of best practice in working collaboratively with people living with dementia.

A detailed overview of the methods applied in this research is provided in the Methodology Chapter (Chapter Three).

1.4 Thesis structure

This thesis is structured into eight chapters. The current chapter (Chapter One; Introduction) is followed by the Literature Review (Chapter Two) and Methodology (Chapter Three). The presentation and discussion of results is undertaken in the following four chapters. Chapter Four: Heritage organisations' perceptions of the barriers and risks of drone technology to engage underserved audiences; Chapter Five: The unique affordances and opportunities of drone technology for the interpretation of and engagement with heritage; Chapter Six, The visual language of empowerment and agency: a framework for analysis, proposes a new approach to examining drone content, and Chapter Seven: Drone technology in cultural democratisation, equity and inclusion, "How can drones empower underserved communities to connect with heritage and culture?" discusses the role of UAVs in facilitating empowerment in heritage and culture.

Finally, Chapter Eight: Conclusions on drone technology empowering underserved audiences to connect with heritage and culture discusses this doctoral project and reflects on the journey and this thesis' contributions to knowledge.

1.4.1 Literature Review

In Chapter Two, the Literature Review considers drone use in civic society, the shift in museums and heritage organisations' relationships with audiences and communities and how experiences of exclusion affect these practices.

Current literature on drone technology explores what Jensen (2016b) refers to as the boomerang effect, a progressive naturalisation from militarised use to a consumer industry. This expansion of the control of the skies mirrors the shift in Heritage and Culture sectors' ideas of knowledge and value. Literature from Museum Studies, used here as a proxy for the Heritage and Culture sectors, discusses the change in museums from producers and holders of knowledge, disseminating the dominant narrative, to social justice advocates facilitating the active engagement of excluded communities with sites and collections.

The changing relationship of heritage organisations with left out groups is central to this research. The silencing, absence and or misrepresentation of their narratives in material and intangible heritage and culture reflects the power imbalances and resulting disparities in civil society. This is further amplified by institutional systems and processes of knowledge production and interpretation.

As it currently stands, these tend not to create space for agency or empowerment for disadvantaged communities, which makes it difficult for them to access and participate in heritage experiences in the same way as dominant groups. This creates an experience of underservice. It is in reference to the latter that this study researches the possibility of using the new agential negotiations afforded by the naturalisation of drones into civil society, to explore opportunities to empower underserved audiences to connect with heritage and culture.

1.4.2 Methodology

Chapter Three introduces this project's research design and methodological approach which were informed by theories including collaborative design, and iterative research. Qualitative methods deployed in this study are flexible and chosen to give participants an opportunity to shape their contribution.

Semi-structured interviews elicit unexpected points of view and ideas from heritage organisations, as do free-text questions in the online survey. These have been used in this research to explore heritage practitioners' views on drone technology and the barriers, risks, and opportunities it offers to engaging underserved audiences and the views of drone users on unique opportunities of UAVs (semi-structured interviews only).

The involvement of people living with dementia as an underserved community is integrated in the research through participatory methods. Employing collaborative design (co-design) as an approach to involve participants living with memory loss is central not only to this group's specific context of exclusion from decision-making, as a result of their diagnosis, but also to investigate what agential possibilities drone technology offers underserved groups.

The research activities with this group use inclusive creative methods, such as collaborative mapping, storymaking, drawing, crafts, and object handling. These techniques were part of the process of supporting participants to shape their drone experience, deepen their connection to the National Trust and SDNP's sites and simultaneously develop trust in the research process, which culminates in a drone livestreaming event.

Positionality is an important element of research, and this study drew on autoethnographic methods to engage with a systematised study of my experience of using a drone, in the context of my identity as a Global Majority, disabled woman. This investigation was part of reflecting on elements to support the understanding of the experiences of the underserved communities the researcher in this study shares characteristics with. Walking practice, filmmaking and creative writing were used in the autoethnographic study, as methods to explore and deliberate.

1.4.3 Project, results, and analysis

The next three chapters present, analyse, and discuss the data captured in this study, examined using inductive and deductive content analysis, thematic coding of qualitative responses and visual content, and descriptive statistical analysis.

Chapter Four establishes how heritage experts' verbalised perceptions of barriers and risks of drone technology to engaging underserved audiences, obscure an institutional concern of loss of power and control.

In Chapter Five, when comparing heritage perceptions of opportunities of UAVs to engage excluded audiences, institutional responses echoed those of drone users. Both groups envisage drones as facilitators of access and accessibility, new perspectives, a greater range of experiences, generators of high-quality digital data, inspiring creativity, and storytelling. Each of these identified opportunities is underpinned by technical capabilities that reframe human interactions, both with their environment (towards broader perspectives and experiences) and with the machine (towards increased access, enhanced opportunities for acquiring data, storytelling and meaning-making).

A comparison between the views of heritage practitioners and the different types of drone users (researchers who use drones in their role, commercial drone users and drone hobbyists) revealed those using drone technology for recreational purposes have stronger relational experiences with drone technology. Consequently, they articulate the drones' capabilities of empowerment in a stronger and clearer manner. To identify how those capabilities are expressed in drone video content, it was necessary to decode visual language in aerial views.

Chapter Six discusses existing theorisations of creating meaning through the use of camera perspectives. Having identified the absence of an appropriate framework to analyse drone videos, the chapter proceeds to draw on the study by Serafinelli and O'Hagan (2022) of drone photography characteristics and their effects, research by Stankov *et al.* (2019) on the effects of drone videos in tourism marketing; the semantics of camera perspectives in Film Studies scholarship and the drone shot taxonomy by Mademlis *et al.* (2019). The above works are used in this project as terms of reference for the proposal of a newly formed analytical framework. This proposed approach integrates contextual information, with theme categories, descriptive data, camera perspectives and effects, and the taxonomy of drone camera shots.

The application of the framework, in Chapter Seven, to the analysis of institutional and drone videos generated by underserved communities, resulted in the identification of specific aesthetic qualities that make the latter distinct from institutional drone videos. This research found that drone video content produced by heritage organisations projects heritage sites and landscapes, as idealised and spectacular. This is what Benjamin (2007) conceptualised as the aura.

Auratic drone views form part of an institutional discourse of power, which underserved communities disrupt by using drone capabilities in non-traditional ways. This disruption was found not only in the pre-existing drone video content analysed in this project, but also in researcher instigated drone content, which includes the drone livestream film generated by people living with dementia, and the autoethnographic drone videos.

Finally, Chapter Eight, concludes the thesis reiterating the key findings in this doctoral project, and their relevance to existing scholarship, reflecting on the research journey, proposing recommendations for heritage organisations, and making considerations on future avenues of research.

1.5 Central argument

The central argument in this project is that drone technology can empower underserved audiences to connect with heritage and culture when used as an empowering tool, in a context where underserved audiences have agency and control over their experience and the

content produced. The use of empowering processes to achieve empowered outcomes (Fetterman *et al*., 2014, p. 193) is the cornerstone of this research project. What this means is that addressing access and accessibility, and the role they play in excluding underserved communities from participation, is not sufficient to correct the power imbalance. This is achieved through empowered outcomes, such as self-efficacy and capacity building.
These are what makes the use of drone technology purposeful in this research project, as UAVs are used to facilitate a remote, live experience of heritage that is shaped and controlled by underserved communities (in this case, people living with dementia).

1.6 Contribution to knowledge

The key outcomes from this investigation, which form its contribution to knowledge, are the design of a framework for analysis of power in the visual language of drone video content, and the conceptualisation of the visual language produced by underserved audiences, when they use the drone as agents. This visual language is composed to create a counternarrative, challenging classic aerial views, and traditional use of drone competencies.

Centrally, through answering its headline research question: “How can drone technology empower underserved audiences to connect with heritage and culture?”, this study brings forth an understanding of the sociotechnical effects (Abbas and Michael, 2023) of drones, or, in other words, of how drones are implicated in changing underserved communities’ experience of heritage (Light *et al*., 2018, p. 408) by facilitating agency and, as a result, disrupting the institutional aesthetics of reverence and power.

Existing research on drones and empowerment has focused on the design of navigation models and systems to facilitate drone control (Krajník *et al*., 2011); (Szafir *et al*., 2017) including research from Mangina *et al.* (2016) on drone livestreaming for mobility access. Nevertheless, there has been little research on how UAVs capabilities to engender freedom and agency might empower underserved communities to shape their connection with heritage and culture beyond accessibility, and the implications of this for visual language in drone content. This is an aspect and approach that has not been explored before this thesis.

1.7 Situating the Project and Researcher

This project is part of the Centre for Doctoral Training in Science and Engineering in Arts, Heritage, and Archaeology (SEAHA). SEAHA is a PhD research training programme supporting research on Heritage Science. It is funded by the Engineering, and Physical Sciences Research Council. It brings together three higher education institutions: the University of Brighton; University College London; and The University of Oxford. Industry partners in the project are The South Downs National Park Authority (SDNPA) and the National Trust (NT).

My relevant background includes a Master of Arts in Museum Studies by the University of Leicester and a professional background in heritage and cultural engagement in Museums, Galleries and Archives and Consultancy in Cultural Equity, Diversity, and Inclusion (EDI) to heritage and cultural organisations, such as the Museums Association, Museum Development England, and the Association for Independent Museums.

As a Global Majority woman, the above situates me as a researcher with lived experience, extensive knowledge in Heritage Management, and ample practice in working with excluded communities. These have included disabled people, schools, Global Majority communities, people with experience of homelessness, young audiences, Age UK, and looked-after children. The researcher's background is important to this research because, according to McCarthy (2015, p. xxxviii) "Much of the academic work on museums has been written by university scholars who may have little experience working in the sector. Unfortunately, not nearly as much has been written about their work by professionals themselves."

Chapter Two: Literature Review

Chapter introduction

This chapter draws on academic and grey literature to discuss the contexts relevant to this investigation's principal research question: How can drone technology empower underserved communities to connect with heritage and culture? To examine this, this study investigates three key themes: drone use in civil society; the role of museums in society; the concept of underserved communities and the affective aspects of being left out. Table 1 below, gives an overview of the authors and ideas central to the examination of the aforementioned three themes.

Table 1 Overview of authors, concepts, and ideas central to the thesis and where in the chapter they are discussed

Author(s)	**Key concepts and ideas**	**Section in chapter where they are elicited**	**How they informed this thesis**
Jensen (2016a) and (2016b)	Naturalisation of drones: the boomerang effect and the democratisation of the aerial space.	2.1 Drones in society	Drones’ expansion of use into civil society
Hildebrand (2020)	Drones exist within a dystopian and utopian dichotomy. The positive side leads to a reimagination of propositions of a more equitable society.		Dystopian and utopian views of drones originate from the same technical capabilities. Reimagining the positive possibilities of drone technology may bring more equitable propositions.
	Drone mindedness.		UAVs foster connectedness with surroundings and a new way of seeing them.
	Drone affordances.		Consideration of drone capabilities as characteristics that make some of the drone behaviours and outcomes possible.
Serafinelli and O’Hagan (2020)	Civic drone visualities signify a new turn in visual culture, as drones produce unanticipated visualisations of the world around us.		UAVs bring new perspectives arising from their capabilities of mobility, verticality, and remote sensing.
Hooper-Greenhill (2000)	Contemporary socio-economic changes in civic society precipitate the questioning of the purpose of museums.	2.2 Museums’ role and the purpose of Heritage and Cultural Organisations in society	Museums’ and heritage organisations’ positioning has changed as they recognise the value of lived experience as a form of expertise. The politics of knowledge production shifts as part of redressing the imbalances of power between institutions and excluded audiences and narratives.

Author(s)	Key concepts and ideas	Section in chapter where they are elicited	How they informed this thesis
Hooper-Greenhill (2000)	The myth of the art museum and the values that underwrite it start to change, and lead to a shift in the politics of knowledge between museums and audiences.	2.2 Museums' role and the purpose of Heritage and Cultural Organisations in society	Museums' and heritage organisations' positioning has changed as they recognise the value of lived experience as a form of expertise. The politics of knowledge production shifts as part of redressing the imbalances of power between institutions and excluded audiences and narratives.
Hooper-Greenhill (2000)	Idea of the museum as an authority with a purpose to civilise the masses, regulating their behaviour and values through exposure to the dominant narrative of the elite.		
	Contrary to modern 19th century practices, post-modern museums are challenged to recognise different voices, narratives, meanings.		
Toth *et al.* (2016)	Narrative control in heritage organisations as a form of power.		
Sandell (1998)	Differences between the concepts of poverty and exclusion. Poverty refers to scarcity or inequality in the access to resources. By contrast, social exclusion is a broader, multi-layered concept, including deprivation, poverty, mostly referring to how individuals relate to institutions, family, community, and civic life, and		Exclusion involves relational aspects that affect how communities interact with institutions. The engagement of underserved communities with heritage and cultural institutions that are not representative or inclusive has an emotional impact, affecting people's expectations of a successful outcome, as a result of that interaction.

Author(s)	Key concepts and ideas	Section in chapter where they are elicited	How they informed this thesis
Sandell (1998)	how these hinder their participation in different activities.	2.2 Museums' role and the purpose of Heritage and Cultural Organisations in society	Exclusion involves relational aspects that affect how communities interact with institutions. The engagement of underserved communities with heritage and cultural institutions that are not representative or inclusive has an emotional impact, affecting people's expectations of a successful outcome, as a result of that interaction.
	Identifies a crucial aspect of the relational dynamics between museums and audiences: museums have, themselves, represented institutional exclusion through their systems and operational models.		
	Pervasive nature of social exclusion and how it interconnects different areas of civic life and institutions.		
Kawashima (2006)	Differentiation between audience development and audience engagement. Outreach as a social engagement practice.		Designing activities to engage people living with dementia considered behavioural, cognitive, and emotional factors as part of heightening the level of involvement with the research. Aiming for engagement is not a sufficiently inclusive approach, as it does not address the agency gap underserved audiences experience. Heritage and cultural programmes must aim for empowerment where underserved audiences have opportunities to make decisions, shape their representation and be visible.

Author(s)	Key concepts and ideas	Section in chapter where they are elicited	How they informed this thesis
Ben-Eliyahu *et al. (*2018)	Engagement as a multifaceted process.	2.2 Museums' role and the purpose of Heritage and Cultural Organisations in society	Designing activities to engage people living with dementia considered behavioural, cognitive, and emotional factors as part of heightening the level of involvement with the research. Aiming for engagement is not a sufficiently inclusive approach, as it does not address the agency gap underserved audiences experience. Heritage and cultural programmes must aim for empowerment where underserved audiences have opportunities to make decisions, shape their representation and be visible.
Stroja (2018); Ben-Eliyahu *et al. (*2018)	Concept of empowerment.		
Benjamin (1969)	Concept of the object of art as having a transcendent quality that elicits reverence for its unattainability. Benjamin calls this the aura. The aura projects the object as having a universally recognised value and meaning. Following this idea, the author considers that means of mechanical reproduction, such as photography, trivialise the art object. They do this by reducing the distance between the object and audiences. Mechanical reproduction removes the objects' unique value placing it in the sphere of the ordinary.	2.2 Museums' role and the purpose of Heritage and Cultural Organisations in society 2.3 Underserved audiences in heritage and cultural engagement	Heritage and cultural institutions use visual language to produce an auratic effect that reiterates their positioning as knowledge authorities. This is a practice that has been intrinsic to how museum objects have been construed as universally unique and of intangible value in themselves, and not in relation to their connection to community narratives.

Author(s)	Key concepts and ideas	Section in chapter where they are elicited	How they informed this thesis
Debord (1994)	Concept of spectacle where reality is presented visually as superior	2.3 Underserved audiences in heritage and cultural engagement	Spectacle is a part of this composition. Aura and spectacle create distance between the object visualised and the viewer.
Gallagher (2012)	Self-efficacy: people’s perceptions of their ability to achieve a purpose or complete an action successfully. Co-design is an empirically supported practice to foster self-efficacy, through empowerment. Mastery, where people attribute the success to their own actions, is the most effective way to develop self-efficacy.		Engagement alone is not sufficient as an aim to develop heritage focused experiences. It is fundamental to consider agential negotiations in the process of underserved communities connecting with heritage. Underserved communities’ experiences of exclusion are marked by an agency deficit which co-production addresses.
Humm, Schrögel and Leßmöllmann. (2020)	The emotional impact of oppression, and its effect on underserved communities’ participation in science and culture. Feelings of not fitting in, and emotional burden when visiting a science museum.		Underservice goes beyond material scarcity and is a result of how institutional processes and systems exclude communities of higher need from accessing services.
Kraehe and Acuff (2013)	Concept of being underserved; processes and systems that create underservice in communities of higher need and in connection with the arts.		

This chapter is structured into three sections. Section 2.1 examines literature on the democratisation of aerial space, in which the widening use of drone technology is situated. Drones' naturalisation from the military to civil society and the public sphere carries inherent renegotiations of power between citizens and institutions. Those renegotiations are also happening in the heritage and culture sectors and are discussed in the next section (2.2). The above form the context for this research. Their exploration in this section involves the consideration of public opinion; the capabilities of drone technology that facilitate its use in society; and examples of current specific uses in different sectors. These inform the interview guides, online survey questions and co-design research.

Section 2.2 focuses on the history of museums, as a proxy for heritage and cultural institutions' changing roles from knowledge authorities, and projections of the thinking of dominant groups, to spaces of social justice. The section explores how heritage and cultural organisations position themselves in relation to publics and overlooked communities, and how digital technologies are used in audience development. The chapter follows to discuss COVID-19 as an important shift in heritage and cultural organisations' approaches to access and understanding of disparities experienced by underserved communities. It considers this project within the context of heritage and cultural organisations' changing perceptions of their role in society and reflects on excluded communities.

Section 2.3 investigates the concept of underserved communities in detail. This involves an exploration of what the idea of underserved groups means, for how heritage and cultural organisations approach engagement practices with communities left out from their narratives. This section also considers how the terminology of underserved[1] reflects a shift in the understanding of exclusion and institutional responsibility to approach community engagement equitably. In this context, the section discusses the concept from the perspective of power imbalances between cultural institutions and underserved communities.

[1] Which are also referred to as left out, groups with experience of exclusion or excluded communities

The section concludes by bringing together the context in which heritage and cultural organisations operate, their strife for relevance and inclusion, what this means for underserved communities, and the role drone technology can play in enhancing equitable practices in the heritage and cultural sectors.

2.1 Drone use in civil society

The growing ubiquity of drone technology in the sky and, as a result, in the public consciousness leads to considerations on public perceptions of UAVs. This is relevant because attitudinal barriers affect the engagement with and use of the technology. The duality of drone technology plays a significant part in the imaginations of use of drones, their acceptance, and embedding into civil society. Concerns over intrusion, privacy and surveillance articulated in the literature appear in the research data (see Chapter Four) as do positive views on drones' opportunities of Mobility, Civic empowerment, and Storytelling (Chapter Five).

This section gives examples of specific uses of drone technology across four sectors: Higher Education; the Environment and natural resources; communities and ordinary life; and culture and media. For Higher Education, the focus is on examples from Geology, Environmental Sciences, Engineering, and Archaeological studies. These relate to both NT and SDNPA, who manage natural resources and collections of geological, engineering, and archaeological interest.

This part of the chapter explores drone uses that engender agency in communities in monitoring and protecting their environment, their identity, and their lands. This includes drone practices in territories reclaimed from a legacy of industrialisation and colonialism, where underserved community groups have a heritage of colonial domination. Finally, the section reflects on how drone technology can harness new possibilities for a democratised use of aerial space.

2.1.1 Drones in society: from military use into ordinary life

According to Mascort-Albea, Ruiz-Jaramillo and Romero-Hernández (2014), UAVs are part of a longstanding human practice of finding ways to capture aerial views. O'Hagan and Serafinelli

(2022, p. 327) consider trans-historic approaches to capturing places and people, comparing historical patterns. This is central to understanding the role and impact of drone technology in contemporary society.

Academic literature (Anderson and Gaston, 2013), (Hildebrand, 2018) agrees drone technology is becoming more mainstream (from the military to civil institutions, to individuals and from experts to non-experts). Jensen describes this as naturalisation (2016b, p. 21), a Foucauldian Boomerang (2016a, p. 20). Choi-Fitzpatrick et al. (2009) state that, in 2012, the non-military use of drones in the United States was higher than its military use. Its growth has been exponential since 2015 (Chabot, 2018).

Currently UAVs are applied in agricultural monitoring (Herwitz *et al*., 2004); ecology, and environmental resources (Paneque-Gálvez *et al*., 2018); education (Krajník *et al*., 2011); fire management (Ambrosia *et al*., 2011); and broader uses which benefit from UAVs' capabilities of mobility, data capture and ubiquity.

The use of these capabilities has broadened into smart cities (Jensen, 2016a); wildlife conservation (Sandvik and Jumbert, 2017); transport and delivery (Barr, 2013); humanitarian aid, falconry (Choi-Fitzpatrick *et al.,* 2009); art (Augenbraun, 2016); heritage management (Stek, 2016); community counter-mapping[2]; social drone clubs (i.e. flying for fun); tourism; ballet (dancing with drones); graffiti (art project where drones are creating graffiti).

Others, from empirical observation, include building information modelling in architecture and built heritage, film, television, visual arts, mapping, theatre, light shows, archaeological surveying. Examining the expansion and different uses of drone technology, particularly at an individual level enables the identification of opportunities and methods to use drones in this project.

[2] This is a term used since the 1990s to refer to the commissioning and development of maps by land, using communities to prevent governments and large corporations from illegally claiming community and indigenous land (Radjawali and Pye, 2017).

The general use of drones by individuals has an important implication. Citizens are no longer passive subjects of drone mobility and verticality; they can deploy those capabilities, which are important affordances both in the domestic and militarised uses of drones. The concept of affordances (Hildebrand, 2021, p. 31) is important in this thesis. Hildebrand identifies relational (2021, p. 4), aero-visual (2021, p. 13), visual-communicative (2021, p. 16), material and immaterial (2021, p. 31), spatial and social affordances (2021, p. 90).

The concept of drone affordances is more effective in conveying the complexity of the range of possibilities engendered by the technical competencies, human interactions, and operations, products, and effects of drone use. This is different from opportunities. In this research, the latter are perceived as referring to possibilities, whilst the concept of affordances relates both to the possibilities and the existing capabilities, operations, and outcomes of drone technology.

In the academic literature reviewed here, civil uses of drone technology are mostly discussed in the context of urban environments. In *Drone city – power, design, and aerial mobility in the age of "smart cities*, Jensen (2016a, p. 1) looks at the drone as a "game changer for urban governance, economics and everyday life." However, the dichotomy between drone use in rural versus urban environments is not greatly explored by research literature.

This is significant because this research's institutional partners are based or care for heritage in the county of Sussex, and the Lovejoy Centre is in Sussex. This County is characterised by a large rural hinterland, affected by isolation, digital and socioeconomic disparities that extend into the urban fringes. Audiences in the urban fringes are an audience engagement priority for NT and SDNPA (National Trust, no date) in the *Changing Chalk* partnership project.

As discussed above, literature shows the use of UAVs currently includes extensive non-violent functions for the benefit of communities. These functions are the reverse side of the affordances that make them effective in militarised use. This duality pervades public opinion. Scholarship raises questions around the perceptions, and assumptions associated with the military origin of drones, that must be addressed, when employing the technology to increase access and participation from communities.

2.1.2 Public perceptions of drones

Engaging audiences is at the centre of this research. Therefore, understanding how public perception of drones affects their use is essential. Public perceptions, both result from and shape the different uses of drones. They interplay with the successful use of the technology as a facilitation tool, with underserved audiences.

As discussed earlier, drone technology exists within a duality: Hildebrand (2020, p. 26) considers the drone "[…] juggles both utopian dreams and dystopian fears." This Manichean view of drones, illustrated in Figure 1 below, encompasses a dichotomy that reflects power disparities correlated with the underserved communities this project focuses on.

Dystopian. Military drone. Warfare, killing, surveillance gaze. Aesthetics of war/military aesthetics. Drones intrinsically connected to disavowal politics (ethnicity, colonialism and new colonialism) Global South.

Utopian. Commercial and play drones. Civil uses (environmental monitoring and conservation, heritage, art, and culture, property, surveying).
Global North.

Power imbalance "Drones are not separable from regimes of power, nor from broader histories of colonial control" (Pong and Richardson, 2024, p 11)

These 2 drone distinctions are geopolitical markers that separate the Global North from the Global South based on normative feelings and conceptions of vulnerability and power roles. (Gaeta, 2024, p 129)

Figure 1 Dichotomy of drone use and its implications on public perceptions and dynamics of power

Surveillance emerges as the reverse coin to new approaches to accessing and collecting data, fears for a new mobile autocracy versus the positive power of autonomous mobility; dereliction of civil liberties and vulnerability to irresponsible or malicious users (Jensen, 2016a, p.4) versus miniaturisation, accessibility and affordability of the technology for a wider range of people (Kim, Kim and Kim, 2016). The capabilities of storing, distributing and receiving data that elicit a sense of vulnerability to hacking are the same competencies, that made the livestream in this project's co-design research study, possible.

The duality of perceptions raises a challenge for governments to respond promptly with appropriate regulatory frameworks (Choi-Fitzpatrick *et al.,* 2009), to bring reassurance to communities over the public anxieties at the root of the psychological resistance (Kim, Kim and Kim, 2016) towards UAVs. Cross-sector collaborative research on how drones "may be (re)shaping and disrupting airspaces and everyday life" (Jackman, 2023) brings together stakeholders from across different industries to discuss with parity, views on drone technology acceptance and regulation.

In summary, the literature shows that drone technology generates a duality that arises because capabilities that elicit negative perceptions are the same that give this technology the aspirational potential to empower communities. Showcasing the possibilities of drone technology to enhance the agency of communities may help change the negative public image of drones.

2.1.3 Key capabilities of drone technology that enhance its civil use

Drone capabilities with the potential to bring positive impacts to communities can highlight the benefits of using the technology. This section discusses how mobility and autonomous access to aerial space lead to empowerment and storytelling, benefitting civil use.

i. Mobility

UAVs' mobility capabilities are at the core of its benign mythology, by which it is meant a utopic view of the drone, as illustrated by Hildebrand (2020, p. 36): "The drone's unique mobile affordances are central to this imagination of future societies and their critical assessment". Mobility enables unique human interactions with air and space. It creates the possibility of control, over space that was previously only accessible to institutions (governments, corporations, the army) and is now available to individuals, extending their reach (Wallace-Wells, 2014). This could mean levelling up power dynamics, not just of the airspace, but of all places where drones navigate, empowering communities with experience of exclusion.

Mobility is a key feature in tourism growth and sustainability (Liu, Yue and Yu , 2022, p. 1). The study of tourism mobility is not limited to arrivals and destinations. It includes "the mobility of people, the mobility of materials, the mobility of ideas (more intangible thoughts and fantasies), and the mobility of technology" (Liu, Yue and Yu , 2022, p. 3). Drones could provide equitable and creative practices to explore digital mobilities, in physical cultural spaces.

COVID-19 brought a marked shift in tourism, due to the "need for technological solutions" (Gretzel *et al.*, 2020, p. 190) that propose new approaches to virtual tourism, reflecting the emergence of mobile lifestyles such as digital nomads (Gretzel *et al.,* 2020, p. 197). Drone technology is one such technological opportunity that, by supporting virtual tourism, can widen access to heritage sites for groups who cannot physically access them.

The mobility capabilities of drones propose a reframing of how societies consider accessing and experiencing places. This happens within a context of agential negotiations between institutions and individuals, which mirror the trend for heritage and cultural organisations to decentre power and involve communities in shaping access to, and narratives of places and spaces.

ii. Autonomous access to aer*ial space leading to civil empowerment*

Literature reflecting on the use of UAVs in civic life focuses on civil empowerment, and reclaimed ownership of community narratives and spaces. Using drones to empower communities could enable underserved communities to reclaim identity narratives and facilitate their ownership of culturally relevant spaces/territories. This informs this project's research activities and is explored further in Chapter Six.

UAVs become a valuable platform to experiment on the framing of civil liberties in increasingly mobile, data-shaped, machine-supported realities. The drone fulfils a similar role to the arts: a speculative tool in (re)imagining past and futures, displacement and placemaking (Tzima *et al.*, 2020, p. 1). Margaritoff (2017, p. 5) agrees: art can often make you think about real-world issues and moral complexities better than a lecture can, and "Donny the Drone[3] has certainly sparked a conversation in my household."

[3] *Donnie the Drone*, 2017. Short film directed by Mackenzie Sheppard about a drone that gains self-awareness.

Drone technology encapsulates the empowering effect of autonomous mobility, arising when individual users perceive themselves as holding a stake in the airspace. This research draws on this perspective to explore how underserved audiences can develop a sense of ownership and agency over space, using new ways of interacting with it afforded by drones. The key for this sense of ownership is to use drone technology to increase the visibility of their narrative.

iii. Storytelling and Narrative Capabilities of drones

Storytelling and narrative are fundamental in Heritage and culture. Interpretation and meaning-making are rooted in the stories that contextualise material culture. Telling a story is not only part of the process of engagement and connection with audiences, but also a tool of representation, knowledge development, and one that engenders a power discourse. This relates to themes of identity and place, and citizen (or civic) agency, as the person who tells the story (or in the case of the camera drone, holds the gaze) controls who is represented and their role in the narrative.

This is also correlated to the concept of drone-mindedness (Hildebrand, 2021, p. 141) which articulates the effect UAVs produce, in how drone users see their surroundings through the eyes of a camera drone. This new way of seeing has implications for how the drone user connects with the environment. It shapes how they make meaning and compose the narratives in their storytelling.

Drone-mindedness opens access to new perspectives and points of view that have been otherwise inaccessible (Kreimer, 2018). UAVs bring new ways of knowing and exploring place, as well as novel forms of experiencing spaces. This relates the technology to other interpretive strategies used by heritage and cultural sectors to explore the human story (Tzima *et al.,* 2020, p. 19) and the potential for its transformation. In art, the drone can become the art object or the performance itself (Brimblecombe, no date), (Augenbraun, 2016). This approach, where the drone becomes the end, not just the means, is radically different to military and civil uses described previously, where the drone is used solely as a tool.

Empowerment arises from the drones' unrestricted movement and legitimised trespassing of the tangible (walls, gates) and intangible (underrepresentation, displacement) barriers through which communities negotiate the connection with a site or a narrative. As Hildebrand states, the figure of the drone promises access, and thus insight into the world, ultimately enhancing his empathetic decision-making (Hildebrand, 2020, p.30).

In *Hidden Britain By Drone* (2016-2018), Tony Robinson uses drones to access buildings of historical, social, and cultural importance, but physically inaccessible to the public for diverse reasons. *Drones in Forbidden Zones* (2015) takes viewers to an abandoned Power Station, a prison, a derelict Maltings. The flights are combined with oral testimonies of people who used, worked, lived near, or played around those spaces.

Although at first sight, the visual experience is shown as disembodied, the synchronicity of the flight path with the reminiscences it elicits connects the experiences with human emotion or memory. In the episode of *Drones in Forbidden Zones*, 'Post Apocalyptic Towers of Power' (2015) the visualisation brought by the mobility of the machine climbing up the inside of a power station tower, whilst we listen to an 85-year-old retelling how they climbed the tower when working at the station, creates an immersive effect, as if we were joining in the acting out of that memory. This creates a temporary switch in positionality.

In summary, UAVs key capabilities of Mobility, Autonomous access to aerial space and Storytelling are central to the benign mythology of the drone. This is as a result of the opportunities these capabilities engender: accessibility, empowered stake over aerial space, leading to new perspectives and narratives. These capabilities are elicited in the data from Heritage and Drone Experts, and they are further explored in Chapter Five.

2.1.4 Specific uses of drone technology in civil society and ordinary life

Birtchnell and Gibson (2015) propose a shift from researching drones as a subject, to investigating their use, as data gathering tools and innovative methods of research. This is reflected in the increasing use of drones in specific sectors in our society.

iv. Drone use in Higher Education teaching, learning and research

Drone technology is becoming part of teaching, learning, and research methods in Higher Education (Krajník *et al.*, 2011; Kumar and Michael, 2012; Birtchnell and Gibson, 2015). This is relevant, as Heritage organisations are research institutions and share with higher education a concern for accuracy, stewardship, and authenticity. Public engagement work in heritage and cultural organisations, in my professional experience, is generally delivered by Education or Learning Departments which cater for schools, universities, lifelong learning, and community groups.

Literature indicates that research practice has been following the direction of travel of the boomerang effect (Jensen, 2016b) expanding from the study of drones from warfare and surveillance applications, to their use as a method for data capture and visualisation in other areas of societal activity. The drone's mobility, autonomy and verticality (Serafinelli and O'Hagan, 2021, para. 4) are at the centre of this change. These capabilities offer access to data. Other tools may not be able to provide the same minimal level of intervention, simplicity of systems, integration of different types of data, or visualisation modes.

Overall, the review of the literature on drone use in Higher Education teaching, learning, and research shifts away from considering the technology as a subject, towards recognising its value as research tool. The potential of drones to facilitate innovative approaches to research and mobile methodologies relates to its potential as a facilitator of engagement, which is explored in later chapters.

v. Drone use for the Environment and natural resources

Drone technology is used in the management of environmental and natural resources, (using terminology from Hackett and Dissanayake, 2014) including the involvement of communities. This is pertinent here, as this project's research partners, the National Trust and the South Downs National Park, are agencies responsible for the management and stewardship of natural heritage.

Research in this field considers drone use in community-led forest monitoring (Paneque-Gálvez *et al.*, 2017) wildlife conservation (Wich and Koh, 2018) and monitoring (Fust and Loos, 2020)[4]. Surveying, monitoring, and surveillance appear to be the most common uses of drones in this context, similar to findings in other academic literature.

[4] At the time the work by Fust and Loos was published, in 2020, there were still limited legal framework and protocols to set out the use of the technology.

The key identified limitations on the use of drones in this field include short battery life, training requirements (pertaining to community led use), legal regulations and obligations prior to a flight. These limitations are also identified by heritage experts and discussed in Chapter Four). Fust and Loos (2020) posit the most important challenge to using drones for wildlife monitoring is the pursuit of UAVs as total replacements for other tools and methodologies. The application of any methodology is dependent on the situation and needs or requirements of the task (Lisein *et al.*, 2013).

As in Birchnell and Gibbons (2015) and Paneque-Gálvez *et al. (*2017, p. 1494), this project focuses particularly on non-expert users such as community groups and the investigation of approaches for capacity building in non-expert users' application of UAVs. Purposeful use is identified by Heritage Experts, in Chapter Four, as essential to impactful, empowered outcomes. Capacity building, leading to self-efficacy in communities (2014, p. 1486) involves considerations on resources, funding (Paneque-Galvez *et al*., 2017, p. 1483), training, digital literacy and poverty (the latter two, elicited in Chapter Four, as barriers to the use of drone technology to engage underserved communities). Automation (Paneque-Galvez et al., 2017), plays an important role in this, as it mitigates the challenges inherent to access to training and limited or absent digital literacy that may affect excluded communities. Further to this, automation helps bridge the skills gap between technology users and non-users.

The COVID-19 pandemic also demonstrated that using technology and benefitting from its affordances (remote learning, remote working, social interaction, cultural engagement) is relative to socio-economic factors, urban and rural geographies, local politics, and institutions.

Overall, the findings from this research are aligned with the considerations in Birchnell and Gibbons (2015) and Paneque-Gálvez *et al*. (2017) that these challenges are balanced with opportunities: better governance; increased community ownership through decentralisation from government and institutions; upskilling communities the ability to physically access areas normally difficult or impossible to access. In the heritage sector, the latter hold a special appeal to audiences, as these spaces are outside of the public sphere.

Conversely, the literature outlines common challenges to the usability of UAV technology: short battery life which affects the duration of activities; legal regulations; administrative processes prior to a flight. Sustainability is also considered: intense training requirements for non-expert heritage and culture users (staff, audiences), as well as funding required to maintain and repair UAVs. Understandings developed from the drone autotechnography by Hildebrand (2022), and my autotechnographic study in this research show the aforementioned factors affect attitudes to deploying the technology. These considerations informed the planning of this project's research activities.

vi. Drone use by Communities and Civic society

As previously mentioned, communities and civil society increasingly use drones to facilitate civil agency; identity and place; and tourism and leisure. In terms of civil agency, UAVs are used as instruments of advocacy, offering leverage, and making discussions with institutions and governments more disputative. This echoes the views of Hildebrand, of the drone as a tool to negotiate the utopia of access to vertical space, mobility and the fears of ubiquitous surveillance and loss of privacy.

Hildebrand (2020, p. 26) explores drones as an utopian method towards elevated visions for an imaginative reconstitution of society. These are interpreted here, as reimagined propositions for a more equitable society. Literature relating to civil drone use mostly documents the use of the technology by business and governmental organisations and experts. There is evidence of a trend for increased individual and community use as discussed in the previous section in relation to Environmental and natural resources. This context includes activism, urbanism, consumer drones, and the individual versus the collective.

Choi-Fitzpatrick *et al*. (2009) acknowledge users in civil society as non-for-profit groups, private individuals, and users who have not identified themselves and employ the devices [...] for their own purposes. (Choi-Fitzpatrick *et al.,* 2009, p. 8), including personal use. Lindley and Coulton (2015) use design fiction to explore the use of drone technology for collaboration between communities and the state. The resulting data is a combination of what is generated by drone systems, including incidental criminal activity, pilots' reflection logs and debrief sessions.

Similarly to Krajník *et al.* (2011), Lindley and Coulton (2015) consider the employment of UAVs as tools integrated in a hybrid data gathering and monitoring practice. Their design fiction methodology (Lindley and Coulton, 2015, p. 618) hypothesizes the use of the technology by non-experts in a reality with diversity and flexibility of approaches. Their methodology rehearses the benefits of using the technology to suit the needs of local communities and geographies.

Sandvik and Jumbert (2017, p. 15) posit that civilian drone mobilities speak to "the growing sense of civil airspaces as both a public good, and as a space that can be personalized" through autonomous mobility. The co-design research (see Chapter Three) in this project is devised as a personalised experience (for non-experts living with dementia), in a collective setting where all participants have control. The approach in this study draws on elements of gamification where participants make decisions on what they view during a live, remote experience of heritage (facilitated by drone).

Lindley and Colton's (2015) gamified[5] methods offer an opportunity to make using drone technology a more relatable and recognisable experience. This method enhances the process of democratisation amongst non-expert communities. Personalisation, mobility, and gamification interplay in increasing the engagement of underrepresented communities with place, heritage, and culture. Mateus *et al*. (2024) consider that gamification promotes a sense of belonging and supports user engagement because it is centred on people making decisions, which, in turn, empowers and fosters ownership. As such, gamification can make the use and presentation of drone technology more inclusive. This is particularly important for underserved audiences, whose experiences of exclusion lead to feelings of not belonging.

Second, drones can be a tool for activism. Paneque-Gálvez *et al. (*2017) explore methodologies to engage, involve, and support communities to use drone technology in a process of grassroots innovation. In this process, ownership of representations of space/territories offers validation of sense of place. Research by Gálvez *et al.* (2017) on Indigenous people's use of drone technologies to reclaim territories from colonialist legacies exemplifies this. Drones become a valuable platform to experiment with the framing of civil

[5] Deterding *et al.* (2011, p. 15) define gamification as the use of game design techniques to encourage and involve individuals in participating in experiences that are not a game.

liberties, in increasingly mobile, data shaped, machine-supported realities. This is relevant to this research as it considers Global Majority narratives.

Drones, as devices that move within places, are locational media. They operate within Geographical Positioning Systems (GPS), and are used to map, document, measure, visualise place. They connect the user with place, which is a central aspect of belonging, offering a theatre-like arena where new identities, experiences, and realities are reconfigured.

Third, drones are increasingly used in tourism. According to Huang *et al*. (2016) the industry has been using 3D virtual technology as an information imparting tool. This is as part of promoting destinations and communicating with target audiences, through engaging them with information-rich virtualized environments. Mirk and Hlavacs (2015) place drone technology as the evolved version of virtual reality tourism, which they designate as Live Virtual Tourism (Mirk and Hlavacs, 2015, p. 46). This is a mixed livestreaming and virtual reality experience, whereby tourists remotely visit a site, in an experience facilitated through UAVs combined with virtual reality.

Martins *et al. (*2017) posit that digital technologies enable consumers to make personalized choices of products, allowing the tourism sector to manage and showcase their offer more efficiently, and the public to be more actively involved in product marketing (through reviews and ratings and, I would add social media content). Investigating the application of UAV technology in architectural heritage, Martins *et al*. (2017) used drones to record data on inspection of the conditions of building, to capture structural and aesthetic detail, unseen, and unpublished views, and survey the interior of buildings.

This data was used beyond its surveying and conservation aims, as the basis for engagement strategies. For instance, it was used to create a virtual presence for the buildings captured and in the development of applications for mobile devices. Gamification and the use of drone products to create Live Virtual Tourism show the opportunities associated with maximising the uses of data collected by drones, beyond functional purposes.

In summary, the literature shows that applications of drone technology in civil society have expanded from businesses, institutions, and the realm of experts to wider communities and individuals. Drone capabilities are used to reframe civil liberties, and challenge current socioeconomic and political architectures of oppression. Literature demonstrates that drone technology harnesses a wide range of possibilities, and opportunities, for community groups and individuals to take ownership of how they position themselves within place, and what that means for making their narrative visible.

Existing research where the technology facilitates access and use by underserved, non-expert users, aligns with the audiences collaborating in this research project. Gamification emerges as a facilitator of decision-making, increasing inclusion and agency of non-expert users. This provides this project with a framework for an approach, which is applied in the co-design research (see Chapter Three).

vii. Drone use in the arts

A consideration of artistic practice that illustrates Hildebrand's (2020) categorisation of the good and bad drone shows that many art works engage with drones, from this perspective. Some artists appear to use drones as a method to problematise the challenges around loss of privacy, disempowerment, omnipresence, and surveillance cultures, such as: *Dronescapes* (Art @ Kathryn Brimblecombe-Fox, no date); or Basus' (2013) *Drones of New York* a web-based collaborative project where artists reimagine armed drones redesigned in colourful patterns, flying the streets of New York and inside museums. Others take advantage of the potential drone mobility offers and propose them as new ways of exploring tridimensionality by combining them with music, light, and space (Augenbraun, 2016). Others yet use them to explore body and movement (ELEVENPLAY x Rhizomatiks Research, 2013) and as tools for artistic empowerment that enable creatives to step through the existing thresholds of human expression.

Drones have also been used in the cinema industry for some time (Crickmar, 2017) to convey the magnitude of human experiences (Weiwei, 2017)[6], to create intimacy and immersion as well as panoramic settings as in Scorsese's *The Wolf of Wall Street*[7] and Lanthimos'[8] *The*

[6] In their film *Human Flow* (*2017), Weiwei shows* the global scale of the refugee crisis, while bringing it closer to the viewers' physical and emotional space.
[7] 2013
[8] 2017

Killing of a Sacred Deer. Drones are present in the imaginations of artists (Pechman, 2013) predominantly in their political context, which is embodied in the anti-drone hoodies, scarves, burkas blocking thermal imaging, designed by artists and on sale in the shop of the New Museum, in New York. Another trend reframes the drone as an art object in itself: for example, *Drone 100 A Magical Experience* (Augenbraun, 2016) creates tri-dimensional light sculptures through flying 100 UAV swarms.

Smethurst and Craps' (2019) article discusses drone fiction, including literature and film. Their subject fiction shows an exploration of drone technology from the point of view of diverse people, including war-stricken populations, drone pilots, drone surveillance subjects, and victims. The drone is the narrative and the narrator. This interrogates the machine-human interaction, and the drone's prosthetic extension of human physicality through embodied sensing. Jensen (2016a, p. 7) refers to Superflux Labs' art projects, where the artwork elicits discussions of airspace ownership, built environment and reframing a geography and politics of space around verticality (Sandvik and Jumbert, 2017).

In conclusion, drones clearly have a presence in artistic production. As in other areas of research, ownership, decentralisation, and citizen empowerment are at the centre of artistic narratives for possible drone futures: drones are used as a method to problematise surveillance, ubiquity, and loss of privacy; to interrogate human machine interaction and explore tri-dimensional space. They explore renegotiations of power through creative approaches to reconfigurations of human geographies. This is considered further in the co-design research methodology (see Chapter Three).

Section conclusion

The examples of drone use in civil society result from its capabilities of mobility, autonomous access to aerial space by communities, and storytelling evidence. Mobility, autonomy, and the capability of capturing data to reconfigure narratives of place are increasingly used across different sectors of civil society, from experts and institutions to non-expert communities and individuals. UAV uses by community groups with experience of exclusion and oppression is of relevance: they relate to practices for challenging hierarchies that disempower these communities. This is a key aim in this research, as the use of drone technology is tested as an instrument of agency for groups who are silenced and left out from heritage narratives, transposing these considerations into the field of heritage and culture.

The widening use of drones in civil society positions UAV technology in a space between two worlds: on the one hand, a tool of hyper surveillance in the dystopian reimagining of a stark future of curtailed privacy and security, and on the other hand, a flagship for the empowerment of individuals to take ownership of identity through space and place. In this context, drones emerge as offering (new) epistemologies of aerial space, and ethnographic possibilities in bringing together the machine, the environment, and humans (Fish, 2020, p. 249). This is the sphere this project falls within.

2.2 Museums' role and the purpose of Heritage and Cultural Organisations in society

Here, the role of museums is considered as an analogy for the place of heritage and cultural organisations in society. This section includes a reflection on the role of museums in society; the values of social inclusion and relevance; concepts of engagement and participation; the use of digital technologies for audience engagement; and the impact of COVID-19. It considers how the purpose of heritage and cultural institutions underpins this research. It continues to make considerations on how these organisations deliver engagement and participation, and the opportunities digital technologies offer to engage audiences. The methods and research activities in this research are informed by these discussions. Finally, the section refers to the shifts brough by COVID-19 in how heritage organisations position themselves towards underserved audiences.

2.2.1 The role of museums in society

Museums originate from people's desire to show and share their collections with the public: they were born from the need to communicate to others. They grow from private cabinets of curiosity into public institutions in the late seventeenth century, with the founding of University Museums where collections are used for teaching and research (Alexander, 1979). Their expansion from private collections' showcases, to formal public spaces to inform and educate is driven by illuminist humanism and sustained by the democratising effects of the Industrial Revolution.

As museums transform, interpretation becomes not only an opportunity to assert the significance of the collector and their contribution to knowledge amongst their peers, but also necessary to the financial survival of organisations now accessible to (and dependent on) the paying public. Interpretation fulfils museums' function of education, dissemination, and research: their "[…] epistemological identity [is] constructed through a range of collection-related disciplines (art, natural history, geology, archaeology, ethnography)." (Hooper-Greenhill, 2000, p. 14).

As European museums, world fairs, and international Great Exhibitions celebrate the disappearing pre-industrial world, and narratives of innovation and technical progress, they also affirm socioeconomic and cultural hierarchies. Heritage and cultural institutions reiterate a narrative of western cultural superiority (Morrell, 2008) based on contemporary ideas of cultural history, and class-based socioeconomic dominance. These guide the strategies of interpretation, communication and object showcase used to capture visitors' attention and interest (Alexander, 1979).

These strategies project collections and institutions as subjects of reverence, with an intrinsic and unequalled value. This is conceptualised in Benjamin's concept of aura (2007, p. 222) central to this thesis. Since their inception as private collections, museums, and heritage collections have been built around the idea of authority rooted on guardianship and gatekeeping of assets, with an unsurpassable value and which institutions construe as having universal meaning.

Surveying the UK's political philosophies and policies between the 1970s and 1990s, Sandell (1998) discusses the conservative economic rationalism of the period, characterised by a reduction of state funding, focus on the economic value of cultural institutions and their potential to drive regeneration, an emphasis on visitor numbers, and increasing pressure for museums to be more accountable and more efficient financial managers. According to Sandell (1998) financial pressures lead some museums to adopt admission charges which, in turn, means that those at more disadvantage, in poverty or deprivation, become excluded from accessing heritage and culture.

Sandell (1998) explores the differences between the concepts of poverty and exclusion. Poverty, they explain in *Museums as agents of social inclusion*, refers to scarcity or inequality in the access to resources. By contrast, social exclusion is a broader, multi-layered concept, including deprivation, poverty, mostly referring to how individuals relate to institutions, family, community, and civic life, and how these hinder their participation in different activities.

Social exclusion becomes a governmental priority for New Labour in the late 1990s (Kawashima, 2006), with the Department for Culture, Media and Sport (DCMS) and a Policy Action Team focusing on how the DCMS portfolio could help tackle the poverty, skills, and training gap. Simultaneously, Britain also witnesses a rise in advocacy and pressure from communities, for equality of access to museums. Cultural practitioners are now expected to develop their audiences through programmes supporting social justice aims. A new museology emerges (McCall and Gray, 2014). The relationship between museums and their audiences is reframed, as is the museum profession, in the change of focus from the object to narratives and voices.

In 2000, the DCMS (2000) publishes policy guidance on social inclusion for DCMS-funded and local authority museums, galleries and archives in England, where social exclusion is highlighted as one of the government's top priorities. Heritage organisations, as stewards of identity and sense of place, have a central role to play in addressing it. Contemporary socio-economic changes in civic society precipitate the questioning of the purpose of museums (Hooper-Greenhill, 2000). What is their contribution and relationship with the socioeconomic structures of society, beyond preserving and caring for material culture? What is the social and economic return of state investment in museums?

The State Secretary's statement (DCMS, 2000) identifies museums, galleries, and archives as agents of social change which, through outreach, can contribute to tackling exclusion. Hooper-Greenhill (2000) argues the myth of the art museum and the values that underwrite it start to change, and lead to a shift in the politics of knowledge between museums and audiences. For the former, communication is no longer centred on transmitting information, and the latter are no longer viewed as passive receptacles.

The concept of museum communication evolves to an exchange, a dynamic interaction and a different power dynamic. The idea of the museum, as an authority with a purpose to civilise the masses, regulating their behaviour and values through exposure to the dominant narrative of the elite (Hooper-Greenhill, 2000; Toth *et al.*, 2016, p. 201) gives place to a new relationship, where audiences are active and their identities differentiable, as are narratives in the museum.

Contrary to modern nineteenth century practices, post-modern museums are challenged to recognise different voices, narratives, meanings (Hooper-Greenhill, 2000). The message of museums is no longer neutral, factual, always tangible, or unchallengeable. Concurrently, the museum sector experiences financial pressures (Hooper-Greenhill, 2000) that result in a drive to increase visitor numbers. In addition to lack of clarity on what separates audience development and social inclusion, in the 1990s, there is very little inquiry and reflection on what it means for museums. What is expected of museum (and cultural) managers? How are they to reframe themselves as organisations, and their relationship with their audiences?

At the time of Hooper-Greenhill's essay (2000), these constructs involve a shift in museum professions to embrace the post-modern museum paradigm. This epistemological change and transformation of sector practices towards representation, decentring and decolonisation remains a challenge (McCall and Gray, 2014) in terms of how it manifests in organisational structures and what it means to the politics of knowledge and its hierarchies. Between then and now, there remains progress to be made.

The literature shows that museums, heritage sites and collections originate from a context of making a direct association between socioeconomic hierarchies and cultural control and authority. Heritage practices (interpretation, marketing, education, recruitment) have been used to nurture ideas of reverence for the historic or art object. Systems communicate collections and sites as auratic (Benjamin, 2007); transcendent, spectacular (Debord, 1994) and universal manifestations of human experiences.

Authentically, these are true only for the dominant group and not the diversity of British society. Museum collections reflect a legacy of power imbalances and politics of disavowal, which heritage and cultural organisations continue, at present, to address to become more relevant and representative of different communities and their narratives. In 2020, following George Floyd's death by the hands of American law enforcement officers, the world witnesses civil unrest and protests around the world.

These events symbolically involve toppling and defacing public monuments that celebrate historical figures with a connection to slavery and the slave trade. They heighten global conversations on racism, colonialist narratives and material culture, the underrepresentation of the voices of people from African, Caribbean, and other ethnic minorities in cultural institutions, as audiences, in leadership, and as workforce. This is the context within which this investigation is situated; it is also the root of its focus.

2.2.2 Social inclusion and relevance

In 2017, Nielsen states that "[...] exhibition, education, or interpretation and a commitment to community have grown to be important aims for the museum in the last century […]" (2017, p. 10). Relevance and connection are a crucial element in the engagement of communities, particularly those left out. To be relevant, museums should adopt popular education strategies, exploring critically events that have affected people's lives, unravelling hidden histories, and offering communities opportunities to "[...] learn, contribute and take action [...]" (Toth *et al*., 2016, p. 212). Toth *et al*. (2016) consider that the critical exploration of events that have affected people's lives is sufficient for heritage content to be relevant. The key issue is, rather, whose people are these stories about? Who is constructing meaning about whom?

Nielsen (2017, p. 442) argues that "The process of meaning-making is the process of making sense of experience, of explaining or interpreting the world to ourselves and others." To be able to interpret it to ourselves we need to understand the context, and connect to the narrative, both as individuals and as part of an "interpretive community" (Hooper-Greenhill, 2000). Meaning-making extends beyond cultural references and ethnicity into socioeconomic contexts, locality, psychosocial and experiential factors, the everyday.

Meaning-making is an internal individual process, as well as part of a collective subconscious of shared experiences (Fish, 1980 cited in Hooper-Greenhill, 2000, p. 26). Fish also notes that interpretive communities are fluid, and that individuals may fluctuate from one to another. They discuss how, if museums communicate from within their own interpretive strategies, their message will be relatable to those who are part of the internal organisation and those will be the only voices in the museum. Therefore, community engagement is crucial to the continued relevance of museums and other cultural and historic sites (Stroja, 2018). Collaboration and involvement of different interpretive communities ensures heritage organisations remain aware and open to becoming more representative and inclusive.

Kawashima (2006) defines Outreach as a method with an exclusively social purpose, while positioning three other types of audience development as: centrally financial (extended marketing); artistic, financial, and educational (taste cultivation); and financial and educational (audience education). This differentiation between the social purposes of audience work, and the financially led public engagement activities also takes place in historic sites, cultural and art venues as they go beyond a strictly collection-based focus to respond to societal issues.

This shift also takes place in heritage, although at a slower speed. Archives, for instance, deliver Learning and Outreach programmes and engage school groups. Exploring the role of museums in tackling social exclusion, Sandell (1998) identifies a crucial aspect of the relational dynamics between museums and audiences: museums have, themselves, represented institutional exclusion through their systems and operational models.

Black Lives Matter advocacy has recently signposted this as a continuing challenge for the cultural sector. For instance, omitting the stories of Black British and other minorities in the interpretation of the history of a place, building, object, or public art, reiterates what can be referred to as the disempowering effect of silence and omission. To combat this, more than creating opportunities for people to learn, it is necessary to create the conditions for underserved communities to make a positive contribution and have agency. This is how the idea of whether drone technology can facilitate agency and empowerment of underrepresented groups is explored in this project.

2.2.3 Engagement and participation

Engagement is a complex and holistic practice (Stroja, 2018). It involves diverse levels of museum operations and staff, from senior leadership to front of house, volunteers, and the groups and networks whose participation is being nurtured and supported. Audience engagement takes place through sustained collaboration with communities, beyond one-off projects or consultation exercises. It means involving groups in museum work, developing thinking partnerships to support empowerment, fostering agency in groups. Methods such as collaborative design are useful here, as they offer an opportunity for communities to actively take part "[...] in the creation and transformation of culture." (Jenkinson, 1994, p. 51 cited in Sandell, 1998, p. 402) and in representing themselves within it.

Ben-Eliyahu *et al.* (2018, p. 87) see engagement "[...] conceptualized as processes that indicate productive participation in learning activities [...]." They find, in their study on schoolchildren engagement with learning activities in school and in a museum, that "in the museum exhibits […], self-efficacy was positively related to overall engagement and performance-approach goal orientations were positively related to behavioural-cognitive engagement." They note that engagement is multifaceted. It involves motivation, emotional connection, cognition, and behaviour; each of these elements relate to each other differently in different environments.

Ben-Eliyahu *et al.* (2018) conclude that, in an informal learning environment such as a museum, where learners have autonomy and anticipate a unique experience, engagement happens because learners perceive positively their ability to handle the information they will be interacting with: they have favourable expectations of their experience and what they will achieve. Therefore, engagement is a response to a context, experience, environment, stimulus. It is not a standalone event, but an output of how the individual (and their interpretive community) perceives the success of their interaction with the object or experience and the outcome of that interaction to be.

This concept of self-efficacy (Gallagher, 2012, p. 314) is a critical consideration in this project's research design and approach, as a determinant of success or failure to involve participants in the research. As illustrated in Figure 2, it is part of the multidimensional process of empowerment as, for individuals and communities to feel empowered, they need to perceive the possibilities of success in the activity they engage in, as realistic (Pinkett and O'Bryant, 2003, p. 193)

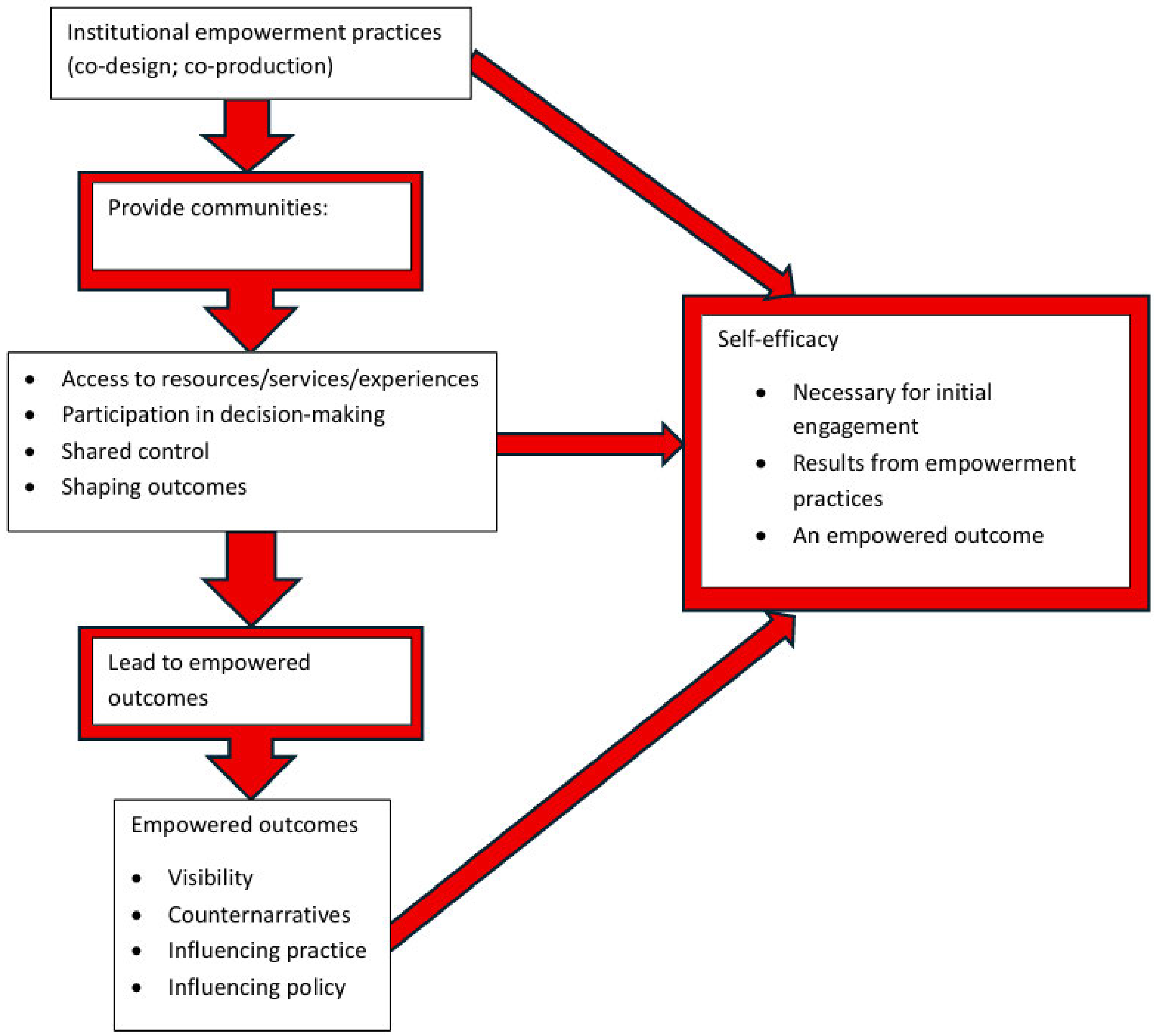


Figure 2 Correlation between empowerment practices, empowerment outcomes and self-efficacy

It means that challenges to the use of drone technology to empower underserved audiences are not limited to negative perceptions of the technology or of the heritage sites. Audience development is centred on the dichotomy between existing audiences and potential future audiences (Kawashima, 2006). The author contrasts audience development with social inclusion, the latter involving cultural institutions working beyond their area of practice to deliver outcomes that are broader than cultural learning.

Nevertheless, the author recognises the validity of Sandell's (1998) approach to the pervasive nature of social exclusion and how it interconnects different areas of civic life and institutions. Cultural organisations are part of this model that perpetuates exclusion. Conversely, McCarthy and Jinnet (2001, cited in Kemp and Poole, 2016, p. 54) consider that audience development concerns practices that appeal to, and meet the needs of, existing and potential new audiences. The authors assert that engagement is an essential component of audience development. In my professional practice I have always considered audience development, within the remit of generic marketing or what Kawashima (2006) categorizes as "extended marketing": reaching people who are interested but have not visited, or people we want to continue to visit (taste cultivation and education segments).

This research focuses on reaching underrepresented communities. The latter is a strategic, tailored, and targeted provision to initiate and sustain relationships with socially excluded communities, using Sandell's (1998) concept. Alexander (1979) and Ben-Eliyahu *et al*. (2018) discuss engagement to tackle the low participation of excluded communities in cultural programmes. To address this, it is essential to consider how cultural institutions compose a traditional top-down, knowledge authority positionality, in which they control whose narratives are valued (Toth *et al.,* 2016).

What this means is that stories of left out communities such as disabled people, LGBTQI+ communities, working class communities, Black, Asian, other ethnic minorities, or women, for example, are excluded, under- or misrepresented in collections. And while their narratives remain invisible, these groups do not have a positive expectation of their interaction with heritage and cultural experiences and are labelled "hard to reach". The expression "hard to reach" communicates the notion that excluded communities are resistant to engaging with heritage and culture.

The challenge for underserved communities is that, as misrepresented or invisible in collections and sites, they have low expectations of experiencing a positive outcome when engaging with heritage. For this to change, heritage and cultural organisations must engage authentically with left out communities. This means adopting practices that go “[...] beyond marketing, accessibility, availability or policy reviews and changes” (Stroja, 2018, p. 304), and embedding them in the culture of the organisation.

These practices should involve close and consistent collaboration with communities, with the aim to facilitate their agency and empowerment, reframing heritage and culture sector approaches into practices where communities make decisions about their own experiences and narratives (Hodges, 2017). This reframed approach is a self-efficacy practice where participants are both consumers and producers. This parallels the dual dimensions of the drone as the narrative and the narrator (see section 2.1 above), as duality fostered by the UAVs’ autonomous capabilities. Those same capabilities harness the power of choice and agency in communities. Agency and power are therefore essential for authentic engagement to happen.

To conclude, increasing participation from underserved communities requires a shift in how heritage and cultural organisations approach engagement. This includes reframing power dynamics between heritage institutions and left-out communities, through collaboration with those communities. This will lead to authentic representation, increase in feelings of belonging and pride, and the expectation of positive experiences of heritage and cultural experiences.

2.2.4 Digital technologies and heritage audience engagement

The shift towards a digital society progresses at a steady pace (MTM London, 2017). The digital mediates how western societies interact with information and manage relationships, from the personal to the political. Naturally, in a sector trying to continuously connect with and engage audiences while (and within that same aim) maximizing knowledge and understanding of collections, the use of digital technologies is growing. For example, Svensson (2016, p. 2) alludes to Jonathan Sterns’ suggestion that “[...] the humanities have a long tradition of engaging with different kinds of materials and on one level, engaging with digital materials is a logical extension of this tradition.”

Kalay, Kvan and Affleck (2007) present digital media as tools that integrate different aspects of heritage to widen access, broaden understanding, and communicate, as well as to enable audiences to manipulate content. They state (2007, p1): "The apparent limitless affordances of digital technologies make them the choice media for the representation, management and dissemination of cultural heritage". As Cameron and Kenderdine (2007) point out, this discourse is rooted in institutional production practices, that shape how the dialogue between the historical/cultural and the digital is framed, and therefore consumed. Digital prompts new definitions of culture, heritage and place.

This research repurposes drone technology, in the context of the sectors' audience engagement practices, to reframe construed ideas of access, production, and participation in culture and heritage. In my opinion, digital technologies can enable cultural venues to replicate how audiences interact with content, to better reach and involve them.

Outside the cultural sphere, digital technologies generally have led the path to democratisation of access to, and participation in different spheres of civic society. Digital technologies are another device, in a longstanding practice of massification of cultural tourism, following the Disneyfied experience (Silberman, 2005). Silberman (2005) relates this phenomenon to the need for the cultural sector to be resilient and sustainable within a context of economic growth, while being engaging and socially purposeful. Kyselaa and Štorková (2015) consider that we benefit from new media (in which they include Augmented Reality) through: its diversity, it being virtual, global, mobile, widely distributed, and collaborative.

Jarrier, Elodie and Bourgeon (2012) conduct a qualitative study to explore museum experiences and evaluate the impact of mediating devices and technologies (including augmented reality) on the behavioural intentions of both visitors and non-visitors to a Fine Art Museum. They conclude mediating technologies enhance emotional engagement, empathy, meaning-making, connection, and contextualisation of self in the narrative. They find the effect of mediating devices on visitor experience correlates with visitor intention for the visit. For instance, visitors to a Fine Art Museum who want to connect and be closer to the artwork, may find the digital mediation disrupts that intention; this may be different in a Science Museum.

This raises the question of "What is gained and what is lost?" (Kalay, Kvan and Affleck, 2007) when using digital technology as a mediation tool, in a museum experience. Could this mean that although, on the surface, digital media seems to be a pinnacle of democratization and widening access, it could be another tool for institutions to impose compositionality? How enabling and participatory are technology mediated experiences?

In their study, Jarrier, Elodie and Bourgeon (2012) report that non-mediated visits elicit both positive and negative feelings, including imagination, inspiration, enjoyment, immersion, disappointment, irritation, fatigue. In digitally mediated experiences, interviewees report that digital devices or technologies support some layers of engagement but not others. Digital mediation reportedly affects the emotional dimension of the experience. A visitor describes: "I don't think I would be more emotional using those devices [...] because when you're in a museum what produces the emotional response are the artworks – it is not seeing them in 3D or virtually." (Jarrier, Elodie and Bourgeon, 2012). In this same study, a non-visitor states they would use mediating technologies, as tools to gather information and acquire knowledge.

The purposeful use of technology as a supporting tool to facilitate interpretation is corroborated by this research's data (see Chapter Five). This means that digital engagement does not always translate as the use of strictly digitally facilitated experiences but can be maximised by combining it with analogue elements (such as those used in this project's co-design research, see Chapter Three). Similarly, the tourism industry (see 2.1.4 above) uses 3D virtual technology as an information-imparting tool, to promote destinations and communicate with target audiences by engaging them with information-rich virtualised environments.

Nevertheless, Ciolfi (2017, para. 4) highlights the importance of sensorial experience, "[…] the sense of being there," as the root of a memorable experience. By implication, experiences that are strictly digital may defeat the purposes of engagement and immersion. Svensson (2016, p. 174) considers that digital technologies, however advanced, are not the answer to all engagement challenges in heritage and culture but an important element of it. They argue that the digital creates a shared space between the traditional and new ways of making; and that digital technology holds an important role in empowering individuals and groups.

This is a valid consideration. Nevertheless, as discussed in Chapter Five, drone technology has important sensorial affordances emerging from its capabilities of mobility, verticality, and remote sensing, which produce cognitive effects that enhance involvement and the sensation of "being there". These affordances, underlined by a live albeit remote experience, have relational and affective implications that make drones a distinctive mediative technology.

Overall, the literature agrees that digital media reframes knowledge and cultural production and empowers individuals to be a part of this process – to have power. For engagement to be meaningful, it must fulfil a purpose; a topic explored further in Chapters Four and Five. In addition, research by Jarrier, Elodie and Bourgeon (2012) shows that emotional connection is a cornerstone of engagement.

Technology may engage visitors with information but, for a holistic experience, the physical presence of objects is essential. But what happens when physical access is not possible, and global communities lose connection to the collective and even to the local? The post-pandemic context, in which this research was undertaken, brought this insight to audience engagement practices and the use of digital technologies.

2.2.5 Covid 19: the global impossibility of access

COVID-19 impacted the cultural sector, leading it to adopt technology in varied ways to respond to the global challenge of the pandemic. This reconfigures engagement and participation and frames this investigation. During COVID-19, digital technologies were the only possible way to access heritage and culture (among other services). What models will museums adopt after the pandemic? This research took place within this context and legacy.

Yan (2020) discusses the extraordinary shift brought by COVID-19. Western societies went from offline interactions, predominantly taking place in the physical world (socialising, elements of healthcare, entertainment, education), to interactions mediated by digital and online platforms. COVID forced providers of cultural experiences to reframe their delivery and develop creative ways of becoming accessible, not because they were fulfilling their commitment to social inclusion but because not reaching out, not being accessible, meant not being sustainable.

The scale of the pandemic meant that it was no longer a minority who was excluded by organisations who did not reach out: it was all their existing audiences and more. Consequently, the social justice aims of the heritage and culture sectors became diluted in a complex agenda involving survival and resilience-building, to remain present and relevant beyond the stewardship of cultural narratives.

COVID brought the culture sector to reflect, review, and transform the delivery of their service. It created an "increased reliance on digital technologies to enable some continuity of social and working lives." (Pratt, 2020, p. 1). If ever there was a time when the culture sector could make a difference to people's lives, harshly impacted on as they were by enforced, extended, and generalised lack of access, impossibility to take part and create, it was during a global pandemic lockdown. As an acute response to an extraordinary situation, museums and cultural organisations started enacting their social role with more visibility and to more audiences than before. "It was evident that those working in the creative sector were providing what many viewed as an essential service, almost as important as those provided by the health care, the food production and food retail industries." (Jeannotte, 2021, p. 1).

The cultural sector was among the most creative in using digital channels to reach audiences Jeannotte (2021). Cervinkova (2020) explores the kind of social experiences virtual access to art may elicit. They looked specifically at the example of a Russian Facebook group with 500,000 members where, daily, individuals selected and recreated a painting using mundane objects at hand. The recreated image was uploaded to the group page with the original artwork. This approach, which meant "the emergence of a new mass form of experiencing visual art and a nascent form of sociality" (Cervinkova, 2020, p. 1) was a response from museums to tackle COVID- associated isolation.

During the pandemic, "having weathered a global storm, this industry [culture] found stability in the one thing that digital technology can't replicate: a communal live experience among other human beings from different backgrounds" (Harrison, 2019 cited in Jeannotte, 2021, p. 4). Although it is a possibility that digital experiences from cultural organisations added to social capital, by bringing people from around the world together, whilst they were isolated and affected by a global and common challenge, the overall stance of Harrison and Jeannotte elicits doubt.

The "communal live experience" was mediated through digital technologies; and although the world may not have previously relied on them before as their only channel for interaction (learning, work, and social engagement), global livestreaming performances and virtual robot tours existed in the cultural offer before COVID-19. The difference is that they were not delivered with the central aim of recreating, and promoting the experience of social contact, which is indeed unique to the pandemic.

Pratt (2020) argued at the time, that cultural producers would require further resilience than they had previously developed, because cultural ecosystems were severely damaged by the pandemic at a global level. Recovery and rebuilding of the sectors' economy post-pandemic, would lead to the need "[...] to address the systemic problems of precarity and injustice in the urban cultural economy and (hopefully) to change it." (Pratt, 2020, p. 1).

Section conclusion

The literature on the role of museums in society, audience engagement, and social inclusion shows that the uses of drone technology in heritage and culture institutions are varied. As they evolved from their original roles, 21st century museums, heritage, and cultural organisations are working towards increasing relevance to, inclusion, and representation of communities that have been underserved so far.

The incursion of UAVs from military into civic life and heritage practices may not be as contrasting as it first appears. Heritage organisations and cultural institutions have long been stages of re-enactment of power distribution and dynamic in their societies. More recently, the Black Lives Matter movement has echoed the oppression and silencing of Black and Brown communities and, even more strongly than before, how communities want to have a voice and, with this, correct their underrepresentation.

Reflecting on this literature, there is potential for UAVs to introduce a new angle from which to access and therefore engage with both material and intangible culture.
Verticality and mobility create the opportunity for new enquiry, challenging and (re)discovering stories beyond the reiteration of the established narratives. Empowerment, ownership, and reclamation of cultural representation are also the values in current heritage

practice, as evident from the *Museums Change Lives* strategy (Museums Association, no date) (which includes a Power to the People framework), and The National Archives' *Archives for Everyone* (2023) strategy, for example.

Here, drones are brought into a conversation that is not without conflict, but where technology can assist with negotiation of power, by broadening access to civic aerial spaces from corporate bodies to communities and individuals. What is meant by negotiation is a dialogue that encompasses the recognition of communities as legitimate stakeholders and users of aerial spaces.

When working towards involving communities, it is crucial to consider that engagement and participation are not the result of a linear process. People's experiences of exclusion affect more than their socioeconomic factors. They impact on how people relate to their communities and the expectations they have of their interaction with heritage and cultural institutions. This is an important factor to integrate into research activities with underserved groups (see Chapters Three and Five).

2.3 Underserved audiences in heritage and cultural engagement

This section focuses on the concept of underserved audiences, and its context in heritage and cultural engagement practice. First, it considers the term underserved, and what it means in the context of the relationship between communities and institutions. This includes a discussion on how the term is used by Heritage partners and how it relates to the legal framework of the (*Equality Act, 2010*). This is followed by a review of the term "exclusion" and how it sits in the context of construed categorisations of groups within processes of unbalancing power.

The above discussions are correlated with terminology used to articulate the idea of groups that are not represented in heritage and cultural institutions. Considerations are made on the term "hard to reach" and how it reflects negative institutional bias, terminology shifts to "underserved audiences" and equivalent phrases. This is supported by a discussion on how heritage and cultural organisations approach equity and concludes with reflections on the key themes elicited by the literature and their relevance to this research.

2.3.1 The term underserved communities

In their strategic documents, both the South Downs National Park Authority and the National Trust either specify the protected characteristics of the audiences they consider a priority, or they use terms such as underrepresented groups (SDNPA, 2018). The term "underserved audiences" is used for the first time in this project, by a Heritage Expert referring to the "audiences and communities that traditionally have been underserved" by their organisation, recognising the role institutional systems and processes play in exclusion.

Both Heritage organisations and academic literature typically use the term underserved audiences in reference to one or more characteristics protected under the *Equality Act* (2010). This Act specifically protects a range of characteristics[9] because there is evidence that groups with these characteristics experience significant level of exclusion and discrimination in accessing employment, education, health, and other goods and services (University of Dundee, 2022).

Access to socioeconomic resources is not currently protected by law but it is also a source of exclusion. Importantly, these characteristics do not work in isolation. Identity is complex and multilayered, and the Fawcett Society campaigns for the recognition of intersectionality and the role it plays in experiences of multiple discrimination (Fawcett Society, 2022) also important for underserved audiences.

Although the Equality Act's (2010) categorisation of groups is socially construed (that is, categories are artificial and result from historical and social processes), Zúñiga and Adams (2016, p. 104) argue that dominant groups project them and perceive them as real. They are passed on through socialisation. Social and institutional practices work to normalise and rationalise those categorisations, establishing social hierarchies as inevitable (Zúñiga and Adams, 2016, p. 106).

They are operationalised to sustain the hegemony of the dominant group and keep it at an advantage over subordinated groups, whilst purporting the system to be just and devoid of

[9] Namely: age, disability, gender reassignment, maternity and pregnancy, marriage and civil partnership, race, religion, sex, and sexual orientation.

bias. This shows how processes, systems, and institutions in society are organised to ensure, on appearance, that they function for everyone; but they are set up to reinforce and amplify the power of those who are white, younger, able bodied, heteronormative, neurotypical, male, and middle or upper class.

Systems in civic society, such as education, healthcare, democratic services, or public safety, are paid for by taxpayers' money and are part of the public service offer, (Jenkinson, 2005, p. 51). If these systems and institutions do not serve everyone, then it is coherent to refer to those they put at disadvantage, as underserved. The concept of "underserved audiences" thus implies a failure by systems and institutions to offer equitable access to the societal mechanisms at the base of a functioning society. It relates inherently to the service associated with cultural organisations (Alexander, Alexander and Decker, 2017, p. 20).

Kraehe and Acuff (2013, p. 295) argue that the concept of underserved audiences is undertheorized. Language can deepen or stifle exclusion so how does the term underserved relate to other terminology? I have in my professional practice found and used the term underserved interchangeably with overlooked, marginalised, minoritized, excluded, left behind, with experience of exclusion, disadvantaged. I emphasise that all these terms are reductive and homogenizing in terms of the diversity and complexities of the experiences of exclusion they refer to, in line with Kraehe and Acuff's findings (2013, p. 296).

Equally, it is perceived that this variety of terms indicates that tackling exclusion and reaching those who are "left out" (Humm, Schrögel and Leßmöllmann, 2020, p. 164) means changing what organisations understand their role to be in power dynamics, and what corrective action they might take to address damage and harm to underserved communities. Addressing power imbalance between communities and heritage and cultural institutions is key to this investigation.

2.3.2 Exclusion and Oppression

Exclusion is a recurrent term in discussions around underserved audiences, but what does it mean and how does exclusion operate within the notion of underserved? As mentioned, some researchers approach underserved audiences from the perspective of characteristics or social categories. Others do so, from the perspective of the narratives of difference and otherization constructed by institutions through their processes, systems, and unwritten rules (Kraehe and Acuff, 2013, p. 295).

These narratives are pervasive, continuous, normalising, hierarchical, internalised, and continuedly reiterated through institutional forces created to sustain attitudes and patterns of behaviour. Kraehe and Acuff (2013, p. 5) define these as oppression. Oppression is based on a model of dominance by a group that positions itself (and its experiences, models of behaviour, communication, knowledge, and understanding) as the norm, relegating all other life experiences to an unequal access to resources and services.

Memmi (2000, p. 17) identifies four measures for oppression. First, iteration of a principle of difference; ascription of a negative value to those differences; extending the characteristics that form differences to a whole group membership; using those negative valuations to legitimise the exclusion of the group, thereby normalising power imbalance and rationalising inequality. Here, the dominant group apportions blame for disparity to the marginalised groups themselves, representing them as un-deserving: "[...] as less than, inferior, and/or deviant."(Bell, 2022, p. 9). Terminology such as "hard to reach" (Freimuth and Mettger, 1990, p. 232) reinforces this discourse. Within a narrative of self-proclaimed hegemony, dominant groups can define, further disempower, and marginalise nondominant groups, by taking away their agency and self-determination. The identities of individuals from oppressed groups become defined by the majority.

Importantly, exclusion goes beyond material resources (Bhalla and Lapeyre, 1997) and extends to relational aspects, such as how people engage with their community. Therefore, emotions associated with being marginalised intersect with emotions elicited by experiences of material deprivation (Humm, Schrögel and Leßmöllmann, 2020). This informs considerations on the design of research activities involving communities, and the process of engagement and empowerment in this doctoral project (see Chapter Three).

2.3.3 The emotional impact of exclusion and oppression

What is the emotional impact of being left out, and what does this mean for connection and engagement? Watermeyer (2013, p. 152) uses the term "emotional oppression" to describe the interplay of material and emotional barriers in disabled people's experiences of exclusion.

This term is also relevant for other underserved audiences. Humm, Schrögel and Leßmöllmann (2020, p. 164) explore the role of feelings and emotions in left-out communities' engagement and participation in science. In their study (2020, p. 120), emotional responses to exclusion are described as fear, frustration, and insecurity in relation to science communication. Insecurity is identified as an intergenerational emotion passed from parents to children and relates to unfamiliarity (Humm, Schrögel and Leßmöllmann, 2020, p. 170).

In this study, ethnic minorities report feeling out of place and "not fitting in" as an emotional burden when they visit places like a science museum, for instance (Humm, Schrögel and Leßmöllmann, 2020, p. 166). These feelings are linked with earlier damaging experiences of schooling (due to institutionalised exclusionary educational practices), which leads these communities to identify places of learning, including informal and cultural learning environments, as places of failure. Drone technology falls within science and technology and engaging these communities with STEM subjects and science communication presents similar barriers to engaging them with heritage and cultural experiences.

When a disabled person visits a museum that is not physically accessible, that space becomes a space of "emotional work" (Hochschild, 1979, p. 561) where individuals are handling their feelings of exclusion into what can be termed as socially acceptable managed responses. The same extends to the experiences of Global Majority communities visiting a WWII exhibition, for instance, which focuses exclusively on the contributions of white soldiers.

The discomfort, frustration, and feelings of "not fitting in" (Dawson, 2019, p. 91) lead to disaffection. Discomfort is elicited not only in the context of heritage onsite, and outreach facilitated experiences. It is also provoked by the interaction with institutional content, in social media, for instance, that projects an auratic (Benjamin, 2007) narrative. This refers to compositions of heritage as a spectacle (Debord, 1994), with a representational value that is construed as absolute and beyond human comprehension.

This idea of heritage and culture as transcendental (auratic) experiences creates a distance between places, sites, narratives and objects and communities that are not part of the dominant narrative. In this context, communities are not only excluded, but they are also denied representation, and therefore visibility. Through auratic representations, institutions propose the dominant narrative as a homogeneous story, in the expectation that it is adopted equally by those who identify with it, and those who are oppressed by it, silencing the latter.

In this context, the aura not only creates distance between the revered object or site and audiences, but it also creates separation between those who recognise the auratic narrative as their own and those who do not, therefore feel they do not fit in (Dawson, 2019, p. 91). Communities with experience of exclusion navigate this, as they interact with civic institutions (heritage, cultural and broadly across society) and public places. This carries an emotional burden that is damaging to underserved communities' sense of belonging and identity and leads to distrust and disengagement.

2.3.4 Towards equitable community engagement by heritage organisations

Considering feelings associated with exclusion is an important starting point to focus organisational change towards equity. Each underserved community has different needs and experiences distinct challenges. This means that organisations must tailor their approach to facilitate their access and participation in heritage, rather than treating everyone the same.

Equitable heritage practice involves listening to communities; respecting and understanding the intricacies of their identities (Garibay, Lannes and González, 2017, p. 3; Humm, Schrögel and Leßmöllmann, 2020, p. 173); and celebrating "the richness of individuals' lived experiences as assets" (Garibay, Lannes and González, 2017, p. 3). This reference to individuals lived experiences as "assets" reflects the National Trust's own language in terms of images of their sites, which are commercialised.

Hunt, Layton and Prince (2015, p. 1) argue that equitable practice and inclusion are not just an ethical and moral imperative. They note (2015, p. 2) that businesses with the most diverse gender or ethnic minority leadership generated 30% higher returns than other companies of similar size. This shows there is a business case for diversity and inclusion – where it is authentic, people have agency, and they contribute to the direction of the organisation.

The same applies to how heritage and cultural organisations approach equity and inclusion. By developing cultural competency and engaging authentically with a process of decentralisation and decolonisation, by sharing power and valuing other ways of knowing (Garibay, Lannes and González, 2017, p. 5), heritage organisations can create conditions that increase the richness of the narratives in their collections.

Empowering excluded communities towards self-representation is key to this (Garibay, Lannes and González, 2017, p. 5) as is, fostering their ownership of their voice and meaning-making processes. When institutions are responsive to communities, opportunities for developing knowledge, bringing added value to collections and sites are created. This, in turn, makes organisations more relevant to a wider range of groups, who will be more open to engage with them, their sites and institutions.

Section conclusion

The literature on underserved audiences demonstrates that exclusion goes beyond the scarcity of material resources. It carries an emotional burden that affects people's wellbeing and relationships with civic institutions and community life. Therefore, when considering underserved audiences, it is important to consider not only the resources groups are left out from, but also the complex emotional aspects that affect these groups' readiness to take part.

Moreover, institutionalised exclusionary educational practices lead to communities experiencing and identifying places of learning, including informal and cultural learning environments, as places of failure. Heritage and cultural organisations must position themselves outside the discourse of knowledge authority to engage with underserved communities. Therefore, heritage and cultural organisations must adopt equitable practices that empower underserved groups, which are based on collaboration.

Chapter conclusion

This review first examines literature on the use of drone technology outside warfare, to understand the use of UAVs in civil society. This informs the planning of the research drone related activities to engage new audiences with cultural experiences. The literature demonstrates that UAVs are used widely across different industries, including heritage and the arts. The scholarship also explains how this use is extending to non-expert users such as community groups. It posits that the legacy of the militarised use of drones continues to affect how they are viewed by society.

Furthermore, the literature shows that drone capabilities constitute both a challenge and an opportunity. This means that the key capabilities that underline their use in conflict, also facilitate empowerment of disadvantaged communities in civil society. In relation to this research, the above indicates that drones not only present opportunities for use to empower communities to connect with heritage but also that they are already deployed by underserved communities to enact their agency and to protect their interests.

This is followed by a discussion on the role of museums in society, to explore how drone technology can help heritage and cultural organisations to engage with underserved audiences and communities in an empowering way. The literature shows that museums (and by extension, heritage and cultural organisations) developed from private collectors' cabinets as symbols for status and wealth. Museums have since worked towards advancing social justice aims, and the 21^{st-} century museum continues to explore how it can reach non-visiting audiences.

The literature highlights that this process of engagement is complex. When successful, it addresses not only exclusion but also its relational impact, as it affects how underserved communities interact with institutions and their expectations of them. In a digital society, people's experiences are predominantly mediated by technology, which is therefore an appropriate engagement tool. Nevertheless, academic writing shows that although digital is a space for place-making and connection, it only enables partial engagement.

Digitally facilitated experiences may have higher impact if delivered in a hybrid combination with analogue activities. The potentials and shortfalls of digital technologies are also highlighted by COVID-19 responses: they make access to heritage and cultural experiences possible, but highlight inequalities brought by digital poverty and digital illiteracy, that are crucial considerations in this project.

Finally, the chapter investigates the concept of underserved audiences, highlighting that in this thesis, communities with experience of exclusion are also referred to as left out, underserved, groups with experience of exclusion or excluded communities. The literature illustrates that the concept of underserved is a fairer terminology for exclusion, rather than hard to reach or non-engaged, as these terms encompass an element of blame to excluded communities.

Scholarship shows that experiences of disparity and exclusion are not just financial, but also relational and therefore emotional. Therefore, for true engagement to happen, approaches to engage these groups must be tailored and specific to their relational and emotional needs.

Overall, academic literature discussed above shows this research is both timely, as drones naturalise into ordinary life, and pertinent within the current strategies of the heritage sector to become more relevant and representative. Their strategic approaches to inclusion involve collaboration with communities; they are underpinned by wider understandings of disparity and inequity; they consider how these understandings impact the process of engagement and subsequent approaches to address it. As current academic writings argue, engagement goes beyond reaching out. It requires agency and a redistribution of power between institutions and communities, which are considered in the following chapters.

Chapter Three: Methodology

Chapter introduction

The methodological approach to this project employed a combination of ethnographic, creative, and participatory qualitative methods to investigate the project's research questions as outlined in the Introduction (Chapter One, page 19). These methods informed by key authors and concepts identified in Table 2 and were rooted in the guiding principles of empowerment, collaboration and agency.

Table 2 Key authors, concepts and ideas that informed research methods

Key authors	Methodological concepts and ideas	How they informed the research
Maxwell (2013)	Iterative research.	Adoption of an iterative approach to the research, as part of a data driven investigation to which an inductive analysis method was applied.
Jenkinson (2005)	Collaborative design as a method that enables communities to own their representation and actively take part in the creation and transformation of culture.	This idea is fundamental to this project's design and decisions in investigating and fostering agency in underserved audiences, in particular. As drone technology is not yet fully accepted and embedded in civil society, and my research explored uses beyond functional practices in heritage it was important to provide opportunities for authentic exploration and creativity in interviews.
Denscombe (2014)	Semi-structured interviews include an element of ownership and power as they create a space for interviewees to make decisions over how they explore the subject and the focus, according to their points of interest and their agenda.	
Denscombe (2014)	Survey as a robust method to use alongside semi-structured interviews.	Online surveys made it possible to gather a wide range of views to be considered alongside the more discursive interviews. As the online survey mirrored the interview questions, they also allowed expansion of data collection from heritage organisations beyond the partners (SDNPA and NT), gathering a wider range of views from organisations with different contexts and sizes. This also helped test the generalisability of interview data.
(Ponto) 2015	Surveys as sampling strategy.	
	Surveys as a suitable method to obtain representative data for a group.	
(Ball) 2019	Online surveys have the benefit of large geographical and industry coverage through email distribution to relevant networks, groups, organisations, who can answer at their own convenience.	

<table>
<tr><th>Key authors</th><th>Methodological concepts and ideas</th><th>How they informed the research</th></tr>
<tr><td rowspan="2">Blumer
(1969)</td><td>Symbolic interactionism: meaning-making as a result of the interactions between humans and their environment.</td><td rowspan="3">Process of meaning-making is relevant to my research, as it focuses on construction of meaning through visual language. The control of/ lack of access to means of production determines visibility and representation or lack thereof.
Whilst investigating the use of drone technology to represent relational and cultural aspects of interaction with natural heritage, considerations were made on how power affected the production of meaning.
To do this in an authentic and respectful way, it was necessary to explore and observe the context being researched, through creative and participatory methods, and to immerse myself in the context, as a member of an underserved community. Hildebrand's (2020) work provided a valid framework of existing academic research to build on, when designing my autotechnographic study.</td></tr>
<tr><td>Importance of empiricism; exploratory studies as direct examination of the empirical social world; of researcher getting inside the worlds of meanings of group researched. Methodology that reflects the respect and understanding of the empirical world.</td></tr>
<tr><td>Hildebrand (2020)</td><td>Autotechnography as a method of research knowledge production.</td></tr>
<tr><td>Springgay and Truman (2018)</td><td>Walking methods as methodologies of enquiry, placemaking, relationship with the environment.</td><td>Using walking as a recognised reflexive method of ethnographic research, as part of immersing myself in the context of the research.</td></tr>
<tr><td>Stankov et al. (2019)</td><td>Effect of drones in video creation and tourism promotion.</td><td rowspan="3">These conceptualisations provide me with a starting point for critical reflection on the examination of power in drone content. They ultimately lead to the production of a new framework for the analysis of drone video content which makes it possible to decode data on visual language and power in drone video content.</td></tr>
<tr><td>Serafinelli and O'Hagan (2022)</td><td>Research on characteristics of drone photography visuals.</td></tr>
<tr><td>Kraft (1987), Ryan and Lenos (2020); Probst, Rothe and Hussmann (2021)</td><td>Media and Film Studies conceptualisations of film camera perspectives and techniques to connect audiences with the narrative.</td></tr>
</table>

Key authors	Methodological Concepts and ideas	How they informed the research
Mademlis *et. al.* (2019)	Taxonomy of UAV cinematographic shot-types.	Identification and use of appropriate terminology when referring to drone shots,

The chapter starts by providing a visual overview of the research design, timeframe, and how the methods link to my research objectives. This is followed by a discussion of the project's three key research stakeholders and a reflection on their stake in the project, their motivations and context.

Next, the chapter presents the data collection methods: qualitative methods (semi-structured interviews, online surveys); participatory methods, including co-design; creative methods, co-mapping and autoethnographic methods (specifically autotechnography,[10] including filmmaking, creative writing and walking practices).

The presentation of each method includes its characteristics, use in academic research, suitability to answer this project's research questions, the aim in using the method and how it is applied in this research. This is followed by methods key challenges and the ethical considerations associated with its use. Finally, the section concludes with a reflection on the methods of data analysis, including the examination of organisational and user generated drone content.

Notably, the methods in this project have not been considered in isolation. The research was designed in what Maxwell (2013, p. 3) defines as an iterative model, where the research activities informed each other.

3.1 Overview of research activities, timeframe, methods, and objectives

Figure 3 is a graphic illustration of the research design, outlining methods and relating the data collecting activities across the Doctoral research timeline, showing how research activities link to objectives and research questions and the resulting data set generated.

[10] An ethnographic study involving the researcher's systematic exploration, analysis, and reflection of their personal experiences with technology - see section 3.5.1 and Glossary.

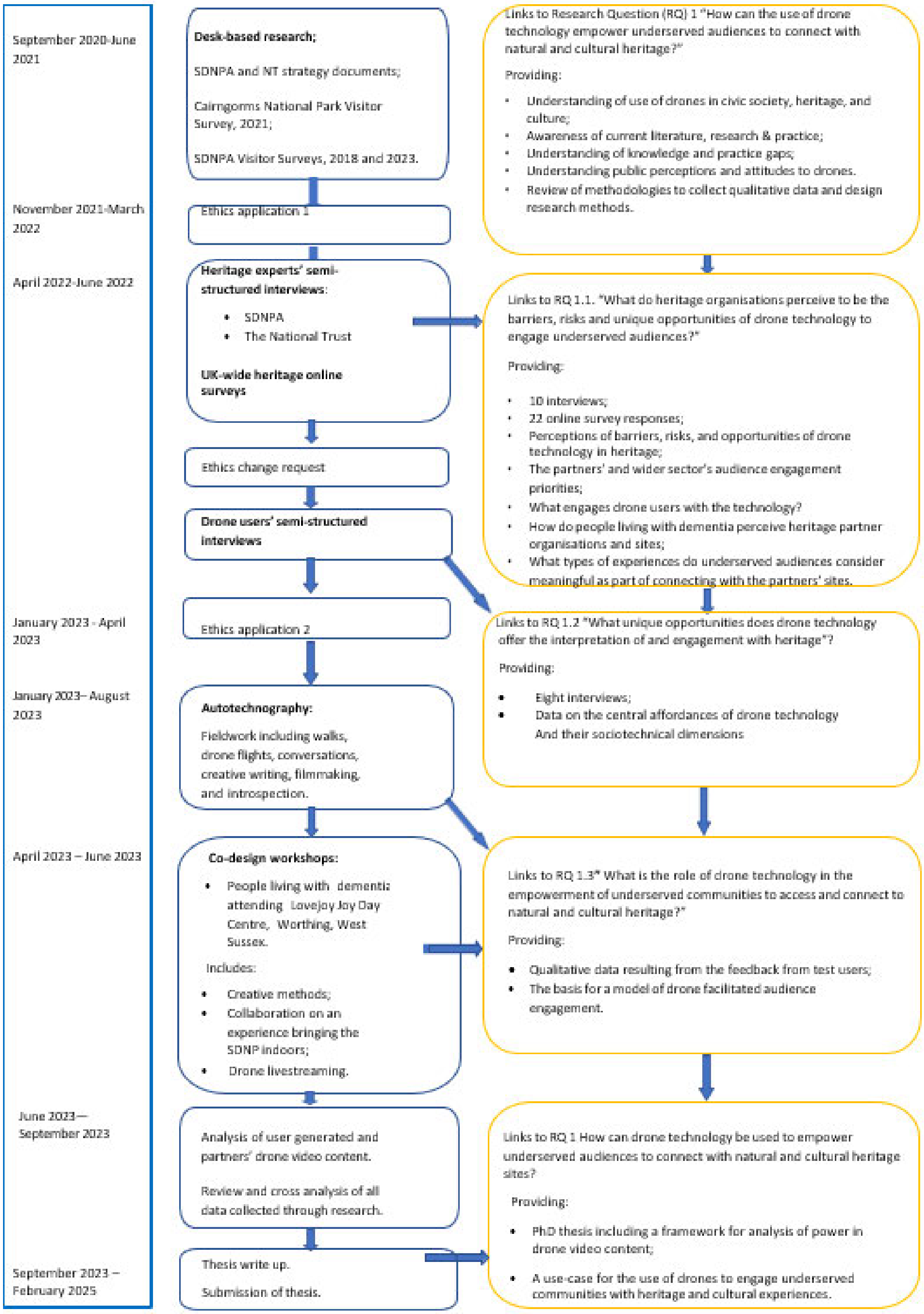


Figure 3 Research design, methods and timeframe

The iterative methodological approach in this project involved the identification of trends in central themes, patterns, and outlying elements in the data. As illustrated by Figure 4 in the next page, this study applied inductive analysis combined with the interpretation of findings from each data set to inform the process of meaning-making and shape the subsequent stage of data collection, refocusing the investigation to respond to its findings. Expert interviews informed the choice of co-research participants and helped define the scope of the co-design workshops. In turn, both sets of data determined the prompts and mind-mapping for the collaborative design workshops, which were bound by the technical opportunities and limitations of the technology and policies of the heritage partners. This integration of methods resulted in a more in- depth understanding of the research problem.

The iteration, continued review of data and refocus of research have led to a shift in the emphasis of the research, from using drones for engagement to empowerment. This redirection highlights the importance of agency in the heritage engagement process, as well as its impact on representation and visibility. Centring empowerment is at the heart of my research decisions including the choice of methods, engagement with the data and autotechnographic explorations.

Central research question: “How can drone technology empower underserved audiences to connect with heritage and culture?”

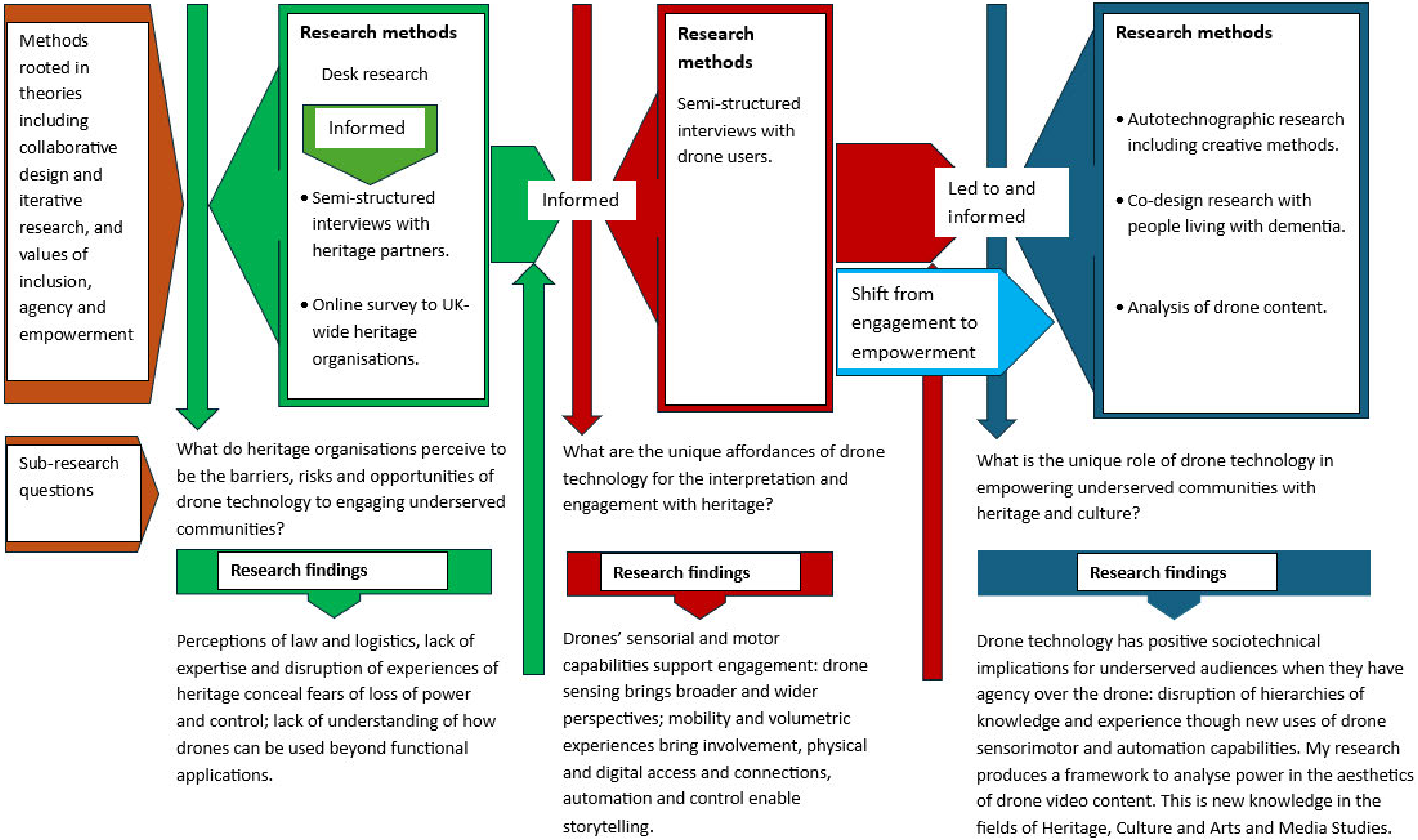


Figure 4 Correlation of methods with research questions and how they integrated leading to research findings

3.2 Project stakeholders

This project has three key stakeholders: the heritage partners and the wider heritage sector; drone technology experts and non-commercial drone users; and overlooked communities (Bolton *et al*, 2019) facing challenges and barriers to accessing the partner sites and assets. Each stakeholder in this research has a different stake and motivations.

Sources of data collection included this project's heritage partners (National Trust and the South Downs National Park Authority); a self-selected group of hobby and professional drone users and a group of people living with dementia. Data was gathered from the latter in the context of co-design research. The next section considers the project stakeholders, their current use of drone technology and how this informed their involvement with the research.

3.2.1 Heritage partners and the heritage sector

Heritage organisations were at the core of this enquiry, as the research was centred on empowering underserved communities' connection *with* heritage. Specifically, the National Trust and the South Down National Park Authority (SDNPA) who were partners in my project. Their engagement policies inform how they engage underserved communities. Generally, heritage and cultural organisations position engagement projects in the context of their efforts to widen their audiences and enhance their visitor experience (Minkiewicz, Bridson and Evans, 2016).

For their audience engagement approach, the National Trust's *Playing our part* (2015) strategy sets out to raise the standard of, and innovate with, the experiences audiences have in their properties, as part of a provision that moves, teaches, and inspires (National Trust, 2015, p. 8). Moreover, the National Trust partnered with the SDNPA and nine other organisations to deliver *Changing Chalk*, a two million pound "landscape-scale partnership connecting nature, people, and heritage on and around the South Downs" (National Trust, no date). This demonstrates both organisations' commitment to diversifying the people who engage and connect with the Downs (National Trust, no date).

The SDNPA's Communications and Engagement Strategy (2020) promotes their aim to be a National Park for all to enjoy and connect with. The strategy encourages new voices, new perspectives, and new ways of doing things.

The National Trust and the SDNPA have both identified similar priority audiences as part of their strategic vision: ethnically diverse communities; urban fringe communities from socioeconomically deprived backgrounds; young people; disabled people. These audiences are underrepresented in their visitor demographics and systematically "left behind" or most deprived (Bolton, Day, and Leach, 2019). In this context, the concept of "left behind," as discussed in Chapter Two, reflects that heritage institutions have underrepresented or omitted narratives and engagement with these communities.

The areas where groups are affected by high socioeconomic deprivation (coastal towns, on the edges of urban areas, in wards on the top 10 % of the Index of Multiple Deprivation) are also marked by experiences of poorer academic, economic and health outcomes, as well as low levels of engagement and participation in public life. This aligns with my reviewed literature's argument (section 2.3) that exclusion from access to resources, and therefore deprivation, is not just a material but also a relational experience of scarcity (Bhalla and Lapeyre, 1997). Overlooked communities are, according to Bolton, Day and Leach (2019), in great need of support to develop confidence and build capacity to make connections with civic assets around their geographical area.

The National Trust's Drone Policy stipulates that drone flying from or over National Trust property is not permitted without licensing (except for Trust staff and contractors meeting the Civil Aviation Authority criteria). An internal National Trust survey on staff attitudes to drone use (Challis, 2022), with a sample size of 143 respondents (where over 57 % hold a role in the Countryside and Rangers job family), shows that drones are used or needed on average once to twice a year.

Vegetation or habitat survey is the most quoted use, followed by general filming and photography. The key emerging themes in the survey's open text comments are access to the technology (internal provision or third party); and how useful/helpful/beneficial it is or has the potential to be. This adds to concerns over deterring drone users from piloting craft from and over National Trust land, as part of safeguarding the integrity of sites and habitats. Issues relating to the safety and privacy of visitors and tenants and protecting the Trust's income generation from image copyright are also elicited as themes.

There is no equivalent report on internal attitudes towards public use of drones in, from, or over The South Downs National Park. The SDNPA has a briefing paper (SDNPA, 2017) and an issues paper, Drone Flying in the South Downs National Park (Alison Farmer Associates, 2018). Both documents highlight SDNPA's concern with the impact of recreational drone use on one of the seven special qualities of the Park, "tranquil and unspoilt spaces" (Alison Farmer Associates, 2018, p. 4). They also recognise the need to balance restrictions, with instances where the technology recognisably meets the purposes of the park, such as the special quality of hosting opportunities for recreational activities and learning experiences (Alison Farmer Associates, 2018).

These considerations shape heritage organisations' interaction with my research. As noted by interviewees (see 3.3.1 below), the National Trust's focus on this research is twofold. First, to understand what opportunities the use of drone technology affords to reaching new audiences. Second, to access examples of how the sector uses drone technology that can be the basis for future exploration and, potentially, decision-making in the long term. Interviews with SDNPA staff, so far, have elicited excitement about the opportunities this research brings to trial novel approaches to engaging underserved audiences with the South Downs National Park, fostering new ways of seeing it and experiencing it.

The engagement of this group involved conducting interviews with a sample of self-selecting experts from project partner organisations to collect qualitative data (see 3.3.1 below). This data was broadened through an online survey capturing perspectives from a wider range of heritage sector organisations. This also tested the generalisability of the heritage partners' interview data (see 3.3.2 and Appendix A).

3.2.2 Drone users

As the recreational and commercial drone use expands, so does the number and diversity of drone users. FPV UK (no date) "The UK association for radio control model and drone flying," showed 1342 pins on its map (accessed on 12 May 2022). These accounted for drone clubs, drone events, proposed clubs, and interested flyers. Empirical observations of drone social media groups and interactions with the drone users interviewed as part of this research showed drone enthusiasts have excellent knowledge of the technology and experience in applying it. Whether as enthusiasts, heritage professionals using the technology in the course of their practice, commercial drone pilots or a combination of any of these,[11] drone users had a significant contribution to make. They were included in the research as part of capturing this groups' insight, interest and knowledge of the technology in the research.

Recruitment of this group was challenging and most drone users who were initially interested in being interviewed following initial contact via social media did not remain engaged after receiving the participant information sheet. One of the potential reasons may have been the fact the National Trust was mentioned as a research partner. There is friction in the institution's relationship with drone users. Given this difficulty, interviewee selection was focused on diversifying recruitment platforms, in order to engage a wider range of drone experts beyond white males.

Although only eight interviews were undertaken, recruitment of interviewees was successful not only in reaching a more ethnically, gender and age-diverse group of interviewees, but also in involving three different types of drone users (heritage sector practitioners and researchers who use drones as part of their role, commercial drone users and hobbyists). Drone user perspectives were integrated with the drone autotechnographic research (see 3.3.1, 3.5 and Appendix C).

3.2.3 Communities underserved by heritage organisations

The focus of this research is to contribute to new approaches to using drone technology, that empower communities with experience of exclusion to connect and develop ownership of heritage and cultural experiences, increasing their access and participation. The decision to involve groups experiencing health inequalities was a result of two key factors. Firstly, the

[11] This is discussed in Chapter Five, in the discussion of the different types of drone users and their attitudes towards UAVs.

disparities uncovered during COVID-19. And secondly, the partners' engagement strategies identification of audiences experiencing health inequalities and disability as priorities.

In 2016, the South Downs National Park commissioned a YouGov Poll, which found that 26% of the 2000 people polled cited "Health issues prevent me" as a key reason for not visiting the SDNP. During COVID-19, these groups were particularly affected by the restrictions of lockdown, which further compounded experiences of exclusion and health inequalities. Not being able to access public spaces equitably creates isolation, impacts on mental wellbeing, sense of place, and identity, compounding experiences of disparity and exclusion. The SDNPA recognises the importance of supporting those who do not access the park by stating "Access for all opportunities" as part of their Learning and Outreach Strategy (SDNPA, 2018). Following COVID, the park also widened its concept of access by adopting the approach to take the Park to people using outreach and digital technologies.

Whilst lockdown restrictions were temporary, for adults who could no longer live independently, the deficit in agency was a familiar experience, particularly for people living with dementia. To recruit participants in this group, recruitment calls were circulated to different services in Sussex. This resulted in the partnership with the Lovejoy Centre.

i. Working with participants from the Lovejoy Centre

For people living with dementia, the diagnosis of memory loss and associated deterioration of cognitive competencies not only affects their sense of identity, independence, and agency but also how society responds to them. In our discussions and in their support letter, Anne Marie Lovejoy, Centre Manager (2022) highlighted this project as an opportunity for participants to make a positive contribution by participating in this research. This was important, as society's focus on the medical diagnosis of people living with dementia leads to limiting perceptions of what they can do and frequently overlooks the value of what they contribute.

This leads to the incorrect notion that a dementia diagnosis removes embodied cultural capital (Bourdieu, 1984). The relevance of this concept in the context of this research, is that it was conceptualised in reference to the distribution of power and therefore inequality. Operating within a principle of public return (Mutibwa, Hess and Jackson, 2020) that the "embodied cultural capital" (Bourdieu, 1984, p. 76) of older adults living with dementia lies in their contribution to heritage and cultural narratives. Those narratives from lived experiences, when made in people's own terms (Vargas-Ramírez and Paneque-Gálvez, 2019), enrich communities' sense of place and belonging.

Lovejoy Centre clients visit the Centre for the day, on a weekly basis, for carer respite or for clients in residential care, to remain connected to the community and have meaningful experiences outside of their institutional setting. The Centre was fully supportive of the research, and the Centre Manager provided a support letter stating the Centre considered the research activities benefited their clients and supported their aims.

Throughout this project, the Centre facilitated access to participants, hosted sessions at the Centre, including them as part of their regular programme, and supported the delivery of activities. They liaised with carers to pursue secondary consent. The Centre Manager and two members of staff supported me in delivering research activities. They knew participants very well and could communicate with them in the most inclusive way. Centre staff involvement was central to the success of my data collection, safeguarding the wellbeing of participants, and offering the support they needed to be empowered to participate.

ii. Organising and delivering co-design research

To make the most of engaging this stakeholder group, several research methods and activities were combined into co-designed research. The study was conducted in collaboration with two groups of six participants each, on average[12]. Sessions were delivered weekly and took place across two months (May and June 2023). Each session was 90 minutes and was structured to start with an introduction about the project, to remind participants and contextualise the activity for the day. Each session included a discussion activity, a sensorial experience, and a creative task (see session plans in Appendix D).

Participants' wellbeing was paramount, and circumstances could change quickly. Therefore, I consulted with the Centre Manager before each session, to gather feedback on the detail of the coming session and applied any necessary changes. The manager's advice provided an improved understanding of how creative methods could become more engaging and

[12] The number of participants in each group was not always the same.

proceeded to design templates for my visual method. Each session was reviewed and redesigned informed by the guidance or events of the previous session. The observation of techniques used by Centre staff, and those promoted by dementia groups, continuously informed research practice and activities (drone livestreaming, approaches to evaluation) and the use of creative methods.

Involving any human subject in research is always complexified by environmental, contextual, psychosomatic variables, and not everything works all the time. Group One sessions operated as the testing ground, and the second group sessions were more successful. Throughout each session, Centre staff was in continued observation mode with each client, to capture changes in mood, difficulties, needs, and respond to those. It was important to pay close attention to staff, for indication on whether to repeat a question to a participant or not draw their attention to the question, on that occasion.

This proved particularly important on one occasion, with a participant who used to be a drone pilot. When a drone was shown to the group in the first session, they engaged with it as well as the others; but afterwards, their family reported they were agitated at home and anxious, thinking they were being spied on by a drone. This prompted an individual risk assessment signed by me, the family, the South Downs National Park Authority Health and Wellbeing officer (who assisted with the delivery of some of the sessions), and the Centre staff.

It meant the participant could not attend the livestreaming (they went on an outing with the Centre Director and rejoined afterwards), but they did take part in all the other sessions. Following the risk assessment, the word “drone” was avoided in the presence of the participant.

iii. Recruitment and consent

As co-research participants lived with dementia, recruitment and consent were iterative. Participants were recruited during visits to the Centre targeting groups suggested by staff. I introduced myself and the project (using an accessible participant information sheet) to explain all the relevant information about the activity and ask individuals about their interest in participating. Participants were informed about the research and their participation orally, as I went through the accessible version of the participation information sheet verbally. This sheet was also made available to each participant in large print.

Centre staff and the SDNPA Health and Wellbeing officer were asked directly about their willingness to share their reflections of the project sessions, explaining this was voluntary. A mixed group of participants, including different stages and types of dementia was recruited; all participants were able to understand what taking part in the project involved.

Consent was reiterated at the start of each activity to check everyone was still happy to participate. The staff and I went through the accessible consent form on a one-to-one basis. Staff were provided with a crib sheet so they could answer questions whilst providing individual support. The consent form covered sound, photography, and publication of research outputs. To comply with GDPR[13] and the Mental Capacity Act 2005, secondary consent was sought from the person who "otherwise than in a professional capacity or for remuneration" (Gov.uk, 2020) was engaged with the participant's care or interested in their welfare. The Centre Manager handed them the form and information sheet.

Participants could withdraw or not participate in the study without giving a reason. All participants were made aware that they could stop taking part at any point in time and engage in any of the other activities easily available at the Centre (puzzles, watching slow TV, drawing, chatting, etc). As the participants were part of a pre-existing group, and individuals had known each other and taken part in group activities for a while, the social atmosphere impacted on their decision to take part. However, as co-design activities include content produced as a group, as well as individual contributions, it was made clear to participants that for practical reasons it would not always be possible to withdraw contributions, particularly if requested later than on the day the content was created. This was reiterated prior to each activity.

iv. Confidentiality, anonymity, and security

The only personal data collected from participants was their name (in the consent forms). Participants' addresses were not collected, nor accessed (families and informal carers were contacted by Centre staff). The Lovejoy Centre had permission from all participants and carers

[13] General Data Protection Regulations, frequently referred to as Data Protection

to video, photograph, and upload the content to social media platforms. Independently from this, participants were asked for consent to film, photograph, and record sound in the sessions, to be used in data analysis. Additional permission was requested for this data to be used in disseminating the research and its outputs in papers, conferences, and by heritage partners through their online platforms.

To meet GDPR regulations I planned to bring to each session, a list with the names of the participants with consent restrictions to their image and/or sound being recorded. This was not necessary, as all participants consented. The individual contributions in the documentary film for the co-design research and in the film of the livestreaming experience were anonymised. There were no instances of lone working with participants. Only the participants' first name was used in the sessions (as these were being recorded) and I signed a confidentiality agreement with the UK-based transcription company.

As the sessions were delivered to a group, internal confidentiality was a possible limitation. However, participants had attended the Centre as a group for a while and had done life story work and other activities that have elicited personal memories, in the past. They were all aware they shared the characteristic of living with memory loss.

v. Data storage and handling

The co-design research generated digital sound and video recordings, co-research sessions transcripts, consent forms. The latter are kept in a locked desk drawer. The camera recorded on to an SD card; files were transferred after each session to the University of Brighton's OneDrive cloud and deleted from the SD card.

The physical products created by participants in co-research sessions have been digitised by me and kept by the participants and the Lovejoy Centre. The film and sound recordings have been handled by me, the supervisory team, and the videographer (who accessed the digital formats only), to create: a film that documents the project; and one film per group to document the co-designed drone experience. Each participant and the Centre received a copy of the relevant film(s). The final visual products were also shared with my heritage partners.

Data will be published in academic papers and this thesis. Results will be accessed by academic publication in reference to the research. The raw data will be deleted when the Doctoral degree is awarded. Anonymised transcripts will be deleted 10 years from the date they were collected, in alignment with the University of Brighton's guidelines.

vi. Benefits and risks

The co-design research benefited my own practice as a researcher and heritage professional, as it helped me develop knowledge and understanding of conducting research involving people living with dementia, and of developing activities facilitated through digital technology to engage this group with heritage content and placemaking. It also enabled me to contribute to knowledge on the use of drone technology as a tool of empowerment to engage people with dementia, as an underrepresented community, to make their own choices and decisions in engaging with heritage and place, supporting their sense of belonging – a key outcome of heritage engagement.

Both heritage partners, in interviews, outlined that this co-design research is as an opportunity for them to safely explore and test their own approach to using drone technology beyond the limitations of their policies, and to consider how drone technology could bring broader benefits that would increase their confidence in using drone assets in a broader context. They also perceive my research as an innovative proposal to change sector thinking and practices on the use of digital technologies to empower audiences and as collaborative tools. Drone experts commented that research and activities such as this project's challenged negative public perceptions of drone technology as malicious, intrusive, and predatory, by showcasing how drones create opportunities that all society can benefit from.

The co-design research benefited the Lovejoy Centre by offering activities delivered using a creative methodology that mirrored their approaches to programme provision. The Centre Director noted:

> "I feel from the participants' perspectives (my service users) and the Centre's it is a wonderful opportunity for us and them to participate and give them a voice and a platform to contribute. It

> helps to dispel myths about people's assumptions of those living with dementia, helps to build and promote dementia friendly communities. […] They are also looking forward to making contributions too." (Anne Marie Lovejoy Bruce-Kerr, 2022)

Furthermore, my research aimed to empower participating Centre users by involving them in creating the experience they wanted, fostering a connection to local community resources, and engaging them with processes that reinforce placemaking and belonging. These were crucial, in the context of memory loss, to invite and celebrate their positive contribution. Each research stakeholder brought diverse data, expertise, and drone practices to this project, and this consideration was made when selecting appropriate groups, and the appropriate methods to engage them. The next section outlines which methods were identified to produce which data.

3.3 Qualitative research methods

Table 3 and Table 4 below, summarise the types and characteristics of the data collected in this project. Table 3 presents the information for the desk research stage, including the publications reviewed, their sources, how they were accessed and analysed. Table 4 gives an overview of data collection methods, identifies participants and sample size, how they were recruited, the purpose of deploying the method, formats of data collected and how it was analysed.

Table 3 Overview of desk research and secondary sources

Data collection methods	**Title of document analysed**	**Author**	**Access**	**Methods of analysis**
Desk research	Key strategic documents			
	• *Learning Outreach Strategic Review 2018-2023* (2018) • *Communications and Engagement Strategy 2021-2025* (2020)	South Downs National Park Authority (SDNPA)	Provided by SDNPA	Thematic analysis
	• *Changing Chalk Area Action Plan* (Internal Document) (no date) • *Strategic Framework for Research 2022-2027* (no date) • *Playing our part: What does the nation need from the National Trust in the 21st century?* (2015) • *National Trust Research Strategy* (2016) • *National Trust Research Opportunities* (2019)	National Trust (NT)	Provided by NT	Thematic analysis

Data collection methods	Title of document analysed	Author	Access	Methods of analysis
	Secondary survey data			
Desk Research	• *Visitor Survey 2018 Final Report* (2019) • *Visitor Survey 2021 Final Report* (2022)	South Downs National Park Authority	Provided by SDNPA	Thematic analysis and coding
	• *Cairngorms National Park Visitor Survey 2021* (2022)	Cairngorms National Park	Cairngorms National Park as a result of receiving the online survey callout	
	• *Attitudes to Drone Use by National Trust Staff* (internal survey report).	National Trust	Provided by National Trust	

Table 4 Summary table of research methods applied, data collected and methods of analysis

Data collection methods	Participants	Number of participants/ respondents	Recruitment/ access	Purposes of deployment	Formats of data collected	Methods of analysis
Semi-structured interviews (60 to 90 mins) (Qualitative) (Please see Appendix B, 1. and 2.)	Selected heritage experts.	10	Via heritage partners	Collecting qualitative data of depth from experts.	Sound files and interview transcripts	Thematic analysis (inductive and deductive) and coding.
	Drone experts	8	Posts on drone pilots' social media groups; referral from the Director of The Drone Project (USA); referrals from heritage partners; networking.			
Online Surveys (Qualitative) (Please see Appendix A)	Invited heritage and cultural organisations' staff.	22	Invitations sent through 17 UK heritage and cultural organisations.	Collecting qualitative data from breadth of sector.	Digital survey reports	Descriptive statistical analysis and thematic coding of qualitative responses.

Data collection methods	Participants	Number of participants/ respondents	Recruitment/ access	Purposes of deployment	Formats of data collected	Methods of analysis
Co-design discussions (Qualitative).	Lovejoy Centre participants (underserved group)	4 staff (four women, two of which from a Global Majority). 9 Lovejoy Centre users.	Access organised through Lovejoy Centre.	Collecting in depth qualitative data to inform co-design process,	Digital sound files, sound files transcriptions, written notes, creative outputs, photographs, and film.	Textual and thematic analysis and coding of session transcripts.
5 co-design workshops per group (Participatory) (Please see Appendix D).				Building trust and co-design the drone livestreaming event.		
• Creative methods (Participatory) • Collaborative mapping • Livestreaming (please see Appendix E)				Building trust; create connection with SDNP and NT as background to the drone livestreaming event.	Video footage, map identifying groups' preferred SDNPA and NT sites.	Drone video content analysis: Visual language analysis using proposed new framework designed for considering power and agency.

Data Collection Methods	Participants/ data source	Number of participants/ respondents	Recruitment/ access	Purposes of deployment	Formats of data collected	Methods of analysis
Autotechnography Walking practices Filmmaking Creative writing (Autotechnographic) (Please see Appendix F section 2.4).	Researcher (belonging to underserved community).	1	N/a	To formalise process of reflection resulting from autotechnographic research.	2 drone video film-poems reflecting on researcher's engagement with heritage using UAVs; autotechnographic observations.	Visual methods – content analysis and thematic coding.
Analysis of institutional drone video content (Qualitative) (Please see Appendix F).	Drone videos produced by SDNPA and NT, hosted on YouTube.	4 videos	Identified by heritage expert interviewees as part of interview.	To analyse how drones are used to produce a visual language, how this correlates with institutional positionality.	Identification of key characteristics of visual language.	Visual methods – content analysis and thematic coding.

Data collection methods	**Participants/ data source**	**Number of participants/ respondents**	**Recruitment/ access**	**Purposes of deployment**	**Formats of data collected**	**Methods of analysis**
Analysis of drone video content produced by underserved audiences (Qualitative) (Please see Appendix F 2.1).	The selection of drone video content produced by underserved audiences resulted from my concern on the lack of diversity in the population of drone user interviewees responding to recruitment call outs, in generic drone pilot groups. Addressing this involve the researching social media platforms specific to marginalised demographics, such as Black drone pilots' and Women Who Drone's Facebook groups, where drone content produced by these communities relating to identity, representation, and visibility, was found.					
	Adagio Dance School (2021) *Road to Freedom!*	7 videos	Video posted by drone pilot in the Black drone pilots' Facebook page.	To analyse how drones are used to produce a visual language; to compare the visual lexicon of institutional drone content with that of drone content created by underserved communities.	Video, video authorship, description, and accompanying text for contextualisation.	Visual methods – content analysis and thematic coding.

Data collection methods	**Participants/ data source**	**Number of participants/ respondents**	**Recruitment/ access**	**Purposes of deployment**	**Formats of data collected**	**Methods of analysis**
Analysis of drone video content produced by underserved audiences (Qualitative) (Please see Appendix F 2.2 and 2.3).	Women Who Drone (2023) International Women's Day 2023	N/a	Video posted by Women Who Drone founder to celebrate International Women's Day 2023.	Analysis of visual language; to compare institutional drone video content with that of underserved communities.	Video, video authorship, description, and accompanying text for contextualisation.	Visual methods – content analysis and thematic coding.
	Attitude Magazine (2022), *Bimini and Dan O'Neill on healing the ocean as LGBTQ people*.	N/a	Video posted by Attitude Magazine.	Analysis of visual language; to compare institutional drone video content with that of underserved communities.	Video, video authorship, description, and accompanying text for contextualisation.	Visual methods – content analysis and thematic coding.

Data collection methods	Participants/ data source	Number of participants/ respondents	Recruitment/ access	Purposes of deployment	Formats of data collected	Analysis methods
Analysis of drone video content produced by underserved audiences (Qualitative) (Please see Appendix F 2.5)	Lovejoy Centre (2023) *Bringing the outdoors, indoors* two drone livestreaming films.	N/a	Produced in co-design research workshops.	To analyse how drones are used to produce a visual language; to compare the visual lexicon of institutional drone content with that of drone content created by underserved communities.	Video of drone livestream session; transcript of conversations during livestream.	Visual methods – content analysis and thematic coding; thematic analysis and coding of conversations.
Analysis of drone video content produced by underserved audiences (Qualitative) (Please see Appendix F 2.4)	Madalena Fine (2023): *These too are footsteps* (*2023a) A room of my own* (*2023b)* "film-poems.	N/a	Produced by researcher as part of autotechnography.	To deepen the understanding of the competencies of drone technology; explore, as a member of an underserved community, how UAVs can empower; to expand the drone video	2 drone video film-poems reflecting on researcher's engagement with heritage using UAVs,	Visual methods – content analysis and thematic coding.

				content analysed from secondary sources into primary source data. Also, to compare the visual lexicon of institutional drone content with that of drone content created by underserved communities.		

Discussion of each method includes their characteristics and validity as an academic research data collection tool, their suitability to answer this project's research questions and to engage specific stakeholders; the purpose of applying it to this research and how the method is used, the challenges of deploying the method and ethical considerations associated with its use.

3.3.1 Semi-structured interviews

Semi-structured interviews (SSIs) were used to capture data from diverse experts in the heritage sector and drone usage. SSIs are a method involving a focused (Silverman and Patterson, 2014, p. 60) conversational interaction between researcher and interviewee, combining closed and open questions, further explored through follow-up questions (Adams, 2015, p. 493). SSIs are well established as a research method frequently used in qualitative research (DeJonckheere and Vaughn, 2019), mostly in social sciences and health research (Magaldi and Berler, 2020).

SSIs allow the researcher to investigate the interviewees' points of view and feelings about a subject (DeJonckheere and Vaughn, 2019) and help address research questions exploring the viability of a proposed offer (Adeyoye-Olatunde and Olenik, 2021, p. 1361). The conversational approach and the flexibility of follow-up questions in semi-structured interviews allow interviewees to expand on their answers with reflective and contextual information. As Whitehead (2005, p. 17) explains: "[...] the interviewer […] attempts to gain a greater understanding of the context and meaning of those responses through various forms of probing." This may elicit unexpected data (Adams, 2015, p. 493) that has the potential to enrich the research.[14] As such, semi-structured interviews are suitable to explore "uncharted territory" (Adams, 2015, p. 494) as is the case of my research project.

The method was applied in this project with staff from heritage partner organisations and drone experts. With heritage experts, the purpose of these interviews is to gain an understanding of heritage partners' perceptions of drone technology, to answer the research question "What do heritage organisations perceive to be the barriers, risks and unique

[14] This is the case when Drone Interviewee Eight introduces the idea of livestreaming drone content from a First-Person View camera drone, into a private You Tube channel. This discussion leads to me investigating the potential of the idea as the approach and designing the co-design research.

opportunities of drone technology to engage underserved, non-visiting audiences?" This method was also applied with drone experts, to capture how they use drone technology themselves, and what they consider to be the unique opportunities UAVs offer to civic society, heritage, and culture. Both sets of interview data supported the research question "What unique opportunities does drone technology offer the interpretation of and engagement with heritage?" and ultimately supported the investigation of drone technology to empower underserved audiences to connect with heritage.

To answer these questions, it was necessary to apply a method that retains the focus of the subject matter while creating opportunities for additional information to emerge and enhance understanding. As discussed by Guest, Namey and Mitchell (2012, p. 31) the interview questions centred on an established topic (using drone technology for audience engagement), which was part of the structure of the interview. Nevertheless, those questions elicited reflection, prompted follow-up questions, and deepened the exploration and understanding of interviewees' points of view and feelings on the topic.

In this project the use of semi-structured interviews and co-design workshops (section 3.3.3 below) was aligned with the guiding principles of empowerment and collaboration. Semi-structured interviews are flexible and can be used developmentally, meaning that interviewees can expand on their ideas and focus on their points of interest within the theme (Denscombe, 2014, p. 187). As such, there is an element of interviewee ownership and power that allows participants to inform how they explore the subject and to present information that reflects the nuances of their own agenda and needs.

Participants were recruited with support from the heritage research partners at the National Trust and at the South Downs National Park Authority. They circulated the Participant Information sheet, attached to a recruitment email, and posted the opportunity on their internal staff communication platforms. Interested participants contacted me directly, via email. Heritage partners also identified colleagues in specific teams, whose perspective was relevant to this research, and either approached them directly or had an initial discussion followed up by myself.

Staff from different departments (public programming, interpretation, communications, learning and engagement, and visitor services) were interviewed, which helped capture the priorities of different departments. Each prospective participant was sent a Participant Information Sheet and offered an opportunity to ask questions and discuss the project with me before agreeing to be interviewed. All interviews were set up as one-to-one and hosted on Zoom, as interviewees were based in different parts of the country (particularly National Trust interviewees) and as mitigation for COVID-19 measures at the time.

The recruitment of drone user interviewees was done through posting call outs on specific Facebook (Meta) group pages: Black Drone Pilots; Women Who Drone. Heritage partner organisations also circulated the Participant Information sheet and recruitment email internally. Again, each prospective interviewee met with me to discuss the project and the interview, before agreeing to participate.

Semi-structured interviews are an involved and time-consuming method, due to their process of recruitment, setting up interviews (Adams, 2015, p. 493), and the extensive data collected and analysed. This considerable outlay means that, although semi-structured interviews allow the exploration of a subject in depth, it is difficult to involve many participants, limiting the sample size for data collection (Adams, 2015, p. 493). Moreover, conducting SSIs warrants researcher focus, lateral thinking, and flexibility (Adams, 2015, p. 493), active listening and sensitivity (DeJonckheere and Vaughn, 2019, p. 14).

Specifically, in this research project, SSIs provided insight into individual attitudes and perceptions combined with departmental and or organisational culture mindsets, and it was difficult to separate both at times. To mitigate this, the interviews were combined with analysis of secondary data, namely the *South Downs National Park Authority Partnership Management Plan 2020-2025* (SDNPA, 2020) and the National Trust's *Research Strategy 2017-2021* (National Trust, 2016), *Playing Our Part* (National Trust, 2015). This was as part of identifying partners' key priorities, challenges and strategies in audience engagement.

In terms of ethical considerations, anonymity is an important topic. Interviews were conducted with heritage partners and, as this partnership is in the public domain, there was a risk that anonymisation and data filtering (replacing the name of the organisation with "a heritage organisation in the South East," for instance) would not suffice to guarantee anonymity and that the organisation or the interviewee were not discoverable. Nevertheless, institutional roles were relevant in this research, as they were part of the data, as indicators of positionality. Eliminating them from the research narrative could have skewed or excluded relevant data. Therefore, the amount of sensitive data collected via research interviews was minimised, but full anonymity could not be guaranteed. Participants were made aware of this in the Participant Information Sheet.

3.3.2 Online surveys

As heritage expert interviews were carried out only with my partner organisations, the data collected was too limited to enable generalisation. Therefore, a national online survey was undertaken, alongside the semi-structured interviews, with the purpose of capturing a snapshot of perspectives from heritage staff, consultants, and volunteers across the United Kingdom. The online survey made it possible to cross reference results, identify patterns and gain a broader overview of the heritage sector's perspectives.

Surveys gather information from individuals, through asking them questions (Ponto, 2015, p. 168) that can be set up in different ways, such as skip logic (Ball, 2019, p. 414) and multiple choice. Surveys are frequently used, particularly in social and psychological research (Ball, 2019, p. 413), to describe and investigate the behaviours of a group of people (Ponto, 2015, p. 168). They are traditionally used to obtain data from a large sample of a population that is representative of the group targeted (Ponto, 2015, p. 168).

According to Denscombe (2014) and Ball (2019, p. 414) a survey is a cost-effective way to undertake research, reaching a wide range of participants (Denscombe, 2014, p. 12). Particularly an automated online survey (Ball, 2019, p. 414), which allows for quick distribution and response returns, whilst providing an understanding of a broad range of views (Greg, Emily and Mitchell, 2004). Online surveys also have the benefit of large geographical and industry coverage, through email distribution to relevant networks, groups, organisations, who can answer at their own convenience (Ball, 2019, p. 414). Research shows that there is little difference between the quality of answers from a survey and those from more traditional methods (Denscombe, 2014, p. 14).

Online surveys were utilised in this research as an opportunity to capture data from a variety of heritage organisations of different sizes, types, and locations. It is also part of a sampling strategy (Pronto, 2015, p. 169) to answer the research question "What do heritage organisations perceive to be the barriers, risks and unique opportunities of drone technology to engage underserved, non-visiting audiences?" By adapting the questions from my semi-structured interviews into a survey, data from heritage partners' expert interviews were tested at a larger scale. The online survey helped gain an understanding of the view of the heritage sector compared to that of this project's partners'. It supported identification of data trends and patterns relating to the perception of drones for heritage engagement. The online survey ran simultaneously with the semi-structured interviews.

Surveys help participants explore ideas (Denscombe, 2014) without the interference of bias or influence arising from the presence of the interviewer (Ball, 2019, p. 414). The aspect of creativity and exploration of ideas was important to elicit potential ideas for the use of drone technology in heritage. For rigour, criteria for the inclusion of participants were established in combination with a specific distribution list (Pronto, 2015, p. 168). Inclusion criteria were: to be a member of staff, trustee, or volunteer in any heritage organisation in the UK, independently from their governance (independent, volunteer run, national). Participating organisations included museums, historic properties, galleries, archives, archaeological sites, National Parks. The survey link was distributed including a Participant Information Sheet, to 17 organisations[15].

The online survey questions were adapted from the heritage expert interviews and hosted on a secure survey platform (Jisc online), under the University of Brighton's account. The survey took on average ten minutes and included a combination of multiple choice, single answer,

[15] Artswork, Association for Independent Museums, CLORE Leadership hub, Group for Education in Museums, Museums Association, Museums Computer Group, Museum Development UK, Museum Development South East, Museum Next, Social History Curators Group, National Lottery Heritage Fund, National Trust, Social History Curators Group, South Downs National Park Authority, The Engine shed, UK. Heritage drones' group, University of Leicester Museum Studies Department

and open text questions. All questions on the survey were mandatory, which meant the survey could not be submitted until all questions were answered. To ensure this would not deter respondents, the open questions offered an option of 'I don't know' and 'Prefer not to say'.

Before being launched, the survey was tested on heritage partners' lead contacts. All their recommendations were implemented, apart from excluding open-text questions. This is because these were central to obtaining more in-depth data from survey respondents, such as the context of their responses. The heritage partners' feedback suggested to me that they were concerned with aspects that could potentially limit the number of completed surveys. The target was to receive between 30 and 50 completed surveys (22 received).

The absence of interpersonal interaction could have meant respondents were more open and authentic in their answers (Hooker and Zuniga, 2017, p. 2), particularly as the survey was completely anonymous. The nearly contactless quality of the survey could also have meant that respondents felt less motivated to invest time and energy in responding to questions in a more detailed way. Also, it did not allow for follow-up (unlike an interview or focus group).

The survey was a limited platform without the interpersonal interaction characteristic of the interview, where the interviewees, by being present, invested their time in the process and has an opportunity to explore, and expand on their answers with minimal effort or the support of the interviewer. Survey respondents drew on their own motivation and interest in completing the task, possibly with minimal disruption to their routine.

This could have led to respondents being more succinct in their responses and focusing specifically on the elements of the answer they valued the most or felt most strongly about. This was understood as both a benefit and a drawback of using online surveys, in comparison with interviews or focus groups. The data from surveys may be more focused and specific, therefore with clearer themes, but it has fewer layers of meaning. It is also important to have an extensive distribution list as response rates are proportionally low.

For ethical considerations, the survey recruitment and participation materials, including the Participant Information Sheet were circulated on my behalf by organisations the survey was sent to. The survey distribution networks were contacted through the University of Brighton's email. When respondents first accessed the survey, a tick box indicated they understood that, by submitting the survey, they gave their consent to take part in the research. This was a required field. The survey was set up so participants could take part anonymously and no personal data was requested. The organisational information collected was generic: for instance, what kind of heritage site or organisation (independent museum or archive). Consequently, no personal data was handled or recorded.

3.3.3 Discussions in co-design research

The co-design workshops were approached by combining aspects inherent to focus groups' research, as this project engaged an ensemble of people (Milena *et al*., 2008) with shared characteristics (Islam, 2022, p. 260), in a conversational exchange (Barbour, 2007, p. 1). Focus groups are a well-established (Luke and Goodrich, 2019, p. 77) widely used, research method to explore how people feel and why, their attitudes, and understandings (Islam, 2022, p. 261). This exploration was an important element in my research. Unlike one-to-one semi-structured interviews, focus group discussions foster the sharing of different ideas, which makes data elicited through the process, rich in quality and diversity of perspectives.

Nevertheless, in alignment with the guiding principles of this project, the research with people living with dementia aimed to empower participants and create opportunities for agency, more than singly capture their knowledge. The researcher facilitated co-design research supported the process of empowerment, by bringing a clearer understanding of people's feelings, ideas, and opinions towards the NT and SDNP, and what led them to think that way (Denscombe, 2014, p. 189). Nevertheless, as this research targeted the involvement of people living with dementia, who may have felt the heritage sites in this project were not a place for them, it was important to go beyond discussions.

Co-design research encouraged Lovejoy Centre clients to be "active participants in the discovery, interpretation and celebration of the cultural heritage of the Downs" (National Trust, 2021). To achieve this, the research combined focus group discussions with participatory activities, including the collaborative design of drone mediated group experiences (see section 3.4).

Working with pre-existing groups is a time and data efficient approach to involving participants. Here, it meant that there was confidence in reaching my recruitment target audience in larger numbers and quicker than via public communications. The co-design research involved working with between nine and twelve participants living with dementia, who attended the Lovejoy Centre (see 3.2.3 above).

Participants were of all genders, aged between 60 and 94, in groups of four to five in each session. The single criterion for participant recruitment was to be a Lovejoy Centre client who used the service Wednesdays and Fridays. These two groups were identified by the Centre manager as more likely to engage proactively with activities and the collaborative design process. Sessions were supported by four Centre and, where possible, one SDNP staff. The high ratio of staff to participants meant it was possible to provide support on a one-to-one basis.

Participatory activities were intercalated with discussions, creative responses, and viewings of drone video content, all as ways to respond to and explore the questions posed (see Appendix D for the co-design research session plans). Participants were led to interact in a conversation where "the moderator pushes the group through discussion of a pre-set list of topics." (Brennen 2017, p. 1). Those topics included: participants' perceptions of National Trust and South Downs National Park sites; past experiences of visiting; social history; archaeology; fauna and flora of the sites; what it felt like to bring the outdoors indoors; and having the power to choose what you view in your remote visit.

Collaborating with community groups involved risks congruent with most community engagement programmes. Participants' health can deteriorate, which can affect the membership of the group and the number of participants. Sessions can deviate from their plan, due to unexpected factors external to the project or participants disinterest in an activity, for instance. Sessions could also have been unsuccessful in gathering the data needed for my research.

Trust is essential for running sessions with community groups. If participants trust that they can share their views safely and comfortably, in an environment where they are respected and treated with dignity, they are more likely to engage with the process. As activities engaged with a pre-existing organised group, trust was already partly established. The meetings took place in a building participants knew, and with the support of their group coordinators.

It was important to take into consideration how the researcher and the project could be perceived by participants, in view of the association with academia and a top-down institutionalised approach to knowledge. To mitigate this, which Denscombe (2014, p. 90) discusses in relation to interviews, but which is relevant to participatory methods with underserved communities, the co-design approach is utilised as a method to further reiterate the value of the expertise of participants living with dementia.

3.4 Participatory research methods

As discussed in 3.3.3, the research with the Lovejoy Centre's participants was designed as a collaborative programme. Co-creative research methods were applied through collaborative mapping and co-production of drone livestreaming with the groups. These methods were deployed in the context of the co-design research (see 3.2.3 above) to generate qualitative data and visual data (drone video footage). The application of participatory methods was part of a strategy of empowerment in the engagement of people living with dementia, considering they are representative of an underserved community.

Participatory methods were the most suitable because they supported stakeholder involvement and collaboration (Fetterman *et al*., 2014, p. 3). These methods were therefore aligned with agency and empowerment, two of the guiding principles in my research. These methods fulfilled a three-fold function in the research: to help participants connect to the heritage of the National Trust and South Downs National Park; to relieve participants from the discussion elements of the sessions which can be cognitively demanding, and to build trust.

The collaborative mapping supported the gathering of data on which two sites, participants wanted to view remotely, and the story making helped establish what participants thought of experiencing place from the ground and from aerial space. The livestream enabled me to test an approach to drone-facilitated empowered access to natural heritage by non-experts, and the films were used to capture the reactions of participants to the livestream, and as a project legacy to share with the Lovejoy Centre.

3.4.1 Collaborative design

Collaborative design, also referred to as co-design (Burkett, no date, p. 4) is a people-centred process that Sanders describes as "[...] discovering possibilities and opportunities, with people, which address their needs and aspirations for experiences" (2000, p. 4). It is an empowering process as defined by Zimmerman (1995, p. 584), as it involves communities in the design, development, and delivery of shared outcomes as equal partners.

Horsfall and Titchen (2009, p. 150) consider creative methods as an intrinsic part of producing "[…] research which is transformatory and democratic." They consider (2009, p. 156) creative methods are an established approach to capture data with excluded groups, and to foster different outcomes on knowledge and power by involving a wider range of voices. According to Burkett (no date, p. 7), co-design benefits organisations by: promoting cross-departmental cooperation; driving focus on users; helping develop the organisations' relationships with their communities; bringing innovation through the exchange of ideas; and leading to changes in practice. It is beneficial to users as it offers opportunities to participate in their service and contribute to shaping it to better meet their needs.

Collaborative design positions "[...] professionals as collaborators instead of authoritative experts," and it values communities' lived experiences and knowledge, as expertise. By designing with people for people (Burkett, no date, p. 4), collaborative design is a democratising approach that helps address the imbalance of power between heritage and academic organisations and disadvantaged communities, as described by Perkins and Zimmerman (1995).

Co-design research was undertaken in this project, as an approach to reducing the disparities built into knowledge practices (Kennemore and Postero, 2021, p. 3) in heritage and culture, which constitute a barrier to engagement. This method was also aligned with the project partner organisations' strategic frameworks and those in the Heritage sector, such as the Museums Association's *Museums Change Lives* Campaign. This campaign advocates community empowerment through its Power to the People framework, a participatory practice approach that fosters partnership and supports museums to forge sustainable relationships with communities (Museums Association, 2012, p. 2).

Zimmerman (1995) identifies the type of participation where people are actively involved, as one of the central elements of the construct of empowerment. This method made it possible to involve participants in making decisions on and shaping their experiences in an equitable way, tailored to participants' needs. This is particularly important in the case of people living with dementia, the group collaborating in this research, as memory loss is associated with loss of identity, agency, and independence. As empowerment and agency are guiding principles in my research, co-design emerges as a central element of empowerment, inclusion, and accessibility.

Co-design underpinned the development and delivery of the research with Lovejoy Centre's participants. Designing the activities collaboratively involved an initial discussion with the Lovejoy Centre Manager and observation of sessions delivered with both participating groups. Following these, a session plan was drafted for each of the five sessions and discussed with the Heritage Partners and the Lovejoy Centre Manager. During the delivery, sessions were adapted according to participants' responses and the staff's feedback.

Co-design focuses on improving outcomes (Burkett, no date, p. 5); yet, from professional experience, managing stakeholder expectations of power and control can be difficult, as they are shared amongst participants. Moreover, outputs and outcomes can be uncertain and unpredictable. The co-design process can elicit very different results from those it proposes to explore. Nevertheless, these challenges are intrinsic to redressing a balance of power. The diversity of voices and experiences informing the process means it is more authentic and likely to be more innovative.

Ethically, when involving underserved audiences, it is essential to consider the emotional burden and complexities of exclusion and plan for inclusive collaborative research (as discussed in section 2.3.3). Within the collaboration with people living with dementia, for whom speech and other cognitive processes are, in some cases, a difficulty, it was important to build sufficient time to re-familiarise participants with the project, and for participants to be able to express their responses in the way that was most accessible to them. An ethics application was submitted specifically to address the ethical risks of working with people living with dementia.

3.4.2 Creative methods

Creative methods are extensively used in different social disciplines such as consumer research, sociology, and psychology (Buckingham, 2009, p. 634) in the form of drawings, videos, or photography, as approaches to empowering participants (Buckingham, 2009, p. 633) through overcoming the limitations of methods that require verbal interaction. Buckingham considers that creative methods (including, but not restricted to, visual methods) are therefore more inclusive and accessible to users with different learning styles and needs. Participants can express themselves using their own means of representation, without privileging one specific way of knowledge production (Horsfall and Titchen, 2009, p. 157) and with less influence from the researcher.

In camera drone technology, the visual is at the centre of the technology's affordances. As my research explores visual media, it was appropriate to draw on visual methods to investigate (Buckingham, 2009, p. 644) and discuss (Buckingham, 2009, p. 648) how participants perceived aerial views, remote viewing, and created their own visual experience of places they wanted to experience virtually (Buckingham, 2009, p. 634).

Creative methods involve diverse ways of knowing, as does drone technology: embodied knowledges that connect people with their environments (Horsfall and Titchen, 2009, p. 151). Taking this into consideration, a combination of visual methods was applied in this research; craft making, object handling; activities that included tact, smell, and auditory experiences. Because of how inclusive creative methods are, and how they encompass a wide variety of ways of experiencing topics (in this case natural heritage), they were inclusive of the needs of collaborators living with dementia.

Creative methods were utilised with three aims: to create an opportunity for participants to bring their voice into the research, sharing their interpretations in a more accessible way that brings enjoyment; as intercalating activities to break down focus group discussions, which have a tiring effect on participants; and to capture and record perspectives, for instance on aerial views. All creative methods used were deployed through co-design. They included photography, object rubbing, collage, story making, reminiscence, collaborative mapping, scenario research, digital media, and museum object handling.

Visuals used included: a collage of the South Downs and National Trust; drawings of handling objects passed around; leaf rubbing using leaves fallen in the South Downs; polaroid photographs of participants with their favourite item from the Worthing Museum and SDNPs handling collections; making a cardboard skylark or barn owl (one of the birds found in the South Downs). Activities also included colouring a line drawing postcard of the livestreaming experience, viewing drone footage.

To engage with the social and natural history, fauna, and flora of the South Downs, participants handled objects from Worthing Museum and Gallery's "The Worthing Story" Education Loan boxes (local archaeology, archive photographs and objects linked to shepherding in the South Downs) and from the SDNPA's Rangers Public Engagement collection (embalmed hare and skylark, antler, and birds' nests). Activities also included viewing video clips of heritage sites in Sussex, touching and smelling seasonal specimens of flora found in the South Downs National Park, listening to sound recordings (a skylark birdsong; Wellington boots splashing during a muddy walk on the Downs; a Skylark feeding its young).

The challenges of using creative methods in research are that they are associated with marketing research (Buckingham, 2009, p. 634) and can therefore be overly shaped by the researcher, who designs the activities the participants engage in. To mitigate this risk, all activities were designed in collaboration with the Lovejoy Centre Director, who drew on their professional expertise, advising on the suitability and effectiveness of activities for working with people with dementia. Activities were also adapted during the delivery of the programme, to reflect participants' preferences and needs. This ensured the creative methods were participant-centred rather than focused on the researcher's aims.

The application of these methods captured data about the places in the South Downs National Park and National Trust the participants were interested in visiting remotely, aspects of the landscape they were particularly attracted to, and any long-term memories and experiences associated with the South Downs and or National Trust sites. From this data it was possible to analyse aspects of existing connections with heritage and participants' responses to aerial views.

3.4.3 Collaborative research: mapping and livestreaming

Participatory or collaborative mapping integrates maps with user data that can represent experiences, relationships, or memories (Cochrane and Corbett, 2018, p. 2). Participatory mapping is well-used to give visibility to the relationship between communities and place, or to identify significant assets in the landscape (Cochrane and Corbett, 2018, p. 2). It is also an approach that involves communities in mapping and deciding what and who is mapped. As a method that problematises cartographic methods of representation, it is part of a discourse of community empowerment (Panek and Netek, 2019, p. 1) and therefore appropriate to produce data with underrepresented communities who experience a deficit of empowerment in their experiences of place and belonging.

The purpose in working with participants on place, using maps, was to situate research; to support participants' decision-making; encourage creativity and imagination; engender a participant-led connection to place; and bring individual and collective together. The use of maps in co-design workshop sessions enabled me to engage the group with the geography of the research (the South Downs National Park). It highlighted resources available to the group and explored their meaning to them, highlighting the value of that relationship and focussing on human assets.

The engagement with the map using a talking points approach (Aran and Reed, 2017, p. 8) elicited participants' perceptions of the SDNP and National Trust sites on which site they felt connected to and how; narratives of belonging and placemaking associated with the South Downs and the National Trust. It also supported a new way of imagining places and human geographies.

During the co-mapping, participants selected a site they wanted to explore through the livestreaming. The purpose of the livestream was to test how drone technology can be used to give agency to people living with dementia, in accessing and connecting to natural heritage experiences. There were technical and operational challenges to the livestreaming. For instance, the wind and drone noise made the live sound unclear. To mitigate this, I connected with the drone pilot via a mobile phone. Moreover, a potential challenge of using co-mapping was that the participatory process would become the focus, rather than the mapping output.

The mapping of assets was a strategy to engage participants with locational data that supported them thereon to frame the livestreaming. Therefore, although the mapping and the livestream were connected, the final outputs did not necessarily represent this visually.

Drone livestreaming, in my research, involved the flying of a drone by a pilot on the location participants selected in the co-mapping activity (Devil's Dyke and Highdown Hill) and the live broadcast of the flight into a YouTube private channel accessed through a device connected to Wi-Fi. Through the live broadcast, participants interacted with image and sound and instructed the drone pilot on the flight path of the drone. This element of participants instructing the pilot is of particular importance as my principal research question is focused on how drone technology can empower underserved audiences to connect with heritage. The process of ensuring participant empowerment and agency was at the centre of this process.

My role as facilitator of the process was enhanced to that of allyship and advocacy for the participants, ensuring their voice was leading the process with minimal interference from the pilot, other than responding to participants' prompts. Whilst the livestream was being broadcast on the Centre's television set, I operated as an amplification device, an antenna, mirroring the drones' own data distribution and reception capabilities. I sent the signal (participants' instructions, questions, reflections and guidance) to the pilot, over mobile phone, and received and relayed the pilot's requests for instruction and clarification to the participants.

The deployment of co-mapping and livestreaming resulted in a film of the livestreaming event. This was analysed in the research, as visual data.

3.5 Autoethnographic methods; autotechnography

Ellis, Adams and Bochner (2010, p. 273) define autoethnography as a method that uses the systematic description and analysis of personal experiences to develop an understanding of cultural experiences. The authors posit that autoethnography makes space for subjective thought, emotions, and the researcher's own place in the research.

Autotechnography is a research method that expands the concept of autoethnography into technological experiences. Blumer (1969) uses the term technography to define the study and writing of the technical structures, that are part of communication processes. The author defines the process of interaction between social objects (humans) and physical objects (drones in the case of this doctoral project), as technography. Josselson and Lieblich (2005) consider that there is evidence of a culture of autotechnography in the media, through blogs, televised reality shows and social media.

In Hildebrand's (2021, p. 3) autotechnographic research, the academic explores their own experiences as a user of consumer camera drones (in combination with interview data, fieldnotes, information available online and policies). They use their personal interactions with the machine to understand the complexities in the human-drone relationship. Taking the self as the starting point, autotechnography is used to support a better understanding of broader social themes. It can inform a "[…] free relationship" (Heidegger, 1977, p. 3) with drone technology, that does not constrain underserved communities to established ways of representing narratives of place and space.

Autotechnographic methods in this project complement the participatory and qualitative interviewing and surveying methods, in that they facilitate a first-hand systematic understanding of the use of the drone as an empowerment tool from my own perspective. As a Global Majority, disabled woman, my intersecting identities engender experiences of being underserved. The resulting positionality significantly informs the rationale and decisions I make in this research. They add an element of knowing through individual introspection, that in turn facilitates a better understanding of the power negotiations the drone facilitates in communities.

Autotechnography was employed in this study through the use of creative writing, filmmaking and walking methodologies (discussed in section 3.5.1), to facilitate the exploration and exposure of the researcher's lived experience. Boylorn (2008, p. 489) defines lived experience as "[...] a representation and understanding of a researcher or research subjects' human experiences, choices, and options and how those factors influence one's perception of knowledge". As the method involves documenting identities of the self, it made researcher's positionality explicit, in the interpretation of data (Denison, 2023, p. 94).

Autotechnographic methods complemented the participatory methods and the qualitative interviewing and surveying methods, in that they facilitated a first-hand systematic understanding of the use of the drone, as an empowerment tool from my own perspective as a Global Majority woman. They added an element of knowing through individual introspection, that in turn facilitated a better understanding of the power negotiations the drone facilitates in communities.

3.5.1 Creative writing and filmmaking

Collaboration, inclusion, agency, and empowerment were guiding values in this research. They shaped my methodological decisions, including the autotechnographic approach which produced two film-poems (English, 2023, p. 207) combining pre-existing and researcher generated drone footage, and original poetry. Tremlett (2021) defines film-poems as film integrating poetry (written as text or voice over). Film poems are also referred to as poem-films, video-poetry (Tremlett, 2021) and other variations. Video-poetry is celebrated at international level with a dedicated International Film Poetry Festival organised by the Institute for Experimental Arts which has been running since 2014.

3.5.2 Walking methodologies

Walking methodologies are defined by Robinson and McClelland (2020) as a research practice that is based on movement and place and is focused on “[…] the practice of locomotion”, to elicit complex and more comprehensive perspectives on how people relate to their environment and vice-versa. They are extensively used in social sciences (Springgay and Truman, 2018, p. 14) and increasingly beyond them (Finlay and Bowman, 2017), particularly in the field of health and disabilities (Butler and Derret, 2014). Springgay and Truman (2018, p. 86) present walking as an appropriate methodology for “situated, relational and material” research in this context, walking offers me a new space “of enquiry and disclosure” (Macpherson, 2016, p. 426).

Walking methodologies build on the importance of place, as well as the sensory enquiry and embodiment (Springgay and Truman, 2018) inherent to placemaking. As such, they are often used to explore people’s exchanges with place (Macpherson, 2016, p. 425) and Pink *et al. (*2010, p. 3) posit that walking is relevant because it is central to how we perceive an environment (Springgay 2017 p. 30). Belkhouri and Laing (2024, p. 68), in their study discussing mobile methods that support grasping the experience of walking in the city and deliberating, consider that through walking, relationships with the city can be deepened, and new perspectives uncovered.

Walking creates a connection formed by intersecting with places (Belkouri and Laing, 2024, p. 66). It facilitates a critical relationship with space (Keinänen and Beck, 2017) bridging the individual experience of space with the collective, bringing awareness (Belkouri and Laing, 2024, p. 62) of broader aspects of placemaking, identity and belonging. Both autotechnography and walking methods constitute forms of knowledge production (Springgay and Truman, 2017, p. 30) created from lived experiences which, elicited through the individual negotiations of space and place, create an understanding of collective experiences.

Kowalewski and Bartłomiejski (2020, p. 61) argue that the right to presence and visibility in heritage sites is unfulfilled by underserved audiences, because the intersection of characteristics that compound exclusion “deprive” communities from visibility in the public space, and the opportunity to move freely. They also argue that underserved communities are limited not just by the availability of the space, but also its accessibility, the social exclusions they experience, and perceptions of safety when accessing spaces in view of those experiences of exclusion.

This means that even when space is available and accessible (open to all, as is the South Downs National Park, or people are offered reduced fee or free tickets, as a mitigation of barriers posed by the cost of membership in the National Trust, for instance) perceptions of reduced self-efficacy and safety prevent communities with experience of exclusion from interacting with place, developing connection and a sense of ownership, and ultimately from having a visible presence.

Walking practices enabled me to experience visibility and presence through walking: what Springgay and Truman (2017, p. 31) identify as the relational-social concept of walking as embodied movement, the interaction of the body with the textures and surfaces of an environment. As these authors argue, relationships established by the interactions with place that are part of embodiment, shape place. Therefore, it is posited that when groups are not able to access place or interact with it in an empowered and equitable way, they are effectively absent and or invisible. Using walking practices enabled me to challenge this.

Heritage organisations play an important role in placemaking, as custodians of narratives of people engaging with places. Heritage sites and collections harness their significance from the value societies that are represented in those sites and collections attribute to place. As sites that are traditionally accessed physically, NT and SDNP were well-suited to walking practices. Walking practices enabled me to investigate the parallels between walking and drone technology as acts of placemaking (O'Neill and Hubbard, 2010) by walking through heritage sites and capturing visual (camera and drone) footage at the same time. Both forms of mobility (walking and drone technology) acted as modes of occupying space and place, and as exercises of my right to presence and visibility.

3.6 Data analysis, review, and interpretation

As illustrated in Table 4, data was examined using thematic analysis, thematic coding of qualitative responses, descriptive statistical analysis, and visual methods.

3.6.1 Thematic analysis

A range of techniques were utilised to critically analyse data, such as the thematic analysis and coding of interviewees' answers per question asked, in order to identify key trends and patterns. The thematic analysis of the data was supported by the use of the qualitative software tool NVivo, which facilitates organisation, analysis and visualisation of data. The analysis combined inductive and deductive approaches, which means the diverse range of data captured was informed by the Literature Review but also driven by the material itself, in a bottom-up approach.

3.6.2 Cultural Engagement analysis

Heritage and culture sectors' evaluation practices focus on impact and outcomes of engagement (Arts Council England's (ACE) *Let's Create* strategy (ACE, 2020) and *Inspiring Learning Toolkit* (ACE, no date). These strategies and frameworks inform the evaluation methodologies deployed by heritage organisations, including the partners in this project. The evaluation of the livestream experience mirrored the South Downs National Park Authority and the Cairngorms National Park visitor surveys. Methods used also drew on participatory practice evaluation.

3.6.3 Video content analysis

Video has been used as a qualitative research tool in Education, workspace studies, clinical research (Erickson, 2011, p. 183; Knoblauch and Schnettler, 2012, p. 334). In this research, drone video was used to explore the research question on the unique opportunities of drone technology to empower underserved groups, in three ways. First, through video content produced as part of the autotechnographic research, as an interpretive method (Knoblauch and Schnettler, 2012, p. 334) to represent researcher experiences. Second, as a primary data source, to show National Trust's and South Downs National Park's drone videos to co-research participants. Third, in my analysis and interpretation of visual language and aesthetics of drone camera videos, comparing pre-existing films produced by heritage organisations, with others generated by underserved communities.

In the process of this research, no frameworks tailored to the analysis of content produced by the new and developing drone technology were found. However, drone technology integrates extensively used models of capture, such as photography and video, into its novel capabilities (identified by heritage interviewees and drone experts such as verticality, mobility, accessibility). Different existing frameworks were combined to analyse drone content: descriptive categories of video content by Stankov *et al.* (2019) and the effects of UAVs in tourism videos; the drone shot taxonomy by Mademlis *et al.* (2019); drone photography characteristics and their effects by Serafinelli and O'Hagan (2022); Film Studies camera perspectives and their semantics.

Chapter conclusion

The methods described in this chapter are well established in academic research practice and formed part of an approach that aimed to develop insights, through engaging stakeholders with the process of development of the research. Using a range of methods allowed me to draw on rich data and to triangulate between different data sets, enhancing the robustness of this study.

As methods were applied to empower people living with dementia, adjustments were made in order to facilitate their involvement through the most inclusive and accessible practice that was possible. The most important and common denominator to the methods used was the creation of an exploratory, organic space of observation, interrogation, and trial that supported the answering of research questions in this project. These organic approaches constituted in themselves, a contribution to equitable research approaches. This is as important for working with audiences, as it is in working with heritage organisations, which are complex, have their own culture, and are shaped by their history, vision, and governance.

Chapter Four: Heritage organisations' perceptions of the barriers and risks of drone technology to engage underserved audiences

Chapter introduction

> "I think, if all of those logistical things were put to one side so if (…) we had landowner permission and we had like the funding to do it and we had the expertise to do it, and we were able to lay the fears of participation that we've just talked about, I can't see a huge element of risk there." (Heritage Expert Two)

This quote outlines the key considerations of participating heritage practitioners on the barriers and risks of drone technology. This chapter addresses the research sub-question:
"What do heritage organisations perceive to be the barriers, risks and unique opportunities of drone technology to engage underserved, non-visiting audiences?" It focuses specifically on perceived barriers and risks. Perceived opportunities are discussed in Chapter Five.

This chapter draws on the results from semi-structured interviews with heritage practitioners working at the National Trust and South Downs National Park Authority (partners in this research), and from an online survey distributed to heritage organisations and practitioners, across the UK. To contextualise the data, the analysis includes a consideration of heritage experts' role or area of heritage practice; organisational background and mission; organisational approach to audience engagement; and organisational approach to drone use.

The data is examined through textual analysis, whereby the choice of words, expressed ideas and concepts articulated in interviews and online survey responses are considered as part of exploring the context, deriving relationships and understandings of interviewees. This is integrated with descriptive statistical analysis of the online survey data and thematic analysis and coding of interview and online survey responses, through the identification of the occurrence of similar ideas across the data considered, to extract key themes and patterns.

To preserve anonymity, when referring to interviewees in relation to their organisation, as the number of interviewees is small, and their organisations are identified in the project, interviewees are not attributed a specific reference number. This is to prevent them from becoming identifiable the number reference used across other instances in the thesis.

Throughout the interviews and online survey, the terminology used refers to the engagement of underserved audiences. This changes to empowerment in Chapter Five, following the integrated analysis of Drone and Heritage Expert data on the opportunities of drone technology. This research's focus is on underserved audiences generally, since the aim is for the considerations of this Doctoral project to apply to different marginalised groups. Therefore, the concerns discussed in this chapter relate to a wide range of non-visiting excluded groups. Specific underserved audiences are considered in Chapter Six.

This chapter is structured into three key sections, each critically reviewing an identified barrier elicited in the data. Each section outlines what the theme elicited in the data encompasses and breaks it down into associated subcategories. Section One (4.1) discusses Usability. This factor includes practical, legal, and technical aspects of deploying drone technology. It goes beyond the necessary knowledge and skills to fly a drone, and it is associated with Governance (who owns the land and the required consent and permissions), by-laws, internal organisational drone management processes, strategies, and policies.

Drawing on organisational practices, this section reflects on how heritage experts' views are shaped by drone policies and broader position on drone use in their organisation. Challenges such as limited funding and lack of resources have been excluded from this discussion, because they reflect a context that is systemic in the heritage sector (Heritage Alliance, 2024, p. 34) and applies broadly to the sector's operations and programmes, rather than specifically to drones.

Section Two (4.2) considers the “Disruption of expected experiences of heritage”. This perception integrates two types of disruption: (i) public opinion of drones informed by irresponsible use and heritage visitors’ anxieties over noise, intrusion, safety, and privacy; and (ii) drone views idealising representations of heritage, creating false expectations of the heritage experience. I reflect on how organisational drone policies and internal attitudes to drone technology inform heritage experts’ points of view on the barriers and risks of UAVs in engaging left-out communities.

Section Three (4.3) discusses “Accessibility”, and how the use of drone technology may amplify limitations to underserved audiences’ access to heritage, therefore becoming a barrier to their engagement. This includes examining socioeconomic factors that influence access to drone technology, the context of digital exclusion and illiteracy, and the need for engagement practices to be tailored to groups’ needs. This section follows to contextualise perceptions of accessibility, as a barrier within heritage organisations’ audience engagement strategies, and their identified priority groups, considering how the characteristics of these groups relate to heritage experts’ views.

The concluding section reflects critically on “Usability”, “Accessibility” and “Disruption” (of experiences of heritage), as perceived barriers and risks to using drone technology to engage underserved audiences. It follows to discuss how these three perceived factors affect heritage organisations’ approaches, and uses of drone technology in audience engagement. This chapter contributes to literature discussing digital technologies in heritage, the naturalisation of drone technology and its use to engage non-visiting, underserved heritage audiences.

4.1 Usability: how practical aspects of deploying drone technology are perceived by heritage experts as limitations to its use in engaging communities

This section considers the wider context of usability as a perceived barrier, by referencing heritage experts’ views back to organisational drone policies, and internal institutional attitudes towards drone technology. The term usability encompasses responses linked to the planning and deployment of Unstaffed Aerial Vehicles and collection of data from their activity. It is defined in this research, as all aspects of drone use relating to planning; organising; understanding; and operating drone technology and its products. Considerations of usability are elicited by all ten Heritage Interviewees. They also emerge in a third of the online survey responses, when participants are asked which three words come to mind when they think of drone technology. One respondent[16] reports: “legality, manageability, use.” Another[17] mentions “permissions, privacy, licensed”.

Three main themes emerge: logistics and regulatory factors; knowledge and technical expertise; and data management. Logistical and regulatory aspects are inherent to planning and organisation before and after deploying a drone, including obtaining the necessary permissions. Knowledge and expertise encompass the technical knowledge and understanding of UAV technology needed to operate it.

Data management issues arose as drones used in heritage create a large quantity of data (as a result of the production of high-resolution photography and video or more complex data collection systems such as LiDAR[18]); this data requires storage, which involves financial concerns, considerations of the longevity of data, legacy software to access it, and the latter’s integration with newer systems. These issues are part of the understanding and operationalisation of drone products.

This research focuses on the perceived barriers above, because they reflect aspects of the positionality of heritage institutions, in relation to drone use, power, and authority. Firstly, how regulations and processes establish who has access to flying drones on site and who does not, and therefore whose drone gaze is represented in institutional narratives. Secondly, what level of skill and knowledge are required to be able to fly drones on heritage sites. Thirdly, which content is valued as data and is stored, given visibility, and who makes this decision.

[16] Survey respondent 95183461
[17] Survey respondent 94677824
[18] Light Detection and Ranging. Remote sensing using laser to calculate distances between the laser and the ground.

4.1.1 Logistics and regulatory factors

Logistics, laws, and regulations are the barrier most referred to by heritage interviewees regarding the use of drone technology. Over 50 % of heritage interviewees consider logistics as a limiting factor to their use of drone technology. Heritage experts highlight the necessary administrative and pragmatic deliberations, such as "[…] practical physical issues of putting the aircraft in the air" (Heritage Expert Four).

These include: the most appropriate drone for the task; times when there is no visitor traffic on site; integrated technologies (cameras, LiDAR); launch location; weather forecast; cloud formation; battery life. Heritage Expert Three argues that "it's difficult because you've got so many different laws, several different considerations […] it is too much of a faff". Here, what the interviewee refers to as "faff" relates to the manageability of the process. Heritage Expert Three's view is echoed by one of the survey respondents[19] in terms of the numerous factors to consider.

Nearly 25% of online survey respondents indicate legal considerations, as the key barrier to using drone technology to engage non-visiting audiences. Interviewees refer to aspects such as consent from landowners (Heritage Expert Two), and internal sign- off (Heritage Expert Seven). Legal frameworks that inform the deployment of UAVs include UK Law; Civil Aviation Authority's Drone and Model Aircraft Code; specific regulations for different flying environments (including airspace restrictions); by-laws; General Data Protection Regulations.

As highlighted by Heritage Expert Four and Heritage Expert Nine, landowners, tenants, residents, businesses, and farmers are stakeholders of the National Trust and SDNP. Most of the land in the latter is privately owned. The South Downs is also the highest populated National Park in England. This means these organisations are required to obtain authorisation from communities or individuals. Furthermore, some National Trust properties in the South-East are themselves in the SDNP, for instance Highdown Hill, near Worthing, West Sussex (the location of one of the co-research livestreaming activities). As such, observance of by-laws, obtaining consent and permissions are part of the administration of drone use and good

---

[19] Respondent 95183461

stakeholder management, which is also a reputational factor for both organisations.

Heritage experts interviewed are mostly commissioners of drone flights, not pilots, and therefore they are less immersed in the practicalities of the technology itself. Only Heritage Expert Three mentioned battery life as a barrier, for instance. It does not mean that this is not a barrier and risk, but that it is less present in heritage experts' minds, as this is within the pilots' remit and not the commissioners'.

The perspectives shared here are an expression of the interviewees' experience, or perception, of the administrative tasks involved in commissioning, and organising drone flights in their organisation. This leads to considerations on how internal organisational attitudes to drone technology, and institutional drone policies, inform interviewees' points of view. Heritage Expert Eight's remarks that "there are some barriers that exist because they're barriers within the National Trust, and they're some of our attitude to drone use in our places." This indicates a perceived co-relation between the organisational attitudes towards drone technology, and staff's perceived barriers to using it to engage left-out communities.

i. The National Trust's organisational attitudes towards drones

The National Trust, who operates a no-drone policy for the public, conducted an independent internal survey, separately from this research: *Attitudes to Drone Use by National Trust Staff* (NT, 2022). This survey was used as a source to explore their organisational culture in relation to the use of drone technology. The survey was sent to all staff, and received 143 responses, across five job families: Countryside and rangers (above 57%); Building survey (over 18%); Wildlife and countryside (over 13%); other: unspecified (5%); Archaeology (over 4%).

The survey reveals that a high proportion of their uses of drone technology are operational. This is consistent with the Trust's restrictions on recreational use of drone technology from or at their properties. It is also consistent with the reportedly perceived risk of showcasing their use of drone technology for public engagement, as an encouragement of drone enthusiasts to deploy drones in National Trust land.

Respondents report they mostly deployed drones to undertake surveys of vegetation or habitat; building or structure; general topographical; archaeological; farm and fauna. The second most frequent use is general filming and photography. When asked about their opinion on how the Trust should manage drone use in future, staff are provided with options for answers that do not include considerations of change of policy or using drone data and assets beyond the purposes of documentation and monitoring.

The answer options focus on possible inhouse models for processes and access to drone equipment and services, including the option of in-house pilots. Free text answers are consistent with this, for instance, "the opportunity to have access to a drone quickly in-house to help with day-to-day work and to capture moments of the work being done would be invaluable." Survey participants either comment on procedural and operational elements of using drone technology, or on the need to retain or amplify the no-drone policy messaging to deter or discourage amateur drone pilots.

The information provided by the survey is found to be limited for four reasons. One, it only includes perspectives on internal use of drone technology. Two, it provides an overview of a very small population in proportion to the number of staff working at the National Trust. Three, the views of the Archaeology Team (just over four percent) and the Nature Data Team (no participants), who make significant direct use of drone technology, are significantly underrepresented. Four, the questions are only focused on practical issues relating to the past use and management of drone technology by the Trust. Therefore, the results project a partial perspective of the use of drones at the National Trust.

Based on this internal staff survey, independent from my research, the National Trust's organisational culture towards drone technology is generally conservative. In the NT survey, nearly 60% of the staff only anticipated using a drone once or twice, in the following year (2023). The survey also shows drone use at the National Trust is focused on purposes that primarily serve the roles of organisational stewardship and custodianship (surveying, monitoring, public engagement content).

My empirical experience of organising drone flights at National Trust sites, including at a property (as part of the autotechnography), reiterates their internal conservative and nervous position towards drone technology. Drone use outside of institutionally prescribed aims is perceived as a risk to NT's role of safeguarding of built and natural heritage. This is described by one of the NT interviewees in this research, as "practical health and safety risks in terms of using the technology […]. There is a risk that it might fail or fall, and that can potentially damage property, hurt people, or animals." Nevertheless, these risks are present independently from the context of use, except for the proximity of non-involved people during a flight.

On the other hand, another NT interviewee argues that "our biggest hinderance would be ourselves to ourselves". Their point of view is that a different mindset will lead to a more open internal discourse. Instead of a no-drone policy, the Trust's internal conversation would be: "Yes, we could do that, there's a bit of a risk, but we can manage that risk, or we can control it in a certain way." They report that, broadly, NT staff are supportive of the no drone policy. Although this interviewee works at the NT, they express different attitudes towards barriers and risks to using drone technology to engage underserved audiences.

This means that organisational context, although important to perceptions of use of drone technology, is not the only factor shaping heritage experts' perspectives: there is nuance in their considerations. Drone Expert Seven highlights this when discussing NT's no drone policy: "it comes from the people who are coming into things like the National Trust and it's, what are their backgrounds." Findings from analysing the data confirm this.

Staff's professional background, role or area of professional practice informs their perception of drones beyond simply working for the organisation. For instance, two of the three interviewees from the National Trust work in a technical area of heritage, one in an area that combines technical and academic elements, with communication of narratives and public engagement. It is the latter interviewee who suggests the Trust creates its own barriers through its culture and mindset towards risk.

ii. The SDNPA’s organisational attitudes towards drones

Unlike the NT, the South Downs National Park Authority does not have a fixed no-drone policy. Prompted by a high number of visitor complaints (mentioned by an SDNPA interviewee) the Park commissioned an issues paper in 2018. This paper recognises the existence of sensitive areas which are of higher risk (areas with a high tourist footfall, areas protected by by-laws where drone flying is banned), and the identification of preferred areas for drone flight.

The paper concludes with a proposal for a “Drone code for the National Park-Responsible visiting” (SDNPA, 2018, p. 21). This recommendation attempts to balance the need to protect the integrity of the most special areas of the park, avoidance of disturbance of wildlife, and protection of the Park’s quality of tranquillity. It also includes guidance for drone flying that extends to Park staff, partner organisations, and recreational users. The paper proposes continued monitoring of the use of drones in the park, internal staff discussions and review, and continued conversations with landowners.

The issues paper (2018) draws on examples from eight other National Parks in the UK; five of them, as reported in 2018, adopt an approach of responsible drone use. The commissioning of this issues paper indicates the SNDPA’s proactive approach to UAVs. In the issues paper, drone flying management is placed at the same level as other recreational activities. Therefore, it is understood as a different way to benefit from the qualities of the park and to enjoy it.

To evaluate how visitors to partner organisations perceive drones I examine Visitor Surveys from the Cairngorms and the South Downs National Parks. In the Cairngorms Visitor Survey, (Cairngorms National Park, 2022) one out of the 274 survey participants comment on the disturbance drone flying causes to peace and tranquillity. In the same comment, they also refer to the Park being Disneyfied, overrun with people. When asked what could have been done to improve their experience of the park, two out of 632 responses mention a ban on drones.

These responses also include preventing people from "[…] playing bagpipes, and shrieking", making noise or using mountain bikes. These public responses to drones from visitors, in the case of Cairngorms National Park, mirror how the South Downs National Park Authority frames drones: in the wider context of diverse ways to experience and enjoy the qualities of the park. Neither of the reports for the South Downs National Park Authority, from 2018 or 2021 mention drone flying in the factors detracting from enjoyment.

iii. Other heritage organisations

Online survey responses in relation to organisational policies and drone use reflect a combination of the NT's and SDNPA's positions. 50% of respondents report that their organisation has used drone technology for functional purposes, but only six percent have used it for audience engagement, specifically within a Public Relations approach (social media, website videos for broader audience engagement). This demonstrates that even when used for audience engagement, drone technology is a tool for reiterating the veneration of the institution and its power – an issue this research aims to challenge. 37.5% of the survey respondents in this research reported their organisation has a drone policy. Those without a drone policy may be, like SDNPA, exploring the current landscape or considering how to reconcile drone use, as a new way to experience heritage, with traditional visits.

iv. Legal and Regulatory frameworks as instruments of gatekeeping

Although both the NT and the SDNPA operate in the same country (UK) and under the same law and regulations (except for some hyperlocal variations), their approach to drone use is markedly different. The National Trust has a restrictive policy, whilst the South Downs National Park Authority explores how to support different ways of experiencing landscape. Nevertheless, both heritage partner organisations deploy drone technology for their own purposes, including public engagement.

Drone Expert Seven is the only drone user who specifically addresses this issue, in their interview. They argue that the National Trust uses their authority to condition access, but that this shows a lack of understanding of the technology. This drone user urges NT to "go out and find out what these things can do". One of this Drone Expert's key points is that heritage institutions, such as the National Trust and English Heritage (now Historic England), do not wield the control they attempt to exercise through their policies, because drone users, and they included themselves in this group, find ways around it: “they are not even gatekeepers anymore, you know, they are not controlling in any way, so there’s a delightful bit of anarchy going on.”

This idea of anarchy is, of course, from the perspective of heritage policies and ideas of control. Drone Expert Seven mentions *Pinned On Places* as an example. They film and use drones in heritage sites to create virtual tours of those places. This is done “despite” (Drone Expert Seven) the organisations. In this example, content creators select the sites and the narratives they are interested in and make them accessible to their audiences. This undermines the role of internal policies and regulations as an exercise of institutional privilege.

A National Trust Interviewee comments that “[w]e get really upset with people fly over our places, even if they’ve taken off somewhere else. And, on the whole, we would rather they just went away and did it somewhere else.” This interviewee shares that visibly using drone technology to attract and engage visitors creates a risk to the enforcement of their no-drone policy, and frames that policy as a gatekeeper privilege: “if we start very obviously using drones, as a, as a kind of an engagement tool or a means to draw people to a place it makes it very hard to, to defend that no drone policy.”

The organisation is aware of that privilege and how adding public engagement uses, to the functional applications of photogrammetry, surveying of fauna and flora, multispectral imaging (listed by a National Trust Heritage Interviewee) dilutes their sense of control, erodes an aspect of their institutional power. The no-drone policy at the National Trust is a reactive approach that uses law and regulations, as a scaffold to sustain an existing mindset.

On the other hand, the SDNPA’s approach to drone policy is responsive: they choose not to impose a policy “and see what would happen in terms of how the public were using drones” (SDNPA Heritage interviewee). This is grounded in the notion that “[w]hat we have to do is find positive ways for them to do that, rather than shutting stuff down and saying you can’t, it’s about promoting good practice.” (SDNPA Heritage Interviewee).

Instead of gatekeeping, the Park considers how drone technology can be a gateway to engaging non-visiting audiences. An SDNPA Heritage Interviewee comments: “I think in some ways, at the moment, now is the time for us to start taking risks, so there may be things that we get wrong but rather get things wrong, than take no action at all, when it comes to addressing balance of use.” Nevertheless, the SDNPA does not own most of the park land. This means drone use is conditioned by the perspectives of the Park’s stakeholders, including the National Trust.

Depending on their position on drones, consent for drone use is either part of the gateway or gatekeeping. This is where regulations can be effective barriers, that can disrupt the Park’s own work negotiating the needs of stakeholders and users. These dynamics are not specific to drone use and are symptomatic of “a base level of intolerance about (…) outsiders coming in and using national parks (…) a base level of intolerance (…) between the people who live in these spaces and how they perceive tourists or other visitors, or day users” (Heritage Expert One). This intolerance and the idea of the other as an outsider reflect the perception of exclusivity in heritage.

This is what is protected by the regulatory frameworks that establish what outsiders’ behaviours should be, in relation to the insiders’, their ownership of land, property, and landscape, and subsequently how these are experienced, narrated, and accessed. Revisiting the consideration of how regulations and processes establish who has access to flying drones on site and who does not, I posit that regulations have another role beyond promoting responsible behaviours. Legal and regulatory frameworks are an expression of privilege in relation to those who have the power to set them. They are used, in this case, as part of a performative gatekeeping artifice that reiterates power. This may be through land ownership, local residence, or asset custodianship, as is the case with the National Trust.

What this means to the research question this chapter responds to is that law and regulations constitute a barrier, not to new uses of drone technology to engage audiences, but to the

National Trust itself and other heritage organisations. They enable the existing organisational culture and are therefore an obstacle to the organisation's own reconfiguration into new models of engaging audiences outside their membership or current visitors. In the case of the NT, this is changing and in the process of this research project, I have been able and supported to test new approaches (discussed in Chapter Six) on National Trust properties and sites, but only working with a certified drone pilot.

4.1.2 Knowledge and technical expertise

In the surveys, the knowledge and specialist training involved in deploying drone technology are the two most elicited barriers (by 40% of respondents) to using drone technology to engage left-out audiences. This is mentioned by only one interviewee. Heritage Expert Three considers: "I will just have to stick to what I know; I lack […] expertise". Currently there are different regulations for flights in different categories. The larger the UAV, the higher the requirements.

As discussed previously, heritage organisations mainly use drone technology for functional purposes. The survey responses evidence this. Of the 63% of respondents who confirm that their organisation has used drone technology, five report using UAVs strictly for surveying, monitoring, preservation. Four of them state that their organisation uses UAVs for both functional and promotional uses (marketing, for instance). Two indicate that their organisations use drone technology for promotional purposes only. Both types of use require high technical specifications, in terms of the data they produce.

This means using larger drones, because these have higher levels of wind resistance and can carry a higher Maximum Take Off Mass (MTOM) – and therefore bigger cameras or other devices that integrate the device with other technologies. Using drones over 250gr requires certification by the Civil Aviation Authority. This involves training and a test by a CAA recognised assessment centre, logged flight hours, obtaining a flier ID and or an operator ID.

One of the key factors here is how much of an appetite there is in organisations for training staff in drone piloting. In the drone survey carried by the National Trust with their staff, 44% of the participants would like internal drone use to be undertaken by external licensed contractors. This denotes an apparent lack of interest in having an inhouse team, or licensed drone pilots within the job families making use of the technology.

As seen in the previous section, Drone Expert Seven considers that gatekeeping is a result of organisations lacking understanding of drone technology. Heritage Expert One reiterates this, when they are asked what comes to mind when they think about digital technology. They consider: "the potential for digital technology, but also the potential focus we need to take on increasing digital literacy as a skill set in the sector." Their comment goes beyond the need for the knowledge and skills necessary to adopting digital technology: it brings into focus, a process of adaptation to the presence of these technologies.

Heritage Expert Three comments that, in their organisation, "technology-averse rangers" would not adopt the technology, unless they were forced to. This highlights that what is central to the sector, in terms of taking advantage of the benefits inherent to new digital technologies, including UAVs, is not necessarily the specialist knowledge to fly a drone, but openness to digital technologies as proposals for new ways of working. This is, more than training and certification, the key barrier: "persuading […] perhaps more traditionally-minded landscape sector professionals, of the potential". The acquisition of technical knowledge and specialist expertise are not the key obstacle, although they are perceived as such on the surface.

In view of the data discussed above, and to answer my research question, the barriers are threefold: the lack of understanding of what drone technology engenders, beyond its functional applications, and what that means for heritage organisations' approaches to engagement. They also include resistance to embracing new ways of working, partly due to risk aversion. The latter is a mindset strongly associated with heritage institutions' reticence in opening themselves to uncertainty and failure. It is worth repeating Heritage Expert One's quote here: "I think in some ways, at the moment, now is the time for us to start taking risks so there may be things that we get wrong but rather get things wrong than take no action at all, when it comes to addressing balance of use."

This is reiterated by some heritage interviewees and in most survey responses. When asked what would encourage them to use drone technology to engage underserved audiences, Heritage Expert Ten would like more confidence in using the technology. Heritage Expert Two suggests “a protocol of how to use it for engagement”, which necessarily would involve a clear understanding of the potential and affordances of the technology.

Heritage Expert Three highlights that, for different uses of UAVs to be approved, Senior Management need to align them with their strategic priorities, so that funding is committed to proposals. This interviewee suggests “we would need an organisation cultural change.” Survey respondents indicate even more clearly that they would be encouraged by gaining clarity on the capabilities of drone technology: 36% of respondents answer: “Better understanding of the technology”, “Being able to access case studies from the sector”, or “Confidence”; while 25% survey respondents answer “Training”, “Trained Staff”, or “Trained volunteers”.

This data suggests that when knowledge is identified as a barrier, this effectively recognises the need for heritage organisations to gain a better understanding of drone technology, in order to feel more confident and take risks. The notion of knowledge as a barrier is predominantly about Heritage organisations feeling in control, and not vulnerable to failure.

4.1.3 Data management

Although “Data management” is mentioned as a barrier only once in the data, it is included here, because it can be central to bringing a shift in empowering underserved audiences. Handling the data produced via drone technology is a challenge, that organisations must consider in deploying the technology. As Drone Expert Seven mentions, with the widespread use of the technology by hobbyists, a large number of videos and photographs are generated and uploaded to social media platforms. For organisations, specifically, this means finding ways to store drone data and sourcing software that will enable continued access to legacy content. This is a barrier to UAV use, due to the associated costs and potential need for staff upskilling.

It is also a barrier, because by empowering communities, heritage organisations promote the generation of more content and new narratives using different aesthetics to the institutional visual language. This challenges the mindset with which heritage and cultural organisations approach drone aesthetics. Although it is a risk, institutions would benefit from content that is more relevant, representative, and impactful. As a result, data assets would likely be reused more consistently and more widely, both for functional and engagement purposes. This means that sustaining and storing UAV-produced data necessitates that their value and importance is recognised, and that they become part of the organisation's operational needs.

Recognising the value of drone visual data generated by sources other than heritage organisations themselves (for instance, hobby drone uses), would go towards addressing the balance of power and the "democratisation of culture" Drone Expert Seven mentions in their interview, and which is discussed in Chapter Five, where the opportunities of drone technology are considered.

Section conclusion

The reviewed data indicates that the perceptions of regulations, as barriers to the use of drone technology to engage left out audiences are predominantly associated with control. Regulatory frameworks enable heritage organisations, and their stakeholders outside of their underserved audiences, to enact a control over who has access, who creates the narrative of Heritage, and how it is composed. They are tools to negotiate levels of exclusivity of heritage, place, and belonging.

Although UAVs are an evolving technology, heritage institutions use it to sustain traditional ways of working; they understand how UAVs integrate with their existing ways of working, making them more efficient. However, when reflecting on the use of drone technology to engage underserved audiences, heritage experts consider knowledge and specialist training as a barrier, because this type of use is not just about using the technology: it is about new ways of working overall. As discussed in the Literature Review, (Section 2.2) the heritage sector is developing approaches that redress the balance of power with underrepresented and excluded communities.

Organisations are still struggling with approaches that reframe their control over the work they do, such as co-design and co-production. Using drone technology for audience engagement is part of this change, in that it presents itself as another form of interrogating institutional control. Organisations are finding it challenging to understand what the full potential of UAVs is in this context of negotiation between having control and sharing it. One interviewee compares the no-drone policy to the no photography signs at the NT which were removed because of the ubiquity of smartphones. Like those signs, the policy is predominantly grounded in a protectionist approach to heritage content: who produces it, who shares it, whose heritage is it? This correlates with institutional decisions on how UAV data is managed: what is stored (and therefore valued) and what is left out (and therefore whom?).

4.2 Disruption of visitors' experiences of heritage

The theme of drone use presenting a disruption to the intended expectations of the heritage experience is elicited in five of the ten Heritage Expert interviews. It is not mentioned in the online surveys. Although second in the order of perceived barriers and risks, this theme is important in my research because it relates to challenging the hierarchies of knowledge and the conventions about what engaging with heritage looks like and means. This theme is elicited in the data in two ways. First, the public opinion of drones is informed by their irresponsible use and heritage visitors' anxieties over intrusion, safety, privacy, and disruption over noise. Second, drone views show idealised representations of heritage, creating false expectations of what the experience of that heritage is.

First, I discuss the perceptions that are a result of drone technology's boomerang effect (Jensen, 2016b) as discussed in the Literature Review (Section One). The acceptance of drone technology by wider civil society is challenged by the drone being a technology of warfare and surveillance, permeating ordinary lives, and by incidents of irresponsible drone use.
These inform people's fears of intrusion of privacy and security. This is not the only factor influencing public opinion. Noise from and visual contact with drones also contribute to drone technology's reputation as altering the quality of experiences of heritage.

To the above follows a discussion of heritage experts' perceptions of the auratic effect of drone content, and how that is disconnected from the actual experience of heritage. This creates a distance between the expectations of the visitor, and the reality of their experience, which further separates viewers from the heritage landscape.

This section concludes with a reflection on what this data means for my research question on the barriers of drone technology to engage underserved audiences with heritage. The tensions between established and new ways to experience heritage are explored, as well as the repositioning of the relationships between institutions, communities, and different types of visitors.

4.2.1 Public opinion of drones

Public opinion refers to the perceptions of drones upheld by members of the public. As a theme, it is mentioned by less than half of interviewees, and only six percent of survey respondents. What the general public think of drones weighs greatly in my own research practice and when undertaking my autotechnography. It also occupies a certain level of concern in interviewees' minds. As reviewed in Chapter Two, there exist several studies on public opinion of drones (Jensen, 2016b; Kim, Kim and Kim, 2016; Tan, Lee and Gao, 2018). Within public opinion, there are two key themes elicited in the data: intrusion of privacy and security, and disturbing visitors.

i. Intrusion of privacy and security

In the online survey one respondent [20], points out that drones "unfortunately suffer from a mixed public image due to their misuse by a minority of ignorant/malicious users." Four heritage interviewees mention negative public opinion as a barrier to using drone technology for engagement of excluded audiences. Heritage Expert Four discusses visitors concerns with "data protection and images of people, privacy" and the "perception of intrusion. That someone's spying on you". This interviewee further refers to the role of Heritage organisations: "we have a duty of care to those people to preserve some degree of privacy for them" (referring to audiences visiting their heritage site). They add that visitors

[20] Respondent 94663288

feel uncomfortable by the potential for drones to take unwanted photos of themselves.

In the online survey, “Risks to privacy and security” are the second most identified risk of using drone technology to engage underserved audiences. Drone users interviewed in my Doctoral project refer to being reproached by members of the public even when they are flying professionally: “I’ve been shouted at for trying to invade other people’s privacy, which I haven’t done but you know it was perceived that I was.” (Drone Expert Five). Public opinion is alluded to by three of the eight drone experts interviewed (three of these are hobby drone users).

Nevertheless, in a digital society where a significant part of sensitive aspects of life is facilitated and mediated by digital technologies (banking, communications, shopping, entertainment, learning) these risks are present in other aspects of life, including interactions with heritage organisations’ systems. Examples include CCTV cameras, downloadable apps, online booking systems, electronic banking. Individuals’ privacy and security have vulnerabilities. Digital footprints that can be misused. Data and devices can be hacked. It is possible for anyone to use a varied type of digital technologies, from the ubiquitous smart phone to camera glasses, maliciously. Issues relating to the safeguarding of personal data are not strictly associated with drones. They are a concern inherent to the vulnerability of digital security in a “digital first” society.

There is, recognisably, and as discussed in the Literature Review, Section One, a psychological resistance that impacts on people engaging positively with UAVs (Kim, Kim and Kim, 2016, para. 1). According to Jensen (2016b) this resistance is a result of the highly computerised nature of UAVs, as they carry large amounts of data, and of their vulnerability to hacking. Furthermore, an unfriendly aesthetics, a history of activities demonstrating the drone operator exercising power over its target and disempowering it, unpredictability, and difficulty to identify who is in control, all add to the factors contributing to what, in the online survey, is referred to as the “bad reputation of drones”. These characteristics appear to be at the core of public anxieties, compounded by fears of ubiquity which pervade collective thinking. Research by Kim, Kim and Kim, (2016, para. 32) finds that people want an element of adorability, some predictability, and some control in companion drones. These characteristics aim to create relatability in interacting with the drone.

One of the survey respondents[21] comments that drone technology is "in a transitory phase, we don't know yet what it will become". This consideration is evident in drone interview data, when Drone Expert Six comments: "it takes people changing their perception of what the technology is first". This is at the centre of the challenges for heritage institutions. They have not reconfigured their perceptions of drones beyond the current applications of monitoring, inspecting and surveying, Drone Expert Four, who is both a commercial and a hobby drone pilot shares how they reassured their neighbours, about their use of drones in the neighbourhood: "I mean luckily, all of my neighbours are very happy, and they know I do drones and so, what I've done is, I've said would you like to have a look and you know, show the screen and say, you know, if you want, I'm flying over your house, I'll take some photographs, you can check your tiles".

Applications such as monitoring and surveying, associated with benign drone use, draw on drones' capabilities of vision, combined with remotely operated mobility and verticality. What separates monitoring, inspecting, and surveying from surveillance and spying? In both, the drone is a tool of control, data capture, and regulation. However, in the latter, it is part of a dystopian (Hildebrand, 2020) imaginary where the drone is a covert tool of control, data capture, and regulation. It is therefore not necessarily the use but the context in which drone technology is used that informs its negative perception.

Drone experts consider that, for the negative perception of drones to change, the potential of drones needs to be showcased. This is reiterated by heritage experts, as they recognise that they need a better understanding of the technology through case studies, for example. Drone Expert Eight suggests: "do kind of open days for communities to actually experience it because they will change their opinion". Drone Expert Six reiterates this by suggesting a mindset change requires taking "the time to educate people": maximising opportunities to showcase a variety of positive uses, and for drone pilots to interact with communities, introduce themselves, talk about how they use drones (Drone Expert Three).

[21] Respondent 94663288

ii. Disturbing visitors

The other aspect that affects public opinion is the idea the presence of drones interferes with visitor enjoyment. There is a perceived tension between drone technology and what users recognise as “traditional” forms of experiencing heritage (Heritage Expert One). This tension has, reportedly, resulted in SDNPA receiving a significant number of complaints from the public, about drone rallies. Tranquillity has been identified by visitors to this site as its most special quality; drones challenge this.

This perspective is shared by a heritage expert from the National Trust, who mentions noise as a consideration in the use of drone technology. Drone Expert Four also comments: “they make that noise, don’t they, they sound like a bee or a swarm of bees”. In my online survey of UK-wide heritage experts, “disturbing and spooking visitors” is the most mentioned risk of using drone technology, referred to in 19 out of 22 responses.

This concern for disturbing visitors is linked to a construed notion of how natural heritage should be enjoyed. Traditionally, visiting the museum, library, art gallery, is a solemn experience where visitors express reverence for the object through their behaviour in the space: being silent, following the flow of an exhibition, not running, or adopting any other behaviour differing from what is expected. The post-modern museum attempts to be interactive, more person-centred and layered to provide different things to different people, as gateways for participation and engagement, including life events such as weddings and children’s parties.

Visitors are more accustomed to the excited school groups or the parent and toddler gallery storytelling. Visitors to natural heritage sites such as National Parks, also engage in ways that differ from a solemn or reverential experience: skydiving, cycling, water sports, weddings, kite flying, dog walking, and now drone flying. I suggest that the idea of drones disturbing visitors being a barrier to their use is associated with heritage sites protecting existing ways of operating and exercising control through regulating visitor behaviour.

4.2.2 Disruption of the experience of heritage

This theme relates to drone technology affecting how audiences engage with heritage. Heritage Experts Ten and Two suggest that drone technology may cause disconnection from landscape, as evident in Heritage Expert Two's quote "there's balance to be had about the engagement with the technology and the actual living in the moment immersive experience of being in that place." These heritage experts articulate the historical construction of heritage and cultural sites and objects, as unique and important in their materiality. As carers of these invaluable assets, heritage and cultural institutions' position themselves as knowledge authorities and protectors of that uniqueness, building an aura around the object and site, and using a projected aesthetics to sustain it.

This evokes Walter Benjamin's theory, that the mechanisation of means of reproduction results in the erosion of the aura (Benjamin, 2007). Drone technology is part of the mechanisation of ways of doing, seeing, and experiencing environments. How might it erode the aura? The auratic (discussed further in Chapter Five) is sustained by creating a distance between the object or subject and the viewer. This is done by projecting the subject as perfection, an unattainable ideal of universal value that is greater than the narratives and experiences of viewers. In heritage, that distance is created through interpretation strategies that elevate the uniqueness and symbolic value of the heritage object or site.

Heritage experts in the online survey and interviews discuss the importance of drone technology being framed within interpretation practices, rather than used as a broad access tool[22]. I consider this argument to be a form of gatekeeping: it retains the heritage organisations' positioning of authority and guardianship instituted since the first private collections, precursors to the museums we know today, were started. The concept of authenticity is a constituent part of Heritage Expert Two's concern with the digital experience not detracting from being in the landscape. This idea of authenticity excludes the experience of the digital object. This reflects an institutionalised view of authenticity and experience, that protects the idea of uniqueness.

---

[22] Respondent 94678072

Two interviewees challenge these archetypes of presentation of heritage and landscape. Heritage Interviewee Eight discusses the potential dilution of the heritage experience when mediated through a drone. Their excitement for the potential of drones is underlined by their concern over an approach, where the idolatry of the drone is placed ahead of its purpose of access and facilitation: "stuff that's about what you're about and stuff that's focused on technology for technology's sake and look at the shiny thing". This quote merges the "identity" of the tool, in this case, the drone, with the subject (the site or collection). Similarly, Heritage Expert Six questions the authenticity of drone content: "[…] drone photography can certainly make things look far more epic than they are."

Certainly, institutions use drone aesthetics to project an idyllic, curated perspective of their sites. Heritage Expert Eight expresses discomfort at the idea of this spectacle: "You create a spectacle and that entertains folks and so they come along but not feeling totally comfortable with that. And, you know, maybe, maybe the greatest thing they can do is, is find the richness rather than the spectacle." This idea of richness Heritage Expert Eight proposes as the way to connect with underserved communities implies that, more than to amaze and impress, drone content should be used to connect. To uncover different ways of seeing and being in heritage, new narratives of place and belonging, that enhance the relevance of heritage and cultural sites. That is where I consider the richness lies.

To conclude here, heritage organisations' perception of the drone, as distorting the experience of heritage by idealising it, is an important barrier to consider in the use of the technology. It is important to note that auratic aerial views from drone content are utilised by institutions, as tools to reiterate authority and project status. Although a small number of heritage experts recognise that presenting heritage as universally spectacular is a misrepresentation of its reality, they do not consider it more critically in the context of composing their brand and reiterating institutional power. This is explored further in Chapter Six.

Section conclusion

Disruption of the expected experiences of heritage is seen as a barrier to using drone technology to engage left out audiences. Concerns with public opinion and disturbance reflect a preoccupation with retaining familiarity for existing users, by protecting their experience of heritage. The Disneyfication of heritage sites has in the past been argued as disruptive, for traditional visitors. Now, interactives, games, screenings, drama are accepted ways for heritage sites to meet the needs of different audiences.

The disruption of visitors' expectations is an idea that reflects the tension between the needs of existing audiences, traditional approaches to experiencing heritage, and the disruptive effect of technologies before they become widely accepted. The great difference between drones and other technical tools is that other tools used in heritage may not exist within a recreational dimension, that collides with the institutional approach. As a survey respondent[23] mentions, it is a new technology, its potential is not yet fully explored, and it is still trying to become mainstream.

Heritage organisations have a role to play in helping change the public mindset towards drones. They need to deconstruct the use of auratic aesthetics and find creative ways to distance the use of drones in heritage, from functional utilisation. They must transfer drones from actors in a theatre of conflict and malicious activity, into facilitators of a greater range of experiences – one of the identified opportunities of drone technology discussed in Chapter Five. In this context, drone technology can widen the perspectives and narratives of landscape and help deconstruct established narrative of heritage as institutions that hold knowledge as a privilege.

4.3 Accessibility

The theme of accessibility considers how communities experiencing inequality can access drone technology. As defined in the field of Education Studies, it relates to the degree to which institutional systems make it possible for groups to participate using institutional resources (van Wee, 2022, p. 1). Accessibility is limited by inequalities created by institutional access systems and processes, that frame and regulate it (Stauber and Parreira do Amaral, 2015, p. 16).

[23] Respondent 94678072

Using the context of access to Education as a reference, institutional systems have specific effects on particular groups, which result in excluding these groups (women, Global Majority and disabled groups for instance). The same happens in heritage, as the narratives of such groups are underrepresented or absent, not everyone can access buildings equitably, and the workforce is not diverse.

40% of heritage experts in the online survey perceive the need for knowledge and specialist training (or lack thereof in their organisations) as a barrier to using drone technology to engage excluded communities. However, this perception is less present when they consider underserved groups. This theme does not emerge in the online surveys, and less than one third of heritage experts interviewed consider accessibility a barrier. As digital poverty is identified as a barrier specifically for underserved communities, it is important to establish who heritage experts are referring to, when discussing this factor as an obstacle to engage those communities with their sites. Two factors linked to digital poverty are mentioned as risks to furthering the experience of exclusion: access to drones; awareness of how to interact with the technology.

4.3.1 Accessibility for whom?

When talking about accessibility as a barrier, whom are heritage experts referring to? In their interviews, SDNPA staff consistently refer to their strategy prioritising "people facing health inequalities, including disabled people; people living in the urban fringe, particularly people in the urban fringe facing socioeconomic barriers" (2022). Specific locations are identified as: "Eastbourne, parts of Worthing, Bognor, Littlehampton, Crawley". The audience priorities of online survey participants partly mirror those of the heritage partners: four groups are priority audiences, with the same number of responses: "People with disabilities"; "Ethnically diverse communities"; "Schools"; "Young people".

To provide further context to the organisational culture that informs approaches to audiences, I provide a brief overview of SDNPA's and NT's history. The National Parks are government-funded public bodies. Accessibility has been at their root from inception: they originated from a campaign for public access to the countryside. The campaign for the National Parks was initiated by a member of Parliament and involved "mass trespasses" by members of the public. The 1949 Act of Parliament established the National Parks, embedding the provision of recreational uses.

NT is a membership organisation with charitable status. It was founded in 1895, with the aim to care and protect the country's "coastline, historic sites, countryside, and green spaces for everyone. Forever." (National Trust, no date). The Trust recognises that "everyone" is limited to those who can afford the visiting experience.

National Trust priorities nationally are "to engage and retain the attention of the current membership base" (a Heritage Expert). In the South East, (National Trust staff interviewed includes national and regional roles) they are addressing this and delivering programmes "around a lot of those urban communities and how we engage those with the properties around them" (a Heritage Expert) to widen access and engagement to those who do not feel NT is for them. One of these projects is *Changing Chalk*, a large-scale landscape project, led by NT in partnership with the SDNPA and other cross sector organisations. This National Heritage Lottery funded programme focuses geographically on communities in the urban fringes, and targets nine audiences, including the four groups identified in the online survey.

There is a consensus among heritage experts that underserved, non-visiting audiences are those communities who experience socioeconomic disparity, are underrepresented, excluded, and disempowered by how institutions provide their services. This means they have less opportunities to and benefits from engaging with heritage. When heritage experts discuss digital poverty as a barrier, they correlate this with communities who experience systematic forms of exclusion. This is important to how excluded communities themselves perceive their interaction with heritage organisations, as discussed in the Literature Review in relation to self-efficacy. Expectations of a positive result when visiting a museum or when using unfamiliar technology are negatively affected by the emotional impact of experiences of exclusion. This conditions how communities experiencing inequalities perceive technologies, and their openness to using them.

4.3.2 Digital poverty: access to drones

Digital poverty is defined by the Digital Poverty Alliance (DPA) as "the inability to interact with the online world fully, when, where, and how an individual wants to." (Digital Poverty

Alliance, 2022, p.2). Accessing the technology, as highlighted by Heritage Experts One, Seven and Eight, is perceived as a barrier because "audiences we are talking about are facing real financial challenges." (Heritage Expert Eight). This recognises that excluded communities face material barriers, that impact on access to basic needs such as food, fuel, water, safe accommodation. These needs must be met before considering anything further, such as purchasing a drone.

In this context, heritage interviewees discuss that drones are, as stated by Heritage Expert Eight, "quite an elitist hobby. Expensive." This interviewee adds that there are ethical considerations relating to using a technology that is expensive and out of reach from excluded communities, to engage them with places, sites, and narratives that they feel are not for them. I agree. Nevertheless, I argue against it as a valid reason to exclude the use of drone technology, to engage underserved audiences. These ethical considerations are not applied consistently, through processes and operations in heritage organisations. Processes are rooted in the assumption that smart technologies are ubiquitous, and that everyone is familiar with how they interact with smart devices used in gallery interpretation; that everyone has a smart phone to access QR codes; and that everyone can afford the items in their cafe menu.

Digital poverty is a legacy of the transference of mechanisms of exclusion and inequality, from an analogic to a digital society. The latter reinforce digital poverty (DPA, 2022, p. 4). This was evidenced during the COVID-19 pandemic. As discussed in the Literature Review (Section 2.2), reliance on digital technologies to ensure access to education and health during COVID-19 amplified inequalities and made social injustice more visible. Those disparities did not start with the digital divide: the divide is a manifestation of them. Similarly, if we consider how heritage organisations operate beyond the use of digital technology, we can find examples where underserved audiences are expected to adopt practices that are out of reach to them.

The example of the membership model at the National Trust shows that drones are not the first example of tools that excluded groups feel further distances them from heritage. As mentioned by a NT interviewee, the model "introduc[es] lower cost memberships and the ability to pay monthly, rather than a single annual direct debit"; but who benefits from this change? Does it address accessibility issues for wider underserved audiences, in the same way as the Trust's adoption of free entry for refugees, for instance? Heritage Expert One also refers to how expensive drones are and considers the possibility of their organisation loaning out UAVs. They recognise risks inherent to this, such as equipment loss and damage, and the associated costs of maintenance and repair. However, they do not consider that equipment access does not address the full complexity of digital poverty.

In the hypothesis that heritage organisations increase accessibility to the technology, by providing access to the equipment (they have used this approach previously, with other technologies, such as smart devices in galleries), this would not fully address digital poverty. Bhalla and Lapeyre (1997) posit that the issues arising from social exclusion are only partly determined by material resources. Lack of access to digital technologies experienced by underserved audiences impacts on their acquisition of skills to use those technologies. This leads to the second factor in digital poverty: digital (il)literacy.

4.3.3 Digital poverty: literacy

Heritage Expert One puts forward the idea that organisations could "build enough skills development, and […] digital awareness" towards their social justice role of inclusivity to underserved audiences. Access to equipment and skills are only one part of a more complex problem. Heritage Expert One touches upon this: "We'd have to think about what does use of drone technology mean to people who may not have that level of digital literacy or technological literacy."

How can excluded communities develop the knowledge, understanding and confidence to use digital technologies such as UAVs if, beyond not being able to afford purchasing them (or the services to use them such as the internet, image storage devices for instance), they are engaging with it to interact with heritage, as set by institutional narratives? There is, as discussed in the Literature Review, Section 2.2, a relational and emotional dimension to exclusion. This affects the level of confidence necessary for excluded groups to engage not only with training but, most and foremost, with heritage organisations. The key question here is, how does the use of drone technology to engage underserved audiences with heritage benefit underserved audiences? And what benefits does it bring?

According to DPA (2022, p. 2), to benefit people and places, approaches to digital poverty must "empower people". It is not sufficient to provide access to resources or to support upskilling. Empowering people involves all these aspects, as well as supporting them to shape the where, when, and how the process happens. It is by doing this, that heritage organisations understand what the use of drone technology means to left out communities. This signifies that what the technology represents in society is as important as the technology itself. By this, I mean that if the technology is used in a way that creates disconnection, social tensions, exclusion, the positive use of its capabilities will be overshadowed by this. If it is used to increase relevance and address the power imbalance between cultural institutions and communities, it will have reverberating impact in the social justice agenda.

Section conclusion

Considering accessibility as a barrier to using drone technology to engage underserved audiences reveals that heritage organisations' understanding of it is limited and superficial. This is because it is considered in the context of dated engagement practices (which are progressively changing, as organisations undertake more collaborative work with excluded communities). In this context, heritage experts are mostly concerned about the use of expensive technology which they themselves cannot understand and which, subsequently, they consider can amplify a sense of disconnection from left-out groups.

Nevertheless, this is not the true barrier. The significant accessibility obstacle lies in heritage organisations' approach to use technology as a conduit of active engagement, which means the institution is in control, and audiences interact with a ready-made experience. For underserved communities to benefit, it is imperative they themselves have control.
This is shown by the example of indigenous communities using drone technology to reclaim their territories and reinstate land boundaries infringed on, by governments in a legacy of imperialism (Section 2.1). These groups have power over the use of the technology and the benefits it harnesses for them. By contrast, heritage institutions consider accessibility a barrier, because it requires them to share control with communities.

This is a challenge, heritage and cultural organisations are continuously trying to address in their social justice agenda of museums by and with people, instead of museums for people. Drones, as digital technology, are impactful when used in a context of empowerment rather than reiteration of institutional power. Relevance is crucial to the process, as evident in the example of community activist groups. I believe drone technology has the potential to facilitate this. This is explored further in Chapter Five, through the consideration of the perceived opportunities drone technology offers to engaging underserved communities.

Chapter conclusion

Considering the data presented in this chapter, to answer the research question: “What do heritage organisations perceive to be the barriers and risks of using drone technology to engage underserved audiences with heritage?” this research concludes that the most important barriers are: fear of loss of control (mitigated through the use of legal and regulatory frameworks); a lack of understanding of drone technology and its potential; and protection of traditional models of operation and resistance to instigate new ways of working, for fear of failure and uncertainty.

Regulatory frameworks (though ineffective) are operationalised to protect gatekeeper privilege and help organisations exercise control over who has access and who is an outsider. However, the SDNPA’s approach to observe drone use and public responses to it, before formulating a policy demonstrates that it is possible to negotiate different interests in a complex governance model, without resorting to a ban.

Although lack of specialist knowledge and skills are identified as a barrier, heritage experts consider that gaining a better understanding of the technology, would encourage them to use drone technology to engage non visiting audiences. This means the real barrier is not the absence of knowledge necessary to fly a drone, but the understanding of the potential of drones outside their current functional uses of, for example, surveying, monitoring, and broad public engagement (publications, social media, marketing). As such, it is challenging for heritage experts to contemplate how they can use UAVs to engage underserved audiences.

What is truly at the root of the perception of practical issues, regulatory frameworks, disruption of visitor experiences, and accessibility as barriers, is that they interrupt and change heritage and cultural organisations' ways of working. Change involves taking risks, possible failure, and uncertainty. Organisations, therefore, perceive themselves as placed in a deficit of power, where they feel they should be leading. Power is indeed a key factor in heritage organisations' positioning in relation to their stakeholders. Heritage experts suggest that digital poverty means the use of drones could alienate underserved audiences.

However, heritage organisations use other expensive technologies inaccessible to underserved audiences in engagement work; for instance, SDNPA rangers use Virtual Reality. I argue that, if drone technology is not as accessible to underserved audiences as it is to those who do not experience disparity, this presents an opportunity for heritage organisations to be the place where underserved audiences engage with digital experiences they would not access otherwise, as part of tackling digital poverty and literacy.

The fundamental point is that drone technology is used to empower underserved audiences, not to reiterate institutional power. Revisiting the concept of engagement discussed in the Literature Review (Section 2.2), heritage organisations must involve underserved communities in the design of approaches to connect them with heritage sites and narratives. It is this empowerment, and not the purchase and loan of equipment, that mitigates the effects of lack of access to digital devices and skills.

Empowering underserved audiences means that they have ownership of the drone products generated. Therefore, these are more realistic representations of natural heritage than the auratic drone content produced by heritage institutions. (This is discussed further in Chapter Six). Consequently, drone videos no longer offer a distorted view of landscape that elicits reverence, but an authentic view, that is inclusive and enriching. Heritage organisations and practitioners find it difficult to negotiate the relationship between the past and present. Drone technology challenges temporality by merging both, as the technology engenders experiences that by showing a broader perspective give visibility to different communities and their narratives.

Chapter Five: The unique affordances and opportunities of drone technology for the interpretation of and engagement with heritage

Chapter introduction

> "[I]f you're trying to communicate, if you're trying to get people to engage in landscape, to feel a connection to it, to feel wonder and awe, if you're trying to recreate that awe around space, drone technology does that, because it gives you those pano-views, it gives you that awe-inspiring sense of space." (Heritage Expert One)

This quote, from Heritage Expert One's interview, illustrates the focus of this chapter, which is to answer the research question: "What unique opportunities does drone technology offer the interpretation of and engagement with heritage?" This is done through the analysis of the interviews undertaken with drone and heritage experts, contextualising them with the heritage practitioners' online survey responses.

The analysis of these is integrated with the review of heritage partners' strategic documents and policies. All the datasets analysed are combined through thematic analysis, in order to capture experts' perceptions of the opportunities and possibilities of drone technology, for engagement with heritage. The chapter is divided in four sections.

Section One (5.1) highlights the contextual aspects informing heritage experts' responses and the background to drone experts' views on the opportunities of drone technology. This is structured by the three types of drone users in the data, based on how they use drone technology: (i) Heritage staff and researchers who use drone technology as part of their heritage activity; (ii) Drone operators who use drones commercially as part of their business; (iii) Hobby drone users. The section concludes with a reflection on how this background information shapes experts' perceptions and attitudes towards what UAVs can offer.

Section Two (5.2), under the umbrella of broader and wider, explores the first two of several themes that emerge from the drone and heritage experts' data, namely "Broader perspectives and experiences" and "A greater range of experiences for wider audiences".

Section Three (5.3) considers the themes of physical and digital access and connections: access and accessibility, "more data of higher quality", "meaning-making", and "control and automation", as elicited in the data. This includes discussing UAVs as ubiquitous digital media, which reframes the human relationship with places through mobility, verticality, and sensing capabilities, and considers agency is regulated in drone technology. The section explores how automation relates to the drone auratic gaze that distances us from landscape, by projecting it as an ideal (auratic), and the associated issues of (in)visibility and representation of underserved audiences.

This chapter concludes (Chapter Conclusion) with a reflection on the importance of heritage organisations developing relationships with the drone hobby community. This would be a prerequisite to leveraging the unique opportunities drone technology offers to empower underserved communities to connect with heritage. The Conclusion also deliberates on what identified opportunities mean, to the process of empowerment and agency. Finally, this section considers the concept of aura, and how empowerment and agency challenge the aesthetics of the idealised landscape, proposing a new drone gaze.

5.1 The contexts behind drone user groups' associated perceptions of the unique opportunities of drone technology

A total of eight interviews were undertaken with drone experts. Four interviewees were referred by four different members of staff from the research partner organisations; two were referred by the Project Director of the *Drone Project*, in the United States; one was contacted directly following their presentation at the conference *Drones in Society: New*

Visual Aesthetics (8 -9 September 2022, Sheffield, UK and online); and the remaining interviewee was recruited through a call out posted on Facebook Drone pilot group pages. This section is centred on the analysis of these eight interviews. The three types and characteristics of drone users who participated in this research are summarised in Table 5, below.

Table 5 Summary of the three types of drone users interviewed and their use and associated perceptions of drone technology

Drone user group type and number of interviewees in this group	**How they use drone technology**	**Crossover with other user types**	**Specific use**	**How they started using drone technology**	**Focus**	**How they perceive drone technology**
Heritage practitioners and researchers who use drones as part of their role [3 interviewees]	As tools for their work in the fields of Archaeology and Academic Research.	None	From photography to integration with LiDAR.[22]	It is a part of their role.	Functionality in relation to data capture, and location.	Drones as a functional tool.
Commercial drone operators [3 interviewees]	Run a drone services business.	2 out of 3 drone users in this group, are also recreational users.	Use of varied capabilities of drone technology, from surveying or measuring, to taking photographs and videos.	Started engaging with drones to solve a practical problem.	Functionalities for the provision of business services.	Drones as a functional tool. Have an affective connection with drones, which is associated with the recreational use, which, in turn, elicits an affective relation.
Drone hobby users [2 interviewees]	Make recreational use of drones for their own enjoyment (and who may also be content creators and have an audience).	As above	To facilitate storytelling; a source of thrill and adventure.	A natural result of their interest in filmmaking and videography for one expert and an interest in learning something new, for the other.	Facilitation of interests in film, exploring, racing.	A source of play. They have an affective connection to what the drone affords them.

[22] Light Detection and Ranging, Remote sensing using laser to calculate distances between the laser and the ground.

5.1.1 Heritage practitioners and researchers who use drones

Practitioners in this group report taking advantage of the varied possibilities drone technology offers to access data. For them, the drone is an instrument to complete tasks in a more efficient, less costly, and quicker way, as this quote from Drone Expert One illustrates: “[T]here’s an acknowledgement that drones have the potential across our organisation to improve our understanding, our deliverables, and our workflows. (…) I do think they’re a tool.”

In this context, the drone features as a progression from previous tools or approaches, as evident in this quote from Drone Expert Two about drones in Archaeology “[I]f you take the drones out of the equation, you can still do the work (…) if you don’t have a drone, you probably need a boat maybe and maybe zoom lens or maybe a helicopter”. This perspective is consistent with individuals operating within organisational policies, and professional frameworks they are naturally obliged to fit within, in their place of work. Nevertheless, in this research, it has been found the majority of interviewees in this group do not fly drones outside their work context.

When asked about this, Drone Expert Three responded: “No, it never occurred with me as hobby, I always saw them as a working tool.” This is interpreted here as a result of the strong association of drone technology, with working life responsibilities and processes. It is corroborated by the same interviewee’s response: “I see a drone, I think of the forms you have to fill in. The hoops I need to go through. The authorisation you need to go in to. I’m, like, no just leave it as it is, I’ll just take a camera and just relax.”

This same sentiment is identified in heritage expert interviews, as illustrated by Heritage Expert Five, when asked which words come to mind when thinking of drones: “I’m just going to go with faff.” This echoes other interviewees who identified logistics as one of the barriers to using drone technology. Even for individuals to whom work is a passion and a successful form of self-actualisation, what is at the centre of the sentiment of self-fulfilment is the context, or final aim of the job the drone is used for

(for example, surveying), and not the use of the drone itself. This reiterates the pragmatic perception of the drone as a tool.

It is therefore congruent that "access and accessibility" are the qualities heritage practitioners and research users in a broader role, deploying the drone as an element of their job, mention most frequently, when considering the unique characteristics of the drone. This is both in the context of accessing images of sites, or areas where it is normally not possible to collect data from, and for people who cannot physically visit and engage with a site.

This is followed by the opportunities of data gathering and visualisation. Drone Experts One and Three, for instance, respond that drones provide a different view, a wow factor, by helping the public to discover sites and understand context, scale, and landscape composition. Drone Expert Two's perspective is dissonant from the views of the other interviewees in this group. This interviewee points out that drone deliverables do not differ from the information previous tools enabled them to gather, and that drone technology, through its data capture capabilities, offers "a platform to build on, in effect."

The views of Drone Expert Two, although the outlier in this context, provide an interesting angle. They consider the data captured by drone technology as a starting point, more than an end in itself. Both perspectives are pertinent. The versatility of the data, and data formats, means it offers opportunities to explore different ways of using, and visualising the data, as for example DRIFT's[25] reconstruction of Gaudi's Sagrada Familia (Lawson-Tancred, 2022).

5.1.2 Commercial drone operators

Two of the three commercial drone operators in this research started engaging with drones to solve a practical problem, as Drone Expert Five illustrates:

> "I was standing up on a chair trying to get a picture of an art balloon design that I had created. It was 10 by 20 feet, so it was fairly large and me taking a picture just on the ground just wasn't

[25] DRIFT is an artist duo who use drones to recreate heritage structures (in ruins or unfinished)

doing it justice. And when I stood up on the chair, someone told me: *Hey, you need a drone!*"

The third interviewee had an interest in video and filming and had previously commissioned drone services, as part of their job in corporate communications. From this, they developed an interest in drone technology and became drone users, subsequently leaving their corporate role and starting their own videography services business. All three interviewees in this group reported property and construction, as the key industries they provide the most services to. The pragmatic approach to drone use from commercial drone users in this research is coherent with the running of a specialist business, and provision of contracted services to clients, which include a final product.

Nevertheless, their relationship with drones is also affective, in relation to what the drone affords them not only professionally, but also in their recreational use. Drone Expert Four's description of the drones' practical uses and capabilities is inscribed in a sense of wonder and awe: "It's just absolutely wonderful to, you know, take measurements off things and find out how big a hole is, find out how deep a lake is, it's just, it's incredible the things you can do with them".

They further describe a flight in their local park, with an empty children's playground and a café, closed at the time, a community hub and local landmark at risk: "So, I flew all around the café but then, after, I flew around the playground as well. I went through the swings, and you know through the climbing frame, all that sort of stuff, all the things I would not do now and all the places I would not fit now." Here, the drone operates as what McLuhan (2013, p. 23) describes as an extension of the human senses.

Drone Expert Four's experience of flying in the playground elicits the interplay between the kinaesthetic "motions, notions and emotions" (Hildebrand 2020, p. 98) and the emotional. The part nostalgic-reliving, part reimagining memories positions the drone as a producer of hybrid imaginaries, by bringing together the re-enacting of fantasies of play, such as a child playing in the swings, and the reimagining of the fantasy as a tangible experience, through human-machine interaction.

By affording new ways of sensing, opening new relationships between "human and data bodies", (Agostinho *et al*., 2020, p. 252) drones bring about a transformation of our sensorial practices. These new ways of experiencing movement while still, mobility while on the ground, connection with nature through the non-human, seeing through your own eyes, reframe the human attachment to its environment. Drone Expert Five describes this as "drones allow you to experience your environment in a greater way"; they enable something that is not naturally possible to the human body.[26]

When talking about UAVs, both Drone Experts Four and Five express a relational dimension to the disembodied flight, in that it facilitates embodied memories, narratives and imaginaries, that are anchored in the emotional complexities of their ideations of play, identity and lived experiences. This is further evident, in how drone technology is positioned within the context of the lived experience of Drone Experts Five and Six. Both drone experts are African-American female pilots, who, as women and Global Majority individuals (people who are not white), possess lived experience of being underrepresented in institutions and decision-making processes, and are part of communities that experience discrimination.

Drone Expert Five is a biologist who, before becoming a drone pilot and instructor, worked in post-secondary education and outreach. Currently, this interviewee teaches drone technology to a wide range of ages, from pre-schoolers to adults, through delivering training to schools, in libraries and to community groups, such as Girl Guides (in the United States and internationally). Their programmes introduce drones in the context of learning about Science, Technology, Engineering, and Maths (STEM). Primary school programmes engage students with building a drone, coding it, and learning to fly it.

Other programmes include projects where students use the flight to create a tri-dimensional model of their school and design a tour of the building, benefitting the school by updating the school's blueprint or integrate their school into the metaverse. For students 16 and over, programmes include preparation to become a licensed drone pilot. Drone Expert Five is reportedly passionate about education. Their educational work with drones coexists with the

[26] This is conceptualised in research by Stankov *et al*. (2019, p. 811) on the effects of drone video, as the elicitation of the ambition of human flight

provision of commercial services.

Their training programmes are delivered as part of a vision that uses drones to help address underrepresentation of women, girls, and ethnically diverse communities in aviation, raise aspirations and help tackle generational poverty, by creating opportunities for gainful employment resulting in opportunities for social mobility: "[…] the focus will be career exposure, then career exploration and then the career training/workforce development."

This interviewee describes a project where secondary school students are tasked with creating a business centred on the use of a drone. One student's idea impressed a parent who works for a well-known commercial aircraft manufacturing company leading to a process of application for a patent. Drone Expert Five's enthusiasm for drone technology has led them to engage their child with drones from the age of three; flow a drone with their centenarian grandmother; and become involved in drone soccer, a game invented in Korea where players build and code their drone and then use it to compete, making any necessary repairs themselves in the course of the game. Drone soccer is sanctioned as a sport by the World Air Sports Federation, and it has a World Cup competition.

There is a strong similarity between the motivations of Drone Experts Five and Six. The latter is originally from a Business Management background and discovered drone technology when studying for a Property Inspection Exam. In addition to their own drone services business, this interviewee is a drone instructor and supports the recruitment and training of pilots to aid police departments and, like Drone Expert Five, they also work extensively in the educational sector. They teach drone technology in public and private secondary schools; develop drone technology training curricula; and work with formal learning establishments to develop tailored drone training programmes and respond to calls from various organisations looking to train girls and women, in technical industries.

Drone Expert Six posits there is great interest in including learning drone technology in education services, "[…] particularly from those that are servicing underrepresented communities or that are looking to work with youth groups that they want to introduce the possibility of a career in drone technology or in any kind of STEM related career field to their students." They refer to the work of organisations introducing drone technology as an educational tool, as an example of successful engagement of underrepresented communities with heritage and culture through drone technology, through the work of organisations training young people from inner city areas that do not have access to developing skills and entering careers in technology. "[L]et's get everyone involved in this to begin that change of course in documenting history. So not necessarily just looking back to uncover history."

That change of course places the drone as an identity archaeologist serving both the present and the past. In this dynamic, drone-sensing brings new methods of data acquisition, and new visions of identity and place for Global Majority communities. It also equips them to build on those methods and visions, creating new stories of entrepreneurship and empowerment over adversity, as evidenced in this quote from Drone Expert Five: "I don't have to check any part of me at the door when I wake up to do my job every single day. I get to be my authentic Black woman, boisterous, enthusiastic, passionate self, you know?".

Both Drone Experts Five and Six are involved in a project training ethnically diverse girls and women to become licensed drone pilots and map local sites relevant to the heritage of Black, Indigenous and People of Colour, leading to the creation of a virtual museum celebrating their uncovered narratives. For both interviewees, drones are a tool beyond its material functionality, that offers opportunities to reach a fairer society through exploring, discovering, recording, and owning your story.

Commercial drone operators running a drone services business emphasise practical aspects of drone user experience, such as the unique and enlarged perspective of a Bird's Eye View, but in relation to the experiential aspect of flying. This position is also reflected in Drone Expert Four's considerations on the agility and mobility of the drone, as unique characteristics. Drone Expert Five considers UAVs broaden day-to-day experiences, particularly for underrepresented communities, by, for instance "[…] giving each kid [they teach] the experience of riding in an aeroplane and being able to look out the window." UAVs also enable disabled students to move in ways and at speeds, their body may not otherwise do.

This underlines the unique value of drone technology in inclusive and accessible practice: a fuller view beyond your own body. Interviewee Six also mentions a quality that no one else referred to directly: the potential of drone technology to promote creativity and innovation. Meanwhile, Drone Expert Four highlights the drone as tool of curiosity and adventure, with which you can explore "[…] what is on the other side of that wall."

To conclude, commercial drone operators have an emotional relationship with the drone and perceive the opportunities of the technology beyond its practical, business-related applications. Differently to most of the heritage practitioners and researcher interviewees, drone technology use pervades their leisure and personal time. Both Drone Expert Four and Five fly drones recreationally, build their own drones and embed drones widely into their life; Drone Expert Four runs social media feeds for their drone content; Drone Expert Five teaches their five-year-old daughter to fly drones and leads Drone Soccer Competitions.

Drone Experts Five and Six, as members of underrepresented communities, are highly motivated to use drone technology to help address the cycle of generational poverty and raise aspirations for girls and women in the aviation industry, and children and young people from disadvantaged backgrounds. To them, drone technology presents opportunities for engagement, that go beyond their functional commercial uses. These opportunities exceed engagement and foster a relational dimension between the user, the machine, and the specific interests it enhances.

5.1.3 Drone hobby users

The critical understanding of recreational drone users is still limited (Hildebrand, 2021, p. 192). To retain a clear distinction between drone user types, this section focuses on two interviewees who are exclusively recreational users of drone technology. These users' considerations of the drone are predominantly relational and affective. As in the case of commercial drone operators, there is a strong element of exploration and discovery, as Drone Expert Eight illustrates: "I love to explore different areas." This interviewee's interests are focused on the experience of flying and they describe the sensation of First Person View (FPV) flights wearing goggles: "So if you put this on your face and you're flying the drone you kind of forget. It looks like you're inside of an aeroplane." A similar comment was made by a member of staff from the Lovejoy Centre in relation to the co-research livestream experience (discussed in Chapter Seven and included in Appendix F).

Drone Expert Eight's perspective centres on an element of thrill and sense of adventure, as their comment about drone racing illustrates "mostly it's kind of challenging each other, like okay this is like a new place. Let's try to explore. If there's like a small gap between the branches, can you go through? Who goes the like fastest? Who goes the furthest? So, it's kind of like big bang playing kids games." This element of challenge also features in First-Person View drone videos, such as Benoit Finck's *Medieval Castle FPV through Mont Saint-Michel*, winner of the 2020 GoPro awards. Finck flies a drone with First Person View capability through World Heritage site Mont Saint-Michel, in France, plunging from the top of the Abbey. The high speed, angles and change of perspectives that emulate the pace of a prison escape film, combined with the visuals of an air show, fuel adrenaline and are compulsive to watch.

Heritage Expert One also establishes a correlation between drone use and gaming environments. Drone Expert Eight considers drone racing "brings video games into the real world." Meanwhile, in one of Drone Expert Seven's Science Fiction film projects, the drone camera stars as a character, in a film where graphic elements are produced to look like a computer game. The affinities between drone use and gaming go beyond the participant being the controller and the protagonist. Each drone flight is a mission, as is each game. Both the drone and the game show the first-person view; as the drone kit is an extension of the human body, so is the console controller that enables your action in the gaming experience; finally, both are a form of storytelling.

Although gamification is outside the remit of this research, it is important to mention, because, as identified in the Literature Review, it relates to how drones afford new perspectives and storytelling: video games are a form of participatory storytelling, where the player collaborates with or competes against the machine. Indeed, considering how they use drone technology in the context of their hobby, Drone Expert Seven considers they have evolved from being a drone pilot and identify themselves as a filmmaker, a storyteller, for whom the drone is a "Swiss army knife of tools". This illustrates what Hildebrand (2021, p vii) describes as the drone user who is also a content creator. Drone Expert Seven publishes their content on their YouTube channel and has a supporting audience, some of which are Patreon[27] members. In fact, across the data in this research, whether uncovering, showing, recording, creating, helping to change or reimagining narratives, drones are elicited as a storytelling instrument.

Drone Expert Seven uses camera drone technology, as a visualisation practice in the process of creating and making place (Hjorth and Pink, 2014, p. 46). They create place through (re)presentations of geographies that were, up until then, unnarrated by them; through the perspectives of mobility and verticality; and by connecting consumers of their content, to familiar and unfamiliar spaces. This illustrates how camera drones fit into what McQuire (2016, p. 2) defines as geomedia.

This concept articulates how media such as camera drones, connect people, the environment, and data by capturing different types of media (ground sound and aerial image); performing in a context of location awareness; being growingly more present in the airspace, visual communications, and the consumer market; and negotiating physical and virtual interactions respectively, through exchanges with bystanders and through feeding and producing live feedback in online platforms.

Drone Expert Seven identifies the drone as "[…] high-definition cataloguing of buildings […] and landscape". This is a valuable reflection on the use of drone technology as an enhanced means of data gathering and the heritage value of digital data assets. They also allude to the drone as an arts democratising device, in two ways: "[…] for the viewer as well as the maker." This reflects this interviewee's reference to drones as a "Swiss army knife of content creation" and filmmaking, offering great opportunities to show places, sites, and narratives.

Drone Expert Eight, as a drone racer, privileges the views afforded by the drone, and incomparable freedom. Like the commercial drone operators, they focus on the experience of

[27] Patreon is a platform which enables fans to financially support content creators and artists through a paid membership scheme.

the flight: “[…] it actually links you to the actual machine. It looks like you’re inside it and you’re actually flying it”. Drone sensing is again elicited as an extension of the human body and the affective relationship with the drone.

Section conclusion

The analysis of the heritage and drone experts’ data shows interviewees’ perceptions of the opportunities of drone technology to engage underserved audiences with heritage are predominantly informed by their field of practice or roles. Heritage experts working in Public Engagement (delivery and strategy), whose practices encompass mostly the use of drone visuals, identify specific opportunities for drones to increase access and inclusion, and widen perspectives.

Interviewees in more technical roles are more reserved about their views of drone technology, as they face an emphasis on the practical administrative aspects of organising the deployment of drone technology. By contrast, drone hobby users have a more affective relationship with UAVs and their opportunities. The views of these users are fundamental to the understanding of what drone technology can offer. This is because their perceptions are elicited from a position of lived experience, rather than that of producers and commissioners, such as the heritage experts interviewed for this project. The views of interviewed content producers/commissioners are focused on the outcomes of drone technology, whilst the views of interviewed users reflect the experience of deployment and content capture itself.

There are distinct perspectives and attitudes within the three categories of drone user group in this research. Researchers and users in a broader role that is not drone specific, focus on the capabilities of the drone to provide data and data visualisation, which can then be used to create audience engagement materials. Their perspective on UAVs as tools is shared by the commercial drone pilots who are also drone operators.

This group values both the technical capabilities of drone technology and the experiential elements of flying a drone. Whereas this last element is prominent for recreational (or hobby) drone users, who articulate a personal interest in flying drones, as a means to access and experience places and geographies. To them, drones extend the abilities of the body and vision; UAVs may be part of a process of democratisation and deinstitutionalisation of aerial space. Nonetheless, all three groups agree UAVs bring something unique to visual culture and the way humans interact with and experience their surroundings.

This points to the diversity of ways in which drone technology can be used purposefully in heritage, for interpretation and engagement and how the outputs framing its purpose can be used to benefit communities. The findings in this section provide context to the themes emerging from drone experts' view on the opportunities brought by drone technology.

5.2 Broader and wider

This section presents the themes – and perceived opportunities: "Broader perspectives including context" and "A greater range of experiences for wider audiences", as identified in the analysis of semi-structured interview data.

5.2.1 Broader perspectives and experiences

Interviewed drone experts consider drone technology offers a broader perspective of the landscape, framing it within its context. This is also articulated by Heritage Expert Nine as "[…] the iconic views, and the species, the wonderful and amazing diversity of species that the South Downs National Park represents." From this point of view, there are several perspectives informed by the interviewee's own narratives.

i. Historic and cultural value of landscape and heritage

The views of Drone Expert One, who uses drone technology as part of their role in natural heritage sector and of Heritage Expert Two, who is a non-drone user with a Public Engagement related role, are very similar. Both consider the perspective of the historic value of the landscape. "If you think about how our ancestors were interacting with the landscape and we can recreate this, like every time you stand at the top of an Iron Age hill or on top of a bronze age burial mound and you see that view, you suddenly realise you're part of something bigger."

Scale and composition inform how these interviewees consider drone views: both the scale of the site captured by the drone, as well as viewers' own sense of their position in the landscape. As Heritage Expert One states, "it flips the narrative of who we are and how we occupy space." This narrative goes beyond physical engagement with place and the ecosystem. It pertains to how drone-mediated new ways of seeing are an entry to connecting with the essence of spaces and transforming our understanding of place. This extends to understanding aerial space through Dark Skies: "Is there a way that drone technology could help a person looking up at the sky understand what they're looking at; […] is there a way that you could create that magic again for people?"

ii. Disrupting our understanding of landscape

Serafinelli and O'Hagan (2022, p. 1) posit drone visuals "[…] produce unexpected perspectives of the world that reveal hidden aspects of our surroundings." In this point of view, drone perspectives are interpreted as "problematising" how we see ourselves within the landscape, and what that means for our identity and belonging. This is central to the research question in this doctoral project: "What unique opportunities does drone technology offer the interpretation of and engagement with heritage?"

One of the interviewed heritage experts highlights that drone footage in programmes such as *Countryfile*[28], produced by their organisation, present the countryside using an aesthetics that is sentimental, nostalgic, and "twee", stating "[…] some of the footage makes your jaw drop. But I don't know how accessible that is to your average person in the town centres." This interviewee articulates Debord's concept of "spectacle": where the real world is presented through images that show it as superior (Debord, 1994, p. 26), that superiority being its true reality, as well as its ultimate aim. Debord (1994) frames "spectacle" within a narrative of economic power, resulting from the commodification of experiences through images.

This extends to cultural value, material, and intangible culture. Heritage narratives focus on the object (both natural and man-made) as "spectacle", used as a platform of control,

[28] *Countryfile* is a British television programme aired on BBC1, dealing with rural, agricultural, and environmental issues.

representing the power of the dominant group (Debord, 1994, p. 13). In fact, Debord posits the spectacle results from the compositionality of the producing organisation (the ways in which organisations arrange elements of aesthetics or narrative to project their desired image or brand).

By representing Heritage as "spectacle", heritage organisations retain and reiterate power by creating an aesthetic that performs regulating effects on behaviours. They homogenise representations as well as unify notions of place, identity, and nationhood, that are omnipresent in institutions and collective consciousness. Using media to project the part, as perfect representations of the whole implies this practice not only reiterates the power of the forces of production, but also amplifies the disempowerment of the consumers, by excluding and devaluing their lived experiences.

iii. Connection, control, and power

The idea of power is central to this study. Empowerment is a guiding principle in this project's co-design research, as it empirically considers how drone technology can create agency, in how people living with dementia connect with and experience natural and cultural heritage. It is important to connect the concept of spectacle with how the heritage sector constructs ideas around its assets. Material and natural heritage objects have traditionally been construed as spectacle, creating a distance between heritage and communities. Heritage Expert Four elicits this, when they comment on the aesthetics of drone footage frequently showcased by the media and organisations.

This consideration is central to interrogating how the unique characteristics and opportunities brought by drone technology can be used to promote equitable access, representation, and agency. For this to happen successfully, it is necessary to explore how to address the issue of aerial views, as an aesthetics of spectacle. New perspectives, in context, is one of the key themes emerging from drone and heritage experts, in identifying what is unique about drone technology. However, the idyllic views of the countryside presented in *Countryfile* and other nature programmes show a unilateral gaze that excludes what the reality of the subject is.

In the case of natural and cultural landscape, this includes the diversity of the communities that are a part of it; and the urban fringes communities impacted by exclusion, socioeconomic deprivation, and challenges to access and participation. Presenting the "spectacle" as the truth of place reflects a fragmented view, the view of the dominant group. This idea is strongly expressed by Heritage Expert One, when they state that: "[…] day- to-day we all live a life where we think the whole world's been created just for ourselves."

It is this separation, which Debord discusses in the concept of "spectacle", that Heritage Expert One explores in relation to natural and cultural heritage, when they contrast pre- historic human behaviour, with how people engage with landscape nowadays. This interviewee argues early societies' interactions with the landscape were about "connectivity" and that there has been a shift from connecting to "[…] controlling space and what that space means to us, as an individual or as a privileged group." This shift is not only about controlling what the space means by and for the privileged individuals or group but, most importantly, about determining what it means for everyone else, as a universal truth.

The aesthetics of the spectacle is a manifestation of the desire of control: a strategy from dominant groups to prevent their own disconnection. In their research on shame and vulnerability, Brene Brown defines shame as "[…] the fear of disconnection—it's the fear that something we've done or failed to do, an ideal that we've not lived up to, or a goal that we've not accomplished, makes us unworthy of connection." (Brown, 2013, p. 68). The nostalgic view of landscape and material culture creates an ideal that excludes those who are not involved in the systems of production of cultural value, therefore disconnecting them from those places and eliciting the feeling that they cannot have ownership of, nor belong to those places.

Visual media responding to the fear of disconnection of dominant groups, inherently creates an aesthetics of displacement, forcing communities out by excluding them from representations and narratives of place. This is relevant to the research question on how drone technology can be used to empower underserved audiences, because the new perspectives of heritage brought through drone technology are still predominantly composed and produced by institutions and, therefore, are a reiteration of their gaze.

They remain informed by a discourse of power and fetishisation of the view they hold, even when “producing unexpected perspectives” (Serafinelli and O’Hagan, 2022, p. 1). This is particularly important in the case of people living with dementia because, for people living with cognitive decline, empowerment is central to retaining a sense of self and belonging. Brown’s consideration (2018, p. 144) that “Only when diverse perspectives are included, respected, and valued can we start to get a full picture of the world.” is echoed in this research.

For drone content to present truly new perspectives that are inclusive and foster connection, it is important to ask: whose perspectives are these? Who is composing the gaze, and controlling where it focuses? This means that, as Heritage Expert Three suggested, rather than an idyllic landscape scenery, a more authentic and inclusive view of a natural landscape would be to include the urban landscape that surrounds it and is part of its context. This would foster in communities living in the urban fringes, the feeling they are a part of that wider narrative, that they belong, have access, and have opportunities to shape how they occupy place. If drones offer “[…] perspectives people don’t normally get” (Heritage Expert Ten) in facilitating their connection with natural and cultural heritage, it is crucial, for that engagement to be authentic and sustained, that these perspectives are not just about a physical point of view, but mostly that communities can see themselves in the gaze that is represented.

5.2.2 A greater range of experiences for wider audiences

Beyond offering broader experiences, interviewees suggest UAV use can provide a greater range of experience for wider audiences. According to the 2021 South Downs National Park Authority’s Visitor Survey (m.e.l. Research, 2022, p. 6) and the Cairngorms 2021 Visitor Survey (Cairngorms National Park Authority, 2021) walking is the most popular activity in these sites. In the Cairngorms National Park Visitor survey (2021), only two out of nearly 300 survey respondents mention drones when asked what could have been improved in their experience of the park.

One respondent[29] calls for a ban on drones and mountain bikes and mentions the need to educate visitors not to shout or make loud noises. Another[30] proposes to stop people from "[…] flying drones, playing bagpipes and shrieking." This respondent also mentions, in their final survey comments, the negative impact of the large number of visitors to the park: they remark negatively again on drone users as well as cyclists, campers, and wedding parties, which, they argue, ruin the tranquillity of the Cairngorms National Park.

Notably, respondents refer to drones as part of a broader group of activities outside of their own ideas on how to enjoy the park. These ideas may correspond to the top three most popular activities in the Cairngorms: walking; bird, wildlife, or nature watching; and using the café or shop. These results are very similar to the SDNPA survey results (m.e.l. research, 2022) where these same activities (walking, including dog walking) are the top three reasons for visiting the park.

Tranquillity is one of the "Special Qualities" of the South Downs National Park (SDNPA, no date). It is highlighted by over 1000 survey participants, in a 2009 public survey undertaken at the time the SDNP was designated a National Park. Respondents to the Cairngorms National Park Survey seem to concur as, when answering "What is the best part of your experience on this visit and why?", the respondent[31] who complains about drones, bagpipes, and shrieking, comments they visit to experience "[…] (the possibility) of peace and quiet." Heritage Expert One discusses the complaints their organisation receives from the public about drones disturbing the tranquillity of visitors, relating them to complaints about off-road vehicle use. They recognise the tensions emerging from diverse groups of people valuing different types of experience from those traditionally viewed as mainstream ways of engaging with place and landscape.

i. Engaging new audiences

These tensions reflect the challenges faced by the heritage and cultural sectors in meeting the

[29] Respondent 12810975014
[30] Respondent 12811035259
[31] Respondent 12811035259

expectations of existing audiences, whilst appealing to a broader range of people and interests. Heritage Expert Two highlights drone technology offers something different which appeals to a different group of people from those who traditionally engage with sites. They state:

> "We are very conscious that when you think about regular recreation opportunities, for example, within this kind of countryside environment, people tend to automatically think of a certain type of person who's got all of the right raincoats and walking boots and wants to go out for really long walks in the countryside with binoculars identifying all of the birds."

They consider drone technology may bring something exciting, and view drone technology as an addition to the breadth of activities different audiences enjoy in natural heritage sites, from kayaking to taking part in events. Framing drone technology in the same way as Cairngorms National Park survey respondents (an untraditional way of experiencing natural landscapes), Heritage Expert Two posits: "I think the more breadth of opportunities we have and the more we promote that breadth of ways of engaging with the landscape, the more different, diverse groups we will have coming into the landscape." Drone Expert Seven also reflects on this possibility of engaging new audiences by welcoming drone users: "You can actually get more people into your areas, if you say yes, you can fly here, we're really happy, please follow these rules, rather than just sticking up a sign saying no."

i. Control and power

Drone Expert Seven's reference to "sticking up a sign" outlines the challenges of institutional regulatory frameworks such as the National Trust's, which not only limit the launch of drones from their land but also tackle issues of income generation and image rights: the National Trust charge a fee for permission to fly from their land and charge for their own photography. The same drone expert relates this to tensions arising between Painting and Photography, at the time the latter emerged and became popular, in the mid- 19th century. They align this perceived conflict between the art of painting and capturing the world through a lens, with the friction between ubiquitous drone technology and landscape photography and film.

They postulate that, by limiting drone flights and photography, the NT limit the production of images of their sites and properties. NT generate income by charging for permission and selling their own photography, which would be compromised by loosened restrictions on UAVs: they would lose control over the proliferation of images (assets) they could be selling, as these are taken by members of the public. Heritage Expert Seven reflects that democratisation does not mean a reduced appreciation of high-quality artistic production, but simply that more people find ways to connect with sites, by using different tools for personal enjoyment and inspiration. This articulates well aspects of institutional power and control at play in public engagement.

More broadly, the views of drone experts who own a drone business and/or fly recreationally are aligned with the perspective that the versatility of drone technology supports the connection with different types of people and can facilitate a variety of ways to connect with narratives. Drone Expert Seven comments: "[…] if you're interested in the arts, if you're interested in photography, film making, map making, geology, you know, just walks, mountain rescue, the drone is a fabulous piece of equipment." They discuss the drone as a "toolkit".

This is interpreted here, as strongly connected to concepts of power and ownership of experiences. This is particularly relevant to this interviewee, a content creator who uploads to YouTube and other social media channels. They frame their content as cultural viewing – a space for niche content production and consumption that is not financially viable for large corporations, such as the BBC[32] to fulfil. They argue people "[…] have an interest, they are exploring that interest, and they are using the technology that's available, and it could just be a phone, it could be some nice kit, it could be a drone, it could be GoPro." At the centre of my principal research question, control and power are equally at the centre of the tensions between heritage institutions and drone users, particularly recreational users.

[32] British Broadcasting Corporation

ii. Drone-mindedness and drone sensing

Campbell (2018, p. 52) argues drone technology extends the effect of physical separation between humans, the world and each other, that digital media creates. For Campbell (2018), this separation stems from the mediation of human experiences through remote ways of seeing. Considering the drone within these ways of seeing (from a distance), relates to drone mindedness and drone sensing, that is, the capabilities of drone technology at the root of perceived opportunities “A broader perspective that provides context” and “A greater range of experiences for wider audiences”.

UAVs are part of a broader portfolio of digital media, in a highly mediatised world. Campbell (2018) situates UAVs at the intersection of a militarised and cinematised society, where events are experienced and wars are fought from a distance, in a continuous media-framed temporality of the now. By this, it is meant society experiences the world through the media stream of events. This stream is generated not only by media businesses, but also by individuals and communities themselves.

It shapes a significant part of western societies’ interactions with their surroundings and activities. What was once unknown, extraordinary, or unique has become mainstream, through the individual production and circulation of personal narratives. Campbell (2018) considers that, as media is omnipresent, so is the cinematic thinking inherent to the media, which frames current human interaction with environment, space, and other humans.

The drone’s assemblage of resources is part of drone sensing. Campbell (2018, p. 53) proposes the drone is “[…] simultaneously a becoming-media of humankind and a becoming-human of media-kind.” This is interpreted as leading to a hybridisation of media as human, doing and seeing, whilst humans turn into media, through the way their own doing and seeing becomes real within the media time and viewscapes. Drone visuals and visualisations play a significant role in how UAV technology reframes the human range of sensorial experiences (Agostinho *et al*., 2020), expanding on them.

The work of Munck Petersen (2016); Garrett and McCosker (2017); Mangold and Goehring (2019); Hildebrand (2019); Serafinelli and O'Hagan (2022) illustrates this. Humans experience the world around them through a mediatic sensibility which, in the case of drones, Hildebrand proposes as "drone-mindedness" (2021, p. 124). Hildebrand describes this concept, as the heightened awareness of geographical surroundings drone users develop, even when not operating their drone.

In my autotechnographic drone work, the aerial views captured by the drone are experienced as unique and beyond the optical, conjuring elements of intangibility and transcendence by bringing two materialities together: mine as a human and that of the machine. Both interacting as collaborators to test and explore volumetric landscapes, that bring new encounters with the place. This exploration of space from the perspective of the drone is the result of humans' mediatic sensibility. In my own experience as a drone pilot, drone-mindedness manifests itself as a mindset through which I, as a camera drone user, continuously scan my surroundings for drone-capture suitability.

In this mindset, I speculate about the processes, systems, and motions of the drone, and the resulting aesthetics of its flight: "This would look lovely from a drone." My drone-mindedness is not restricted to top-down, or panoramic views. I find myself often considering points of view that are unusual for drones, and which subvert the preconceptions around their use and capabilities, such as ground Close-up flights following the detail of feet walking on grass, for example, or slow First-Person View film shots.

Drone-mindedness is reinforced by the theme of "A broader perspective that includes context". This is made possible by drone technology's affordances for mobility and verticality. Both facilitate an aerial point of view, which is not naturally available to the human body in the absence of intermediary devices. The movement of drones in different directions, diagonally and vertically, creates a unique experience of space that is three- dimensional and volumetric (Jensen, 2020, p. 418).

Drones open up experiences of place (facilitating the opportunity of "A greater range of experiences") that challenge boundaries of power and control. Drone Expert Four, when asked what is unique about drone technology, comments: “Accessibility. I think that is the main thing. Because you can – I think about it all the time, but many other people who don’t fly drones everyday don’t think about – but what is on the other side of that wall – you’ve got no idea, you can send a drone up.” This extension of the capabilities of the body broadens the concept of “Access and accessibility”: interpreted here as framed within the possibilities drone technology offers to access and explore places which cannot be physically accessed through the use of the human body alone.

5.3 Physical and digital access and connections

“Broader physical and digital access and connections” are discussed in the semi-structured interviews data. These include perceived opportunities for “access and accessibility”, “more data of higher quality” and “meaning-making”.

5.3.1 Access and accessibility

The themes of “Access and accessibility” are explored by drone experts in close connection with themes of inclusion, and engagement of audiences. Although the two latter concepts are not explicitly named by interviewees, both drone and heritage experts reflect on how drone technology can engage people. Drone Expert Six and Heritage Experts One and Seven mention that one opportunity for drone technology to engage non-visiting audiences is that the technology proposes a different and new way to connect with places; therefore, it will appeal to groups of people different from existing audiences.

i. Engaging young audiences

Both Drone Experts Five and Six, who, in addition to running a drone services business, work extensively on drone education programmes, operationalise the drone as a levelling tool. Drone Expert Five reflects: “I can’t give every kid the experience of riding in an aeroplane and being able to look out the window. I’m able now to allow other individuals to get that view when they may not have gotten it in any other way, shape or form.” They consider the Bird’s Eye View as a perspective that is not easily accessible to children and young people from deprived backgrounds, as there is significant underrepresentation of these communities in aviation. Also, because some of these children and young people never leave their city to access different landscapes or to fly. This reflects the heritage expert's view of the drone showing "[…] perspectives that people do not normally get", which can be due to socio-economic inequities.

Engaging young people with the South Downs National Park and the National Trust through drone technology emerged as a strong theme in the heritage experts' interviews, particularly from those who work in roles involving public engagement. Heritage interviewees highlight drones present an opportunity to engage young people on the periphery of the South Downs National Park, or those living in London, by promoting knowledge and understanding of the landscape, educating them about the Downs, and helping them develop new skills. Like Drone Experts Five and Six, Heritage Experts One, Two, Seven and Ten also mention the opportunity to use drones to upskill young people, who could create drone assets used to benefit nature or other people to: "[…] use that technology to create responses to landscape, that can then be used to engage people facing health inequalities.".

ii. Familiarisation

Heritage experts in both the interviews and online survey highlight the use of drone content, as a resource to help people familiarise themselves with sites they have not visited before: to help someone who is considering or planning a visit. One respondent[33] notes: "[…] people can plan specific monument areas to visit prior to travelling to the site." In the SNDPA's 2018 Visitor Survey (2019 p. 19), 21% of visitors used the internet to plan their visit. This could seem a low proportion of visitors, but according to the report, 71% of the visitors using other resources knew the location, were not first-time visitors, or resided/had resided in the area.

Heritage experts also discussed the use of drone content to help manage feelings of nervousness and anxiety for people who are weary of going to the countryside. This could help groups who do not feel they belong or feel intimidated, or it could meet the needs of neurodivergent visitors for reassurance, encouragement, and knowing what to expect. In these circumstances, a drone video can provide "[…] real-vision experience." Drone Expert Five reflects: "I wanted to share with them […] the way that drones have made a space for

[33] Respondent 99644983

those who maybe have a learning disability."

The connection drone technology facilitates is not accessed only through the direct products of drone technology, such as videos and photographs. Heritage and drone expert interviewees emphasise that the technology presents opportunities to build on drone assets created as part of monitoring and surveying activities: using Sketchfab[34], Photogrammetry[35], Virtual Reality, or LiDAR to enhance the ways in which drone technology can help the understanding, awareness of, and access to natural and heritage sites. Heritage Expert Seven reflects on accessibility as an area of practice offering opportunities for development and creativity "[…] that we need to be almost testing ourselves as well as coming up with new ideas."

iii. Accessing the inaccessible

Heritage Experts Four and Seven reflect on how drone technology can be part of a set of mobile digital technology tools to facilitate "[…] access to the inaccessible." This is the case, for instance, of National Trust sites inaccessible to the public, because they are let by tenants in long-term rentals; or sites that cannot be physically accessed due to health and safety, the architecture of the space, or the physical integrity of the collections.

The NT's Shell Gallery, at Basildon Park, is such an example. This room is elaborately decorated in seashells; the space is fragile and at risk of irreparable damage from even a slight brush past. One heritage interviewee, a video producer, mentioned this room, as an example of where remote technology is already used: "That is one area where we do already use remote control cameras, so people can explore that space, but they are fixed position […] I can foresee a future where some more mobile technology will allow us to get into those spaces."

Both Drone Expert Two and Heritage Expert Seven also discuss access for people who cannot physically visit, due to accessibility issues or because they live outside Britain. Heritage Expert Seven suggests that a drone-captured photogrammetry model can offer those visitors an

[34] This is software used to publish 3D content.
[35] This is a technique that overlaps three-dimensional photographs to produce 3D models of objects and landscapes.

experience of the site. It is central to this discussion that drones, or any other technology, are not used as a replacement for the fulfilment of corporate responsibility in making sites inclusive and accessible.

Digital technologies, and drone technologies specifically, can nevertheless support people who are "housebound" (Jubilee Medical Group, no date), living with conditions that mean they do not or cannot access public spaces at all, or independently. Heritage Expert Two posits: "[…] it sort of enables us to connect people in a very direct way, without them physically having to visit themselves, so it broadens that opportunity for participation."

Heritage Expert One also relates this to people experiencing poor health outcomes: "[…] how can we create assets that would create a meaningful sense of space for people that can't physically come to the park, and potentially when we say that people who can't, you know, who will not necessarily even physically leaving a building?" Likewise, Drone Expert Seven: "A lot of my subscribers are people who have a disability so they can't actually go out. […] So, they live vicariously through me and through my drone." The opportunity identified here is not just that viewers connect with places they have not or cannot visit by viewing them via drone footage. Also, the drone, by providing a perspective that is not part of the mainstream portfolio of human physical mobility, offers an experience that is not a standalone replacement of a visit, but rather a unique way of engaging with those places, even for those who can visit.

There is an intangible aspect to this unique perspective fostered by the drone, which Drone Expert Six discusses: drone technology brings into play a new configuration of past and present. As heritage organisations predominantly explore accounts of the past, drones present a perspective that places our present selves into historic narratives, as Drone Expert Six suggests when they say "[d]rone technology is allowing this progressive timeline of reality capture to be created and held, and shared and not just locally; it can be uploaded to a virtual museum. It can be added to a database of information."

5.3.2 More data of higher quality

Indeed, one of the most used capabilities of drone technology in heritage is data capture (monitoring, surveying, photographing, filming, measuring). The improved quality of this data is a key theme emerging from the data in this research.

i. Preservation by Record

As drones capture and generate "high quality images and video footage", they facilitate what Heritage Expert Seven defined as "preservation by record." They consider this, particularly in relation to the climate emergency and the impact of coastal change on local communities. They postulate: "[…] we know that as a heritage sector, we're not heavy with the answers but what we could be ready with is, at least, a way of saying we've recorded a moment in time enough that we've at least done something."

This perspective of preservation by record is also illustrated by Drone Expert Seven, as they consider the archival value of their drone content: "[…] for future archivists this is great! They get to see the state of a castle that I've filmed in 2020, and that then becomes a record, you know, if there's damage or if there's a change, you've got this kind of massive archive." This interviewee speaks of the high-definition cataloguing of buildings and landscapes, which, being potentially geotagged, allows people to look back. This reflection brings the activities and data captured by drone pilots into the realm of citizen science, and I argue it could be interpreted in the context of cultural democratisation, as individuals curate their own digital archive of landscapes, buildings, and stories.

ii. Archival Challenges

One of the challenges of digital data archiving is discoverability. Heritage Expert Seven uses an example of drone photography capturing details of a historic property, in the course of structural work. The drone assets produced are used to create CAD (Computer Aided Design) drawings, which supports the interpretation of the site. However, those drawings can only be accessed by people who know how to find them, which means they are not widely accessible to a broader segment of the public.

Heritage Expert Seven considers that their organisation should look for "[…] opportunities to use those (their data) to actually engage with people. […] Just showing some of those different viewpoints would be something that we need to consider, really." Drone Expert Two, an archaeologist, posits that: "[…] once we start moving to the digital, a drone photograph of a site can become almost like an editable notebook for someone recording archaeology." They argue that the key benefit of using drone technology is that it is a cost-effective approach providing "abundant data" that you can use, however it suits your aims, including as a platform to build on. They highlight that the development of drone technology is directly correlated to the evolution of software packages that increase the potential for manipulation of drone outputs (elevation models, orthorectified photographs and CAD drawings, model reconstruction and three-dimensional models on Sketchfab and other platforms for instance).

Relatedly, Heritage Expert One mentions citizen science, as an opportunity to use drone technology for working with communities outside "[…] the demographics of people that are traditionally seen as an audience for national parks." It is currently used in the application "I-Naturalist", which SDNPA and other organisations use to encourage the public to record data from the natural environment. The data collected with "I-Naturalist" becomes part of a Big Data set used to monitor biodiversity.

Heritage Expert One argues that this approach is successful in engaging previously non-visiting audiences and states that there is opportunity to deploy drones to document climate change related events such as coastal erosion, for instance. Drone Expert Two highlights the challenges intrinsic to "[…] the long-term effects of collecting one terabyte of data per site": the costs of data storage and legacy data, when systems or formats become obsolete. Drone Expert Seven supports this point of view referring to a time in the next 500 years, when archivists care for "[…] millions of SD cards" and other formats, that contain all the data but can no longer be accessed.

Digital preservation is a current concern in the heritage sector, and more specifically in the Archives sector. Technology has changed significantly in the last 50 years. We are unable to predict how long technologies we use to record and store data will survive.

iii. Spatial mobility

Mobility and verticality are two intrinsic characteristics of UAVs. Drones are efficient mobile sensing platforms, that easily navigate different environments, as a result of their high levels of mobility (Szafir, Mutlu and Fong, 2017, p. 514). Grémillet *et al.* (2012, p. 53) state that the capacity of drones to "move more or less anywhere, flying, gliding, or hovering" combined with their ease of use, are what make UAVs popular.

Mobility, however, is not just about motion. It frames how we perceive, focus on, and act in our environment (Hafermalz *et al*., 2020, p. 21). Drone mobility, therefore, fosters drone-mindedness. Hafermalz *et al.* (2020, p. 21) go as far as to argue that our attention to, and perception of our surroundings depends on movement. This corroborates that drone motion provides a unique way to connect with our environments, which goes beyond diversifying the visualisation of spaces.

Drone mobility creates an effect of immersion, that leads to a sense of full involvement in the environment. Relatedly, vertical views increase mental simulation. In other words, top-down views from drones, according to grounded cognition theory, create mental imagery that imitates a hypothetical experience and its sensorial processes (Royo-Vela and Black, 2020, p. 23). In the case of drones, this hypothetical experience is that of flying. The viewer sees themselves in the activity (Royo-Vela and Black, 2020, p. 22). This imitative experience engenders the generation of stories (Royo-Vela and Black, 2020, p. 24).

Both mobility and verticality are therefore essential to the process of engagement, as they stimulate a sense of involvement – viewers feel they have a stake in their experience. Aerial mobility underpins “Accessibility”: it brings in a different dimension to the way communities interact with place, beyond the sensorial, into a cognitive and relational level, where agency is broadened beyond institutional and commercial systems, into communities.

This is important to the investigation of how drone technology can engage underserved audiences, whose narratives are underrepresented in heritage and culture: it points to the importance of fostering, in using the technology, a space for human creativity within the production of richer drone products.

5.3.3 Meaning-making

Meaning-making emerges as a theme, firstly as a storytelling tool, and secondly, more broadly, around creativity. The theme of storytelling emerging from this research data encompasses meaning-making and connection, and inclusivity. It is elicited in discussions of the purposeful use of technology to meet a specific aim, such as interpretation, rather than the technology being the end in itself, a panacea.

Two heritage survey respondents[36] reflect this consideration, as illustrated by this quote: "The point of departure for any technology/media is your interpretation – what are you saying, to whom, and why?" Heritage Expert Two considers that drone technology can help connect young people with the landscape story: how the landscape has been created and its position in the bigger narrative of the region.

They highlight a Brighton project, where a group of young filmmakers with learning disabilities use drone technology to explore what the landscape means to them. As Heritage Expert One posits, drones can help people make meaning in different ways connected to their personal narrative. Relating connection and inclusivity to meaning-making is true for other groups, as well as young people, particularly communities with experience of exclusion. Drone views can offer them an opportunity to explore their own place in the narrative of landscape.

This means that raising awareness and promoting knowledge and understanding of heritage sites is key for supporting people to connect with those places. Heritage Expert One adds that, for left-out communities, drone technology can help tell a story about the South Downs National Park, showing "[what it] is like and helping to build that bridge between where people are coming from and where we're suggesting they might like to visit." This makes it particularly important to reflect on the aesthetics of drone assets, who it excludes, institutional power, and how organisations compose and project their place within the narrative of landscape.

[36] Respondents 996449 and 94678072

Drone Expert Five emphasizes the value of diverse lived experiences: "[…] you and I could look at any given drone and have a different perspective of the best use of that drone as well. And it's going to be based off of you know, needs, morals, values, dreams, aspirations." Those individual experiences and perspectives are present and visible in the western media landscape where people access and create their own stories, as Drone Expert Seven, a content creator on YouTube and other social media channels, comments.

Heritage Expert Seven discusses how heritage organisations condition what is considered important or relevant. They reflect that a change in organisations' mindsets is needed to better connect with visitors and give them a voice: "[…] telling those stories using some of our visitors, or how people might see the landscape." This is a crucial point, and I identify a marked difference between drone content produced by institutions, and drone content created by underserved communities exercising agency, in the following chapters. The understanding that people not only see the landscape differently but also articulate that perspective using visual language that is distinct to that of institutions is central to the discussion on the unique opportunities of drone technology.

i. Creative, creativity, and innovation

The complex processes associated with flying a drone (referred to as a "faff" by Heritage Expert Three) emerge in the data, as a challenge and are mentioned specifically by Heritage Expert Nine as hampering creativity. Nonetheless, creativity is seen overall as a key affordance of UAVs. Drone Expert Four agrees, positing that the possibilities are endless. Heritage Expert One used these same words when referring to drone technology. Interestingly, "creativity" emerges in heritage expert interviews, whilst in most drone expert data the word "creative" is more prominent. This is an indicator of the positioning of interviewees.

For the majority, heritage organisations appraise drone content as a product, whilst drone pilots place themselves within the output, as content producers who also use the device as a tool for self-expression, as referred to by Drone Expert Five (as a biochemist, this interviewee comments "[…] there is not a lot of creativity when it comes to laboratory testing"). Drone Expert Four considers that the upside of an excessive production of footage is that more people will access it, and creative people will devise more increasingly unique and "[…]

amazing […]" content.

There is consensus among interviewees, that drone technology can foster innovation in engaging different communities, not only because of the new perspectives of places it provides, but also because it can support new approaches to branding, which can showcase organisations as more inclusive and help dispel their more traditional image. For instance, for the National Trust, as discussed by one Heritage Expert: "There's probably an inbuilt perception from people, when they hear the word National Trust or what they see in the press there's always a… Oh this is what the National Trust does and potentially it's not for me." This interviewee proposes that just trying something different may help connect with visitor profiles different to the NT's current demographics.

Interviewees also consider creative responses and connections. Heritage Expert Ten suggests that, if organisations are creative about how drone technology is used, they may inspire people to find new ways of showcasing and interpreting heritage themselves. Similarly, Heritage Expert One expresses a strong interest in using drone technology to "produce creative responses." They also suggest that the use of drone technology in the creative sector is underexplored. The heritage sector already works closely with creative industries: they use different artforms to bring collections to life through gallery drama, film, crafts, demonstrations, and others.

Both Heritage Experts Eight and One explore the new experiences and opportunities UAVs afford to connect creatively with heritage and culture. Heritage Expert One is the only interviewee who broaches how artists themselves interact with drone technology: "What is it they're trying to do with drones, when it comes to perspective and space?" Reportedly, this interviewee's interests lie in the processes, rather than the results of using drones to explore places or narratives, and how drone technology brings new potential to the process of reimagining places and belonging, changing the existing narrative.

They add that recreational drone use is still treated as niche, which means there is significant, but untapped, potential for their creative use. They query how drone technology can engage new audiences by using a model that replicates bringing communities together with artists, to work collaboratively on new interpretations of place and narratives. This highlights the importance of the context within which drone technology is deployed. It points to UAVs facilitating the discovery of new interpretations, if communities are involved in the process of meaning-making.

In summary, within the theme of physical and digital access and connections elicited by the data, "access and accessibility" are central to the considerations of equitable institutional practices. Drone technology is perceived to facilitate them physically, to a high standard through its vertical mobility and through more complex ways associated with challenging existing ways of connecting to place including who connects where, and reconfiguring identities through volumetric geographies. These new ways of relating to place elicit new narratives which, in turn, are a gateway to reimagined experiences of belonging.

5.3.4 Control and automation

Automation has a specific definition, in the context of UAVs. Autonomous or automated drone operations (Department for Transport and Ministry of Defence, 2016, p. 14) take place when the aerial vehicle plans and controls tasks, to goals specified by humans (Kumar and Michae, 2012, p. 1286). This requires sensing, planning and control software and algorithms (Kumar and MichaeL, 2012, p. 1286). Drone Expert Six, for example, refers to their moment of realisation of what drone automation made possible "This is such an amazing tool. So, I immediately switched gears, I put my inspection test prep aside and I started studying instead for my FAA Part 107 certification".

Automation is therefore an integral part of drone navigation systems (D'Andrea, 2014, para. 16). Automation processes have become more sophisticated in recent years (Kumar and Michael, 2012, p. 1286) and are still evolving. It is not within the scope of this research project to discuss in depth the different levels of automation and their implications in terms of efficiency and results. The focus is therefore on the effects of automation on vehicle control and human agency, in achieving desired goals.

i. Automation and engagement

Strictly from the point of view of task allocation, there are different levels of drone automation, from a high level where the machine executes all aspects of the task independently, to low automation where all decisions and actions are taken by a human (Szafir, Mutlu and Fong, 2017, p. 516). It is this element of human agency that I consider relevant to the discussion of control and automation in the process of making meaning.

As a starting point, Szafir, Mutlu and Fong (2017, p. 518) posit that robotic tasks and activities are ultimately human activities, facilitated through computational processes. Drone Expert Four's comment illustrates this "[I]f you want it to be exciting, you use a drone, because like, you know, you can go underneath things, over things, through things, you know if you want to make it a bit thrilling, you know, get people excited about it". This also relates to the purposeful use of drone technology and that the technology should be suitable in relation to the aim.

However, Campbell (2018, p. 7) states that increasing levels of sophistication in automated systems removes human power and control, over robotic processes. This is not only because there is a reduction of human shaping of the task, but also because autonomy makes systems more complex and therefore more impervious to human understanding (Szafir, Mutlu and Fong, 2017, p. 517). High levels of automation also carry the risk of making humans more passive (Mondschein, 2021 cited in Belkouri and Laing, 2024, p. 66).

Paneque-Gálvez *et al.* (2014, p. 1486) argue that in community-based forest monitoring, high levels of automation mean that communities require less specialised training (to operate UAVs), than they would, if using fully manual control. This means a faster learning curve for drone users. I argue it may also enhance the operator's confidence, in that some automated systems deal with aspects that could be challenging to non-expert drone users, such as obstacle avoidance, recalibrating in response to weather conditions, automatic return to home if signal is lost.

Automation may also enhance the quality of the data captured, through automated pre- and post-processing. These elements mean automation can be an empowering capability for underserved communities, bridging potential gaps in knowledge, confidence, and sense of self-efficacy (their expectation of success when using a drone). Therefore, automation, although requiring lower levels of direct control from humans, may provide a level of empowerment.

Nevertheless, full automation brings specific challenges. If, for instance "you have like a predetermined route that the drone would do, it would just keep to that trail" and if an aspect of data collection requires alteration of the route during navigation, the pilot needs to have the skills to override automation to achieve the aims of their flight. Johnson (2014) argues that high levels of automation have detrimental implications, because they obfuscate elements of the diversity of backgrounds, experiences, and ideas that underpin creativity in human problem-solving (Johnson, 2014).

This is an important consideration to meaning-making, which is enriched by the diversity of interpretive communities. Johnson (2014) proposes that instead of striving for full automation, we should look at a coactive model (Johnson *et al,* 2014) which encompasses a relationship of interdependence, as in the context of teamwork. This is particularly true in drones built by drone racers.

Drone Expert Eight refers to this ability to control automation levels: "you adjust a gyroscope, so you kind of know what sort of commands you're giving to the drone, through your remote." This notion is aligned with the idea of the drone, as a prosthetic to the human body, and the relational aspect that is elicited by hobby drone users, in their interaction with their drones. This is particularly evident on First Person View as described by Drone Expert Eight "you're like, oh I'm almost hitting someone, and you completely forget that that someone is you. Because it actually links you to the actual machine. It looks like you're inside it and you're actually flying it."

Hildebrand discusses the way hobby drone users engage with their drone, as a companion (2021, p. 152) which they name and talk to. This relationship is different for drone racers, where the technology is simple and does not include cameras and "if you lose it, you lose it" (Drone Expert Eight). This form of human engagement with the drone is beyond utilitarian: it is affective. This is evident in the DJI[37] consumer drone adverts. For instance, the DJI Mini 4 Pro's (DJI, 2023) YouTube promotional clip shows the drone as part of fun, informal, relaxed activities with family and friends.

Combining the capabilities of smartphones and autonomous robotics, consumer drones such as the DJI Miniseries have progressively included more automated features, which show it to be more intuitive, easier to control, more skilled, and producing more data of higher quality – in short: more accessible and with better results. Automation capabilities therefore contribute to the democratization of UAVs and their inclusion in the mainstream mobile media market aiming to widen the use of drone technology in ordinary life (DJI, 2023).

ii. Control and automation

Control is elicited in the online survey, by three respondents. One[38] considers "visitor control of drone to choose where to explore", as an opportunity for drone technology to engage non-visiting audiences. Another respondent[39] refers to the importance of interaction in user engagement and identifies "operation/control" as one of the challenges to feeling encouraged to using drone technology to engage non-visiting audiences.

Finally, a third respondent[40] considers that engagement (with young people) can potentially happen, if they are able to actively use the technology and not passively consume content. These heritage survey respondents point out to what Szafir, Mutlu and Fong (2017, p. 516) define as shared control "in which a robot can support direct teleoperation, by attempting to predict user intentions while augmenting received inputs in a process that is invisible to the user".

This is the level of automation (semi-automation) that best articulates a collaborative

[37]D-Jiang Innovations
[38]Participant 95192171
[39]Participant 94678072
[40]Participant 95949409

interaction between humans and drones, because it enhances agency and empowerment, giving the human agent control over the storytelling process. Full drone automation, here, means standardisation and massification of representations of heritage and culture that project them as epic.

Revisiting Walter Benjamin's concept of aura, I agree that under circumstances where there is very little control from the human user, drone products create a distance between the captured and the viewer, because they show an idealised narrative and generalise its representational value. Heritage Expert Six highlights this risk of using drone technology: "So, drone photography can certainly make things look far more epic than they are, and its expectations versus the reality of that moment".

Conversely, semi-automation facilitates agency and equity. It supports the user in achieving their goal by making adjustments necessary to its achievement, without increasing required expertise or the financial investment in training. For example, in my autotechnographic work, I looked for drone models with automation capabilities that enabled me to achieve a specific aesthetic, such as the Circle capability, which I could have not carried with the same stability and visual effect, through manual operation.

The semi-automated capabilities of the drone empowered me to achieve my goal, without the need for a professionalised drone pilot. Automation and control play an important role in agential negotiations in drone use. For the drone to be successful in facilitating meaning-making that articulates representative narratives of underserved audiences, it is essential that the human-machine interaction is one of collaboration (what Johnson (2014) refers to as coaction), and that the level of control is adjusted to the needs and choices of interpretive community users.

Automation plays an important role in drone use. It is mentioned by all drone users as a key element in drone technology and in some instances, it is articulated as the catalyst to interviewees adopting the technology. Nevertheless, automation is also discussed in the Literature Review (Section One) and in Chapter Four (Section Two), as a factor leading to negative perceptions of the technology due to concerns of privacy and security - ultimately because automation restricts the control to the pilot and takes it away from those concerned with being intruded on. Given the relational aspect of UAVs, a collaborative, purposeful approach to the technology's automation is the most empowering approach, as it regulates control to suit the user.

Chapter conclusion

The conclusion on the unique opportunities drone technology offers to the interpretation of and engagement with heritage, encompasses a reflection on how the perceived unique opportunities, effects, and affordances of drone technology interplay in the configuration of UAVs as a medium with sociotechnical impacts. It also considers how the combination of opportunities, effects, and affordances bring possibilities to mainstream audiences, that remain exclusionary to underserved audiences, because they amplify the perspectives of dominant groups, through an auratic view.

It follows to discuss how the drone gaze can project an idealised view of place, increasing the engagement gap between underserved audiences and heritage, and considers how auratic views relate to exclusion, power, and agency. Finally, it proposes that it is essential to identify how these aspects manifest themselves in drone content, to be able to establish how drone technology can empower underserved audiences to connect with heritage and culture.

The drone competencies of drone mindedness and drone sensing, spatial mobility and verticality, control and automation are significant factors in the process of drone-facilitated engagement with heritage and culture. They are the key affordances that make the perceived opportunities possible. Drone-mindedness, an increased awareness of our surroundings and environments informed by drone sensing proficiencies, enables drone users to propose new perspectives and experience place in new ways. These new narratives are informed by people's lived experiences and backgrounds.

Spatial mobility enables access to the inaccessible. This widens the range of digital data that can be captured and heritage sites that are recorded to a high standard. Subsequently, if content is discoverable, it means that more people can engage with heritage sites captured. Its most important affordance is that movement provokes deeper involvement in drone content, because physical motion and top-down views create mental simulation, fostering meaning-making (Schendan and Ganis, 2012, p. 14) and therefore new narratives.

Storytelling and creativity are nevertheless only enabled in drone content, if users can exercise agency over the drone devices. This is regulated by the level of automation of the drone. Semi-automated drones operate in equitable collaboration, creating a human-machine interaction that is affective, leaves space for human creativity, is enhanced by the diversity of backgrounds and lived experiences of users.

The perceptions of heritage and drone experts on the opportunities of drone technology to engage non-visiting audiences are informed by their role within the sector. Heritage experts work from a commissioner's perspective, rather than that of a user, whereas drone experts relate to the technology from a working knowledge position. Drone commercial users who are also recreational users of UAVs consider that UAVs' have opportunities to have a wider socioeconomic impact; they elicit drone technology as a tool of authenticity, self-actualisation and democratisation.

Additionally, heritage experts talk about co-production, upskilling young people through drone piloting training; and survey respondents consider aspects of people having agency, rather than being passive consumers. Heritage and drone experts' qualitative data highlights six opportunities for UAVs to engage non-visiting audiences. These encompass new ways to view (broader perspectives), experience (a wider range of experiences), access (better, broader access), record (more data of higher quality), retell (meaning-making) and recreate (control and automation) visualisations of place, that are supported by the drones' technical capabilities and affordances. Heritage and Drone Expert participants consider these have a generally positive outcome, but what do they mean for underserved audiences who are underrepresented and disempowered by institutional systems?

Drone technology can enable underserved communities to generate their own narratives in relation to heritage. Through UAVs' effect of mental simulation, people can see themselves in the speculative reality they are viewing images of, leading to the articulation of self-generated narratives. These stories result from an enhanced awareness of place and space, and the ability to shape new perspectives. (The latter, in turn, is a result of the increased accessibility enabled by drone sensing and its inherent automation capabilities.) However, as discussed in the Literature Review (Section 2.2.4), authentic engagement encompasses both the expectation of a successful experience or outcome, and active involvement.

These necessitate that people have a stake on shaping their experience. It is therefore about empowerment and agency. Although, fundamentally, these reflect processes of cultural democratisation, only Drone Expert Seven makes this connection: "So absolutely, democracy of the arts works in two dimensions, two ways: it's for the viewer as well as the maker." This quote is important: it explores the positioning of the content producer and consumer, in relation to each other. This dynamic is at the centre of how heritage organisations relate to communities and the negotiations of power at play.

The above represents a new perspective on the use of drone technology in heritage and culture. It indicates that drone technology holds the necessary tools to empower underserved communities to connect with place. However, those tools are not yet fully used to empower underserved audiences. Empowering communities through collaboration involves what Heritage Expert Seven describes as "fostering confidence in people and providing the space and freedom for them to try something". To create this environment, heritage organisations and practitioners will benefit from adopting an open attitude towards change, what Nancy Kline (2001) defines as a "growth mindset".

This presents an opportunity for heritage organisations to collaborate with hobby drone users. Recreational drone pilots have a relationship with drone technology, that goes beyond the functional: it recognises the power and impact of mental simulation, as a form of self-efficacy. That increased positive interaction from heritage organisations with hobby drone users can help heritage and cultural organisations develop new approaches to the use of drone technology, that are more evocative of its relational effects. This can help heritage and cultural institutions grow more confident to venture into less conservative drone policies, as well as to consider using drone technology as an innovative way to empower underserved audiences to connect with heritage and culture.

The question now for both drone users and heritage and cultural organisations is: how is empowerment expressed in drone content created by underserved audiences? Currently, drone content from heritage organisations uses drones as a novel medium to reiterate the existing power dynamic that idealises place. This auratic view of place construes it as a spectacle and replicates the terrestrial architecture of hierarchies in the aerial space.

To identify what differentiates this, from content created by underserved audiences, where they have agency, the next chapter draws on theoretical terms of reference to create a framework of drone content analysis. This framework is used to compare heritage institutional drone videos and UAV visual content created by underserved audiences. Furthermore, I test the proposition by creating conditions where the power between institutions and underserved communities is levelled up, through the analysis of the visual output of my autotechnographic research, and the film resulting from the co-design research with people living with dementia.

Chapter Six: The visual language of empowerment and agency: a framework for analysis

Chapter introduction

> "[…] maybe it's a change of presentation style […] or-I think actually what I do notice that's different is that that its always footage of landscapes that are pretty but there's never any connection to the people that probably set the drones off in the first place […] or where they come from." (Drone Expert Three)

This quote from Drone Expert Three epitomizes the focus of this chapter on my research question exploring drone technology, as a facilitator of empowerment of underserved audiences to connect to and access heritage. It identifies the need for organisations to change their practice, to approach drone aesthetics in a way that it is representative of the different communities who are part of the landscape narrative but are invisible in institutional drone aesthetics.

The narrative in this research project starts with interrogating how drone technology can support the engagement of underserved audiences. This investigation draws on the Literature Review (Section Two) for the concept of engagement, which is worded as "productive involvement" (Ben-Eliyahu *et al*., 2018, p. 87), and "active participation" (Stroja, 2021, p. 310). Power and control have been identified early on in this research, in Chapter Four, as the fundamental challenge for heritage organisations in using drone technology with underserved audiences. Organisations find it difficult to reframe themselves in a new power dynamic, through engagement practices that give underserved communities agency.

Continuing to use the word engagement or expressions by Ben-Eliyahu *et al*. (2018) and Stroja (2021), as equivalent terms for empowerment to define what is a new relationship between heritage and cultural institutions and communities means a continuation of the existing mindset and extends the current architecture of hierarchies. While the sector continues to talk about engagement, institutions continue to position themselves in control and the shift remains metaphoric. The language that engenders the necessary shift in practice is empowerment- where communities are involved in making decisions over aspects that affect them. This leads to considerations of this research from, the perspective of the use of drone technology to facilitate agency and empowerment, in underserved communities' experiences of heritage and culture.

To answer the research question discussed in this Chapter, it is necessary to analyse the discourse of power, in the visual language used in drone content. This is a challenge because currently, no studies examining power and agential negotiations in drone videos, exist. This chapter reviews existing theorisations of the characteristics and effects of drone technology, to identify the aspects that make it possible to critically consider agential negotiations, through the use of camera drone content. These theorisations are used to inform the design of a new conceptual framework for the analysis of video content, from the perspective of empowerment.

This chapter is structured in two sections: Section One (6.1) reviews existing frameworks for the analysis of drone content. Section Two (6.2) outlines how these theorisations are applied in the design of a new conceptual framework for the analysis of video drone content. The chapter concludes with a reflection on the framework findings and what they mean for drone technology as an empowerment tool for underserved audiences engaging with heritage.

The above involves a critical consideration of the terms of reference informing my framework for the analysis of camera drone videos, and their usefulness to answer research question ("What is the role of drone technology in the empowerment and engagement of underserved communities to connect to heritage?"). Finally, the chapter concludes with a discussion on how empowered drone use by underserved communities is manifested in the visual language used by underserved communities.

6.1 Existing frameworks for analysing drone photography

This section assesses four existing theorisations of film and drone camera effects for analysis of drone video content. The purpose of this is to further investigate the affordances and unique opportunities drone technology provides to empower underserved audiences, as discussed in Chapter Five. In that chapter, in the cross-analysis of drone users' interviews, heritage partners strategic documents, heritage organisations' interviews and online survey results, six themes were identified. These themes summarise heritage organisations' and drone users' perceptions of the unique capabilities of drones, to bring new and more diverse ways of seeing and interpreting natural and cultural heritage. These are:

i. A broader perspective that provides context.
ii. Access and accessibility.
iii. More data of higher quality.
iv. A greater range of experiences for wider audiences.
v. A storytelling tool.
vi. Creativity.

The identified opportunities reiterate the value of drone technology in heritage and cultural environments, beyond their functional uses of monitoring and surveying. They also enunciate the possibilities for drones to facilitate agency in audiences. Nevertheless, these opportunities, which are a result of what drones can offer through their mobility, verticality, and remote sensing, are predominantly engendered within principles of institutional compositionality. This is evident in the National Trust's drone video discussed in Chapter Seven, *How to spot a Capability Brown Landscape* (National Trust, 2016).

The identified opportunities are prevalent in any drone content produced or commissioned by heritage organisations, where the latter are in control. What do these opportunities mean, when left out communities have control? To analyse what devices drone technology offers to empower communities to "flip" the narrative (Heritage Expert One), I draw on four analysis frameworks:

- Serafinelli and O'Hagan (2022): research on characteristics of drone photography visuals.

- Stankov *et al.* (2019): effect of drones in video creation and tourism promotion.
- Kraft (1987); Ryan and Lenos (2020); Probst, Rothe and Hussmann (2021): Media and Film Studies' conceptualisations of film camera perspectives and techniques to connect audiences with the narrative.
- Mademlis *et. al.* (2019): Taxonomy of UAV cinematographic shot-types.

Each framework starts with the identification of the method of the framework and its proposed themes or characteristics. This is followed by a review of those characteristics, for their suitability to analyse issues of power and agency: each framework is considered from the point of view of the resources camera drones make available to construct narratives; present new perspectives; and offer access to a greater range of experiences. This analysis leads to the conclusion that a new framework is needed. The new framework, incorporating elements from all the theorisations in this Chapter is a bespoke tool to analyse the visual language of power and agency, in camera drone-generated material.

6.1.1 Serafinelli and O'Hagan's characteristics of drone photography

Work by Serafinelli and O'Hagan (2022), summarised below in Table 6, explores the semiotic choices in drone photography, its context (institutional or user-generated, narrative of community of interest) and their potential meanings and possible sociocultural effects. Their framework is therefore useful to expand the analysis in this research, into more complex aspects of constructing meaning, and representational strategies through drone visuals. They propose ten categories. Although the characteristics discussed here are in relation to drone photography, they can be usefully extrapolated to video footage.

Table 6 Summary of characteristics of drone photography based on Serafinelli and O'Hagan (2022)

Theme/ Category	**Purpose**	**Corroborated by the data in this research**	**Useful for analysing power and agency**
Top-down views	Shot categorisation	N/a	Y
360 degrees panoramic views	Shot categorisation	N/a	Y
Classic landscape	Shot categorisation	N/a	Y
Access to inaccessible places	Affordance	Y	Y
Defamiliarising the Familiar	Affordance	Y	Y
Shadow Play	Theme – imagery	N/a	
Extreme Close-up Shots	Shot categorisation	N/a	Y
Power of weather	Theme – imagery	N/a	
Animal interactions	Theme – Imagery	N/a	

Serafinelli and O'Hagan (2021) categories of "Shadow play", "Power of weather", "Animal interactions" are considered in this research, as themes that describe the imagery at a denotative level. All other categories refer to the technological modality (Rose, 2016, p. 66). Although several of the categories by Serafinelli and O'Hagan (2022) correspond with some of the key affordances of drones identified in Chapter Five, they do not all fit the purpose in analysing power and agency in drone visual footage.

Serafinelli and O'Hagan's (2022, p. 231) understanding of "Access to inaccessible places" is considered here reductive in acknowledging drone technology's potential to increase access. "Top-down views", "360-degree panoramic views", " Classic landscape perspectives" and

"Extreme close-ups" are interpreted in this research, as qualifiers for the theme “A broader perspective that provides context” elicited in this project: they relate to the ways in which drone technology widens ways of looking.

The drone gaze gives access to the landscape, in ways that are not possible from the ground, but also, because it does so, as Drone Expert Nine posits: “You can see the top and the edge of the cliff, you can see the coast, and then you can see the sea.” This is what is conceptualised as a volumetric perspective (Jensen, 2020, p. 418) whereby drone volumes enable the viewer to go beyond the flatness of maps accessing spaces in-between and visualising them as tridimensional volumes. This is also articulated by Drone Expert Seven: “You know so if you’ve got fast moving ground in the foreground, then the mountains in the background appear to be moving fast, so a more dynamic use of this three-dimensional space.“

Serafinelli and O’Hagan (2022, p. 2) postulate the drone’s ability to take photographs from an unusual viewpoint has the potential to change the way people see the world, in the same way that point and shoot cameras were thought to do. These cameras, particularly the Leica, were used in USSR’s social realism period, where those new ways of looking were employed to promote a novel socio-political discourse.

Rodchenko took advantage of the point and shoot camera to produce revolutionary perspectives, from up above as in his photograph *At the Telephone* (1928), from down up, etc. Through them, the artist promoted state power within the collective consciousness (Glebova, 2022). This propagandistic approach is currently used by organisations to promote institutional narratives, including heritage sites. This is a strategy of control – and, as evidenced in the examples of use by activist communities discussed in the Literature Review, is also used to challenge political narratives (and reclaim agency).

Serafinelli and O’Hagan (2022) also focus on UAVs’ ability to innovate storytelling practices, a strong theme in heritage expert interviews and survey data. In fact, the authors consider this has driven the use of drone technology in the cultural and creative industries, as well as recreationally. The potential for relevant and meaningful storytelling is extended by UAV technology's ability to produce "More data of higher quality", as identified in the themes in this thesis.

Data previously only easily accessible to institutions and corporations is not only consumed but, above all, produced and distributed by a wider range of people. This, in turn, widens the opportunities for new stories and meanings. Innovative storytelling is important in considering the agency of underserved audiences to connect with heritage and culture. Eliciting new narratives that have been hidden or silenced is central to empowering communities and any proposed approaches to using the technology require this to be recognised and implemented.

The angles and the composition of the camera drone vision through the production of "Top-down views", "360-degree panoramic views", " Classic landscape perspectives" and "Extreme close-ups" are constituents of the opportunities of camera drones. They inform the process of change in how people view and express themselves, in relation to their surroundings. They are characteristics that contribute to camera drone offering unique opportunities to support the telling of the landscape story and, it is their use in a context of disembodied mobility and verticality, that produces the shift in points of view.

Although there are common aspects between the drone affordances identified in this project's research data and drone photography categories by Serafinelli and O'Hagan (2022), the latter are based on visual content and ethnographic research of drone users. Consequently, their categories are particularly centred on the visual product. Conversely, the data in this research is gathered from heritage expert and drone users interviews and surveys.

The data from these stakeholders is used in this research to identify competencies in drone technology, that go beyond the results of using drones and the visual and focus instead on the dynamic of human-machine-environment interactions and identity negotiations. Therefore, although their work is useful to this analysis, it is insufficient to make considerations on the empowerment and agency of underserved communities in drone video footage.

6.1.2 Stankov *et al.*: effects of user-generated drone videos

Research by Stankov *et al*. (2019) considers the impact of drone videos in promoting tourism activity. This relates to heritage as a touristic cultural offer. Moreover, as discussed in Chapter Two, virtual tourism is increasing due to the emergence of mobile lifestyles (Gretzel *et al*., 2020, p. 197). Stankov *et al*. (2019) focus on user-generated drone videos, which also relates to the use of drone technology to empower communities that may not be expert users. Finally, the authors examine drone videos from the point of view of their impact on video creation. This is pertinent to the investigation of the role of drone technology in empowering underserved communities, as it is an approach focused on impact and outcomes.

As summarised in Table 7, Stankov *et al.* (2019) outline the effects of drone videos (which they designate as categories) as: "Titles and descriptions of videos", "Length and date of creation", "Locations", and their "Population density". It is important to highlight that one of the aims of this research is to identify the impact of drone videos, on travel destination marketing and therefore, the results, interpretation and categorisations are informed by this. They also analyse the descriptions of user generated drone videos given in titles, blurbs, locations, and the categories to which they are related.

Table 7 Review summary of visual framework, based on Stankov et al. (2019)

Theme/ category	**Purpose**	**Corroborated by the data**	**Useful for analysing power and agency**
New perspectives through the capture of aerial views, including Bird's Eye View	Effect	Y	Y
Access to previously hard or impossible to film locations	Affordance	Y	
Elicitation of the ambition of human flight	Effect		Y
Facilitation of access to the creation of aerial video	Effect		Y

Theme/ category	**Purpose**	**Corroborated by the data**	**Useful for analysing power and agency**
A new type of aerial filming	Effect		Y
Access to a multiplicity of modes of data capture	Affordance		Y
Location analysis	Method of thematic categorisation	Y	
Description of contents	Method of thematic categorisation		Y

These are useful in categorising drone videos thematically. Heritage organisations' video can often fit in several themes, including "Education", "Science" and "Technology" as well as the three most frequent themes identified by the authors for user-generated YouTube drone videos ("People and blogs", "Travel and events", and "Film and Animation)". While these thematic categories are used in the new framework, I add Heritage as another, to identify specific content produced by heritage organisations beyond broader aspects of identity and culture. Similarly to Serafinelli and O'Hagan (2022), research by Stankov *et al*. (2019) reflects the themes elicited in the data from heritage experts and drone users.

The drone video effect of "New perspectives through the capture of aerial views, including Bird's Eye View" in Stankov *et al*. is elicited in the data, through "A broader perspective that gives context", considering angles and shots as central to the new perspective. Access to previously hard or impossible to film locations" is implied in "Access and accessibility. "A new type of aerial filming" and "Access to a multiplicity of modes of data capture" are synonymous with "More data of higher quality".

Reflecting on the latter two aforementioned effects of drone video (Stankov *et al.,* 2019) - "Access to a multiplicity of modes of data capture" and "A new type of aerial filming"-, these are here interpreted as producing a dual outcome. First, they are part of widening the perspective through new ways of seeing and telling stories. Second, they are themselves interconnected, as "A new type of aerial filming" is perceived here, as a proposal for different

ways of capturing data not only on places, but aerial space in particular.
Aerial space, and the human perception of the boundaries of physical limitations, in terms of mobility and vertical space, are important for the discourse on empowerment. They are articulated in the work of Stankov *et al.* (2019, p. 811), through the "Elicitation of the ambition of human flight". Although this theme has not appeared directly in any of the data investigated in this doctoral project, it is considered here inherent to articulations of "empowerment and freedom". This is because the "Elicitation of the ambition of human flight" (2019, p. 811) adds an intangible dimension to the aspect of "Empowerment and freedom", which goes beyond the physical operation of the drone.

As argued in Chapter Five (Section 5.3), the position from Royo-Vela and Black (2020, p. 3) is that aerial views enhance the process of mental simulation. Stankov *et al*. (2019, p. 811) agree and consider that aerial drone videos evoke and enable a realistic mental simulation of the longstanding human aspiration to fly. According to Royo-Vela and Black (2020), within the process of mental simulation, people imagine themselves in the place, activity, or experience. This provides an anchor for developing knowledge and, in turn, connecting to their surroundings and narratives.

In this context, the drones' top-down views engage with the fulfilment of the human imaginary and ambitions. It is posited here that this enhances empowerment because it construes a sense of overcoming a human limitation, enhancing self-efficacy, a concept referred to in Chapter Two in relation to individuals' expectations of succeeding in having a positive outcome as a result of an experience, interaction, or activity. As discussed, exclusion affects sense of self-efficacy negatively, resulting in low levels of aspiration in underserved communities. Aerial views effect of simulating the fulfilment of a human aspiration may therefore contribute to empowering people.

The effect of "Access to previously hard or impossible to film locations" is partly aligned with the theme of "Access and accessibility". Nevertheless, the focus is not only on drones facilitating access to places but, more importantly, new ways of visualizing them and recording them. This is articulated also in drone videos providing "Access to a multiplicity of modes of data capture". The latter is elicited in the data, in the context of drone technology producing "More data of higher quality" and "Digital archiving". This is particularly important for heritage organisations in terms of monitoring, preservation, and documentation. It is equally significant in the context of underserved audiences adding their voices with the legitimized authority from lived experience, that addresses the power imbalances of silenced or misrepresented narratives, in natural and cultural heritage.

6.1.3 Film visual language (Media and Film Studies)

As the data in this research indirectly refers to camera perspectives and the technological modality (under the themes "A broader perspective that includes context" and "Access and accessibility"), it is necessary to inform the new conceptual framework, with concepts relating to the use of camera perspectives as part of signifying practices in film. This means drawing on Media and Film Studies' theorisations.

The Film Language framework draws on the research of Kraft (1987), Ryan and Lenos (2020), (2020) Probst, Rosse and Hussman (2021), (Savardi *et al.*, 2023), Yilmaz *et al.* (2023). As illustrated in Table 8, it essentially considers film camera perspectives, which includes camera levels, movement and orientation, angles and shot sizes. Each of the camera perspective categories has its own modalities and affects the process of visual storytelling. This is crucial in investigating how drone visual language can construct new narratives for underserved communities.

Table 8 Review summary of visual framework based on visual language of film cameras, drawn from Media and Film Studies

Theme/category	**Purpose**	**Corroborated by the data**	**Useful for analysing power and agency**
Camera Levels/Orientation	Shapes storytelling	Indirectly	Y
Camera Movement	Engagement with narrative	Indirectly	Y
Camera Angles	Meaning-making	Indirectly	Y
Shot Sizes	Conveys emotions	Indirectly	Y

Camera levels refer to the orientation of the camera (Savardi *et al.*, 2023, p. 79) in a lateral and longitudinal axis. Levels are outlined as Aerial, Eye, Shoulder, Hip, Knee, and Ground (Savardi *et al.*, 2023, p. 81). Cameras are positioned at different levels, as part of shaping storytelling through the control of the viewers' point of view of the captured landscape or character (Savardi *et al.,* 2023, p. 79). This affects viewers' emotional responses towards characters and events being depicted.

Camera movement relates to the technique leading to different types of motion of the camera on screen (Yilmaz *et al*., 2023, p. 10). Film Studies' literature identifies Pan, Tilt and Zoom as the key camera movements. The motion of the camera regulates viewing audiences' physical engagement and immersion in the film and how they experience it (Yilmaz *et al*., 2023, p. 1). As a result, camera movement techniques bring the audience closer to the narrative (Morgan, 2016 cited in Yilmaz *et al*., 2023, p. 2) extending the audiences' perception of physical sensing into embodied experience (Yilmaz *et al*., 2023, p. 1).

Camera angles are what Savardi *et al*. (2023, p. 79) identify as the positions of the camera. These can be Overhead, High, Neutral, Low, and Dutch angles (Savardi *et al.,* 2023, p. 81). According to Kraft, camera angles are "visual adjectives" (1987, p. 301). As such, they have a significant effect on viewers' construction of meaning from a picture and in the process of remembering the narrative (1987, p. 291), including when a scene does not include humans (Savardi *et al.*, 2023, p. 79). Gruber *et al*. (2023) suggest there is a multilayered relationship between camera use, camera angles, and perceptions of power, because camera angles establish a power dynamic which Savardi, *et al.* (2023) posit to be established between the characters in the film.

This dynamic extends to the relationship between the depicted and the viewer, be the latter human or not. The view of Savardi *et al.* (2023) echoes the theory (Kraft, 1987, p. 305) that camera angles represent real life learned meanings and are presented as their equivalent: for instance, a Low Angle Shot places the viewer looking up from below or the floor in a position of vulnerability; whilst both the subject and the viewer at the same level communicates equal levels of power between the depicted. Conversely, the increase in angle height is associated with heightening of power. Savardi *et al* (2023, p. 79) recognise that camera angles can be used in stylistic analysis as they support a deeper understanding of visual language in film. Shot sizes designate the field of view determined by the distance between the camera, and the subject it is capturing (Yu, 2023, para. 4). They also encompass other elements, such as the length of the focus (Probst, Rosse and Hussman, 2021, p. 179). Each of the six shot size modalities (Extreme Close-up; Close-up; Point of View; Medium Shot; Long Shot and Extreme Long Shot) fulfils a different function in the narrative process of the film.

Shot sizes are considered an essential aspect of the language of representation in film (Hanmakyugh, 2020; Probst, Rosse and Hussman, 2021; Yu, 2023), partly due to the increased use of mobile media (Yu, 2023). They vary in meaning, significantly (Ryan and Lenos, 2020, p. 47). The way they are used affects how audiences engage with the film. This is because shot sizes convey emotional aspects of the narrative eliciting different emotions in audiences (Probst, Rosse and Hussmann, 2021, p. 185).

Camera perspectives are elicited indirectly in the data in this research, under the theme "A broader perspective that includes context" and "Access and accessibility": aerial views that are all encompassing of the landscape are the perceived dominant perspective in drone visuals, that all respondents refer to when mentioning specific perspectives. The categorisation of camera perspectives and their employment as signifying practices in the Film Studies discipline supports the analysis of camera drone video visuals, beyond the distance of aerial mobilisation and into more nuanced meanings communicated through the size of the shots, levels, and angles.

This is important to deconstruct what the broader perspectives of landscape are, and what narratives they produce. It is from this point of view, that the opportunities of "Storytelling" and "Creativity" in the data are related to this framework, as it categorises camera perspectives, identifying their role in shaping meaning-making and engagement with a narrative.

This theorisation of camera perspectives in cinematography confers a formal term of reference to identify camera shots, angles and levels that are perceived as specific to film cameras, that are applicable to both film cameras and drone cameras and that those are unique to camera drone technology. It provides my research with an academically recognised matrix for the identification and interpretation of visual information in drone content.

However, to consider empowerment and agency, this framework lacks contextualisation. This is because it does not consider new approaches to filmmaking, such as drone videos. This means that the conceptualisations used in Film and Media Studies can be extrapolated to drone content, but only to the extent that some ground camera perspectives are transferable into the drones' camera. This is the case of low and high angles, for instance. As a result, in order to be able to utilise the semiotic theorisations of Film and Media Studies in this research, it is necessary to integrate them with other approaches, into my newly designed framework.

6.1.4 Mademlis *et al.*: UAV cinematography shot-type taxonomy

Film and media studies theorisations of camera perspectives and their meanings include aerial views. Nevertheless, the reference to aerial views produced through drone, or any other aircraft, is not explored further to include the wide range of shots drone technology enables. As such, the taxonomy of drone shots (Mademlis *et al.,*2019) provides an overview of drone camera perspectives in cinematographic applications, including a description of the movement that characterises the shots. This taxonomy is limited in exploring the meaning of different drone shots and perspectives. This is one of the reasons why it is insufficient to analyse drone video content.

Research by Mademlis *et al*. (2019) has a teaching element and aims to support non-experts to engage with cinematographic applications of drone technology, in a structured way. The work by Mademlis *et al*. (2019) provides a systematic categorisation of drone shot types and a tutorial to identify them. As it formalises current industry practices in UAV cinematographic use, research by Mademlis *et al.* (2019) enables the identification of drone shot types when analysing drone content with more accuracy and rigour.

To develop a framework of analysis of drone content in the process of meaning-making, this taxonomic framework complements the previously discussed work of Stankov *et al.* (2019), and Serafinelli and O'Hagan (2022). The latter explore elements of camera drone content from the point of view of impact and effects, including ethnographic and semantic aspects of camera drone outputs.

The taxonomy by Mademlis *et al.* (2019) does not consider effects, but it proposes a structured overview of form that is specific to camera drones. This allows me to expand my analysis into drone shots that are not shared with film cameras. This formalisation of cinematographic UAV shot types expands on the catalogue of camera perspectives considered in Film Visual Language. The categories relevant to my research relate to camera drone framing shot types (i); the subject focus (target or scene oriented) (ii); drone movement or inertia (iii). These are presented in Table 9, below, as identified by Mademlis *et al*. (2019):

Table 9 Review summary of visual framework, based on Mademlis et al. (*2019)*

Theme/category	**Purpose**	**Corroborated by the data**	**Useful for analysing power and agency**
Framing shot types - Extreme Long Shots - Long Shots - Medium Shots - Medium Close-up	Transferred from traditional film camera ground and aerial shots and therefore connected to Film Visual Language.		Y
- Two-shot/Three-shot - Over the Shoulder			
Scene and target shots - Scene shots: landscapes - Target shots: people, animals, activities	Relate to captured subject	Y	Y

Theme/ category	Purpose	Corroborated by the data	Useful for analysing power and agency
Drone movement: static - Static shots of scenes - Static shots of still targets - Static shots of moving targets - Static Aerial Pan - Static Aerial Tilt -	Categorises drone movement		Y
Drone movement: dynamic scene shots - Moving Aerial Pan - Moving Aerial Tilt - Pedestal or Elevator Shot - Bird's Eye Shot - Moving Bird's Eye Shot - Survey Shot - Flythrough -	Categorises drone movement in relation to scene		y
Drone movement: dynamic target shots - Constrained lateral tracking shots - Pedestal/Elevator Shot with target - Reveal Shot	Categorises drone movement in relation to target		Y

In this framework, the taxonomy extends to a technical description and illustrations of what each of the shots looks like. This reflects the author's aim to engage non-experts. This supports accuracy, in my analysis and conclusions on the aesthetics of drone content based on the use of camera perspectives. Above all, this is a practical framework, focusing on objective aspects of camera drone perspectives, which bridges drone expert knowledge and practices with traditional Film Studies camera uses and makes it possible to recognise innovative perspectives in drone cinematography. This framework also enhances my literacy of drone camera perspectives and capture techniques. However, this taxonomy is technical and does not consider meaning-making processes. Therefore, the connection between my data and the taxonomy is a direct result of a process of data analysis and interpretation, and not of the taxonomy itself.

The categorisation of drone shot types in cinematographic applications by Mademlis *et al.* (2019) confers a formalised technical component to my proposed framework for the analysis of drone content. The taxonomy helps the understanding of the effects of technical configurations, in drone produced cinematographic content and, subsequently, makes it possible to discuss agency and empowerment, as sociotechnical outcomes of drone technology. The taxonomy is used in the next chapter, specifically to discuss the presentation of drone visuals and deconstruct the artifices that facilitate the opportunities of drone technology.

One of the key elements of the taxonomy is that it differentiates between static and dynamic capture and, within that, between scene- and target-focused drone shots. Although this research explores engagement with natural heritage sites, which implicates scene shots, this project has a specific focus on investigating the visibility of underserved communities and elements of human presence in drone content, and these are elicited through target shots.

The taxonomy by Mademlis *et al.* (2019) relates to single drone cameras which extends my framework beyond elementary aerial view categories into a wider range of perspectives within the aerial space. This taxonomy recognises the complexity of drone vision and remote sensing, types of movement, and the captured subject; these are not as explicit in the other theorisations analysed in this chapter. It supports the identification of camera drone perspectives, with more accuracy. However, it does not concern itself with meaning-making or semantic intentions, other than identifying dynamic Aerial Pan and Tilt shots as context setters. This taxonomy is not a stand-alone tool for analysis of camera drone storytelling practices and therefore, not sufficient to make considerations on drone perspectives in the context of power and agency.

Section conclusion

To conclude, none of the conceptualisations and categorisations presented in this section facilitate a systematic basis, for the analysis of power in drone video content. They indicate aspects to consider generally, which are operationalised into the framework proposed in this research (as identified in Table 6, Table 7, Table 8 and Table 9) and some categories also reiterate themes that have emerged in the data from heritage and drone experts.
The considerations of power by Serafinelli and O'Hagan (2024, p. 233) in relation to the observer and the observed: the view from above and below and considerations on drones effecting the unequal power dynamics between producer and viewers (Ahmad, 2012 cited in Serafinelli, 2024 p. 229) emphasize how the drone's assemblage of capabilities and sensorial competencies integrate power and visibility (2024, p. 230). These reflections are also elicited in the data, as discussed earlier in this section.

Not all features of the frameworks discussed above are reflected in the heritage and drone experts' data, in this research. Categories of themes and locations by Stankov *et al.* (2019), for instance, are not directly replicated in the data collected in this project. This is possibly because experts made their considerations based on the use of drone technology to connect with heritage and this takes the place of the theme and locations captured in their diverse materializations (natural heritage, historic sites, built environment, narratives).

The taxonomy by Mademlis *et al.* (2019) is not articulated within the considerations of heritage or drone experts, again because for the former, the focus is on the final product but also, because as identified in Chapter Four, heritage practitioners indicate that knowledge and understanding of drone technology is one of the barriers to their use and empirically, technical understandings are a part of this. For drone experts, the central aspects of the use of drone technology are associated with the broader aspects of mobility and verticality and there is no mention in the data of specific shot types. Rather, experts discuss practical and relational affordances of the technology.

The theorisations of meaning through the use of camera perspectives in Film and Media Studies provide the most structured terms of reference, for the analysis of power in drone video content. Their limitations reside with the fact that these terms of reference are mostly specific to film cameras, except for perspectives that are shared across both film and drone cameras. This is the case of aerial levels (Bird's Eye View, Classic Aerial landscapes), Pan and Tilt movements.

The above summarising arguments lead to the conclusion that a new framework of analysis is necessary to answer research question "What is the role of drone technology in the empowerment and engagement of underserved communities to connect to heritage and culture?". This is because without examining drone content produced under different conditions (produced by institutions and produced by underserved communities in a position of agency) it is not possible to clarify whether there is a role for UAVs and what that role is.

The analysis of drone content requires a systematic and structured approach to the identification and categorisation of visual language in drone content, in order to ascertain how empowerment manifests itself. Section 6.2 proposes a new framework for analysis of empowerment in drone video content. This framework is applied to a selection of 11 videos (four by heritage institutions and seven by underserved communities) in Chapter Seven.

6.2 A new framework

The conceptual framework proposed in this thesis integrates descriptive data included in Stankov *et al.* (2019): title, description and themes. This data is combined with four more indicators, which are author/content producer, "Heritage" as a theme option, language and additional context (produced collaboratively, for instance). These categories elicit data associated with the conditions of production of drone videos. Examination of these is the first stage of analysis proposed in this new framework.

This project's proposed framework also combines theorisations on the effects of camera perspectives from Film and Media Studies, with the typology of drone shots by Mademlis *et al*. (2019) (including their definitions and construed effects) and drone photography characteristics and their effects, by Serafinelli and O'Hagan (2022, p.13): Bird's Eye View and Classic landscapes (2022, p. 235).

The second stage of the framework involves the identification of the drone shots used in the drone video content being examined and a consideration of the meanings they produce. It is important to note that every attempt has been made to identify drone shots (in pre-existing drone content) correctly using Mademlis *et al.* (2019) as reference. The identification of drone shots in existing video content is challenging and it is not possible to be completely accurate.

Finally, the analysis is completed by correlating the contextual data with the camera perspectives, to draw conclusions on elements of empowerment, agency, and involvement with heritage. The roadmap for this process is illustrated in Figure 5.

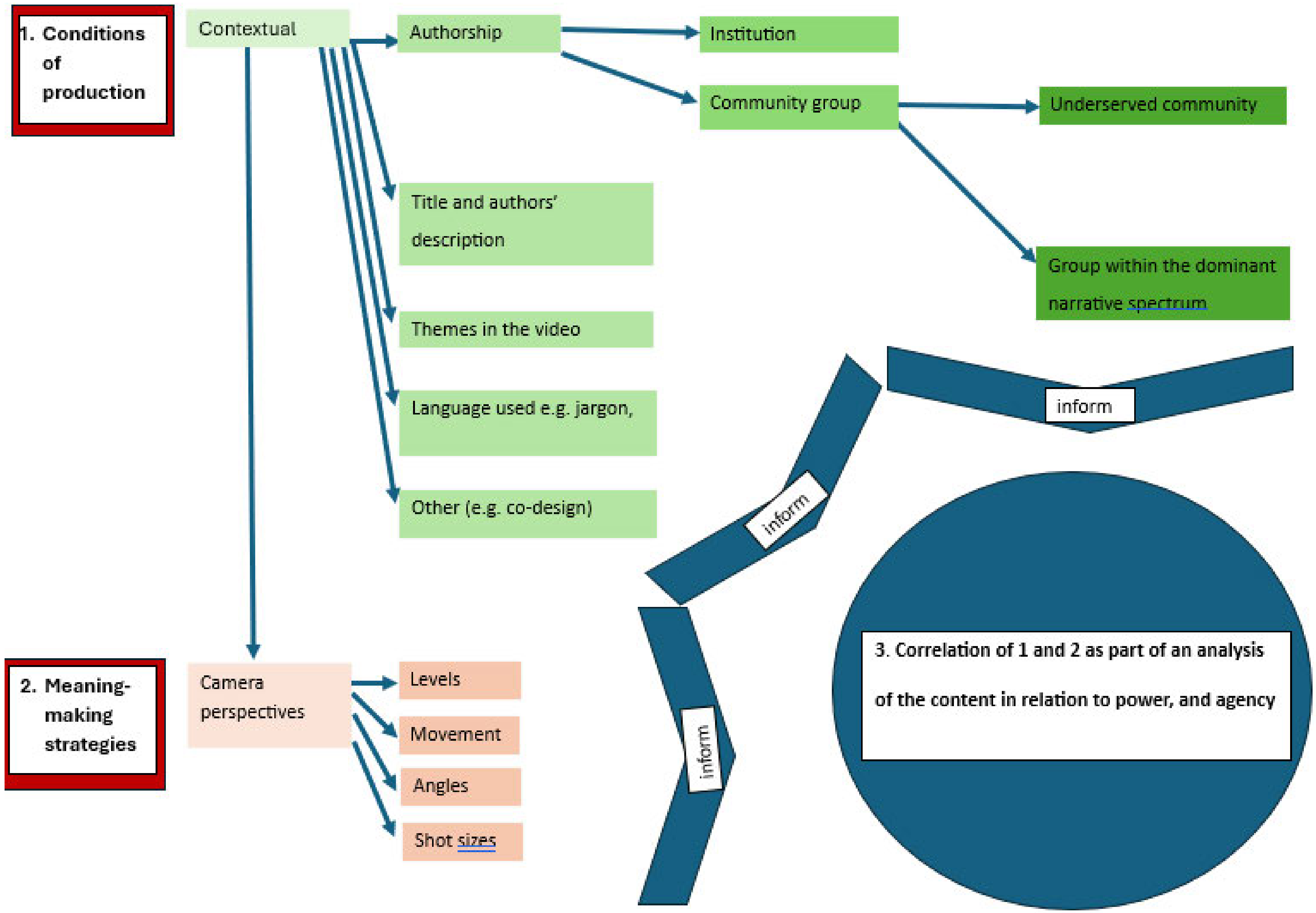


Figure 5 Road map for framework of analysis of power in drone content

This newly proposed framework differs from the academic work presented in section 6.1 as follows. Whilst Serafinelli and O'Hagan (2022) investigate the process through which drone photography affects, influences, transforms visual culture and society's readings of it, this research tests and discusses how communities operationalise drone technology to affect, influence and transform their identity narratives increasing their visibility, and as a result creating a new approach to visual language and meaning-making processes.

Stankov *et al.* (2019) consider constituents of camera drone videos as well as their effects, as part of discussing the impact of drone videos in the promotion of tourism. This approach is itself anchored in power, as tourism promotion is part of the commodification of mobility, culture, and place. Their work, therefore, does not make a distinction between the language of drone content production, and that of drone content consumption. They are entangled within the same considerations. This would be suitable if producers and consumers of drone content coincided with each other to a full extent. This is not the case.

Drone technical specification titles and content categories are more reflective of content producers' considerations on how to attract views and interest from fellow drone pilots. Fulfilling the aspiration of human flight is a significant sociocultural factor in the effect of drone videos, nevertheless this is combined with a wide range of other effects, that are disconnected from the cognitive effects of drone technology.

Overall, what primarily distinguishes the theorisation proposed in this research and the work of Stankov *et al*. (2019), Mademlis *et al. (*2019) and Serafinelli and O'Hagan (2022) are the focus. The latter are primarily concerned with visual media and media consumption effects, whilst my framework is centred on the sociotechnical implications (Fischer and Herrmann, 2011) of UAVs. These are the social effects, disruptions and new configurations resulting from the human-drone interactions. These disruptions are articulated both by expert and non-expert drone users, through signifying practices in drone videos.

The framework I propose facilitates the examination of aspects of power, authority, or disruption thereof, by integrating conditions of production with camera perspectives and examining how these interplay, to articulate or disrupt a narrative, and in the processes of harnessing agency and ownership. Further to this, the framework also facilitates an understanding of what distinguishes drone content produced by heritage institutions is distinct from that generated by underserved communities, when they have agency and control of the drone competencies and opportunities discussed in Chapter Five (section 5.2).

By combining these four approaches to drone and film camera perspectives the framework integrates into one theorisation, the different aspects that are relevant to discuss the shifts in power considered here as sociotechnical impacts (Fischer and Herrmann, 2011, p. 5) of drone technology. This proposed framework goes beyond visual culture effects and effects on other forms of content production. It also approaches drone technology from a more complex level than the nomenclature and technical specification of drone shots.

Drawing closely on the work from Film and Media studies, my framework of analysis proposes an equally systematic consideration of gazes mediated through drones, as scholarship upholds for film cameras. It recognises that drone technology's transformational effect on visual language can be extensively more elaborate than the established classic, auratic, breathtaking views or the activist functional drone capture video or photography captured by environmental activists or oppressed groups reclaiming colonialised territories, as discussed in the Literature Review (Section One).

This new framework also engenders the recognition there is a language of dissent and disruption that challenges idealised institutional and commercial compositions of landscape and place. By using it, this thesis asserts that the disruptions of drone technology are far deeper and more complex, than at the level of visual culture. They subvert existing conventions of the drone as an extension of the human body, into an instrument of visibility that interrogates the established archetypes and hierarchies of belonging, power, and representation.

Visibility, power negotiations and agency are at the centre of the framework. I have designed it for the purpose of analysing and identifying empowered language and institutional control strategies. I have done this because as someone from a Global Majority background and as a woman, I am interested in how drone technology can serve me in articulating my relationship with belonging in a way heritage organisations do not yet do. This framework is a tool that can be utilised by heritage and cultural organisations. Simultaneously, it is a tool that functions as a standalone facilitator of placemaking narratives.

6.2.1 Analysing power and empowerment in drone footage

The four frameworks analysed in this section are all useful in analysing drone videos. They all talk to the interview and survey data in this research: the taxonomies and categories they propose relate broadly to the affordances of drone technology identified in the previous chapter. For instance, Serafinelli and O'Hagan (2022) propose characteristics in drone photography considered to be epitomized by the drone technology affordance of a contextualised and broader perspective of landscape, as identified in the data, by heritage experts and drone users. These wider perspectives go beyond a visual context into a socio-cultural dimension, because they allow us to "see ourselves as part of the story" (Heritage Expert One).

Notwithstanding, the wider perspectives emerge from the study of what is a predominantly visible and dominant demographic in western knowledge production systems: significantly high proportion of male, white and middle-aged drone users (Serafinelli and O'Hagan, 2022, p. 227). Furthermore, the categories from Serafinelli and O'Hagan's work (2022), that are drawn on in this research to inform the proposed framework of analysis of drone videos created by underserved audiences ("top-down views", "360-degree panoramic views", "classic landscape perspectives" and "extreme close-ups") are mostly centred on visual culture impacts.

Serafinelli and O'Hagan (2021, para. 4) consider empowerment in drone technology from the point of view of how verticality and mobility, as technical competencies, democratise access to aerial space: they constitute an exercise of power. According to the authors, drone technology provides visual experiences, that go beyond the visualisations of space and places humans are accustomed to. They postulate that these new ways of seeing bring power and visibility together (Serafinelli and O'Hagan, 2021, para. 3). This is because, beyond the ability to conquer aerial space, drones enable the witnessing of the renegotiations of control of aerial space.

Those negotiations are manifested through new visual practices, which are forms of "reaction, redistribution and resistance" (Pauschinger and Klauser, 2020 cited in Serafinelli and O'Hagan, 2021, para. 3). These authors discuss power, as the result of drone technical competencies (physically the drone enables to go beyond human capacity) and see it leading to new hegemonies. Instead, in this research, it is posited that drones are a technology where power and agency emerge within a sociotechnical configuration (the drone is a technical agent of social outcomes and impacts). This is where the drone's potential for disruption lies. Where the production of content becomes positioned as action, not reaction.

Serafinelli and O'Hagan (2021) consider that reimaginations of place resulting from the power inherent to the drone capabilities of verticality and mobility have a de-territorialising effect. I partly disagree with this as the co-research data shows that aerial views can lead to a reacquainting with place, reappropriating it by expanding on its meaning through a process of revisiting past memories and narratives, and building new ones. The de-territorialising effect, conversely, may arise from aerial views enabling (non-local) others to relate to a place in an empowered away, developing a sense of connection that is independent to local belonging, and linking to the universal aspiration of human flight (Stankov *et al.*, 2019, p. 811).

This project focuses on the opportunities for these reimaginations of territory to foster reconnection, to support belonging, and the widening of placemaking strategies.
Serafinelli and O'Hagan (2021a) argue that movement, range, and autonomy challenge limitations of the human senses. I agree and add that those capabilities of drone technology expand on the potential of the human senses –verticality and mobility are interpreted here, as expanding the potential of sensorial experiences of place through the new re-imaginings of place and landscape. In turn, Stankov *et al.* (2019) by investigating the effect of user generated drone videos, in the promotion of tourism and in video creation, are attempting to

capture the auratic effect of camera drone videos on consumers' decisions to engage with places.

The aura, an influencing factor that attracts people by turning a place into spectacle, creates a distance between the subject and the experience. It excludes people from the narrative presented by drone footage, by placing the experience out of the reach of the experiencer. This is one of the elements conducive to exclusion. The auratic view places the experience in a position that is visible, touted as universal, although in essence, that universality is not true for everyone: underserved audiences are not reflected in the projected spectacle. By identifying the "Elicitation of the ambition of human flight" as one of the major effects of drones on video creation, Stankov *et al. (*2019, p. 811) position drone technology on the threshold of social disruption. As a result of the mental simulation afforded by vertical drone views discussed in Chapter Five (Section 5.2.2), this adds an intangible dimension to drone visuals beyond the physical operation of the drone.

To conclude, this framework is centred on the consideration of the audience as generator, in a modality of underserved consumer with agency, which is an approach reflective of the increased use of mobile media (Yu, 2023, para. 1) and mobile media practices. Differently, the four theorisations that inform the new proposed framework, approach content predominantly from the point of view of content producer and consumer as distinct entities.

These theorisations are centred on the effects of drone technology on visual culture. Their approach considers some elements of power, by including the affordances that enable access and accessibility and widening perspectives, but they do not contemplate the disruptive power of drone technology, beyond the outcomes of verticality and mobility, and what those might mean for unsettling established knowledge production models in heritage and cultural organisations.

6.2.2 A new conceptual framework for the analysis of the visual language of power and agency in drone footage

Table 10, in the next page, provides a simplified overview of the newly proposed conceptual framework. It has been included here to provide a more immediate engagement with the key components of the framework, signposting the consideration of aspects associated with the conditions of production, as the starting point for the analysis, followed by the examination of camera perspectives, as meaning-making strategies. The analysis of power in drone video content is the result of the integration of both the aforementioned aspects.

Table 10 Summary of framework of analysis of power in drone video content

<table>
<tr><td rowspan="6">Contextual information (which exposes conditions of production)</td><td>Author</td><td colspan="8"></td></tr>
<tr><td>Title and author's 'description</td><td colspan="8"></td></tr>
<tr><td rowspan="2">Themes</td><td>People and blogs</td><td>Travel and events</td><td>Film and Animation</td><td>Entertainment</td><td>Science and Technology</td><td>Sports</td><td>Autos and Vehicles</td><td>Education</td></tr>
<tr><td>Comedy</td><td>Gaming</td><td>How to and style</td><td>Music</td><td>Non-profits and activism</td><td>News and politics</td><td>Pets and animals</td><td>Heritage</td></tr>
<tr><td>Language used</td><td colspan="8"></td></tr>
<tr><td>Other contextual details</td><td colspan="8">Collaboration:
Human presence:</td></tr>
<tr><td>Shots in content analysed (refer to full framework)</td><td colspan="9"></td></tr>
<tr><td>Camera perspectives</td><td colspan="4">Source: Film camera</td><td colspan="5">Source: Drone camera</td></tr>
<tr><td rowspan="4">Levels</td><td colspan="4" rowspan="3">Aerial</td><td colspan="2" rowspan="3">Bird's Eye View</td><td colspan="3">Bird's Eye Shot</td></tr>
<tr><td colspan="3">Moving Bird's Eye Shot</td></tr>
<tr><td colspan="3">Static Bird's Eye Shot</td></tr>
<tr><td colspan="9">Classic Aerial Landscape</td></tr>
<tr><td rowspan="5">Movement</td><td colspan="4">Eye Level</td><td colspan="5" rowspan="5">Over the shoulder</td></tr>
<tr><td colspan="4">Shoulder</td></tr>
<tr><td colspan="4">Hip</td></tr>
<tr><td colspan="4">Knee</td></tr>
<tr><td colspan="4">Ground</td></tr>
</table>

<table>
<tr><th>Camera perspectives</th><th>Film camera</th><th colspan="2">Drone camera</th></tr>
<tr><td rowspan="12">Movement</td><td rowspan="2">Pan Shot</td><td colspan="2">Static Aerial Pan</td></tr>
<tr><td colspan="2">Moving Aerial Pan</td></tr>
<tr><td rowspan="2">Tilt Shot</td><td rowspan="2">Moving Aerial Tilt</td><td>Upwards Reveal</td></tr>
<tr><td>Downwards Reveal</td></tr>
<tr><td>Arc Shot</td><td colspan="2">Circle</td></tr>
<tr><td colspan="3">Zoom</td></tr>
<tr><td rowspan="6"></td><td colspan="2">Pedestal/Elevator Shot</td></tr>
<tr><td colspan="2">Survey Shot</td></tr>
<tr><td colspan="2">Flythrough</td></tr>
<tr><td colspan="2">Chase Shot</td></tr>
<tr><td colspan="2">Reveal Shot</td></tr>
<tr><td colspan="2">Ascent</td></tr>
<tr><td rowspan="5">Angles</td><td colspan="3">Overhead</td></tr>
<tr><td colspan="3">High</td></tr>
<tr><td>Neutral</td><td rowspan="3" colspan="2"></td></tr>
<tr><td>Low</td></tr>
<tr><td>Dutch</td></tr>
</table>

Camera perspectives	**Film camera**	**Drone camera**
Shot sizes	Extreme Long Shot	
	Long Shot	
	Medium Shot	
	Point of View	First Person View
	Close-up	Close-up
		Medium Close-up
		Extreme Close-up

Below, Table 11 shows the framework in a detailed version. This illustration includes the definitions and effects, that facilitate the interpretation of camera perspectives, as meaning-making devices in drone video content. It its full format, the framework operates both as guidance and a reference tool to be applied to each video analysed. When definitions or effects are not specified in this framework, it is because appropriate references or definitions could not be found in the literature reviewed.

Table 11 Detailed version of the proposed framework for the analysis of power in video drone content

<table>
<tr><td rowspan="6">Contextual information (which exposes conditions of production)</td><td>Author (s)</td><td colspan="7"></td></tr>
<tr><td>Title and authors’ description</td><td colspan="7"></td></tr>
<tr><td rowspan="2">Themes</td><td>Travel and events</td><td>Film and Animation</td><td>Entertainment</td><td>Science and Technology</td><td>Sports</td><td>Autos and Vehicles</td><td>Education</td></tr>
<tr><td>Gaming</td><td>How to and style</td><td>Music</td><td>Non-profits and activism</td><td>News and politics</td><td>Pets and animals</td><td>Heritage</td></tr>
<tr><td>Language used</td><td colspan="7"></td></tr>
<tr><td>Other contextual details</td><td colspan="7">Collaboration:

Human presence:</td></tr>
</table>

Summary of camera perspectives in drone content analysed (please refer to perspectives outlined below)		

<table>
<tr><th>Media source
Camera perspectives</th><th>Film camera</th><th colspan="3">Drone camera</th></tr>
<tr><td rowspan="3">Levels

Definition: Height of the camera in the scene in relation to subject being framed (Savardi et al., 2023, p. 81).
Effect: Shape storytelling through controlling viewers' point of view.</td><td colspan="2" rowspan="3">Aerial

Definition:
"An extreme high angle shot "from a helicopter or an airplane." (Ryan and Lenos, 2020, p. 268).

Effect: Provides viewers with a reference in space, time or reality (Savardi et al., 2023, p. 79).</td><td rowspan="3">Bird's Eye View
(Mademlis et al., 2019, p. 14) (Serafinelli and O'Hagan, 2022, p. 229).

Effect: Flattens, Defamiliarises the familiar.</td><td>Bird's Eye Shot
Definition: UAV is slowly flying up or down with constant speed. Camera axis is facing vertically down, and the target is directly underneath the drone. Altitude decreases and increases. (Mademlis et al., 2019, p. 18).
Effect: None other specified in literature.</td></tr>
<tr><td>Moving Bird's Eye Shot
Definition: Camera gimbal remains stable; UAV slowly flies in parallel to the terrain with constant speed. Camera is facing vertically down. (Mademlis et al., 2019, p. 18).
Effect: None other specified in literature.</td></tr>
<tr><td>Static Bird's Eye Shot
Definition: UAV flies vertically up while the camera stays stable and focused vertically down. (Mademlis et al., 2019, p. 14).
Effect: None other specified in literature.</td></tr>
</table>

Media source / Camera perspectives	Film camera	Drone camera		
			Classic Landscape (Serafinelli and O'Hagan, 2022, p. 13). **Effect:** None other specified in literature.	
	Eye Level **Definition:** Shot taken at eye level Effect: visual parity (Kraft, 1987, p. 305); neutral (Savardi *et al.*, 2023, p. 79).			
		Over the shoulder (Mademlis *et al.*, 2019, p 11) **Effect:** None specified in literature		
	Shoulder			
	Hip			
	Knee			
	Ground Rarely appears in movies (Savardi *et al.*, 2023, p 81).			
Movement **Definition:** Camera and lens motion in relation to the scene or characters (my own definition) **Effect:** Regulates viewers' immersion and involvement in the film.	**Pan Shot** Definition: A shot in which the camera rotates on its pivot or axis to follow or anticipate action. **Effect:** Used to make connection (Ryan and Lenos, 2020, p. 56).	**Static Aerial Pan** (Mademlis *et al.*, 2019, p. 16) Effect: None specified in literature		
		Moving Aerial Pan **Definition:** Camera gimbal rotates slowly with respect to the yaw axis,		

Media source / Camera perspectives	Film camera	Drone camera		
(cont.) Adds inflection and valuation to events or characters, (Ryan and Lenos, 2020, p 55). Movement elicits response to the execution of action as well as the observation of execution of action. (Heimann *et al.*, 2019, p. 2). Lens and camera movement are one of the most powerful devices to physically engage audiences with film and camera movement activates sensorimotor cortex. (Heimann *et al.*, 2019) Mirror mechanism (Heimann *et al.*, 2019, p. 1) Action perception link. (Heimann *et al.*, 2019, p. 1).		while the UAV is slowly flying at a steady trajectory with a steady speed. (Mademlis et al., 2019, p. 17). **Effect:** None specified in literature.		
	Tilt Shot **Definition:** Angling the camera up or down. (Ryan and Lenos, 2020, p. 269)	**Moving Aerial Tilt** Camera gimbal rotates slowly with respect to the pitch axis, while the UAV is slowly flying at a steady trajectory with a steady speed. (Mademlis *et al.*, 2019, p. 17). **Effect:** None specified in literature.	**Upwards Reveal**	
			Downwards Reveal	
		Static aerial tilt (Mademlis *et al.*, 2019, p. 17)		
	Arc Shot **Definition:** A continuous take that moves the camera in a circle around a character or an action (Ryan and Lenos, 2020, p. 268). **Effect:** None specified in literature.	**Circle** **Definition:** as the arc shot but often automated in a drone. **Effect**: None specified in literature.		
		Pedestal/Elevator Shot		

Media source / Camera perspectives	Film camera	Drone camera	
Adds inflection and valuation to events or characters (Ryan and Lenos, 2020, p. 55) Camera movement activates sensorimotor cortex (Heimann *et al.*, 2019).		**Definition**: UAV is slowly flying up or down with constant speed. Camera axis is parallel to the ground plane. Target moves identically to the UAV. (Mademlis *et al.*, 2019, p. 18). **Effect:** None specified in literature.	
		Survey Shot **Definition:** Camera gimbal remains stable, UAV slowly flies in parallel to the terrain with constant speed. Camera is facing ahead. (Mademlis *et al.*, 2019, p. 18). **Effect:** None specified in literature.	
		Flythrough **Definition**: Camera gimbal remains stable, with camera facing ahead and the UAV flying forward and through an opening/gap/hole with constant velocity. Small form UAVs. (Mademlis *et al.*, 2019, p. 19). **Effect**:	

Media source / Camera perspectives	Film camera	Drone camera	
		None specified in literature	
		Chase Shot **Definition:** UAV follows or leads a moving target maintaining a steady distance and remaining focused on the target. **Effect:** "simulation" of the target's motion within its environment while the target is fully visible. (Mademlis *et al.*, 2019, p. 14).	
		Reveal Shot **Definition:** Camera gimbal is stable, the target initially out of frame. UAV flies at a steady trajectory with constant speed until target is fully visible. **Effect:** None specified in literature.	
		Ascent **Definition**: Camera gimbal slowly rotates to always	

Media source / Camera perspectives	Film camera	Drone camera	
		keep the target properly framed, as the UAV linearly backs away from the target, increasing altitude (Mademlis *et al.*, p. 21). **Effect:** None specified in Literature.	
		Descent **Definition:** Camera gimbal slowly rotates to always keep the target properly framed as the UAV linearly intercepts the target, decreasing altitude. (Mademlis *et al.*, p. 21). **Effect:** None specified in Literature.	
		Fly over **Definition:** Camera gimbal is slowly rotating to keep the target properly framed. UAV intercepts target from behind, at a steady altitude and with constant velocity flies above it. (Mademlis *et al.*, p. 21). **Effect:** None specified in literature	

Media source / Camera perspectives	Film camera	Drone camera	
		Fly-by **Definition:** Camera gimbal is slowly rotating to keep the target properly framed. UAV intercepts target from the front and left/right at a steady altitude, passes it and keeps on flying at a linear trajectory with camera still focusing on receding target. (Mademlis *et al.*, 2019, p. 21). **Effect:** None specified in literature.	
	Zoom **Definition:** Zooms magnify and demagnify the image by varying the focal length (Ryan and Lenos, 2020, p. 269). **Effect**: Zooms focus (magnification) and move away (demagnification) from subject (Ryan and Lenos, 2020, p. 61).		
Angles **Effect:** Used to establish a power dynamic. The meaning of angle shots is not uniform (Ryan and Lenos, 2020, p. 53)	**Overhead** **High** Definition: When camera looks down at object. Ryan and Lenos, 2020, p. 268. **Effect:** Fosters an impression of domination as when someone looks down on someone else or suggest someone has lost power (Ryan and Lenos, 2020, p. 52). Allows viewer to look down on the actors placing viewer in place of visual dominance (Kraft, 1987, p. 305) Communicates weakness and vulnerability of the subject (Savardi *et al.*, p. 79). Intimidation, powerlessness, inferiority (Hanmakyugh, 2020, p. 106).		

<table>
<tr><th>Media source
Camera perspectives</th><th>Film camera</th><th colspan="3">Drone camera</th></tr>
<tr><td rowspan="3">As angles change down people seem taller, stranger, more unafraid, bolder, more aggressive (Kraft 1987, p. 296). Actors in a position of visual dominance (Kraft, 1987, p. 305).</td><td>Neutral
Effect: Parity (Kraft, 1987, p. 305)</td><td colspan="2" rowspan="3"></td><td rowspan="3"></td></tr>
<tr><td>Low
Effect: Attribute power to someone by looking up, augment the size of characters and increase the visual mass of objects (Ryan and Lenos, 2020, p52). Low Angle photographs judged more active, more potent than high angle (Kraft, 1987, p. 292).</td></tr>
<tr><td>Dutch Angle
Definition: When the cameras is tilted to make the object filmed or world appears at an angle, Ryan and Lenos, 2020, p. 269. Effect: None specified in literature.</td></tr>
<tr><td rowspan="2">Shot sizes

Definition:
Effect: Elicit different emotions in audiences.</td><td colspan="2">Extreme Long Shot
Definition: A shot from a great distance from the action, so that the ratio between figure and ground (landscape or set) is extreme (Ryan and Lenos, 2020, p. 269). Effect: Establishes a setting that orientates the audience (Hanmakyugh, 2020, p 109) Extreme Long Shot (Mademlis et al., 2019, p. 11).</td><td></td><td></td></tr>
<tr><td colspan="2">Long shot</td><td></td><td></td></tr>
</table>

<table>
<tr><th>Media source
Camera perspectives</th><th>Film camera</th><th colspan="3">Drone camera</th></tr>
<tr><td rowspan="5"></td><td colspan="2">Definition: A distant view of the action that shows the spatial relationships between the subject and the environment Ryan and Lenos, p. 268. (Mademlis et al., 2019, p. 11). Effect: Impersonality and objectivity Ryan and Lenos, p. 48. Public space distance (Probst et al., p. 185).</td><td rowspan="2"></td><td rowspan="2"></td></tr>
<tr><td colspan="2">*Medium Shot
Definition: Shot from a medium distance. Ryan and Lenos, p. 268 (Mademlis et al., 2019, p. 11). Effect: Translates into personal distance (Probst et al., p. 185).</td></tr>
<tr><td>Point of view
Definition: Shot through the eyes of a character (Ryan and Lenos, 2020, p. 269). Effect: Audience feels in the mind of the character Objectivity (Hanmakyugh, 2020, p. 107).</td><td>First Person View</td><td rowspan="2"></td><td rowspan="2"></td></tr>
<tr><td rowspan="2">Close-up
Definition: When the camera is close to the object so that a significant detail of the object takes up most of the space within the frame (Ryan and Lenos, p. 47). Effect: Intimacy (Probst et al., p. 185) Draws the attention of the viewer to an area of interest (Hanmakyugh,</td><td>Medium Close-up (Mademlis et al., 2019, p. 11).</td></tr>
<tr><td>Close-up (Mademlis et al., 2019, p. 11).
Effect: Intimacy. It gives the viewer an opportunity to access detailed information (Bowen and Thompson, 2017, p. 8).</td><td></td><td></td></tr>
</table>

Media source / Camera perspectives	Film camera	Drone camera		
	2020, p. 110). Give small movements prominence (Hanmakyugh, 2020, p. 110). Enables filmmaker to exercise control over what the viewer sees (Hanmakyugh, 2020, p. 110) What does a Close-up exclude? **Effect:** Conveys intimacy. Eliminates setting from view.	**Extreme Close-up Shots** (Serafinelli and O'Hagan, 2021, para. 6). **Effect:** A detail shot. It removes the captured subject from its context (Bowen and Thompson, 2017, p. 19)		

Chapter Conclusion

Following the review of conceptualisations of the characteristics, effects, taxonomies, and film camera perspectives in drone technology, it is concluded that these, used independently, are not suitable to analyse power and agency negotiations, in video drone content. This is because they do not consider the techno-social effects of drone technology or how drones interplay with hierarchies of meaning-making or invisibility in relation to left out communities and their contributions to the new tenets of Visual Culture produced through the drone gaze. This leads me to conclude that it is necessary to develop and refine frameworks such as this one that investigate the impact of empowerment in the generation of content by underrepresented communities.

I draw further conclusions on the analysis framework in Chapter Seven, after applying it to institutional video drone content and drone videos generated by underserved communities. This is because it is only after applying it that I can conclude on its suitability for answering my principal research question on how drone technology can empower underserved audiences to connect with heritage and culture.

Chapter Seven: Drone technology in cultural democratisation, equity and inclusion "How can drones empower underserved communities to connect with heritage and culture?"

Chapter introduction

> "[…] it goes back to - it allows people not only to - I mentioned about how drones allow you to experience your environment in a greater way - but it can just allow you to experience yourself, you know, in greater way (...) I get to be my authentic Black woman, boisterous, enthusiastic, passionate self." (Drone Expert Five)

This quote epitomizes the focus of this chapter, on answering the research question on how drone technology can facilitate empowerment of underserved audiences to connect to and access heritage.

This chapter deliberates on the findings from the application of the framework of analysis of power and agency in drone video content (presented in Chapter Six, Section Two). To identify the characteristics of a visual language of empowerment, in drone video content generated by underserved audiences the framework is applied to four drone institutional videos by the National Trust and the South Downs National Park Authority (summarised in Table 12) by underserved audiences (summarised in Table 13). A full in-depth analysis of the selected drone content is available in Appendix F. The discussion also integrates data from the heritage experts' online survey responses and heritage and drone experts' interviews.

Table 12 Overview of the institutional heritage drone videos analysed and key aspects of their production

Video title and author	Drone shot use	Source/access	Conditions of production/context	Theme	Reasons for selection
How to spot a Capability Brown Landscape, National Trust (2016)	Drone video illustrating the characteristics of Capability Brown's landscape work.	YouTube link sent by an interviewee from NT.	Institutional product with an educational purpose (described as a bluffer's guide).	Education, Heritage, History, Landscape, Historic Buildings.	Proposed by an NT interviewee, in reference to a project in which drone technology is used effectively in the Media and Video team.
Iron Age hillfort - Cissbury Ring video trail, London and South East National Trust (LSENT) (2021)	Drone video combined with ground camera video, about the Iron Age Hillfort at Cissbury, South Downs, Sussex.	YouTube link sent by an interviewee from NT.	Institutional product.	Landscape, Archaeology, History, Natural Heritage.	Proposed for analysis by an NT interviewee, as an example of work the organisation does using drone technology for audience engagement.
Dewponds of the South Downs (SDNP, 2021)	Drone shots combined with ground shots.	Retrieved by researcher.	Institutional product produced as part of a landscape specific Festival.	Heritage, Education, Landscape	Signposted by an SDNPA interviewee.
Shoreham Cement Works- 360° views, SDNP (2022)	360-degree views captured by a drone with a GoPro camera, where the viewer is in control of their experience.	Retrieved by researcher.	Institutional product with a community involvement aim.	Heritage; Community Consultation	Suggested by an SDNPA interviewee.

Table 13 Overview of the drone videos generated by underserved audiences analysed in this research, including conditions of production and reasons for selection

Video title and author	**Drone shot use**	**Source/access**	**Conditions of production/context**	**Theme**	**Reasons for selection**
Road to Freedom! Adagio Dance School (2021)	Drone video of a dance performance inside a derelict building.	Black Drone pilots Facebook private group page.	British Virgin Islands drone pilot and dance school.	Identity, self-determination and problematization of the legacy of enslavement.	Presents a counternarrative of the history of enslavement. Discusses identity and self-determination in the historical past of enslavement and the present reflections on the identity in the British Virgin Islands (BVI) in view of the legacy of slave trade and the political governance of BVI.
Bimini and Dan O'Neill on healing the ocean as LGBTQ people, Attitude Magazine (2022)	Ground camera video which includes drone shots.	YouTube	Produced by a LGBTQI+ publication. People featured in the video are members of LGBTQI+ communities and fashion, arts and media and environment professionals, including a drone pilot.	Identity, visibility, how the climate crisis affects LGBTQI+ communities.	Combines fashion, problematization of the dominant aesthetics and discussions of experiences of exclusion. Considers the role of nature in healing. Inclusion and empowerment as part of healing.
International Women's Day 2023, Women Who Drone founder (2023)	Drone shots produced showing drone being launched ascending and descending.	Women Who Drone Facebook private group page	Video created by a woman who is a drone pilot, to celebrate International Women's Day.	Celebrating International Women's Day 2023.	Reflects specifically on women's underrepresentation in drone pilots and more broadly in leadership. Presents a discourse on the visibility of women.

Video title and author	Drone shot use	Source/access	Conditions of production/context	Theme	Reasons for selection
Bringing the outdoors, indoors, two drone livestreaming events and videos Lovejoy Centre (2023)	Combination of drone livestream footage, with ground camera shots of participants engaging with the experience at the Lovejoy Centre.	Collaboration with Lovejoy Centre clients.	Co-design research	Remote visit to National Trust/SDNPA sites	Conditions of production- co-design research with people living with dementia where they shaped and controlled the drone facilitated experience.
These too are footsteps, Fine (2023a) and *A room of my own,* Madalena Fine (2023b)	Combination of preexisting drone shots, drone shots author captured, drone shots author choreographed, and which were filmed by a pilot. Inclusion of ground camera shots.	Researcher generated.	Autotechnographic research	Machine-human interaction, visibility, identity, gender	Understanding of the affordances, competencies and limitations of drone technology as a facilitator of empowerment for underserved audiences.

The application of the framework leads to the identification of three central themes, that distinguish drone content generated by heritage institutions, from that produced by underserved audiences. The chapter is structured around these interconnected themes, presented in the order below.

The subversive use of camera perspectives is discussed in Section One (7.1), which considers how underserved communities use existing camera drone techniques differently in their narrative and how, by doing so, they challenge existing drone video practices, as well as the construed approaches to using the drone's mobility and verticality. These are compared to the Classic Aerial Views and landscape-centric drone visuals produced and used by heritage organisations.

Next, Section Two (7.2) reflects on the theme of narrative ownership. This focuses on how drone video content creators position themselves, in relation to the narrative. Their distinctive use of camera perspectives leads to them becoming an agent and, therefore, they own their story. This is contrasted with institutions' use of the drone to propose a collective narrative centred on objectivity and evidence, where the drone is treated as witness and not a form of witnessing.

This section concludes with a critical reflection on how drone technology can empower underserved audiences to connect with heritage and culture and what this means for how heritage organisations use UAVs and the aesthetics they produce.

7.1 Problematising the aura of iconic views

This section focuses on discussing how underserved audiences use existing camera drone techniques differently in their narrative and how, by doing so, they challenge existing drone video practices, as well as the construed approaches to using the drone's mobility and verticality. Three themes emerge when considering the composition of distinct drone aesthetic practices by underserved audiences.

First, the subversive use of camera perspectives that are distinct from the classic aesthetics of drone camera videos. Second, proxemics as a strategy of control that enables refamiliarisation with the defamiliarized. Third, the effect of movement and pace, in the disruption of the auratic, towards connection and involvement, and how different approaches to drone aesthetics create new forms of witnessing. To address these themes, this research refers to the camera perspective's elements in the newly proposed analysis framework, as well as the content of the drone videos.

7.1.1 Reframing use of camera perspectives

As discussed in Chapter Four (Section Two), some of the heritage experts interviewed articulate that epic views of drone visualisations can create a distorted idea of heritage and landscape, which people will not recognise when they visit. Nevertheless, the majority of heritage interviewees and survey respondents exalt drone visuals for their "iconic views" (Drone Expert Nine), spectacular[41], powerful images[42]. Academic literature discusses drone visuals' most frequent characteristics as "Top-down views", "360-degree panoramic views" and "Classic landscape perspectives" (Serafinelli and O'Hagan, 2024, p. 223). These features afforded through drone mobility and verticality are at the centre of what heritage and drone experts identify as unique opportunities of drones, to engage left out audiences.

"A broader perspective that includes context", one of the themes discussed in Chapter Five is, as perceived by drone and heritage experts, generally a result of the aerial and panoramic views that create the sense of embodied omniscience. There is, nevertheless, an outlier perspective in the heritage expert interviews and survey data articulated by one interviewee, and which is central to the discussion on the challenging of the aura of iconic views. Heritage Expert Three discusses institutional drone aesthetics as "that epic 50-year- old way of looking at the countryside" and they suggest that this aesthetics needs to change.

The findings from the application of this research's proposed framework of analysis reiterate Heritage Expert Three's point of view. The application of the framework to four heritage institutional videos and seven (both the Lovejoy Centre and the autotechnographic study produced two videos, each pair analysed as a set) drone videos created by underserved

[41] Respondent 99850533
[42] Respondent 94720856

audiences (framework available in Chapter Six, Section Two) reveals that drone video content created by underserved audiences challenges the classic aesthetics of aerial sights. This is a result of the use of proximity through approaches to shot sizes, camera movement and pace. This is found to be the case in all seven videos analysed. Both approaches are discussed in the order below, and they are contrasted with the approach in institutional drone videos.

i. Use of proximity as a strategy of control

The use of close shots in drone videos generated by underserved audiences contrasts with the classic aerial drone views Drone Expert Nine describes as "your 360° view all at once with the drone technology showing you the cliff, showing you how high it is, showing the coast, and showing the sea at the same time." According to my framework, close shots create a sense of personal space distance (Probst, Rothe and Hussmann, 2021, p. 185) and intimacy.

The best example of this use in the drone videos analysed is the Women Who Drone's "Happy International Women's Day 2023!" post illustrated in Figure 6. It starts with a Medium Close-up, Moving Bird' Eye View shot of the pilot holding a drone controller and looking directly on to the drone camera. The Close-up draws the eye of the viewer to the pilot creating a sense of complicity between them and the viewer (Hanmakyugh, 2020, p. 110), reiterated in this video by the gesture of affection from the pilot. The Close-up shot is a strategy of control (Hanmakyugh, 2020, p. 110), where the pilot defines what is seen and unseen (Hanmakyugh, 2020, p. 111) and therefore the flow of the narrative.

Figure 6 "Happy International Women's Day!" (Women Who Drone) International Women's Day Instagram post showing a Medium Close-up Shot of pilot blowing a kiss

Another example of the use of proximity, as part of gaining control of the narrative is the co-research's livestream. In the study, participants living with dementia are encouraged to instruct the pilot on the path of the drone. In both livestreams, participants use the Moving Aerial Survey Shots that offer a panoramic view of the landscape, as orientation. These panoramic views, as established in the analytical framework, give participants the opportunity to have an overview of the surroundings.

I observe participants' use the panoramic views to provide context, focusing on details, for instance when a participant asks about what they are seeing, as illustrated in Figure 7 where a participant identifies a point of interest in the livestream "What's the dot on the-The sheep are up- "(Lovejoy Centre, 2023a, 03:26).

Figure 7 "What's the dot?" Lovejoy Centre participant on the right enquires about a detail

This is a technique to address the effect of aerial views of "Defamiliarising the familiar." (Serafinelli and O'Hagan, 2022, p. 229). In my framework this response of curiosity for detail is identified as "Refamiliarising with the defamiliarized", referencing the effect identified by

Serafinelli and O'Hagan: "Defamiliarising the familiar" in drone photography. It is important to make this distinction, as part of recognizing that the defamiliarisation resulting from aerial views has a disorienting effect, which disrupts sense of control and, therefore, agency. As such, refamiliarization with the defamiliarized reacquaints the viewer/content generator with the landscape and reorientates them. This interpretation is supported by Lovejoy Centre Staff Two's comment: "Seeing something from a distance, you can't really make out what it is and then being able to zoom in (…) when you got closer, you could see the gardens and the swimming pools. It's like coming to land on an airplane, isn't it, when you've been on holiday".

In the livestream narrative, there is an intercalated pattern of visuals that alternates between orienting, Figure 8 illustrates the drone capture following Lovejoy Centre Participant One's comment during the Panoramic Shot: "Nice little village down there" (Lovejoy Centre, 2023a, 10:23). In Appendix D, the number and types of prompts that refer to closer shots are identified as part of a discussion on how participants exercise agency during the livestream experience.

Figure 8 "Nice little village down there". Medium Close-up of village, 'Bringing the outdoors, indoors' (Lovejoy Centre)

Further along in the session, following a zoom in on the Brighton seafront through a Moving Aerial Pan, at the request of Lovejoy Centre staff, the group was asked “Do you want him (the pilot) to go a bit further up, a bit left, show all around again?” Lovejoy Centre Participant Three answers “Oh, yeah, yes please” (Lovejoy Centre, 2023a, 18:02). This resulted in the view illustrated in Figure 9 (Lovejoy Centre, 2023a, 18:29).

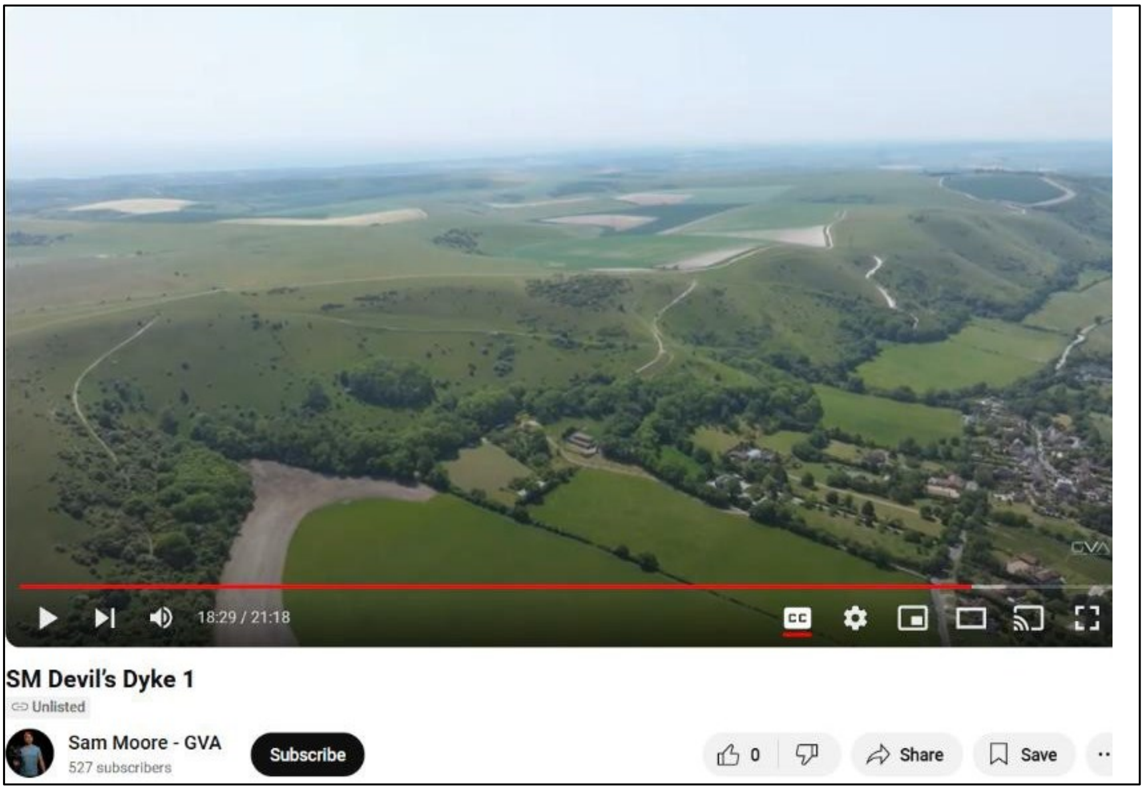


Figure 9 Panoramic view of Devil’s Dyke, ‘Bringing the outdoors, indoors’ (Lovejoy Centre)

The above-described interactions to capture participants’ interests and desires over their experience elicited a key finding, in terms of power, in the co-research design. The aspect of empowerment was anchored in the notion that its exercise was manifested through the participants proactively making specific requests relating to the direction of or focus of the drone. This conceptualisation of empowerment was based on what personal power meant to me and therefore biased.

Empowerment is relative to the communities within which it is fostered. Any ascribed views of empowerment that are outside of the lived experience of a group cannot be prescribed to that group, because they reflect external needs to those of that group. In the co-research, empowerment is expressed though engaging with the context of the viewing experience through the Panoramic Aerial views. The prompts for change of direction or focus on a detail emerge through the conversations and curiosity the viewing experience fosters, not through directives.

It is noticeable that, in institutional drone videos, the views are predominantly created through panoramic, Moving Aerial Survey/Bird's Eye View/Pan and Tilt, Long or Extreme Long Shots, as illustrated by Figure 10, National Trust's *How to spot a Capability Brown Landscape* (2016, 00:07) presents a good example of this. This video combines a sequence of Long and Extreme Long Survey, Moving Aerial Tilt and Pan Shots, that offer cinematographic views of an extensive National Trust property.

Figure 10 Survey Shot of Capability Brown landscape, 'How to spot a Capability Brown landscape?' (NT)

Through the application of my framework, I posit the use of these shots creates distance and has an effect of objectivity and impersonality (Ryan and Lenos, 2020, p. 48). This is because of the defamiliarizing effect of the aerial view, as discussed earlier in this section. Objectivity is one of the notions that heritage organisations have promoted extensively in the past, as part of an image of neutrality. In reality, museum and heritage organisations have reiterated and amplified the over valorisation of western narratives of heteronormative maleness, power and wealth. The current discussions in the heritage sector are focused on addressing this imbalance and this is the context behind the research and the principal research question on the use of drone technology to empower underserved audiences.

As determined in the framework, the use of distant shots, as in Figure 11 (NT, 2016, 00:02) replicates the public space distancing and frames the institution as a knowledge authority.

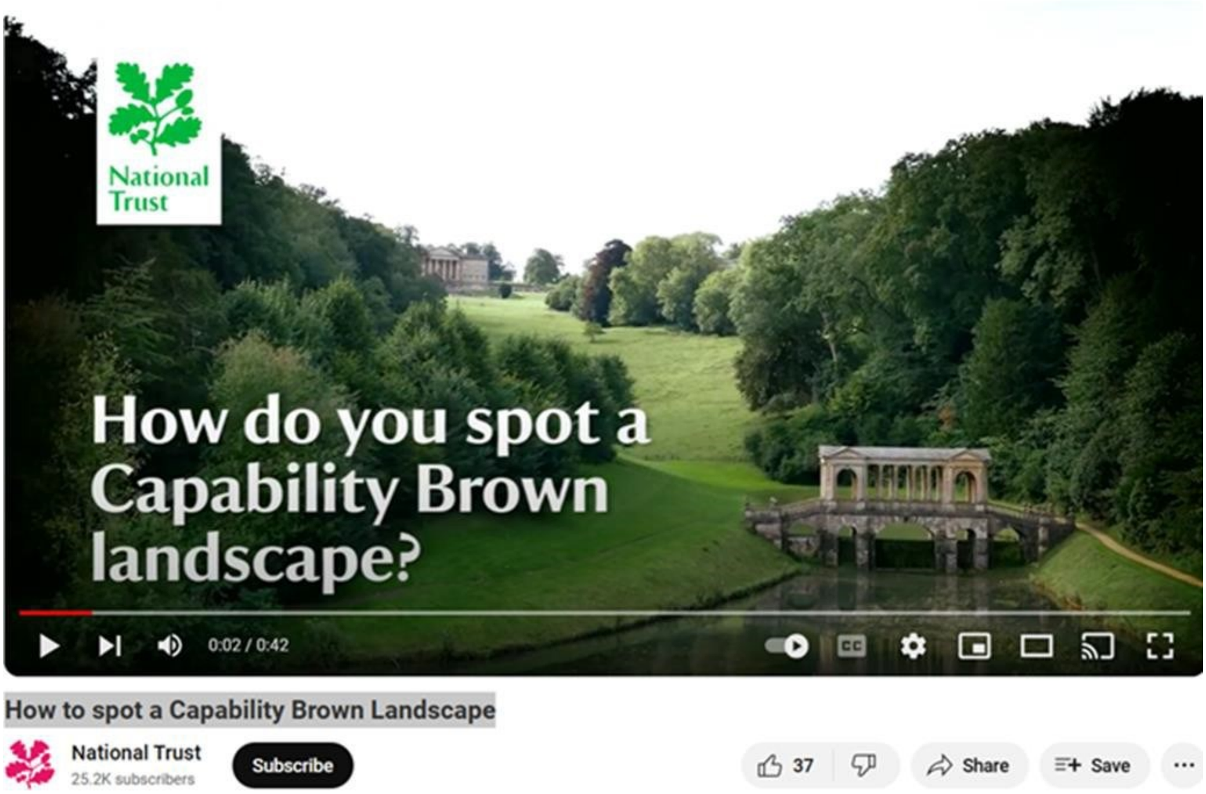


Figure 11 Introductory shot, 'How to spot a Capability Brown landscape?' (NT)

This is further reiterated by the choices on what is shown and what is not shown. In the National Trust video (2016) exploring the work of Capability Brown, the Establishing Shot uses a distant vantage point whereby the building is partly obscured and difficult to see, due to the use of a Long Shot.

The withholding of elements in the landscape, which as outlined in my framework is a form of control, creates an atmosphere of mystery (Bowen and Thompson, 2017, p. 65). It produces a division between knowledge shared and knowledge withheld, and who has the power to decide. This correlates with the gatekeeping culture discussed in Chapter Four, when considering the perception of regulatory frameworks, as barriers to using drone technology to engage left out groups.

As it is a National Trust branded property, the viewer is aware that although unclear where and what it is, it is a place of significance. To be revered. The drone views reiterate this, as in the shot, the drone descends, positioning the viewer at a Lower Angle, in relation to the estate which takes the position of the dominant entity. Contrarily to what is established in the framework, in this video, the closer shot shown in Figure 12, below, (NT, 2016, 00:29) is used not to primarily refamiliarise as we only see one shot of the planting scheme, but to show detail on a spot the difference approach to illustrating Brown's dislike for formal planting. This is a strategy for imparting knowledge, to educate.

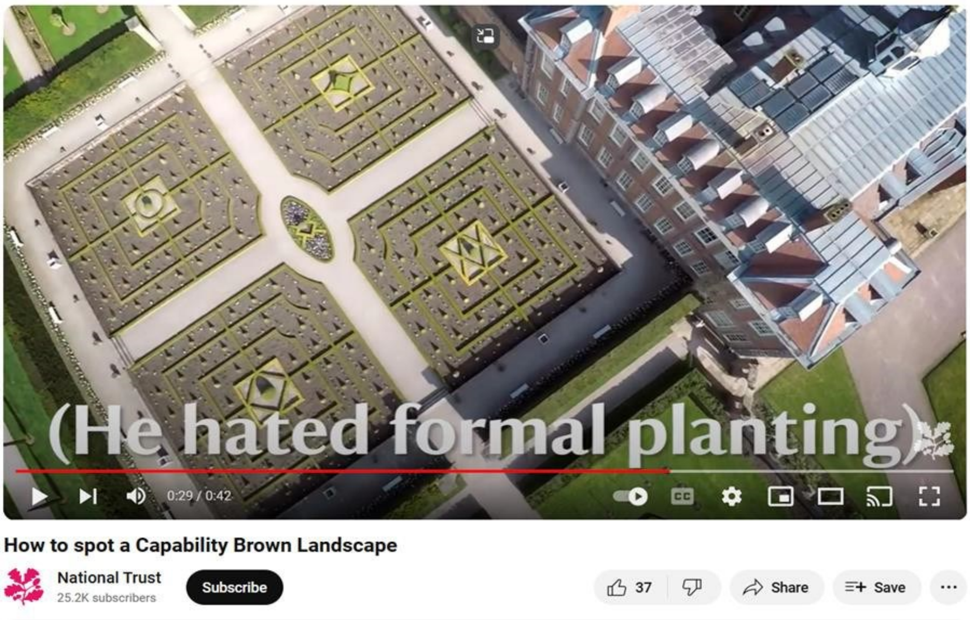


Figure 12 Bird's Eye View Shot of Capability Brown landscape, 'How to spot a Capability Brown landscape?' (NT)

An NT interviewee describes this drone video as conveying an innovative approach, different to the usual photography views used by the NT to capture and showcase their sites. This approach is coherent with the Trust's attempt to "engage audiences differently" (an NT Heritage Expert). This interviewee's comment reiterates the recognition that institutional drone aesthetics needs a new approach, in order to create connection with audiences.

In summary, following the guidance in my framework, excluded communities use close shots as a strategy to deconstruct the aesthetics of auratic aerial views, which create distance between the visualiser and the visualised. Through the close shots, drone content generators regain control and agency, over the drone mediated experience of landscape. It is in the context of this use, that drone technology facilitates empowerment, as it confers underserved audiences the ability to “flip the narrative” (Heritage Interviewee One).

ii. Movement and pace

Mobility is a central capability of drone technology, and it is considered in my framework. It plays a fundamental role in involving viewers in the experience of landscape, as articulated by Lovejoy Centre Participant Three in response to the livestream “It just feels good to be walking there”. This quote highlights the effect of what in my framework is outlined as mental simulation, an outcome of drone mobility (as discussed in Chapter Five), and how it elicits involvement in the experience of viewing, as an experience of doing, particularly as participants in the co-research have agency over their experience.

In four of the seven drone videos generated by underserved audiences there is a distinctive approach to pace, through using different combinations of drone movement and perspective. This is best illustrated in *Road to freedom!* (Adagio Dance School, 2021, 01:03) where the use of Flythrough Shots produced with a First-Person View drone adds an adrenaline-fueled viewing experience, that immerses the viewer in the emotional tension of the narrative of enslavement and self-determination.

The Flythrough Shots are combined with Moving Aerial Bird’s Eye Shots, Low Angle Shots, Close-up Shots. In the use of different perspectives, the content producer creates a different pace to that of institutional heritage drone videos. The same is true for the autotechnography videos where, particularly in *A room of my own* (Fine, 2023b, 03:48) an FPV Flythrough in the Monks House Garden, illustrated here in Figure 13, in combination with a rock music instrumental backtrack, intentionally challenge the traditional pace movement and sound that are part of the National Trust’s drone aesthetics and visual branding.

Figure 13 First-person view Flythrough Shot, 'A Room of my Own' (Fine)

The Flythrough Shots epitomise the key themes in both videos: first person voice, owning one's narrative, agency over one's journey. These themes are adjoined to the functions and capabilities of the drone used to create the content: a First-Person View Drone (FPV). As a device, FPVs are used in drone racing and their capabilities are associated with speed, acrobatics, adrenaline and therefore power.

The use of the Flythrough Shot is not the only approach affecting the pace of drone content. The alternation between Panoramic Views and Close-up Shots in the livestream film, also results in a more dynamic pace than that experienced in institutional drone videos. It creates movement in itself, and it brings the experience closer to the physical act of visiting a natural heritage site, where the visitor takes in both the surrounding views and the detail, because they are in control of their experience. In contrast, in the heritage institutional videos, the movement and pace are slower and more homogeneous. This is the case in three of the four institutional videos. The fourth video that is partly outside of this characterisation is the 360-degree video, as viewer interaction determines the movement and pace.

The National Trust's *Iron Age Hillfort- Cissbury Ring video trail* (LSENT, 2021) illustrated in Figure 14 (2021, 00:30), is the best example of the movement and pace that are characteristic of heritage organisations' drone content.

Figure 14 Survey Shot of Cissbury Ring, 'Iron Age Hillfort- Cissbury Ring video trail' (LSENT)

In combination with aerial perspectives, movement and pace in institutional content are designed to elicit feelings of witnessing a spectacle, inspiring awe in relation to "the beauty, (…) the realness of it." (Heritage Expert Nine). Heritage Expert Three describes drone content in their organization saying, "it's always got the same soundtrack, if you imagine this kind of like hippy on a guitar playing some nice sweet folk music".

This reference reflects the use of slow Surveying Shots. A Heritage Expert comments the videos in their organization are construed to show "the assets we've got in the South Downs from above", which surveying shots are suited to, as part of the process of itemisation, measurement or documentation of heritage resources institutions are custodians of. This is part of an aesthetics of power and control.

In conclusion, there is a clear distinction between movement and pace in the drone content generated by underserved audiences when compared to institutional drone content. This is as much a result of the use of different drone speeds such as racing drone Flythroughs in combination with non-racing drones, as it is the effect from using intercalated long and close shots and the background music the video is set to.

7.1.2 Forms of witnessing: dualities of presence, absence, and time

Mobile media, such as drones, produce technology-based forms of testimony (Wang, Lee and Wang, 2013, p. 492) and the different approaches to the drone gaze are part of a complex discourse connecting the aesthetics drones produce, autonomous mobility and visual forms, into systems of witnessing (Richardson, 2018, p. 7). These systems create new ways of "mediation, representation and experience" (Frosh and Pinchevski, 2014, p. 594) which differ, according to the content generators' positionality on their narrative, and the viewers' context.

Drone technology engenders a duality of presence and absence and here and then temporalities. This is because as drone content consumers, viewers are simultaneously here and there (Uzunogullari, 2023, p. 134), as they travel vicariously and simultaneously in their present time and the time framed within the experience of the drone content. Vicarious witnessing is particularly important, in the case of heritage places that are physically inaccessible or not open to public access.

In this context, the drone witnessing is not only mediating access. Its gaze becomes a gateway and not a form of gatekeeping. This is pertinent to consider, in the context of institutional power. It is again, as discussed in the analysis of the drone video content, an expression of control, that is articulated in the perceived opportunity of "A broader range of experiences to wider audiences". These experiences equate to a wider range of witnessing events as a result of opening up access or restricting it. The latter itself producing effects of awe (accessing back of the scenes, repository rooms, the making of).

In the same way as the effect of the livestreams' showing of the pilot launching the drone, in my autotechnographic film *These too are footsteps*, I include shots that disrupt the idea of back of the scenes, as an idealized view, as is the experience curated in using the drone to provide access to places outside of public areas. I do this by including a reflection on witnessing, as I capture the drone, capturing me, performing an automated circle movement around a point of interest (which is myself), as I observe the resulting image. This is depicted below, in Figure 15 (Fine, 2023a, 00:42).

Figure 15 Medium Close Shot from behind, 'These too are footsteps' (Fine)

In doing this, I discuss the complexity of the witnessing event, within a narrative of placemaking. I also explore more complex aspects of self-surveillance and curating own narrative, within the tensions of a public space, under human witnessing and machine testimony. In a way, the drone here shows me the back of the scenes, parts of myself I am unable to see without being mediatically captured by someone else. My internal "behind the scenes", in my relationship with the drone is articulated though the pixelated, loss of control of the drone event (shown in Figure 16) (Fine, 2023a, 01:10).

Figure 16 Shot of collision with tree, 'These too are footsteps' (Fine)

These instances are both part of my drone film because, integrated with the professional drone footage, they constitute testimony of my process of developing, reflecting on, experiencing and representing my counternarrative and are therefore part of witnessing.

Starting with the notion of presence and absence and how that affects witnessing (Peters, 2001, p. 707) by and through media (Frosh and Pinchevski, 2014, p. 596), this is elicited in my research, through the co-research's drone livestreams. These two events where participants living with dementia experienced the Devil's Dyke and Highdown Hill National Trust sites live, and directed the drone gaze, framed participants as producers and consumers of a witnessing experience (Frosh and Pinchevski, 2014, p. 608). This experience constitutes a type of witnessing in itself, which Peters (2001, p. 708) defines as televisual witnessing.

The difference between this form of witnessing in the co-research livestream and on a media live broadcast is that in the livestream, the participants/producers have an equal level of agency over the media event. This also means there is parity in the relationship between the viewers themselves, and between the viewers and the content as they are producing it using

their shared and individual understandings, as active witnesses.
Although the experience is repeatable, as it becomes a record of the witnessing of a live event, it is in itself a singular experience (Frosh and Pinchevski, 2014, p. 596). This singularity harnesses a form of power, as a result of synchronous witnessing: authenticity (Peters, 2001, p. 709). In the case of the co-design study, participants do not only have a "privileged proximity to events" (Peters, 2001, p. 707) they shape them. I argue this position further amplifies that power.

This duality of presence and absence also happens in the experience of viewing institutional drone videos or any other drone content as it is intrinsic to drones as media. Nevertheless, only the South Downs National Park Authority's video, *Shoreham Cement Works- 360°* (2022) elicits similar engagements with presence and absence, as the livestream events. This is because, similarly to the livestream events, in the 360° videos, viewers have agency to shape their viewing experience.

In the remaining three videos there is a concentric assemblage of testimony and witnessing. Organisations use drone aesthetics to sustain historical witnessing that takes place in physical site visits, reiterating the idea of objectivity in heritage narratives. Witnessing is an important aspect in the heritage and cultural narrative, where it is anchored on evidence from the past and the drone is, by extension, witness relaying evidence.

When viewing the heritage drone content, viewers witness the drone's testimony of the here and now whilst also witnessing the drone both recording and witnessing the past. This means that the way the drone is used enables viewers to become witnesses to an institutional representation of claimed narratives of authenticity, that are taken as truthful. The institution positions themselves as having privileged access to accurate information, to authenticity (Peters, 2001, p. 709) and the viewer is put in the position of trusting the institution.
For example, in the video *Iron Age Hillfort-Cissbury Ring video trail* (LSENT, 2021) the viewer witnesses the landscape as source of evidence. Only the drone view is a first-hand source of evidence into the scale and context of Cissbury. It enhances the understanding of its meaning and relevance to place, through the illustrative visualisation of the site and its surroundings. This has a similar effect to a map, and in effect, the next drone shot in the video, also a Survey

Shot, presents the drone landscape visuals as a cartographic exercise.
Maps are produced from aerial surveying shots. That relationship between place and data is elicited by this. Drone technology as a method of data capture emerges in the research findings relating to the opportunities of drone technology, under the theme of "More data of higher quality (and digital archiving)". The volumetric (Hildebrand, 2021, p. 135) experience of the drone shots in the video is triangulated with the secondary sources of narration by a specialist and an ordnance survey map.

It is important to consider that the relationships between live landscape and mapping reflect deliberations on land ownership, wealth and access and therefore, maps are instruments of power. This is evident in the examples discussed in the Literature Review (Chapter Two, section 2.1.5) where indigenous communities whose land has been appropriated by government use drone technology to document their original land ownership and reclaim their territories from governments and corporations.

By comparison, in *Road to Freedom!* (Adagio Dance School, 2021)[43] the drone witnesses the then in the narrative of slavery shown through the plantation house in ruins and its surroundings, a memorialization of the past and therefore a form of asynchronous (non-institutional) "historycity witnessing", (Peters, 2001, p. 721). Historicity, because it is part of a collective, memorialized past in which witnesses can be physically present in space, but are absent in time (Peters, 2001, p. 721).

The drone capture of the dancers' performance, symbolizing the agency and self-determination of enslaved ancestors through their choreography, produces a different form of witnessing. To the viewer, it is a recorded representation of the islanders' negotiations of identity and belonging in their homeland of the British Virgin Islands, still a British-ruled territory (Cohen, 1998, p. 190). To the dancers and drone pilot, it is the being "here and now" creating a testimony, the drone itself witnesses.

[43] Please see Appendix F for an in-depth analysis of this video.

Section conclusion

In conclusion, my framework enables me to compare the use of camera perspectives by underserved audiences and heritage institutions and interpret their respective meanings and the narrative they construct. In the comparison of drone video content, I find that underserved audiences use camera perspectives, movement and pace in ways that challenge the auratic views of institutional drone content, disrupting them.

This is principally achieved through the use of closer shots that project a physical personal distance, creating intimacy. This enables the viewer and content creator to refamiliarize themselves with the landscape following the defamiliarizing effect of Long Shot Aerial views utilized as part of orientation within the bigger context of the landscape. The different perspectives contribute to a distinct movement and pace in drone content that projects underserved communities' exercise in agency and control of the narrative.

Through their aesthetic strategies, heritage institutions and underserved audiences propose different forms of witnessing. This is a practice that is between experience and agency (Ashuri and Pinchevski, 2009, p. 133). The forms of witnessing in content produced by heritage institutions and prompted by drone content generated by underserved audiences reflect that difference. Witnessing in heritage institutions is predominantly "historicity" (Peters, 2001, p. 721) as viewers are absent in time and present in place (telepresent in the case of drone facilitated events). Peters considers witnessing raises problems of truth (2001, p. 712) and this is particularly the case, in heritage media witnessing. This is because viewers have to rely on the trustworthiness of institutional truth and authenticity. These have both, traditionally, been construed as the root of the power of heritage institutions.

When used by heritage organisations, drones become institutional witnesses. This is the context within which heritage experts consider drones offer access and broaden perspectives. This positions viewers as spectators. Peters argues that cameras show events as they happen without "embellishments" (2001, p. 708). Nevertheless, from the analysis of heritage organisations' drone content, their auratic perspectives of the landscape constitute an embellishment in itself. The auratic is, as one Heritage Expert argues, a distortion of the truth of place. This contradicts the idea of truth and authenticity, heritage organisations' power is anchored in. The heritage spectator experiences representations of institutional truth.

In the case of drone content generated by underserved audiences, the drone is both witness and testimony as it documents a record that makes negotiations of identity visible. Considering witnessing and testimony result from how drone affordances interact to mediate, represent and provide ways of experiencing (Frosh and Pinchevski, 2014, p. 594) Drone witnessing is not repeatable. As in a theatre experience, each flight in the same location, at the same time of day, will be different. There will be different events (birds flying, people and dogs walking by).

In the livestreaming, it is not only the fact that participants are directing the gaze of the drone but that they are doing it live, in a real time telepresent witnessing experience (Peters, 2011, p. 719). Real time has in itself power associated with authenticity. This is mostly done through witnessing by record, which is a fluid but fleeting form of testimony, when compared to the mass memorialisation of heritage sites and museums. (Peters, 2001, p. 720)

In relation to my research question, the competencies of drone technology, such as mobility and verticality empower underserved audiences because, by operationalising them to create their testimonies, underserved audiences challenge current models of communication and production of meaning. In doing this, they contribute to redressing the balance of power for the benefit of excluded communities and the reconfiguration of existing socioeconomic and cultural hierarchies.

7.2 Ownership of narrative

Telling one's own story is a form of reclaiming authority (Shuman, 2015, p. 53). It is immersed in complex considerations of relationships, ethics and power about who owns the story, who can retell it, what can be told and what remains untold, the source of knowledge the story is elicited from. A grounding principle for narrative ownership is lived experience (Shuman, 2015, p. 41). This is an expertise that constitutes a form of authority recognized in heritage practices through oral testimony and more recently, in my experience, in the last two decades, through collaborative design and production of projects, programmes, interpretation and other activities, with communities of interest.

In this section, I refer to my analysis framework where it relates to discuss how underserved communities, as a result of shaping the aesthetics of drone content they generate, position themselves as an agent. I consider the categories of “title”, “description” and “human presence” categories, to analyse narrative ownership, as part of reflecting on the ways in which narrative ownership is articulated in drone content and how this differs from heritage institutional drone content.

In my discussion of how underserved communities exercise narrative ownership, I consider how they propose a counternarrative, project identity and cultural visibility in drone content narratives. I compare each one of these with institutional drone content narrative composition. I conclude the section with considerations about what approaches to narrative through drone use mean for the role of drones in empowering excluded communities to connect with heritage and culture.

7.2.1 Proposing a counternarrative

All the seven videos generated by underserved audiences draw on themes that are relevant to their communities. They use the drone video content to deconstruct dominant narratives that reproduce and amplify discrimination and reframe them into more authentic and relevant representation, which constitutes a counternarrative (Saldanha *et al.*, 2023, p. 478). This is part of communities’ strategies to negotiate their experiences of exclusion and create social change (Saldanha *et al*., 2023, p. 478).

Dominant narratives ascribe blame to excluded communities, reinforce stereotypes (Miller *et al.*, 2021, p. 18) and protect the *status quo*. I discussed this in the Literature Review when considering the terminology used to refer to underserved audiences as disengaged, for example. Terms like these fail to recognize the unfairness of institutional processes and systems, they do not take account of the experiences of excluded communities, and they do not acknowledge how exclusion is ingrained in societal structures (Miller *et al.*, 2021). It is important to recognize the role of heritage institutions such as museums, in the construction and maintenance of normative cultural frames (Saldanha *et al*., 2023, p. 479) shaping, sustaining and amplifying the master narratives (Saldanha *et al*., 2023, p. 481). The research question I am answering in this chapter reflects how heritage organisations are working to address under and misrepresentation of communities experiencing disparity, by finding ways to enhance opportunities for collaboration and empowerment.

As Miller *et al.* (2021) and Saldanha *et al.* (2023) note, counternarratives are under-researched. To discuss how underserved audiences shape their counternarratives, I draw on the themes in the videos captured, through their titles and descriptions, as well as the content. *Bimini and Dan O'Neill on healing the ocean as LGBTQ people* (Attitude Magazine, 2022) is a good example of a counternarrative. This video is not a drone video *per se,* but it includes drone content that is used meaningfully in the video. The video combines it with ground film of a photoshoot of Drag Race Queen Bimini, wildlife presenter Dan O'Neill, climate activist Noga Levy-Rapoport and wildlife filmmaker Jasmine Qureshi and their interviews.

By using Bimini's and Dan O'Neill's names in the descriptive title, *Attitude* gives visibility to and creates representations of environmental activism in the Lesbian, Gay Bisexual, Transgender, Queer and Intersex (LGBTQI +) community who, due to experiences of institutional exclusionary practices in health, employment and housing are significantly affected by climate change (Goldsmith and Bell, 2022)[44]. The narratives in the video discuss the experiences of oppression, in institutional and corporate environments and composes an aesthetics of authenticity and vulnerability. As part of this, environmentalists featured in the video articulate how the ocean is part of their wellbeing, their considerations on ways to protect the ocean and how communities can succeed in building a better future through collaboration.

All the Attitude Magazine drone shots in this video are from a Bird's Eye View, both static and moving, as the still in Figure 17 shows. These drone shots of the ocean are used to illustrate

[44] This is because these communities experience significant exclusion which puts them at greater risk of homelessness and displacement and challenges in asserting their rights within the political system.

(2022, 02:56) or metaphorise (2022, 07:15) the discussions in the video, and to articulate personal narrative (2022, 01:58).

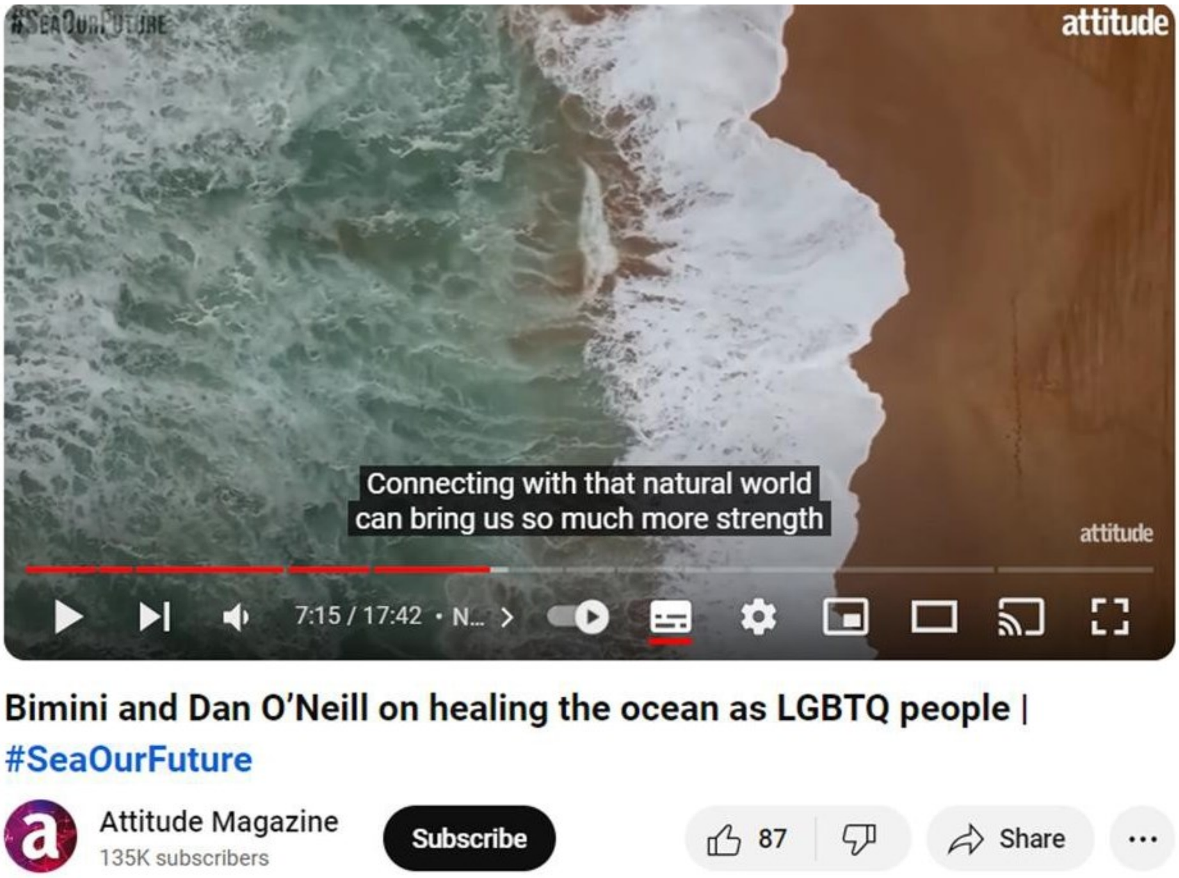


Figure 17 Static Bird's Eye View Shot, 'Bimini and Dan O'Neill on healing the ocean as LGBTQ people' (Attitude Magazine)

The Bird's Eye perspective produces emotional responses (Royo-Vela and Black, 2020, p. 36) in viewers. In Figure 18 (Attitude Magazine, 2022, 01:58) a Bird's Eye perspective is used as the background to Dan O'Neill's sharing of mental health struggles, as a result of discrimination. Oppression and exclusion are emotional subject matters as is the climate crisis. In describing the video as "bridging the separation between people and the ocean" (Attitude Magazine, 2022) the narrators signpost viewers to what the conversations in the video are focused on: the recognition of the need for socio-political actions and subjects (Brighenti, 2010, p. 188) to create change towards an inclusive and equitable society, that addresses challenges collaboratively.

Figure 18 Bird's Eye View of the ocean, 'Bimini and Dan O'Neill on healing the ocean as LGBTQ people' (Attitude Magazine)

By contrast, heritage organisations' drone content is composed to reiterate the institutional voice. I use the *Dewponds of the South Downs* (SDNP, 2021) as an example. The Establishing Shot (SDNP, 2021 00:00), illustrated by Figure 19, shows the title (dewponds) although this approach assumes everyone knows what a dewpond is, and what it looks like. From the first eight seconds the viewer is exposed to the branding and a specialist theme.

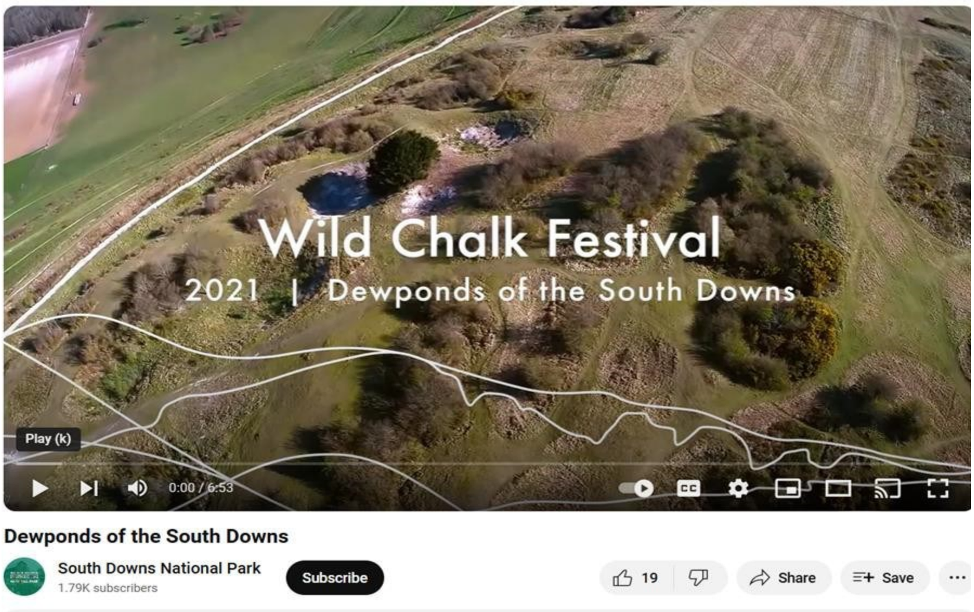


Figure 19 Establishing shot of Dewponds of the South Downs (SDNP)

The majority of the shots used in this video are Survey Shots, which are functional and are often used to undertake the heritage monitoring and surveying activities, heritage interviewees and survey participants reported. The narrative is mainly historical, as in the Iron *Age Hillfort- Cissbury Ring video trail* (LSENT, 2021) and educational as in the *How to spot a Capability Brown landscape* (NT, 2016).

Following the establishing shot, the landscape is unwrapped by a Reveal Shot illustrated by Figure 20 (SDNP, 2021, 0:03/0:04). I bring attention to this shot, because it composes the narrative as the witnessing of something important, to be in awe of. The drone shots in this video reinforce the place, and subsequently the institution, as spaces that command reverence through their breathtaking views (Millerson, 1994).

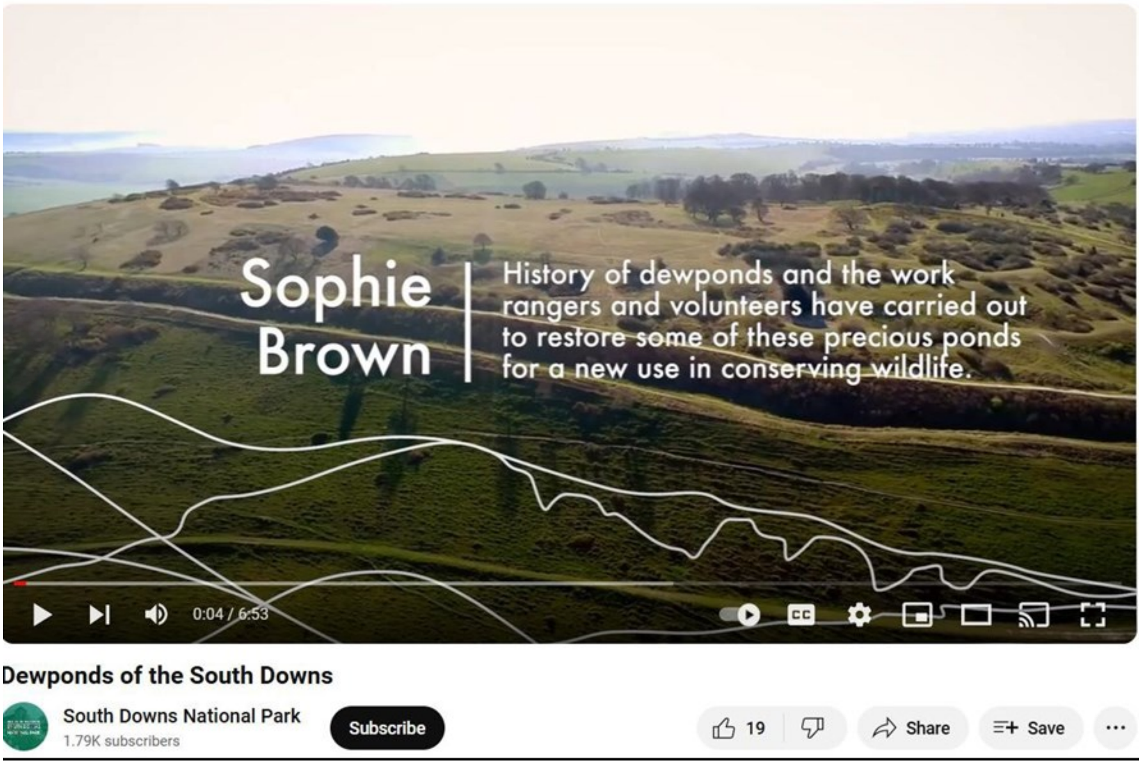


Figure 20 Reveal Shot of the South Downs, 'Dewponds of the South Downs' (SDNP)

The drone shots contribute to the notion of historical objectivity, and they are used to create an experience of authentic witnessing. They directly illustrate the information given verbally by the SDNPA Ranger, as in Figure 21 (SDNP, 2021, 00:36) when Ranger Sophie Brown talks about the decline of dewponds which is combined with a Moving Aerial Tilt Shot showing landscape where dewponds, are no longer visible. Authority of knowledge and narrative control are a dominant part of the master narrative of heritage organisations, which they are working to change. For instance, the *Iron Age Hillfort-Cissbury Ring video trail* (LSENT, 2021) ends with a question that indicates knowledge uncertainty.

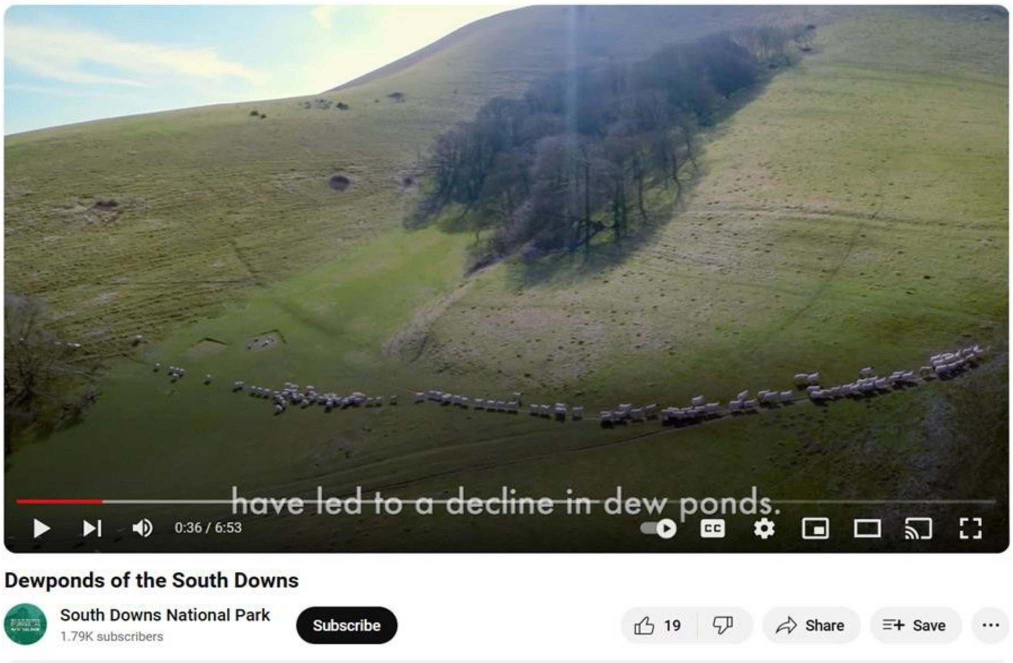


Figure 21 The decline in dew ponds. Moving Aerial Shot, 'Dewponds of the South Downs' (SDNP)

This is a form of vulnerability and shows a level of institutional openness to consider different possibilities for narratives, left to the viewer to reflect on and discuss. Nevertheless, who are the audiences for these videos. Who does this composition, theme, terminology attract? Who is it relevant to? Both videos include human presence, but as institutional representation and neither reflect directly on narratives that can be relevant to urban fringe communities. If they were to do so, what would that look like?

In summary, in my analysis of drone content produced by underserved groups, including my own autotechnographic drone film-poems, I consider that counternarratives overall propose an exploratory approach to identity stories and not a fixed narrative. This gives space for individual lived experience accounts, to be included in bringing change, as they illustrate how representative the counternarrative is of different variants in which oppression manifests itself in that community. This also means individuals feel they have agency and a voice, as their individual story is valued (Miller *et al*., 2021, p. 20). Attitude Magazine's video illustrates this well by bringing the stories of four different individuals together to address systemic challenges.

A key aspect of counternarratives is that they are focused on efficacy and are aspirational, looking to achieve positive outcomes. (Miller, 2021). Drone content by heritage organisations reiterates the narrative of power, authority and control, although there are subtle signs of change, but not in the aesthetics of the drone video content. The Heritage sector is negotiating ways to work within a culture of care and empowerment where they approach power and authority differently. This is not yet visible in how heritage organisations compose drone content.

Nevertheless, in relation to my principal research question, drone technology can empower underserved communities by facilitating their construction of a counternarrative. For this to happen successfully, in the context of a collaboration with heritage organisations, it is necessary first, for institutions to truly understand drone technology and its agential competencies in narrative reframing. Second, that the sector develops the same insight as Heritage Expert Three stated and a better understanding of drones and of themselves, in terms of the legacies that have centralised the narrative and interpretive power on them, and the journey of change ahead.

7.2.2 Projecting identity and cultural visibility

In Chapter Five, drone users and heritage experts agree one of the unique opportunities for drone technology to engage underserved audiences is through its capability to elicit "A broader perspective that includes context". A Heritage Expert posits drone technology has a role in "connecting young people, in particular with understanding the landscape". The drone video *Road to Freedom!*[45] (2022), which showcases a school dance performance evoking Tortola's past of colonialist subjugation and slavery challenges this perspective. It reframes it within a context where young people (the young women and girls performing in

[45] Drone video produced by Adagio Dance School (2021) located in the capital island of Tortola, the largest of the British Virgin Islands (BVI) (Cohen and Mascia-Lees, 1993, p. 132).

the video) interrogate, rather than conform to the landscape story, and the role it has played in their identity and history.

One of the ways in which drone content generated by underserved audiences creates visibility is through human presence, in the drone facilitated video visualisations. This is a category in my framework of analysis, and I draw on it to note that presence in six of the seven videos analysed.[46]

Road to freedom! (Adagio Dance School, 2021) which explores the connections between landscape and identity, discusses the historic legacy of slavery which is still alive in the current identity of the BVI as British ruled territories. It does not comply with the discourse that privileges the landscape "showing why that area is important is really valuable" (Heritage Expert Five). Instead, it predominantly focuses on the human aspect, through the use of the First Person- View drone, which reflects the reclaiming of the narrative by those with firsthand generational experience of the impact of colonialism and lived experience of imperialism.

In this video, as the drone starts with a Flythrough (Figure 22) (Adagio Dance School, 2021, 00:09) and performs it across the building and the dancers, the latter become part of the volumetric composition of the narrative space. The composition of the video, using black and white instead of colour, employs Low Angles that show the enslaved people, personified by the dancers, as powerful agents.

[46] In the video *Bimini and Dan O'Neill on healing the ocean as LGBTQ people* (Attitude Magazine, 2022) the visibility of human presence element is strongly represented through the predominance of ground camera footage of the people featuring in the video immediately before and after the drone shots.

Figure 22 Establishing shot: Low Angle Flythrough, 'Road to freedom!' (Adagio Dance School)

Also, the Close-up Shots that help the viewer connect with the narrative are part of a visual choreography of narrative ownership. Nevertheless, this narrative is not proposed as a fixed story. It is an instance of witnessing (Frosh and Pinchevski, 2014, p. 597) whilst reiterating this right to presence and visibility in the public space (O'Neill, 2018, p. 74) and to authentic representation. This is true for all the drone content created by excluded groups I analyse in this thesis.

In the livestream events, visibility is negotiated through the inclusion of the drone pilot, who is at the external site, and their interaction with the participants. This is done through the pilot establishing a connection with the Lovejoy Centre participants by waving (shown in Figure 23), (2023b, 00:35) introducing themselves and saying hello. In this group, the inclusion of human presence is an important intervention. This may be different for different groups.

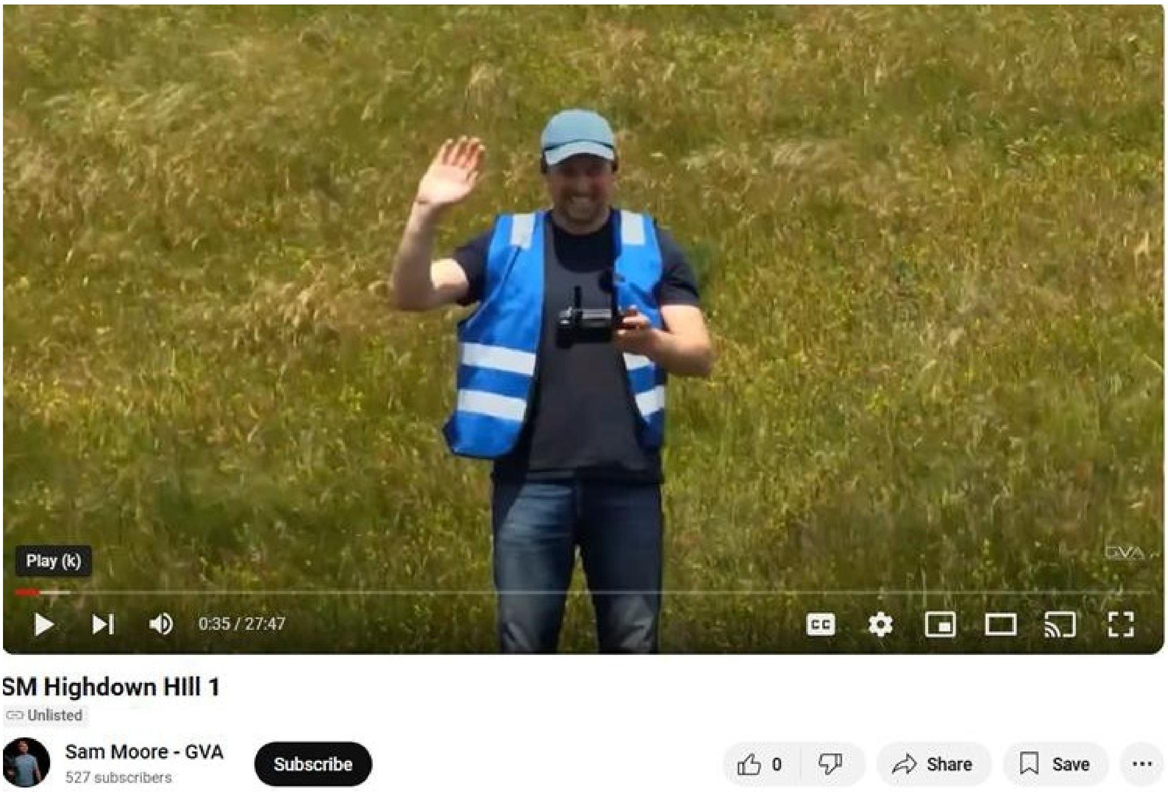


Figure 23 Medium Close-up to eye level, pilot waving, 'Bringing the outdoors, indoors' (Lovejoy Centre)

The visibility of the group themselves is produced through their ownership of the drone gaze and reiterated in the post produced film of the livestream, which combines filming of the group at the centre and the livestream from the National Trust locations.

The participants mention the pilot in the course of the livestream "That man is usually sitting down" (Lovejoy Centre Participant One) associated with visualisations of the pilot during the livestream as shown in Figure 24 (2023a, 00:27). Making the pilot visible created a layer of relatability and familiarity in the livestream experience, that may be an indicator of participants' positive associations with the project.

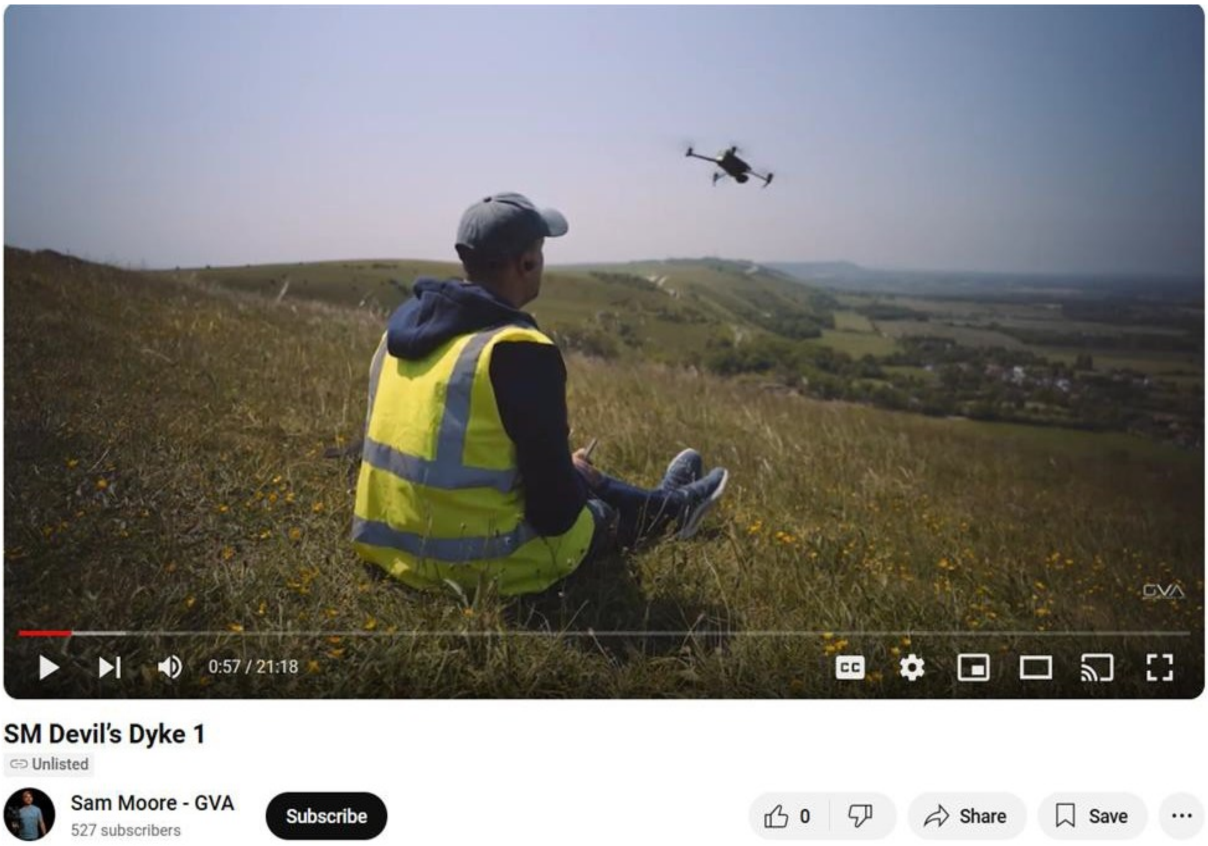


Figure 24 Medium Close-up of pilot from behind, 'Bringing the outdoors indoors' (Lovejoy Centre)

At the end of the livestream, the pilot reappears to wave goodbye. This establishes a social relationship between the participants and the pilot that makes them visible to each other. This is particularly important for participants living with dementia, as underserved audiences. In this context, their visibility emerges from the conversations they generate as they engage with the drone gaze, and which are their tool of agency to shape their experience of landscape through the drone. The details participants choose to view are connected to the participants' personal narratives. Where they live, the elements in the landscape that intrigue them "Someone walking there, look!" (Lovejoy Centre Participant One, 2023b, 03:45), what they recognise "I've got some lavender in my garden" (Lovejoy Centre Participant Three, 2023b, 18:04), "More creatures there" (Lovejoy Centre Participant One, 2023b, 22:31).

As discussed in the Literature Review (Section One) and in Chapter Four (Section Two), anxieties about intrusion, surveillance and privacy populate the imaginary of the negative public opinion of drones. Heritage Expert Three reflects on how people feel about being captured by a drone “they were doing some filming with the drone off Southsea seafront. And you know, you can hear all the people complaining about it because they’ve got bikinis and shorts on, and all this in the nice weather.” This is a concern pertaining to non-involved people and the perceived difficulty in controlling the drones’ activities. It informs perceptions of barriers from heritage organisations, in the context of using UAVs in engagement of left out communities.

None of the four institutional drone videos analysed here include human presence in the drone facilitated visualisation of landscape. Two videos include members of staff as presenters which I interpret as part of the composition of the institutional voice. I recognise the requirement for compliance with General Data Protection Regulations. These requirements apply equally to ground video and photography.

As discussed in the Literature Review and in Chapter Four, there are psychological factors affecting the acceptance of the technology nevertheless, visibility can be composed in different ways. Heritage Expert Three highlighted institutional drone content is not relatable or relevant to communities who do not live, work or have a business in the landscape shown because it excludes them from the narrative. The *Iron Age Hillfort- Cissbury Ring video trail* (LSENT, 2021) is a good example of this.

In minute 0:23 (National Trust, 2021), illustrated in Figure 25, an urban settlement is visible within the range of an Extreme Long Shot, Worthing. This is a priority engagement area for the National Trust (National Trust, 2021, p. 32). The video makes no reference to the area and its relationship to the natural landscape. Referring to it in the context of the landscape story would be a form of making it relevant to those communities and start to address the absence of their voices.

In this sense, the institutional drone gaze amplifies the hierarchies between those who view and those who are viewed (Stahl, 2013, p. 663) and who is seen, the insiders and outsiders (Heritage Expert One). This separation is based on the identity, class, gender and disability of specific communities excluded from public visibility (Kowalewski and Bartłomiejski, 2020, p. 61).

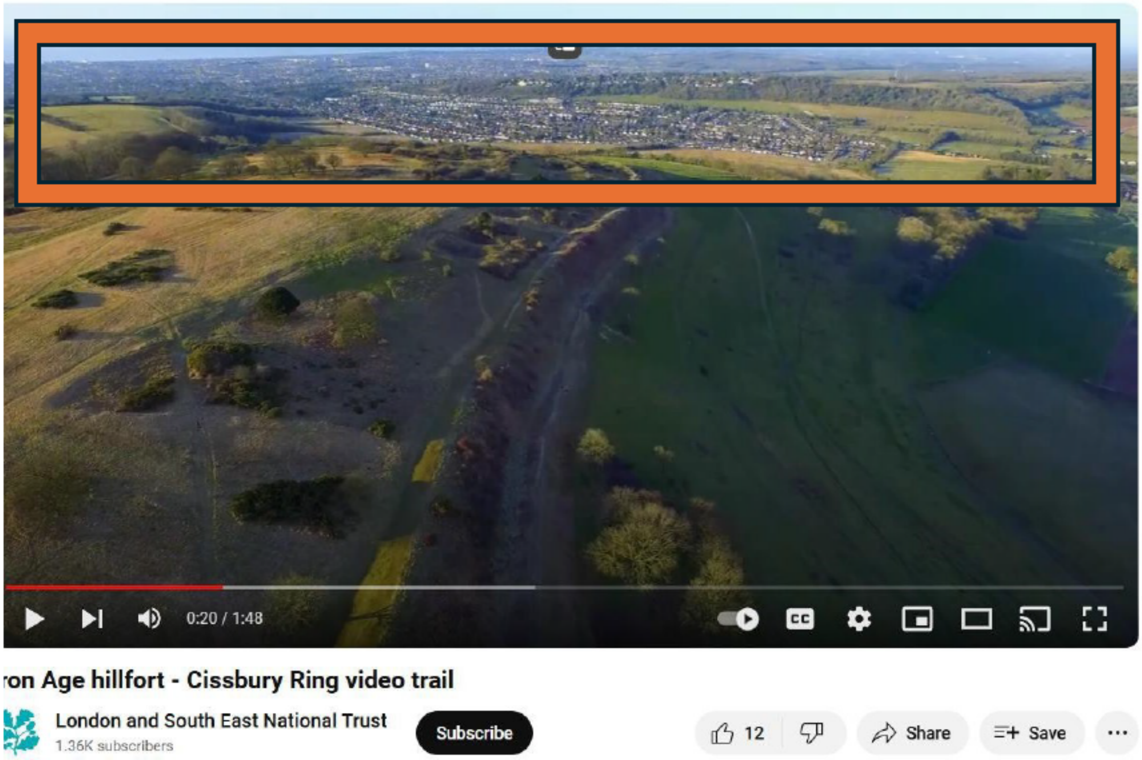


Figure 25 Area within square shows Worthing. Moving Aerial Pan shot of Cissbury Ring,' Iron Age Hillfort- Cissbury Ring video trail' (LSENT)

Visibility is at the centre of mapping. Histories of underserved communities are frequently not represented in cartographic documents and traditional drone visuals are affected by this issue. Heritage Expert Three argues that as a result of this, drone visuals are less relatable and lose some of their power. Visibly acknowledging communities as part of landscape is an element of establishing a connection with contemporary human geographies and bridging past and present.

Section conclusion

The application of the framework of analysis has provided me with a systematic approach to discussing the divergent approaches to narrative building, between drone content generated by underserved audiences and done content from heritage organisations. Underserved communities use drone technology competencies as a form of resistance to the master narrative.

The counternarratives in the drone videos and content created by underserved audiences use narrative not just as authentic representation but, more importantly as what I designate a critical exercise in visibility. By this, I mean the use of the drone competencies facilitates a reflexive process of negotiations of internalised identities, external projections and aggregates of knowledge, emotional labour and lived experiences. The counternarrative exposes those negotiations and the hope and aspirations for change that it elicits.

In terms of heritage, as discussed in Chapter Four, heritage organizations' biggest barrier to the use of drone technology for engagement is loss of power, control and authority compounded by the fear of taking risks and failing. This is consistently reiterated through my data and the analysis of institutional drone content. This is not necessarily in equal measure for both the National Trust and the South Downs National Park. In line with the discussion on their organizational culture and origin story in Chapter Four, each organization is at a different stage of the journey in terms of taking risks, the SDNPA being more comfortable with this prospect as articulated by a Heritage Expert.

As organisations work to adapt their internal processes and systems to more equitable practices, they are less attentive to making this change of culture visible in their visual identities and aesthetics. As such, their drone content still reflects reiterations of their positionality as gatekeepers of knowledge.

What this means for my research question is that UAVs not only empower but they enhance the possibilities for underserved communities to challenge dominant narratives and propose a reframed story shaped by themselves and which they own. It does this by diversifying the devices of storytelling and the meaning-making mechanisms to include aerial space, which creates a volumetric visibility. The human presence in public space is not only visible but it becomes part of the tridimensionality created by the drone gaze.

With drone technology as a new media of identity narrative building, it is important that heritage organisations consider what the technology brings to the politics and discourse of representation, particularly as they engage in the process of increasing relevance and adding the voices of diverse lived experiences that have been silenced in the past.

Chapter conclusion: Drone technology as a facilitator of empowerment of underserved audiences

Here, the findings from applying my analysis framework to institutional videos with drone content and videos with drone content generated by underserved communities are discussed. I draw on those findings to make considerations that support the answer to the principal research question: "What is the role of drone technology in empowering underserved communities to connect with heritage?".

My findings in relation to the differences between drone content by heritage organisations and drone content created by underserved audiences are that there are two key aspects that distinguish them from each other: problematisation of the aura of iconic views and ownership of narrative. I summarise my findings in Table 14.

Table 14 Comparison of characteristics of drone video content

Produced by heritage and cultural organisations	**Drone content produced by underserved audiences**	**New aesthetics in the use of drone technology**
Camera perspectives Use of shots that: • Are centred on showing context (moving aerial survey, pan and tilt shots). • Are a legacy from functional drone use such as surveying and monitoring. • Project a public space distance (Probst, Rothe and Hussmann, 2021, p. 184). • Help the viewer orientate themselves. • Present classic landscape views (Serafinelli and O'Hagan, 2022, p. 226), with an auratic effect that projects landscape as "spectacle" (Benjamin, 2017, p. 232). • Offer an objective view of space, framing the viewing experience as a witnessing event.	Camera perspectives Use of shots that: • Present perspectives that challenge the traditional classic landscape views (Serafinelli, and O'Hagan, 2024, p.226) through shots that: ○ Create a sense of personal space distance (Probst, Rothe and Hussmann, 2021, p. 185) (Medium and Close Shots) that create intimacy with the landscape, help regain control. ○ As a result of the aforementioned characteristic enable refamiliarisation with the defamiliarised (Serafinelli and O'Hagan, 2024, p. 230) by the aerial perspective.	Reframed use of camera perspectives • Refamiliarizes the defamiliarized. • Movement and pace disrupt institutional rhythms. • Drone as a witness and as a testimony in dual temporality.

Produced by heritage and cultural organisations	Drone content produced by underserved audiences	New aesthetics in the use of drone technology
Narrative • Institutional compositionality that reiterates the organisation's branding as a knowledge production authority. • Predominantly descriptive forms that are driven by historic information. • Themes have a didactic tone.	**Narrative** • Themes of identity and belonging. • Exploratory narrative. ○ Gives communities physical and emotional visibility, including them in the visualizations of the narrative. ○ Uses movement and pace differently.	**Ownership of narrative** • Proposing a counternarrative ○ It does not propose a description or a fixed narrative. ○ Challenges a fixed narrative. ○ Visual language is predominantly metaphoric and figurative. ○ Uses a combination of elements, such as title; non- traditional visual language.
		Identity and cultural visibility • Includes human presence. • Representation as a critical exercise in visibility.

The findings above lead to the conclusion that both the National Trust and the South Downs National Park Authority use the analysed drone videos, as a form of reaffirming their role as repositories of knowledge. This is particularly evident in National Trust videos *How to spot a Capability Brown landscape* (2016) and *Iron Age hillfort - Cissbury Ring video trail* (2021). In both videos there is a marked use of Classic aerial views through Long Survey, Moving Aerial Pan Shots which, referring back to my framework, offer a broader perspective, an opportunity identified by heritage and drone experts in my interviews and survey. The use of these shots, while providing context also creates intellectual distance, which conjures objectivity and a public space experience.

This is also the case for the SDNP's *Dewponds of the South Downs*. All three aforementioned videos have a didactic tone, which is aligned with heritage institutions' master narrative of knowledge authority. In the *Iron Age hillfort - Cissbury Ring video trail* (2021) and *Dewponds of the South Downs* (2022) the human presence, welcoming and friendly but potentially not representative of underserved communities confers further visibility to the institutional brand of expertise and knowledge production.

The two videos that stand out in this analysis, because they share the purpose of audience engagement, but offer diametrically different approaches to significant effect in terms of empowerment, are the National Trust's *How to spot a Capability Brown landscape* (2016) and the SDNP's *Shoreham Cement Works- 360°* (2022). The former excels in the projection of the National Trust institutional culture: the use of cinematographic views, Reveal Shots and auratic views of an extensive estate that shows visible wealth and grandeur, whilst being used as the showpiece in a crib sheet of the characteristics of Capability Brown's gardens.

The tone is playful and interactive. Nevertheless, it draws on a discourse of knowledge as a privileged asset, as discussed in Chapter Four and again the dynamic between the insiders and the outsiders. The idea of a cheat sheet does not only elicit ideas of examination and judgement of levels of knowledge but also, more importantly, that there is a knowledge level required to be able to enjoy the heritage presented.

The *Shoreham Cement Works- 360° views* (2022), on the contrary, is part of an institutional narrative of accessibility and equity. It has been created to provide local communities with an opportunity to explore the site and participate in consultation about its use. Although the views are of Long Shots, the fact it is a 360-degree virtual experience allows communities to direct their own experience to the level of detail they wish.

The cement works' structure disrupts the potential for the views to become idealised. The movement of cars can be seen at a distance. This video does not present a fixed narrative or propose new camera perspectives, beyond the limitations of the data captured by the drone and reproduced as a virtual experience. It offers communities an opportunity to revisit their own narratives, when interacting with the site. The visibility is fleeting and personal, rather than public. It becomes so, when people choose to share their views.

All the videos produced by underserved communities (summarised in Table 13, page 248) share, as a characteristic, the presentation of a counternarrative. For instance, for Lovejoy's co-research participants it is the ability to make decisions, have power over their experiences and shape them. By using camera perspectives, movement, titles and descriptions in a way that is relevant to them, underserved communities challenge the aesthetics of the auratic. Here, the assets to be revered are communities' determination to create change and bring social justice. They take control as agents.

It is in that position they have ownership of the narrative. Communities use this agency to reframe themselves within a counternarrative, that reflects the issues that affect, interest and inform their social and cultural identities. The videos in Table 13 offer a new form of witnessing which is not historical, as in the case of institutional videos, but about visibility and representation.

As mentioned above, the SDNP's 360° of Shoreham cement works is different to the other institutional drone content analysed, as it hands over the control to the viewer, who is able to determine, to a certain extent (within the limitations of what is captured), the angle, level and shot size of what they view. This is closer to an empowered experience of landscape, that partly integrates the roles of viewer/consumer and creator/producer. Partly, because the content is curated, preset by someone else. Further to this, 360° viewing does not create visibility beyond an effect of self-memorialisation, that remains within the private sphere of the viewer.

What does this mean for the use of drone technology to empower underserved audiences to connect with heritage and culture? Power is at the centre of this question (and this thesis) not only because it investigates empowerment but, primarily, because fear of loss of power and control constitutes a fundamental barrier to heritage organisations engaging with drone technology and using it to foster agency for underserved communities. This is also because, ultimately, empowerment of underserved communities in the context of heritage organisations means organisations must take risks and try new approaches.

Drone technology can empower underserved communities, but only if they are agents. This does not mean they must physically control the drone. It means they must be in a position where they enact agency over the perspectives, movement and gaze of the drone, constructing a narrative that represents their experiences and interests. With this, they create a presence in the public space, their visibility disrupting existing notions of hierarchies of knowledge and authenticity.

Chapter Eight: Conclusions on drone technology empowering underserved audiences to connect with heritage and culture

This doctoral project aimed to investigate how drone technology can be used to empower underserved audiences to connect with heritage and culture. To answer this research question, it also explored what heritage organisations perceive to be the barriers and risks of drone technology to engage underserved, non-visiting audiences; what unique opportunities drone technology does offer for the interpretation of and engagement with heritage, and what the role of drone technology is in the empowerment of underserved communities to access heritage.

Drone technology presents itself as the platform *par excellence* to explore changes in the distribution of power in the heritage and culture industries, as these embrace and work to embed practices that engender agency in underserved groups. The naturalisation of drones in civil society symbolises the power shifts in the control of aerial space from corporations, power regimes, governments, and dominant groups to the individual consumer economy.

Further to the agential negotiations drones operationalise, they offer six key affordances: drone mindedness, drone sensing, spatial mobility, verticality, control, and automation. These affordances underpin broader perspectives for users, a greater range of experiences for wider audiences; increased access and accessibility, capture of “more data of higher quality” and “meaning-making opportunities” for communities to engage with heritage, as identified by heritage and drone experts.

In the course of the research process, it became clear that to address the experience of underservice in excluded communities, it is crucial to go beyond “active engagement” and “involvement”, terms that are characteristic of heritage and culture audience development discourses. Both terms are often correlated with empowerment, semantically and in practice.

The institutional drone visual language in heritage organisations is composed to project an aesthetics of power, through auratic aerial views that idealise the object or landscape, presenting it as an absolute, to be revered. Characteristics which, by implication, confer the same status to the institution. Benjamin's concept of aura (2007) is at the centre of the agential negotiations between underserved communities and institutional narratives. Benjamin considers that technologies of reproduction such as photography and film erode the transcendental quality of the art object and material culture, which before occupied a space of reverence, presiding over human narratives of belonging and placemaking.

Auratic views are a device used in drone films by heritage institutions, to project the authority and knowledge in which their power is rooted. The data reveals that whilst working towards being more representative of excluded narratives, heritage and cultural organisations are concerned with the loss of that power and the inherent loss of control over the systems of knowledge production and validation. These concerns are behind heritage organisations' perceived barriers and risks of using UAVs to reach audiences.

As the aura is central to the visual language of institutional power, using drone technology to address power imbalances has involved the investigation of what drone content looks like, when underserved groups have control of the drone. What is the alternative to auratic views, and how does it articulate the agency of the producer? The co-research with people living with dementia, an underserved community which, as a result of their diagnosis, experience a deficit in agency and decision-making, aimed to explore this.

The affordances of drone mobility and verticality were central to this research method. Drone movement fosters involvement, and top-down views generate an effect of mental simulation. As a result, participant's cognitive processes operate as if the participant is experiencing first-hand, what they are viewing. The live-site experience (Agostinho, Maurer and Veal, 2020, p. 255) creates a "liveness" and a sense of control (Hildebrand, 2020, p. 83) that are absent in preset content (virtual reality experiences, 360° aerial videos or the choices of institutions or pilots).

All the above, are key factors in establishing the drone livestream approach as a valid empirical method of documenting the agential negotiations facilitated by drone technology, and drone's unique role in fostering empowerment in this way.

8.1 Research findings

Chapter Four finds that public opinion is not perceived by heritage experts as the top barrier to using drone technology to connect underserved audiences with heritage, although there is extensive literature on public perception of drones (see Literature Review Section 2.1.2, page 5). Instead, the three key concerns from heritage organisations relate to usability (which include logistics, regulations, specialist knowledge and data management) followed by disruption of heritage visitors' experience of heritage and accessibility. From the analysis of the data, there is a fundamental factor that is common to these three aspects: fear of loss of power and control as a result of change in the ways of working.

One of the key findings in this research is that heritage organisations draw on regulatory frameworks as part of gatekeeping practice, to retain control. This is because although, as discussed in the Literature Review and in the data, they use drone technology for functional purposes, heritage organisations do not have a full understanding of the potential of drone technology beyond LiDAR, measuring, surveying, for example around audiences. Using drone technology beyond functional purposes, namely, to reach and empower audiences, means engaging with new ways of working.

Heritage organisations are rather averse to uncertainty and failure. A new proposal for using drones differently thus places them in a vulnerable position, which they perceive to endanger their role and authority, as knowledge repositories. The analysis in this research moves beyond this barrier, and towards reconfiguring new approaches as an opportunity. As drone technology and culture are continuously developing, and the regulatory frameworks still new and being tested, this is arguably the appropriate time to take risks and develop opportunities for novel drone use.

Chapter Five finds that heritage experts' perceptions of the opportunities of drone technology to engage non-visiting audiences are informed by their role, whereas drone experts' perceptions are informed by the context in which they use drones. Both heritage organisations' and drone experts' perceptions of the unique opportunities for drone technology to engage underserved audiences are aligned around two key areas: "Broader and wider perspectives" - including the themes "Broader perspectives that include context" and "A greater range of experiences for wider audiences"; physical access and connection - including themes "Access and accessibility", "More data of higher quality" and "Meaning-making: storytelling and creativity".

These themes primarily reflect the technical competencies of drone technology: drone mindedness and drone sensing, spatial mobility and verticality, control, and automation. These capabilities, as discussed in the Literature Review, make it possible for UAVs to provide the opportunities heritage and drone experts perceive them to offer, such as the reimagination of possible futures. There is a significant difference between how these capabilities are framed in the literature and in the data. The literature projects the competencies of mobility, automation, and new narratives as features of empowerment, whilst heritage experts, and the majority of drone experts perceive these competencies as a product of the use of UAVs.

It is important here, to reiterate the importance of the specific nature of empowerment, in relation to engagement. There are different levels of engagement, from passive encounters with narratives and sites (visiting an exhibition, for instance) to active participation and contributions to interpretation (the case of oral history testimonies). Engagement is a part of the process of empowerment. Nevertheless, authentic representation happens when communities have agency. When they can control or shape their experience or contribution in a way that meets their needs, supporting their self-efficacy.

Drone technology does hold the necessary tools to empower underserved communities to connect with heritage and culture. Its capabilities facilitate broader and wider perspectives and experiences. But whose? They increase access and accessibility, but from whose point of view? They enhance storytelling, but for whose story? Following the identification of issues around retaining institutional power and control in Chapter Four, it is important to observe and discuss under what conditions, drone technology does facilitate these opportunities. And what does that mean for underserved audiences in expressing their authenticity or becoming producers and consumers (aspects highlighted by drone experts in context of the drone as a democratising tool)? This is the reflection that led me to conclude that it is necessary to reconsider this project's research questions from engagement to empowerment.

The Literature Review discusses the change in the role of museums in society leading to museums, heritage and cultural organisations being active in addressing social exclusion. In the literature, this is framed within a new practice of active engagement. This research concludes that the terminology does not embody the required reconfiguration in the approaches from heritage organisations, which will contribute to fulfilling social justice aims. The central problem in achieving social justice is that there is an imbalance of power.

Heritage organisations (as evidenced in Chapter Four, are fearful of losing power and control). For drone technology to bring innovative approaches that help connect underserved audiences with heritage and culture, rather than sustaining existing practices that amplify the status of institutions, it is necessary to ask what their role is in the power dynamic between heritage and cultural organisations. That involves thinking beyond active engagement and being clear in conveying that good practice means a shift in the power dynamics, by referring to it as empowerment. As a result, the research narrative shifts from interrogating the possibilities of the drone as a facilitator of engagement of underserved audiences, to exploring how it can empower these communities. This wording encompasses structural institutional change, as well as a new approach to aesthetics and messaging.

In Chapter Six, the review of conceptual frameworks from drone research, media, and film studies leads to the conclusion that none of the frameworks can be successfully used on their own to analyse the evidence of power and agency in drone video content. This is because the frameworks analysed focus on the effects of drone video content, its categorisations, the characteristics and themes of drone photography and the meanings of ground camera perspectives. In the literature, drone researchers recognise that the socio-cultural impacts of drone technology are underexplored, which justifies the absence of a framework that supports the analysis of sociotechnical effects of drone technology. It is informed by these insights that this project involves the design of a new framework for the analysis of power, in drone video content produced by heritage institutions and underserved audiences.

Chapter Seven focuses on the application of the newly proposed framework to the analysis of pre-existing drone video content produced by heritage organisations and by underserved communities, and researcher instigated drone video content created by underserved groups. The latter includes the co-designed content generated by people living with dementia, and the autotechnographic films. Both test the central concept of the characteristics of content shaped by excluded groups.

The thesis concludes that drone technology can empower underserved audiences, if they are in control of the drone gaze. This means that they do not necessarily have to physically control the drone, although there has been research on integrating drone technology with generic and assistive gaming technology, but that audiences must have power over the gaze of the drone. This study finds that when this is the case, underserved audiences produce an aesthetics that challenges the auratic views characteristic of institutional drone video content. They do this by using verticality and mobility differently to the ways that produce classic aerial views, which results in a reframing of camera perspectives.

Instead of the dominance of Long Shots and panoramic views, they use close shots that foster a refamiliarization with the defamiliarized. This not only affords control and intimacy, but mostly, it disrupts the idea of drone visuals as auratic views that idealise place and, therefore, create a distance between the experience of landscape and its visual projection. In doing this, underserved audiences become an agent who produces their own counternarrative as a form of resistance.

The Literature Review shows there is a growth in individuals using drones for their own purposes, as well as communities. It discusses examples of indigenous peoples using drone technology, as a countermapping exercise to reclaim territories that have been usurped in the context of colonialism (Paneque-Gálvez *et al.,* 2017, p. 17). The aesthetics of the counternarrative includes not only the defiance of the classic idealised views through perspectives, movement, pace, and forms of witnessing, but also through engendering visibility.

To articulate the interplay of these components and the aesthetics they create in drone videos, the project introduces, as part of its contribution to knowledge, the term aesthetics of critical representation. This aesthetics both proposes and forms, in itself, a method of self-efficacy and resilience, through which underserved communities take back control and disrupt the master narrative. It is a critical exercise because it does not establish a fixed story. On the contrary, it is enriched by the diversity of individual lived experiences in the generating community, and it is rooted in reflection, offering a counterpoint to existing dominant stories that do not represent and are not authentic of the community in question.

The Literature Review discusses the emotional impact of exclusion and oppression (Watermeyer, 2013, p. 152). It considers exclusion's effect of emotional oppression and how the latter creates an emotional burden that is damaging and leads to disaffection that is passed on from generation to generation. Counternarratives, discussed earlier on in this section as a form of resistance, propose change and therefore become not only places of critical reflection and agency, but offer themselves to be places of emotional healing.

What this means for heritage and cultural institutions is that, although challenging in view of the factors affecting the sector in England, such as under-resourcing and underfunding, institutions must accept that change only happens if current structures and mindsets change. That leadership means taking risks, and pushing forward through resistance, hesitation, uncertainty, to reconfigure existing hierarchies of knowledge and valued expertise, in a way that gives excluded groups parity in power. This is certainly a process that both the National Trust and the South Downs National Park are engaging with.

8.2 Contributions to knowledge

This research contributes new empirical and conceptual thinking in several ways. The findings from this research are the result of a considered approach to the application and integration of methods. The methodological approach in this project was rooted in the use of empowered processes to achieve empowered outcomes.

Reflecting the distinction between engagement and empowerment practices that steered this research, the methods in this study were designed to go beyond reaching participants and documenting their views. They were selected for their flexibility and participatory nature, which fostered exploration and empowerment in contributors. This was reinforced by an iterative approach, where the results from each stage of data collection shaped the next, enabling participants to shape the direction of the research.

The co-designed research provides empirical validation of the crucial role of collaboration with underserved communities as part of a process of facilitation of agential negotiations between the latter and heritage institutions. Co-design supports self-efficacy as a result of the empowerment of participants to be agents. As actors with control, participants associate the success of their experience with their own actions or decisions. This changes the conditions of production, as communities are both creators and consumers. It is in these circumstances, that drone affordances translate into opportunities for drone users from underserved communities, facilitating new perspectives and stories.

Collaborative methods are therefore essential to the power shift in heritage and cultural organisations. This is because it requires the design of new forms of knowledge production towards authentic and representative narratives, such as this project's drone livestream, created with people living with dementia. Specifically, the research provides evidence of how user-centred co-design of drone content and use can facilitate agency, in ways that support empowerment of diverse audiences resulting in drone footage that challenges "classic institutional heritage approaches to aerial views".

The application of visual methods to investigate the use of camera perspectives to produce meaning, elicited both empirical and conceptual considerations. To make it possible to analyse signifying practices in drone video content creation, it was necessary to design a new analytical framework bringing together theorisations from drone and film and media studies scholarship.

The operationalisation of this framework to examine drone content produced by heritage institutions and underserved communities led to the identification and conceptualisation of two distinct drone aesthetics. An institutional aesthetics characterised by the auratic, where the visual language projects heritage as a spectacle (Debord, 1994), and underserved communities aesthetics of critical representation, which presents a counternarrative and uses drone competencies to challenge classic aerial views. Whilst the former is an exercise in reiteration of power and authority, the latter is an exercise of visibility.

Existing research acknowledges the sociocultural impacts of drone technology are under researched. The approach to drone use in this study's co-design research with people living with dementia demonstrated that drone technology brings changes that affect both how we perceive the agential interactions of people living with memory loss with technology and the preconceptions of configurations of use of technology, by underserved communities, in the context of material and relational disparities. This understanding has sociotechnical implications. Meaning it influences perceived configurations of the human and the technical dimensions of the drone, which can foster change to future uses.

8.3 Reflections on the research journey

This project adopted an iterative approach to the methodology. As the understanding of the central research problem developed, the focus was adjusted, and the research was reframed and refined to increase the visibility and empowerment of underserved audiences. The iterative process allowed space for growth and change. It supported researcher knowledge development, and it helped increase confidence in the vision for the project.

The creative methods applied to the collaborative research with people living with memory loss were the most significant risk taken in this project. Not only were the expectations of recruiting a community group open to collaborating in the research exceeded, but also the Lovejoy Centre's[47] leadership and staff were excited by the research and committed to being involved. They are driven by the same values as those guiding the research:
a user-centred approach fostering clients' empowerment, agency, and self-efficacy. The relationship with the Lovejoy Centre staff and clients was extremely positive. Participants' welcoming behaviour on arrival (smiling and expressions of affection) indicated they made positive associations with the project.

The relationship with the heritage partners (National Trust and South Downs National Park Authority has been equally positive and supportive. Partners organised access to the sites, advice, encouragement, access to resources from other organisations they collaborate with such as Worthing Museum, and a significant portion of their time and availability. Initially, the possibility of undertaking field work within the National Trust seemed unlikely, due to the organisations' conservative position on drones. Nevertheless, this changed, and the Trust embraced the activities in this research, despite the no drone policy. This signalled a level of openness towards a different use of drones to the Trust's reported applications of monitoring, surveying, LiDAR.

There have also been challenges, some of them common to the wider sector, in terms of staffing and resourcing and at the point of delivery, a difficulty to differentiate between an organisational project and academic research and to understand how disruption of who collects the data and how this affects the integrity of the data and methodology. This is reflective of aspects of organisational power and control discussed in the thesis. It is also this factor, which is at the centre of the most significant challenge in the context of the work with the partners. This refers to the restriction on making the livestream events publicly available beyond the Lovejoy Centre due to the fact it undermines the National Trust's no drone policy.

[47] The Lovejoy Centre is a day centre based in Worthing, providing activities for people living with dementia. The co-design research was undertaken in partnership with this centre (including the drone livestreaming events to National Trust sites, Highdown Hill and Devil's Dyke).

This limits the dissemination of the research. The films of the livestream and the autotechnographic films will be available at a University of Brighton repository.

Collaborating with a group of people living with memory loss, as a non-expert memory loss practitioner was a challenge. The barriers experienced as a researcher were not barriers from participants, but barriers resulting from how the co-design research was planned. The groups were mixed ability and varied, as some participants only attended the Centre once a fortnight. Not all participants lived with dementia; some participants lived with mental illness. There were different types of dementia and different stages in the group. All these elements added complexity to considerations of delivering the sessions.

The co-design research elicited strong responsibility towards the participants and their wellbeing. The co-design sessions were informed by a process of continuous consultation with the Centre Manager to adapt the approach to the research activities with the participants. Not everything worked. It proved challenging to capture data to evaluate participants' views on the sessions. This was as a result of the application of evaluation models generally used with groups who do not live with memory loss. This led to tensions between the necessity to capture research data and ending the session without asking further questions to avoid upsetting or frustrating participants.

In terms of researcher practice, the process of designing sessions with the group led to the realisation that in the heritage sector, content is often intellectualised and considered sequentially, as a linear thread. This is part of the idea of categorising and structuring which is ultimately about power and control. For participants, presenting activities in a sequential way is possibly irrelevant as short-term memory is more challenging, so the most important approach was to be within a theme.

On reflection, this research project would have benefitted from a longer period of researcher non-participant observation and planning of co-research activities with the Lovejoy Centre, as part of developing a deeper understanding of different types of dementia and better working knowledge of evaluation strategies with people living with memory loss. This indicates new approaches to future iterations of the design of collaborative research with people living with dementia. This may include engaging a dedicated memory loss academic researcher or practitioner, in addition to the group lead and setting up a focus group including Lovejoy Centre staff and clients, dementia researchers and museum practitioners working with people living with memory loss. These could address Lovejoy Centre's limited capacity, despite staff's commitment and good will. Feedback from the Centre staff indicated that they would have preferred the delivery of fewer sessions, with a focus on the livestream events.

Undertaking a Doctorate is a complex and isolating journey. In hindsight I would have made more time to read or listen to blogs and podcasts reflecting on the journey, as this would have helped me internalise that many of the challenges I faced were part of the journey. Nevertheless, a significant part of the challenges I experienced were unexpected outside of the academic journey itself. Life events such as instances of loss and bereavement, physical and mental health, experiences of racism and violence significantly disrupted my flow of energy. The latter were particularly relevant in the context of researching empowerment and agency of underserved audiences. My research has been parallel to my personal struggle against being silenced, oppressed and asserting a counternarrative of truth and authenticity.

Unfortunately, this is a topical theme. There is a resurgence in political rhetoric that victimises and demonises ethnically diverse communities; President Trump repudiates diversity and inclusion policies with the argument they equate to the recruitment of incapable and unskilled staff (blaming the latter for an aviation accident). As these considerations are made, I revisit the discussion in the Literature Review, concerning the 21st century activist museum and its social justice responsibility. How might heritage and cultural organisations respond to political discourse of hatred and oppression? What forms of allyship may they adopt?

This is the context in which this research's findings are relevant. They indicate that heritage and cultural organisations play an important role in empowering underserved audiences to develop and amplify their counternarratives, as strategies of resistance and resilience.

8.4 Recommendations for heritage and cultural institutions

Considering the findings and reflections from this research, the recommendations for heritage and cultural organisations are for them to recognise the importance of user-centred collaborative design and production methods, in involving underserved communities with the aim of fostering their power and agency. Following on from this, this study has also highlighted the opportunities associated with developing a collaborative relationship with drone hobbyists. This will support heritage organisations' identified need to develop a deeper understanding of drone technology and, by implication, of its sociotechnical impacts, which organisations will maximise as part of employing drone technology to facilitate agency when working with disadvantaged communities.

The final recommendation for heritage organisations is that they consider their use of visual language in the range of institutional practices observed when addressing power imbalances and representation of underserved groups. This includes deepening their understanding of how power is articulated and reiterated through the auratic aesthetics of the drone content they produce.

8.5 Limitations of the research

This research is constrained by the lack of an abundance of prior studies, on the use of drone technology to empower underserved audiences to connect with heritage and culture. This means there are no existing theoretical foundations to answer this study's research questions. This required the development of a new conceptual framework for the analysis of drone content based on agential positioning.

It is not within scope to conduct an extensive analysis of drone content and therefore the sample size of drone content is small. This is a factor that could be addressed by further research. The co-design research also has a small sample size, best practice when working with people living with dementia.

This research does not include an evaluation of participants' and heritage partners' responses to the products resulting from the drone activity, such as the recorded livestream, the 360 - degree experience of the sites where the livestreaming took place, or any other product. This is outside the scope of this research, which focuses on the engagement through the live element of the livestreaming, within the context of providing a stimulating experience, where participants are empowered to shape the drone vision they engage with.

This study is not concerned with exploring different types and sizes of drones or integration of drones with other technologies. The drone consumer market offers a wide variety of products, and this is more relevant when the research is centred on the specificities of the technical competencies and differentiations between devices. This research is user-centred, and the focus is on camera drones and their use in facilitating the experiences communities wish to engage with. The user preference determines the capabilities of the equipment used.

8.6 Future avenues of research

Current studies of drone photography and video identify general characteristics and effects of drone content, but do not discuss the aspects of power in underserved communities. Having developed and tested the conceptual findings from this research across researcher instigated, and pre-existing drone video content produced by different underserved communities, it is apposite to investigate how other types of research design, or different modes of sampling might affect the results of this study. For instance, testing the analytical framework, through a large-scale analysis of researcher instigated drone content generated by underserved communities.

This project also presents an opportunity to test the framework of analysis of drone video content, by replicating the study with a larger sample of content. Another potential area for future research is a longitudinal study of the cognitive impact of participant-led drone livestreamed experiences of heritage, in people living with memory loss and other cognitive differences.

Glossary

Access

The ability or opportunity to use, experience, acquire, participate in, or benefit from.

Accessibility

The concept of accessibility, as defined in Education Studies, relates to how institutional systems and processes either facilitate or constitute a barrier to left out communities, in accessing the services those institutions provide. Accessibility is therefore related to institutional power negotiations. Equitable practices increase accessibility. In turn, accessibility is restricted by the inequalities institutional systems create or amplify.

Aesthetics of empowered critical representation

I created this term to identify the distinctive visual language of drone video content generated by underserved communities, where they have agency and power to shape the composition of the content and therefore how it represents the content generators. This aesthetic has five key characteristics, as described in Chapter Six.

Affordances (drone)

Characteristics of drone technology, which make it possible for the drone to complete certain actions (mobility enables it to access locations not physically possible to the human body) or achieve certain outcomes (aerial photography). The key affordances identified in the thesis are drone sensing, mobility, verticality and automation.

Agency

Individuals and communities' ability or opportunity to exercise control, influence or make decisions about activities or events that affect them.

Autotechnography

Ethnographic study involving the researcher's systematic exploration, analysis, and reflection of their firsthand experiences with technology (drones, in the case of this research project). The study is undertaken with the aim of deepening the understanding of the experiences of groups the researcher identifies with.

Drone experts

Professional and recreational drone users.

Empowerment

In this thesis, empowerment is associated with agency. It refers to the process of strengthening the confidence, sense of control and power of communities to be involved in shaping their experiences of heritage.

Heritage and cultural organisations

This includes museums, archives, galleries, national parks, archaeological sites, historic properties, historic environment, community cultural centres, festivals.

Heritage experts

Practitioners in heritage organisations, including paid staff, freelancers, and volunteers.

Self-efficacy

In this thesis, this concept relates to an individual's perception of their ability to have a successful and positive outcome when visiting, participating in, or engaging with heritage and cultural experiences.

Underserved communities and audiences

Communities who experience exclusion because of institutional and societal systems and practices. The term “underserved” highlights how the way institutions provide their services extends inequity and sustains the disparities that affect those communities. In this thesis I use this term interchangeably with the terms: left out, excluded, with experience of discrimination, underrepresented.

Visibility

This word is used in reference to underserved communities being seen because of their authentic representation. It is a result of these communities having agency and being empowered to shape their voice and presence, through composing their own narrative.

List of references

Abbas, R. and Michael, K. (2023)' Socio-Technical Theory: A review'. In S. Papagiannidis (ed.), *TheoryHub* Book. Available at: https://open.ncl.ac.uk/theory-library/socio-technicaltheory.pdf ISBN: 9781739604400 (Accessed 10 February 2025).

Adagio Dance School (2021) *Road to Freedom!* 16 September [Video online] Available at: https://www.facebook.com/adagiodance.bvi/videos/4006655662748813/ (Accessed 15 August 2023).

Adams, W. (2015) 'Conducting semi-structured interviews', in *Handbook of Practical Program Evaluation*. Wiley, pp. 492–505. Available at: https://doi.org/10.1002/9781119171386.ch19

Adeyoye-Olatunde, O. A. and Olenik, N., (2021) 'Research and Scholarly Methods', *Journal of the American College of Clinical Pharmacy*, 4 (10), pp. 1358-1367. Available at: https://doi.org/10.1002/jac5.1441

Agostinho, D., Maurer, K. and Veel, K. (2020) 'Introduction to The Sensorial Experience of the Drone', *The Senses and Society*, 15(3), pp. 251–258. Available at: https://doi.org/10.1080/17458927.2020.1820195

Alexander, E. P. (1979). *Museums in Motion: an Introduction to the History and Functions of Museums* (First). Nashville, Tenn.: American Association for State and Local History.

Alexander, E.P., Alexander M., Decker, J. (2017) *Museums in Motion: An Introduction to the History and Functions of Museums,* Third Edition. Edited by M. Alexander and J. Decker. Lanham: Rowman & Littlefield Publishers.

Alison Farmer Associates (2018) *Drone Flying in the South Downs National Park* (Issues Paper). Provided by the SDNPA.

Ambrosia, V. G., Wegener, S., Zajkowski, T., Sullivan, D. V., Buechel, S., Enomoto, F., Lobitz, B., Johan, S., Brass, J., and Hinkley, E. (2011). 'The Ikhana unmanned airborne system (UAS) western states fire imaging missions: From concept to reality (2006-2010)', *Geocarto International*, *26*(2), 85–101. https://doi.org/10.1080/10106049.2010.539302

Anderson, K. and Gaston, K.J. (2013) 'Lightweight unmanned aerial vehicles will revolutionize spatial ecology', *Frontiers in Ecology and the Environment*, 11(3), pp. 138–146. Available at: https://doi.org/10.1002/9781119171386.ch19

Angelou, M. (1994) 'Still I rise'. *The complete collected poems of Maya Angelou,* Random House, New York, pp. 273.

Aran, E. and Reed, P. (2015) 'Talking points collaborative mapping-Forest Service Public Participation GIS Application', *Esri User Conference*. United States Forestry Service. Available at: https://proceedings.esri.com/library/userconf/proc15/papers/890_503.pdf (Accessed 25 March 2023).

Arts Council England (2020) *Let's Create (Strategy 2020-30)*. Available at: https://www.artscouncil.org.uk/lets-create/strategy-2020-2030/strategy (Accessed: 2 December 2022).

Arts Council England (no date). *Inspiring Learning -detailed framework*. Available at: https:www.artscouncil.org.uk/sites/default/files/S2D12_Detailed_framework.pdf (Accessed 25 March 2024).

Ashuri, T., Pinchevski, A. (2009). 'Witnessing as a Field' in: Frosh, P., Pinchevski, A. (eds.) *Media Witnessing*. Palgrave Macmillan, London. Available at: https://doi.org/10.1057/9780230235762_7

Attitude Magazine (2022) *Bimini and Dan O'Neill on healing the ocean as LGBTQ people.* 20 April. [Video online]. Available at: https://www.youtube.com/watch?v=5kD-zs4aFeA. (Accessed 25 July 2023).

Augenbraun, E. (2016) 'Making Art with Drones', *Scientific American Blog Network*', *Scientific American*, 14th January. Available at: https://blogs.scientificamerican.com/observations/making-art-withdrones/ (Accessed: 28 February 2020).

Ball, H. (2019) 'Conducting Online Surveys', *Journal of Human Lactation*, 35(3), pp. 413–417. Available at: https://doi.org/10.1177/0890334419848

Barr, A. (2013) 'Amazon testing delivery by drone, CEO Bezos says', *USA Today*. Available at: https://eu.usatoday.com/story/tech/2013/12/01/amazon-bezos-drone-delivery/3799021/ (Accessed: 10 July 2020).

Basu, R. (2013) *Drones of New York*. [Webpage] Available at: https://www.rajeevbasu.com/projects/dronesofnewyork (Accessed: 17 July 2022)

Belkouri, D. and Laing, R. (2024) 'Walking through the abstract[ed] city and co-creating urban space', *Applied Mobilities*, 9(1), pp. 63–83. Available at: https://doi.org/10.1080/23800127.2024.2309742

Bell, L.A. (2022) 'Theoretical Foundations for Social Justice Education', *Teaching for Diversity and Social Justice*. New York: Routledge, pp. 3–25. Available at: https://doi.org/10.4324/9781003005759-2

Ben-Eliyahu, A., Moore, D., Dorph, R. and Schunn, C. D. (2018) 'Investigating the multidimensionality of engagement: Affective, behavioral, and cognitive engagement across science activities and contexts', *Contemporary Educational Psychology*, 53, pp. 87–105. Available at: https://doi.org/10.1016/j.cedpsych.2018.01.002

Benjamin, W. (1969). *The Work of Art in the Age of Mechanical Reproduction*.

Benjamin, W. (2007) The Work of Art in the Age of Mechanical Reproduction. H. Arendt (ed.) *Illuminations: Essays and reflections*. New York: Schocken Books, pp. 217–251.

Bhalla, A. and Lapeyre, F. (1997) 'Social Exclusion: Towards an Analytical and Operational Framework', *Development and Change*, 28(3), pp. 413–433. Available at: https://doi.org/10.1111/1467-7660.00049

Birtchnell, T. and Gibson, C. (2015) 'Less talk more drone: social research with UAVs'. *Journal of Geography in Higher Education*, 39(1), pp. 182–189. Available at: https://doi.org/10.1080/03098265.2014.1003799

Blumer, H. (1969) 'The Methodological Position of Symbolic Interactionism' in *Symbolic Interactionism: Perspective and Method*. Los Angeles: University California Press, pp. 1–60.

Blumer, H. (1973) 'A Note on Symbolic Interactionism', *American Sociological Review*, 38(6), p. 797. Available at: https://doi.org/10.2307/2094141

Bolton, M, Day, R. and Leach, M. (2019) 'England's overlooked neighbourhoods: Defining, understanding and regenerating "left behind" communities', *Journal of Urban Regeneration and Renewal*, 13(2), pp. 116–123. Available at: https://hstalks.com/article/5392/englands-overlooked-neighbourhoods-defining-unders (Accessed: 14 March 2022).

Bourdieu, P. (1984) *Distinction. a Social Critique of the Judgement of Taste.* London: Routledge.

Bowen, Christopher J., and Thompson, R., T. (2017) *Grammar of the Shot*. Taylor & Francis Group Taylor & Francis Group ProQuest Ebook Central. Available at: https://doi.org/10.4324/9780080927053

Boylorn, R. M. (2008). Lived experience. *The Sage encyclopedia of qualitative research methods* (pp. 489–490). Thousand Oaks: Sage.

Brennen, B.S. (2017) *Qualitative Research Methods for Media Studies*. Second edition. New York: Routledge, 2017. Available at: https://doi.org/10.4324/9781315435978

Brighenti, A.M. (2010) 'Visibility and Social Theory' in *Visibility in Social Theory and Social Research*. London: Palgrave Macmillan UK, pp. 37–70. Available at: https://doi.org/10.1057/9780230282056_2

Brimblecombe, K. (no date). *Art @ Kathryn Brimblecombe-Fox: DRONESCAPES*. Available at: https://kathrynbrimblecombeart.blogspot.com/p/dronescapes.html (Accessed: 28 February 2020).

Brown, B., (2013), *Daring greatly: how the courage to Be Vulnerable Transforms the Way We Live, Love, Parent and Lead*, Penguin Random House.

Brown, B. (2018), *Dare to lead: brave work, tough conversations, whole hearts,* Vermillion.

Buckingham, D. (2009) 'Creative visual methods in media research: possibilities, problems and proposals', *Media, Culture & Society*, 31(4), pp. 633–652. Available at: https://doi.org/10.1177/0163443709335280

Burkett, I. (no date), *An Introduction to co-design*. University of New South Wales Centre for Social Impact (Knode). Available at: https://www.yacwa.org.au/wp-content/uploads/2016/09/An-Introduction-to-Co-Design-by-Ingrid-Burkett.pdf (Accessed 23 December 2023).

Butler, M., Derrett, S. (no date). 'The walking interview: an ethnographic approach to understanding disability', *Internet J. Allied Health Sci. Practice*, *12*(3), 1–8.

Cairngorms National Park Authority (2021) *Cairngorms National Park Visitor Survey* (report). Grantown on Spey, Scotland: CNPA. Provided by the Cairngorms National Park Authority.

Calçada Portuguesa (2023) *Calçada Portuguesa - Lisbon Stories.* 26 July [Video online]. Available at: https://www.youtube.com/watch?v=Ypcj_54bk7Q (Accessed 23 August 2023).

Cameron, F. and Kenderdine, S. (Eds.) (2007): *Theorizing Digital Cultural Heritage : a Critical Discourse*. The MIT Press.

Campbell, R. (2018) 'Drone Film Theory: The Immanentisation of Kinocentrism', *Media Theory*, 2(2), pp. 52–78. Available at: https://journalcontent.mediatheoryjournal.org/index.php/mt/article/view/57 (Accessed: 31 July 2023).

Campbell-Stephens, R. (2022) *Global Majority; Decolonising the language and Reframing the Conversation about Race*, p. 7. Available at: https://www.leedsbeckett.ac.uk/-/media/files/schools/school-of-education/final-leeds-beckett-1102-global-majority.pdf. (Accessed: 5 June 2024).

Cervinkova, H, (2020) 'Citizenship after Covid-19: Thoughts from Poland', *Social Anthropology/Anthropologie Sociale*, 28 (2). Available at: https://doi :10.1111/1469-8676.12864

Chabot, D. (2018) 'Trends in drone research and applications as the *Journal of Unmanned Vehicle Systems* turns five', *Journal of Unmanned Vehicle Systems*, 6(1), pp. vi–xv. Available at: https://doi.org/10.1139/juvs-2018-0005

Challis, Keith (2022) *Attitudes to Drone Use by National Trust Staff* (Internal Survey report). Provided by the National Trust.

Chichester Festival Theatre (2020). *Crave*. Available at: https://www.cft.org.uk/archive/crave (Accessed 12 September 2020).

Choi-Fitzpatrick, A., Chavarria, D., Cychosz, E., Dingens, J.P., Duffey, M., Koebel, K., Siriphanh, S., Tulen, M. Y., Watanabe, H., Juskauskas, T., Holland, J. and Almquist, L., (2009). *Up in the Air: A Global Estimate of Non-Violent Drone Use 2009-2015* (report). https://digital.sandiego.edu/gdl2016report/1 (Accessed: 28 February 2020).

Ciolfi, L. (2017) *Physical-Digital Technologies for Culture and Heritage*. Digital Leaders. Available at: https://digileaders.com/physical-digital-technologies-culture-heritage/ (Accessed 17 July 2025).

Cochrane, L. and Corbett, J. (2018) 'Participatory Mapping' in *Handbook of Communication for Development and Social Change*. Singapore: Springer Singapore, pp. 1–9. Available at: https://doi.org/10.1007/978-981-10-7035-8_6-1

Cohen, C.B. (1998) 'This is De Test: Festival and the Cultural Politics of Nation Building in the British Virgin Islands', *American Ethnologist*, 25(2), pp. 189–214. Available at: https://doi.org/10.1525/ae.1998.25.2.189

Cohen, C. B., and Mascia-Lees, F.E. (1993) 'The British Virgin Isles as Nation and desti-nation. Representing and Siting Identity in a Post- colonial Caribbean', *Social Analysis: The International Journal of Social and Cultural Practice*, 33, pp. 130–151. Available at: http://www.jstor.org/stable/23163043

Crickmar, R. (2017) *Repurposing the drone. Ai Weiwei's Human Perspective, Filmhouse cinema* [online]. Available at: https://www.filmhousecinema.com/latest/repurposingdrone-ai-weiweis-human-perspective-shift (Accessed: 1 May 2021).

D'Andrea, R. (2014) 'Guest Editorial Can Drones Deliver?', *IEEE Transactions on Automation Science and Engineering*, 11(3), pp. 647–648. Available at: https://doi.org/10.1109/TASE.2014.2326952

Dawson, E. (2019). *Equity, Exclusion and Everyday Science Learning. The Experiences of Minoritised Groups.* Routlege.

Debord, G. (1994) *The Society of the Spectacle*. Translated by Nicholson-Smith, D. Zone Books. Available at: https://doi.org/10.2307/j.ctv1453m69

De Cruz (2024) *Wonderstruck: How Wonder and Awe Shape the Way We Think*, Princeton, NJ: Princeton University Press, pp. 232 Available at: https://doi.org/ 10.2307/jj.7249506

DeJonckheere, M. and Vaughn, L.M. (2019) 'Semi-structured interviewing in primary care research: a balance of relationship and rigour', *Family Medicine and Community Health*, 7(2), p. e000057. Available at: https://doi.org/10.1136/fmch-2018-000057

Denison, S. (2023) 'Putting Ourselves in the Picture. An Autoethnographic Approach to Photography Criticism', in J. Goode, K. Lumsden, and J. Bradford (eds.) *Crafting Autoethnography Processes and Practices of Making Self and Culture*. London: Routledge. Available at: https://doi.org/10.4324/9781003309239

Denscombe, M. (2014) *Good Research Guide: For Small-Scale Social Research Projects*. 5th edition. McGraw-Hill Education. Available at: https://www.researchgate.net/file.PostFileLoader.html?id=582a0dbf217e20276533f5a5&assetKey=AS:428404664213506@1479151039119

Department for Culture, Media and Sport (DCMS) (2000) *Centres for Social Change: Museums, Galleries and Archives for All* [Preprint]. Available at: https://openlibrary.org/books/OL38563248M/Centres_for_social_change_museums_galleries_and_archives_for_all (Accessed 21 January 2021).

Department for Transport and Ministry of Defence, (2016) *Public dialogue on drone use in the UK*. Available at: https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/579550/drones-uk-public-dialogue.pdf

Department for Transport, Home Office, Ministry of Defence, and T.R.H.C.G.M. (2019) *Drones: consultation response, 8 January 2019 - GOV.UK*. Available at: https://www.gov.uk/government/speeches/drones-consultation-response-7-january-2019 (Accessed: 12 February 2020).

Deterding, S., Dixon, D., Khaled, R., Nacke, L. (2011). 'From Game Design Elements to Gamefullness: defining Gamification', *Proceedings of the 15th International Academic MindTrek Conference: Envisioning Future Media Environments*, 9–15. Available at: https://doi.org/10.1145/2181037.2181040

Digital Poverty Alliance (2022). *UK Digital Poverty Evidence Review 2022*. Available at: https://digitalpovertyalliance.org.uk/wp-content/uploads/2022/06/UK-Digital-Poverty-Evidence-Review-2022-v1.0-compressed.pdf (Accessed 22 July 2025).

DJI (2023) [Website] Available at: https://www.dji.com (Accessed 18 November 2023)

ELEVENPLAY x Rhizomatiks Research (2013) *24 drones*. Available at: http://www.daito.ws/en/work/elevenplay-rhizomatiks-research-3-dancers-and-24drones.html (Accessed: 11 January 2021).

Ellis, C., Adams, T.E. and Bochner, A. (2010) 'Autoethnography: An Overview', *Forum: Qualitative Sozialforschung/Qualitative Social Research*, 12(1), Available at: https://doi.org/10.171169/fqs-12.1.1589

English, L. (2023) 'Finding a new approach; the use of visual images in spoken word poetry film' in Vorrath, M. and Korecka, W. (eds.), *Poetry and contemporary visual culture/Lyric und zeitgenossische Visuelle Kultur. Poetry in the digital age* (Vol. 3, pp. 207–222). De Gruyter. Available at: https://librar.oapen.org/bitstream/handle/20 (Accessed 2 July 2025)

Equality Act (2010) Available at: https://www.legislation.gov.uk/ukpga/2010/15/contents (Accessed 12 July 2025).

Erickson, F. (2011) 'Uses of video in social research: a brief history', *International Journal of Social Research Methodology,* 14:3, 179-189, 14(3), 179–189. Available at: https://doi.org/10.1080/13645579.2011.563615

Fawcett Society (2022), *Make Equality Law Fit For The 21st Century*. Available at: https://www.fawcettsociety.org.uk/make-equality-law-21st-century (Accessed: 28 November 2022).

Fetterman, D., Rodríguez-Campos, L., Wandersman, A. and O'Sullivan, R. (2014) ' Collaborative, Participatory, and Empowerment Evaluation', *American Journal of Evaluation*, 35(1), pp. 144–148. Available at: https://doi.org/10.1177/1098214013509875

Fine, M. (2023a) *These too are footsteps* [film-poem], unpublished.

Fine, M. (2023b) *A room of my own* [film-poem], unpublished.

Finlay, J M. and Bowman, J. A. 'Geographies on the Move: A Practical and Theoretical Approach to the Mobile Interview', *The Professional Geographer*, 69 (2), 263-274. Available at: https://doi.org/10.1080/00330124.2016.1229623

Fischer, G. and Herrmann, T. (2011) 'Socio-technical Systems: A Meta-Design Perspective', *International Journal of Sociotechnology and Knowledge Development*, 3(1), 1-33, January-March 2011 *Available* at: https://doi.org/10.4018/jskd.2011010101

Fish, A. (2020) 'Visual Anthropology from the Air', in *Handbook of Ethnographic Film and Video*. Routledge.

FPV UK (no date) *Drone Club and Event Map* [Webpage] Available at: https://www.fpvracingworld.com (Accessed: 12 May 2022).

Freimuth, V. S., and Mettger, W. (1990) 'Is there a hard-to-reach audience?', *Public health reports*, 105(3), pp. 232–238. Available at: https://pubmed.ncbi.nlm.nih.gov/2113680/

Frosh, P., and Pinchevski, A. (2014). 'Media witnessing and the ripeness of time', *Cultural Studies*, 28(4), 594–610. Available at: https://doi.org/10.1080/09502386.2014.891304

Fust, P. and Loos, J. (2020) 'Development perspectives for the application of autonomous, unmanned aerial systems (UASs) in wildlife conservation', *Biological Conservation*. Elsevier Ltd. (Vol. 241) Available at: https://doi.org/10.1016/j.biocon.2019.108380

Gallagher, M.W. (2012) 'Self-Efficacy', in Ramachandran *et. al.* (eds.) *Encyclopedia of Human Behavior.* Elsevier, pp. 314-320. Available at: https://doi.org/10.1016/B978-0-12-375000-6.00312-8

Glebova, A. (2022) *Aleksandr Rodchenko: Photography in the time of Stalin*. New Haven: Yale University Press, 256 pp (Accessed 31 January 2025).

Garibay, C., Lannes, P., González, J.; (2017) GENIAL: *Latino Audiences: Embracing Complexity*, 8pp. Available at: https://www.researchgate.net/publication/362680273_GENIAL_Latino_Audiences_Embracing_Complexity (Accessed 19 July 2025).

Garrett, B.L. and McCosker, A. (2017) 'Non-human Sensing: New Methodologies for the Drone Assemblage', *Refiguring Techniques in Digital Visual Research*. Cham: Springer International Publishing, pp. 13–23. Available at: https://doi.org/10.1007/978-3-319-61222-5_2

Goldsmith, L. and Bell, M.L. (2022) 'Queering Environmental Justice: Unequal Environmental Health Burden on the LGBTQ+ Community', *American Journal of Public Health*, 112(1), pp. 79–87. Available at: https://doi.org/10.2105/AJPH.2021.306406

GOV.UK (2016) *Public dialogue on drone use in the UK*. Available at: https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/579550/drones-uk-public-dialogue.pdf (Accessed: 30 April 2020).

Grémillet, D., Puech, W., Garçon, V., Boulinier, T., and Maho, Y. Le. (2012) 'Robots in Ecology: Welcome to the machine', *Open Journal of Ecology*, *02*(02), 49–57. Available at: https://doi.org/10.4236/oje.2012.22006

Gretzel, U., Fuchs, M., Baggio, R., Hoepken, W., Law, R., Neidhart, J., Pesonen, J., Zanker, M and Xiang, Z. (2020) 'e-Tourism beyond COVID-19: a call for transformative research', *Information Technology & Tourism*, 22(2), pp. 187–203. Available at: https://doi.org/10.1007/s40558-020-00181-3

Gruber, M. E., Galliano, M. S., Brosnihan, A. P., and Hancock, P. A. (2023). 'The impact of camera usage and angle on perceptions of power during videoconferencing', *Computers in Human Behavior Reports*, 10 (September 2022), 100285. Available at: https://doi.org/10.1016/j.chbr.2023.100285

Guest, G., Namey, E. E. and Mitchell, M. L. (2012) *Collecting Qualitative Data: A Field Manual for Applied Research*. SAGE Publications Inc.

gustavofrazao (2017) *Aerial View of Monument to the discoveries, Lisbon, Portugal.* 19 December. istock item-875673258.mov, Available at: https://www.videvo.net/video/aerial-view-of-monument-to-the-discoveries-lisbon-portugal/864470/ (Accessed 31 June 2023).

Gusterson, H. (2017) 'Drone warfare', *AIP Conference Proceedings*. American Institute of Physics Inc. Available at: https://doi.org/10.1063/1.5009234

Hackett, S. and Dissanayake, S.T.M. (2014) *Environmental and Natural Resources Economics*. Routledge. Available at: https://doi.org/10.4324/9781315704586

Hafermalz, E., Johnston, R., Hovorka, D. and Riemer, K. (2020) 'Beyond "mobility": A new understanding of moving with technology', *Information Systems Journal*, 30(4), pp. 762–786. Available at: https://doi.org/10.1111/isj.12283

Hanmakyugh, T. (2020) 'The psychology of camera shots: a determinant for audience perception of film images', *Journal of Theatre and Cultural Studies*, 14(1), pp. 104–113. Available at: https://ezenwaohaetorc.org/journals/index.php/AMA/article/view/2306 (Accessed: 28 July 2023).

Hastings Contemporary (2020) *Robot Tours*. [Video online] Available at: https://www.hastingscontemporary.org/robot-tours/. (Accessed 10 June 2020).

Heidegger, M. (1977). *The question concerning technology and other essays*. Harper and Row. New York and London.

Heimann, K., Uithol, S., Clbi, M., Umiltà, M.A., Guerra, M., Fingerhut, J. and Gallese, V. (2019) 'Embodying the camera: An EEG study on the effect of camera movements on film spectators' sensorimotor cortex activation', *PLoS One*. March 13, 14(3) Available at: https://doi.org/10.1371/journal.pone.0211026

Heritage Alliance (2024) *On the Brink: Heritage in the Cost of Living Crisis- Heritage Alliance Insight Report*, published May 2024. Available at: https://www.theheritagealliance.org.uk/our-work/publications/cost-of-living-report (Accessed 22 July 2025).

Herwitz, S.R., Johnson, L.F., Dunagan, S.E., Higgins, R.G., Sullivan, D.V., Zheng, J., Lobitz, B.M., Leung, J.G., Gallmeyer, B.A., Ayoagi, M Slye, R.E. and Brass, J.A. (2004) 'Imaging from an unmanned aerial vehicle: Agricultural surveillance and decision support', *Computers and Electronics in Agriculture*, 44(1), pp. 49–61. Available at: https://doi.org/10.1016/j.compag.2004.02.006

Hidden Britain by Drone (2016-2018), [TV series], Prime Video, Channel 4.
Hildebrand, J.M. (2018) 'Situating Hobby Drone Practices', *Digital Culture & Society*, 3(2), pp. 207–218. Available at: https://doi.org/10.14361/dcs-2017-0212

Hildebrand, J.M. (2019) 'Consumer drones and communication on the fly', *Mobile Media & Communication*, 7(3), pp. 395–411. Available at: https://doi.org/10.1177/2050157919850603.

Hildebrand, J.M. (2020) 'Drone-topia as method', *Mobilities*, 15(1), pp. 25–38. Available at: https://doi.org/10.1080/17450101.2019.1663079

Hildebrand, J.M. (2021) *Aerial Play*. Singapore: Springer Singapore. Available at: https://doi.org/10.1007/978-981-16-2195-6

Hjorth, L. and Pink, S. (2014) 'New visualities and the digital wayfarer: Reconceptualizing camera phone photography and locative media', *Mobile Media & Communication*, 2(1), pp. 40–57. Available at: https://doi.org/10.1177/2050157913505257

Hochschild, A.R. (1979) 'Emotion work, Feeling Rules, and Social Structure', *American Journal of Sociology*, 85(3), pp. 551–75. Available at: https://www.jstor.org/stable/2778583 (Accessed : 16 July 2025).

Hodges, S. (2017) *Visitors of colour*. [Tumblr post]. August 4. Available at: https://visitorsofcolor.tumblr.com/image/163774348237 (Accessed: 23 October 2024)

Hong, B., Zhang, L and Sun, H. (2019) 'Measurement of the Vertical Spatial Metaphor of Power Concepts Using the Implicit Relational Assessment Procedure', *Frontiers in Psychology*, 10, Article 1422 Available at: https://doi.org/ 10.3389/fpsyg.2019.01422

Hooker, C.M. and De Zúñiga, H.G. (2017) 'Survey Methods Online' in *The International Encyclopaedia of Communication Research Methods*. Wiley, pp. 1–7. Available at: https://doi.org/10.1002/9781118901731.iecrm0244

Hooper-Greenhill, E. (2000) 'Changing Values in the Art Museum: rethinking communication and learning', *International Journal of Heritage Studies*, 6(1), pp. 9–31. Available at: https://doi.org/10.1080/135272500363715

Horsfall, D. and Titchen, A. (2009) 'Disrupting edges – opening spaces: pursuing democracy and human flourishing through creative methodologies', *International Journal of Social Research Methodology*, 12(2), pp. 147–160. Available at: https://doi.org/10.1080/13645570902752324

Huang, Y. C., Backman, K. F., Backman, S. J.; Chang, L. L. (2016) 'Exploring the Implications of Virtual Reality Technology in Tourism Marketing: An Integrated Research Framework', *International Journal of Tourism Research* 18 (2), 116--128. https://doi.org/ 10.1002/jtr.2038

Humm, C. and Schrögel, P. (2020) 'Science for All? Practical Recommendations on Reaching Underserved Audiences', *Frontiers in Communication*, 5. Available at: https://doi.org/10.3389/fcomm.2020.00042

Humm, C., Schrögel, P. and Leßmöllmann, A. (2020) 'Feeling Left Out: Underserved Audiences in Science Communication', *Media and Communication*, 8(1), pp. 164–176. Available at: https://doi.org/10.17645/mac.v8i1.2480

Hunt, V., Layton, D and Prince, S. (2015) *Why diversity matters*. Available at: https://www.mckinsey.com/capabilities/people-and-organizational-performance/ourinsights/why-diversity-matters (Accessed: 12 September 2024).

Islam, M., (2022) 'Focus Groups in Tourism Research' in *Encyclopedia of tourism management and marketing*, pp. 261-264, Eddward Elgar Publishing. Available at: https://doi.org/ 10.4337/9781800377486.focus.groups.in.tourism.research

Jablonowski, M. (2020) 'Beyond drone vision: the embodied telepresence of first-person view drone flight', *The Senses and Society*, 15(3), pp. 344–358. Available at: https://doi.org/10.1080/17458927.2020.1814571

Jackman, A. (2023). *Diversifying drone stories, 2021-2023.* [Data collection]. UK Data Service. https://doi.org/10.5255UKDA-SN-856753

Jarrier, Elodie and Bourgeon, D. (2012) 'Impact of Mediation Devices on the Museum Visit Experience and on Visitors Behavioural Intentions', *International Journal of Arts Management*, 15(1), pp. 18–29 (Accessed 6 October 2024).

Jeannotte, M. S. (2021) 'When the gigs are gone: Valuing arts, culture and media in the COVID-19 pandemic', *Social Sciences & Humanities Open*, *3*(1), 100097. Available at: https://doi.org/10.1016/j.ssaho.2020.100097

Jenkinson, P. (2005) *Museum Provision and Professionalism*. G. Kavanagh (ed.), Routledge. Available at: https://doi.org/10.4324/9780203974469

Jensen, O.B. (2016a) 'Drone city - power, design and aerial mobility in the age of "smart cities" in *Geographica Helvetica*, 71(2), pp. 67–75. Available at: ttps://doi.org/10.5194/gh71-67-2016

Jensen, O. (2016b) 'New Foucauldian Boomerangs: Drones and Urban Surveillance', *Surveillance & Society*, 14(1), pp. 20–33. Available at: https://ojs.library.queensu.ca/index.php/surveillance-and-society/article/view/boomerang

Jensen, O.B. (2020) 'Thinking with the drone – visual lessons in aerial and volumetric thinking', *Visual Studies*, 35(5), pp. 417–428. Available at: https://doi.org/10.1080/1472586X.2020.1840085

Johnson, M., (2014) *Coactive Design: Designing Support for Interdependence in Human-Robot Teamwork*. PhD, Delft University of Technology. Available at: https://researchgate.net/publication/267393898-Coactive_Design_Designing_Support_for_interdependence_in Human-Robot_Teamwork (Accessed 23 July 2023).

Johnson, M, Bradshaw, M., Feltovitch, P., Jonker, M., van Riemsdijk, M. and Sierhuis, M., (2014) 'Coactive Design: Designing Support for Interdependence in Joint Activity', *Journal of Human-Robot Interaction.* Available at: https//doi.org/10.5898/JHRI.3.1.Johnson

Josselson, R. and Lieblich, A. (2005) *Carolyn Ellis, The Ethnographic I: A Methodological Novel about Autoethnography Walnut Creek,* California: Altamira Press, 2004, 384 pages. Available at: https://doi.org/10.1080/10408340308518298

Jubilee Medical Group (no date), What is the definition of 'housebound' (webpage). Available at: https://www.jubileemg.nhs.uk (Accessed 7 July 2022).

Kalay, Y., Kvan, T. and Affleck, J. (eds.) (2007) *New Heritage: New Media and Cultural Heritage*, London, Routledge.

Kawashima, N. (2006) 'Audience Development And Social Inclusion In Britain', *International Journal of Cultural Policy*, 12(1), pp. 55–72. Available at: https://doi.org/10.1080/10286630600613309

Keinänen, M. and Beck, E.E. (2017) 'Wandering intellectuals: establishing a research agenda on gender, walking, and thinking', *Gender, Place & Culture*, 24(4), pp. 515–533. Available at: https://doi.org/10.1080/0966369X.2017.1314940

Kemp, E. and Poole, S.M. (2016) 'Arts Audiences: Establishing a Gateway to Audience Development and Engagement', *The Journal of Arts Management, Law, and Society*, 46(2), pp. 53–62. Available at: https://doi.org/10.1080/10632921.2016.1150929

Kennemore, A. and Postero, N. (2021) 'Collaborative ethnographic methods: dismantling the “anthropological broom closet”? ', *Latin American and Caribbean Ethnic Studies*, 16(1), pp. 1–24. Available at: https://doi.org/10.1080/17442222.2020.1721091

Kim, H.Y., Kim, B. and Kim, J. (2016) 'The Naughty Drone', *Proceedings of the 10th International Conference on Ubiquitous Information Management and Communication - IMCOM 16*. New York, New York, USA: ACM Press, pp. 1–6. Available at: https://doi.org/10.1145/2857546.2857639

Kline, N. (2001) *Time to Think.* Cassell.

Knoblauch, H. and Schnettler, B. (2012) 'Videography: analysing video data as a "focused" ethnographic and hermeneutical exercise', *Qualitative Research*, 12(3), pp. 334–356. Available at: https://doi.org/10.1177/1468794111436147

Kowalewski, M. and Bartłomiejski, R. (2020) 'Is it research or just walking? Framing walking research methods as "non-scientific", *Geoforum*, 114, pp. 59–65. Available at: https://doi.org/10.1016/j.geoforum.2020.06.002

Kraehe, A.M. and Acuff, J.B. (2013) 'Theoretical Considerations for Art Education Research with and about "Underserved Populations", *Studies in Art Education*, 54(4), pp. 294–309. Available at: https://doi.org/10.1080/00393541.2013.11518904

Kraft, R.N. (1987) 'The influence of camera angle on comprehension and retention of pictorial events', *Memory & Cognition*, 15(4), pp. 291–307. Available at: https://doi.org/10.3758/BF03197032

Krajník, T., Vonasek, V., Fišer, D. and Faigl, J. (2011) 'AR-drone as a platform for robotic research and education', *Communications in Computer and Information Science*. Springer, Berlin, Heidelberg, pp. 172–186. Available at: https://doi.org/10.1007/978-3-642-21975-7_16

Kreimer, B. (2018) 'Drone Journalism: Storytelling from a New Perspective: Data, Visual Analytics and Innovative Methodologies in International Reporting', *Digital Investigative Journalism* (pp. 91–102).

Kumar, V.Y. and Michael, L., N. (2012) 'Opportunities and challenges with autonomous micro aerial vehicles', *International Journal of Robotics Research*, pp. 1279–1291. Available at: https://doi.org/10.1177/0278364912455954

Kyselaa, J. and Štorková, P. (2015) 'Using Augmented Reality as a Medium for Teaching History and Tourism', *Procedia-Social and Behavioral Sciences*, 174, pp. 926–931. Available at: https://doi.org/10.1016/j.sbspro.2015.01.713

Lawson-Tancred, J. (2022) 'See how the Dutch collective DRIFT is using illuminated drones to rebuild the Sagrada Familia, the Colosseum, and other landmarks', *Artnet News*. Available at: https://news.artnet.com/art-world/see-how-the-dutch-collective-drift-uses-drones-andlights-to-digitally-rebuild-the-colosseum-and-other-major-landmarks-2199804 (Accessed: 26 October 2024).

Light, B., Bagnall, G., Crawford, G. and Gosling, V. (2018) 'The material role of digital media in connecting with, within and beyond museums', *Convergence: The International Journal of Research into New Media Technologies*, 24(4), pp. 407–423. Available at: https://doi.org/10.1177/1354856516678587

Limb, P. (2023) 'Where the river flows out to the sea', in Goode, J., Lumsden, K. and Bradford, J. (eds.) *Crafting Autoethnography*. London: Routledge. Available at: https://doi.org/10.4324/9781003309239

Lindley, J. and Coulton, P. (2015) 'Game of drones', *CHI PLAY 2015 - Proceedings of the 2015 Annual Symposium on Computer-Human Interaction in Play*. New York, NY, USA: ACM, pp. 613–618. Available at: https://doi.org/10.1145/2793107.2810300

Lisein, J., Pierrot-Deseilligny, M., Bonnet, S. and Lejeune, P. (2013) 'A photogrammetric workflow for the creation of a forest canopy height model from small unmanned aerial system imagery', *Forests*, 4(4), pp. 922–944. Available at: https://doi.org/10.3390/f4040922

Liu, J., Yue, M., Yu, F. and Tong, Y. (2022) 'The contribution of tourism mobility to tourism economic growth in China', *PLoS ONE*, *17*(10). Available at: https://doi.org/10.1371/journal.pone.0275605

London and South East National Trust (2021) *Iron Age hillfort - Cissbury Ring video trail.* 9 July. [Video online] Available at: https://youtu.be/Wk8vo4eIGi8 (Accessed: 17 July 2023).

Lovejoy, A. M, (2022) *Letter of support provided for Ethical Application for research with older adults living with dementia,* [letter], 24 December 2022.

Lovejoy Centre, (2023) *Bringing the outdoors Indoors: Devil's Dyke*, (unreleased co-research livestream, online video), Ed: Visual Air .

Lovejoy Centre, (2023) *Bringing the outdoors Indoors: Highdown Hil,* (unreleased co-research livestream, online video), Ed: Visual Air.

Luke, M. and Goodrich, K.M. (2019) 'Focus Group Research: An Intentional Strategy for Applied Group Research?', *The Journal for Specialists in Group Work*, 44(2), pp. 77–81. Available at: https://doi.org/10.1080/01933922.2019.1603741

Luvaas, B. (2019) 'Unbecoming: The aftereffects of autoethnography', *Ethnography*, 20(2), pp. 245–262. Available at: https://doi.org/https://doi.org/10.1177/14661381177426

m.e.l. Research (2022) *South Downs National Park Visitor Survey 2021*. Available at: https://www.southdowns.gov.uk/wp-content/uploads/2023/01/South-Downs-VisitorSurvey-2021-Final-report.pdf (Accessed: 6 July 2020).

Macpherson, H. (2016) 'Walking methods in landscape research: moving bodies, spaces of disclosure and rapport', *Landscape Research*, 41(4), pp. 425–432. Available at: https://doi.org/10.1080/01426397.2016.1156065

Mademlis, I., Nikolaidis, N., Tefas, A., Pitas, I., Wagner, T. and Messina, A. (2019) 'Autonomous UAV Cinematography: A Tutorial and a Formalized Shot Type Taxonomy', *ACM Computer Surveys* (September 2019). Available at: https://doi.org/10.1145/3347713

Magaldi and Berler, (2020) 'Research and scholarly methods: Semi-structured Interviews' in Zeigler-Hill, V., Shackelford, T.K. (eds.) *Encyclopedia of Personality and Individual Differences*, Springer, Cham. Available at: https://doi.org/10.1007/978-3-319-24612-3-857

Mangina, E., Keeffe, E., Eyerman, J, G. (2016) 'Drones for Live Streaming of Visuals for People with Limited Mobility', *Proceedings of the 2016 International Conference on Virtual Systems and Multimedia, VSMM 2016*. Institute of Electrical and Electronics Engineers Inc. Available at: https://doi.org/10.1109/VSMM.2016.7863162

Mangold, E.B. and Goehring, C. (2019) 'The visual rhetoric of the aerial view: From surveillance to resistance', *Quarterly Journal of Speech*, 105(1), pp. 25–41. Available at: https://doi.org/10.1080/00335630.2018.1553305

Margaritoff, M. (2017). *Watch Sci-Fi Short Donny the Drone Starring Guy Pearce*. The Drive.

Martins, J, Gonçalves, R., Branco, F., Barbosa, L.; Melo, M. and Bessa, M. (2017) 'A multisensory virtual experience model for thematic tourism: A Port wine tourism application proposal', *Journal of Destination Marketing & Management* 6 (2), pp. 103–109. https://doi.org/10.1016/j.jdmm.2017.02.002

Mascort-Albea, E.J., Ruiz Jaramillo, J. and Romero-Hernández, R. (2014) 'Analizando el Patrimonio Cultural a través de vuelos no tripulados: nuevas estrategias de investigación de espacios rurales y urbanos', *Virtual Archaeology Review*, 5(11), p. 55. Available at: https://doi.org/10.4995/var.2014.4181

Mateus, C, Campis, R., Jabba,, D, Erazo, A.M., Romero., V. (2024). 'Gamification as a tool for Inclusion, Education and Human Development' in Fahriye, A. and Zehra, A. (eds.), *Intellectual and Learning Disabilities-Inclusiveness and Contemporary Teaching Environments*. IntechOpen. Available at: https://doi.org/10.5772/intechopen.113229

Maxwell, J.A. (2013) 'A Model for Qualitative Research Design' in *Qualitative Research Design*, pp. 1–22. Available at: https://us.sagepub.com/sites/default/files/upm-binaries/5055_Maxwell_Chapter_1.pdf (Accessed 10 June 2021).

McCall, V. and Gray, C. (2014) 'Museums and the "new museology": Theory, practice and organisational change', *Museum Management and Curatorship*, 29(1), pp. 19–35. Available at: https://doi.org/10.1080/09647775.2013.869852

McCarthy, C. (2015) 'Grounding Museum Studies: Introducing Practice', *The International Handbooks of Museum Studies*. Wiley. Available at: https://doi.org/10.1002/9781118829059.wbihms200

McLuhan, M. (2013) *Understanding Media: The extensions of man*. Gingko Press.

McQuire, S. (2016). *Geomedia: Networked cities and the future of public space.* Polity. Available at: https://www.amazon.co.uk/Geomedia-Networked-Cities-Future-Public-ebook/dp/B01LWTZOOV?asin=0745660762&revisionId=&format=4&depth=4 (Accessed 12 June 2024).

Memmi, A. (2000) *Racism*. University of Minnesota Press.

Mental Capacity Act 2005 Available at: https://www.legislation.gov.uk/ukpga/2005/9/contents. (Accessed 15 May 2023).

Mental Capacity Act Code of Practice (2007) Available at: https://assets.publishing.service.gov.uk/media/5f6cc6138fa8f541f6763295/Mental-capacity-act-code-of-practice.pdf (Accessed 12 March 2019).

Milena, Z., Grundey, D. and Stancu, A. (2008) 'Qualitative Research Methods: a comparison between focus-group and in-depth interview', *Annals of Faculty of Economics*, 4, pp. 1279–1283. Available at: https://www.researchgate.net/publication/46533517_QUALITATIVE_RESEARCH_METHODS_A_COMPARISON_BETWEEN_FOCUS-GROUP_AND_IN-DEPTH_INTERVIEW (Accessed 12 June 2021).

Miller, T., Volmert, A., Rochman, A. and Aassar, M. (2021) *Talking About Poverty: Narratives, Counter-Narratives, and Telling Effective Stories,* Washington, DC, FrameWorks Institute. Available at: https://www.frameworksinstitute.org/appuploads/2021/09/Talking-about-poverty.pdf (Accessed 28 October 2024).

Millerson, G. (1994) *Video camera techniques*. 2nd edition. Oxford: Focal.

Minkiewicz, J., Bridson, K. and Evans, J. (2016) 'Co-production of service experiences: insights from the cultural sector', *Journal of Services Marketing*, 30(7), pp. 749–761. Available at: https://doi.org/10.1108/JSM-04-2015-0156

Mirk, D. and Hlavacs, H. (2015) 'Virtual Tourism with Drones', *Proceedings of the First Workshop on Micro Aerial Vehicle Networks, Systems, and Applications for Civilian Use*. New York, NY, USA: ACM, pp. 45–50. Available at: https://doi.org/10.1145/2750675.2750681

Morrell, A. (2008) 'Museums After Modernism: Strategies Of Engagement By Griselda Pollock And Joyce Zemans (eds.) in *The Art Book*, 15(3), pp. 45–46. Available at: https://doi.org/10.1111/j.1467-8357.2008.00975_15.x

MTM London. (2017). *Digital Culture*. Available at: https//www.nesta.org.ukreportdigital-culture-2017/ 46pp (Accessed 8 October 2023).

Munck Petersen, R. (2016) 'Quiet Areas: Outer experiences and inner sensations – a qualitative approach using film and drones', *Inter.noise 2016.* Hamburg. Available at: https://www.academia.edu/28414243/Quiet_Areas_Outer_experiences_and_inner_sensations_a_qualitative_approach_using_film_and_drones. (Accessed 8 November 2024)

Museums Association. (no date) *Museums Change Lives*. Available at: https://museumsassociation.org/campaigns/museums-change-lives/ (Accessed 15 July 2025).

Mutibwa, D.H., Hess, A. and Jackson, T. (2020) 'Strokes of serendipity: Community co-curation and engagement with digital heritage', *Convergence: The International Journal of Research into New Media Technologies*, 26(1), pp. 157–177. Available at: https://doi.org/10.1177/1354856518772030

Nasrabadi, A.T., Samiei, A., Mahzari, A., McMahan, R., Prakash, R., Farias, M.C.Q. and Carvalho, M. M. (2019) 'A taxonomy and dataset for 360° videos', *Proceedings of the 10th ACM Multimedia Systems Conference*. New York, NY, USA: ACM, pp. 273–278. Available at: https://doi.org/10.1145/3304109.3325812

National Parks and Access to the Countryside Act 1949 c 97 (Regnal.12-13 and 14) Available from: https://www.legislation.gov.uk/ukpga/Geo6/12-13-14/97/part/IV/crossheading/creation-of-new-public-rights-of-way/enacted (Accessed 23 July 2023).

National Trust (no date) *Strategic Framework for Research*. Available at: https://www.nationaltrust.org.uk/who-we-are/research/strategic-framework-for-research. (Accessed 18 January 2022).

National Trust (no date) *Changing chalk partnership*. [Webpage] Available at: https://www.nationaltrust.org.uk/visit/sussex/the-changing-chalk-partnership (Accessed: 18 April 2024).

National Trust (no date) *Changing Chalk Area Action Plan* (Internal Document). Provided by the National Trust.

National Trust (2015) *Playing our part: What does the nation need from the National Trust in the 21st century?* Available at: https://www.readkong.com/page/playing-our-part-nationaltrust-7850492 (Accessed 18 January 2022).

National Trust (2016) *How to spot a Capability Brown landscape* [Video online]. Available at: https://youtu.be/md2_2v7D6Oc (Accessed 17 July 2023)

National Trust (2017) *Research Strategy 2017-2021* (Internal Document). Provided by the National Trust.

National Trust (2019) *Research Opportunities 2019-2021* (Internal Document). Provided by the National Trust.

National Trust (2020) *575 Wandsworth Road: Virtual Tour Trials 2020 - Unlocking the Potential of Digital Technologies to Improve Access* (Report). Provided by the National Trust.

NCVO. (2024). *National Inclusion Week: Why the language we use matters in building belonging.* Available at: https://www.ncvo.org.uk/news-and-insights/news-index/why-language-matters-in-building-belonging/ (Accessed 8 July 2025).

Nielsen, J.K. (2017) 'Museum communication and storytelling: articulating understandings within the museum structure', *Museum Management and Curatorship*, 32(5), pp. 440–455. Available at: https://doi.org/10.1080/09647775.2017.1284019

O'Hagan, L.A. and Serafinelli, E. (2022) 'Transhistoricizing the Drone: A Comparative Visual Social Semiotic Analysis of Pigeon and Domestic Drone Photography', *Photography and Culture*, 15(4), pp. 327–351. Available at: https://doi.org/10.1080/17514517.2022.2116899

O’Neill, M. (2018) 'Walking, well-being and community: racialized mothers building cultural citizenship using participatory arts and participatory action research', *Ethnic and Racial Studies*, 41(1), pp. 73–97. Available at: https://doi.org/10.1080/01419870.2017.1313439

O'Neill, M., and Hubbard, P. (2010) 'Walking sensing, belonging: ethno-mimesis as performative praxis', *Visual Studies*, *25*, 46–58.

Painter, N.I. (2016) 'Slavery: A Dehumanizing Institution', *Oxford University Press Blog (OUPblog)*. 14th February, Available at: https://blog.oup.com/2006/02/slavery_a_dehum/ (Accessed: 24 July 2023).

Panek, J. and Neték, R. (2019)' Collaborative Mapping and Digital Participation: A Tool for Local Empowerment in Developing Countries', *Information*, 10 (8), 255. Available at: https://doi.org/10.3390/info10080255

Paneque-Gálvez, J., McCall, M.K., Napoletano, B.M. and Wich, S. (2014) 'Small drones for community-based forest monitoring: An assessment of their feasibility and potential in tropical areas', *Forests*, 5(6), pp. 1481–1507. Available at: https://doi.org/10.3390/f5061481

Paneque-Gálvez, J., Pérez-Llorente, I., Luz, A. C., Guèze, M., Mas, J., Macía, M. J., Orta-Martínez, M., and Reyes-García, V. (2017) 'Grassroots Innovation Using Drones for Indigenous Mapping and Monitoring', *Land*, 6(4), p. 86. Available at: https://doi.org/10.3390/land6040086

Paneque-Gálvez, J., Pérez-Llorente, I., Luz, A.C. and Reyes-Garcia, V. (2018)' High overlap between traditional ecological knowledge and forest conservation found in the Bolivian Amazon', *Ambio*, 47(8), pp. 908–923. Available at: https://doi.org/10.1007/s13280-018-1040-

Pauwels, L. (2010) 'Visual Sociology Reframed: An Analytical Synthesis and Discussion of Visual Methods in Social and Cultural Research', *Sociological Methods & Research*, 38(4), pp. 545–581. Available at: https://doi.org/10.1177/0049124110366233

Pechman, A. (2013) 'The Art Worlds Game of Drones', *Artnews.com*. Available at: https://www.artnews.com/art-news/news/drones-as-art-2343/ (Accessed: 28 February 2020).

Perkins, D.D. and Zimmerman, M.A. (1995) 'Empowerment theory, research, and application', *American Journal of Community Psychology*, 23(5), pp. 569–579. Available at: https://doi.org/10.1007/BF02506982

Peters, J. (2001) 'Witnessing', *Media Culture and Society*, 23(6), 707–723. Available at: https://doi.org/10.1177/016344301023006002

Pink, S. (2011) *Doing Visual Ethnography,* Sage Research Methods, Sage Publications limited. Available at: https://dx.doi.org/10.4135/9780857025029

Pink, S. and Morgan, J. (2013) 'Short-term ethnography: Intense routes to knowing', *Symbolic Interaction*, 36(3), pp. 351–361. Available at: https://doi.org/10.1002/symb.66

Pink, S., Hubbard, P., O'Neill, M. and Radley, A. (2010) 'Walking across disciplines: from ethnography to arts practice' *Visual Studies*, 25 (1), 1-7. Available at: https://doi.org/10.1080/14725861003606670

Pinkett, R. and O'Bryant, R. (2003) 'Building Community, Empowerment and Self-sufficiency', *Information, Communication & Society*, pp. 187–210. Available at: https://doi.org/10.1080/1369118032000093888

Ponto, J. (2015) 'Understanding and Evaluating Survey Research', *Journal of Advanced Practical Oncology*, 6(2), pp. 168–171. Available at: https://www.researchgate.net/publication/286445115_Understanding_and_Evaluating_Survey_Research (Accessed 11 April 2024).

'Post Apocalyptic Towers of Power' (Episode 2), 31 March 2015. *Drones in Forbidden Zones,* 2015, [TV series]. Available at: All 4 (Accessed 12 August 2023).

Pratt, A.C. (2020) 'COVID – 19 impacts cities, cultures and societies', *City, Culture and Society*. Elsevier Ltd, p. 100341. Available at: https://doi.org/10.1016/j.ccs.2020.100341

Probst, P.C., Rothe, S. and Hussmann, H. (2021) 'Camera Distances and Shot Sizes in Cinematic Virtual Reality', *ACM International Conference on Interactive Media Experiences*. New York, NY, USA: ACM, pp. 178–186. Available at: https://doi.org/10.1145/3452918.3458804

Proulx, M., Ross, L., Macdonald, C., Fitzsimmons and Smit, M. (2021) 'Indigenous Traditional Ecological Knowledge and Ocean Observing: A Review of Successful Partnerships', *Frontiers in Marine Science*, 8. Available at: https://doi.org/10.3389/fmars.2021.703938

Radjawali, I. and Pye, O. (2017) 'Drones for justice: inclusive technology and river-related action research along the Kapuas', *Geographica Helvetica*, 72(1), pp. 17–27. Available at: https://doi.org/10.5194/gh-72-17-2017

Richardson, M. (2018) 'Drones eyeview: affective witnessing and technicities of perception', T. Schankweiler, K, Straub, V, and Wendl (ed.) *Image Testimonies: Witnessing in Times of Social Media.* New York: Routledge.

Robinson, J.S. and McClelland, A. G. (2020) 'Troubling Places: Walking the "troubling remnants" of post conflict space', *Area,* 52 (3), 654-662. Available at: https://doi.org/10.1111/area.12616

Rodchenko., A (1928) *At the Telephone* [Gelatin silver print photograph]. Available at; https://www.khanacademy.org./humanities/art-1010/cubism-early-abstraction/photography-early-20th-century/alexander-rodchenko-at-the-telephone (Accessed: 19 July 2025)

Rose, G. (2016) *Visual Methodologies. An introduction to researching with visual materials*. 4th edn. SAGE Publications Ltd.

Royo-Vela, M. and Black, M. (2020) 'Drone images versus terrain images in advertisements: Images verticality effects and the mediating role of mental simulation on attitude towards the advertisement', *Journal of Marketing Communications*, 26(1), pp. 21–39. Available at: https://doi.org/10.1080/13527266.2018.1425896

Ryan, M. and Lenos, M. (2020). *An Introduction to Film Analysis: Technique and Meaning in Narrative Film* (2nd ed.). Bloomsbury Publishing.

Saldanha, F. P., Pozzebon, M., Mailhot, C., and Le Puil, D. (2023) 'Counter-Narratives Mobilized by Deprived Communities Through Theatre Interventions: Deconstructing and Reframing Master Narratives', *Management Communication Quarterly* 37(3), 478–507. Available at: https://journals.sagepub.com/doi/full/10.1177/08933189221121345

Sandell, R. (1998) 'Museums as agents of social inclusion', *Museum Management and Curatorship*, 17(4), pp. 401–418. Available at: https://doi.org/10.1016/s0260-4779(99)00037-0

Sanders, E.B. (2000) 'Generative Tools for Co-designing', *Collaborative Design*. London: Springer London, pp. 3–12. Available at: https://doi.org/10.1007/978-1-4471-0779-8_1

Sandvik, K. and Jumbert, M.G. (2017) *The Good Drone*. London: Routledge. Available at: https://www.taylorfrancis.com/books/edit/10.4324/9781315553405/good-drone-mariajumbert-kristin-sandvik (Accessed: 12 February 2020).

Savardi, M., Kovacs, A., Signoroni, A., Benini, S. (2023) 'Recognition of camera angle and level in movies', *ACM International Conference on Interactive Media Experiences Workshops. (IMXw 23). June 12-15, 2023*. Nantes: ACM, pp. 79–85. Available at: https://doi.org/https://doi.org/10.1145/3604321.3604334

Schendan, H.E. and Ganis, G. (2012) 'Electrophysiological Potentials Reveal Cortical Mechanisms for Mental Imagery, Mental Simulation, and Grounded (Embodied) Cognition' *Frontiers in Psychology*, 3. Available at: https://doi.org/10.3389/fpsyg.2012.00329

Seb the Nomad (2023) ‘Lisbon/Almada (Santuário de Cristo Rei)’, *Drone Shots – Portugal (4k Travel Videos.* 3 February. Available at: https://m.youtube.com/watch?v=Fb8tCW7Bp3U (Accessed 18 June 2023).

Seligmann, L.J. and Estes, B.P. (2020) 'Innovations in Ethnographic Methods', *American Behavioral Scientist*, 64(2), pp. 176–197. Available at: https://doi.org/10.1177/0002764219859640

Serafinelli, E. and O’Hagan, L. (2021) 'Views from the blue: a glimpse in drone photography', *Pint of Science*. White Rose University Consortium. Available at: https://eprints.whiterose.ac.uk/173952/3/Pint (Accessed: 15 July 2023).

Serafinelli, E., and O’Hagan, L.A. (2022) 'Drone views: a multimodal ethnographic perspective', *Visual Communication*, 23(2), pp. 223–243. Available at: https://journals.sagepub.com/doi/full/10.1177/14703572211065093 (Accessed: 17 February 2022).

Serafinelli, E. and O'Hagan, L.A. (2024) 'Drone views: a multimodal ethnographic perspective', *Visual Communication*, 23(2), pp. 223–243. Available at: https://doi.org/10.1177/14703572211065093

Shuman, A. (2015) 'Story Ownership and Entitlement' in De Fina, A. & Georgakopoulou, A. (eds.) *The Handbook of Narrative Analysis*, Available at: https://doi.org/10.1002/9781118458204.ch2

Silberman, N.A. (2005): *Beyond Theme Parks and Digitized Data: What Can Cultural Heritage Technologies Contribute to the Public Understanding of the Past?* Available at: https://works.bepress.com/neil_silberman/39/ (Accessed 17 July 2020)

Silverman and Patterson, (2014), *Qualitative Research Methods for Community Development*, New York, Routledge, 138pp.

Smethurst, T. and Craps, S. (2019) 'Towards an Understanding of Drone Fiction', *Journal of War & Culture Studies*, 12(1), pp. 85–102. Available at: https://doi.org/10.1080/17526272.2018.1514759

South Downs National Park Authority (2016) *South Downs National Park Authority Visitor Survey 2015: Final Report* (Report). Provided by the SDNPA.

South Downs National Park Authority (2017) *Briefing Paper on Drones* (Internal Document). Provided by the SDNPA.

South Downs National Park Authority (2018) *Learning and Outreach- strategic review 2018-2023*. (Internal document) Midhurst, West Sussex: SDNPA. Provided by the SDNPA.

South Downs National Park Authority (2019) *2018 Visitor Survey. Final Report.* (Internal document) Midhurst, West Sussex: SDNPA. Provided by the SDNPA.

South Downs National Park Authority (2020) *Communications and Engagement Strategy 2021-2025* (Internal document) Midhurst, West Sussex: SDNPA. Provided by the SDNPA.

South Downs National Park Authority (2020) *The South Downs by air* [Video online] Available at: https://youtu.be/Zm02qlzvAdw. (Accessed 12 May 2023).

South Downs National Park (2021) *Wild Chalk Festival – Dewponds of the South Downs.* 26 August. [Video online]. Available at: https://www.youtube.com/watch?v=mVSkrdJO3yg. (Accessed 15 June 2023).

South Downs National Park (2022) *Shoreham Cement Works – 360° Views.* 16 May. [Video online] Available at: https://youtu.be/ALPJOS_i-d8 (Accessed 12 April 2023).

South Downs National Park (2025) *In their footsteps* (version 9.0.381-prod) [mobile app] Available at: Google Play and Apple Play (Downloaded: 5 June 2023).

Sparkes, A.C. (2020) 'Autoethnography: accept, revise, reject? An evaluative self reflects', *Qualitative Research in Sport, Exercise and Health*, 12(2), pp. 289–302. Available at: https://doi.org/10.1080/2159676X.2020.1732453

Springgay, S. and Truman, S.E. (2017) *Walking methodologies in a more-than-human world: WalkingLab*. London: Routledge. Available at: https://doi.org/10.4324/9781315231914

Stahl, R. (2013) 'What the drone saw: the cultural optics of the unmanned war', *Australian Journal of International Affairs*, 67(5), pp. 659–674. Available at: https://doi.org/10.1080/10357718.2013.817526

Stankov, U., Kennell, J., Morrison, A. M. and Vujičić, M. D. (2019). 'The view from above: the relevance of shared aerial drone videos for destination marketing', *Journal of Travel and Tourism Marketing*, 36(7), 808–822. Available at: https://doi.org/10.1080/10548408.2019.1575787

Stauber, B. and Parreira do Amaral, M. (2015) 'Access to and Accessibility of Education: An Analytic and Conceptual Approach to a Multidimensional Issue', *European Education*, 47(1), pp. 11–25. Available at: https://doi.org/10.1080/10564934.2015.1001254

Stek, T.D. (2016) 'Drones over Mediterranean landscapes. The potential of small UAVs (drones) for site detection and heritage management in archaeological survey projects: A case study from Le Pianelle in the Tappino Valley, Molise (Italy)', *Journal of Cultural Heritage*, 22, pp. 1066–1071. Available at: https://doi.org/10.1016/j.culher.2016.06.006

StoryVerse Collective (2023) 'Still I rise', *Maya Angelou|Live and Unplugged*. 27 June. [Video online]. Available at: https://m.youtube.com/watch?v=LyxTcyFGfxM (Accessed: 7 August 2025).

Svensson, P. (2016) 'Making Digital Humanities' in *Big Digital Humanities: Imagining a Meeting Place for the Humanities and the Digital*. University of Michigan Press, pp. 172–221. Available at: https://doi.org/10.2307/j.ctv65sx0t.9

Svensson, J. (2018) 'My history, your history, our history: Developing meaningful community engagement within historic sites and museums', *Queensland Review*, 25(2), pp. 300–321. Available at: https://doi.org/10.1017/qre.2018.29

Szafir, D., Mutlu, B. and Fong, T. (2017) 'Designing planning and control interfaces to support user collaboration with flying robots', *The International Journal of Robotics Research*, 36(7), pp. 514–542. Available at: https://doi.org/10.1177/0278364916688256

Tan, H., Lee, J. and Gao, G. (2018) 'Human-Drone Interaction', *Proceedings of the 19th International ACM SIGACCESS Conference on Computers and Accessibility - DIS 18*. New York, New York, USA: ACM Press, pp. 183–187. Available at: https://doi.org/10.1145/3197391.3205433

TheDmel (no date) *TheDmel* [YouTube channel]. Available at: https://www.youtube.com/user/TheDmel?feature=watch (Accessed: 10 July 2020).

The Killing of a Sacred Deer, 2017, [Film] Directed by Y. Lanthimos, Ireland, United Kingdom, United States, A24.

The National Archives. (2023). *Archives for Everyone*. Available at: https://www.nationalarchives.gov.uk/aboutourroleplans-policies-performce-and-projects/our-plans/archives-for-everyone/ (Accessed 19 July 2025).

The Wolf of Wall Street, 2013, [Film] Directed by M. Scorsese, United States, Paramount

Toth, E. R., Guttman, S., Kimura, A., and Quinn, T. (2016) 'Expanding the boundaries of museum studies: Popular education through engagement with hidden histories of organising and activism', *Journal of Adult and Continuing Education*, 22(2), pp. 199–215. Available at: https://doi.org/10.1177/1477971416672326

Tremlett, S. (2021). *The poetics of Poetry film; Film poetry, Videopoetry, Lyric Voice, Reflection, Intellect.* Bristol, UK and Chicago USA p.8 Available at: https://doi.org./10.2307/j.ctv36xw4c0 (Accessed on 02 June 2025).

Tzima, S., Styliaras, G., Bassounas, A., and Tzima, M. (2020). (2020) 'Harnessing the potential of storytelling and mobile technology in intangible cultural heritage: A case study in early childhood education in sustainability', *Sustainability (Switzerland)*, 12(22), pp. 1–22. Available at: https://doi.org/10.3390/su12229416

UNESCO, Institute for Statistics (2009) *UNESCO Framework for Cultural Statistics*. Available at: https://uis.unesco.orgenglossary-termcultural heritage (Accessed 14 July 2025).

University of Dundee (2022) *The protected characteristics explained* [Webpage]. Available at: https://www.dundee.ac.uk/guides/protected-characteristics-explained (Accessed: 28 November 2022).

Uzunogullari, S. (2023) 'Armchair travel through video games: stories from elsewheres and elsewhens', *Leisure Studies*, 42(1), pp. 133–146. Available at: https://doi.org/10.1080/02614367.2022.2153904

Vannini, P. and Vannini, A. (2008) 'Of Walking Shoes, Boats, Golf Carts, Bicycles, and a Slow Technoculture', *Qualitative Inquiry*, 14(7), pp. 1272–1301. Available at: https://doi.org/10.1177/1077800408322708

van Wee, B. (2022) 'Accessibility and equity: A conceptual framework and research agenda', *Journal of Transport Geography*, 104, p. 103421. Available at: https://doi.org/10.1016/j.jtrangeo.2022.103421

Vargas-Ramírez, N. and Paneque-Gálvez, J. (2019) 'The Global Emergence of Community Drones (2012–2017)', *Drones*, 3(4), p. 76. Available at: https://doi.org/10.3390/drones3040076

Vindrola-Padros, C. (2020) *Rapid ethnographies: A practical guide.* Cambridge, UK: University of Cambridge Press.

Wallace-Wells, B. (2014, October) 'Drones and everything after', *New York Magazine*. Available at: https://nymag.com/intelligencer/2014/10/drones-the-next-smartphone.html

Watermeyer, B. (2013). *Towards a contextual psychology of disablism*. Routledge.

Watts, A.C., Ambrosia, V.G. and Hinkley, E.A. (2012) 'Unmanned aircraft systems in remote sensing and scientific research: Classification and considerations of use', *Remote Sensing*, 4(6), pp. 1671–1692. Available at: https://doi.org/10.3390/rs4061671

Whitehead, T. L. (2005). *Basic Classical Ethnographic Research Methods* (Ethnographically Informed Community and Cultural Assessment and Cultural Assessment Research Systems (EICCARS) Working Paper Series). University of Maryland, CEHC.

Wich, S.A. and Koh, L.P. (2018) *Conservation Drones: Mapping and Monitoring Biodiversity*. Oxford University Press. Available at: https://doi.org/10.1093/oso/9780198787617.001.0001

WomenWhoDrone and elenabuenrostro (2023) *Happy International Woman's Day!* [Instagram Reel] Available at: https://www.instagram.com/reel/Cpi2vpesCCz/ (Accessed: 25 July 2023).

Woolf, V. (1953) Edited by Leonard Woolf, *A Writer's Diary. Being Extracts from the Diary of Virginia Woolf*. Available at: https://share.google/kzg37z6L8Mc5hHFol (Accessed 9 August 2025).

Yan, Z. (2020) 'Unprecedented pandemic, unprecedented shift, and unprecedented opportunity', *Human Behavior and Emerging Technologies*, 2(2), pp. 110–112. Available at: https://doi.org/10.1002/hbe2.192

Yilmaz, M. B., Lotman, E., Karjus, A., and Tikka, P. (2023) 'An embodiment of the cinematographer: emotional and perceptual responses to different camera movement techniques', *Frontiers in Neuroscience* (Vol. 17). Available at: https://doi.org/10.3389/fnins.2023.1160843

Yu, Z. (2023) 'Shot Size Narration in the Mobile Media Era: Characteristics and Evolution', *Journal of Research in Social Science and Humanities*, 2 (11). Available at: https://doi.org/10.56397/JRSSH.2023.11.04

Zimmerman, M.A. (1995) 'Psychological empowerment: Issues and illustrations', *American Journal of Community Psychology*, 23(5), pp. 581–599. Available at: https://doi.org/10.1007/BF02506983

Zúñiga, X and Adams, M. (2016) 'Core concepts for Social Justice Education', *Teaching for Diversity and Social Justice*, by Adams, M. *et. al.* (ed.) London: Routledge, pp. 95–130.

Bibliography

Abbas, R. and Michael, K. (2023) 'Socio-Technical Theory: A review', in Papagiannidis, S. (ed.), *TheoryHub* Book. Available at: https://open.ncl.ac.uk/theory-library/socio-technicaltheory.pdf ISBN: 9781739604400 (Accessed 10 February 2025).

Abd el Salam, H., McCormack, D. and Martin, B. (2017) 'Connecting People with Their Museum', *Journal of Museum Education*, 42(4), pp. 385–395. Available at: https://doi.org/10.1080/10_598650.2017.1371521

Abtahi, P., Zhao, D.Y., L.E., J and Landay, J.A. (2017) *Drone Near Me: Exploring Touch-Based Human-Drone Interaction*. Available at: https://doi.org/10.1145/3130899

Achterbergh, I. (2016). *Arts & culture memberships as contemporary gatekeepers of cultural goods: selection and symbolic production practices*. Master of Arts, Erasmus University, Rotterdam. Available at: https://www.semanticscholar.org/paper/Arts-%26-culture-memberships-as-contemporary-of-and-Achterbergh/b0c4403cfe9d4a4e777c87a70a80c324e81674a5 (Accessed 29 October 2024).

Acorn Tourism Consulting Ltd and Natural Values (2012) *South Downs National Park Visitor Survey 2012: Environment Element* (Report). Provided by the SDNPA.

Adagio Dance School (2021) *Road to Freedom!* 16 September [Video online]. Available at: https://www.facebook.com/adagiodance.bvi/videos/4006655662748813 (Accessed 15 August 2023).

Adams, W. (2015) 'Conducting semi-structured interviews', in *Handbook of Practical Program Evaluation*. Wiley, pp. 492–505. Available at: https://doi.org/10.1002/9781119171386.ch19

Adams, M. and Bell, L.A. (ed.)(2016) *Teaching for Diversity and Social Justice*. Routledge. Available at: https://doi.org/10.4324/9781315775852

Adams, T. E., Jones, S. H and Ellis, C. (2015). *Autoethnography*. Oxford University Press.

Adams, T.E., Jones, S.H. and Ellis, C. (2021) *Handbook of Autoethnography*. New York: Routledge. Available at: https://doi.org/10.4324/9780429431760

Adeyoye-Olatunde, O. A. and Olenik, N., (2021) 'Research and Scholarly Methods', *Journal of the American College of Clinical Pharmacy*, 4 (10), pp. 1358-1367. Available at: https://doi.org/10.1002/jac5.1441

Aerial View of Monument to the discoveries, Lisbon, Portugal, istock-875673258.mov, Available at: https://www.videvo.net/video/aerial-view-of-monument-to-the-discoveries-lisbon-portugal/864470/ (Accessed 31 June 2023).

Agius, C. (2017) 'Ordering without bordering: Drones, the unbordering of late modern warfare and ontological insecurity', *Postcolonial Studies*, 20(3), pp. 370–386. Available at: https://doi.org/10.1080/13688790.2017.1378084

Agostinho, D., Maurer, K. and Veel, K. (2020) 'Introduction to The Sensorial Experience of the Drone', *The Senses and Society*, 15(3), pp. 251–258. Available at: https://doi.org/10.1080/17458927.2020.1820195

Ahmed, A., Kaushal, A., Wang, K., and Kobuse, Y. (2020) 'Why is there a higher rate of impostor syndrome among BIPOC?', *Across The Spectrum of Socioeconomics*, 1(2). Available at: https://doi.org/https://doi.org/10.5281/zenodo.4310477

Alda, K. (2017) 'Walking as an Ethnographic Method: Sensory Memories and Intimacy of Place in a Post-socialist Czech Neighbourhood (breakout presentation)', *Journal of Transport & Health*, 7, pp. S74–S75. Available at: https://doi.org/10.1016/j.jth.2017.11.121

Alexander, E. P. (1979) *Museums in Motion: an Introduction to the History and Functions of Museums* (First). Nashville, Tenn.: American Association for State and Local History.

Alexander, Edward P., Mary Alexander, J.D. (2017) *Museums in Motion: An Introduction to the History and Functions of Museums*. Third Edition. Edited by M. Alexander and J. Decker. Lanham: Rowman & Littlefield Publishers.

Alison Farmer Associates (2018) *Drone Flying in the South Downs National Park* (Issues Paper). Provided by the SDNPA.

Alzheimer's Society (no date) *Ideas to help a person with dementia enjoy nature*. [Webpage]Available at: https://www.alzheimers.org.uk/get-support/publications-and-factsheets/dementia-together/ideas-enjoy-nature-dementia (Accessed: 28 July 2023).

Ambrosia, V. G., Wegener, S., Zajkowski, T., Sullivan, D. V., Buechel, S., Enomoto, F., Lobitz, B., Johan, S., Brass, J., and Hinkley, E. (2011) 'The Ikhana unmanned airborne system (UAS) western states fire imaging missions: From concept to reality (2006-2010)', *Geocarto International*, 26(2), 85–101. https://doi.org/10.1080/10106049.2010.539302

Amoroso, L. (2023) 'Walking as Knowing, Healing, and the (Re)making of Self', Goode, J., Lumsden, K. and Bradford, J. (eds.) *Crafting Autoethnography Processes and Practices of Making Self and Culture*. Routledge. Available at: https://doi-org.libproxy.ucl.ac.uk/10.4324/9781003309239

Anderson, L. (2006) 'Analytic Autoethnography', *Journal of Contemporary Ethnography*, 35(4), pp. 373–395. Available at: https://doi.org/10.1177/0891241605280449

Anderson, A. and Mileham, M.A. (2020) 'Welcome to the Museum: Reflecting on Representation and Inclusion in Museum Evaluation', *Curator: The Museum Journal*, 63(4), pp. 597–603. Available at: https://doi.org/10.1111/cura.12382

Anderson, K. and Gaston, K.J. (2013) ‘Lightweight unmanned aerial vehicles will revolutionize spatial ecology’, *Frontiers in Ecology and the Environment*, 11(3), pp. 138–146. Available at: https://doi.org/10.1890/120150

Anderson, A. M., Brading, L., Swaithes, L., Evans, N., Fedorowicz, S. E., Murinas, D., Atkinson, E., Moult, A., Yip, T., Ayub, P., Dziedzic, K., and Conaghan, P. G. (2024). ‘Building trust and inclusion with under - served groups: a public involvement project employing a knowledge mobilisation approach’, *Research Involvement and Engagement* Nov 11; 10(1). Available at: https://pubmed.ncbi.nlm.nih.gov/39529198/

Angelou, M. (1994) ‘Still I rise’, *The complete collected poems of Maya Angelou,* Random House, New York, pp. 273.

Angliss, R.P., Ferguson, M.C., Hall, P., Helker, V., Kennedy, A. and Sformo, T. (2018) ‘Comparing manned to unmanned aerial surveys for cetacean monitoring in the Arctic: methods and operational results’, *Journal of Unmanned Vehicle Systems*, 6(3), pp. 109–127. Available at: https://doi.org/10.1139/juvs-2018-0001

Appel, L., Ali, S., Narag, T., Mozeson, K., Pasat, Z., Orchanian-Cheff, A. and Campos, J. (2021) ‘Virtual reality to promote wellbeing in persons with dementia: A scoping review’, *Journal of Rehabilitation and Assistive Technologies Engineering*, 8, p. 205566832110539. Available at: https://doi.org/10.1177/20556683211053952

Aran, E. and Reed, P. (2015) ‘Talking points collaborative mapping-Forest Service Public Participation GIS Application’, *Esri User Conference*. United States Forestry Service. Available at: https://proceedings.esri.com/library/userconf/proc15/papers/890_503.pdf (Accessed 25 March 2023).

Archer, Zach, Simpson, A. (2017) ‘Combining autoethnography and multimodal social semiotics: potentials for theory and method’, *Social Semiotics, 27:5, 656-670,* 27(5), pp. 656–670. Available at: https://doi.org/10.1080/10350330.2016.1264689

Arts Council England (no date) *Inspiring Learning - detailed framework*. Available at: https:www.artscouncil.org.uk/sites/default/files/S2D12_Detailed_framework.pdf (Accessed 25 March 2024).

Arts Council England and Shared Intelligence (2017) *Stand by me, The contribution of public libraries to the well-being of older people.* (Report) Available at: https://www.artscouncil.org.uk/sites/default/files/download-file/Combined%20older%20people%20report%2017%20July.pdf

Arts Council England (2020*) Let's Create (Strategy 2020-30).* Available at: https://www.artscouncil.org.uk/lets-create/strategy-2020-2030/strategy (Accessed: 2 December 2022).

Ashmolean Museum Oxford (no date) *History of the Ashmolean*. Available at: https://www.ashmolean.org/history-ashmolean (Accessed: 20 November 2022).

Ashuri, T., Pinchevski, A. (2009). Witnessing as a Field. In: Frosh, P., Pinchevski, A. (eds.) *Media Witnessing*. Palgrave Macmillan, London. Available at: https://doi.org/10.1057/9780230235762_7

Assor, N. (2014) *Lessons of leaving your body*. United States. Available at: https://youtu.be/K1ZDX5Z78NE

Attitude Magazine (2022) *Bimini and Dan O'Neill on healing the ocean as LGBTQ people.* 20 April. [Video online]. Available at: https://www.youtube.com/watch?v=5kD-zs4aFeA. (Accessed 25 July 2023).

Augenbraun, E. (2016) 'Making Art with Drones', *Scientific American Blog Network*', *Scientific American*, 14th January. Available at: https://blogs.scientificamerican.com/observations/making-art-withdrones/ (Accessed: 28 February 2020).

Avila Soto, M., Funk, M., Hoppe, M., Boldt, R., Wolf, K. and Henze, N., (2017) 'DroneNavigator: Using Leashed and Free-Floating Quadcopters to Navigate Visually Impaired Travelers', *Proceedings of the 19th International ACM SIGACCESS Conference on Computers and Accessibility*. New York, NY, USA: ACM, pp. 300–304. Available at: https://doi.org/10.1145/3132525.3132556

Ball, H. (2019) 'Conducting Online Surveys', *Journal of Human Lactation*, 35(3), pp. 413–417. Available at: https://doi.org/10.1177/0890334419848

Barber, J.F. (2013) 'Walking-talking Soundscapes', *Flâneurs*, and the Creation of Mobile Media Narratives. J. Farman (ed.) *The Mobile Story: Narrative Practices with Locative Technologies*. New York: Routledge.

Barbour, R. (2007) *Doing Focus Groups*. London: SAGE Publications. Available at: http://methods.sagepub.com.libproxy.ucl.ac.uk/book/doing-focus-groups

Bard College (no date) *Other Research* [Webpage]. Available at: https://dronecenter.bard.edu/projects/other-projects/ (Accessed: 30 April 2020).

Bard College (2017) *Portfolio: Dronescapes by Kathryn Brimblecombe-Fox*. Available at: https://dronecenter.bard.edu/portfolio-dronescapes-by-kathryn-brimblecombe-fox/ (Accessed: 28 February 2020).

Barr, A. (2013) *Amazon testing delivery by drone, CEO Bezos says*. *USA Today*. Available at: https://eu.usatoday.com/story/tech/2013/12/01/amazon-bezos-drone-delivery/3799021/ (Accessed: 10 July 2020).

Barsalou, L.W. (2009) 'Simulation, situated conceptualization, and prediction', *Philosophical Transactions of the Royal Society B: Biological Sciences*, 364(1521), pp. 1281–1289. Available at: https://doi.org/10.1098/rstb.2008.0319

Basu, R. (no date) *Drones of New York*. [Webpage]. Available at: https://www.rajeevbasu.com/projects/dronesofnewyork (Accessed: 17 July 2022).

Bazazzadeh, H. (2020) 'Truth of sincerity and authenticity or lie of reconstruction. Which do the visitors of cultural heritage trust?' in J. Battaino at al., (eds.) *Defining the architectural space-the truth and lie of architecture*, pp. 7–18. Available at: https://doi.org/10.23817/2020.defarch.9-1

Bazeley, P. (2020) *Qualitative Data Analysis- Practical Strategies*. Western Sydney University, Australia.

Becker, H.S. and Richards, P. (2008) *Writing for Social Scientists.* Third edition. Chicago: University of Chicago Press.

Bekele, M.K., Pierdica, M.K., Frontoni, E, Savina Malinverni, E. and Gain, J., (2018) 'A Survey of Augmented, Virtual, and Mixed Reality for Cultural Heritage', *Journal on Computing and Cultural Heritage*, 11(2), pp. 1–36. Available at: https://doi.org/10.1145/3145534

Bell, F., Fletcher, G., Greenhill, A., Griffiths, M. and McLean, R. (2013) 'Science fiction prototypes: Visionary technology narratives between futures', *Futures*, 50, pp. 15–24. Available at: https://doi.org/10.1016/j.futures.2013.04.004

Bell, L.A. (2022) 'Theoretical Foundations for Social Justice Education', *Teaching for Diversity and Social Justice*. New York: Routledge, pp. 3–25. Available at: https://doi.org/10.4324/9781003005759-2

Bell, P., Lewenstein, B., Shouse, A. and Feder., A (2009) *Learning Science in Informal Environments: People, Places, and Pursuits*. Edited by C. on L.S. in I. Environments and N.R. Council. National Academies Press. Available at: http://makepuppet.org/after_school/learningScience_inInforalSetting.pdf.

Belkouri, D. and Laing, R. (2024) 'Walking through the abstract[ed] city and co-creating urban space', *Applied Mobilities*, 9(1), pp. 63–83. Available at: https://doi.org/10.1080/23800127.2024.2309742

Ben-Eliyahu, A., Moore, D., Dorph, R. and Schunn, C. D. (2018) 'Investigating the multidimensionality of engagement: Affective, behavioral, and cognitive engagement across science activities and contexts', *Contemporary Educational Psychology*, 53, pp. 87–105. Available at: https://doi.org/10.1016/j.cedpsych.2018.01.002

Benford, S., Greenhalg, C., Hopkins, G. and Brown, C. (1998) 'Understanding and constructing shared spaces with mixed-reality boundaries', *ACM Transactions on Computer-Human Interaction*, 5(3), pp. 185–223. Available at: https://doi.org/10.1145/292834.292836

Benita, F., Bansal, G. and Tunçer, B. (2019) 'Public spaces and happiness: Evidence from a large-scale field experiment', *Health & Place*, 56, pp. 9–18. Available at: https://doi.org/10.1016/j.healthplace.2019.01.014

Berghs, M. (2017) 'Practices and discourses of ubuntu: Implications for an African model of disability?', *African Journal of Disability*, 6. Available at: https://doi.org/10.4102/ajod.v6i0.292

Benjamin, W. (2007) 'The Work of Art in the Age of Mechanical Reproduction', H. Arendt (ed.) *Illuminations: Essays and reflections*. New York: Schocken Books, pp. 217–251.

Berkowitz, R. (2014) 'Drones and the Question of "The Human", *Ethics & International Affairs*, 28(2), pp. 159–169. Available at: https://doi.org/10.1017/S0892679414000185

Bernarda, J., Ferreira, A.M., Silva, C. S. and Queiroz, I. (2017) 'Design as a process tool of collaborative and multidisciplinary learning in society', *The Design Journal*, 20(sup1), pp. S900–S914. Available at: https://doi.org/10.1080/14606925.2017.1353035

Berni, J.A.J., Zarco-Tejada, P. and Suarez, L. (2009) 'Thermal and narrowband multispectral remote sensing for vegetation monitoring from an unmanned aerial vehicle', *IEEE Transactions on Geoscience and Remote Sensing*, 47(3), pp. 722–738. Available at: https://doi.org/10.1109/TGRS.2008.2010457

Berry, D.M. (2012) *Understanding digital humanities*, *Understanding Digital Humanities*. Palgrave Macmillan. Available at: https://doi.org/10.1057/9780230371934

Bhalla, A. and Lapeyre, F. (1997) 'Social Exclusion: Towards an Analytical and Operational Framework', *Development and Change*, 28(3), pp. 413–433. Available at: https://doi.org/10.1111/1467-7660.00049

Birtchnell, T. and Gibson, C. (2015) 'Less talk more drone: social research with UAVs', *Journal of Geography in Higher Education*, 39(1), pp. 182–189. Available at: https://doi.org/10.1080/03098265.2014.1003799

Black, G. (2012) *Transforming Museums in the 21st Century*. Hoboken: Taylor and Francis.

Blight, L.K., Bertram, D.F. and Kroc, E. (2019) 'Evaluating UAV-based techniques to census an urban-nesting gull population on Canadas pacific coast', *Journal of Unmanned Vehicle Systems*, 7(4), pp. 312–324. Available at: https://doi.org/10.1139/juvs-2019-0005

Blumer, H. (1969) 'The Methodological Position of Symbolic Interactionism', *Symbolic Interactionism: Perspective and Method*. Los Angeles: University California Press, pp. 1–60.

Blumer, H. (1973) 'A Note on Symbolic Interactionism', *American Sociological Review*, 38(6), p. 797. Available at: https://doi.org/10.2307/2094141

Blythe, M. (2014) 'Research through design fiction', *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. New York, NY, USA: ACM, pp. 703–712. Available at: https://doi.org/10.1145/2556288.2557098

Bolter, J.D., MacIntyre, B., Gandy, M., and Schweitzer (2006) ‘New Media and the Permanent Crisis of Aura’, *Convergence: The International Journal of Research into New Media Technologies*, 12(1), pp. 21–39. Available at: https://doi.org/10.1177/1354856506061550

Bolton, M., Day, R. and Leach, M. (2019) ‘England’s overlooked neighbourhoods: Defining, understanding and regenerating “left behind” communities’, *Journal of Urban Regeneration and Renewal*, 13(2), pp. 116–123. Available at: https://hstalks.com/article/5392/englands-overlooked-neighbourhoods-defining-unders (Accessed: 14 March 2022).

Bourdieu, P. (1984) *Distinction. a Social Critique of the Judgement of Taste.* London: Routledge.

Bourgois, R. (no date) *The Power of Collective Voice: Culture, Identity, and Empowerment* (online). Villa Albertine, the French Institute for Culture and Education, a division of the French Embassy in the United States. Available at: https://villa-albertine.org/va/magazine/the-power-of-collective-voice-culture-identity-and-empowerment/

Bovaird, T. and Loeffler, E. (2012) ‘From Engagement to Co-production: The Contribution of Users and Communities to Outcomes and Public Value’, *VOLUNTAS: International Journal of Voluntary and Nonprofit Organizations*, 23(4), pp. 1119–1138. Available at: https://doi.org/10.1007/s11266-012-9309-6

Bowen, Christopher J., and R.T. (2017) *Grammar of the Shot*. Taylor & Francis Group Taylor & Francis Group ProQuest Ebook Central. Available at: https://doi.org/10.4324/9780080927053

Boylorn, R.M. (2008) ‘Lived experience’ in *The Sage encyclopedia of qualitative research methods*. Thousand Oaks: Sage.

Bracken-Roche, C. (2016) ‘Domestic drones: The politics of verticality and the surveillance industrial complex’, *Geographica Helvetica*, 71(3), pp. 167–172. Available at: https://doi.org/10.5194/gh-71-167-2016

Bratchford, G. and Zuev, D. (2020) ‘Aerial visibilities: towards a visual sociology of the sky’, *Visual Studies*, 35(5), pp. 402–416. Available at: https://doi.org/10.1080/1472586X.2020.1843283

Brennen, B.S. (2017) *Qualitative Research Methods for Media Studies*. Second edition. New York: Routledge, 2017. Available at: https://doi.org/10.4324/9781315435978

Brighenti, A.M. (2010) ‘Visibility and Social Theory’, *Visibility in Social Theory and Social Research*. London: Palgrave Macmillan UK, pp. 37–70. Available at: https://doi.org/10.1057/9780230282056_2

Brimblecombe, K. (no date). *Art @ Kathryn Brimblecombe-Fox: DRONESCAPES*. Available at; https://kathrynbrimblecombeart.blogspot.com/p/dronescapes.html (Accessed: 28 February 2020).

Brooks, M. (2012) ‘Welcome to the personal drone revolution’, *New Scientist*, 216(2894), pp. 42–45. Available at: https://doi.org/10.1016/S0262-4079(12)63142-X

Brown, B., (2013), *Daring greatly: how the courage to Be Vulnerable Transforms the Way We Live, Love, Parent and Lead*, Penguin Random House.

Brown, B. (2018). *Dare to lead: brave work, tough conversations, whole hearts*, Vermillion.

Buckingham, D. (2009) ‘Creative visual methods in media research: possibilities, problems and proposals’, *Media, Culture & Society*, 31(4), pp. 633–652. Available at: https://doi.org/10.1177/0163443709335280

Building Bridges Art Exchange. (2020) *Robotix: Intersections of Art & Technology*. Available at: https://www.buildingbridgesartexchange.org/robotix

Burke, V., Jørgensen, D. and Jørgensen, F.A. (2020) 'Museums at Home: Digital Initiatives in Response to COVID-19', *Norsk museumstidsskrift*, Vol.6, Iss.2. Available at: https://doi.org/10.18261/issn.2464-2525-2020-02-05

Burkett, I. (no date) *An Introduction to co-design*. University of New South Wales Centre for Social Impact (Knode). Available at: https://www.yacwa.org.au/wp-content/uploads/2016/09/An-Introduction-to-Co-Design-by-Ingrid-Burkett.pdf (Accessed 23 December 2023).

Burrows, J. (2004) 'Textual Analysis' in Schreibman, S. *et. al.* (eds.) *A Companion to Digital Humanities.* Blackwell Publishing Ltd, pp. 323–347. Available at: https://doi.org/10.1002/9780470999875.ch23

Butler, M., Derrett, S. (no date). 'The walking interview: an ethnographic approach to understanding disability', *Internet Journal of Allied Health Sci. Practice*, *12*(3), 1–8.

Butters, K. (2016) *Storytelling and dementia*. Available at: https://www.sfs.org.uk/articles/2016/09/storytelling-and-dementia (Accessed: 8 December 2022).

Cairngorms National Park Authority (no date) *Cairngorms' 2021 Visitor Survey Responses* (Dataset). Provided by the Cairngorms National Park Authority.

Cairngorms National Park Authority (2021) *Cairngorms National Park Visitor Survey* (report). Grantown on Spey, Scotland: CNPA.

Calçada Portuguesa (2023) *Calçada Portuguesa - Lisbon Stories.* 26 July [Video online]. Available at: https://www.youtube.com/watch?v=Ypcj_54bk7Q (Accessed 23 August 2023).

Caliandro, A. (2017) ‘Digital Methods for Ethnography: Analytical Concepts for Ethnographers Exploring Social Media Environments’, *Journal of Contemporary Ethnography*, p. 089124161770296. Available at: https://doi.org/10.1177/0891241617702960

Cameron, F. and Kenderdine, S. (eds.) (2007): *Theorizing Digital Cultural Heritage: a Critical Discourse*: The MIT Press.

Campbell, R. (2018) ‘Drone Film Theory: The Immanentisation of Kinocentrism’, *Media Theory*, 2(2), pp. 52–78. Available at: https://journalcontent.mediatheoryjournal.org/index.php/mt/article/view/57 (Accessed: 31 July 2023).

Campbell-Stephens, R. (2022) *Global Majority; Decolonising the language and Reframing the Conversation about Race*, p. 7. Available at: https://www.leedsbeckett.ac.uk/-/media/files/schools/school-of-education/final-leeds-beckett-1102-global-majority.pdf. (Accessed: 5 June 2024).

Carousel (no date) *My inner calmness* [Video online]. Available at: https://www.youtube.com/watch?v=XcBFGHmz0lI

Cascone, K. (2000) ‘The aesthetics of failure: “Post-digital” tendencies in contemporary computer music’, *Computer Music Journal*, 24(4), pp. 12–18. Available at: https://doi.org/10.1162/014892600559489

Cassidy, C.A., Fabola, A., Miller, A., Well, K., Urbina, S., Antas, M. and Cummins, A., (2018) ‘Digital Pathways in Community Museums’, *Museum International*, 70 (1–2), pp. 126–139. Available at: https://doi.org/10.1111/muse.12198

Castro, T. (2013) ‘Aerial Views and Cinematism, 1898–1939’ in Dorrian, M. and Pousin, F. (eds.) *Seeing from Above: The Aerial View in Visual Culture*, pp. 118–133. Available at: https://www.academia.edu/4090955/Aerial_Views_and_Cinematism_1898_1939

Cauchard, J.R., L.E., Jane, Zhai, K., and Landay, J. A. (2015) *Drone & Me: An Exploration Into Natural Human-Drone Interaction*. Available at: https://doi.org/10.1145/2750858.2805823

CB Insights (2020) *38 Ways Drones Will Impact Society: From Fighting War To Forecasting Weather, UAVs Change Everything*, Available at: https://www.cbinsights.com/research/drone-impact-society-uav (Accessed: 28 February 2020).

Celestial (no date) *Shows* [Website]. Available at: https://celestial.show/shows/

Centre for Economics and Business Research (2019) *Skills gap/needs in the Heritage Sector. A report for Historic England* (Report). Available at: https://historicengland.org.uk/content/heritage-counts/pub/2019/skill-gaps-needs-in-the-heritage-sector-pdf/

Cervinkova, H, (2020) 'Citizenship after Covid-19:Thoughts from Poland', *Social Anthropology/Anthropologie Sociale*, 28 (2). Available at: https://doi:10.1111/1469-8676.12864 (Accessed 30 January 2021).

Ch'ng, E., Cai, S., Leow, F-T. and Zhang, T., (2019) 'Adoption and use of emerging cultural technologies in China's museums', *Journal of Cultural Heritage*, 37, pp. 170–180. Available at: https://doi.org/10.1016/j.culher.2018.11.016

Ch'ng, E., Cai, S., Leow, F-T. and Zhang, T., (2019) 'Datasets from the evaluation of the adoption and use of digital technologies in China museums', *Data in Brief*, 25. Available at: https://doi.org/10.1016/j.dib.2019.104067

Chabot, D. (2018) 'Trends in drone research and applications as the *Journal of Unmanned Vehicle Systems* turns five', *Journal of Unmanned Vehicle Systems*, 6(1), pp. vi–xv. Available at: https://doi.org/10.1139/juvs-2018-0005 (Accessed: 28 February 2020).

Chabot, D. and Bird, D.M. (2012) 'Evaluation of an off-the-shelf Unmanned Aircraft System for Surveying Flocks of Geese', *Waterbirds*, 35(1), pp. 170–174. Available at: https://doi.org/10.1675/063.035.0119

Chabot, D. and Bird, D.M. (2015) 'Wildlife research and management methods in the 21st century: Where do unmanned aircraft fit in?', *Journal of Unmanned Vehicle Systems*, 3(4), pp. 137–155. Available at: https://doi.org/10.1139/juvs-2015-0021

Challis, K. (2022) *Attitudes to Drone Use by National Trust Staff* (Survey report). Provided by the National Trust.

Chapman, A. (2014) 'It's okay to call them drones', *Journal of Unmanned Vehicle Systems*, 02(02), pp. iii–v. Available at: https://doi.org/10.1139/juvs-2014-0009

Chee, F.Y. (2023) 'Microsoft president goes to Europe to shape AI regulation debate', *Reuters (online)*. Available at: https://www.reuters.com/technology/microsoft-president-goes-europe-shape-ai-regulation-debate-2023-06-29/ (Accessed: 4 July 2023).

Cheesbrough, A.E., Garvin, T. and Nykiforuk, C.I.J. (2019) 'Everyday wild: Urban natural areas, health, and well-being', *Health & Place*, 56, pp. 43–52. Available at: https://doi.org/10.1016/j.healthplace.2019.01.005

Chen, R.H. and Chen, C. (2022) *Artificial Intelligence. An Introduction for the Inquisitive Reader*. Boca Raton: Chapman and Hall/CRC. Available at: https://doi.org/10.1201/9781003214892

Cheng, E. (2015) *Aerial Photography and Videography Using Drones*. San Francisco: Peachpit Press.

Chisel, P. by (no date) *Find yourself in Sydney's iconic cultural spaces* [Video online]. Australia. Available at: https://youtu.be/Vmsfzw3NRX8

CHO, Renee (2017) *How Drones are Advancing Scientific Research*. Available at: https://blogs.ei.columbia.edu/2017/06/16/how-drones-are-advancing-scientific-research/ (Accessed: 28 February 2020).

Choi-Fitzpatrick, A., Chavarria, D., Cychosz, E., Dingens, J.P., Duffey, M., Koebel, K., Siriphanh, S., Tulen, M. Y., Watanabe, H., Juskauskas, T., Holland, J. and Almquist, L., (2009) *Up in the Air: A Global Estimate of Non-Violent Drone Use 2009-2015* (report). Available at: https://digital.sandiego.edu/gdl2016report/1 (Accessed: 28 February 2020).

Ciolfi, L. (2017) *Physical-Digital Technologies for Culture and Heritage*. Digital Leaders. Available at: https://digileaders.com/physical-digital-technologies-culture-heritage/ (Accessed 17 July 2025).

Çizel, B. and Ajanović, E. (2018) 'Virtual Reality for Cultural Heritage Tourism', *Proceedings of the 4th International Scientific Conference - SITCON 2018*. Belgrade, Serbia: Singidunum University, pp. 131–134. Available at: https://doi.org/10.15308/Sitcon-2018-131-134

Clark, L.C. (2018) 'Grim reapers: ghostly narratives of masculinity and killing in drone warfare', *International Feminist Journal of Politics*, 20(4), pp. 602–623. Available at: https://doi.org/10.1080/14616742.2018.1503553

Clarke, R. (1988) 'Information technology and dataveillance', *Communications of the ACM*, 31(5), pp. 498–512. Available at: https://doi.org/10.1145/42411.42413 (Accessed: 11 June 2020).

Cochrane, L. and Corbett, J. (2018) 'Participatory Mapping', *Handbook of Communication for Development and Social Change*. Singapore: Springer Singapore, pp. 1–9. Available at: https://doi.org/10.1007/978-981-10-7035-8_6-1

Cohen, C.B. (1998) ‘This is De Test: Festival and the Cultural Politics of Nation Building in the British Virgin Islands’, *American Ethnologist*, 25(2), pp. 189–214. Available at: https://doi.org/10.1525/ae.1998.25.2.189

Cohen, C. B., and Mascia-Lees, F.E. (1993) ‘The British Virgin Isles as Nation and desti-nation. Representing and Siting Identity in a Post- colonial Caribbean’, *Social Analysis: The International Journal of Social and Cultural Practice*, 33, pp. 130–151. Available at: http://www.jstor.org/stable/23163043

Coletta, C. (2018) ‘Scott McQuire 2017: Geomedia: Networked Cities and the Future of Public Space’, *International Journal of Urban and Regional Research*, 42(6), pp. 1162–1164. Cambridge: Polity. Available at: https://doi.org/10.1111/1468-2427.12723

Collins, Y.A. (2019) ‘Colonial residue: REDD+, territorialisation and the racialized subject in Guyana and Suriname’, *Geoforum*, 106, pp. 38–47. Available at: https://doi.org/10.1016/j.geoforum.2019.07.019

Colomina, I. and Molina, P. (2014) ‘Unmanned aerial systems for photogrammetry and remote sensing: A review’, *ISPRS Journal of Photogrammetry and Remote Sensing*. Elsevier B.V., pp. 79–97. Available at: https://doi.org/10.1016/j.isprsjprs.2014.02.013

Constable, G., Rimalovski, F. and Fishburne, T. (2014) *Talking to Humans: Success Starts with Understanding Your Customers*. Giff Constable (self-published).

Constantino, P.A.L. (2015) ‘Dynamics of hunting territories and prey distribution in Amazonian Indigenous Lands’, *Applied Geography*, 56, pp. 222–231. Available at: https://doi.org/10.1016/j.apgeog.2014.11.015

Convery (2012) *Making Sense of Place*, *Making sense of place: multidisciplinary perspectives*. Woodbridge, Suffolk: Boydell Press. Available at: https://doi.org/info:doi/

Cooper, R. and Lilyea, B. (2022) ‘I’m Interested in Autoethnography, but How Do I Do It?’, *The Qualitative Report* [Preprint]. Available at: https://doi.org/10.46743/2160-3715/2022.5288

Cornelio, G.S. and Ardévol, E. (2011) ‘Practices of place-making through locative media artworks’, *Communications*, 36(3). Available at: https://doi.org/10.1515/comm.2011.016

Cox, P. (2020) ‘Theorising infrastructure: A Politics of Spaces and Edges’, Cox, P. and Koglin, T. (eds.) *The politics of cycling infrastructure: Spaces and (in)equality*. 1st edition. Bristol: Bristol University Press (Spaces and (in)equality), pp. 15–34. Available at: https://doi.org/10.2307/j.ctvvsqc63.6

Crampton, J.W. (2016) ‘Assemblage of the vertical: Commercial drones and algorithmic life’, *Geographica Helvetica*, 71(2), pp. 137–146. Available at: https://doi.org/10.5194/gh-71-137-2016

Creech, C. and Shriner, W. (2020) ‘DIY ecology class: Transitioning field activities to an online format’, *Ecology and Evolution*, 10(22), pp. 12437–12441. Available at: https://doi.org/10.1002/ece3.6656

Crenshaw, K. (1991) ‘Mapping the Margins: Intersectionality, Identity Politics, and Violence against Women of Color’, *Stanford Law Review*, 43(6), p. 1241. Available at: https://doi.org/10.2307/1229039

Crickmar, R. (2017) *Repurposing the drone. Ai Weiwei’s Human Perspective, Filmhouse cinema* [Webpage]. Available at: https://www.filmhousecinema.com/latest/repurposing-drone-ai-weiweis-human-perspective-shift (Accessed: 1 May 2021).

Cui, Y., He, Y., Xu, X., Zhou, L., Nutakor, J.A. and Zhao, L. (2024) ‘Cultural capital, the digital divide, and the health of older adults: a moderated mediation effect test’, *BMC Public Health*, 24(1), p. 302. Available at: https://doi.org/10.1186/s12889-024-17831-4

D'Andrea, R. (2014) 'Guest Editorial Can Drones Deliver?', *IEEE Transactions on Automation Science and Engineering*, 11(3), pp. 647–648. Available at: https://doi.org/10.1109/TASE.2014.2326952

Da, N.Z. (2019) 'The computational case against computational literary studies', *Critical Inquiry*, 45(3), pp. 601–639. Available at: https://doi.org/10.1086/702594

Dal Falco, F. and Vassos, S. (2017) 'Museum Experience Design: A Modern Storytelling Methodology', *The Design Journal*, 20(sup1), pp. S3975–S3983. Available at: https://doi.org/10.1080/14606925.2017.1352900

David, E.J., Gutiérrez, J., Outrot, A., Perreira da Silva, M and Le Callet, P., (2018) 'A dataset of head and eye movements for 360° videos', *Proceedings of the 9th ACM Multimedia Systems Conference*. New York, NY, USA: ACM, pp. 432–437. Available at: https://doi.org/10.1145/3204949.3208139

David, E.J., Lebranchu, Perreira da Silva, M. and Le Callet, P. (2022) 'What are the visuo-motor tendencies of omnidirectional scene free-viewing in virtual reality?', *Journal of Vision*, 22(4), p. 12. Available at: https://doi.org/10.1167/jov.22.4.12

Dawson, E. (2019). *Equity, Exclusion and Everyday Science Learning. The Experiences of Minoritised Groups.* Routlege.

Debord, G. (1994) *The Society of the Spectacle*. Translated by Nicholson-Smith, D. Zone Books. Available at: https://doi.org/10.2307/j.ctv1453m69

Decker, J. (ed.) (2015) *Engagement and access: innovative approaches for museums* (Innovative approaches for museums; no. 2). Rowman & Littlefield Publishers.

De Cruz (2024) *Wonderstruck: How Wonder and Awe Shape the Way We Think*, Princeton, NJ: Princeton University Press, pp. 232 Available at: https://doi.org/ 10.2307/jj.7249506

DeJong, K., Love, B. (2016) ‘Youth and elder oppression’, *Teaching for diversity and social justice*, Adams *et. al.* (eds.), pp. 339–368. New York: Routledge.

DeJonckheere, M. and Vaughn, L.M. (2019) ‘Semistructured interviewing in primary care research: a balance of relationship and rigour’, *Family Medicine and Community Health*, 7(2), p. e000057. Available at: https://doi.org/10.1136/fmch-2018-000057

Deller, R. (2011) ‘Twittering on: Audience research and participation using Twitter’, *Participations, vol. 8 Issue 1* [Preprint]. Available at: https://www.participations.org/08-01-11-deller.pdf

De Luca, G., Dastgerdi, A. S., Francini, C. and Liberatore, G. (2020) ‘Sustainable cultural heritage planning and management of overtourism in art cities: Lessons from atlas world heritage’, *Sustainability* (Switzerland), 12(9), p. 3929. Available at: https://doi.org/10.3390/su12093929

DeMirjyn, M. (2018) ‘The Goddess on Wheels: Maria R. Palacios’, *Sexualities*, Volume 21(3), pp. 255–499.

Denison, S. (2023) ‘Putting Ourselves in the Picture. An Autoethnographic Approach to Photography Criticism’ in Goode, J., Lumsden, K., and Bradford, J. (eds.) *Crafting Autoethnography Processes and Practices of Making Self and Culture*. London: Routledge. Available at: https://doi.org/10.4324/9781003309239

Denscombe, M. (2014) *Good Research Guide: For Small-Scale Social Research Projects*. 5th edition. McGraw-Hill Education. Available at: https://www.researchgate.net/file.PostFileLoader.html?id=582a0dbf217e20276533f5a5&assetKey=AS:428404664213506@1479151039119 (Accessed: 19 June 2024).

Department for Culture, Media and Sport (2000) *Centres for Social Change: Museums, Galleries and Archives for All* [Preprint]. Available at: https://openlibrary.org/books/OL38563248M/Centres_for_social_change_museums_galleries_and_archives_for_all. (Accessed 19 June 2020).

Department for Levelling up, H. and C. (2012) *National Planning Policy Framework*. Available at: https://www.gov.uk/guidance/national-planning-policy-framework/15-conserving-and-enhancing-the-natural-environment (Accessed 19 June 2020).

Department for Transport (2017) 'Small Remotely Piloted Aircraft Systems (drones) Mid-Air Collision Study', *Department of Transport*, (July), p. 18. Available at: http://www.nationalarchives.gov.uk/doc/open-government-licence/version/3/ (Accessed: 12 February 2020).

Department for Transport and Ministry of Defence, (2016) *Public dialogue on drone use in the UK*. Available at: https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/579550/drones-uk-public-dialogue.pdf (Accessed 19 June 2020).

Department for Transport, Home Office, Ministry of Defence, and T.R.H.C.G.M. (2019) *Drones: consultation response, 8 January 2019 - GOV.UK*. Available at: https://www.gov.uk/government/speeches/drones-consultation-response-7-january-2019 (Accessed 12 February 2020).

Derry, S. J., Pea, R.D., Barron, B., Engle, R.A., Erickson, F., Goldman, R., Hall, R., Koschmann, T., Lemke, J.L., Sherin, M.G. and Sherin, B.L. (2010) 'Conducting Video Research in the Learning Sciences: Guidance on Selection, Analysis, Technology, and Ethics', *The Journal of the Learning Sciences*, 19(1), pp. 3–53. Available at: https://doi.org/10.1080/10508400903452884

Deterding, S., Dixon, D., Khaled, R., Nacke, L. (2011). 'From Game Design Elements to Gamefullness: defining Gamification', *Proceedings of the 15th International Academic MindTrek Conference: Envisioning Future Media Environments*, 9–15. Available at: https://doi.org/10.1145/2181037.2181040

Díaz-Delgado, R. and Mücher, S. (2019) 'Editorial of Special Issue Drones for Biodiversity Conservation and Ecological Monitoring', *Drones 2019, Vol. 3, Page 47*, 3(2), p. 47. Available at: https://doi.org/10.3390/DRONES3020047

Digital Poverty Alliance (2022) *UK Digital Poverty Evidence Review 2022*. Available at: https://digitalpovertyalliance.org.uk/wp-content/uploads/2022/06/UK-Digital-Poverty-Evidence-Review-2022-v1.0-compressed.pdf (Accessed 22 July 2025).

Dinhopl, A. and Gretzel, U. (2016) 'Conceptualizing tourist videography', *Information Technology & Tourism*, 15(4), pp. 395–410. Available at: https://www.semanticscholar.org/paper/Conceptualizing-tourist-videography-Dinhopl-Gretzel/9e1705cf6f3bfd1f650831d9166e83224bae932c (Accessed: 19 June 2024).

DiSalvo, C., Louw, M., Holstius, D., Nourbakhsh, I. and Akin, A. (2012) 'Toward a Public Rhetoric Through Participatory Design: Critical Engagements and Creative Expression in the Neighborhood Networks Project', *Design Issues*, 28(3), pp. 48–61. Available at: https://doi.org/10.1162/DESI_a_00161

DJI (2023) [Website]Available at: https://www.dji.com (Accessed 18 November 2023).

Dolničar, V. and Fortunati, L. (2014) 'Exploring and Conceptualizing Empowerment: Introduction to the Special Issue on Media and Empowerment', *The Information Society*, 30(3), pp. 165–168. Available at: https://doi.org/10.1080/01972243.2014.896672

Donny the drone. 2017. [Film] Directed by Mackenzie Sheppard. Japan, Bangladesh, India, United States.

Drones Direct (no date) *The UK Drone Users Report*. Available at: https://www.dronesdirect.co.uk/files/pdf/dronesreport.pdf (Accessed: 14 February 2022).

Drones in Forbidden Zones, 2015, [TV series], All 4, 31 March 2015.

Dujon, A.M. (2019) 'Citation patterns of publications using unmanned aerial vehicles in ecology and conservation', *Journal of Unmanned Vehicle Systems*, pp. 1–10. Available at: https://doi.org/10.1139/juvs-2019-0013

Dwight, R., Hansen, C., Rudolph, E. (2002) 'Thermal State of Permafrost and Active-layer Monitoring in the Antarctic: Advances During the International Polar Year', *Periglacial Research. South African Journal of Science*, 98(2), pp. 182–197. Available at: http://www.draganfly.com/news/2008/08/24/introduction-to-unmanned-aerial-vehicles-uavs/.*20/05/ (Accessed: 23 September 2021).

East Sussex County Council (2021) *Beachy Head Style Guide*. Provided by East Sussex County Council.

ELEVENPLAY x Rhizomatiks Research (2013) *24 drones*. Available at: http://www.daito.ws/en/work/elevenplay-rhizomatiks-research-3-dancers-and-24-drones.html (Accessed: 11 January 2021).

Ellingson, L. (2008) 'Embodied Knowledge' in *The Sage encyclopaedia of qualitative research methods*. Thousand Oaks: Sage.

Ellis, C., Adams, T.E. and Bochner, A. (2010) 'Autoethnography: An Overview', *Forum: Qualitative Sozialforschung/Qualitative Social Research*, 12(1), Available at: https://doi.org/10.171169/fqs-12.1.1589

English, L. (2023) Finding a new approach; the use of visual images in spoken word poetry film. Vorrath, W and Korecka, M. (eds.), ‘Poetry and contemporary visual culture/Lyric und zeitgenossische Visuelle Kultur’, *Poetry in the digital age* (3), p. 2027-222, De Gruyter. Berlin. Available at: librar.oapen.org/bitstream/handle/20P (Accessed 2 July 2025).

Equality Act 2010. Available at: https://www.legislation.gov.uk/ukpga/2010/15/contents (Accessed 12 July 2025).

Equality and Human Rights Commission (2022) *Gender Reassignment Discrimination* [Webpage]. Available at: https://www.equalityhumanrights.com/en/advice-and-guidance/gender-reassignment-discrimination (Accessed 28 November 2022).

Ericksen, E.G. (1980) *The territorial experience: human ecology as symbolic interaction*. Edited by H. Blumer. Austin: University of Texas Press. Available at: https://doi.org/10.7560/780385.

Erickson, F. (2011) ‘Uses of video in social research: a brief history’, *International Journal of Social Research Methodology*, 14(3), pp. 179–189. Available at: https://doi.org/10.1080/13645579.2011.563615

Eyerman, J., Goodman, L. and Mangina, E. (2016) *Drones as enabling devices: mobile robotics for extreme user access*, pp. 5118–5118. Available at: https://doi.org/10.21125/inted.2016.2303

Farman, J. (ed.) (2013) *The Mobile Story: Narrative Practices with Locative Technologies*. ProQuest E. Taylor & Francis Group. Available at: https://ebookcentral-proquest-com.ezproxy.brighton.ac.uk/lib/ubrighton/detail.action?docID=1377510

Fawcett Society (2022) *Make Equality Law Fit For The 21st Century*. Available at: https://www.fawcettsociety.org.uk/make-equality-law-21st-century (Accessed: 28 November 2022).

Feenstra, S., Begeny, C. T., Ryan, M.K., Rink, F.A., Stoker, J. and Jordan, J. (2020) 'Contextualising the impostor syndrome', *Tomorrow's Challenges* [Preprint]. Available at: https://doi.org/10.3389/fpsyg.2020.575024

Felderhof, L. and Gillieson, D. (2012) 'Near-infrared imagery from unmanned aerial systems and satellites can be used to specify fertilizer application rates in tree crops', *Canadian Journal of Remote Sensing*, 37(4), pp. 376–386. Available at: https://doi.org/10.5589/m11-046

Fennessy, J., Woods, S., Johnson, H., Rivas, C., Norbury, D., Almeida-Harvey, I., Moriarty, J., Wimpenny, K., Bushell, K. and Blake, P. (2019) ‚Carnival of invention', *Forum Qualitative Sozialforschung*, 20(2). Available at: https://doi.org/10.17169/fqs-19.2.3260

Ferdinand, S. (2015) 'Seeing from above: the aerial view in visual culture', *Visual Studies*, 30(1), pp. 112–113. Available at: https://doi.org/10.1080/1472586X.2014.887320

Fernandez, S. (2006) More than Just an Arts Festival: Communities, Resistance, and the Story of Desh Pardesh. *Canadian Journal of Communication*, Vol. 31(Issue 1), pp. 17–34.

Fetterman, D., Rodríguez-Campos, L., Wandersman, A. and O'Sullivan, R. (2014) 'Collaborative, Participatory, and Empowerment Evaluation', *American Journal of Evaluation*, 35(1), pp. 144–148. Available at: https://doi.org/10.1177/1098214013509875

Finch, David; Tinson, A. (2022) *The continuing impact of COVID on health inequalities*. Available at: https://www.health.org.uk/publications/long-reads/the-continuing-impact-of-covid-19-on-health-and-inequalities (Accessed: 19 June 2023).

Fine, M. (2023a) These too are footsteps [film-poem], unpublished.

Fine, M. (2023b) A room of my own [film-poem], unpublished.

Finlay, J M., and Bowman, J. A. 'Geographies on the Move: A Practical and Theoretical Approach to the Mobile Interview', *The Professional Geographer*, 69 (2), 263-274. Available at: https://doi.org/10.1080/00330124.2016.1229623

Fish, A. (2020) 'Visual Anthropology from the Air' in *Handbook of Ethnographic Film and Video*. Routledge.

Fischer, G. and Herrmann, T. (2011) 'Socio-technical Systems: A Meta-Design Perspective', *International Journal of Sociotechnology and Knowledge Development*, 3(1), 1-33, January-March 2011, Available at: https://doi.org/10.4018/jskd.2011010101

Fletcher, M. (2020) 'Casualties of Britain's Culture Wars: The Museum Exhibits You Can No Longer See', *Telegraph Magazine*, 17 October. Available at: https://www.telegraph.co.uk/news/0/casualties-britains-culture-wars-museum-exhibits-can-no-longer/ (Accessed: 30 January 2021).

Flick, C., Zamani, E. D., Stahl, B. C. and Brem, A. (2020) 'The future of ICT for health and ageing: Unveiling ethical and social issues through horizon scanning foresight', *Technological Forecasting and Social Change*, 155, p. 119995. Available at: https://doi.org/10.1016/j.techfore.2020.119995

Focus (2014) *The Meon Valley: a journey of discovery*. [Video online] Available at: https://www.youtube.com/watch?v=dUQKlP26Jbs (Accessed 22 January 2024).

Fogg-Rogers, L. A. and Hickman, M. (2014) 'Collected Thoughts 2014: Essays inspired by the annual Science Communication Conference' in *The people, the people, the people: Engaging under-served audiences*. British Science Association, pp. 1–5. Available at: https://uwe-repository.worktribe.com/output/813957 (Accessed: 15 May 2024).

Forceville, C. (1999) 'Educating the eye? Kress and Van Leeuwen's Reading Images: The Grammar of Visual Design (1996)', *Language and Literature: International Journal of Stylistics*, 8(2), pp. 163–178. Available at: https://doi.org/10.1177/096394709900800204

Foucault, M. (2007) 'Questions on Geography' in *Space, Knowledge and Power. Foucault and Geography*. Edited by J.W.C. and S. Elden. Abingdon: Routledge.

FPV UK (no date), *Drone Club and Event Map*, [Webpage] Available at: https://www.fpvracingworld.com (Accessed: 12 May 2022).

Fraser, J. (2020) 'A Time of Reckoning', *Curator: The Museum Journal*, 63(4), pp. 479–481. Available at: https://doi.org/10.1111/cura.12393

Freeman, C., Waters, D.L., Buttery, Y. and van Heezik, Y. (2019) 'The impacts of ageing on connection to nature: the varied responses of older adults', *Health & Place*, 56, pp. 24–33. Available at: https://doi.org/10.1016/j.healthplace.2019.01.010

Freimuth, V. S., and Mettger, W. (1990) 'Is there a hard-to-reach audience?', *Public health reports*, 105(3), pp. 232–238. Available at: https://pubmed.ncbi.nlm.nih.gov/2113680/ (Accessed: 18 November 2023).

Frenkel, V. (2007) 'A Place for Uncertainty: Towards a New Kind of Museum' in Pollock, G. and Zemans, J. (eds.) *Museums After Modernism: Strategies of Engagement*. Wiley, pp. 119-130. Available at: https://doi.org/10.1002/9780470776636.ch6

Frith, Jordan and de Souza e Silva, A. (2013) 'Re-narrating the city through the presentation of location' in J. Farman (Ed.) *The Mobile Story: Narrative Practices with Locative Technologies*. New York: Routledge.

Frosh, P., and Pinchevski, A. (2014) 'Media witnessing and the ripeness of time', *Cultural Studies*, 28(4), 594–610. Available at: https://doi.org/10.1080/09502386.2014.891304

Fust, P. and Loos, J. (2020) 'Development perspectives for the application of autonomous, unmanned aerial systems (UASs) in wildlife conservation', *Biological Conservation*. Elsevier Ltd. Available at: https://doi.org/10.1016/j.biocon.2019.108380

Gallagher, M.W. (2012) 'Self-Efficacy', Ramachandran *et. al.* (eds.) *Encyclopedia of Human Behavior.* Elsevier, pp. 314-320. Available at: https://doi.org/10.1016/B978-0-12-375000-6.00312-8

Garibay, C., Lannes, P., González, J.; (2017). GENIAL: *Latino Audiences: Embracing Complexity*, 8pp. Available at: https://www.researchgate.net/publication/362680273_GENIAL_Latino_Audiences_Embracing_Complexity. (Accessed 19 July 2025).

Garrett, B.L. and McCosker, A. (2017) 'Non-human Sensing: New Methodologies for the Drone Assemblage', *Refiguring Techniques in Digital Visual Research*. Cham: Springer International Publishing, pp. 13–23. Available at: https://doi.org/10.1007/978-3-319-61222-5_2

Gauld, C. (2017) 'Democratising or privileging: the democratisation of knowledge and the role of the archivist', *Archival Science*, 17(3), 227–245. Available at: https://doi.org /10.1007/s10502-015-9262-4

Getzin, S., Wiegand, K. and Schöning, I. (2012) 'Assessing biodiversity in forests using very high-resolution images and unmanned aerial vehicles', *Methods in Ecology and Evolution*, 3(2), pp. 397–404. Available at: https://doi.org/10.1111/j.2041-210X.2011.00158.x

Gibbs, R.W. (2006)' Metaphor Interpretation as Embodied Simulation', *Mind and Language*, 21(3), pp. 434–458. Available at: https://doi.org/10.1111/j.1468-0017.2006.00285.x

Glebova, A. (2022) *Aleksandr Rodchenko: Photography in the time of Stalin*. New Haven: Yale University Press. (Accessed 31 January 2025).

Goggin, G. and Hjorth, L. (eds.) (2014) *The Routledge Companion to Mobile Media*. London: Routledge (Routledge companions). Available at: https://doi.org/10.4324/9780203434833

Goldsmith, L. and Bell, M.L. (2022) 'Queering Environmental Justice: Unequal Environmental Health Burden on the LGBTQ+ Community', *American Journal of Public Health*, 112(1), pp. 79–87. Available at: https://doi.org/10.2105/AJPH.2021.306406

Gössling, S. (2021) 'Tourism, technology and ICT: a critical review of affordances and concessions', *Journal of Sustainable Tourism*, 29(5), pp. 733–750. Available at: https://doi.org/10.1080/09669582.2021.1873353

GOV.UK (2016) *Drone use: dialogue conducted to understand public attitudes*. Available at: https://www.gov.uk/government/publications/drone-use-in-the-uk-public-dialogue (Accessed: 12 February 2020).

GOV.UK (2016) *Public dialogue on drone use in the UK*. Available at: https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/579550/drones-uk-public-dialogue.pdf (Accessed: 30 April 2020).

GOV.UK (2017) *Drones update*. Available at: https://www.gov.uk/government/speeches/drones-update-27-november-2017 (Accessed: 12 February 2020).

GOV.UK (2019) *Drone legislation: use, restrictions and enforcement*. Available at: https://www.gov.uk/government/consultations/drone-legislation-use-restrictions-and-enforcement (Accessed: 12 February 2020).

GOV.UK (2019) *Drones: national campaign and extended no fly zone around airports*. Available at: https://www.gov.uk/government/speeches/drones-national-campaign-and-extended-no-fly-zone-around-airports (Accessed: 12 February 2020).

GOV.UK (2020) Mental Capacity Act Code of Practice. Available at: https://www.gov.uk/government/publications/mental-capacity-act-code-of-practice

Government of the Virgin Islands (2019) *Our history* [Webpage]. Available at: https://bvi.gov.vg/content/our-history#:~:text= (Accessed: 24 July 2023).

Government of the Virgin Islands (2023) *Virgin Islands Culture & Heritage Policy & Strategy 2023 – 2028 (DRAFT) Building a Nation, Centring Culture and Heritage*, p. 89. Available at: https://bvi.gov.vg/sites/default/files/resources/final_draft_-_virgin_islands_culture_heritage_policy_strategy_-_14_feb_2023_1.pdf (Accessed: 24 July 2023).

Grafton, Q., Adamowicz, W., Dupont, D., Nelson, H., Hill, R. J. and Renzetti, S. (2008) *The Economics of the Environment and Natural Resources*. John Wiley & Sons.

Graham, S., and Hewitt, L. (2012) 'Getting off the ground: On the politics of urban verticality', *Progress in Human Geography*, 37(1), pp. 72–92. SAGE Journals. Available at: https://journals.sagepub.com/doi/abs/10.1177/0309132512443147

Grand, S. and Wiedmer, M. (2010) 'Design Fiction: A Method Toolbox for Design Research in a Complex World', *Design and Complexity*. Montreal. Available at: https://doi.org/10.21606/drs.2008.47

Gray, A. (2003) *Research Practice for Cultural Studies*. 6 Bonhill Street, London, UK, SAGE Publications Ltd. Available at: https://doi.org/10.4135/9780857024596

Green, D. P., Rose, M., Bevan, C., Farmer, H., Cater, K. and Stanton Fraser, D. (2021) 'You wouldn't get that from watching TV!: Exploring audience responses to virtual reality non-fiction in the home', *Convergence: The International Journal of Research into New Media Technologies* 27(3), 805-829. Available at: https://doi.org/10.1177/1354856520979966

Greene, D. (2015) ‘Drone Vision’, *Surveillance & Society*, 13(2), pp. 233–249. Available at: https://doi.org/10.24908/ss.v13i2.5346

Greffe, X., Krebs, A. and Pflieger, S. (2017) ‘The future of the museum in the twenty-first century: recent clues from France’, *Museum Management and Curatorship*, 32(4), pp. 319–334. Available at: https://doi.org/10.1080/09647775.2017.1313126

Greg, M.L.G., Emily E.N. and Mitchell (2004) *Qualitative research. Nephrology nursing journal: journal of the American Nephrology Nurses Association*. SAGE Publications Inc. Available at: https://doi.org/10.1044/1058-0360(2003/062)

Gregory, D. (2011) ‘The everywhere war’, *Geographical Journal*, 177(3), pp. 238–250. Available at: https://doi.org/10.1111/j.1475-4959.2011.00426.x

Grémillet, D., Puech, W., Garçon, V., Boulinier, T., and Maho, Y. Le. (2012) ‘Robots in Ecology: Welcome to the machine’, *Open Journal of Ecology*, 02(02), pp. 49–57. Available at: https://doi.org/10.4236/oje.2012.22006

Gretzel, U., Fuchs, M., Baggio, R., Hoepken, W., Law, R., Neidhart, J., Pesonen, J., Zanker, M and Xiang, Z. (2020) ‘e-Tourism beyond COVID-19: a call for transformative research’, *Information Technology & Tourism*, 22(2), pp. 187–203. Available at: https://doi.org/10.1007/s40558-020-00181-3

Grovesnor, I. (2007) From the “Eye of History” to “a Second Gaze”: The Visual Archive and the Marginalised in the History of Education. *History of education*, 36(4–5), pp. 607-622. Available at: https://doi.org/10.108000467600701496948

Gruber, M. E., Galliano, M. S., Brosnihan, A. P., and Hancock, P. A. (2023). ‘The impact of camera usage and angle on perceptions of power during videoconferencing’, *Computers in Human Behavior Reports*, 10 (September 2022), 100285. Available at: https://doi.org/10.1016/j.chbr.2023.100285

Guest, G., Namey, E. E. and Mitchell, M. L. (2012) Collecting Qualitative Data: A Field Manual for Applied Research. SAGE Publications Inc.

gustavofrazao (2017) Aerial View of Monument to the discoveries, Lisbon, Portugal. 19 December, istock item-875673258.mov, Available at: https://www.videvo.net/video/aerial-view-of-monument-to-the-discoveries-lisbon-portugal/864470/ (Accessed 31 June 2023).

Gusterson, H. (2017) 'Drone warfare', *AIP Conference Proceedings*. American Institute of Physics Inc. Available at: https://doi.org/10.1063/1.5009234

Hackett, S. and Dissanayake, S.T.M. (2014) *Environmental and Natural Resources Economics*. Routledge. Available at: https://doi.org/10.4324/9781315704586

Hafermalz, E., Johnston, R., Hovorka, D. and Riemer, K. (2020) 'Beyond "mobility": A new understanding of moving with technology', *Information Systems Journal*, 30(4), pp. 762–786. Available at: https://doi.org/10.1111/isj.12283

Hamer, C. (2016) 'A collaborative methodology between photography and performance in ethnographically informed research photography and performance in ethnographically informed research', *Critical Arts, 30 (3),* pp. 341-356. Available at: https://doi.org/10.1080/02560046.2016.1205321

Hanmakyugh, T. (2020) 'The psychology of camera shots: a determinant for audience perception of film images', *Journal of Theatre and Cultural Studies*, 14(1), pp. 104–113. Available at: https://ezenwaohaetorc.org/journals/index.php/AMA/article/view/2306 (Accessed: 28 July 2023).

Hanrahan, L. (2018) 'Philip George "Pip" Rudkin: a Memoriam (2018)', *Journal of Unmanned Vehicle Systems*, 6(1), pp. iii–v. Available at: https://doi.org/10.1139/juvs-2018-0004

Hansen, A., Cottle, S., Negrine, R. and Newbold, C. (2009) 'Analysing visuals. Still and moving images', *Mass Communication Research Methods*. SAGE Publications, pp. 227–254. Available at: https://doi.org/10.1007/978-1-349-26485-8_8

Harrison, T. and Waite, K. (2015) 'Impact of co-production on consumer perception of empowerment', *The Service Industries Journal*, 35(10), pp. 502–520. Available at: https://doi.org/10.1080/02642069.2015.1043276

Hastings Contemporary (2020) *Robot Tours*. [Video online] Available at: https://www.hastingscontemporary.org/robot-tours/. (Accessed 10 June 2020).

Hazelwood, E., Lawson, R. and Aitken, R. (2009) 'An essential guide to audience development', *Marketing Intelligence and Planning* [Preprint]. Available at: https://doi.org/10.1108/02634500910988681

Heidegger, M. (1977). *The question concerning technology and other essays*. Harper and Row. New York and London

Heimann, K., Uithol, S., Clbi, M., Umiltà, M.A., Guerra, M., Fingerhut, J. and Gallese, V. (2019), 'Embodying the camera: An EEG study on the effect of camera movements on film spectators' sensorimotor cortex activation', *PLoS One*. March 13, 14(3) Available at: https://doi.org/10.1371/journal.pone.0211026

Hennink, M, Hutter I, B.A. (2020) *Qualitative research methods.* London: SAGE Publications.

Hensley, N.K. (2018) 'Drone form: Mediation at the end of empire', *Novel*. Duke University Press, pp. 226–249. Available at: https://doi.org/10.1215/00295132-6846084

Heritage Alliance (2024) *On the Brink: Heritage in the Cost of Living Crisis. Insight Report*, published May 2024. Available at: https://www.theheritagealliance.org.uk/our-work/publications/cost-of-living-report (Accessed 22 July 2025).

Herrman, A. (2017) ‘Focus Groups’ in *The SAGE Encyclopaedia of Communication Research Methods*. Thousand Oaks: SAGE Publications. Available at: https://doi.org/10.4135/9781483381411.n208

Herwitz, S.R., Johnson, L.F., Dunagan, S.E., Higgins, R.G., Sullivan, D.V., Zheng, J., Lobitz, B.M., Leung, J.G., Gallmeyer, B.A., Ayoagi, M., Slye, R.E. and Brass, J.A. (2004) ‘Imaging from an unmanned aerial vehicle: Agricultural surveillance and decision support’, *Computers and Electronics in Agriculture*, 44(1), pp. 49–61. Available at: https://doi.org/10.1016/j.compag.2004.02.006

Hidden Britain by Drone, 2016-2018, [TV series], Prime Video, Channel 4.

Hildebrand, J.M. (2018) ‘Situating Hobby Drone Practices’, *Digital Culture & Society*, 3(2), pp. 207–218. Available at: https://doi.org/10.14361/dcs-2017-0212

Higgins-Desbiolles, F. (2008) ‘Justice Tourism and Alternative Globalisation’, *Journal of Sustainable Tourism*, 16(3), pp. 345–364. Available at: https://doi.org/10.1080/09669580802154132

Hildebrand, J.M. (2018) ‘Over-Extended Media: Hashtag Hatred and Domestic Drones’, Cirucci, A.M. *et. al.* (eds.) *Black Mirror and Critical Media Theory*. Lexington Books, pp. 171-183. Available at: https://www.researchgate.net/publication/328170557_Over-Extended_Media_Hashtag_Hatred_and_Domestic_Drones (Accessed: 6 July 2020).

Hildebrand, J.M. (2018) ‘Situating Hobby Drone Practices’, *Digital Culture & Society*, 3(2), pp. 207–218. Available at: https://doi.org/10.14361/dcs-2017-0212

Hildebrand, J.M. (2019) ‘Consumer drones and communication on the fly’, *Mobile Media & Communication*, 7(3), pp. 395–411. Available at: https://doi.org/10.1177/2050157919850603

Hildebrand, J.M. (2020) ‘Drone-topia as method’, *Mobilities*, 15(1), pp. 25–38. Available at: https://doi.org/10.1080/17450101.2019.1663079

Hildebrand, J.M. (2021) *Aerial Play*. Singapore: Springer Singapore. Available at: https://doi.org/10.1007/978-981-16-2195-6

Hinkley, E.A. and Zajkowski, T. (2011) 'USDA forest service-NASA: Unmanned aerial systems demonstrations - pushing the leading edge in fire mapping', *Geocarto International*, 26(2), pp. 103–111. Available at: https://doi.org/10.1080/10106049.2011.555823

Hirt, I. and Lerch, L. (2013) 'Cartographier les territorialités indigènes dans les Andes boliviennes: enjeux politiques, défis méthodologiques.', *Cybergeo*, 2013. Available at: https://doi.org/10.4000/cybergeo.25843

Historic England (no date) *Drones for Heritage uses*. Available at: https://historicengland.org.uk/research/methods/airborne-remote-sensing/drones/

Hjorth, L. and Pink, S. (2014) 'New visualities and the digital wayfarer: Reconceptualizing camera phone photography and locative media', *Mobile Media & Communication*, 2(1), pp. 40–57. Available at: https://doi.org/10.1177/2050157913505257

Hochschild, A.R. (1979) 'Emotion work, Feeling Rules, and Social Structure', *American Journal of Sociology*, 85(3), pp. 551–75. Available at: https://www.jstor.org/stable/2778583 (Accessed: 16 July 2025).

Hodges, S. (2017) *Visitors of colour*. [Tumblr post]. August 4. Available at: https://visitorsofcolor.tumblr.com/image/163774348237 (Accessed: 23 October 2024)

Holm, A., Lepp, M. and Ringsberg, K.C. (2005) 'Dementia: involving patients in storytelling - a caring intervention. A pilot study', *Journal of Clinical Nursing*, 14(2), pp. 256–263. Available at: https://doi.org/10.1111/j.1365-2702.2004.01042.x

Holton, A.E., Lawson, S. and Love, C. (2015) 'Unmanned Aerial Vehicles', *Journalism Practice*, 9(5), pp. 634–650. Available at: https://doi.org/10.1080/17512786.2014.980596

Holzman, P.S. (1969) 'On Hearing and Seeing Oneself', *The Journal of Nervous and Mental Disease*, 148(3), pp. 198–209. Available at: https://doi.org/10.1097/00005053-196903000-00002

Hong, B., Zhang, L and Sun, H. (2019) 'Measurement of the Vertical Spatial Metaphor of Power Concepts Using the Implicit Relational Assessment Procedure', *Frontiers in Psychology*, 10, Article 1422 Available at: https://doi.org/ 10.3389/fpsyg.2019.01422

Hooker, C.M. and De Zúñiga, H.G. (2017) 'Survey Methods Online', *The International Encyclopaedia of Communication Research Methods*. Wiley, pp. 1–7. Available at: https://doi.org/10.1002/9781118901731.iecrm0244

Hooper-Greenhill, E. (Ed.) (1994) *The Educational Role of the Museum*. London: Routledge.

Hooper-Greenhill, E. (2000) 'Changing Values in the Art Museum: rethinking communication and learning', *International Journal of Heritage Studies*, 6(1), pp. 9–31. Available at: https://doi.org/10.1080/135272500363715

Horsfall, D. and Titchen, A. (2009) 'Disrupting edges – opening spaces: pursuing democracy and human flourishing through creative methodologies', *International Journal of Social Research Methodology*, 12(2), pp. 147–160. Available at: https://doi.org/10.1080/13645570902752324

Horsti, K. (2016) 'Visibility without voice: Media witnessing irregular migrants in BBC online news journalism', *African Journalism Issues*, Vol 37 (1). Available at: https://doi.org/10.1080/23743670.2015.1084585

Hotel Solutions (2014) *South Downs National Park Visitor Accommodation Review: Report of Key Findings* (Report). Provided by the SDNPA.

House of Commons (2023) *Debate on Artificial Intelligence*. London. Available at: https://researchbriefings.files.parliament.uk/documents/CDP-2023-0152/CDP-2023-0152.pdf (Accessed 17 May 2024).

Hua, D., Wei, H. and Blevis, E. (no date) 'MemoryPin', *Proceedings of the 2018 ACM Conference Companion Publication on designing interactive systems*. ACM (DIS 18 Companion), pp. 253–258. Available at: https://doi.org/10.1145/3197391.3205445

Huang, Y. C., Backman, K. F., Backman, S. J.; Chang, L. L. (2016) 'Exploring the Implications of Virtual Reality Technology in Tourism Marketing: An Integrated Research Framework', *International Journal of Tourism Research* 18 (2), 116--128. https://doi.org/10.1002/jtr.2038

Hubbard, S., Pak, A., Gu, Y. and Jin, Y. (2017) 'UAS to Support Airport Safety and Operations: Opportunities and Challenges', *Journal of Unmanned Vehicle Systems* [Preprint]. Available at: https://doi.org/10.1139/juvs-2016-0020

Humm, C. and Schrögel, P. (2020) 'Science for All? Practical Recommendations on Reaching Underserved Audiences', *Frontiers in Communication*, 5. Available at: https://doi.org/10.3389/fcomm.2020.00042

Humm, C., Schrögel, P. and Leßmöllmann, A. (2020) 'Feeling Left Out: Underserved Audiences in Science Communication', *Media and Communication*, 8(1), pp. 164–176. Available at: https://doi.org/10.17645/mac.v8i1.2480

Hunt, V., Layton, D and Prince, S (2015) *Why diversity matters*. Available at: https://www.mckinsey.com/capabilities/people-and-organizational-performance/our-insights/why-diversity-matters (Accessed: 12 September 2024).

Iphofen, R, and Tolich, M. (eds.) (2018) *The SAGE Handbook of Qualitative Research Ethics*. London, England: SAGE Publications Ltd.

Islas, O. (2016) ‘Understanding Media: The Extensions of Man (1964), the foundations of Marshall McLuhan’s tetrad’, *Explorations in Media Ecology*, 15(1), pp. 81–91. Available at: https://doi.org/10.1386/eme.15.1.81_1

Islam, M., (2022) ‘Focus Groups in Tourism Research’ in *Encyclopedia of tourism management and marketing*, pp. 261-264, Eddward Elgar Publishing. Available at: https://doi.org/10.4337/9781800377486.focus.groups.in.tourism.research

Ivaldi, E. and Ciacci, A. (2021) ‘Social Deprivation’, *Wiley StatsRef: Statistics Reference Online*. Wiley, pp. 1–8. Available at: https://doi.org/10.1002/9781118445112.stat08105

Ivanova, M., Buda, D.M. and Burrai, E. (2021) ‘Creative and disruptive methodologies in tourism studies’, *Tourism Geographies*, 23(1–2), pp. 1–10. Available at: https://doi.org/10.1080/14616688.2020.1784992

Jablonowski, M. (2020) ‘Beyond drone vision: the embodied telepresence of first-person-view drone flight’, *The Senses and Society*, 15(3), pp. 344–358. Available at: https://doi.org/10.1080/17458927.2020.1814571

Jackman, A. (2023). *Diversifying drone stories, 2021-2023*. [data collection]. UK Data Service. Available at: https://doi.org/10.5255UKDA-SN-856753

Jackson, S., Tan, P.L. and Nolan, S. (2012) ‘Tools to enhance public participation and confidence in the development of the Howard East aquifer water plan, Northern Territory’, *Journal of Hydrology*, 474, pp. 22–28. Available at: https://doi.org/10.1016/j.jhydrol.2012.02.007

Jansen, K. and Vellema, S. (2011) ‘What is technography?’, *NJAS: Wageningen Journal of Life Sciences*, 57(3–4), pp. 169–177. Available at: https://doi.org/10.1016/j.njas.2010.11.003

Jarrier, Elodie and Bourgeon, D. (2012): (2012) 'Impact of Mediation Devices on the Museum Visit Experience and on Visitors Behavioural Intentions', *International Journal of Arts Management.*, 15(1), pp. 18–29. (Accessed 6 October 2024).

Jeannotte, M.S. (2021) 'When the gigs are gone: Valuing arts, culture and media in the COVID-19 pandemic', *Social Sciences & Humanities Open*, 3(1), p. 100097. Available at: https://doi.org/10.1016/j.ssaho.2020.100097

Jenkinson, P. (2005) *Museum Provision and Professionalism*. Edited by G. Kavanagh. Routledge. Available at: https://doi.org/10.4324/9780203974469

Jensen, O.B. (2016a) 'Drone city - power, design and aerial mobility in the age of "smart cities"', *Geographica Helvetica*, 71(2), pp. 67–75. Available at: https://doi.org/10.5194/gh-71-67-2016

Jensen, O. B (2016b) 'New "Foucauldian Boomerangs": Drones and Urban Surveillance', *Surveillance & Society*, 14(1), pp. 20–33. Available at: https://doi.org/10.24908/ss.v14i1.5498

Jensen, O.B. (2020) 'Thinking with the drone – visual lessons in aerial and volumetric thinking', *Visual Studies*, 35(5), pp. 417–428. Available at: https://doi.org/10.1080/1472586X.2020.1840085

Jin, L., Xiao, H. and Shen, H. (2020) 'Experiential authenticity in heritage museums', *Journal of Destination Marketing & Management*, 18(September), p. 100493. Available at: https://doi.org/10.1016/j.jdmm.2020.100493

Jockers, M.L. (2013) *Macroanalysis*. University of Illinois Press. Available at: https://doi.org/10.5406/illinois/9780252037528.001.0001

Jóhannesdóttir, G.R. (2016) 'Phenomenological Aesthetics of Landscape and Beauty in Phenomenology and the Environment' in Bryan Bannon (ed.) *Nature and Experience: Phenomenology and the Environment*. Rowman & Littelfield, pp. 187–214.

Johnson, M., (2014) *Coactive Design: Designing Support for Interdependence in Human-Robot Teamwork*. PhD, Delft University of Technology Available at: https://researchgate.net/publication/267393898_Coactive_Design_Designing_Support_for_interdependence_in_Human-Robot_Teamwork (Accessed 23 July 2023).

Johnson, M., Bradshaw, M., Feltovitch, P., Jonker, M., van Riemsdijk, M. and Sierhuis, M., (2014) 'Coactive Design: Designing Support for Interdependence in Joint Activity', *Journal of Human-Robot Interaction.* Available at*:* https//doi.org/10.5898/JHRI.3.1.Johnson (Accessed 30 January 2025).

Joinson, S. (2021) *My Downs My Home South Downs National Park Authority Covid-19 Recovery Fund End of Project Report* (Internal document). Provided by the SDNPA.

Jones, G.P., Pearlstine, L.G. and Percival, H.F. (2006) 'An Assessment of Small Unmanned Aerial Vehicles for Wildlife Research', *Wildlife Society Bulletin*, 34(3), pp. 750–758. Available at: https://doi.org/10.2193/0091-7648(2006)34[750:aaosua]2.0.co;2

Jones, O. (2015) 'Not Promising a Landfall …: An Autotopographical Account of Loss of Place, Memory and Landscape', *Environmental Humanities* 6, pp. 1–27. Available at: https://www.researchgate.net/publication/304074137_Not_Promising_a_Landfall_An_Autotopographical_Account_of_Loss_of_Place_Memory_and_Landscape

Josselson, R. and Lieblich, A. (2005) *Carolyn Ellis, The Ethnographic I: A Methodological Novel about Autoethnography Walnut Creek,* California: Altamira Press, 2004, 384 pages. Available at: https://doi.org/10.1080/10408340308518298

Jubilee Medical Group (no date), What is the definition of 'housebound' (webpage). Available at: https://www.jubileemg.nhs.uk (Accessed 7 July 2022).

Kalay, Yehuda; Kvan, Thomas; Affleck, J. (eds.) (2007) *New Heritage: New Media and Cultural Heritage*. London: Routledge.

Kardasz, P. and Doskocz, J. (2016) ‘Drones and Possibilities of Their Using’, *Journal of Civil & Environmental Engineering*, 6(3). Available at: https://doi.org/10.4172/2165-784x.1000233

Kawashima, N. (2006) ‘Audience Development And Social Inclusion In Britain’, *International Journal of Cultural Policy*, 12(1), pp. 55–72. Available at: https://doi.org/10.1080/10286630600613309

Keinänen, M. and Beck, E.E. (2017) ‘Wandering intellectuals: establishing a research agenda on gender, walking, and thinking’, *Gender, Place & Culture*, 24(4), pp. 515–533. Available at: https://doi.org/10.1080/0966369X.2017.1314940

Keith, L. and Griffiths, W. (2021) ‘SCENE: A novel model for engaging underserved and under-represented audiences in informal science learning activities’, *Research for All*, 5(2). Available at: https://doi.org/10.14324/RFA.05.2.09

Kelaher, B.P., Peddemors, V.M., Hoode, B., Colefax, A. and Butcher, P.A. (2019) ‘Comparison of sampling precision for nearshore marine wildlife using unmanned and manned aerial surveys', *Journal of Unmanned Vehicle Systems*, pp. 1–14. Available at: https://doi.org/10.1139/juvs-2018-0023

Kemp, R. and Lieblich, A. (2005) ‘Carolyn Ellis, The Ethnographic I: A Methodological Novel about Autoethnography’, *Life Writing*, 2(2), pp. 199–206, Walnut Creek, California: Altamira Press, 2004, 384 pages, ISBN 0759100500,
Available at: https://doi.org/10.1080/10408340308518298

Kemp, E. and Poole, S.M. (2016) ‘Arts Audiences: Establishing a Gateway to Audience Development and Engagement’, *The Journal of Arts Management, Law, and Society*, 46(2), pp. 53–62. Available at: https://doi.org/10.1080/10632921.2016.1150929

Kennedy, A. and Malde, S. (2018) *Let's Get Real 6. Connecting digital practice with social purpose.* (Culture 24 report) Available at: https://www.keepandshare.com/doc/8226734/let-s-get-real-6-culture-24-rgb-single-page-pdf-10-5-meg?da=y (Accessed: 15 March 2019).

Kennemore, A. and Postero, N. (2021) 'Collaborative ethnographic methods: dismantling the "anthropological broom closet"?', *Latin American and Caribbean Ethnic Studies*, 16(1), pp. 1–24. Available at: https://doi.org/10.1080/17442222.2020.1721091

Khoo, S.L. (2020) 'Towards an inclusive creative city: How ready is the Historic City of George Town, Penang?', *City, Culture and Society* [Preprint]. Available at: https://doi.org/10.1016/j.ccs.2020.100367

Kim, H.Y., Kim, B. and Kim, J. (2016) 'The Naughty Drone', *Proceedings of the 10th International Conference on Ubiquitous Information Management and Communication - IMCOM 16*. New York, New York, USA: ACM Press, pp. 1–6. Available at: https://doi.org/10.1145/2857546.2857639

King, L., Stark, J.F. and Cooke, P. (2016) 'Experiencing the Digital World: The Cultural Value of Digital Engagement with Heritage', *Heritage & Society*, 9(1), pp. 76–101. Available at: https://doi.org/10.1080/2159032X.2016.1246156

Klauser, F. and Pedrozo, S. (2017) 'Big data from the sky: Popular perceptions of private drones in Switzerland', *Geographica Helvetica*, 72(2), pp. 231–239. Available at: https://doi.org/10.5194/gh-72-231-2017

Kline, N. (2001) *Time to Think.* Cassell.

Kljun, M., Pucihar, K.Č. and Lochrie, M. (2015) *StreetGamez: A Moving Projector Platform for Projected Street Games*. Available at: https://clok.uclan.ac.uk/13105/1/13105_2015-08-06-StreetGamez.pdf

Kreimer, B. (2018) ‘Drone Journalism: Storytelling from a New Perspective: Data, Visual Analytics and Innovative Methodologies in International Reporting’, *Digital Investigative Journalism* (pp. 91–102).

Kyselaa, J. and Štorková, P. (2015) ‘Using Augmented Reality as a Medium for Teaching History and Tourism’, *Procedia-Social and Behavioral Sciences*, 174, 926–931. Available at: https://doi.org/10.1016/j.sbspro.2015.01.713

Knoblauch, H. and Schnettler, B. (2012) ‘Videography: analysing video data as a “focused” ethnographic and hermeneutical exercise’, *Qualitative Research*, 12(3), pp. 334–356. Available at: https://doi.org/10.1177/1468794111436147

Knowles, S.E., Allen, D., Donnelly, A., Flynn, J., Gallacher, K., Lewis, A., McCorkle, G., Mistry, M., Walkington, P. and Drinkwater, J. (2021) ‘More than a method: trusting relationships, productive tensions, and two-way learning as mechanisms of authentic co-production’, *Research Involvement and Engagement*, 7(1), p. 34. Available at: https://doi.org/10.1186/s40900-021-00262-5

Knutz, E., Lenskjold, T.U. and Markussen, T. (2016) ‘Fiction as a resource in participatory design’, *DRS2016 Research Papers*. Available at: https://doi.org/10.21606/drs.2016.476

Ko, Y. and Noh, W. (2021) ‘A Scoping Review of Homebound Older People: Definition, Measurement and Determinants’, *International Journal of Environmental Research and Public Health*, 18(8), p. 3949. Available at: https://doi.org/10.3390/ijerph18083949

Koh, A. (2014) ‘Niceness, building, and opening the genealogy of the digital humanities: Beyond the social contract of humanities computing’, *Differences*, 25(1), pp. 93–106. Available at: https://doi.org/10.1215/10407391-2420015

Koubaa, A. and Qureshi, B. (2018) ‘DroneTrack: Cloud-Based Real-Time Object Tracking Using Unmanned Aerial Vehicles Over the Internet’, *IEEE Access*, 6, pp. 13810–13824. Available at: https://doi.org/10.1109/ACCESS.2018.2811762

Kowalewski, M. and Bartłomiejski, R. (2020) ‘Is it research or just walking? Framing walking research methods as “non-scientific"’, *Geoforum*, 114, pp. 59–65. Available at: https://doi.org/10.1016/j.geoforum.2020.06.002

Kracauer, S. (1952) ‘The challenge of qualitative content analysis’, *The Public Opinion Quarterly*, 16(4), pp. 631–42. Available at: http://www.jstor.org/stable/2746123 (Accessed: 12 August 2020).

Kraehe, A.M. and Acuff, J.B. (2013) ‘Theoretical Considerations for Art Education Research with and about “Underserved Populations”’, *Studies in Art Education*, 54(4), pp. 294–309. Available at: https://doi.org/10.1080/00393541.2013.11518904

Kraft, R.N. (1987) ‘The influence of camera angle on comprehension and retention of pictorial events’, *Memory & Cognition*, 15(4), pp. 291–307. Available at: https://doi.org/10.3758/BF03197032

Krajník, T., Vonasek, V., Fišer, D. and Faigl, J. (2011) ‘AR-drone as a platform for robotic research and education’, *Communications in Computer and Information Science*. Springer, Berlin, Heidelberg, pp. 172–186. Available at: https://doi.org/10.1007/978-3-642-21975-7_16

Kreps, S.E. and Kaag, J. (2012) ‘The Use of Unmanned Aerial Vehicles in Contemporary Conflict: A Legal and Ethical Analysis’, *SSRN Electronic Journal* [Preprint]. Available at: https://doi.org/10.2139/ssrn.2023202

Kress, G. and van Leeuwen, T. (2020) *Reading Images*. Third edition. London; New York: Routledge, 2021.: Routledge. Available at: https://doi.org/10.4324/9781003099857

Kress, Gunther; Adami, E. (2009) *Multimodality A Social Semiotic Approach to Contemporary Communication*. G. Kress (ed.). London: Routledge, pp. 184–197. Available at: https://doi.org/10.4324/9780203970034

Kumar, A. *et. al.* (2020) 'Mixed Reality Simulation of High-Endurance Unmanned Aerial Vehicle with Dual-Head Electromagnetic Propulsion Devices for Earth and Other Planetary Explorations', *Applied Science, Special Issue on Multi-Robot Systems*. Available at: https://www.mdpi.com/2076-3417/10/11/3736 (Accessed: 8 June 2020).

Kumar, V.Y. and Michael, L., N. (2012) 'Opportunities and challenges with autonomous micro aerial vehicles', *International Journal of Robotics Research*, pp. 1279–1291. Available at: https://doi.org/10.1177/0278364912455954

Kunstmuseum Liechtenstein (2015) *Alexander Rodchenko: Photography and Design* (exhibition catalogue). Available at: https://kunstmuseum.li/bilder/5685.pdf (Accessed 8 November 2024).

Landriscina, F. (2014) 'The role of mental simulation in understanding and in creating scientific concepts' in Corni, F. (ed.) *Proceedings from the 3d Conference on Innovation in Science Education in Primary School and Kindergarten*. University of Modena and Reggio Emilia, November, 21-22, 2014., pp. 21–22.

Landscape Institute (2017) *UAVs-Unmanned Aerial Vehicles in Landscape Practice* (Technical Information Notice). Available at: https://doi.org/10.13140/RG.2.2.14058.00963

Larsen, J. (2006) 'Geographies of Tourism Photography: Choreographies and Performances' in J.A. Falkheimer, J. (ed.) *Geographies of Communication: The Spatial Turn in Media Studies*. Nordicom, pp. 243–261. Available at: https://www.semanticscholar.org/paper/Geographies-of-Tourism-Photography%3A-Choreographies-Larsen/95957586f6c171a5df6aea180edd28eb0c449400 (Accessed 12 March 2024).

Lawson-Tancred, J. (2022) ‘See how the Dutch collective DRIFT is using illuminated drones to rebuild the Sagrada Familia, the Colosseum, and other landmarks’, *Artnet News*. Available at: https://news.artnet.com/art-world/see-how-the-dutch-collective-drift-uses-drones-and-lights-to-digitally-rebuild-the-colosseum-and-other-major-landmarks-2199804 (Accessed: 26 October 2024).

Lazarus, S., Taliep, N., and Naidoo, A. (2017) ‘Community asset mapping as a critical participatory research method’ in Seedat, M., Suffla, S. and Christie, D.J. (eds.) *Emancipatory and participatory methodologies in peace, critical, and community psychology*. Cham: Springer International Publishing, pp. 45–49.

Learning and Work Institute (U.K.); WorldSkills UK (2021) *Disconnected? Exploring the digital skills gap*. Available at: www.enginuity.org (Accessed: 16 November 2023)

Ledin, P. and Machin, D. (2020) ‘Introduction to Multimodal Analysis’, *Visual Communication Quarterly*. Bloomsbury Academic. Available at: https://doi.org/10.5040/9781350069176

Leiper, N. (1979) The framework of tourism. Towards a definition of tourism, tourist, and the tourist industry. *Annals of Tourism Research*, 6(4), pp. 390–407. Available at: https://doi.org/10.1016/0160-7383(79)90003-3

Lentink, D., Haselsteiner, A.F. and Ingersoll, R. (2015) ‘In vivo recording of aerodynamic force with an aerodynamic force platform: From drones to birds’, *Journal of the Royal Society Interface*, 12(104). Available at: https://doi.org/10.1098/rsif.2014.1283

Levine, P. (2013) ‘On common ground Here as There’, Farman, J. (Ed.) *The Mobile Story: Narrative Practices with Locative Technologies*. London: Routledge, pp. 143–158.

Levine D (2016) ‘Design Fiction’, *Digital Experience Design (Medium)*. Available at: https://medium.com/digital-experience-design/design-fiction-32094e035cd7 (Accessed: 14 June 2020).

Light, B., Bagnall, G., Crawford, G. and Gosling, V. (2018) 'The material role of digital media in connecting with, within and beyond museums', *Convergence: The International Journal of Research into New Media Technologies*, 24(4), pp. 407–423. Available at: https://doi.org/10.1177/1354856516678587

Limb, P. (2023) 'Where the river flows out to the sea', Goode J., K. Lumsden, and Bradford, J. (eds.) *Crafting Autoethnography*. London: Routledge. Available at: https://doi.org/10.4324/9781003309239

Lindley, J. and Coulton, P. (2015) 'Back to the future', *Proceedings of the 2015 British HCI Conference*. New York, NY, USA: ACM, pp. 210–211. Available at: https://doi.org/10.1145/2783446.2783592

Lindley, J. and Coulton, P. (2015) 'Game of drones', *CHI PLAY 2015 - Proceedings of the 2015 Annual Symposium on Computer-Human Interaction in Play*. New York, NY, USA: ACM, pp. 613–618. Available at: https://doi.org/10.1145/2793107.2810300

Lisein, J., Pierrot-Deseilligny, M., Bonnet, S. and Lejeune, P. (2013) 'A photogrammetric workflow for the creation of a forest canopy height model from small unmanned aerial system imagery', *Forests*, 4(4), pp. 922–944. Available at: https://doi.org/10.3390/f4040922

Liu, J., Yue, M., Yu, F. and Tong, Y. (2022) 'The contribution of tourism mobility to tourism economic growth in China', *PLoS ONE*, *17*(10). Available at: https://doi.org/10.1371/journal.pone.0275605

Lo Turco, M. and Giovannini, E.C. (2020) 'Towards a phygital heritage approach for museum collection', *Journal of Archaeological Science: Reports*, 34. Available at: https://doi.org/10.1016/j.jasrep.2020.102639

London and South East National Trust (2021) *Iron Age hillfort - Cissbury Ring video trail.* 9 July. [Video online] Available at: https://youtu.be/Wk8vo4eIGi8 (Accessed: 17 July 2023).

Long, A. (2017) *The challenges of research in the care home setting*. Available at: https://www.bgs.org.uk/blog/the-challenges-of-research-in-the-care-home-setting (Accessed: 9 June 2022).

Longley, A. and Duxbury, N. (2016) 'Introduction: Mapping cultural intangibles', *City, Culture and Society*, 7(1), pp. 1–7. Available at: https://doi.org/10.1016/j.ccs.2015.12.006

Louv, R. (2008) *Last Child in the Woods: Saving Our Children from Nature-Deficit Disorder.* Chapel Hill: Algonquin Books.

Lovejoy, A. M, (2022) *Letter of support provided for Ethical Application for research with older adults living with dementia,* [letter], 24 December 2022.

Lovejoy Centre (2020) *What we do*. [Webpage] Available at: https://lovejoycentre.wordpress.com/what-we-do/ (Accessed: 28 July 2023).

Lovejoy Centre, (2023a) *Bringing the outdoors Indoors: Devil's Dyke* [Co-research livestream video]. Ed: Visual Air, unpublished

Lovejoy Centre, (2023b) *Bringing the outdoors Indoors: Highdown Hill*. [Co-research livestream video]. Ed: Visual Air, unpublished.

Luke, M. and Goodrich, K.M. (2019) 'Focus Group Research: An Intentional Strategy for Applied Group Research?', *The Journal for Specialists in Group Work*, 44(2), pp. 77–81. Available at: https://doi.org/10.1080/01933922.2019.1603741

Lumb D. (2022) *Google Maps Is Adding a 3D Immersive View That's Like Flying a Drone From Afar*. Available at: https://www.cnet.com/tech/services-and-software/google-maps-adding-3d-immersive-view-thats-like-flying-a-drone-from-afar/.

Lumsden, K. (2023) 'Reflections and Confessions on the Making of a Performative Autoethnography in University Performance Development Reviews and the Academic Self' in

Goode, J., Lumsden, K. and Bradford, J. (eds.) *Crafting Autoethnography*. London: Routledge. Available at: https://doi.org/10.4324/9781003309239

Luvaas, B. (2019) 'Unbecoming: The aftereffects of autoethnography', *Ethnography*, 20(2), pp. 245–262. Available at: https://doi.org/https://doi.org/10.1177/14661381177426

Lynch, B. (2002) 'If the Museum is the Gateway, who is the Gatekeeper?' *engage review* 11 – Winter 2001, S. 12-21. Available at: https://engage.org/wp-content/uploads/2019/03/Engage11_IftheMusuemistheGateway_BernadetteLynch.pdf (Accessed: 20 October 2024).

m.e.l. Research (2022) *South Downs National Park Visitor Survey 2021*. Available at: https://www.southdowns.gov.uk/wp-content/uploads/2023/01/South-Downs-VisitorSurvey-2021-Final-report.pdf (Accessed: 6 July 2020).

MacDonald, S. ed. (1998) *The Politics of Display. Museums, Science, Culture*. (Heritage: Care-Preservation-Management). Routledge, London. Available at: https://doi.org/10.43249780203838600

Macdonald, C. (2012) 'Understanding Participatory Action Research: A Qualitative Research Methodology Option', *Canadian Journal of Action Research*. Available at: https://doi.org/10.33524/CJAR.V13I2.37

Mack, N., Woodsong, C., Macqueen, K., Guest, G. and Namey, E. (2005) *Qualitative Research Methods: A Data Collectors Field Guide*. Family Health International.

Mackenzie, J., Tan, P., Hoverman, S. and Baldwin, C. (2012) 'The value and limitations of Participatory Action Research methodology', *Journal of Hydrology*, 474, pp. 11–21. Available at: https://doi.org/10.1016/j.jhydrol.2012.09.008

Macpherson, H. (2016) ‘Walking methods in landscape research: moving bodies, spaces of disclosure and rapport’, *Landscape Research*, 41(4), pp. 425–432. Available at: https://doi.org/10.1080/01426397.2016.1156065

Mademlis, I., Nikolaidis, N., Tefas, A., Pitas, I., Wagner, T. and Messina, A. (2019) ‘Autonomous UAV Cinematography: A Tutorial and a Formalized Shot Type Taxonomy’, *ACM Computer Surveys* (September 2019). Available at: https://doi.org/10.1145/3347713

Magaldi and Berler, (2020) ‘Research and scholarly methods: Semi-structured Interviews’ in Zeigler-Hill, V., Shackelford, T.K. (eds.) *Encyclopedia of Personality and Individual Differences*, Springer, Cham. Available at: https://doi.org/10.1007/978-3-319-24612-3-857

Malizia, A. (2019) *Design fictions to mitigate social injustice in possible futures*. Available at: https://blog.ubiquity.acm.org/author/malizia/ (Accessed: 2 June 2020).

Mangina, E., Keeffe, E., Eyerman, J, G. (2016) ‘Drones for Live Streaming of Visuals for People with Limited Mobility’, *Proceedings of the 2016 International Conference on Virtual Systems and Multimedia, VSMM 2016*. Institute of Electrical and Electronics Engineers Inc. Available at: https://doi.org/10.1109/VSMM.2016.7863162

Mangold, E.B. and Goehring, C. (2019) ‘The visual rhetoric of the aerial view: From surveillance to resistance’, *Quarterly Journal of Speech*, 105(1), pp. 25–41. Available at: https://doi.org/10.1080/00335630.2018.1553305

Margaritoff, M. (2017). *Watch Sci-Fi Short Donny the Drone Starring Guy Pearce*. The Drive.

Marino, M.C. (2013) ‘Mobilizing cities Alternative Community Storytelling’ in Farman, J. (ed.) *The Mobile Story: Narrative Practices with Locative Technologies*. Taylor & Francis, pp. 290–303. Available at: https://ebookcentral-proquest-com.ezproxy.brighton.ac.uk/lib/ubrighton/detail.action?docID=1377510# (Accessed: 16 January 2025).

Martín-Béjar, S., Claver, J., Sebastián, M.A. and Sevilla, L. (2020) 'Graphic Applications of Unmanned Aerial Vehicles (UAVs) in the Study of Industrial Heritage Assets', *Applied Sciences*, 10(24), p. 8821. Available at: https://doi.org/10.3390/app10248821

Martins, J, Gonçalves, R., Branco, F., Barbosa, L.; Melo, M. and Bessa, M. (2017) 'A multisensory virtual experience model for thematic tourism: A Port wine tourism application proposal', *Journal of Destination Marketing & Management* 6 (2), pp. 103–109. https://doi.org/10.1016/j.jdmm.2017.02.002

Mascort-Albea, E.J., Ruiz Jaramillo, J. and Romero-Hernández, R. (2014) 'Analizando el Patrimonio Cultural a través de vuelos no tripulados: nuevas estrategias de investigación de espacios rurales y urbanos', *Virtual Archaeology Review*, 5(11), p. 55. Available at: https://doi.org/10.4995/var.2014.4181

Mastai, J. (2007) 'There is No Such Thing as a Visitor', *Museums After Modernism*. Oxford, UK: Blackwell Publishing Ltd, pp. 173–177. Available at: https://doi.org/10.1002/9780470776636.ch10

Mateus, C, Campis, R., Jabba,, D, Erazo, A.M., Romero., V. (2024) 'Gamification as a tool for Inclusion, Education and Human Development' in Fahriye, A. and Zehra, A. (Eds.) *Intellectual and Learning Disabilities-Inclusiveness and Contemporary Teaching Environments*. IntechOpen. Available at: https://doi.org/10.5772/intechopen.113229

Maxwell, J.A. (2013) 'A Model for Qualitative Research Design', *Qualitative Research Design*, pp. 1–22. Available at: https://us.sagepub.com/sites/default/files/upm-binaries/5055_Maxwell_Chapter_1.pdf (Accessed 10 June 2021).

McCall, V. and Gray, C. (2014) 'Museums and the "new museology": Theory, practice and organisational change', *Museum Management and Curatorship*, 29(1), pp. 19–35. Available at: https://doi.org/10.1080/09647775.2013.869852

McCarthy, C. (2015) ‘Grounding Museum Studies: Introducing Practice’, *The International Handbooks of Museum Studies*. Wiley. Available at: https://doi.org/10.1002/9781118829059.wbihms200

McClanahan, Lauren (2020) ‘More Than Just a Middle School Film Festival: Encouraging Voice, Self-Expression, and Empowerment through Film’, *Voices from the Middle*, Volume 27, Issue 4, May 2020, p. 35 – 43. Available at: https://doi.org/10.58680/vm202030703

McCosker, A. (2015) ‘Drone Media: Unruly Systems, Radical Empiricism and Camera Consciousness’, *Culture Machine*, 16. London: Open Humanities Press, pp. 1–21.

McCosker, A. (2015) ‘Drone Vision, Zones of Protest, and the New Camera Consciousness’, *Media Fields* [Preprint], (9). Available at: http://mediafieldsjournal.org/drone-vision-zones-of-protest/2015/8/21/drone-vision-zones-of-protest-and-the-new-camera-consciousne.html

McIntosh, I. and Wright, S. (2019) ‘Exploring what the Notion of “Lived Experience” Offers for Social Policy Analysis’, *Journal of Social Policy*, 48(03), pp. 449–467. Available at: https://doi.org/10.1017/S0047279418000570

McLuhan, M. (2013). *Understanding Media: The extensions of man*. Gingko Press.

McPherson T (2012) *Why are the Digital Humanities So White? Or Thinking the Histories of Race and Computation*. Available at: https://doi.org/10.5749/MINNESOTA/9780816677948.003.0017

McQuire, S. (2016). *Geomedia: Networked cities and the future of public space.* Polity. Available at: https://www.amazon.co.uk/Geomedia-Networked-Cities-Future-Public-ebook/dp/B01LWTZOOV?asin=0745660762&revisionId=&format=4&depth=4 (Accessed 12 June 2024).

Meijer, A. (2012) 'Co-production in an Information Age: Individual and Community Engagement Supported by New Media', *VOLUNTAS: International Journal of Voluntary and Nonprofit Organizations*, 23(4), pp. 1156–1172. Available at: https://doi.org/10.1007/s11266-012-9311-z

Memmi, A. (2000) *Racism*. University of Minnesota Press.

Mental Capacity Act 2005 Available at: https://www.legislation.gov.uk/ukpga/2005/9/contents. (Accessed 15 May 2023).

Mental Capacity Act Code of Practice (2007). Available at: https://assets.publishing.service.gov.uk/media/5f6cc6138fa8f541f6763295/Mental-capacity-act-code-of-practice.pdf (Accessed 12 March 2019).

Message, K. (2006) 'The New Museum', *Theory, Culture & Society*, 23(2–3), pp. 603–606. Available at: https://doi.org/10.1177/0263276406023002110

Microdrones (no date) *Drone Surveying Equipment and Software for Education, Science and Academic Research* [Webpage]. Available at: https://www.microdrones.com/en/industry-experts/science-and-academic-research/ (Accessed: 28 February 2020).

Milena, Z., Grundey, D. and Stancu, A. (2008) 'Qualitative Research Methods: a comparison between focus-group and in-depth interview', *Annals of Faculty of Economics*, 4, pp. 1279–1283. Available at: https://www.researchgate.net/publication/46533517_QUALITATIVE_RESEARCH_METHODS_A_COMPARISON_BETWEEN_FOCUS-GROUP_AND_IN-DEPTH_INTERVIEW (Accessed 10 June 2021).

Miller, T., Volmert, A., Rochman, A. and Aassar, M. (2021) *Talking About Poverty: Narratives, Counter-Narratives, and Telling Effective Stories,* Washington, DC, FrameWorks Institute. Available at: https://www.frameworksinstitute.org/appuploads/2021/09/Talking-about-poverty.pdf (Accessed 28 October 2024).

Millerson, G. (1994) *Video camera techniques*. 2nd edition. Oxford: Focal.

Minkiewicz, J., Bridson, K. and Evans, J. (2016) 'Co-production of service experiences: insights from the cultural sector', *Journal of Services Marketing*, 30(7), pp. 749–761. Available at: https://doi.org/10.1108/JSM-04-2015-0156

Mirk, D. and Hlavacs, H. (2015) 'Virtual Tourism with Drones', *Proceedings of the First Workshop on Micro Aerial Vehicle Networks, Systems, and Applications for Civilian Use*. New York, NY, USA: ACM, pp. 45–50. Available at: https://doi.org/10.1145/2750675.2750681

Mohdin, A. (2023) 'How the fall of Edward Colston's statue revolutionised the way British history is told', *The Guardian (online)*, 5 May. Available at: https://www.theguardian.com/news/2023/may/05/how-the-fall-of-edward-colstons-statue-revolutionised-the-way-british-history-is-told (Accessed: 12 August 2024).

Montsho, G. (2020) *Making Museums Accessible to Those With Disabilities*. Available at: https://www.museumnext.com/article/making-museums-accessible-to-those-with-disabilities/ (Accessed: 28 April 2020).

Morrell, A. (2008) 'Museums After Modernism: Strategies Of Engagement By Griselda Pollock And Joyce Zemans (eds.) in *The Art Book*, 15(3), pp. 45–46. Available at: https://doi.org/10.1111/j.1467-8357.2008.00975_15.x

MTM London. (2017). *Digital Culture*. Available at: https//www.nesta.org.ukreportdigital-culture-2017/ 46pp (Accessed 8 October 2023).

Mules, W. (2020) Figures in Flight: the Camera Movement Effect in Film. *School of Arts and Social Sciences Research Seminar Series*, Southern Cross University. Available at: https://www.academia.edu/52070493/Figures_in_Flight_the_Camera_Movement_Effect_in_Film (Accessed 6 July 2023).

Munck Petersen, R. (2016) Quiet Areas: Outer experiences and inner sensations – a qualitative approach using film and drones. *Inter.noise 2016.* Hamburg. Available at: https://www.academia.edu/28414243/Quiet_Areas_Outer_experiences_and_inner_sensations_a_qualitative_approach_using_film_and_drones. (Accessed 8 November 2024).

Munck Petersen, R. (2020) The dispatched drone and affective distance in fieldwork. *The Senses and Society*, 15(3), pp. 311–328. Available at: https://doi.org/10.1080/17458927.2020.1820188

Munck Petersen, R. (2021) Drone affect: Folded points of view as a co-affection method for empathy and care. *Emotion, Space and Society*, 41, p. 100842. Available at: https://doi.org/10.1016/j.emospa.2021.100842

Munro E. (2013) People just need to feel important, like someone is listening: Recognising museums community engagement programmes as spaces of care. *Geoforum* 48, pp.54-62. Available at: https://www.sciencedirect.com/science/article/abs/pii/S0016718513000808 (Accessed: 24 January 2021).

Museums Association. (no date). *Museums Change Lives*. Available at: https://museumsassociation.org/campaigns/museums-change-lives/ (Accessed 15 July 2025).

Museums Libraries and Archives Council (no date) *Generic Social Outcomes: Indicator Bank for museums, libraries and archives* (report). Available at: https://www.artscouncil.org.uk/sites/default/files/S3D29_GSO_Indicator_Bank.pdf (Accessed: 20 November 2022).

Muthyala, J. (2019) Drones and Surveillance Cultures in a Global World. *Digital Studies/Le champ numérique*, 9(1). Available at: https://doi.org/10.16995/dscn.332. (Accessed: 12 February 2020).

Mutibwa, D.H., Hess, A. and Jackson, T. (2020) Strokes of serendipity: Community co-curation and engagement with digital heritage. *Convergence: The International Journal of Research into New Media Technologies*, 26(1), pp. 157–177. Available at: https://doi.org/10.1177/1354856518772030

NAPA (no date) *Arts in Care Homes* [Webpage]. Available at: https://napa-activities.co.uk/arts-in-care-homes

NAPA (2021) *Free Resources*, *Website*. Available at: http://www.napa-activities.com/membership/free-resources#activityideas (Accessed: 28 July 2023).

Nasrabadi, A.T., Samiei, A., Mahzari, A., McMahan, R., Prakash, R., Farias, M.C.Q. and Carvalho, M. M. (2019) A taxonomy and dataset for 360° videos. *Proceedings of the 10th ACM Multimedia Systems Conference*. New York, NY, USA: ACM, pp. 273–278. Available at: https://doi.org/10.1145/3304109.3325812

National Parks and Access to the Countryside Act 1949 c 97 (Regnal.12-13 and 14) Available at: https://www.legislation.gov.uk/ukpga/Geo6/12-13-14/97/part/IV/crossheading/creation-of-new-public-rights-of-way/enacted (Accessed 23 July 2023).

National Trust (no date) *Strategic Framework for Research*. Available at: https://www.nationaltrust.org.uk/who-we-are/research/strategic-framework-for-research. (Accessed 18 January 2022).

National Trust (no date) *Changing chalk partnership*. [Website] Available at: https://www.nationaltrust.org.uk/visit/sussex/the-changing-chalk-partnership (Accessed: 18 April 2024).

National Trust (no date) *Changing Chalk Area Action Plan* (Internal Document). Provided by the National Trust.

National Trust (2015) *Playing our part: What does the nation need from the National Trust in the 21st century?* Available at: https://www.readkong.com/page/playing-our-part-nationaltrust-7850492 (Accessed 18 January 2022).

National Trust (2016) *How to spot a Capability Brown landscape* [Video online]. Available at: https://youtu.be/md2_2v7D6Oc (Accessed 17 July 2023).

National Trust (2017) *Research Strategy 2017-2021* (Internal Document). Provided by the National Trust.

National Trust (2019) *Research Opportunities 2019-2021* (Internal Document). Provided by the National Trust.

National Trust (2020) *575 Wandsworth Road: Virtual Tour Trials 2020 - Unlocking the Potential of Digital Technologies to Improve Access* (Report). Provided by the National Trust.

NCVO. (2024). *National Inclusion Week: Why the language we use matters in building belonging.* Available at: https://www.ncvo.org.uk/news-and-insights/news-index/why-language-matters-in-building-belonging/ (Accessed 8 July 2025).

Negrón-Blanco, L., de Pedro-Cuesta, J., Almazán, J, Rodríguez-Blázquez, C., Franco, E and Damián, J. (2016) Prevalence of and factors associated with homebound status among adults in urban and rural Spanish populations, *BMC Public Health*, 16(1), p. 574. Available at: https://doi.org/10.1186/s12889-016-3270-z

Nelson, K. (no date) From the trenches: education goes unplugged, museums plug into interactives, and drones are wireless video sources, too, *Systems contractor news*, 22(6), p. 38. Bath: Future Publishing Ltd.

NESTA; Gov.uk (2019) Drones: consultation response. Available at: https://www.gov.uk/government/speeches/drones-consultation-response-7-january-2019 (Accessed: 12 February 2020).

Nielsen, J.K. (2017) Museum communication and storytelling: articulating understandings within the museum structure. *Museum Management and Curatorship*, 32(5), pp. 440–455. Available at: https://doi.org/10.1080/09647775.2017.1284019

Niles, S.; Hanson, S. (2003) A new era of accessibility? *Journal of the Urban and Regional Information Systems Association*, 15 (Special Issue: Access and Participatory Approaches I), pp. 35–41. Available at: https://www.researchgate.net/publication/272791390_A_New_Era_of_Accessibility

Nisi, V., Dionisio, M., Barreto, M and Nunes, N. (2018) A Mixed Reality neighborhood tour: Understanding visitor experience and perceptions, *Entertainment Computing*, 27, pp. 89–100. Available at: https://doi.org/10.1016/j.entcom.2018.04.002

Nowak, M.M., Dzióbi, K. and Bogawski, P. (2019) Unmanned Aerial Vehicles (UAVs) in environmental biology: A review, *European Journal of Ecology*, 4(2), pp. 56–74. Available at: https://doi.org/10.2478/eje-2018-0012

Nyumba, T. O., Wilson, K., Derrick, C.J. and Mukherjee, N. (2018) The use of focus group discussion methodology: Insights from two decades of application in conservation, *Methods in Ecology and Evolution*. Edited by D. Geneletti, 9(1), pp. 20–32. Available at: https://doi.org/10.1111/2041-210X.12860

O'Hagan, L.A. and Serafinelli, E. (2022) Transhistoricizing the Drone: A Comparative Visual Social Semiotic Analysis of Pigeon and Domestic Drone Photography. *Photography and Culture*, 15(4), pp. 327–351. Available at: https://doi.org/10.1080/17514517.2022.2116899

O'Hagan, L.A. and Serafinelli, E. (2023) Rethinking verticality through top-down views in drone hobbyist photography. *Visual Studies*, pp. 1–14. Available at: https://doi.org/10.1080/1472586X.2023.2201239

O'Neill, M. (2018) Walking, well-being and community: racialized mothers building cultural citizenship using participatory arts and participatory action research. *Ethnic and Racial Studies*, 41(1), pp. 73–97. Available at: https://doi.org/10.1080/01419870.2017.1313439

O'Neill, M., and Hubbard, P. (2010). Walking sensing, belonging: ethno-mimesis as performative praxis. *Visual Studies*, *25*, 46–58.

Olphert, L.D. and A.J.M. (2005) Towards digital inclusion – engaging older people in the "digital world". *Accessible Design in the Digital World Conference 2005*. Available at: https://doi.org/10.14236/ewic/AD2005.17

Onwuegbuzie, A.J., Dickinson, W.B., Leech, N.L. and Zoran, A. G., (2009) A Qualitative Framework for Collecting and Analyzing Data in Focus Group Research. *International Journal of Qualitative Methods*, 8(3), pp. 1–21. Available at: https://doi.org/10.1177/160940690900800301

Oppegaard, B. and Grigar, D. (2013) The interrelationships of mobile storytelling Merging the Physical and the Digital at a National Historic Site. J. Farman (ed.) *The Mobile Story: Narrative Practices with Locative Technologies*. Taylor & Francis Group, pp. 17–33. Available at: https://www.researchgate.net/publication//259292830_The_Interrelationships_of_Mobile_Storytelling_Merging_the_Physical_and_the_Digital_at_a_National_Historic_Site (Accessed 12 September 2024).

Painter, N.I. (2016) *Slavery: A Dehumanizing Institution*, *Oxford University Press Blog (OUPblog)*. 14th February. Available at: https://blog.oup.com/2006/02/slavery_a_dehum/ (Accessed: 24 July 2023).

Pallud, J. (2017) 'Impact of interactive technologies on stimulating learning experiences in a museum', *Information & Management*, 54(4), pp. 465–478. Available at: https://doi.org/10.1016/j.im.2016.10.004

Panek, J. and Neték, R. (2019) 'Collaborative Mapping and Digital Participation: A Tool for Local Empowerment in Developing Countries', *Information*, 10 (8), 255. Available at: https://doi.org/10.3390/info10080255

Paneque-Gálvez, J., McCall, M.K., Napoletano, B.M. and Wich, S. (2014) 'Small drones for community-based forest monitoring: An assessment of their feasibility and potential in tropical areas', *Forests*, 5(6), pp. 1481–1507. Available at: https://doi.org/10.3390/f5061481

Paneque-Gálvez, J., Pérez-Llorente, I., Luz, A. C., Guèze, M., Mas, J., Macía, M. J., Orta-Martínez, M., and Reyes-García, V. (2017) 'Grassroots Innovation Using Drones for Indigenous Mapping and Monitoring', *Land*, 6(4), p. 86. Available at: https://doi.org/10.3390/land6040086

Paneque-Gálvez, J., Pérez-Llorente, I., Luz, A.C. and Reyes-Garcia, V. (2018) 'High overlap between traditional ecological knowledge and forest conservation found in the Bolivian Amazon', *Ambio*, 47(8), pp. 908–923. Available at: https://doi.org/10.1007/s13280-018-1040-0

Parks, L. and Kaplan, C. (eds.) (2017) *Life in the Age of Drone Warfare*. Duke University Press. Available at: https://doi.org/10.1215/9780822372813

Paskaleva-Shapira, K., Azorín, J. and Chiabai, A. (2008) 'Enhancing digital access to local cultural heritage through e-governance: innovations in theory and practice from Genoa', Italy. *Innovation: The European Journal of Social Science Research*, 21(4), pp. 389–405. Available at: https://doi.org/10.1080/13511610802568031

Paterson, B.L., Bottorff, J.L. and Hewat, R. (2003) 'Blending Observational Methods: Possibilities, Strategies, and Challenges', *International Journal of Qualitative Methods*, 2(1), pp. 29–38. Available at: https://doi.org/10.1177/160940690300200103

Pauschinger, D. and Klauser, F. (2020) 'Aerial Politics of Visibility: Actors, Spaces, and Drivers of Professional Drone Usage in Switzerland', *Surveillance & Society*, 18(4), pp. 443–466. Available at: https://doi.org/10.24908/ss.v18i4.13434

Pauwels, L. (2010) 'Visual Sociology Reframed: An Analytical Synthesis and Discussion of Visual Methods in Social and Cultural Research', *Sociological Methods & Research*, 38(4), pp. 545–581. Available at: https://doi.org/10.1177/0049124110366233

Pease, J.L. (2015) 'Parks and Underserved Audiences: An Annotated Literature Review', *Journal of Interpretation Research*, 20(1), pp. 11–56. Available at: https://doi.org/10.1177/109258721502000103

Pechman, A. (2013) *The Art Worlds Game of Drones*, *Artnews.com*. Available at: https://www.artnews.com/art-news/news/drones-as-art-2343/ (Accessed: 28 February 2020).

Perkins, D.D. and Zimmerman, M.A. (1995) 'Empowerment theory, research, and application', *American Journal of Community Psychology*, 23(5), pp. 569–579. Available at: https://doi.org/10.1007/BF02506982

Peron, A. (2014) 'The "Surgical" Legitimacy of Drone Strikes? Issues of Sovereignty and Human Rights in the Use of Unmanned Aerial Systems in Pakistan', *Journal of Strategic Security*, 7(4), pp. 81–93. Available at: https://doi.org/10.5038/1944-0472.7.4.6

Peters, J. (2001) ‘Witnessing’, *Media Culture and Society*, 23(6), 707–723. Available at: https://doi.org/10.1177/016344301023006002

Petrétei, A. (2016) ‘Securing Sámi livelihoods - Does mining undermine traditional ways of living?’, *Studies in International Minority and Group Rights*. Brill Nijhoff, pp. 155–172. Available at: https://doi.org/10.1163/9789004314399_011

Photoflight Aerial Media. (2021) *Hansel & Gretel Aerial Droning Project – Art installation by Ai Weiwei* [Webpage]. Available at: https://www.photoflightam.com/project/hansel-gretel-art-installation-by-ai-weiwei-at-new-yorks-park-avenue-armory/ (Accessed 15 July 2025).

Pick, D., Hallsworth, M. and Marks, S. (2021) ‘Hidden persuaders on film: Exploring young peoples lived experience through visual essays’, *Research for All*, 5(2). Available at: https://doi.org/10.14324/RFA.05.2.13

Pink, S. (2011) *Doing Visual Ethnography,* Sage Research Methods, Sage Publications limited. Available at: https://dx.doi.org/10.4135/9780857025029

Pink, S. (2015) ‘Going Forward Through the World: Thinking Theoretically About First Person Perspective Digital Ethnography’, *Integrative Psychological and Behavioral* Science vol. 49, pp. 239–252. Available at: https://doi.org/10.1007/s12124-014-9292-0

Pink, S. and Morgan, J. (2013) ‘Short-term ethnography: Intense routes to knowing’, *Symbolic Interaction*, 36(3), pp. 351–361. Available at: https://doi.org/10.1002/symb.66

Pink, S., Hubbard, P., O’Neill, M. and Radley, A. (2010) ‘Walking across disciplines: from ethnography to arts practice’, *Visual Studies*, 25 (1), 1-7. Available at: https://doi.org/10.1080/14725861003606670

Pinkett, R. and O'Bryant, R. (2003) ‘Building Community, Empowerment and Self-sufficiency’, *Information, Communication & Society*, pp. 187–210. Available at: https://doi.org/10.1080/1369118032000093888

Pollastri, S., Cooper, R., Dunn, N. and Boyko, C. (2016) ‘Visual conversations on urban futures. Participatory methods to design scenarios of liveable cities’, *DRS2016 Research Papers*. Available at: https://doi.org/10.21606/drs.2016.436

Pollock, G. and Zemans, J. (eds.) (2007) *Museums after modernism strategies of engagement.* Malden, MA: Blackwell (New interventions in art history).

Ponto, J. (2015) ‘Understanding and Evaluating Survey Research’, *Journal of Advanced Practical Oncology*, 6(2), pp. 168–171. Available at: https://www.researchgate.net/publication/286445115_Understanding_and_Evaluating_Survey_Research (Accessed 11 April 2024)

Popan, C. (2016) *Reframing drone methodologies*. Lancaster University Centre for Mobilities Research. Available at: https://www.lancaster.ac.uk/cemore/reframing-drone-methodologies/ (Accessed: 29 May 2021).

'Post Apocalyptic Towers of Power' (Episode 2), 31 March 2015. *Drones in Forbidden Zones,* 2015, [TV series]. Available at: All 4 (Accessed 12 August 2023).

Postill, J.S.P. (2012) ‘Social media ethnography: the digital researcher in a messy web’, *Media International*, (145), pp. 123–134 United Kingdom: SAGE Publications, Ltd. Available at: https://methods.sagepub.com/book/doing-visual-ethnography

Powell, K. (2010) ‘Making Sense of Place: Mapping as a Multisensory Research Method’, *Qualitative Inquiry*, 16(7), pp. 539–555. Available at: https://doi.org/10.1177/1077800410372600

Pratt, A.C. (2020) ‘COVID – 19 impacts cities, cultures and societies’, *City, Culture and Society*. Elsevier Ltd, p. 100341. Available at: https://doi.org/10.1016/j.ccs.2020.100341

Prior, S.D. and Scanlan, J. (2013) Civilian drones UK: BBC Radio 4. Available at: https://eprints.soton.ac.uk/356541/ (Accessed: 10 July 2020).

Probst, P.C., Rothe, S. and Hussmann, H. (2021) ‘Camera Distances and Shot Sizes in Cinematic Virtual Reality’, *ACM International Conference on Interactive Media Experiences*. New York, NY, USA: ACM, pp. 178–186. Available at: https://doi.org/10.1145/3452918.3458804

Proulx, M., Ross, L., Macdonald, C., Fitzsimmons and Smit, M. (2021) ‘Indigenous Traditional Ecological Knowledge and Ocean Observing: A Review of Successful Partnerships’, *Frontiers in Marine Science*, 8. Available at: https://doi.org/10.3389/fmars.2021.703938

Radjawali, I. and Pye, O. (2017) ‘Drones for justice: inclusive technology and river-related action research along the Kapuas’, *Geographica Helvetica*, 72(1), pp. 17–27. Available at: https://doi.org/10.5194/gh-72-17-2017

Ragni, M. and Stolzenburg, F. (2015) ‘Higher-Level Cognition and Computation: A Survey’, *KI - Künstliche Intelligenz*, 29(3), pp. 247–253. Available at: https://doi.org/10.1007/s13218-015-0375-y

Rahaman, A. and Sasse, M.A. (2010) ‘A framework for the lived experience of identity’, *Identity in the Information Society*, 3(3), pp. 605–638. Available at: https://doi.org/10.1007/s12394-010-0078-3

Ramirez, R., Mukherjee, M., Vezzoli, S. and Kramer, A.M. (2015) ‘Scenarios as a scholarly methodology to produce “interesting research”’, *Futures*, 71, pp. 70–87. Available at: https://doi.org/10.1016/j.futures.2015.06.006

Randall, J. (2015) *Technically older: an update on digital arts and creative ageing.* (Report) The Baring Foundation. Available at: https://baringfoundation.org.uk/wp-content/uploads/2015/10/TechnicallyOlder.pdf (Accessed 17 July 2024).

Reino, S., Mitsche, N. and Frew, A.J. (2007) *Information and Communication Technologies in Tourism*. Springer. Available at: http://eresearch.qmu.ac.uk/229/Thepublishedversionisavailableonlineat:http://www.springerlink.com/content/p62041333g032731/http://eresearch.qmu.ac.uk/policies.htmlhttp://eresearch.qmu.ac.uk (Accessed: 6 July 2020).

Rhee, J. (2018) *The Robotic Imaginary*. University of Minnesota Press. Available at: https://doi.org/10.5749/j.ctv62hh4x

Richardson, M. (2018) 'Drones eyeview: affective witnessing and technicities of perception', Schankweiler T., Straub, K. and Wendl, V. (eds.) *Image Testimonies: Witnessing in Times of Social Media.* New York: Routledge.

Risam, R. (2018) *New Digital Worlds*. Northwestern University Press. Available at: https://doi.org/10.2307/j.ctv7tq4hg

Robinson, J.S. and McClelland, A. G. (2020) 'Troubling Places: Walking the "troubling remnants" of post conflict space', *Area,* 52 (3), 654-662. Available at: https://doi.org/10.1111/area.12616

Roca-Riu, M. and Menendez, M. (2019) *Logistic deliveries with drones State of the art of practice and research*. Available at: https://doi.org/10.3929/ethz-b-000342823

Rodchenko., A (1928) *At the Telephone* [Gelatin silver print photograph]. Available at; https://www.khanacademy.org./humanities/art-1010/cubism-early-abstraction/photography-early-20th-century/alexander-rodchenko-at-the-telephone (Accessed: 19 July 2025).

Rodríguez, A., Negro, J., Mullero, M., Rodríguez, C., Hernández-Pilego, J. and Bustamante, J. (2012) 'The Eye in the Sky: Combined Use of Unmanned Aerial Systems and GPS Data Loggers for Ecological Research and Conservation of Small Birds', *PLoS ONE*, 7(12). Available at: https://doi.org/10.1371/journal.pone.0050336

Rose, G. (2016) *Visual Methodologies. An introduction to researching with visual materials*. 4th edn. SAGE Publications Ltd.

Rosello E. (2017) *Design Fiction: Prototyping Desirable Futures*. CCCB LAB. Available at: http://lab.cccb.org/en/design-fiction-prototyping-desirable-futures/ (Accessed: 14 June 2020).

Rossi, S., De Simone, F., Frossard, P. and Toni, L. (2019) 'Spherical Clustering of Users Navigating 360° Content', *ICASSP 2019 - 2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, pp. 4020–4024. Available at: https://doi.org/10.1109/ICASSP.2019.8683854

Ruíz, A.A.B. (2015) *Evaluación del comportamiento agronómico de ocho variedades de Malanga (Colocasia Esculenta) en las condiciones edafoclimáticas, Finca Buena Vista, comunidad El Tepeyac; departamento de Matagalpa, I Semestre 2015*. Universidad Nacional Autonoma de Nicaragua. Available at: http://repositorio.unan.edu.ni/2986/1/5624.pdf (Accessed: 18 May 2020).

Rowe, G. and Frewer, L.J. (2005) 'A Typology of Public Engagement Mechanisms', *Science, Technology, & Human Values*, 30(2), pp. 251–290. Available at: https://doi.org/10.1177/0162243904271724

Royo-Vela, M. and Black, M. (2020) 'Drone images versus terrain images in advertisements: Images verticality effects and the mediating role of mental simulation on attitude towards the advertisement', *Journal of Marketing Communications*, 26(1), pp. 21–39. Available at: https://doi.org/10.1080/13527266.2018.1425896

Ryan, M. and Lenos, M. (2020) *An Introduction to Film Analysis: Technique and Meaning in Narrative Film* (2nd ed.). Bloomsbury Publishing.

Salazar, N.B., Elliot, A. and Norum, R. (2017) 'Introduction', *Methodologies of Mobility*. Berghahn Books, pp. 1–24. Available at: https://doi.org/10.2307/j.ctvw04gfd.5

Saldanha, F. P., Pozzebon, M., Mailhot, C., and Le Puil, D. (2023) 'Counter-Narratives Mobilized by Deprived Communities Through Theatre Interventions: Deconstructing and Reframing Master Narratives', *Management Communication Quarterly* 37(3), 478–507. Available at: https://journals.sagepub.com/doi/full/10.1177/08933189221121345

Salguero, B. (2020) 'Defining the Museum: Struggling with a New Identity', *Curator: The Museum Journal*, 63(4), pp. 591–596. Available at: https://doi.org/10.1111/cura.12383

Sandell, R. (1998) 'Museums as agents of social inclusion', *Museum Management and Curatorship*, 17(4), pp. 401–418. Available at: https://doi.org/10.1016/s02604779(99)00037-0

Sanders, E.B. (2000) 'Generative Tools for Co-designing', *Collaborative Design*. London: Springer London, pp. 3–12. Available at: https://doi.org/10.1007/978-1-4471-0779-8_1

Sanders, E.B. and Stappers, P.J. (2008) 'Co-creation and the new landscapes of design', *CoDesign*, 4(1), pp. 5–18. Available at: https://doi.org/10.1080/15710880701875068

Sandvik, K. and Jumbert, M.G. (2017) *The Good Drone*. London: Routledge. Available at: https://www.taylorfrancis.com/books/edit/10.4324/9781315553405/good-drone-mariajumbert-kristin-sandvik (Accessed: 12 February 2020).

Sannö, A., Öberg, A. E., Flores-Garcia, E. and Jackson, M. (2019) 'Increasing the Impact of Industry–Academia Collaboration through Co-Production', *Technology Innovation Management Review*, 9(4), pp. 37–47. Available at: https://doi.org/10.22215/timreview/1232

Sanz Fernández, J. (2017) 'Snapchat, más cerca de tener su propio drone para hacer selfies', *Cinco dias*. Available at: https://cincodias.elpais.com/smartlife/2017/08/02/gadgets/1501663239_140848.html (Accessed: 15 May 2020).

Sardà-Palomera, F., Bota, G., Viñolo, C., Pallarés, O., Sazatornil, V., Brotons, L., Gomáriz, S. and Sardà, F. (2012) 'Fine-scale bird monitoring from light unmanned aircraft systems', *Ibis*, 154(1), pp. 177–183. Available at: https://doi.org/10.1111/j.1474-919X.2011.01177.x

Saska, M., Vonásek, V. and Preucil, L. (2011) 'Roads sweeping by unmanned multi-vehicle formations', *Proceedings - IEEE International Conference on Robotics and Automation*. IEEE, pp. 631–636. Available at: https://doi.org/10.1109/ICRA.2011.5979998

Saunders, P. (2011) 'Measuring deprivation', *Down and Out*. Bristol University Press, pp. 115–152. Available at: https://doi.org/10.46692/9781847428400.006

Savardi, M., Kovacs, A., Signoroni, A., Benini, S. (2023) 'Recognition of camera angle and level in movies', *ACM International Conference on Interactive Media Experiences Workshops. (IMXw 23). June 12-15, 2023*. Nantes: ACM, pp. 79–85. Available at: https://doi.org/https://doi.org/10.1145/3604321.3604334

Scanlan, J. (2019) *How drones are changing our lives.* [Webpage]. University of Southampton. Available at: https://www.southampton.ac.uk/news/2016/07/drone-technology.page (Accessed: 28 February 2020).

Scheible, J., Funk, M., Pucihar, K.C., Kljun, M., Lochrie, M. and Egglestone, P. (2017) 'Using Drones for Art and Exergaming', *IEEE Pervasive Computing*, 16(1), pp. 48–56. Available at: https://doi.org/10.1109/MPRV.2017.4

Schendan, H.E. and Ganis, G. (2012) 'Electrophysiological Potentials Reveal Cortical Mechanisms for Mental Imagery, Mental Simulation, and Grounded (Embodied) Cognition', *Frontiers in Psychology*, 3. Available at: https://doi.org/10.3389/fpsyg.2012.00329

Schneegass, S, Alt, F., Scheible, J. and Schmidt, A. (2014) ‘Midair Displays: Concept and First Experiences with Free-Floating Pervasive Displays’, *Proceedings of The International Symposium on Pervasive Displays*. New York, NY, USA: ACM, pp. 27–31. Available at: https://doi.org/10.1145/2611009.2611013

Schnepf, J.D. (2017) ‘Domestic aerial photography in the era of drone warfare’, *MFS - Modern Fiction Studies*. Johns Hopkins University Press, pp. 270–287. Available at: https://doi.org/10.1353/mfs.2017.0022

Schoellig, F.A. and R.D. (2010) ‘A Platform for Dance Performances with Multiple Quadropters’, *IEEE/RSJ International Conference on Intelligent Robots and Systems -* Workshop on Robots and Musical Expressions, pp. 1–8. Available at: https://www.researchgate.net/publication/228836805_A_platform_for_dance_performances_with_multiple_quadrocopters (Accessed: 6 August 2023).

Schroeder, N. M., Panebianco, A., González Musso, R. and Carmanchahi, P. (2020) ‘An experimental approach to evaluate the potential of drones in terrestrial mammal research: a gregarious ungulate as a study model’, *Royal Society Open Science*, 7(1), p. 191482. Available at: https://doi.org/10.1098/rsos.191482

Schulman, R. (2014) *What is engagement, and when is it meaningful?* [Webpage]. Available at: https://museumquestions.com/2014/08/20/what-is-engagement-and-when-is-it-meaningful/. (Accessed: 29 August 2023).

Seb the Nomad (2023) ‘Lisbon/Almada (Santuário de Cristo Rei)’, *Drone Shots – Portugal (4k Travel Videos)*. 3 February. Available at: https://m.youtube.com/watch?v=Fb8tCW7Bp3U (Accessed 18 June 2023).

Seligmann, L.J. and Estes, B.P. (2020) ‘Innovations in Ethnographic Methods’, *American Behavioral Scientist*, 64(2), pp. 176–197. Available at: https://doi.org/10.1177/0002764219859640

Serafinelli, E. and O'Hagan, L. (2021) 'Views from the blue: a glimpse in drone photography', *Pint of Science*. White Rose University Consortium. Available at: https://eprints.whiterose.ac.uk/173952/3/Pint (Accessed: 15 July 2023).

Serafinelli, E. and O'Hagan, L.A. (2021) 'The view from above: a drone's perspective on the world', *Futurum Careers* [Preprint]. Available at: https://doi.org/10.33424/FUTURUM215.

Serafinelli, E., and O'Hagan, L.A. (2022) 'Drone views: a multimodal ethnographic perspective', *Visual Communication*, 23(2), pp. 223–243. Available at: https://journals.sagepub.com/doi/full/10.1177/14703572211065093 (Accessed: 17 February 2022).

Serafinelli, E. and O'Hagan, L.A. (2024) 'Drone views: a multimodal ethnographic perspective', *Visual Communication*, 23(2), pp. 223–243. Available at: https://doi.org/10.1177/14703572211065093

Seyfang, G. and Smith, A. (2007) 'Grassroots innovations for sustainable development: Towards a new research and policy agenda', *Environmental Politics*, 16(4), pp. 584–603. Available at: https://doi.org/10.1080/09644010701419121

Shahbazi, M., Théau, J. and Ménard, P. (2014) 'Recent applications of unmanned aerial imagery in natural resource management', *GIScience and Remote Sensing*, 51(4), pp. 339–365. Available at: https://doi.org/10.1080/15481603.2014.926650

Shakir, M.Z. and Ramzan, N. (eds.) (2020) *AI for Emerging Verticals: Human-robot computing, sensing and networking*. Institution of Engineering and Technology. Available at: https://doi.org/10.1049/PBPC034E

Sharma, V. (2019) 'Advances in Drone Communications, State-of-the-Art and Architectures', *Drones*, 3(1), p. 21. Available at: https://doi.org/10.3390/drones3010021

Sharma, A., Vanjani, P., Paliwal, N., Basnayaka, C. M. W., Jayakody, D. N.K., Wang, H. and Muthuchidambaranathan (2020) 'Communication and networking technologies for UAVs: A survey', *Journal of Network and Computer Applications*. Academic Press, p. 102739. Available at: https://doi.org/10.1016/j.jnca.2020.102739

Shuman, A. (2015) 'Story Ownership and Entitlement', *The Handbook of Narrative Analysis*, ed. De Fina, A. & Georgakopoulou, A. Available at: https://doi.org/10.1002/9781118458204.ch2

Silberman, N.A. (2005) *Beyond Theme Parks and Digitized Data: What Can Cultural Heritage Technologies Contribute to the Public Understanding of the Past?* Available at: https://works.bepress.com/neil_silberman/39/ (Accessed 17 July 2020).

Siebert, S. and Teizer, J. (2014) 'Mobile 3D mapping for surveying earthwork projects using an Unmanned Aerial Vehicle (UAV) system', *Automation in Construction*, 41, pp. 1–14. Available at: https://doi.org/10.1016/j.autcon.2014.01.004

Silverman and Patterson, (2014) *Qualitative Research Methods for Community Development*, New York, Routledge, 138pp.

Sletto, B. (2009) 'We Drew What We Imagined: Participatory Mapping, Performance, and the Arts of Landscape Making', *Current Anthropology*, 50(4), pp. 443–476. Available at: https://www.journals.uchicago.edu/doi/10.1086/593704

Smethurst, T. and Craps, S. (2019) 'Towards an Understanding of Drone Fiction', *Journal of War & Culture Studies*, 12(1), pp. 85–102. Available at: https://doi.org/10.1080/17526272.2018.1514759

Sodero, S. (2017) *Drone-topia? A mobile utopia experiment*. [Webpage] Available at: https://stephaniesodero.weebly.com/experiment-blog/drone-topia-a-mobile-utopia-experiment. (Accessed 8 November 2022).

Sodero, S. and Rackham, R. (2020) 'Blood drones: using utopia as method to imagine future vital mobilities', *Mobilities*, 15(1), pp. 11–24. Available at: https://doi.org/10.1080/17450101.2019.1673034

Soja, E; Dufaux, F; Gervais-Lambony, P; Buire, C; Desbois, H (2011) 'Spatial Justice and the Right to the City: an Interview with Edward Soja', *Justice Spaciale, Spatial Justice*, 3.

Soren, B.J. (2009) 'Museum experiences that change visitors', *Museum Management and Curatorship*, pp. 233–251. Available at: https://doi.org/10.1080/09647770903073060

South Downs National Park Authority (no date) *BME Inclusion and Access to the South Downs National Park* (Internal Document). Provided by the SDNPA.

South Downs National Park Authority (no date) *Health and Well-being Strategy* (Internal document). Provided by the SDNPA.

South Downs National Park Authority (no date) *Partnership Management Plan 2020-2025* (Internal Document). Provided by the SDNPA.

South Downs National Park (2015) *Sustainable Tourism Strategy 2015-20* (Internal Document). Provided by the SDNPA.

South Downs National Park Authority (2016) *South Downs National Park Authority Visitor Survey 2015: Final Report* (Report). Provided by the SDNPA.

South Downs National Park Authority (2017) *Briefing Paper on Drones* (Internal Document). Provided by the SDNPA.

South Downs National Park Authority (2018) *Learning and Outreach- strategic review 2018-2023*. (Internal document) Midhurst, West Sussex: SDNPA. Provided by the SDNPA.

South Downs National Park Authority (2019) *2018 Visitor Survey. Final Report.* (Internal document) Midhurst, West Sussex: SDNPA. Provided by the SDNPA.

South Downs National Park Authority (2020) *Communications and Engagement Strategy 2021-2025* (Internal document) Midhurst, West Sussex: SDNPA. Provided by the SDNPA.

South Downs National Park (2020) *The South Downs by air* [Video online] Available at: https://youtu.be/Zm02qlzvAdw. (Accessed 12 May 2023).

South Downs National Park (2021) *Wild Chalk Festival – Dewponds of the South Downs*. 26 August. [Video online]. Available at: https://www.youtube.com/watch?v=mVSkrdJO3yg. (Accessed 15 June 2023).

South Downs National Park (2022) *Shoreham Cement Works – 360° Views.* 16 May. [Video online] Available at: https://youtu.be/ALPJOS_i-d8 (Accessed 12 April 2023).

South Downs National Park (2025) *In their footsteps* (version 9.0.381-prod) [mobile app] Available at: Google Play and Apple Play (Downloaded: 5 June 2023).

Sparkes, A.C. (2020) 'Autoethnography: accept, revise, reject? An evaluative self reflects', *Qualitative Research in Sport, Exercise and Health*, 12(2), pp. 289–302. Available at: https://doi.org/10.1080/2159676X.2020.1732453

Springgay, S. and Truman, S.E. (2017) *Walking methodologies in a more-than-human world: WalkingLab*. London: Routledge. Available at: https://doi.org/10.4324/9781315231914

Sprusansky, D. (2014) 'Protests target drone exhibit at National Air and Space Museum', *The Washington report on Middle East affairs*, 33(8), p. 58. Available at: https://www.wrmea.org/events-blog/protests-target-drone-exhibit-at-national-air-and-space-museum.html (Accessed 6 October 2021).

Srinivasan, R., Boast, R., Furner, J. and Becvar, K.M. (2009) 'Digital Museums and Diverse Cultural Knowledges: Moving Past the Traditional Catalog', *The Information Society*, 25(4), pp. 265–278. Available at: https://doi.org/10.1080/01972240903028714

Stahl, R. (2013) 'What the drone saw: the cultural optics of the unmanned war', *Australian Journal of International Affairs*, 67(5), pp. 659–674. Available at: https://doi.org/10.1080/10357718.2013.817526

Stankov, U., Kennell, J., Morrison, A. M. and Vujičić, M. D. (2019) 'The view from above: the relevance of shared aerial drone videos for destination marketing', *Journal of Travel and Tourism Marketing*, 36(7), 808–822. Available at: https://doi.org/10.1080/10548408.2019.157578

Stankov, U., Vasiljević, Đ., Jovanović, V., Kranjac, M., Vujičić, M. D., Morar, C. and Bucur, L. (2019) 'Shared Aerial Drone Videos — Prospects and Problems for Volunteered Geographic Information Research', *Open Geosciences*, 11(1), 462–470. Available at: https://doi.org/10.1515/geo-2019-0037

Stauber, B. and Parreira do Amaral, M. (2015) 'Access to and Accessibility of Education: An Analytic and Conceptual Approach to a Multidimensional Issue', *European Education*, 47(1), pp. 11–25. Available at: https://doi.org/10.1080/10564934.2015.1001254

Stek, T.D. (2016) 'Drones over Mediterranean landscapes. The potential of small UAVs (drones) for site detection and heritage management in archaeological survey projects: A case study from Le Pianelle in the Tappino Valley, Molise (Italy)', *Journal of Cultural Heritage*, 22, pp. 1066–1071. Available at: https://doi.org/10.1016/j.culher.2016.06.006

STEM Women (2022) *Women in STEM Statistics*. Available at: https://www.stemwomen.com/women-in-stem-percentages-of-women-in-stem-statistics (Accessed: 1 December 2022).

Stewart, H. (2023) ‘We have to flip the AI debate towards hope: Labour’s techno-optimist, Darren Jones’, *The Guardian (online)*, July. Available at: https://www.theguardian.com/technology/2023/jul/04/we-have-to-flip-the-ai-debate-towards-hope-labours-techno-optimist-darren-jones (Accessed 5 September 2024)

Stoecker, R. (1999) ‘Making Connections: Community Organizing, Empowerment Planning, and Participatory Research in Participatory Evaluation’, *Sociological Practice*, Vol. 1(3), pp. 209–231. Available at: https://doi.org/10.1023/A.1022874507194

Stone, A. (2008) ‘Being, Knowledge, and Nature in Novalis’, *Journal of the History of Philosophy*, 46(1), pp. 141–64.

Stone, R. and Ojika, T. (2000) ‘Virtual heritage: what next?’, *IEEE Multimedia*, 7(2), pp. 73–74. Available at: https://doi.org/10.1109/93.848434

StoryVerse Collective (2023) ‘Still I rise’, *Maya Angelou|Live and Unplugged*. 27 June.[Video online]. Available at: https://m.youtube.com/watch?v=LyxTcyFGfxM (Accessed: 7 August 2025).

Stroja, J. (2018) ‘My history, your history, our history: Developing meaningful community engagement within historic sites and museums’, *Queensland Review*, 25(2), pp. 300–321. Available at: https://doi.org/10.1017/qre.2018.29

Suhonen, R., Stolt, M. and Leino-Kilpi, H. (2013) ‘Older people in long-term care settings as research informants’, *Nursing Ethics*, 20(5), pp. 551–567. Available at: https://doi.org/10.1177/0969733012463722

Sumartojo, S. and Pink, S. (2018) *Atmospheres and the Experiential World*. London: Routledge. Available at: https://doi.org/10.4324/9781315281254

Superflux (2015) *Drone Aviary*. [Webpage] Available at: https://superflux.in/index.php/work/drones/# (Accessed: 9 September 2023).

Svensson, J. (2018) 'My history, your history, our history: Developing meaningful community engagement within historic sites and museums', *Queensland Review*, 25(2), pp. 300–321. Available at: https://doi.org/10.1017/qre.2018.29

Svensson, P. (2016) 'Making Digital Humanities', *Big Digital Humanities: Imagining a Meeting Place for the Humanities and the Digital*. University of Michigan Press, pp. 172–221. Available at: https://doi.org/10.2307/j.ctv65sx0t.9

Swain, J. and King, B. (2022) 'Using Informal Conversations in Qualitative Research', *International Journal of Qualitative Methods*, 21. Available at: https://doi.org/10.1177/16094069221085056

Swain, J. and Spire, Z. (2020) 'The Role of Informal Conversations in Generating Data, and the Ethical and Methodological Issues They Raise', *Forum: Qualitative Research*, 21(1). Available at: https://doi.org/10.17169/fqs-21.1.3344

Sweetow, S. (2017) 'Mobile Video Production and Delivery: Smartphones, Tablets, Drones, and Microcameras', *Corporate Video Production*. Routledge, pp. 313–324. Available at: https://doi.org/10.4324/9781315689883-27

Szafir, D., Mutlu, B. and Fong, T. (2017) 'Designing planning and control interfaces to support user collaboration with flying robots', *The International Journal of Robotics Research*, 36(7), pp. 514–542. Available at: https://doi.org/10.1177/0278364916688256

Tan, H., Lee, J. and Gao, G. (2018) 'Human-Drone Interaction', *Proceedings of the 19th International ACM SIGACCESS Conference on Computers and Accessibility - DIS 18*. New York, New York, USA: ACM Press, pp. 183–187. Available at: https://doi.org/10.1145/3197391.3205433

Tang, L. and Shao, G. (2015) ‘Drone remote sensing for forestry research and practices’, *Journal of Forestry Research*, 26(4), pp. 791–797. Available at: https://doi.org/10.1007/s11676-015-0088-y

Tang, L., Shao, G. and Dai, L. (2009) ‘Roles of digital technology in China’s sustainable forestry development’, *International Journal of Sustainable Development and World Ecology*, 16(2), pp. 94–101. Available at: https://doi.org/10.1080/13504500902794000

Tavabie, J.A. and S.A.T. (2015) ‘The Patient Liaison Officer in UK General Practice Co-ordinating Care for Housebound Patients’, *Quality in Primary Care*, 23(5), pp. 286–291. Available at: https://www.primescholars.com/articles/the-patient-liaison-officer-in-uk-general-practice-coordinating-care-for-housebound-patients-100336.html (Accessed:25 June 2024).

TEAM Tourism Consulting (2018) *South Downs National Park Visitor Market Proposition Final Report* (Report). Provided by the SDNPA.

TheDmel (no date) *TheDmel* [YouTube channel]. Available at: https://www.youtube.com/user/TheDmel?feature=watch (Accessed: 10 July 2020).

The Killing of a Sacred Deer, 2017, [Film] Directed by Y. Lanthimos. Ireland, United Kingdom, United States, A24.

The National Archives. (2023). *Archives for Everyone*. Available at: https://www.nationalarchives.gov.uk/aboutourroleplans-policies-performce-and-projects/our-plans/archives-for-everyone/ (Accessed 19 July 2025).

The Prince’s Trust (2021) *Confidence Crossroads: The path ahead for young people*. Available at: https://www.princes-trust.org.uk/Document_Censuswide-report-2021.pdf (Accessed: 30 October 2022).

The Wolf of Wall Street, 2013, (film), Directed by M. Scorsese, United States, Paramount.

Themistocleous, K. (2020) *The Use of UAVs for Cultural Heritage and Archaeology*., pp. 241–269. Available at: https://doi.org/10.1007/978-3-030-10979-0_14

Tiidenberg, K. (2018) ‘Post Selfie?’ in *Selfies: Why We Love (and Hate) Them*. Emerald Publishing Limited, pp. 101–130. Available at: https://doi.org/10.1108/978-1-78754-357-720181005

Tourism South East (2013) *South Downs Visitor & Tourism Economic Impact Study* (Technical Report). Provided by the SDNPA.

Toth, E. R., Guttman, S., Kimura, A., and Quinn, T. (2016) ‘Expanding the boundaries of museum studies: Popular education through engagement with hidden histories of organising and activism’, *Journal of Adult and Continuing Education*, 22(2), pp. 199–215. Available at: https://doi.org/10.1177/1477971416672326

Townsend, P. (1987) ‘Deprivation’, *Journal of Social Policy*, 16(2), pp. 125–146. Available at: https://doi.org/10.1017/S0047279400020341

Tremlett, S. (2021). *The poetics of Poetry film; Film poetry, Videopoetry, Lyric Voice, Reflection, Intellect.* Bristol, UK and Chicago USA p.8 Available at: https://doi.org./10.2307/j.ctv36xw4c0 (Accessed on 02 June 2025).

TSE Research Services (no date) *South Downs National Park Visitor Travel Survey: Report of Findings* (Report). Provided by the SDNPA.

Tzima, S., Styliaras, G., Bassounas, A., and Tzima, M. (2020) ‘Harnessing the potential of storytelling and mobile technology in intangible cultural heritage: A case study in early childhood education in sustainability’, *Sustainability (Switzerland)*, 12(22), pp. 1–22. Available at: https://doi.org/10.3390/su12229416

Underwood T. (2019) *Distant Horizons: Digital Evidence and Literary Change.* University of Chicago Press. Available at: https://doi.org/10.7208/CHICAGO/9780226612973.001.0001

UNESCO, Institute for Statistics (2009). *UNESCO Framework for Cultural Statistics*. Available at: https://uis.unesco.orgenglossary-termcultural heritage (Accessed 14 July 2025).

UNESCO (2016) *Improving Cultural Asset Sustainability*. Available at: https://en.unesco.org/creativity/policy-monitoring-platform/improving-cultural-asset (Accessed: 29 November 2022).

United Nations (no date) *The impact of digital technologies* (Webpage/ PDF). Available at: https://www.un.org/en/un75/impact-digital-technologies (Accessed: 27 May 2021).

University of Dundee (2022) *The protected characteristics explained* [Webpage]. Available at: https://www.dundee.ac.uk/guides/protected-characteristics-explained (Accessed: 28 November 2022).

University of Reading (no date) *Drone Research* [Webpage]. Available at: https://research.reading.ac.uk/drone-geographies/research/ (Accessed: 18 November 2024)

Uzunogullari, S. (2023) 'Armchair travel through video games: stories from elsewheres and elsewhens', *Leisure Studies*, 42(1), pp. 133–146. Available at: https://doi.org/10.1080/02614367.2022.2153904

van Laer, T., Feiereisen, S. and Visconti, L.M. (2019) 'Storytelling in the digital era: A meta-analysis of relevant moderators of the narrative transportation effect', *Journal of Business Research*, 96, pp. 135–146. Available at: https://doi.org/10.1016/j.jbusres.2018.10.053

van Wee, B. (2022) 'Accessibility and equity: A conceptual framework and research agenda', *Journal of Transport Geography*, 104, p. 103421. Available at: https://doi.org/10.1016/j.jtrangeo.2022.103421

Vannini, P. and Vannini, A. (2008) ‘Of Walking Shoes, Boats, Golf Carts, Bicycles, and a Slow Technoculture’, *Qualitative Inquiry*, 14(7), pp. 1272–1301. Available at: https://doi.org/10.1177/1077800408322708

Vargas-Ramírez, N. and Paneque-Gálvez, J. (2019) ‘The Global Emergence of Community Drones (2012–2017)’, *Drones*, 3(4), p. 76. Available at: https://doi.org/10.3390/drones3040076

Vas, E., Lescroël, A., Duriez, O., Boguszewski, G. and Grémillet, D. (2015) ‘Approaching birds with drones: First experiments and ethical guidelines’, *Biology Letters*, 11(2). Available at: https://doi.org/10.1098/rsbl.2014.0754

Vassenden, A. and Jonvik, M. (2019) ‘Cultural Capital as a Hidden Asset: Culture, Egalitarianism and Inter-Class Social Encounters in Stavanger’, Norway, *Cultural Sociology*, 13(1), pp. 37–56. Available at: https://doi.org/10.1177/1749975518782244

Vindrola-Padros, C. (2020) *Rapid ethnographies: A practical guide.* Cambridge, UK: University of Cambridge Press.

Visit England (2017) *Discover England Fund Research Summary Report: Making Great memories in England’s National Parks and countryside* (Report) Provided by the SDNPA.

Visitorsofcolor (no date) *Visitors of Color* [Tumblr account]. Available at: https://visitorsofcolor.tumblr.com/post/163774348237/i-was-lucky-enough-to-grow-up-in-a-home-and (Accessed 10 January 2024).

Voorberg, W.H., Bekkers, V.J.J.M. and Tummers, L.G. (2015) ‘A Systematic Review of Co-Creation and Co-Production: Embarking on the social innovation journey’, *Public Management Review*, 17(9), pp. 1333–1357. Available at: https://doi.org/10.1080/14719037.2014.930505

Vujičić, M. D., Kennell, J., Stankov, U., Gretzel, U., Vasiljević, Đ. A., and Morrison, A. M. (2022) 'Keeping up with the drones! Techno-social dimensions of tourist drone videography', *Technology in Society*, 68 (December 2021). Available at: https://doi.org/10.1016/j.techsoc.2021.101838

Wakkary, R., Desjardins, A., Hauser, S. and Maestri, L. (2013) 'A sustainable design fiction', *ACM Transactions on Computer-Human Interaction*, 20(4), pp. 1–34. Available at: https://doi.org/10.1145/2509404.2494265

Wallace-Wells, B. (2014, October) 'Drones and everything after', *New York Magazine*. Available at: https://nymag.com/intelligencer/2014/10/drones-the-next-smartphone.html

Waltl, C. (2006) 'Museums for visitors: audience development–a crucial role for successful museum management strategies', *Intercom* [Preprint]. Available at: https://www.researchgate.net/publication/265145508_Museums_for_visitors_Audience_development_-A_crucial_role_for_successful_museum_management_strategies (Accessed 2 October 2024).

Wang, B. Y., Lee, F. L. F., and Wang, H. (2013) 'Technological Practices, News Production Processes and Journalistic Witnessing: Hong Kong journalists in the 2011 Japan earthquake', *Journalism Studies* vol14(4). Available at: https://doi.org/10.1080/1461670X.2012.739315

Ward, L., Barnes, M. and B.G. (2012) *Well-being in old age: findings from participatory research*. Available at: https://www.brighton.ac.uk/_pdf/research/ssparc/wellbeing-in-old-age-full-report.pdf (Accessed: 12 March 2024)

Watkins, S., Burry, J., Mohamed, A., Marino, M., Prudden, S., Fisher, A., Kloet, N., Jakobi, T. and Clothier, R. (2020) 'Ten questions concerning the use of drones in urban environments', *Building and Environment*, 167, p. 106458. Available at: https://doi.org/10.1016/j.buildenv.2019.106458

Watermeyer, B. (2013). *Towards a contextual psychology of disablism*. Routledge.

Watts, A.C., Perry, J.H., Smith, S.E., Burgess, M.A., Wilkinson, B.E., Szantoi, Z., Ifju, P.G. and Percival, H.F. (2010) 'Small Unmanned Aircraft Systems for Low-Altitude Aerial Surveys', *Journal of Wildlife Management*, 74(7), pp. 1614–1619. Available at: https://doi.org/10.2193/2009-425

Watts, A.C., Ambrosia, V.G. and Hinkley, E.A. (2012) 'Unmanned aircraft systems in remote sensing and scientific research: Classification and considerations of use', *Remote Sensing*, 4(6), pp. 1671–1692. Available at: https://doi.org/10.3390/rs4061671

Weller, T. (2013) A Changing Field. T. Weller (ed.) *History in the Digital Age*. Oxford, UK: Routledge, pp. 195–205. Available at: https://doi.org/10.4324/9780203093443

West Sussex County Council (no date) *The life you want to lead: Adult social care in West Sussex (2022-25)*. Available at: https://www.westsussex.gov.uk/media/17120/adult_social_care_strategy_2022-25.pdf (Accessed: 12 September 2024).

West Sussex County Council (2019) 'West Sussex Joint Strategic Needs Assessment Briefing. Indexes of Deprivation 2019', Available at: https://jsna.westsussex.gov.uk/assets/core/Briefing-West-Sussex-IMD-2019.pdf (Accessed 19 May 2022).

Whitehead, T. L. (2005) *Basic Classical Ethnographic Research Methods* (Ethnographically Informed Community and Cultural Assessment and Cultural Assessment Research Systems (EICCARS) Working Paper Series). University of Maryland, CEHC.

Whitehead, K. and Hugenholtz, C.H. (2014) 'Remote sensing of the environment with small unmanned aircraft systems (UASs), part 1: a review of progress and challenges', *Journal of Unmanned Vehicle Systems*, 02(03), pp. 69–85. Available at: https://doi.org/10.1139/juvs-2014-0006

Whitworth Art Gallery (2015) *A handbook for cultural engagement with older men.* The Baring Foundation and the University of Manchester. Available at: https://documents.manchester.ac.uk/DocuInfo.aspx?DocID=24251 (Accessed 12 December 2022).

Wich, S.A. and Koh, L.P. (2018) *Conservation Drones: Mapping and Monitoring Biodiversity*. Oxford University Press. Available at: https://doi.org/10.1093/oso/9780198787617.001.0001

Wilken, R. (2014) *Teletechnologies, Place, and Community*. London: Routledge. Available at: https://doi.org/10.4324/9780203807026

Williams, D.R. (2014) 'Making sense of "place": Reflections on pluralism and positionality in place research. *Landscape and Urban Planning'*, 131, pp. 74–82. Available at: https://doi.org/10.1016/j.landurbplan.2014.08.002

Wing, M.G, Burnett, J., Sessions, J., Brungardt, J., Cordell, V., Dobler, D and Wildon, D. (2013) 'Eyes in the Sky: Remote Sensing Technology Development Using Small Unmanned Aircraft Systems', *Journal of Forestry*, 111(5), pp. 341–347. Available at: https://doi.org/10.5849/jof.12-117

Wing, M.G., D. Burnett, J. and Sessions, J. (2014) 'Remote Sensing and Unmanned Aerial System Technology for Monitoring and Quantifying Forest Fire Impacts', *International Journal of Remote Sensing Applications*, 4(1), p. 18. Available at: https://doi.org/10.14355/ijrsa.2014.0401.02

Winn, E., Kissane, M., Merriel, S., Brain, T., Silverwood, V., Whitehead, O., Howe, L. D., Payne, R. and Duncan, P. (2023) 'CHiP study protocol: using the primary care academic CollaboraTive to explore the characteristics and healthcare use of older housebound patients in England', *BJGP Open*, p. BJGPO.2023.0114. Available at: https://doi.org/10.3399/BJGPO.2023.0114

Women Who Drone and elenabuenrostro (2023) *Happy International Woman's Day!* [Instagram Reel] Available at: https://www.instagram.com/reel/Cpi2vpesCCz/ (Accessed: 25 July 2023).

Woolf, V., (2025) *A room of one's own*, Royal Masterpiece Classics. USA.

Woolf, V. (1953) Woolf L. (ed.) *A Writer's Diary. Being Extracts from the Diary* of *Virginia Woolf*. Available at: https://share.google/kzg37z6L8Mc5hHFol (Accessed 9 February 2024).

Woven Films (2019) *SS Great Britain Poem Drone Showreel.* Vimeo, [Video online]. Available at: https://wovenfilms.co.uk/video/ss-great-britain-poem/ (Accessed 12 March 2023).

Xu, H., Zhao, Q., Ma, Y., Wang, S., Yan, C. and Dai, F. (2023) 'Free-Viewpoint Navigation of Indoor Scene with 360° Field of View', *Electronics*, 12(8), p. 1954. Available at: https://doi.org/10.3390/electronics12081954

Yan, Z. (2020) 'Unprecedented pandemic, unprecedented shift, and unprecedented opportunity', *Human Behavior and Emerging Technologies*, 2(2), pp. 110–112. Available at: https://doi.org/10.1002/hbe2.192

Yilmaz, M. B., Lotman, E., Karjus, A., and Tikka, P. (2023) 'An embodiment of the cinematographer: emotional and perceptual responses to different camera movement techniques', *Frontiers in Neuroscience* (Vol. 17). Available at: https://doi.org/10.3389/fnins.2023.1160843

Yu, Z. (2023) ‘Shot Size Narration in the Mobile Media Era: Characteristics and Evolution’, *Journal of Research in Social Science and Humanities*, 2 (11). Available at: https://doi.org/10.56397/JRSSH.2023.11.04

Yuan, C., Zhang, Y. and Liu, Z. (2015) ‘A survey on technologies for automatic forest fire monitoring, detection, and fighting using unmanned aerial vehicles and remote sensing techniques’, *Canadian Journal of Forest Research*, 45(7), pp. 783–792. Available at: https://doi.org/10.1139/cjfr-2014-0347

Zak, D. (2014) ‘Unmanned future: Drones to clean your gutter, tend to crops and take aerial selfies?’, *The Washington post*. Available at: https://www.washingtonpost.com/lifestyle/style/unmanned-future-drones-to-clean-your-gutter-tend-to-crops-and-take-aerial-selfies/2014/03/24/75e1d1f4-b376-11e3-8cb6-284052554d74_story.html (Accessed: 3 July 2020).

Zarco-Tejada, P.J., Diaz-Varela, R., Angileri, V and Loudjani, P. (2014) ‘Tree height quantification using very high resolution imagery acquired from an unmanned aerial vehicle (UAV) and automatic 3D photo-reconstruction methods’, *European Journal of Agronomy*, 55, pp. 89–99. Available at: https://doi.org/10.1016/j.eja.2014.01.004

Zeffiro, A., Hildebrand, J., Frith, J., Hjorth, L and McGrane, C. (2020) ‘Locative-Media Ethics: A Call for Protocols to Guide Interactions of People, Place, and Technologies’, *Journalism & Mass Communication Quarterly*, 97(1), pp. 13–29. Available at: https://doi.org/10.1177/1077699019898773

Zhang, J., Hu, J., Lian, J., Fan, Z., Ouyang, X. and Ye, W. (2016) ‘Seeing the forest from drones: Testing the potential of lightweight drones as a tool for long-term forest monitoring’, *Biological Conservation*, 198, pp. 60–69. Available at: https://doi.org/10.1016/J.BIOCON.2016.03.027

Zhou, Q., Zink, M., Sitaraman, R, Nahrstedt, K. (2023) ‘360° TripleView: 360-Degree Video View Management System Driven by Convergence Value of Viewing Preferences’, *arXiv:2306.08089* [Preprint]. Available at: https://doi.org/https://doi.org/10.48550/arXiv.2306.08089

Zimmerman, E. (1994) ‘Flying high: L.A. artist Renee Lusano is putting even our best #selfies to shame with her secret soaring weapon – a drone’, *Marie Claire,* New York, 23(3), p. 292.

Zimmerman, M.A. (1995) ‘Psychological empowerment: Issues and illustrations’, *American Journal of Community Psychology*, 23(5), pp. 581–599. Available at: https://doi.org/10.1007/BF02506983

Zuev, D. and Bratchford, G. (2020) ‘Seeing Like a Drone’, *Visual Sociology*. Cham: Springer International Publishing, pp. 119–140. Available at: https://doi.org/10.1007/978-3-030-54510-9_5

Zúñiga, X and Adams, M. (2016) ‘Core concepts for Social Justice Education’, *Teaching for Diversity and Social Justice*, by Adams, M. *et. al.* (eds.) London: Routledge, pp. 95–130.

Appendix A: Online Survey Questions

Online Anonymous Survey

This survey aims to capture how heritage organisations, staff and volunteers view the use of drone technology to engage new audiences and what they think are the opportunities, challenges and barriers to its successful use.

The survey is voluntary and anonymous. Your personal data (name, email etc.) will not be captured or recorded.

Please tick this box to indicate your understanding that, by submitting the survey, you are giving your consent to taking part in the research.

1. What type of heritage place do you work or volunteer for?

   Archive Museum Historic Property
   Other (please tell us more)

2. What is your relationship with the organisation?

   Volunteer Contractor (freelancer/consultant)
   Paid staff

3. In which department(s) do you work for or volunteer with?

   Senior Management Community Engagement Formal Learning
   Curatorial Marketing Visitor services/visitor experience
   Interpretation Other (please tell us more)

4. Does your organisation have an Audience Development Plan?

Yes No I don’t know

5. If yes, is digital engagement included in your organisations’ audience development plan or strategy?

Yes No I don’t know

6. Which words come to mind when you think of digital technologies?

7. Which communities are in your organisations’ audience engagement priorities?

Ethnically diverse	Disabled	LGBTQI+	Working class
Families	Young people	Rural area	Older people
I don’t know			

8. Has drone technology been used by your organisation (in general)?

Yes If yes, how? No I don’t know

9. And for audience engagement?

Yes If yes, how? No I don’t know

10. Which words come to mind when you think of drones?

11. What are your views on drone technology?

12. In what ways do you see drone technology being used to engage new audiences with collections, narratives and sites?

13. Have you come across any interesting drone activities/projects inside or outside the heritage and cultural sectors?

14. What would you see as the 3 main barriers to using drone technology to engage non-visiting audiences?

Expensive technology	Drone's bad reputation	Need for technical knowledge
Need for specialist training	Legal considerations	Other (please tell us more)

15. What are the 3 main risks?

Damage to site or collections	Risks to privacy and security	Loss of control of the vehicle
Spooking other visitors	Other (please tell us more)	

16. What are the 3 main opportunities?

17. What would encourage you to use the technology to engage non-visiting audiences?

Funding opportunities	Senior leadership direction	Better understanding of the technology
Training	Access to equipment	Trained staff
Trained volunteers	Other (please tell us more)	

18. Is there anything else you would like to say about using drone technology to engage non-visiting audiences?

Appendix B: Expert Interview Questions

1. Heritage Experts' Interview Guide

These questions were adapted according to the role of the interviewee.

1. Please could you tell me your name, job title and a few words about your role in the organization?
2. Which words come to mind when you think of digital technologies?
3. What words come to mind when you think of drones?
4. What is your view on drone technology?
5. Is digital engagement included in your organisation's audience development action plan or strategy?
6. What are your organisations' audience engagement priorities?
7. Has drone technology been used by the organisation (in general)? How?
8. And for audience engagement? How?
9. In what ways do you think the technology could be used to engage non-visiting audiences with collections, narratives and sites?
10. What would you see as the main barriers to using drone technology to engage non- visiting audiences?
11. What are the risks?
12. What are the opportunities?
13. What would encourage you to use the technology to engage non-visiting audiences?
14. Is there anything else you would like to say about using drone technology to engage non-visiting audiences?

2. Drone Experts' Interview Guide

These questions were adapted for drone hobbyists as appropriate.

1. Could you please tell me your name and role title?
2. Could you please describe what you do?
3. How do you/your organisation currently use drone technology? For what purpose?
4. Which industries, in your experience, make the most use of drone technology and for what types of outcomes?
5. If respondent not sure, ask instead; Do you have examples of projects or activities where the technology has been used successfully to engage the public off-site with a subject (an exhibition, landscape, story, artwork)?
6. What is unique about drone technology?
7. What can drones bring to someone's experience of historic buildings, nature spaces or astronomy for instance, that isn't offered already through other media?
8. What are the possibilities or opportunities you see for the use of drones that we don't yet explore?
9. How could these be better explored?
10. How could drone technology be used to support virtual tourism, for those who can't travel due to limited mobility, for instance?
11. In what ways do you think it possible to champion drones as a beneficial technology to the general public?
12. How do drones integrate with other technologies?
13. What do you think is the future for this technology (in heritage and culture)?
14. Is there anything else you would like to say about the use of drone technology to create better access to places by non-drone users?

Appendix C: Autotechnographic research

I developed the brief and planning for the autotechnographic research, when applying for the *Shifting The Gaze* Bursary scheme, a writers' Research and Development programme funded by the South Downs National Park Authority and delivered by Writing Our Legacy.[48] The programme *Shifting The Gaze* invites ethnically diverse writers to apply for funding to research and develop work focused on experiences of the South Downs.

I use the bursary application as the framework to map my autotechnographic data collection methods, leading to my production of creative work – namely a film-poem titled *These too are Footsteps* (Fine, 2023a*)*. The data collected about my interaction with drone technology involves the combination of traditional ethnographic methods, such as participant observation, with approaches Seligmann and Estes (2020, p. 177) identify as innovative. In alignment with Seligmann and Estes' considerations (2019, p. 177) my study is a short ethnography, which is multisite (there is more than one fieldwork space as part of exploring issues of power); the data is interpreted and represented in my film-poem. The spaces of fieldwork include nature walks (Limb, 2023), informal conversations (Seligmann and Estes, 2019, p. 177), video (Pink, 2011), photography and poetry (Luvaas, 2019).

The National Trust Research Opportunities framework proposes to explore relevance and representation (National Trust, 2019, p. 13). For them, non-visiting groups facing barriers to accessing their sites may include diverse ethnic communities, who are underrepresented in collections and sites. Using autoethnography reflects current concerns with decolonising and

[48] Writing Our Legacy is an arts organisation based in Brighton, UK, supporting ethnically diverse authors and creative practitioners.

decentralising knowledge production systems; it also valorises lived experiences that are excluded.

With autotechnography, I aim to investigate the construed nature of drone technology and its aesthetic, by capturing, reflecting on, and writing about how drone technology can shape "[…] material and symbolic substance" (Vannini and Vannini, 2008, p. 1299) of place and narrative in a cultural and socio-political context. I follow Hildebrand's (2021) autotechnographic approach to frame my interaction with the technology to access heritage sites. Specifically, I draw on my identity and positionality as a cisgender Global Majority female researcher (Campbell-Stephens, 2022) to capture and document my experiences and reflections on using drone technology to facilitate my engagement with Monks House, a National Trust property in the South Downs National Park.

Having never used a drone previously, I design my practice to help me develop drone flying skills through my own practice and through engaging with other people's experiences. I document, analyse, and reflect on events such as how I select a drone, how I familiarise myself with the machine, my hesitations on flying it alone, how the affordances of the drone inform my relationship with my locality, weather events and where I might go out.

I include in my fieldwork: notes and reflections from informal conversations with other drone academics; interactions with people who approach me when I am flying the drone; collaboration with the commercial drone pilot capturing livestreaming and the film-poem drone shots at Monks House; conversations with the staff at Monks House; joint work with multidisciplinary artists and filmmakers; and mentoring writing sessions with a drone pilot who is also a creative practitioner.

I use creative methods to capture my interaction with my environment (Horsfall and Titchen, 2009, p. 151). I also use a representation method (Horsfall and Titchen, 2009), my experimental autotechnographic film, *These too are footsteps*, as a critical reflection on the use of drone technology to empower me to connect with heritage and culture. I investigate and reflect on my use of competencies, resources, understandings, information, and approaches in my engagement with hobby camera drones. I triangulate these with walking and writing, video, photography, and conversations to capture my own interactions with camera drones.

I not only draw on my skills and knowledge but also on my identity and positionality as a researcher. Rapid Ethnographies (Vindrola-Padros (2020) provides me with a structured framework designed with me, the researcher, at the centre of the fieldwork, participating in brief periods of immersion (Pink and Morgan, 2013) to create specific instances of observation and reflection. I also investigate the entanglements through which knowledge is produced (Pink and Morgan, 2013), and how it evolves at the intersection between theoretical frameworks for the study of drone vision, mobilities, drone aesthetics, agency, power and identities, and the experiential components of flying an unstaffed aerial vehicle. My purpose is to uncover aspects that otherwise remain hidden or are dismissed.

Conducting autoethnographic research incurs challenges. With this method, self-exposure makes me, the researcher, vulnerable (Sparkes, 2024, p. 128). Ethical challenges (Sparkes, 2024, p. 111) are inherent to my existence within a context of relationships and interactions that emerge as I tell my story. This has implications for narrative ownership. Moreover, autotechnography, like autoethnography, is subjective and open to interpretations: this can challenge the premises of rigour and accuracy intrinsic to research.

Autotechnography is proposed as a representation of the experiences of an excluded community. However, the autotechnographer belongs to themselves, so there is a risk of creating or amplifying stereotypes of that community (Boylorn, 2017, p. 17). This can happen in the process of perceived homogenisation of the socio-cultural complexities of the identities and experiences of that group. Stereotyping, generalisation, and cultural homogenisation are part of the tools of misrepresentation and further exclusion, and silencing of narratives of underserved communities. Therefore, I need to consider my autotechnographic data within this context.

Walking practices

As a *Shifting The Gaze* grant recipient (the grant and mentoring programme funding ethnically diverse writers to research and develop work inspired by, or based on the South Downs National Park), I access walks and guided visits to heritage sites in the South Downs, provided by the programme to facilitate the connection of ethnically diverse groups, with the natural and cultural heritage of the Downs, and empower them to become visible (using Kowalewski and Bartłomiejski's terminology, 2020, p. 61). I walk in the South Downs as a facilitation tool for introspection, capture of footage, and in combination with poetry writing, in a process of creative exploration.

This supports my reflections on the research question examining the use of drone technology to empower underserved communities to connect with heritage and culture. Before undertaking the walks, I read excerpts from Virginia Woolf's essay *A room of one's own* (Woolf, 2025), where the author uses a spatial metaphor (Hong, Zhang and Son, 2019, p. 1) to reflect critically on the socio-political context of being a woman (Alexander, 2010, p. 273).

I also listen to "In their footsteps" (SDNPA, 2023), an audio-guided walk with visual cues following Virginia Woolf's walking routes. Virginia Woolf writes of a connection between walking and a disembodied experience of space in her diaries, which mirrors the combination of walking and flying a drone: "Oh the joy of walking! I've never felt it so strong in me […] the trance like, swimming, flying through the air; the current of sensations and ideas." (Woolf, 1953, p. 226).

Through my walking practice, I explore how camera drones parallel the affordances of walking practices, for instance deepening my relationship with space and eliciting new perspectives (Belkouri and Laing, 2024, p. 68). I also investigate the unique forms of engaging with space they provide, specific to aerial mobilities, as identified by (Serafinelli and O'Hagan, 2022, np): experiencing place as art, vertical mobility (such as top-down view) and fulfilling a human interest in conquering aerial space.

I undertake eight walks in the South Downs National Park, capturing visual footage each time:

- Three walks in the South Downs Link, near Henfield, West Sussex, including using a drone.
- One walk from Monks House, Rodmell, East Sussex, to the river Ouse, using a GoPro camera.
- One circular walk with a drone pilot, from Southease Railway Station, East Sussex, to Monks House, using a GoPro camera and a drone.
- Participation in one group walk led by Worthing Museum Curator through Highdown Hill, Worthing, West Sussex.
- One walk in Highdown Hill, Worthing, alone.
- One walk in Highdown Hill, Worthing, West Sussex with South Downs National Park Authority Health and Wellbeing officer.

The walks are not time bound, and their length depends on my physical mobility limitations: how far I am able to walk and how long for (for the South Downs Link); whether I complete the route, in the case of replicating Virginia Woolf's walking routes from Monks House (the writer's home) to the Ouse and from Southease station to Monks House. Walks take place when I am available. Except for the guided walk to Highdown Hill and accompanied walks (with the drone pilot and SDNPA Health and Wellbeing Officer), they are not pre-organised.

Each walk encompasses an element of digital capture. I write notes on feelings experienced during the walk, poetry inspired by the walk, and any inspiring observations in nature. I also capture the walk in 360° degrees, using a Go-Pro camera or a drone, to explore elements of immersion and vertical mobility (although the latter is limited when using the camera, rather than the drone; I carry the camera on a tripod positioned at shoulder height).

Human beings move in different ways, so walking is not a fully identical activity to everyone. Factors that affect how humans move include bodies, culture, religion, gender, disability. Moreover, interaction with the environment is an internal process, unique to each individual.

Data elicited through this method (walking) is shaped by these aspects, and therefore subjective. Moreover, walking outdoors increases endorphin levels, which could distort the research value of knowledge and reflections elicited from this activity.

Walking research methodologies can, therefore, be perceived as lacking rigour and repeatability. Nonetheless, Macpherson (2016, p. 426) considers that walking methodologies combined with audiovisual methods of documentation such as photography, video, and other artistic ways of documenting, amplify underrepresented and silenced voices. Creative practices, as seen in the previous section (3.5) are also shaped by the ethnographers' personal experiences and meaning-making processes (Pink, 2011, p. 22) and reflect people's socio-political contexts.

Walking practices involve the physical activity of the researcher in public and carry associated risks. To mitigate risks, I undertake walks including a stop over to fly my drone after six o'clock in the afternoon: this is a quieter time, and it is also safer to pilot the drone away from bystanders; I am less likely to encounter people who may react negatively to the drone. I also make a conscious decision to only fly my drone when accompanied by my white British, male partner, to mitigate the risk of gender and racial bias-informed responses from walkers-by.

Autotechnographic and walking practices methods help me understand my relationship with drone technology and how it can facilitate empowerment as a non-expert member of an underserved community. They lead me to produce two film-poems in response to my experience of heritage: *These too are footsteps* (Fine, 2023a) and *A room of my own* (Fine, 2023b). I draw on these to consider how my experiences and understandings relate to those of underserved communities, who I share characteristics with (women and Global Majority groups). I analyse visual data from my films in Chapter Seven.

Appendix D: Co-design research with the Lovejoy Centre

Session Plans for *Bringing the outdoors, indoors* (co-research project title decided with participants)

Session times: 1.30pm Wednesdays and Fridays

Duration: 90 mins

Structure: Each session combines one discussion activity with one hands-on activity.

Staffing: Each session is delivered by Isilda Almeida and supported by Kate Drake (SDNPA Health and Wellbeing Officer) and Lovejoy Centre staff (4 people).

NB: After delivering the first two sessions, activities were adapted to be more accessible. The Centre Director suggested I provide a printed version of the questions (printed on yellow background, to be more readable) to participants so they could better process the questions and be able to respond. Moreover, sessions were restructured to break up the hour discussion into two 30-minute sessions, intercalated with a 30-minute activity.

Resources

- A laptop
- Projector or large tv screen
- Access to Wi-Fi
- Multidirectional recording device
- Activity guidance sheet for each member of staff (1 sheet per session)
- Large print name badges for IA and KD (dementia friendly)
- Magnifying glasses
- Printed images of:

    - South Downs National Park, park wildlife and plants
    - Images of where the Park connects with urban areas
    - Images of historic landmarks
    - Images inside historic landmarks (National Trust)

- Handling samples:

    - Foliage
    - Bark - dried (for bark rubbings)
    - Herbs
    - Earth in a zip bag so people can still feel it
    - Leaves, holly and berries for ppl to look at
    - Branches, gorse flowers, bluebells, snowdrops etc

- Sounds of birds and other animals you can find in the park
- Herb-scented oils (appropriate for reminiscence sessions)
- Yes and No post-it notes
- Empty cards/post-it notes
- Crayons or coloured pencils
- Glue
- Wildflower seeds (to make wildflower seed balls that participants can take home and plant) -Bringing the outside indoors.
- Large print out of a map of the South Downs Park
- Sound tile or sound tins to add sound to collage.

Lovejoy Centre has a bird's nest and a birdsong CD.

Landmark selection: Clarify in sessions that choice of landmarks is limited, because we need permission from organisations where we will be flying the drone. Explain that if the group doesn't all agree on one landmark that we will choose based on the number of votes.

1. Session One: Setting the context

i Introductions

What are we doing today? Talk about project and activity for the day

vii. Group discussion:

1. Have you always lived in Worthing?
2. What do you like about living in Worthing?
3. What do you dislike about living in Worthing?
4. Where do you go when you go out?
5. Have you visited the Downs? (Give examples. Places like Highdown Gardens, Rodmell). Depending on the group we will select two locations from the above and these: Stanmer Park, Devil's Dyke, Park, Ditchling Beacon, Pulborough Brooks, Cissbury Ring.
6. Which is your favourite place or landmark in the Downs where you have been for walks, picnics, to the pub? (Participant or staff assisting writes their name and the name of the landmark on a travel tag)?
7. Pin it on the map of the South Downs. (Participant or person assisting writes the name of the place on a travel tag, finds it on the map of the South Downs and pins the travel tag on the map in its -approximate- location).

viii. Alternatively use a storymaking technique:

1. Have 3 or 4 photos of four characters and work with group to create a story
2. Let's imagine their day out:

They live in Worthing and want to go out on the Downs for the day with family and friends. Where could they go?

What can they do? If there are children, do they do something different to the adults? Did they go there for a picnic, to visit a museum, a walk, family outing?

What can they see? Did they see any flowers? Did they see or hear any animals?

What can they smell?

What can they touch? Do they touch any flowers, trees?

What are they talking about?

How are they feeling about their day?

ix. Hands on activity: Talking points mapping and object handling

1. What do these places/landmarks make you think of? Have you been there on a date, did you go to a wedding, a family picnic, walking, taken the dogs out, met friends? Participant or staff helping add this to the landmark tag and pin it on the map of the South Downs in its -approximate- location).
2. Provide participants with magnifying glasses for them to look through.
3. Pass around items from the park (archaeological handling collection, bark, foliage, soil (through a bag), branches.
4. Ask prompt questions: what patterns can you see? What does it feel like?
5. Participants do bark rubbing, drawing around foliage and branches and paint their drawing.

2. Session Two: Mapping

x. Introductions

What are we doing today? Talk about project and activity for the day (Anne Marie will have done this.)

xi. Mapping talking points

Show the map of the Downs. Ask about their preferred landmark. Pin the tag on the map.

xii. Being in the Downs

1. Ask participants to close their eyes and describe the sounds they can hear in their current location.

   Where are these sounds coming from?
   What are the sounds of (a car driving by)?
   Are the sounds loud or faint?

2. Repeat the above exercise but asking participants to imagine they are standing in the middle of the Downs or at the National Trust. Ask participants to close their eyes.

   Imagine they are in the middle of the Downs.
   Describe the sounds they would hear if they were in the Downs.
   Where are these sounds coming from?
   Are these sounds loud or faint?
   Play sounds: Wind on leaves
   Sheep and lambs
   Horses trotting
   Skylark feeding chicks
   People walking in muddy woods
   Wind on leaves and squirrel calls

3. The Downs are a busy place with a lot of history.
   (Pass on hare, skylark, lemon balm leaves, wild garlic, pinecones, bluebells, soft toy sheep, archaeology).

4. Project name selection

xiii. Hands on activity: collage

Participants use cut outs of images of the Downs, nature, NT properties, historical objects and urban areas, samples of plants, smells and sounds found in nature, to create a collage. Cut out words can be added too.

Can be finished in other sessions.

Ask participants to pick from a group of word cards placed on their table, including the below in large print yellow cards:

1. Which of these two words make you think about the Downs (or being in nature)? (using words from the SDNPA's Visitor Surveys)

   - Tranquillity
   - Scenic landscape
   - Breath-taking views
   - Unspoilt spaces
   - Historical places
   - Nature
   - Wildlife
   - Variety of the landscape
   - Inspiration
   - Pride
   - Walks
   - Blank word cards (participants add their own)

2. Which of these two words make you think about the Downs (or being in nature)? (using words from the SDNPA's Visitor Surveys)

   - Dog mess

- Litter
- No toilets
- Crowded
- Noisy
- Unsafe
- Boring
- Quiet
- Stressful
- Too far to walk
- Lonely
- Blank cards (participants add their own)

xiv. Before next session:

In preparation for session 3, ask participants (Centre will write a letter to the care home/carer) to bring in an object or a photograph they link to their experiences in the Downs.

NOTE: some participants don't have people involved to help (for instance if they are in a care home). IA to bring items that have a connection and that people can choose from, which resonates with them.

3. Session Three: Exploring interest in Bird's Eye View

xv. Introductions

What are we doing today? Talk about project, last week's and today's activity.

1. Ask Carmen (attending for the first time) which is her favourite place in the South Downs.
2. Voting.

xvi. Bird's Eye View exploration

If you were a bird, which bird would you be?

What food would you eat?

What would you do/where would you fly?

We are going to visit the South Downs as if we were a bird flying over it. Show a part of this clip https://www.youtube.com/watch?v=Zm02qlzvAdw It shows images of:

- Cuckmere Haven
- Seven Sisters
- Wilmington
- Black Down
- Truleigh Hill
- Devil's Dyke
- Falmer
- Stedham Common
- Ditchling
- Lewes
- Harting Down

1. What can you see?
2. Do you like what you see?
3. What do you like about seeing a place from a Bird's Eye View?
4. Is there anything you don't like?
5. What else would you like to see or hear in this film? (people chatting, people's stories of visits to the Downs, birds singing, animals, flowers)

xvii. Polaroid Photos

1. Make available objects and leaves used before and ask participants to choose their favourite (hare, bird, antler, nest, axe head, sheep, owls, or any of the plants)
2. Ask why it is their favourite.
3. Another group member can take a photo of participant holding the item with Polaroid.

4. After doing this, bring the collage out and ask group to add the Polaroid and ask them to choose the words in the yellow cards they think describe what they think about the South Downs and National Trust and glue them to the collage. If no time, do this in session 4.

xviii. Hands on activity: Wildflower seedball (led by SDNPA)

Make a wildflower seedball and take it home to blossom.

4. Session Four: Comparing aerial and ground photo and video

xix. Introductions

What are we doing today? Talk about project, last two weeks and today's activity

xx. Storymaking with photos

1. *Talk about the bird – using Lovejoy Centre's plastic bird collection.
2. Participants use hare template to make up a leaping hare, using fabric and glue.
3. Give participants a ground photo of the same area.
4. The hare and the bird both live in the Downs. One day they meet and become friends. They go on a journey through the Downs together. Use cards to ask questions below:

- What does the bird say to the hare?
- What does the hare say to the bird?
- What kinds of things does the bird spot from the air?
- What kinds of things does the hare spot on the ground?
- Which journey is more exciting?
- Which one (bird or hare), would you like to be?

xxi. Comparing ground views and aerial views

Show a clip of an aerial drone view of the Downs/NT and of the same place, as a view from the ground and discuss with participants:

1. Show *Pinned On Places*, Lewes Priory https://www.youtube.com/watch?v=s-GVDV365DI 11:51 mins (show mins 8:00 to mins 10:11)

- Do you like this film?
- What do you think and feel when you see this film?
- What do you like about it?
- What are the things you didn't like about it?
- If you were a bird flying over that place, where would you like to go?

2. Show Walking the South Downs National Park https://youtu.be/-kjqtnWtwGQ for hare view, mins 2:06 to 2:12.

- Do you like this film?
- What do you think and feel when you see this film?
- What do you like about it?
- What are the things you didn't like about it?
- What is different between the two?
- If you were a hare leaping in that place, where would you go?
- Which one do you prefer and why?

xxii. Hands on activity: collage

Continue the collage. Ask group to add the Polaroid photos. Ask the group to choose the words in the yellow cards they think describe what they think about the South Downs and National Trust and glue them to the collage.

5. Session Five: Wednesday 7th June Devil's Dyke, 1pm

xxiii. Introductions

1. Connect to YouTube channel
2. Call Sam on the phone. Put on loudspeaker.

What are we doing today? Talk about project, remind group of what we have done before.

Explain today's activity. Today we are visiting Devil's Dyke. Has anyone been to Devil's Dyke before?

Explain to participants that they will give the pilot directions, as if they were in a car directing someone. Give each participant a set of direction cards ⬅ left, right ➡ etc.

xxiv. Livestreaming

1. Pilot introduces himself.
2. Staff introduce themselves and participants.
3. Sam will tell us where he is, take off and will and show the surrounds.
4. As part of giving context, I will mention 3 brief historical information nuggets throughout the livestreaming.

- Devil's Dyke is the longest, deepest and widest dry valley in the UK.
- It was used as a hillfort.
- Hundreds of years ago, there was an amusement park in Devil's Dyke (cable car, rides and a trainline)

5. Ask questions:

- What can you see?
- What can you hear?
- What do you think and feel about seeing this place from the sky?
- Where do you want to go next? Ask participants to use their cards or their voice to give the pilot directions.

6. Repeat questions as prompts or let conversation flow.

xxv. Evaluation

1. What do you think and feel about seeing Devil's Dyke from the air?
2. What do you think and feel about being able to choose what the camera shows?
3. Did you like the experience?

4. What did you like about it?
5. Is there anything you didn't like about this experience?
6. Would you like to do this again?
7. If not, why not?
8. Which words would you choose to describe this experience?

xxvi. Hands on activity: collage

Bring the collage out and ask group to add the Polaroid photos. Ask: what do you think and feel about seeing and touching objects from the past and present?

6. Session Five: Friday 9th June Highdown Hill, 1.30pm

xxvii. Introductions

1. Connect to YouTube channel.
2. Call Sam on the phone. Put on loudspeaker.

What are we doing today? Talk about project, remind group of what we have done before.

Explain today's activity. Today we are visiting Highdown Hill. Has anyone been before?

Explain that they will give the pilot directions like if they were in a car directing someone. Give each participant a set of direction cards ← left, right → etc

xxviii. Livestreaming

1. Pilot introduces himself
2. Staff introduce themselves and participants
3. Sam will tell us where he is, take off and will and show the surrounds.
4. As part of giving context, I will mention 3 brief historical information nuggets throughout the livestreaming.

- John Olliver was a miller in the area, over 300 years ago. Rumour says he was involved in smuggling and used the sails of the windmill to signal when the customs police were not around.
- These trees were planted here, in the time of Queen Victoria, by the Henty family. So they could see the trees from their house. When men were digging the ground, they found this was a cemetery.
- The farming field used to be a Roman bathhouse.

5. Ask questions
   - What can you see?
   - What can you hear?
   - What do you think and feel about seeing this place from the sky?
   - Where do you want to go next? Ask participants to use their cards or their voice to give the pilot directions.
6. Repeat questions as prompts or let conversation flow.

xxix. Evaluation

1. What do you think and feel about seeing Devil's Dyke/Highdown Hill from the air?
2. What do you think and feel about being able to choose what the camera shows?
3. Did you like the experience?
4. What did you like about it?
5. Is there anything you didn't like about this experience?
6. Would you like to do this again?
7. If not, why not?
8. Which words would you choose to describe this experience?

xxx. Hands on activity: collage

Bring the collage out and ask group to add the Polaroid photos. Ask: what do you think and feel about seeing and touching objects from the past and present?

7. Session Five: Contingency (if flight is not possible)

xxxi. Introductions

What are we doing today? Talk about project, remind group of what we have done before and today's activity. Explain the change of plan.

xxxii. Hands on activity: postcards (colouring in)

1. Colouring in an aerial view of the place they had the livestreaming experience from (captured prior to the day).
2. Show pre-recorded content on the landmark they had chosen to focus on.
3. Ask questions such as:
    - What can you see?
    - What can you hear?
    - What would you like to see more of?
    - What have you done when you have been there/if you were there?

8. Follow up visit

Show what has been created as part of the activities with the group and collect group feedback:

This could include:

- YouTube film with activity cards.
- A 35-to-60-piece puzzle.
- Printed postcards.
- Each participant, and the Centre, will be given a copy or version of everything created as part of the research for instance, if it is a YouTube video, they will be given a DVD of it.

We are also doing puzzles and looking at photos.

- Thay- seafront aerial image puzzle
- Vivien – Romell aerial view puzzle
- Eileen- Highdown aerial view puzzle
- Alfie-Cissbury Ring aerial view puzzle
- Delia- Devil's Dyke aerial view
- Shirley- Pullborough Brooks aerial view puzzle
- Sheila- Tangmere aerial view puzzle
- David -Ditchling Beacon aerial view
- Maggie, Eileen- Highdown Gardens aerial view- Maggie might prefer to look at a group of photos.
- Worthing aerial view
- Thay- Pier aerial view

Appendix E: Drone Livestream Participant Engagement Data Table

Table 15, below, aggregates instances of engagement with/from participants and staff, during the two livestream sessions of the co-design research. This data complements the visual drone footage captured during these sessions to record the co-design element of the sessions.

Table 15 Drone livestream participant engagement data

Description	**Livestream 1[49] (LV1)**	**Livestream 2 [50] (LV2)**	**Observations**
Number of invitations to direct prompts	11	12	Similar number of prompts in LV1 and LV2
Number of instances participants' response follows researcher's/staff's suggestion	4	4	Same number of times participants align with researcher's /staff's suggestion
Number of times participants respond "I don't know/I am not sure"	2	1	Livestream 1: 50% higher number of times participants respond "I don't know/I am not sure"
No response from participants	3	4	Livestream 1: fewer instances of no response from participants
Number of instances participants ask about or bring up a detail in the landscape they are viewing	9	7	Livestream 1: Nearly 2/3rds more requests for detail from participants
Number of instances participants ask to continue to view what they are seeing	1	0	Livestream 1: One more request to stay with view

[49] Devil's Dyke livestream
[50] Highdown Hill livestream

Description	**Livestream 1**	**Livestream 2**	**Observations**
Number of instances staff respond to researcher invitation prompts	1	6	Six times higher number of instances when staff directly respond to researcher invitation prompts
Direct requests from staff to view detail	1	2	Livestream 2: 50% more staff directly request a specific detail
Staff ask questions to participants about the landscape or reiterate researcher invitation prompts	3	7	More than double the questions from staff to participants

Appendix F: Visual Analysis of drone videos

This appendix contains the in-depth visual analysis of the selected drone videos selected for this project. This analysis was produced as the background to Chapter Seven of this thesis.

Contents

Visual analysis of drone videos

Each video analysed includes a summary of the conditions of production, a full list of the drone shots in the video and a list of the shots analysed in this research. In videos where there are more than ten shots analysed, time stamps have been added to the respective shot.

1. Institutional drone videos

1.1 Context to National Trust's *Iron Age Hillfort- Cissbury Ring video trail*

- Video titled *Iron Age Hillfort- Cissbury Ring video trail* (LSENT, 2021).
- Proposed for analysis by a Heritage Expert, as an example of work the organisation does using drone technology for audience engagement.
- Produced by London and South East National Trust and hosted by NT in their YouTube channel at https://youtu.be/Wk8vo4eIGi8 .
- Using theme categories by Stankov *et al*. (2019, p. 813), it is considered to feature under Education and Heritage.
- The National Trust describe this one minute and forty-eight seconds-long video as a short clip about the Iron Age Hillfort at Cissbury, South Downs, Sussex. It is presented by a member of staff, Archaeologist James Brown, who introduces the site and describes the surrounding views within a historical context.

xxxiii. Full list of shots

1. Survey Shot
2. Survey Shot
3. Moving Aerial Tilt or Elevator Shot
4. Moving Aerial Pan shots showing morphological details of the hillfort.
5. Id.
6. Survey Shot

7. Elevator Shot as closing shot.

The clip starts with a static image of the Hillfort (with the National Trust logo and showing the National Trust plaque, both part of the institutional branding and identity) in Figure 26 (LSENT, 2021, 00:01).

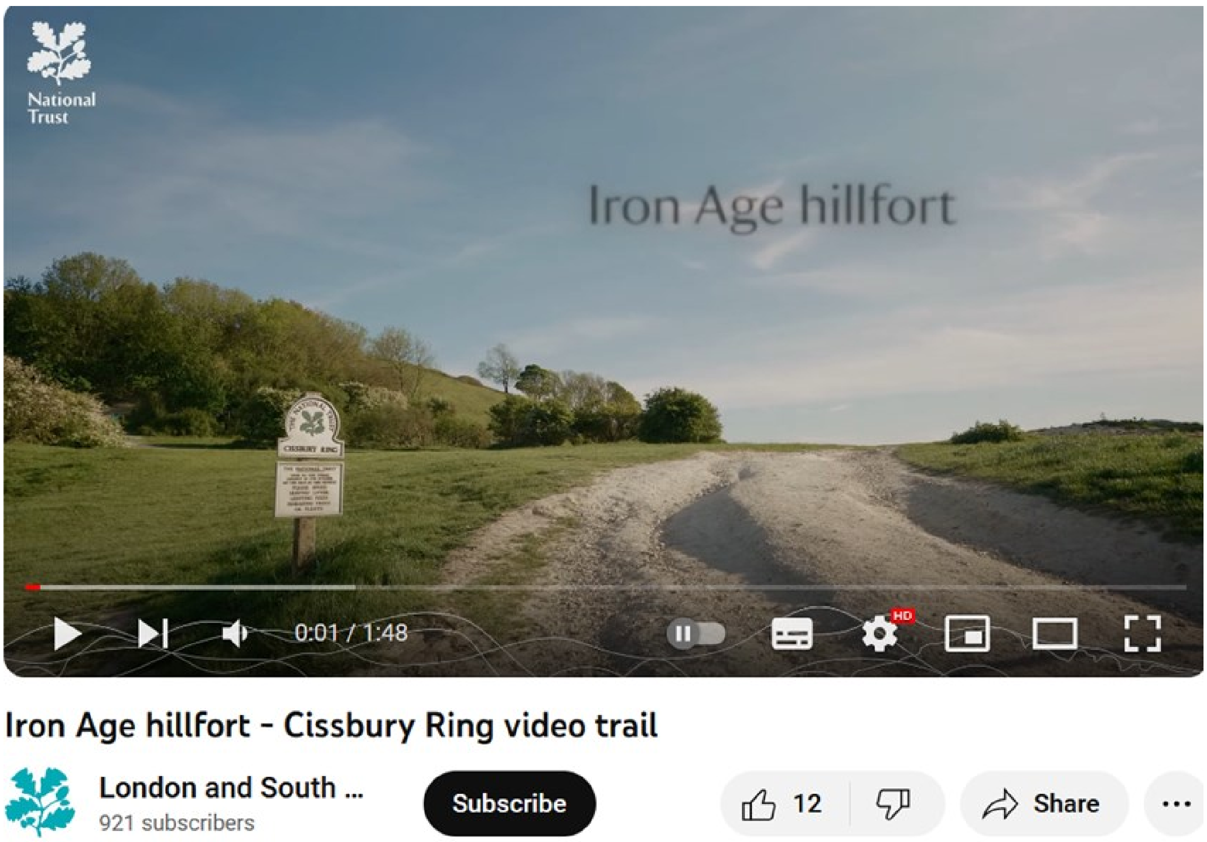


Figure 26 Introductory shot

This sets out the tone for the video as the drone content is predominantly made of Survey Shots, as illustrated in Figures 27 a) (LSENT, 2021,00:30) and b) (LSENT, 2021, 01:34), which are generally associated with functional activities of inspecting and monitoring.

Figures 27a) (left) and b) (right) Survey Shots

These shots may be the legacy of such operations included in the organisational responsibilities of custodianship as illustrated by survey respondent's[51] example of use of drone content for public engagement: "The products of drone survey have been used to illustrate a variety of internal and external communications, whether vlogs, blogs, newsletters etc, but also academic research etc".

Combined with Moving Aerial Pan and Tilt shots, Survey Shots present classic (Serafinelli and O'Hagan, 2022, p. 7) perspectives of the landscape and are part of a traditional approach to viewing and experiencing it which is summarised in Heritage Expert Three's observation that these visuals "kind of got this little England kind of vibe."

The functional aim behind the capture of these shots also informs how they are framed within the video. They are employed to provide context to the heritage experts' considerations. As such, they fulfil one of the opportunities of drone technology, identified by heritage experts from the NT and SDNPA in their interviews: providing "A broader perspective of the landscape that includes context". The drone shots are used in this video to help put the bigger narrative together. For example, presenter James' description of the morphology of the site, referring to the use of chalk excavated and thrown onto the top to create banks as part of a palisade, is

[51] Survey respondent 99754555

combined with a panning shot, illustrated in Figure 28 (LSENT, 2021, 01:13) across the landscape site.

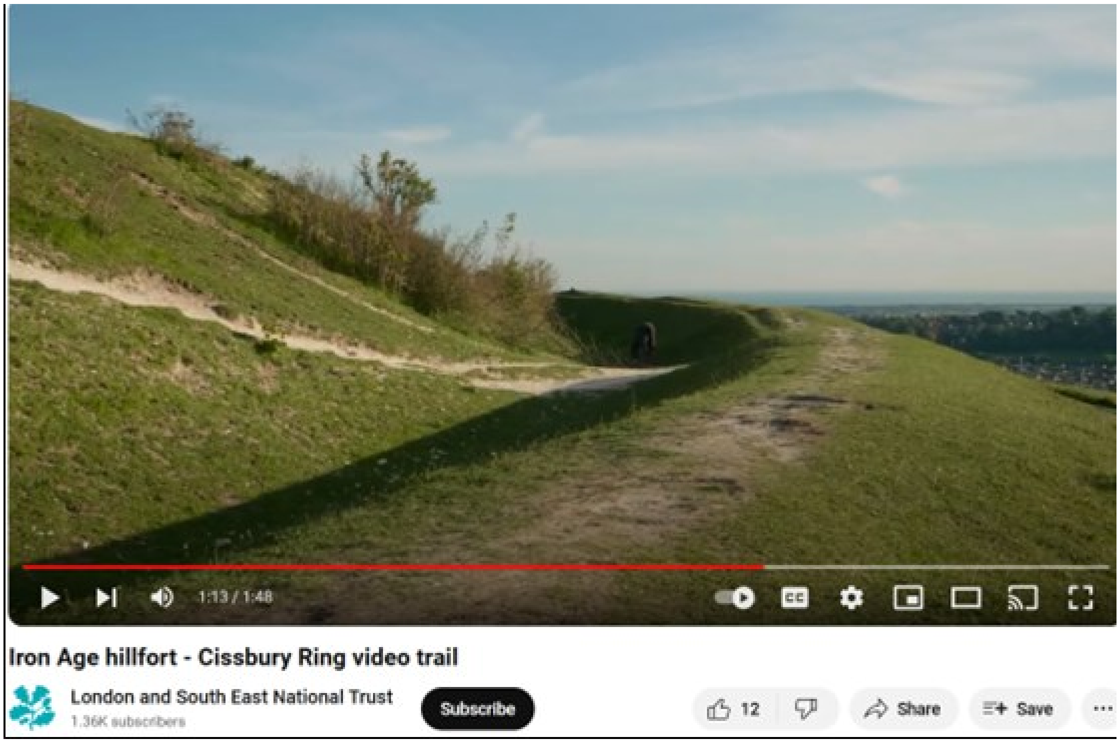


Figure 28 Moving Aerial Pan shot of Cissbury Ring as Archaeologist describes morphology of the site

This illustrates James' point and confers a layer of objectivity to the statement. The viewer witnesses the landscape as source of evidence. Only the drone view is a first-hand source of evidence into the scale and context of Cissbury, enhancing the understanding of its meaning and relevance to place, through the illustrative visualisation of the site and its surroundings. This is a similar effect to a map, and in effect, the next drone shot in the video, also a Survey Shot, presents the drone landscape visuals as a cartographic exercise.

Maps are produced from aerial surveying shots. That relationship between place and data is elicited by this video. Drone technology as a method of data capture emerges in the research findings relating to the opportunities of drone technology, under the theme of "More data of higher quality (and digital archiving)". The volumetric (Hildebrand, 2021, p. 135) experience of the drone shots in the video is triangulated with the secondary sources of narration by a heritage specialist and an ordnance survey map. It is important to consider the relationships between live landscape and mapping reflect deliberations on land ownership, wealth, and access and therefore, maps are instruments of power. This is evident in the examples discussed in the Literature Review (Chapter Two, section 2.1.5) where indigenous communities, whose land has been appropriated by government, use drone technology to document their original land ownership and reclaim their territories from governments and corporations.

Visibility is at the centre of mapping. Histories of underserved communities are frequently not represented in cartographic documents and traditional drone visuals are affected by this issue. Heritage Expert Three considers that the jaw dropping views of the countryside brought through drone technology, exclude the urban landscapes and communities that are a part of its narrative. As a result of this, they argue drone visuals are less relatable and loose some of their power. In minute 00:20 (LSENT, 2021) illustrated by Figure 29, an urban settlement is visible within the range of an Extreme Long Shot.

Figure 29 Urban settlement (Worthing) visible from the Extreme Long Shot

This is a part of the landscape that is not visibly acknowledged as part of offering an opportunity for connection with contemporary human geographies and bridging past and present.

The populated area in drone sight, Worthing, is one of the urban fringe areas with a high level of disengagement with the National Trust (2021, p. 32). By excluding those connections, the video narrative turns invisible the wider context of place and lived experiences (Debord, 1994). Stahl (2013, p. 663) considers drone vision is part of a longstanding imperial looking position that creates hierarchies between those who look and those who are looked at. This idea is separated here from the context of military hostility and surveillance and considered within the field of Media and Culture Studies.

Here, the civic use of drone technology exposes political and socioeconomic hierarchies that are manifested through who hold gaze and who is invisible or left out from it. In this context, those or what is viewed are a part of the narrative of power. The National Trust has a conservative position on the use of drone technology at and from their properties with the exception of internal use. As discussed in Chapter Five, this policy is echoed in the views of some of NT's staff (Challis, 2022).

Although aerial views are considered in my data to widen the context of narratives they explore, in this video, the accessibility of place is affected by the language, some of it technical, used to bring the views presented, into their historical context, for example "ramparts" (LSENT, 2021, 00:14), "gleaming" (2021, 00:16), "palisade" (2021, 00:20), "univallate" (2021, 00:24). The vocal description is illustrated by static images of the sites being described: "rolling Downland" (2021, 00:12), "and the Weald to the North" (2021, 00:13 to 00:14). This assumes a level of education and proficiency in English language, where viewers recognize the specialist terms and read the visual information in the morphological features that are being talked about, when illustrated by the drone shots. This aspect contributes to widening the gap of accessibility, rather than bringing heritage closer to underserved audiences.

The video concludes with an impactful perspective of Cissbury (LSENT, 2021, 01:42) shown below in Figure 30 which, as it emerged in the heritage and drone expert interviews and heritage online survey, provides a form of witnessing (Richardson, 2018, p. 7) historical evidence.

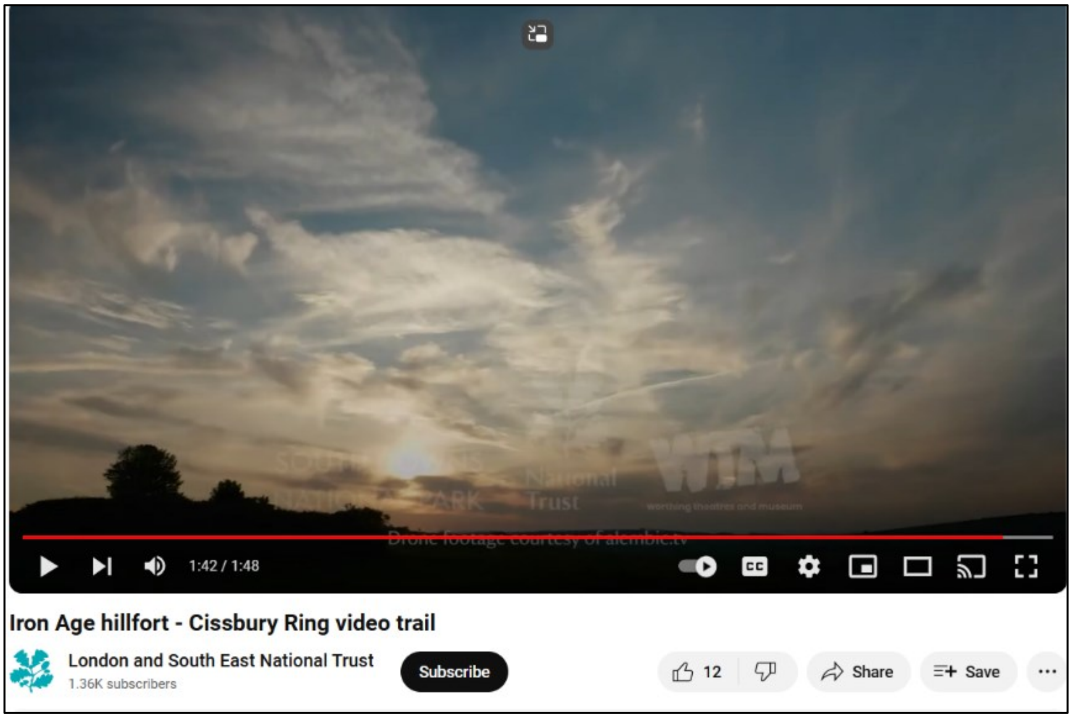


Figure 30 Impactful closing shot of Cissbury Ring

1.2 Context to National Trust's *How to spot a Capability Brown Landscape*

- Video titled *How to spot a Capability Brown Landscape* (National Trust, 2016).
- Proposed for analysis following a suggestion from an NT interviewee, in reference to a project in which drone technology is used effectively in the Media and Video team.
- Produced and hosted by NT in their YouTube channel at https://youtu.be/md2_2v7D6Oc
- Using theme categories by Stankov *et al*. (2019, p. 813) it is considered to feature "How to and style", "Education", and "Heritage".
- The National Trust describe this forty-two second video as a "Bluffers guide" (National Trust, 2016).

NT's description of this video shows an intention to make the subject more accessible and appealing to those who may not be familiar with Capability Brown's extensive legacy of garden design, and what it entails. This type of content is part of a growing trend whereby influencers and a wide range of content creators share their insights and experience, helping the wider public to do it themselves. It is a disruption in the knowledge economy, as it opens up access to expertise. The video is a crib sheet to help people identify or recognise the key characteristics of Capability Brown's Garden design work, as part of celebrating Capability's 300th anniversary.

The video is a combination of drone shots in succession accompanied by written captions. It does not include voice or any physical human presence.

xxxiv. Full list of shots

1. Descent is the Establishing Shot which sets the context: a National Trust property showing the expanse of land, a large building in the shot.
2. Moving Aerial Pan.
3. Survey Shot- "Rivers that aren't rivers".
4. Survey Shot- "He loved thin lakes".
5. Moving Bird's Eye Shot- "He hated formal planting", contrasting with the previous static camera image of lawns up to the front door. Illustrating what formal planting looks like.
6. Moving Aerial Tilt. Reveal Upwards from Medium Close-up to Long Shot.

As in the previous National Trust video, the majority of drone camera perspectives used in this content are Survey Shots and Moving Aerial Tilt and Pan Shots. This, again, illustrates the reuse of drone technology legacy shots for public engagement purposes. Recognisably, the effect of these types of shots is to show the bigger picture and provide context to the narrative. In the case of this video, these shots are organised to present a three-part narrative.

Expanding on each of the above, the Establishing Shot (Figure 31) (NT, 2016, 00:02) provides a significantly more cinematographic introduction to the video than the static National Trust plaque, in the *Iron Age Hillfort- Cissbury Ring video trail*. The drone shot is positioned as the background to the title of the video and the logo of the organisation.

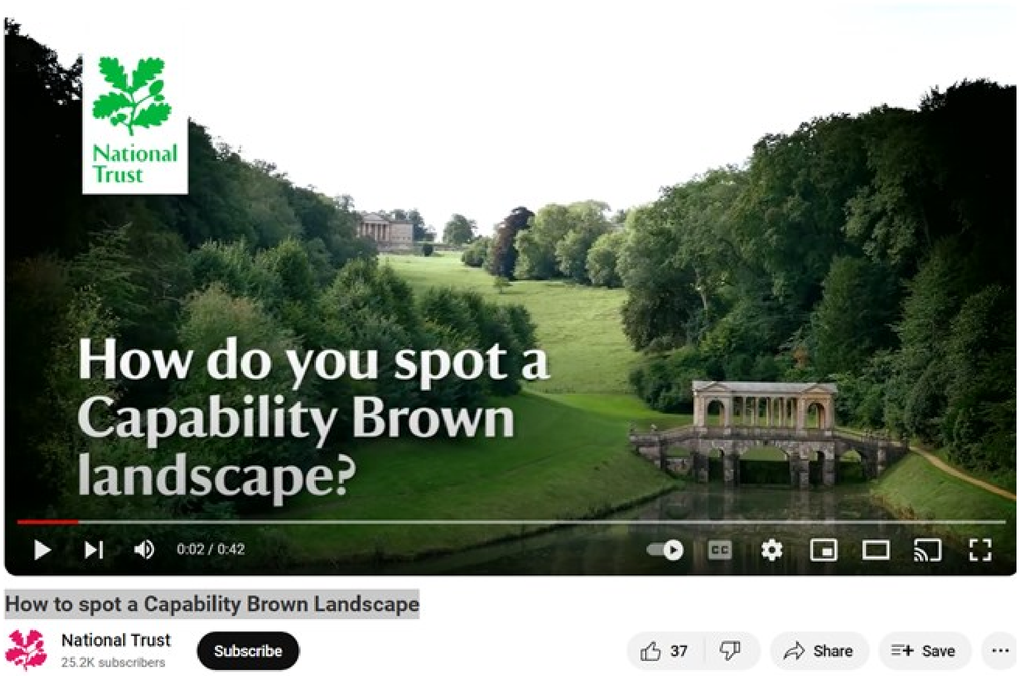


Figure 31 Establishing Shot showing the estate

This first drone shot shows a large area of space from a distant vantage point, where the building is partially obscured (Bowen and Thompson, 2017, p. 65), which creates an atmosphere of mystery. In the frame, there are two subjects: the main house in an Extreme Long Shot and the bridge structure at medium distance. The "classic" landscape perspective (Serafinelli and O'Hagan, 2022, p. 13) presented through the Long Shot of the drone, reinforces the place, and subsequently the institution, as spaces of reverence.

The perspective offered by the descent, positions the viewer into a Lower Angle which inscribes a higher value (Ryan and Lenos, 2020, p. 55) to the estate (building and landscape). Through the movement and angle brought by the Descent Shot, the viewers' eyes are drawn up from the bridge in Medium Shot, to the house in the Long Shot. This results, in my interpretation, in an emphasis on the estate's large size and expanse, an indication of wealth and power.

This composes an auratic gaze, creating a significant contrast in the dynamics of power between the site and those who visit. This is a classic (Serafinelli and O'Hagan, 2022) way of presenting landscape, as a "chocolate box", using Heritage Expert Three's words. Although movement fosters involvement (Ryan and Lenos, 2020, p. 55), the Establishing Shot distances the viewer from the view. This is because the auratic effect projects the lifestyle and experiences symbolised in the landscape views which are disconnected from those of the ordinary person and particularly from audiences with experience of exclusion and whose narratives are invisible.

By framing the house in the Long Shot frame, which becomes more visible as the drone descends, the clip embeds in the frame, the public space distance (Probst, Rothe and Hussmann, 2021, p. 185) resembling the in-person visiting experience. The use of the Long Shot also projects a sense of objectivity (Ryan and Lenos, 2020, p. 48) that I link to the concepts of historical research, knowledge authority and accuracy that are part of heritage branding and specifically that reflect the positionality of the estate's custodian, the National Trust. This initial shot of grandeur, wealth and power projects that authority and dominance.

The Descent Shot is followed by a Moving Aerial Pan in Figure 32 (NT, 2016, 00:05), which brings the viewer and the scenery closer, creating a connection (Ryan and Lenos, 2020, p.56).

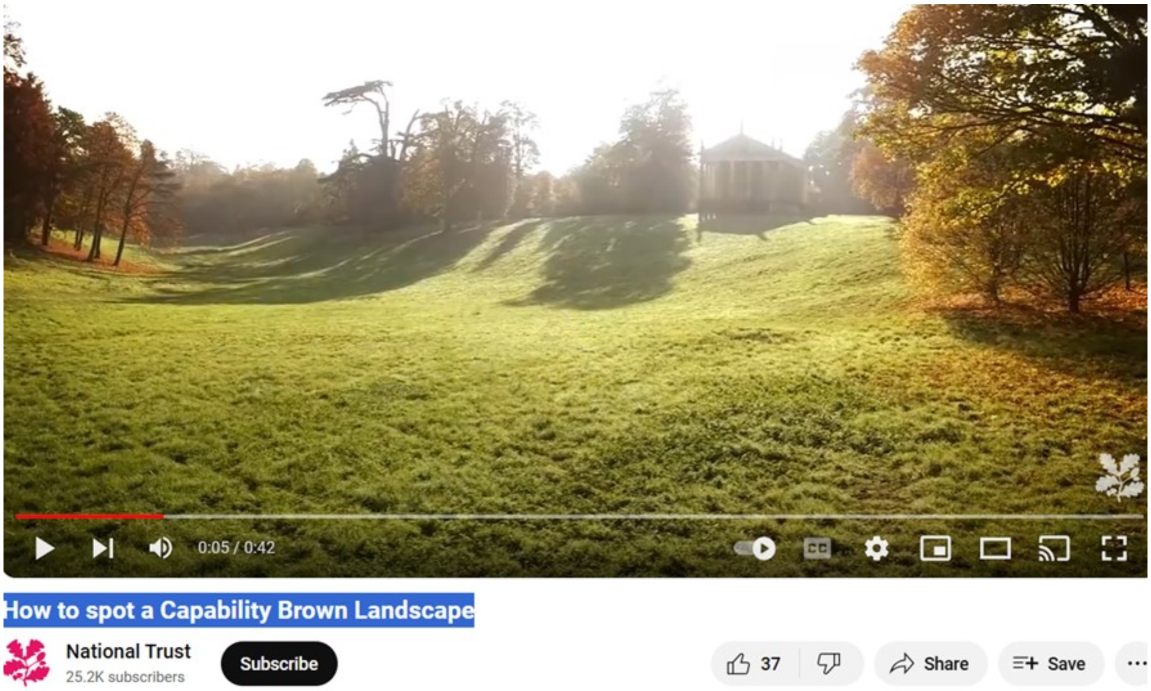
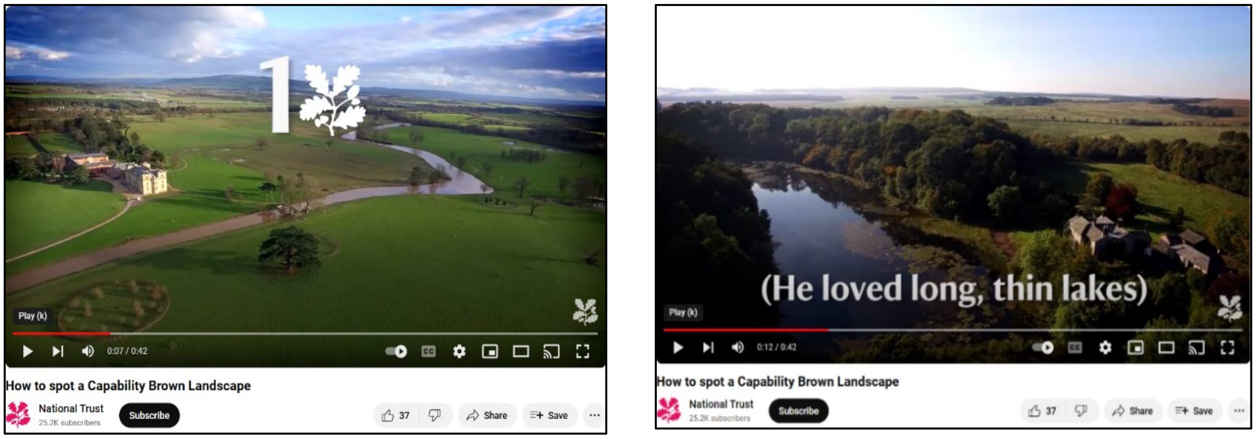


Figure 32 Moving Aerial Pan Shot

Nevertheless, this is still maintaining a significant distance and articulating a similar power dynamic as before, as the house is at a higher angle than the viewer. Introducing the film with this perspective is not coincidental and, according to Bowen and Thompson, (2017) it is a way to establish the place, in its grandeur, and set the disposition of the audience. It can be argued this also has the effect of setting the position of the audience in terms of power and authority, in relation to the institutions'. Although the shots used have a predominantly contextual effect, their execution shows a cinematographic intention that means they have been captured to create feelings of awe, rather than as a result of functional use.

The next three drone shots (Figures 33 a) (NT, 2016, 00:07) and b) (NT, 2016, 00:12) and Figure 34) (NT, 2016, 00:29) also reflect intentionality rather than functionality, by being part of what Heritage Expert Four, an NT interviewee, describes as conveying an innovative approach, different to the usual photography views used by the NT to capture and showcase their sites.



Figures 33 a) (left) and b) (right) Two sequential Survey Shots integrated playfully

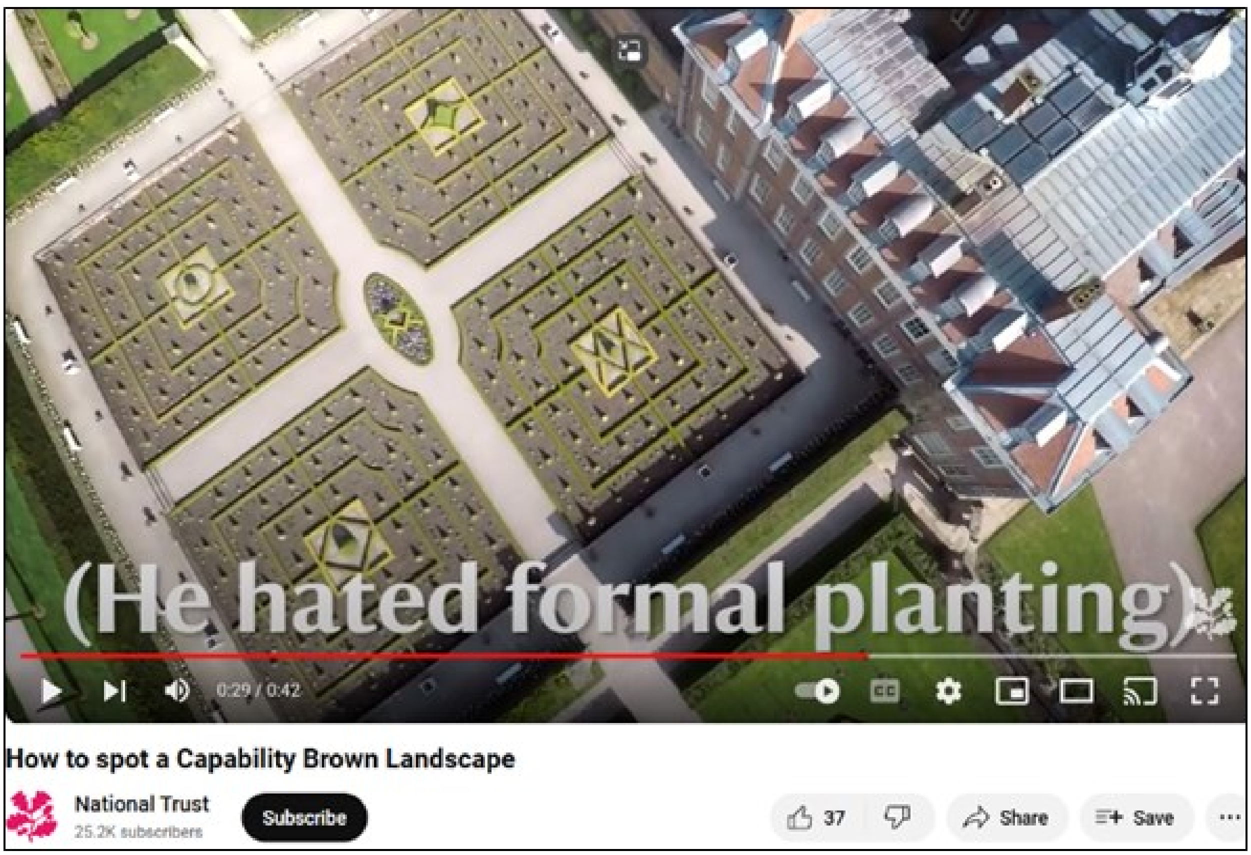


Figure 34 Moving Aerial Tilt Shot of a formal landscape contrasting with the caption

This approach is coherent with the Trust's attempt to "engage audiences differently" (Heritage Expert Four). It is also consistent with the Trust's work to appeal to audiences that are different to their membership demographics, as recognised by Heritage Expert Eight: “We really want to diversify our membership away from that, not because those people aren’t valued but because they represent a tiny fraction of the population of our incredibly rich and diverse country.”

The landscape partnership project *Changing Chalk*, led by NT, aims to reach and involve underserved audiences. “Downs from above” is one of the programmes included in *Changing Chalk*, focused on creating aerial photographs and LiDAR[52] surveys of parts of the South Downs. It has also been used to record coastal heritage at risk in East Sussex and as part of creative work inspired by visits to the Downs. Only my research uses drone content created or directed by underserved audiences.

The playful approach in Figures 33 a) and b) and Figure 34 uses compare and contrast techniques, commonly used in formal learning (school teaching). They each illustrate a design feature of Capability Brown’s design and are accompanied by a descriptive caption, for instance, as shown in Figures 33: a) shows what seems to be a river from a Long Shot perspective followed by a shot with a caption "Rivers that aren’t rivers". This is followed by b) a Survey Shot, medium size, showing a serpentine with a caption "He loved long, thin lakes".

Survey Shots are used as part of discussing morphological features in Capability Brown’s landscapes, as a technique that connects the footage with ordnance survey maps and elevates the drone footage to a digital archiving method, fulfilling one of the six opportunities of drone technology elicited in my data from both heritage and drone experts: “More data of higher quality”.

[52] Light Detection and Ranging, Remote sensing using laser to calculate distances between the laser and the ground.

The use of the compare and contrast technique gives place to a spot the difference format. The Bird's Eye View shot of what, at first sight, looks like a formal garden, in Figure 34 captioned "(He hated formal planting)" encourages the viewer to further scrutiny of the shot, to spot that each area of planting is, in detail, different from the other three.

This is almost imperceptible when viewed from the ground, which is one of the aspects mentioned by Heritage Interviewee Four, when describing the video. It is considered here, the Bird's Eye Shot offers a duality of perspectives: the movement shows it as a dynamic experience alluding to the walk through the structured paths of the geometry of a formal garden layout. The top-down view flattens it (Serafinelli and O'Hagan, 2024, p. 233) making it dull. This forms a contrast with the static image shown previously, of lawns up to the front door of the house.

The top-down view re-perspectivises (Serafinelli and O'Hagan, 2022, p. 10) the subject, whilst flattening it and making it static, enabling it to be read like a map (Jablonowski, 2020, p. 346). Again, the cartographic aspect is present in this video and with it, the aspects of power are further highlighted. Maps contain information that, to be understood, needs to be decoded. This same experience is proposed to the viewer through the compare and contrast approach, which involves audiences in an exercise of decoding their landscape. This has entertainment value for existing heritage audiences and illustrates Serafinelli and O'Hagan's (2022) considerations on drone technology turning landscapes into artistic views, bringing new forms of witnessing this historical figure's work.

Nevertheless, the compare and contrast approach to this landscape further separates the audience from the landscape, in the case of non-engaged groups who are left out. The potential unfamiliarity and or relevance of the theme (Capability Brown's landscapes) and the defamiliarizing effect of aerial views (Serafinelli and O'Hagan, 2022) combined with the required decoding of the landscape, have a compounded effect of disempowerment to audiences who feel excluded by this narrative of wealthy estates and grandiose gardens.

The final shot in Figure 35 (NT, 2016, 00:32) is a Reveal of a historic property. It creates a cinematic aesthetics, with the drone pulling back from the building at a steady speed, progressively showing more of the setting. Again, this perspective shows an epic point of view of the landscape, with the large and majestic building at the centre of the landscape. The view is reminiscent of cinematic views in television period dramas such as *Bridgerton* or *Downton Abbey*, and which are used to mark the wealth and abundance in a narrative. The pulling away motion, which is associated in this analysis with exiting the site, is also in my interpretation, a way to involve the viewer in the journey, marking its end with a sense of awe.

Figure 35 Moving Bird's Eye Shot, final frame in the video

The final cinematic shot of the historic property is part of a composition of institutional power, metaphorized through the architecture, size, and historical context of the property the landscape shown in previous shots, envelops. As Campbell (2018, p. 59) asserts, cinematic views bring visibility and "[T]o be most cinematic, you must adopt the capitalist logic of having and producing symbols of wealth."

This video constitutes a reiteration of the dominant narrative of wealth and power that is associated with National Trust sites. This is not only as a result of the compositional elements of the video referred to above, but also, its purpose of constituting a crib-sheet for identification of Capability Brown's work. The video is framed within a pedagogical approach, using a didactic methodology to explore the theme, through the use of teaching strategies such as comparing and contrasting, for instance. The playful use of visual information and a lightly gamified spot the difference game in a jovial tone, less formal than what the NT is associated with, conveys the idea of innovation and creativity that viewers may associate with the use of drone technology. This works as a rebranding strategy.

The idea of a guide to knowledge emerges from the authoring organisation, the National Trust, positioning themselves as the knowledge authority, using visual language to elicit thoughts of reverence and awe. Differently to the South Downs National Park video on the *Dewponds of the South Downs*, this video does not include human presence, or include any references, links or signposting to content on the lives of those who worked for Capability Brown.

The narratives in the video are those traditionally associated with the National Trust and their reputation of "middle aged and fuddy duddy and, you know, not, not, nothing to interest anybody under 50." (a Heritage Interviewee). This means that, in my interpretation, although the purpose of the video, may be to increase understanding of heritage and landscape, its instructional nature, the classic views of landscape and the pedagogical approach may result in underserved communities considering that there is a significant knowledge and understanding requirement to visit a National Trust property, without which they may find it a challenging experience.

Further to this, it may also lead to considerations around relevance in terms of the narratives to which those communities can relate to. This video is particularly strong on the drone affordances of constituting high-quality data and digital archiving and widening the range of experiences people can have of the National Trust, as it enables a new way of seeing the Trust's places and understanding them. It is also an innovative way of helping the understanding of the story, giving a broader perspective. Subsequently this does increase access to the heritage the National Trust is a custodian of.

The drone shots are purposefully used and fulfil their didactic function, providing a broader perspective that shows Capability Brown's work in context. It widens the understanding of the landscape artist's work, and it is used in an illustrative way as well as humorous. Nevertheless, the views and camera perspectives starting with the Reveal of the area, in the first shot, set the identity of the NT as a place of large estates, power, wealth, and magnitude. All perspectives and shots communicate this. They show a traditional institutional image from a different viewpoint, but one that is still classic and auratic, positioning the institution in a place of authority. It does bring forth the drone technology affordances of widening the range of experiences of heritage, it is creative but does not foster a reimagining of different stories in the landscape.

It widens access through enabling new ways of seeing landscape and its narratives in a way that is not physically possible to do with such an overview of the landscape, broader perspectives and new ways of presenting the data, as well as data recorded differently from archival maps and plans, for instance or photographs. Nevertheless, the element of empowerment is lacking. The audience is being educated, given a crib sheet to join the curatorial understanding of place.

1.3 Context to South Downs National Park's *Shoreham Cement Works- 360° views*

- Video titled *Shoreham Cement Works-360° views* (SDNP, 2022).
- Proposed for analysis by an SDNPA interviewee.
- Produced and hosted by SDNPA on their website and on YouTube, at https://youtu.be/ALPJOS_i-d8
- Using the approach to theme categories by Stankov *et al.* (2019, p. 813), it is considered to feature under Heritage.
- This video has been commissioned by the South Downs National Park Authority as part of the SDNPA's public consultation on the Shoreham Cement Works, for its Area Action Plan.

The video was published in 2022, and it provides access to the now decommissioned Shoreham cement works' site, in Sussex, through 360-degree views captured by a drone with a GoPro camera. In this video, the viewer is in control of their experience to a great extent, within the limitations of the content captured and made available.

The site where the video was filmed is privately owned and except for the chimney, which can be seen from the A Road 283, it is mostly secluded from public view (Heritage Interviewee Six). As such, the SDNPA's choice to show the site through a 360-degree film perspective makes it possible for the site to be accessible, without complex issues around ownership, staffing, people traffic, highway systems, health, and safety. These would have, from my perspective made in person visits impossible.

The choice of 360-degree video facilitates user interaction and control, as the user can choose where to navigate and what to view, using the directional arrows (illustrated by Figure 36).

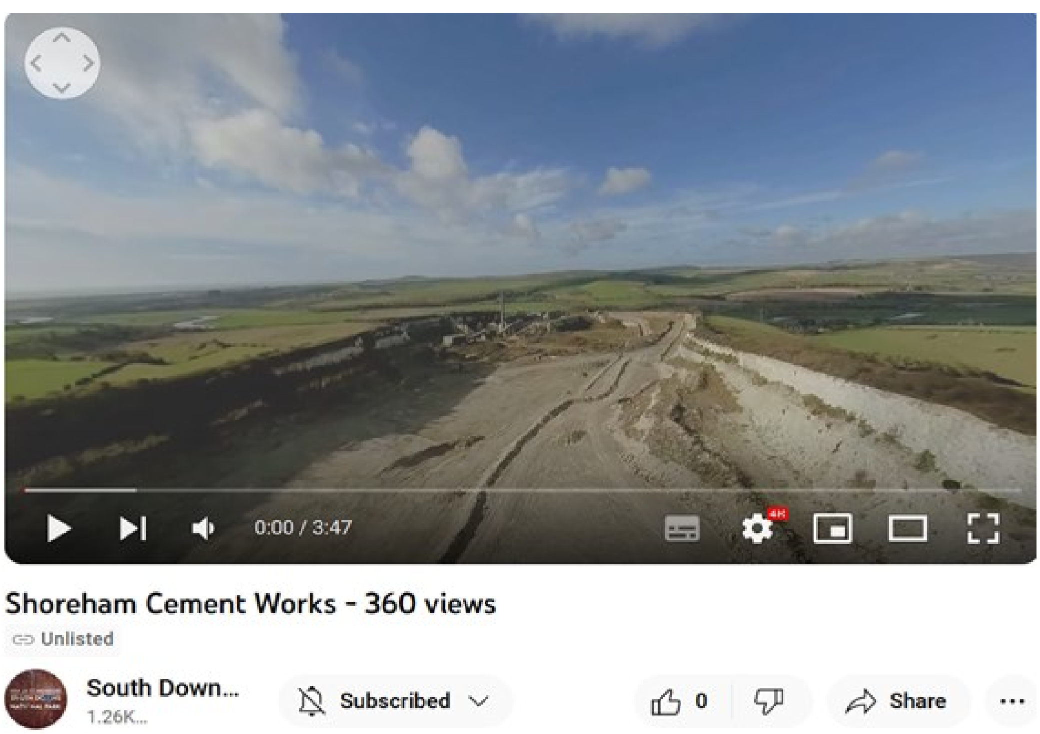


Figure 36 Still of 360° video of Shoreham cement works showing directional arrows used to control the viewing experience

It is not only a replacement for an in-person visit, but it also widens access, offering an accessibility that a walk-in visit would not be able to fulfil, by providing a view of the subject (the cement works), its setting and context, which includes, to an extent, the surrounding areas where key stakeholders are.

The immersive view enables the viewer to gain a better understanding of the relationship of the site with local communities, whilst manually controlling their visual experience. Applying the taxonomy of 360-degree videos by Nasrabadi *et al.* (2019, p. 275), the view is filmed in a horizontal camera motion, without immediately perceptible moving targets, until the viewer zooms in on the chimneys, which are close to the road, where there is the movement of road traffic.

This area of the video is the area I recognise when driving by and which I use, when watching the video, as a point of reference to understand the remainder of the visual information, in a process of orientation of my imagination of being present. Although I am not the drone pilot flying a First-Person View craft, as a 360-degree viewer I have manual control of my experience. Drone panoramas foster a democratising performative cartography, engaging the viewer as a “co-maker” (Serafinelli, 2024, p. 234).

In this context, I negotiate a combination of sensorial registers that create an experience of embodied immersion. Agostinho, Maurer and Veel (2020, p. 255) consider drone filming extend the body’s senses and materiality as a “prosthetic interface that mimics immersive real-site experiences.” My experience as a viewer reflects this. As posited by Jablonowski (2020, p. 351) drones are a “technology of telepresence.”

This video combines different aspects that Stankov *et al*. (2019, p. 811) identify as having brought significant changes to video creation. The 360 degree or panoramic view offers a range of potential viewing directions, which allow viewers to prioritise what is important to them. This level of agency is important in a process of consultation, as there are different communities with different interests and priorities in relation to the future of the site and how decisions impact them. This is associated here with a level of democratisation of spaces and image production. Drone Experts Four and Eight both expressed their enthusiasm for exploring abandoned areas or areas not usually open to public access. The subject of the film features within the interests of drone hobby and commercial users and publics who may previously not consumed drone imagery.

The video, provides an exploration of human air mobility through its simulation as the viewer selects different angles and directions, including top-down views, Long Shots, Close-up Shots through the zooming facility. It gives access to a space that is otherwise boundaried and excluded from public access and is therefore generally not accessible to being captured. Finally, this video engenders as does all drone filming, a new media genre. Through their ubiquity and proliferation in the civic realm, drones, are, as mobile phones, a composite of mediatic possibilities and are therefore more than a mobility technology, a new form of mobile media (Hildebrand, 2019, p. 396).

1.4 Context to South Downs National Park Authority's *Dewponds of the South Downs*

- Video titled *Dewponds of the South Downs* (SDNP, 2021)
- Proposed for analysis by SDNPA interviewee
- Produced and hosted by SDNPA on their YouTube channel at https://youtu.be/mVSkrdJO3yg
- Using theme categories by Stankov *et al.* (2019, p. 813), it is considered to feature under Education and Heritage.
- This six minutes and fifty-three seconds video clip is part of the 2021 Wild Chalk Festival, and it focuses on the restoration of dew ponds as part of wildlife conservation. It is presented by an SDNPA ranger who discusses the historical and landscape context of dew ponds and the work to restore them.

xxxv. Full list of shots

1. Survey Shot

2. Reveal
3. Survey Shot
4. Moving Aerial Pan
5. Survey Shot

Consistently with the first two institutional videos, (potentially) legacy Survey Shots are predominant in this videos drone content. It is found in this research there is a similar approach to the use of drone content, between this video and the formerly analysed *Iron Age Hillfort-Cissbury Ring video trail*. In both, drone shots are used to contextualise and directly illustrate the oral content.

The initial eight seconds of the video (Figure 37) project the identity of the organisation. The Establishing Shot (SDNP, 2021, 00:00) is a Medium Shot Moving Aerial Pan of the South Downs, showing four dry dewponds. This sets the context for the video. This is a branding shot which identifies the clip as part of the Wild Chalk Festival in 2021.

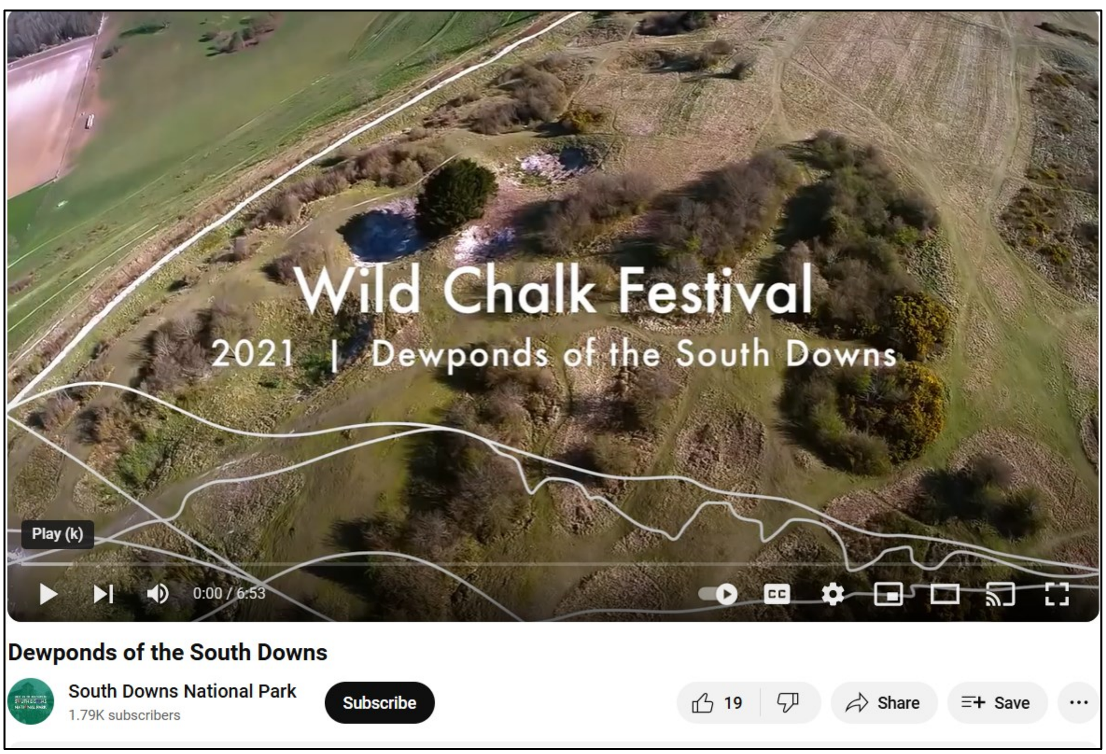


Figure 37 Introductory frame to the video

That shot is followed by a Wide Reveal Shot of the landscape, which in visual language presents the viewing experience as the unwrapping of a present. This communicates that what we are about to see is special and unique: the auratic perspective.

The following two Long Survey Shots reveal the expanse of the area, which in my perspective, has the effect of a proxy to communicate the extent and qualities of the places that make up the South Downs National Park, as in the case of the National Trust's video *How to spot a Capability Brown landscape* (National Trust, 2016, 00:32). Similarly, these two shots reinforce the place, and subsequently the institution, as spaces that command reverence through their breathtaking views (Millerson, 1994). These shots also introduce the subject (dew ponds) in its context, fulfilling an illustrative purpose.

Next, (Figure 38) (SDNP, 2021, 00:04) illustrates the video as it introduces Sophie Brown, an SDNPA ranger, who presents the video and channels the institutional voice. The approachability of the Ranger, their verbal acknowledgement that most people do not live in a setting such as the one in the video clip (in an expanse of land and with nature on their doorstep) and the comprehensive explanation and contextualisation of dew ponds make the video more accessible, relatable and inclusive.

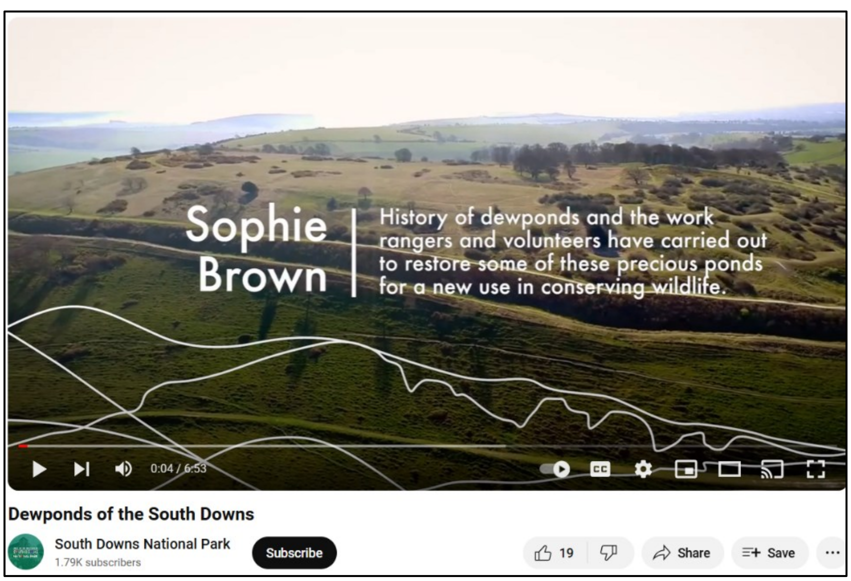


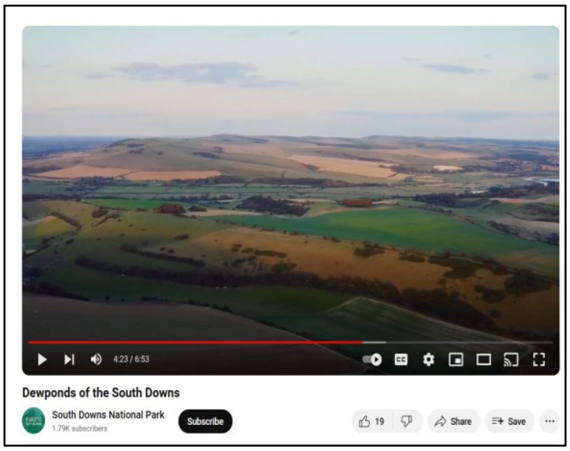
Figure 38 Survey Shot as the background to introducing broader institutional context of the video

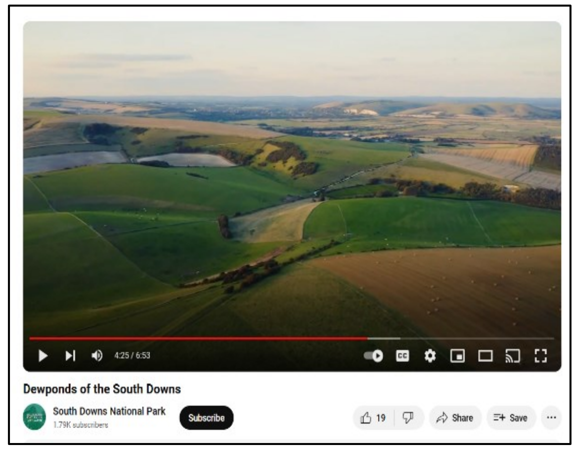
As in the establishing shot, the branding (a scaled back version of the SDNPA's logo design) is layered on the image and is combined with the introduction of the speaker and a written description of the content. On the foreground, the widening Reveal Shot shows the breadth of the South Downs landscape.

Figure 39, below, illustrates the next drone shot (SDNP, 2021, 00:36) is a Moving Aerial Tilt that stabilises height into a Survey Shot, as the speaker talks about the decline in the use of dewponds, witnessed in the surveying shot showing sheep in the Downs and no dewponds in use. The Surveying Shot not only complements the Moving Aerial Pan, providing an overview of the landscape, but also adds a sense of monitoring and data gathering where the viewer, as witness can observe and survey the landscape themselves. The Survey Shot emerges as a primary source of historical interpretation.

Figure 39 From Moving Aerial Tilt to Survey Shot provides context to the information imparted by the Ranger

The next two drone shots side by side in Figures 40 a) Moving Aerial Pan, (SDNP, 2021, 04:23) and b) Survey Shot (SDNP, 2021, 04:25) reconnect the viewer with the landscape of the South Downs (Ryan and Lenos, 2020, p. 56) by providing emplacement into the narrative. Both are, in terms of shot size, Long Shots, showing the spatial relationships between the viewer and the environment (Ryan and Lenos, p. 268).





Figures 40 a) Moving Aerial Pan (left) Shot followed by b) Survey Shot (right)

Although Long Shots are considered to have an effect of creating distance, objectivity, and impersonality (Ryan and Lenos, p. 48), in this instance, their use is interpreted as part of orienting the viewer within the landscape and connecting the detail of the shots of dewponds and that specific narrative, with the bigger picture.

The drone shots in this video fulfil one of the affordances of drone technology, which is providing broader perspectives that contextualise the narrative and illustrate the theme of the video clip. As in the previously analysed videos, some of the frames may have been primarily sourced from legacy drone camera data created as part of organisational surveying and monitoring activities. Again, the length of the drone frames appears too brief for the movement to engross the viewer in the viewing experience, in an embodied way.

The clip is created in the context of viewer engagement and education, raising awareness of the work of the South Downs National Park Authority, the authoring organisation. There is not a connection between the landscape in this video and the urban communities on its fringes, which would make it more relatable.

2. Drone video content generated by underserved communities

2.1 Context to Adagio Dance School's *Road to Freedom!*

- Video titled *Road to Freedom!* (Adagio Dance School, 2021)
- Discovered by researcher in Black Drone Pilots' Facebook Group.
- Posted by pilot and produced in collaboration with Adagio Dance School in https://www.facebook.com/adagiodance.bvi/videos/4006655662748813/
- Using theme categories by Stankov *et al.* (2019, p. 813), it is considered to link to the themes of Heritage, Entertainment, Not for profits and Activism.
- This two minute and thirty-eight seconds clip features dancers from Adagio Dance School, located in the capital island of Tortola, the largest of the British Virgin Islands (BVI) (Cohen and Mascia-Lees, 1993, p. 132)

The video showcases a school dance performance evoking Tortola's past of colonialist subjugation and slavery, possibly also reflecting on the present legacy as BVI are a British Overseas Territory (Cohen, 1998, p. 190). The title may be construed as a reference to where the clip is filmed, and a reflection on the idea of road as a journey. I consider this video is the latter).

xxxvi. Full list of shots

1. Out of focus Close-up Ground Flythrough (00:05)
2. Close-up Ground Flythrough (00:15)
3. Flythrough (Medium, Close Shot) (00:23)
4. Moving Aerial Pan or Fly By (00:33)
5. Ascending Shot? (00:42)
6. Flythrough (01:03)
7. Moving Aerial Tilt (01:08)
8. Ascending Shot (01:09)
9. Moving Aerial Pan (01:13)

10. Moving Aerial Pan and 360° (01:34)
11. Dutch Angle Moving Aerial Pan or Fly By (01:37)
12. Flythrough (01:44)

The first eight seconds, illustrated in Figure 41, set the context, starting with a blurred fast moving Low Angle Flythrough Shot (Adagio Dance School, 2021, 00:06), which refocuses as it passes the threshold of the building, increasing altitude through a Moving Aerial Tilt Flythrough, into where the Caribbean dancers are, girls and women, in two parallel single files, dressed in long white dresses and moving their feet in a slow rhythmic pattern, swaying from side to side.

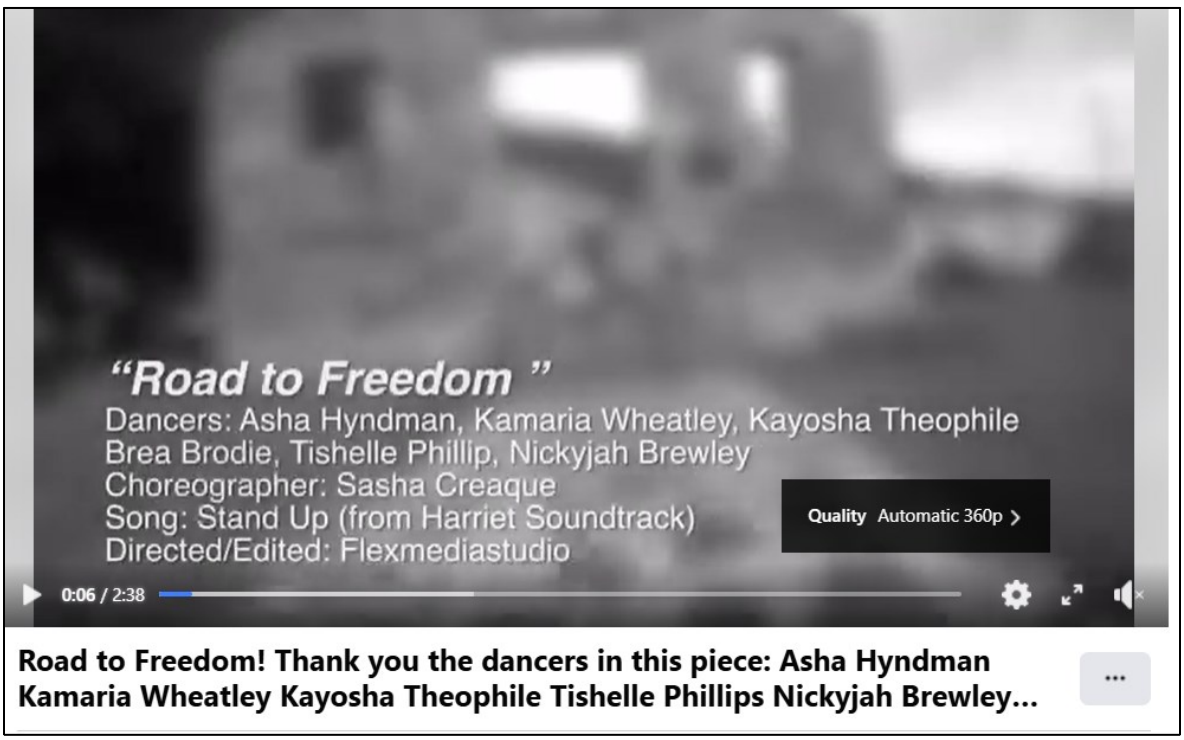


Figure 41 Establishing Shot is a blurred fast-moving Flythrough Shot

The initial blurred shot, monochromatic photography, and unusual proximity to the ground, create a very different aesthetics to the aerial visuals most commonly produced and shared in the drone social media channels I am a member of. The adoption of visuals that disrupt common drone aesthetics, through subverting the use of verticality and mobility, are read here as a metaphor for the self-determination of Caribbean enslaved people. Particularly the resilience of women and girls, as the blurred image becomes in focus (Figure 42) (2021, 00:09) when the dancers are in the shooting frame. This is also a way to emphasize the focus on people, not buildings or land, as a counternarrative to the plantation economics which dehumanises people (Painter, 2016), by commodifying them as assets.

Figure 42 Dancers in focus after the initial blurred shot

Bowen (2017) associates the Low Camera Angle, below eye level, with an effect of power, which in the video is conferred to the slave characters portrayed by the dancers. The Low Angle Shot, busts the myth of powerlessness projected by the forced demeanour of enslavement. As Painter (2016) remarks, slave owners demanded enslaved people display a joyful and submissive attitude. Painter (2016) adds that in advertisements for enslaved people who had escaped, they are commonly described as always looking down. The use of the Low Angle reframes the enslaved people from oppressed to holding personal and collective agency which, during colonialism, manifested through the frequent enslaved people’s revolts (Government of the

Virgin Islands, 2019. In this video, Flythrough Shots are a dominant feature, as Figure 43 (Adagio Dance School, 2021, 00:15) and Figure 44 (2021, 01:23) illustrate.

Figure 43 Flythrough Shot

Figure 44 Another example of a Flythrough Shot

They epitomise the themes in the video: first person voice, owning one's narrative, agency over one's journey. These themes are adjoined to the functions and capabilities of the drone used to create the content: a First-Person View Drone (FPV). As a device, FPVs are used in drone racing and its capabilities are associated with speed, acrobatics, adrenaline. The combination of these with the cinematographic use of the drone through the use of the Dutch Angle, for instance (Figure 45) (Adagio Dance School, 2021, 01:37) creates an aesthetic that is edgy, eliciting the tensions and lyricism of the theme.

Figure 45 Dutch Angle Shot

The video is shot in a location near the coast, inside a building in ruins and without a roof, in Tortola's Road Town, the capital city of Tortola, near the seafront. The clip is monochrome, which creates a composition that is referential to the historical past.

The next drone shot is a repeat of the establishing Low Angle Shot Flythrough drone visual, reiterating the idea of agency and power. After the fixed camera shows the dancers breaking up the shackles, in a movement where they stamp their feet and arch their arms towards the sky. The drone shows a partial Bird's Eye View of the dancers. This perspective is, again, a disruption of the conventions found in photographic image composition, as although the drone shows a Bird's Eye View, which Serafinelli and O'Hagan (2022) define as a classic of photographic perspective, it challenges the rule of three thirds. It splits the frame into two: the building that constrains and oppresses and a part of the group of dancers, reaching to the sky, where the viewer, watches in a God-like way.

It is possible that this approach reflects the dichotomies in British Virgin Islands' identity. A place still immersed in the commodified romance of a colonial past elicited through tourism marketing (Cohen and Mascia-Lees, 1993, p. 131) and the political dependency on Britain, whilst women reclaim their own representation (Cohen and Mascia-Lees, 1993, p. 134) and the Government and communities invest in reframing the country's identity between honouring "[T]he legacy of ancestors, sacrifices, way of life" and forging nation building (Government of the Virgin Islands, 2023, p ii).

Notions of power are central to this video, and they are reflected, in my opinion, in the choice of drone camera angles and movement. The drone movement patterns subvert the "classic landscape views" (Serafinelli and O'Hagan, 2022) that form the traditional expectation of aerial visuals. This is achieved through the ground views that produce Low Angles, projecting a sense of power onto the dancers, as part of changing the narrative on enslaved people and also, by using the aerial views to contrast the sense of personal power with the oppression of the context and the freedom of thought that is metaphorized in the absence of an obstacle between the characters and the sky, populated with their thoughts of freedom and authentic identity.

The use of the First-Person View compounds this sense of the narrated having ownership of the narrative, first person voice. This perspective, which technically allows for the FPV drone to produce Flythrough views, amplifies the sense of intimacy that dominates the video. Through the movement of the Flythrough, the film involves the viewer in the tensions intrinsic to the narrative.

2.2 Context to Women Who Drone's "Happy International Women's Day 2023!"

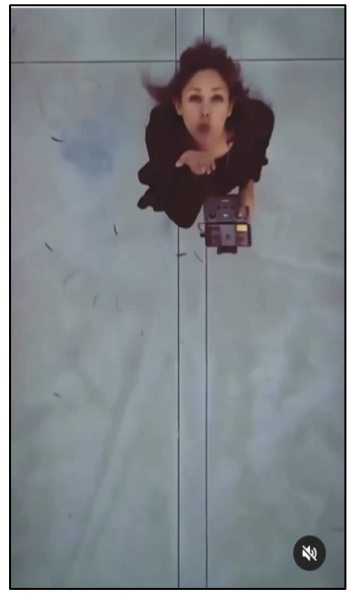
- Video titled "Happy International Women's Day 2023" (Women Who Drone, 2023).
- Researcher retrieved it from the WWD Facebook Group page.
- Posted on WWD's Instagram feed by the group founder, on the 8th of March 2023, as an International Women's Day 2023 reel. Reposted on Facebook Group page.
- Using theme categories by Stankov *et al*. (2019, p. 813), it is considered to feature under Non-profits and Activism, and Education.
- The twenty-two second video reel is accompanied by a written post which starts with the statement "Happy Women's Day! The power of a woman is so beautiful."
- WWD is an online community to empower women and girls through drone technology. I became a member of this group after hearing about it from Drone Expert Five. It was the highlight story in the group's Facebook page on International Women's Day.

xxxvii. Full list of shots

1. Moving Aerial Bird's Eye View
2. Medium Close-up
3. Medium Shot
4. Long Shot
5. Long Shot, Dutch Angle
6. Long Shot
7. Medium Shot
8. Medium Close-up

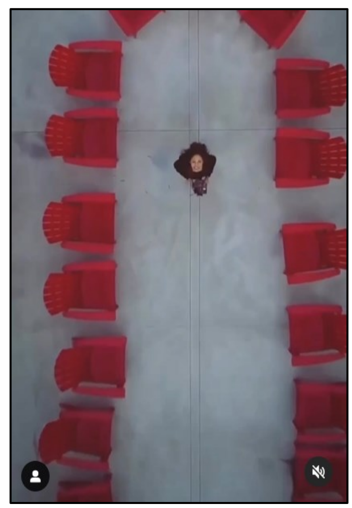
This Instagram reel is exclusively made of one continued Bird's Eye View Shot executed at different shot sizes and a Dutch Angle. In the first visual (Figures 46 a) 2023) and b), the Moving

Bird's Eye View Shot shows a Medium Close-up Shot of the pilot holding a drone controller and looking directly on to the drone camera.



In cinema, the Bird's Eye View or God-like vision (McCosker, 2015, p.13) reveals a character's movement in the environment. This effect is fulfilled when the pilot waves and blows a kiss at the viewer. The use of the Bird's Eye View also has the effect of holding viewers' attention through revealing a new perspective that creates an emotional response in the viewer (Agostinho, *et al*., 2020, p. 25).

The Close-up Shot focuses the attention of the viewer (Hanmakyugh, p. 110) on the female pilot, creating intimacy (Bowen, 2017; Probst, Rothe and Hussmann, 2021, p. 185) between the viewer and the subject (Hanmakyugh, 2020, p. 110). The Medium Close-up also spotlights the pilot's initial small gesture of blowing a kiss to the audience. Close-up Shots allow the filmmaker to control what the viewer sees (Hanmakyugh, 2020, p. 110) including and excluding what they want to show or hide (Hanmakyugh, 2020, p. 111).



Figures 46 a) First shot (top), Moving Bird's Eye View ascending to b) to show a broader, symbolic visualisation (bottom)

As the drone ascends, the drone view shows the pilot is surrounded by 21 red empty chairs (representing the date of the 21st) set in an oblong boardroom shape, a metaphor for women in leadership. The pilot is inside the shape.

The upwards motion of the camera shifts from focusing on the pilot to, as it ascends, widening the field of vision, to progressively show the context, as illustrated by Figure 47, the roof terrace of an urban building. The red chairs position the drone pilot as the focus point and become part of a practice of resistance through visibility (Brighenti, 2010, p. 188).

Figure 47 Continuous Ascending Shot showing moving from the individual to the collective

Visibility is part of the process of negotiations of power, and power is rooted in the management of visibilities (Brighenti, 2010). Drone Expert Five discusses this in their interview: “So the history, when it comes to aviation, not to give a lesson but if you didn’t know a pilot, if you weren’t related to a pilot, if you didn’t know a pilot or if you weren’t in the military, chances of you becoming a pilot or doing anything with that or having an aviation career was really slim. So particularly with drone education, I do believe it is a gateway.”

The pilot’s visibility, here, epitomises the underrepresentation of women in the aviation industry and it disrupts the genderised perception of drone users as male. There is one single instance of obvious gender bias in my data, in a heritage expert survey response to the question: In what ways do you think drone technology could be used to engage non-visiting audiences with collections, narratives and sites? Participant 95192171 answered: “My instinct is that it may help to encourage an interest from boys and younger men who are attracted by new technology.”

The ascension effect can also be interpreted as the pilot taking other women up with her, on this rising journey to the top. The Extreme Long Shot widens this context establishing the setting (Hanmakyugh, 2020, p. 109), the top of a building in an urban area surrounded by other executive looking buildings. Before reversing the journey, as illustrated in Figure 48, as the drone ascends, it turns 180 degrees into a Dutch Angle and the square flattened view of the building becomes a lozenge shape, further defamiliarizing the view of an everyday structure, and subsequently captivating the viewer (Serafinelli and O'Hagan, 2022, p. 8*).*

Figure 48 Dutch Angle showing a flattened view

The use of the Bird's Eye View, which is considered to position the viewer in a God's point of view (Serafinelli and O'Hagan) and a position of power and dominance over the subject here, is used partly in a subversive way. As previously mentioned in section 5.3, in relation to "The drone gaze and auratic views," vertical angles are associated with divine omnipresence in Judaeo-Christian thinking (Serafinelli and O'Hagan, 2022, p. 7) reflecting an unequal power dynamic between the powerful creator and the viewer being observed and judged.

In this video, the initial Top-down Shot, positions the drone pilot as both the viewer and the creator, challenging the established hierarchies and power dynamics of a patriarchal society. In the Women Who Drone (WWD) Facebook page, members discuss the abuse and aggression they receive from male group members when they post on other drone groups. The Top-down view is considered to generate emotional responses (Royo-Vela and Black, 2020) which is consistent with the purpose of the video, celebratory of women and the central aims of the WWD group: to create a safe space for people who identify as women; whilst challenging gender bias in Science, Technology, Engineering and Mathematics, inspiring and empowering women.

The Descent depicted in Figure 49 reverses the initial trajectory and takes the viewer back to the beginning, from the collective and broader narrative to the individual. The Close-up Shot is used as an opportunity to make a connection with viewers and establish the narrative for International Women's Day. It allows viewers to relate to the pilot. It is not often that pilots feature themselves in drone videos and this is, in itself, a different and empowering approach, that is about visibility of an underrepresented group. The Close-up is used to establish intimacy, express affection, and solidarity.

Figure 49 Reverse Shots (descent)

The connection and relatability of the Close-up bring viewer and content creator together and, as the drone ascends, both are in a God's view position, rather than just the viewer holding power over the subject. The ascension into a Long Shot has the effect of broadening the narrative from an individual perspective to the collective, beyond personal (Ryan and Lenos, 2020, p. 48) which in this project is interpreted as the public arena (Probst, Rothe and Hussmann, 2021, p. 185).

The element of control, in the sense of personal power, is strongly present in the control of the drone vision itself, through the different shot sizes and angles in sequence, in a very short amount of time, without straying from its route or loosing focus.

2.3 Context to Attitude Magazine's *Bimini and Dan O'Neill on healing the ocean as LGBTQ people*

- Video titled *Bimini and Dan O'Neill on healing the ocean as LGBTQ people* (Attitude Magazine, 2022).

- This film was found as a result of searching for drone content created by Jasmine Qreshi, a transgender writer, drone pilot, wildlife filmmaker and marine biologist who was assigned as my mentor, as part of the *Shifting The Gaze*[53] programme.
- This is hosted on YouTube at https://www.youtube.com/watch?v=5kD-zs4aFeA
- Using theme categories by Stankov *et al*. (2019, p. 813), it is considered to feature under Activism.
- This seventeen minute and forty-two seconds video is produced by Attitude Magazine, a magazine of LGBTQ news. It is part of the #SeaOurFuture collaboration, bridging the separation between people and the ocean.

As some of the communities most affected by institutional discrimination leading to the experience of health, employment and housing disparities, the Lesbian, Gay, Bisexual, Transgender, Queer and Intersex (LGBTQI +) population is significantly impacted on by harmful environmental impacts (Goldsmith and Bell, 2022)[54]. This video engenders practices of visibility of LGBTQI+ communities and their environmental activism, from both an optical point of view and the perspective of recognition of socio-political actions and subjects (Brighenti, 2010, p. 188).

xxxviii. Full list of shots

1. Static Bird's Eye Shot
2. Moving Bird's Eye Shot
3. Static Bird's Eye Shot
4. Static Bird's Eye Shot
5. Static Bird's Eye Shot
6. Moving Tilt Bird's Eye Shot
7. Moving Bird's Eye

[53] This programme is part of the National Trust led *Changing Chalk* project. It is a partnership between the SDNPA and Brighton-based arts organisation Writing our legacy. *Shifting The Gaze* provides funding and mentoring to ethnically diverse writers to research and develop their writing inspired by the South Downs National Park.
[54] This is because these communities experience significant exclusion which puts them at greater risk of homelessness and displacement; challenges in asserting their rights within the political system.

8. Ascending Moving Bird's Eye Shot, from Close Shot to Medium Shot

All the identified drone shots in this video are Bird's Eye Shots that are either static or dynamic. As discussed in the Women's International Day 2023 Instagram reel, the Bird's Eye View captures the subject's movement within the environment (McCosker, 2015, p.13). This is pertinent to this film piece as it discusses human interactions with the environment, the changes they create and the need for communities to move differently within the environment, in order to preserve its equilibrium.

Royo-Vela and Black (2020) consider the Bird's Eye View perspective produces emotional responses and the climate crisis is an emotional subject matter. This is also the case for the discrimination experienced by LGBTQI+ communities excluded from the conversations on the environment, whilst being part of the narrative of groups who are the most vulnerable to its impacts, as a result of institutional exclusionary systems and attitudes.

In the video, the drone shots are used in two different ways: to illustrate or metaphorize the discussions in it and to articulate personal narrative. The illustrative function of drone images is applied in different ways:

i. As a direct illustration of verbal discourse (2022, 02:56).
ii. A metaphor for an idea communicated in the verbal speech in the video (2022, 07:15).
iii. A reflection on personal experience (2022, 01:58).

An example of i. is when Dan talks about the land and the ocean being interconnected and interdependent (Attitude Magazine, 2022, 02:56). The Moving Bird's Eye View shown in Figure 50 illustrates this in a linear way.

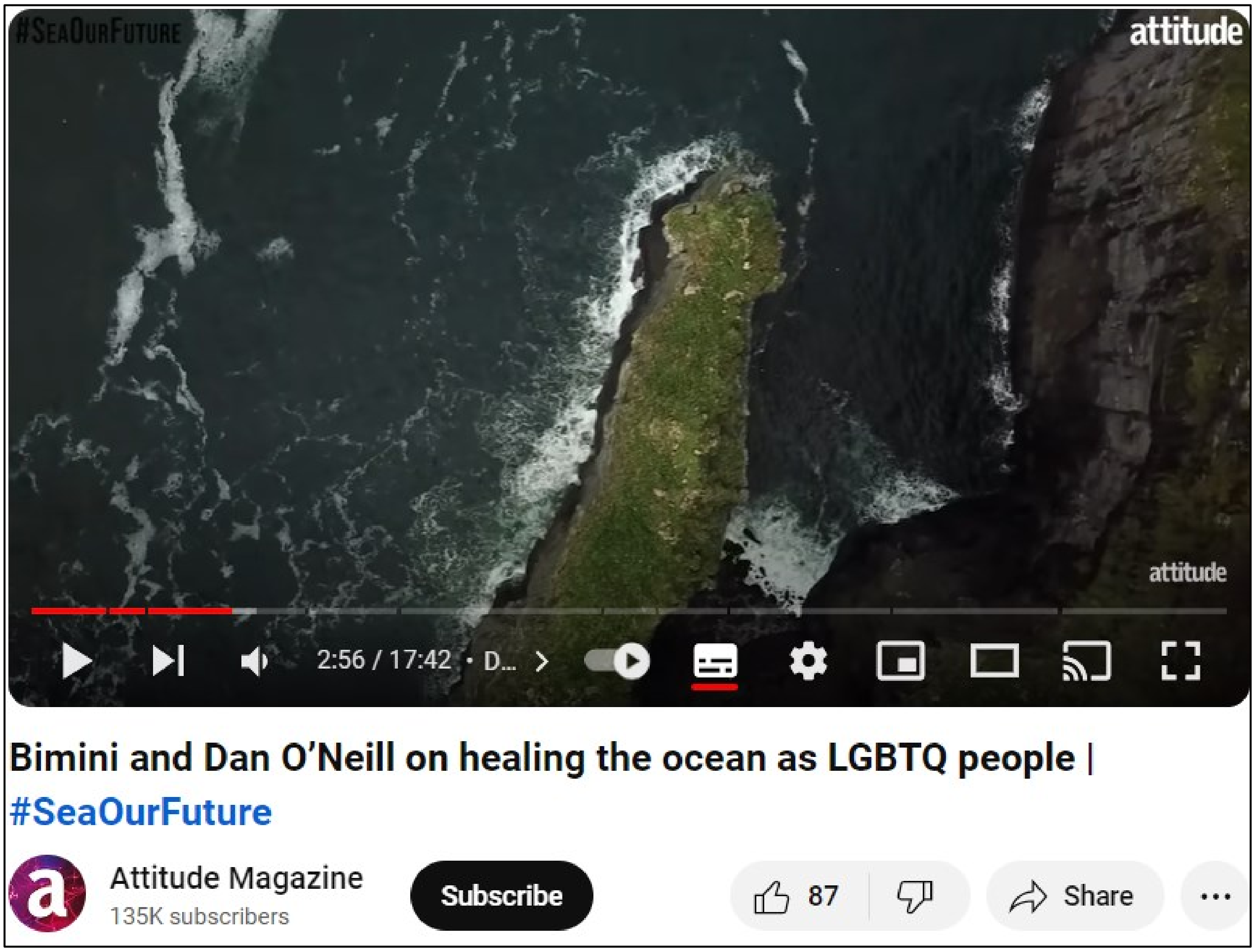


Figure 50 Moving Bird's Eye View of the ocean

The next shot, illustrated by Figure 51 (Attitude Magazine, 2022, 07:15) is an example of ii.: the use of the drone visual as a metaphor, as Noga Levy-Rapoport talks about their fears of the climate crisis. Whilst they discuss their concern about human disconnection from the environment, the drone visual shows a unique view of how the sea and the earth, both have their place in our ecosystem, and both are connected.

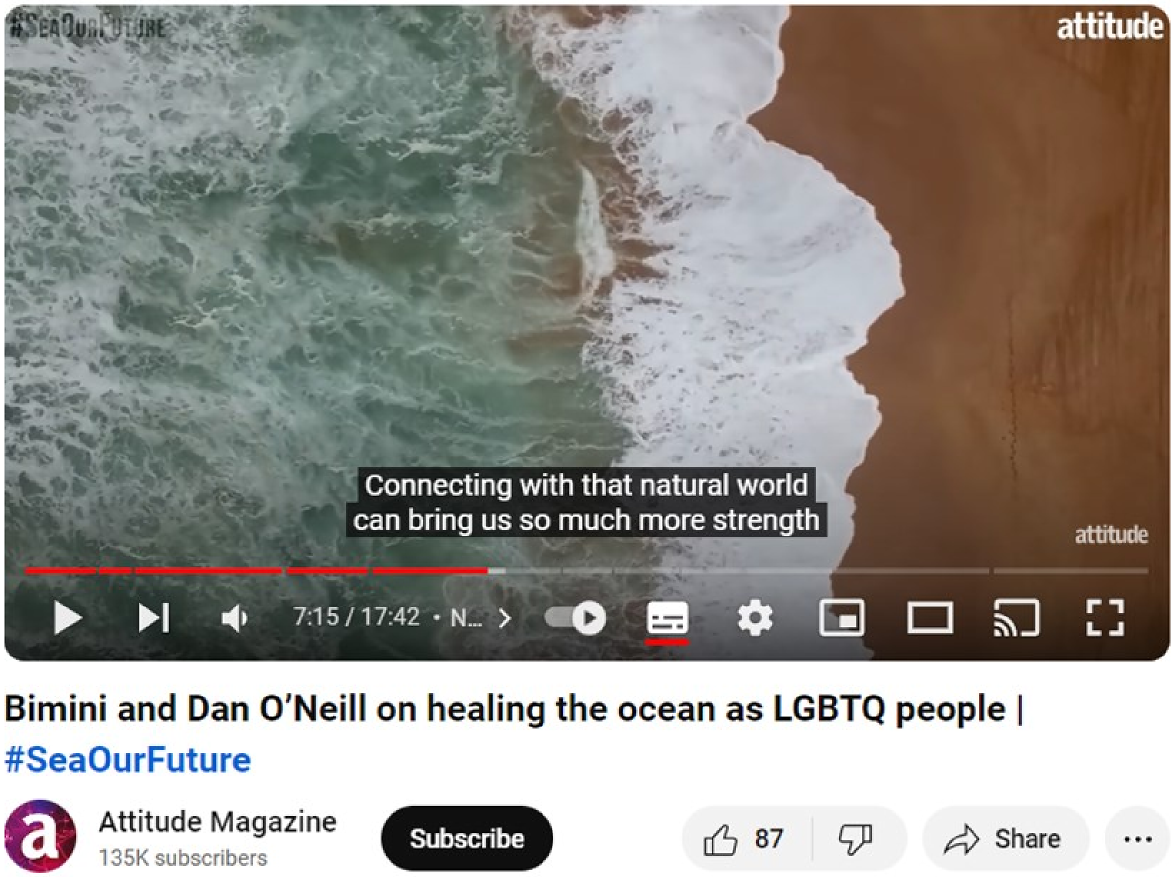


Figure 51 Bird's Eye View of the ocean showing the connection between the ocean and land

The static aspect of this Bird's Eye View puts the focus of the viewer on to the ocean, as the main character in the scene, as it webs on to the beach. The ocean is both powerful, as part of the forces of nature and vulnerable to the power of human action elicited through the top-down position of the viewer observing and acting on the ocean. This power is disrupting our ecology. The fear of disconnection engenders shame which, in turn, silences people's authentic selves (Brown, 2011).

This is particularly important for underserved communities, such as LGBTQI+ groups represented in the video. The oppressing and silencing inherent to experiences of exclusion affect connection and belonging. By being vulnerable and authentic, the activists featured in the video also become visible. This happens as a result of them harnessing their personal power and capacity to create change, which is ultimately central to collectively address the climate crisis.

Finally, as an example of iii., the Bird's Eye View of the ocean in Figure 52 (Attitude Magazine, 2022, 01:58) following a fixed camera Close-up of Dan O'Neill, continuing the intimacy that frames Dan's sharing of the exclusion they experienced as a zoologist who is gay.

Figure 52 Medium Close-up Bird's-Eye View of the ocean

The Close-up Shot of the waves is shown through a slow, backward, Medium Shot, moving against the current, in a Reveal effect. As the drone moves against the tide, showing the waves bursting, the viewer hears Dan describe the meaning and role of the ocean to them, as a place of healing from experiences of homophobic microaggressions, whilst performing their role as wildlife presenter.

The view of the ocean is used to illustrate the relational experience, which the viewer is invited to witness by entering the personal space and intimacy shared between Dan and the ocean (Serafinelli and O'Hagan, 2022, p. 10) eliciting an emotional response from the viewer (Serafinelli and O'Hagan, 2022, p. 9). The slow backward motion shot is a metaphor for defiance of the power of societal pressures to conform to the dominant aesthetics and ways of being, established by a heteronormative hierarchy.

Simultaneously, the flattened view of the waves, hints at the power of nature seen from a unique perspective that includes proximity, impossible to access in any other way, as the size and agility of the drone enable it to be captured without interfering with it, as a helicopter, for instance, would by creating ripples in the water, if it flew that close to the ocean. Further to this, the flattening of the ocean through the Top-down view cartographises the ocean, eliciting its importance as a way of knowing (Proulx *et al*., 2021, p. 2). As Proulx *et al. (*2021) argue, observing and reading the ocean are central to understanding climate change and the preservation of the marine environment.

In this frame, Figure 52, the top-down views of the ocean are used as a vehicle to establish territorial and human hierarchies, that continue to be amplified in power imbalances and exclusionary practices. These silence marginalised communities, such as the LGBTQI+ population, ethnically diverse groups, disabled people, working class communities, women and all the intersections of identities that further compound experiences of exclusion. Proulx *et al. (*2021, p. 2) state: “We cannot achieve equity and inclusion in the management of marine resources, without an understanding of the ocean that includes learning to respect other ways of knowing and working together”. This is relevant because this video clip is focused on making the voices of LGBTQI+ environmental experts and campaigners, part of the bigger conversation.

This video clip is about visibility of LGBTQI+ communities and the intersectional environmentalism - those experiencing exclusion are most affected by ocean health; the importance of diversity in environmentalism, connection to the ocean and empowerment of excluded communities to speak up and be involved in the conversations. The drone shots are not at the centre of the video, the people are. The shots are used to illustrate the conversations, as well as show different perspectives of the ocean.

Bird’s Eye Shots dominate the use of drone shots and although these have a flattening effect (Serafinelli and O’Hagan, 2022) in this case, the use of predominantly Moving Bird’s Eye Shots creates an effect of focus on the ocean and a positioning of the viewer as an observer of the ocean, a spectator in a gallery viewing the dynamic artwork changing, being fluid, freedom boosting. Movement in the Bird's Eye View highlights the value (Ryan and Lenos, p. 55) of the ocean, the theme of the video.

This is then enhanced by the use of Medium and Close Shots that establish a connection between the viewer and the images and theme which is about people connecting with the ocean and taking part in collective action. Medium and close shots are also creating visibility- the same visibility that is discussed in the video in terms of excluded communities. The use of the shots is more illustrative than orienteering and contextual. The context here is connection and the use of drone shots enhances and fulfils this.

The use of drone shots in a film that is not a drone video, has the effect of offering viewers a reference to reality, the theme in focus (Savardi *et al*., 2023, p. 79). The use of Medium and Close Shots rather than Long Shots, not only fosters connection, in terms of visual language, it is also a way to emphasize how humans are in control of what happens to the ocean, the use of the Bird's Eye View is associated here with the meanings of the High Camera Angle. their dominance and power of the God-like point of view, also highlighting the ocean's vulnerability (Savardi *et al*., 2023, p. 79) to human action and choices.

2.4 Context to Madalena Fine's (2023), *These too are footsteps* and *A room of my own*: film-poems

- Two film-poems integrated into one video: *These too are footsteps* (2023a) *A room of my own* (2023b).
- Autotechnographic research.
- Not publicly available.
- Using theme categories approach by Stankov *et al.* (2019, p. 813), it is considered to feature under Heritage.
- Both films test approaches to using drone technology to develop a film-poem to explore my experience of the South Downs National Park, in the context of reframing my identities as a writer, migrant, ethnic minority, and disabled cisgender woman. The films

explore the role of writing in nature as part of my process of negotiating my sense of self, placemaking, healing from displacement and finding home and family as places within myself.

These films are a discussion on my relationship with the drone, embedded in more general aspects of my identity as a woman, researcher, and Global Majority individual. The second video, *A room of my own* (2023b), also explores identity and creative practice within that, drawing on Virginia Woolf's essay *A room of one's own* (2025) to reflect on identity, representation, and visibility of ethnic minorities. I primarily analyse *These too are footsteps* (2023a) and draw from *A room of my own*'s (2023b) salient points to avoid repetition.

The Research and Development work has been funded by *Shifting The Gaze*, an SDNPA programme in partnership with Writing Our Legacy, to support Global Majority writers to create work resulting from their engagement with the South Downs. This programme is part of the bigger National Trust-led project "Changing Chalk" (National Trust, no date). This work has also received funding from the University of Brighton School of Arts and Media's Centre for Digital Cultures and Innovation, to work in collaboration with a filmmaker and a drone pilot to produce the ideas resulting from the *Shifting The Gaze* Research and Development activity.

xxxiii. Full list of Shots

These too are footsteps…

1. Moving Aerial Pan
2. Moving Aerial Pan
3. Survey Shot
4. Moving Aerial Tilt
5. Survey Shot
6. Top-down view "Sodade", Dutch Angle
7. Circle Arc Drone
8. Ground Close-up Shot. Static Drone Shot

9. Survey Shot of the Downs
10. Drone Ground Static Close-up
11. Split screen of my feet. Replicating shot above
12. Drone Survey FPV of Calçada Portuguesa like a runway
13. Drone Ascent. Out of focus. Collision with a tree
14. Drone Ascent over the Downs
15. Moving Aerial Pan of the Downs
16. Close-up of Padrão dos Descobrimentos- Elevator Shot downwards
17. Close-up, side view, Ascent
18. Survey Shot of Downs showing Monks House
19. Survey Shot
20. Moving Aerial Pan
21. Elevator Ascent

A room of my own

1. Top-down view Ascending. Medium Shot to Long Shot
2. Monks House walk through
3. Moving Aerial Tilt
4. Moving Aerial Tilt
5. Circle Arc Medium Shot
6. Survey Shot of South Downs
7. Fly by
8. Moving Aerial Pan or Circle
9. Ascent
10. Medium Shot of Monks House. Elevator Shot
11. FPV Flythrough with music
12. Top-down views of Calçada Portuguesa Ascent
13. Ascent
14. Survey Shot and Ascent

xxxiv. Selection of shots analysed

These too are footsteps

- Moving Aerial Pan (00:15)
- Moving Aerial Pan (00:18)
- Survey Shot (00:22)
- Top-down view "Sodade", Dutch Angle (00:34)
- Circle/ Arc Drone. Close-up (00:41)
- Ground Static Close-up drone Shot (00:49)
- Ground Static Close-up (01:00)
- Split screen of my feet. Replicating shot above (01:04)
- Drone Survey FPV of Portuguese Historic Pavement (01:10)
- Drone Ascent. Out of focus (01:15)
- Close-up side view Ascent (01:46)
- Survey Shot of Downs showing Monks House (01:54)
- Moving Aerial Pan (02:13) (end of *These too are footsteps*)

A room of my own

- Top-down view Ascending. Medium Shot to Long Shot (02:37)
- Circle Arc Medium Shot (03:08)
- Medium Shot of Monks House. Elevator/Moving Aerial Tilt (03:36)
- FPV Flythrough with music (03:48)
- Survey Shot and Ascent (03:59)

The storyboard for the films were created prior to shooting. Both films include a combination of drone footage I have captured or requested the drone pilot I worked with to capture, and drone footage from third parties (with their consent). Although the latter have not been created by myself or under my specification, they are the shots that I would have captured if I had been able to shoot on location, in Lisbon.

The shots are relevant, and pertinent to be included in this analysis alongside the original material analysed, because they have been identified and selected in the context of articulating my narratives of personal power, identity and placemaking. My analysis includes a selection of the three sources of drone shots, based on their relevance to the construction of the film narrative. It is important to mention, the films were created before engaging in the analysis of the other five drone content videos discussed earlier, in this section of the Appendix.

Although not consciously, I note the narrative is constructed through the alternation of broad long overview shots such as Moving Aerial Pan and Survey Shots, which I use to provide context to the narrative. The first drone shot, illustrated by Figure 53[55] (Fine, 2023a, 00:15) shows a Moving Aerial Pan in a Medium to Long Shot of the national monument to the Discoveries, in Lisbon, my city of birth. This shot situates me and my reflections on identity, within the film.

Figure 53 Medium to Long Shot of the Padrão dos Descobrimentos (national monument to the Discoveries)

[55] Sourced from Seb the Nomad (2023, 03:37)

The monument is a symbol of the Portuguese colonising endeavour into Africa, which I am a result of (part of my genetic roots are Cabo Verdean). The monument is located in Belém, by the river Tagus, the unchangeable view of the diasporas. Mine, by plane (illustrated by the Moving Aerial Pan of the Cristo-Rei statue in Figure 54[56] (Fine, 2023a, 00:18); my mother's by ship (Survey Shot of a ship on the river) (Fine, 2023a, 00:22). The drone shots are here used to discuss the complex relationships the airspace embeds. Trade, migration, Catholicism, the return of the white Portuguese after the independence of Portuguese colonies, the incoming Africans, and the departing Portuguese, like me.

Figure 54 Moving Aerial Pan of Christ The King landmark, Lisbon

The drone engenders my own sense of place within that identity of being a migrant. I am neither here, not there. Like the drone, I hover in the middle point between homesick and home-shy. The Top-down Medium Shot view of the traditional Portuguese pavement pattern in Lisbon, spelling

[56] Sourced from Seb the Nomad (2023, 06:32)

Sodáde" in Figure *55*[57] (Fine, 2023a, 01:23) epitomises this. "Sodáde" is the Cabo Verdean (Creolo) spelling of "Saudade" (which means to miss something or someone).

Figure 55 Top-down Medium Shot view of "sodáde"

This shot is also included as a reference to an underexplored element of my identity, a reference both to Cabo Verde and singer Cesaria Évora which the post imperial country both celebrates and racializes. The shot is at a slight Dutch Angle which creates a sense of unsettlement, an unfinished journey. The next drone perspective is less refined than the other shots in the video. Their inclusion is intentional, and it aims to bring the narrative into the present time, where I am visible and am exploring my sense of place and identity using the drone.

The drone capability "Circle" is used as part of utilising the drone as a form of introspection and exploration of self and, possibly, self-surveillance. As it moves under my control, the drone shows

[57] Sourced from Calçada Portuguesa (2023, 01:26)

a part of me, that is normally hidden to myself. I watch the drone seeing me (Figure 56) (Fine, 2023a, 00:42) seeing myself being seen through the drone's eye. This Medium Close Shot creates intimacy with the subject (both myself and the drone). This continues with the next shot (Fine, 2023a, 00:49), where the static drone captures my bare feet in a Close-up Shot, as I walk on the grass.

Figure 56 Circle capability: the self in the drone's eye

I am interested in subverting the drone gaze from Long Shots to Close-up Shots and from aerial to ground (Figure 57) (Fine, 2023a, 00:49), as part of taking control through an untraditional way of showing myself. This is my way to explore the connections across the different spaces I occupy, but do not often truly see. This is reiterated in the following two shots where the screen multiplies that shot into eight, in a split screen.

Figure 57 Drone shot from the ground

Truly, I realise my interaction with the drone, launching it in the air, is a contrived experience. I hesitate whenever a person is in sight (even if distant), and I only take the drone out when accompanied by my White British husband. The drone on the ground is safe, inconspicuous. I avoid the vulnerabilities of exposure to confrontation experienced by all drone users at one point or another, independently from their identities and here compounded by my gender and ethnicity in particular.

The above affects my relationship with the drone. It is both empowering, because I persevere, and disempowering, because it feels like settling on a lower goal. It stops me from rising high. The chaotic collision with the tree shot (Figure 58) (Fine, 2023a, 01:18) is a result of this, illustrated by the accompanying verse:

> "These too are footsteps,
> said no one
> of my slips and trips.

My hands are my body's guests:
misplaced and disconnected." (Fine, 2023a, 01:18)

Figure 58 Drone crashing

To balance this, the film continues with a Moving Aerial Pan of the South Downs (Fine, 2023, 01:37) showing its expanse and magnificence followed by a return to Lisbon, in a Close-up Moving Aerial Tilt shot of the Padrão dos Descobrimentos[58] (Figure 59) (Fine, 2023a, 01:47). In the shot, the monument to the Discoveries is at eye level with the viewer, suggesting a closer examination of this narrative, with parity between the coloniser and colonised. This is a critical reflection on the history of the Portuguese Empire, and the celebration of the "entrepreneurial" men who "expanded" the Portuguese nation, overtaking the silenced oppression and dismissal of the communities oppressed by it.

[58] Sourced from gustavofrazao (2017, 00:33)

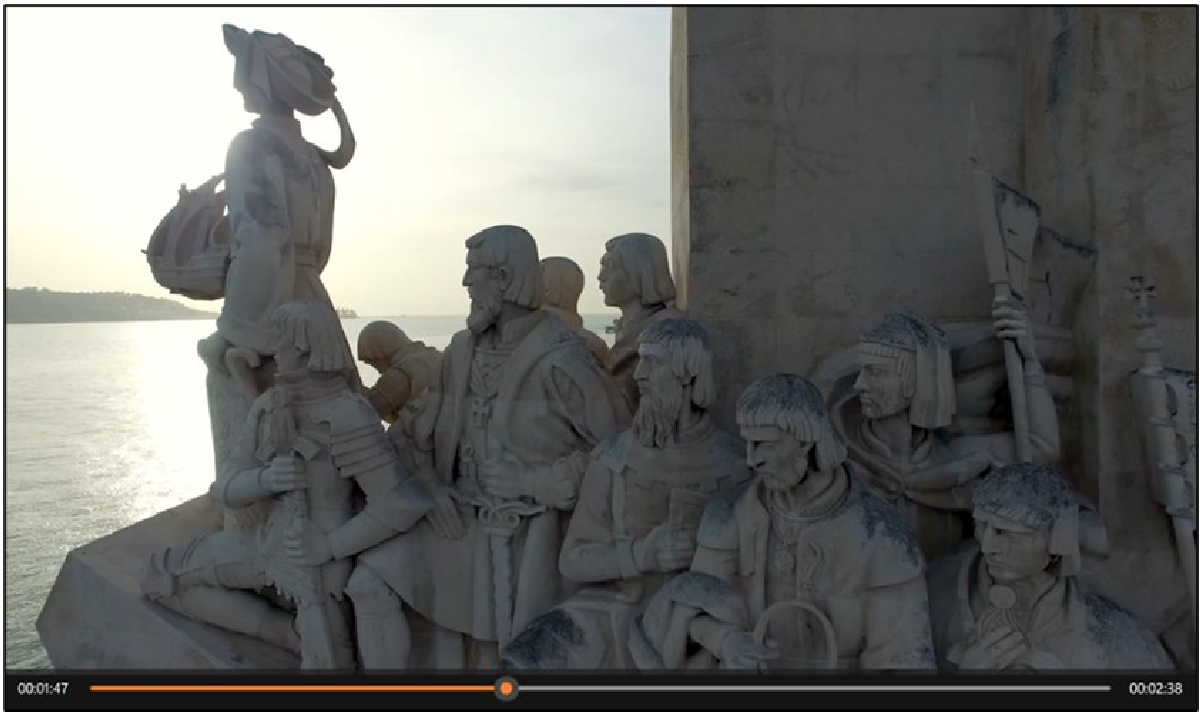


Figure 59 Moving Aerial Tilt of the Padrão dos Descobrimentos

The poem reads: “I too want to be glorious/ to conjure thought cantatas/and word ballets.” (Fine, 2023a, 01:44). The drone ascends and this view, pushes the fifteenth century to the past in favour of the recognition of the agency I have as an African descendant and the value I can bring. The use of drone shots to explore identity and placemaking following a pattern of alternation between Wide and Close-up Shots continues in the second film: *A room of my own* (Fine, 2023b) This film takes *These too are footsteps* (2023a) further, by considering Virginia Woolf’s essay *A room of one’s own* (2025) as a starting point to expand on Woolf’s social justice discussions, into the space of Global Majority women, through my own point of view.

What stands out in this film is the use of the drone to challenge the established image and expectations from a National Trust property. The Ascending view of Monks House, seen from a High-level Angle (Figure 60) (Fine, 2023b, 03:36), places the viewer in a position of power over the house, whilst showing it in all its context, in the village where it is located (Southease).

Figure 60 Ascending view of Monks House

The silence and slow pace of landscape-gazing from the ground are disrupted by an unexpected First-Person View Flythrough of the garden (Figure 61), with high beat music (Fine, 2023b, 03:41). I intend, with this, to disrupt the construct of heritage experiences and the inherent aesthetics that amplify them, as discussed by Heritage Expert Three when they relate the drone content produced by their institution, to that of *Countryfile'*s.

Figure 61 Flythrough of Monks House's Garden

Though recognising the views are impactful, they characterise it as depicting the 21st century countryside through a nineteen fifties filter. They include the soundtrack in the projection of the nostalgia "And it's always got the same soundtrack of; you imagine this kind of like hippy on a guitar playing some nice sweet folk music." The FPV sequence feels radical, contrary, and misaligned from the traditional audiences and behaviours in NT properties. The speed, angles and rupturing through the air, represent my use of the drone to takeover and challenge those parameters. As a heightened expression of drone mobility, FPV drones challenge traditional ways of connecting with place, as do kayaking, skydiving or motor racing (Cairngorms National Park, 2021).

Further to this, the FPV is also a projection of my own voice, enacting my personal power and desire to be seen. This is reinforced in the next visual, a Survey Shot where the drone ascends over me, as I walk towards the house (Figure 62). I choose the Long, High-level Shot, whence as the viewer I hold a position of power and I gaze down at another part of myself as the captured part of the landscape, emplacing myself in its narrative. Through this camera perspective, combined with Dr Maya Angelou's voice reciting a verse of their poem *Still I rise* (StoryVerse

Collective, 2023, 00:32) “Just like hopes springing high, /Still I rise” (Angelou, 1994, p. 163). This is a discussion on the duality of the body, on the vulnerability of its materiality.

Figure 62 Long Shot of the walk towards Monks House

Both film-poems are relevant to my research, because they are a critical reflection on power, using drone technology as a creative practice tool to explore identity, belonging and the importance of natural heritage in my process of placemaking. Ground shots are used as part of subverting ideas on drone mobility and verticality, and the affordances they bring. I do this through playing with the capabilities of drone vision to explore notions of visibility, and interrogate the relationship between viewer and viewed, consumer and producer. This opens a discussion on the premises for narrative control and negotiation.

Relating these films with the themes discussed in the heritage interviews and surveys and drone expert interviews, “Creativity” and “storytelling” are the drone technology opportunities that are particularly salient. These are a strong element in the creation of the content. However, when analysing it beyond the surface, the intention behind utilising those opportunities is to present “A broader perspective" introducing new contexts and, by doing this, proposing "A greater range of experiences" of the South Downs National Park and Monks House to other visitors and, above all, myself. This way of experiencing these sites, makes them more accessible, physically, culturally, and intellectually.

Power is a ubiquitous and conspicuous theme in these videos. It is present in the writing and the metaphoric use of camera perspectives. This is not only true for the instances where I create my own aesthetic language beyond that which is expected from drones (aerial views and Long Survey, Pan or Tilt Shots) by using static, Ground Close-up Shots and views of myself, by myself. It is further unravelled in the use of a First-Person View sequence in a National Trust space.

The metaphoric potential of the FPV drone, is in itself a strategy to harness agency. The technology of this mode of drone, enables the simulation of a fast-paced meandering through the garden, at Monks House, in a way that is not possible to the human body, but still enhancing its capabilities, through simulating the extension of these abilities in a way that it feels to me exhilarating.

My use of drone technology in these two film-poems is a practice for regaining control, discussing what it means and subverting classic drone aesthetics.

2.5 Context to Lovejoy Centre's *Bringing the outdoors, indoors* (2023), two drone livestreaming videos

- Livestream videos titled *Bringing the outdoors, indoors* (2023a) (2023b). The title of the project and videos has been suggested and agreed by the group of service users and staff.
- Produced as a result of co-design research with Lovejoy Centre clients, post-produced by the drone pilot (this involved integrating film of the experience at the Lovejoy with film of the livestream broadcast from the drone).
- Not publicly available due to restrictions arising from NT's drone policies.

- Using the approach to theme categories by Stankov *et al.* (2019, p. 813), it is considered to feature under Heritage.
- The title reflects what stands out mostly for the group, access to the outdoor spaces in their local area, in a way they would otherwise not be able to experience independently, without difficulty in terms of mobility, and the need for support.

Differently to the institutional and community generated drone video content analysed previously in this chapter, the livestream films (Lovejoy Centre, 2023a and 2023b) are not edited by the people generating the content. They have been post-produced by the drone pilot, who combined the drone capture with the film of the session. This makes it possible to view the participants' responses and the drone content they are responding to, instead of the drone footage alone. This is a new element in relation to the videos previously analysed in this chapter. This is also a decision based on visibility of the community which is, as previously discussed in the Literature Review, an element of empowerment for underserved communities.

The drone is controlled on site by the drone pilot, as it is in any of the other drone video content analysed in my research. The element of agency is present in that the participants make the decisions on the route the drone takes and what it captures and, subsequently, what they view. The pilot enacts the participants' instructions.

In the analysis of previous drone video content, the study was structured around the identification of drone camera perspectives and their use in meaning-making and storytelling. The livestreamed drone videos are different to the institutionally produced drone content and the drone videos created by underserved communities discussed earlier in the chapter.

Three characteristics distinguish them from the latter:

- The viewers/consumers are also the creators/producers.
- Both the producers and the process of production and negotiations leading to the final content are visible and exposed through the content.

- The camera perspectives in the videos are not the result of a conscious cinematographic treatment from the producers, but a direct result of how they negotiate their interaction with place, landscape, and their own narratives.

The latter aspect impacts on the approach to analyse the livestreaming videos. As agency, power and control are central elements to the co-research and the research, the analysis considers the independent responses of participants living with dementia, to the camera drone perspectives they are seeing. By independent, it is meant responses to views other than those pointed out or suggested by the drone pilot or the centre staff. Table 15 provides a summary of the types of responses and resulting observations.

As staff also wanted to participate and they are in their own right representative of underserved communities (women - white and Global Majority), their responses to the drone livestreaming experience are analysed separately and compared with the participants' responses. As part of this analysis, three key aspects are identified:

1. The number of prompting questions I ask participants inviting their instruction on where they want the drone to travel ("So where do you want Sam to go now? What would you like to see?").
2. The types of participant responses to questions inviting guidance on the direction of the drone, including the number of instances:

- Participants living with dementia respond to the invitation, by agreeing to the suggestion made by myself, the pilot, or staff ("What would you like to see more of? "All around?")
- Participants answer: "Don't know/I am not sure".
- Participants spontaneously refer to a specific detail in a Medium or Close Shot.
- Participants ask to stay with the view they are seeing.

3. The modes of staff engagement with the experience through the instances when:

- Staff respond to the researcher invitation to direct the drone themselves.
- Staff take the initiative to request to zoom on a specific element in the landscape.
- Staff refer to or ask about a specific detail in a medium or close shot.
- Staff directly ask questions about the landscape to participants or reiterate questions asked by the researcher.

Table 15 Drone livestream participant engagement data is repeated below for ease of cross referencing.

Description	**Livestream One (LV1)**	**Livestream Two (LV2)**	**Observations**
The number of prompting questions researcher asks participants to invite their instruction	11	12	Similar number of prompts in LV1 and 12 in LV2
Number of instances participants' response follows researcher/staff suggestion	4	4	Same number of times people respond and go with researcher /staff suggestion
Number of instances participants' response is "I don't know/I am not sure"	2	1	Livestream 1: Response is "I don't know/I am not sure" by 50% more instances
No response from participants	3	4	Fewer instances of "No response"
Number of instances participants take initiative to ask about or bring up a detail in the landscape they are viewing	9	7	Livestream 1: Nearly 2/3rds more requests for detail from participants in LV 1
Number of instances participants ask to continue to view what they are seeing	1	0	Livestream 1: Once more to stay with view

Description	Livestream One (LV1)	Livestream Two (LV2)	Observations
Number of instances staff respond to researcher invitation prompts	1	6	Livestream 2: Six times more instances when staff directly respond to researcher invitation prompts
Direct requests to view detail from staff	1	2	Livestream 2: 50% more instances when staff directly request a specific detail
Staff ask questions to participants about the landscape or reiterate researcher invitation prompts	3	7	Livestream 2: More than double the questions from staff to participants

It is noticeable in the first livestreaming video, that there is less involvement from centre staff and that there are nearly two thirds more instances when participants take the initiative to ask about or bring up a detail in the landscape they are viewing. The level of staff initiative, based on the number of instances they directly respond to my invitation prompts or make direct requests to view detail are also lower in the first livestreaming session. This may be due to the fact the Centre Manager was present only in the second livestreaming session, participant fatigue (the first livestream session was on a Wednesday afternoon at 1pm and the second livestreaming session was on Friday afternoon at 1.30pm, that week) or other factors.

Based on the tabled data, it is considered here participants living with dementia exercise more agency in the first livestreaming session, at the Devil's Dyke. The focus is primarily on the analysis of the data from this session. Due to the total number of active participants, including staff (five people), and the length of the video, there is a much larger number of drone shots. Only the drone shots that have resulted from comments from participants living with dementia are analysed. Specifically comments on what they are seeing (details they mention, elicited conversations as a result of the viewing experience. They are time stamped below and highlighted in bold. The initial and last shots in the video are always included as they relate to power dynamics.

xxxv Full list of shots

1. Medium Shot of pilot
2. Moving Aerial Pan
3. Ascending Shot
4. Moving Aerial Pan
5. Moving Aerial Pan
6. Moving Aerial Pan
7. Moving Aerial Pan
8. Survey Shot
9. Moving Aerial Tilt
10. Survey Shot
11. Survey Shot
12. Moving Aerial Pan
13. Moving Aerial Pan or Survey Shot
14. Moving Aerial Pan
15. Survey Shot
16. Moving Aerial Pan
17. Zoom and Moving Aerial Tilt in a Medium Shot
18. Moving Aerial Pan
19. Survey Shot
20. Moving Aerial Pan/ Survey Shot
21. Moving Aerial Pan on a Medium Shot
22. Survey Shot
23. Moving Aerial Tilt
24. Moving Aerial Pan
25. Moving Aerial Pan

26. Moving Aerial Pan

27. Moving Aerial Pan

28. Moving Aerial Pan

29. Moving Aerial Pan

30. Extreme Long Shot

31. Medium Close-up

Xxxvi Selection of Shots analysed

1. Medium Close-up (2023a, 00:56)
2. Medium Shot of pilot (2023a, 01:17)
3. Moving Aerial Pan (2023a, 01:44)
4. “What are those dots at the top?” (2023a, 03:23). Leads to Medium Shot of landscape in relation to where trees are
5. Moving Aerial Pan (2023a, 03:41)
6. Lovejoy Centre Participant Three “I don’t know what that white is” (2023a, 05:12) Leads to Survey Shot to explain white patterns on landscape (2023a, 05:17)
7. Moving Aerial Pan (2023a)
8. “Nice little village” (2023a, 10:20) “It is a nice little village” (2023a, 11:09) “People must live there” (2023a, 12:21) and a Zoom to Medium Shot and Moving Aerial Tilt to show houses (2023a, 10:48)
9. Moving Aerial Pan “Someone walking there” (2023b, 14:48)
10. Lovejoy Centre Participant Three asks: “All round” and Moving Aerial Pan (2023a, 04.12)
11. “Hundreds of sheep or whatever they are” (2023a, 18:12) and Moving Aerial Pan
12. Moving Aerial Pan from Extreme Long Shot to Medium Close-up and Pilot waves (2023a, 20:34)

Again, in this video, shots that show the broader context, such as Moving Aerial Tilt and Moving Aerial Pan Shots are intercalated with a focus on detail through camera Zoom Shots, on specific aspects of the landscape or conversations that focus on landscape details. The interest in detail has not been consistently manifested in Medium Close-up Shots in the livestream, because during the broadcast, the details elicited by participants are not initially zoomed in on. The Zoom is used more predominantly as a refamiliarization strategy in the Highdown Hill livestream.

Both livestream videos start with an interaction with Sam, the drone pilot. This is part of framing the livestreaming experience as one of human engagement. Participants see a Medium Close-up of the pilot showing them launching the drone (Figure 63) (Lovejoy Centre, 2023a, 00:57) which establishes a narrative of witnessing. I organised this with the pilot, as part of involving participants in elements of the drone flight, that are behind the scenes.

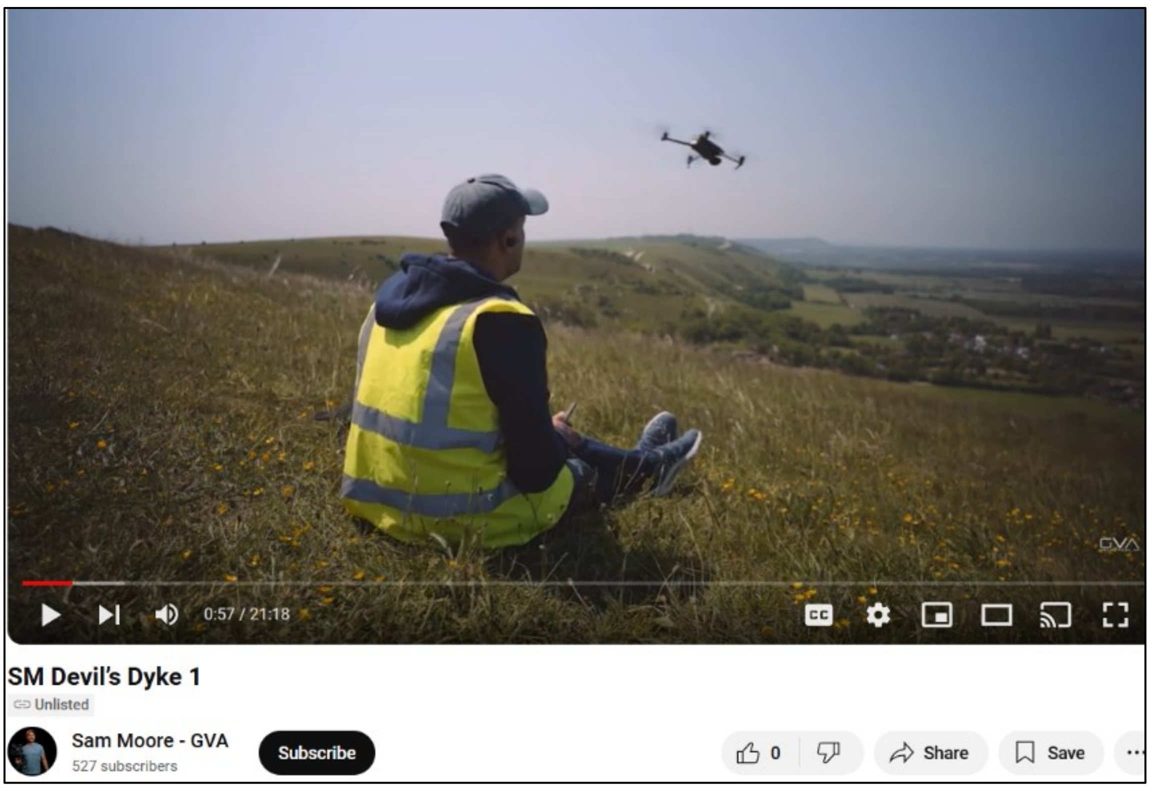


Figure 63 Pilot launching the drone at the start of the livestream

Taking participants behind the scenes is an intentional approach to making them part of the content generation process. Nothing is hidden from them. This extends to highlighting all the technical issues that happen throughout the experience.

In a subsequent shot, Sam exchanges ‘hellos' with the participants at the Lovejoy Centre (Figure 64) (Lovejoy Centre, 2023a, 00:35). He waves at the group in a Medium Shot close to Eye Level (a slightly higher level for the viewer).

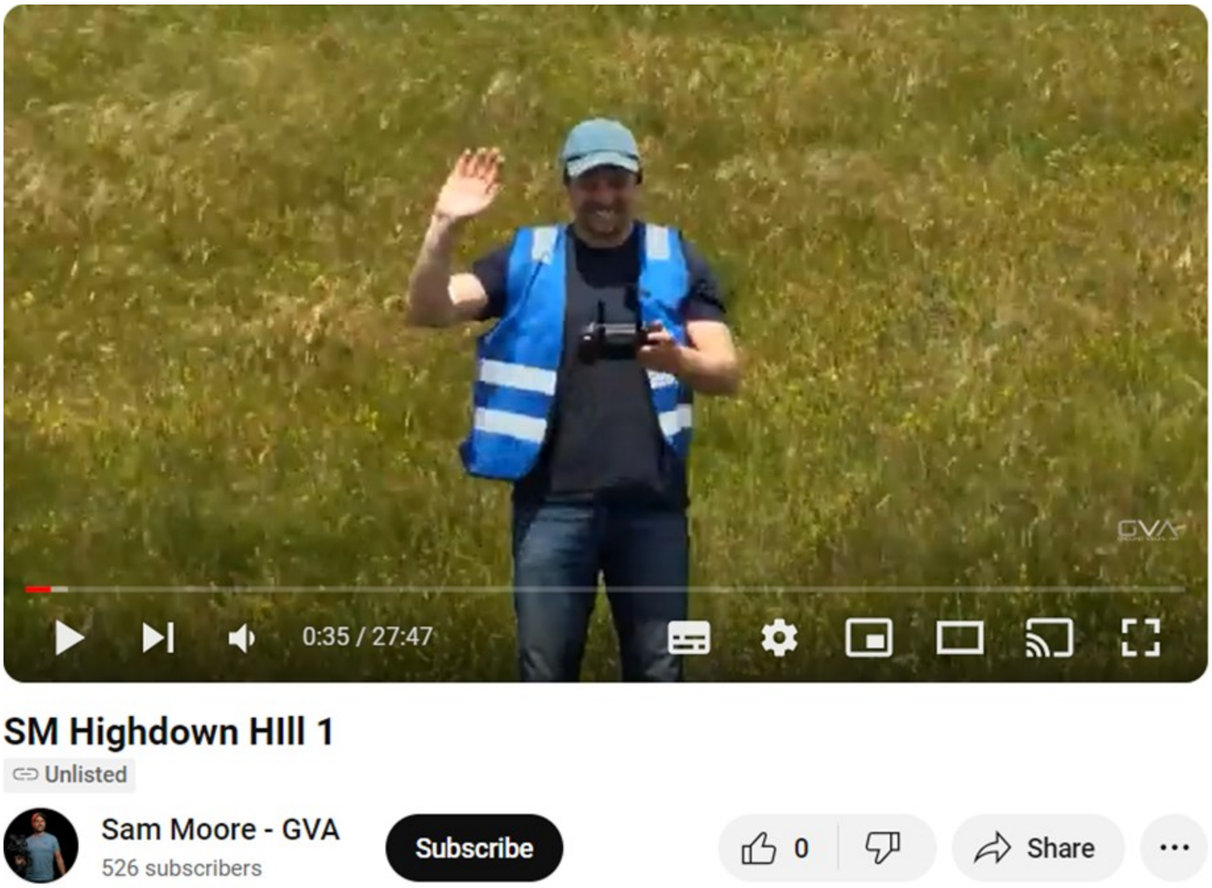


Figure 64 Pilot waves hello to participants at the Lovejoy Centre

Participants respond by waving back shown in Figure 65 (Lovejoy Centre, 2023a, 00:43). This conveys parity between pilot and participants. It establishes a level of complicity and intimacy, inherent to the emulation of personal distance through the Close-up shot. This same effect is seen in the Women Who Drone’s video (2023), which starts with a Close-up of the pilot, who waves and blows a kiss at the audience.

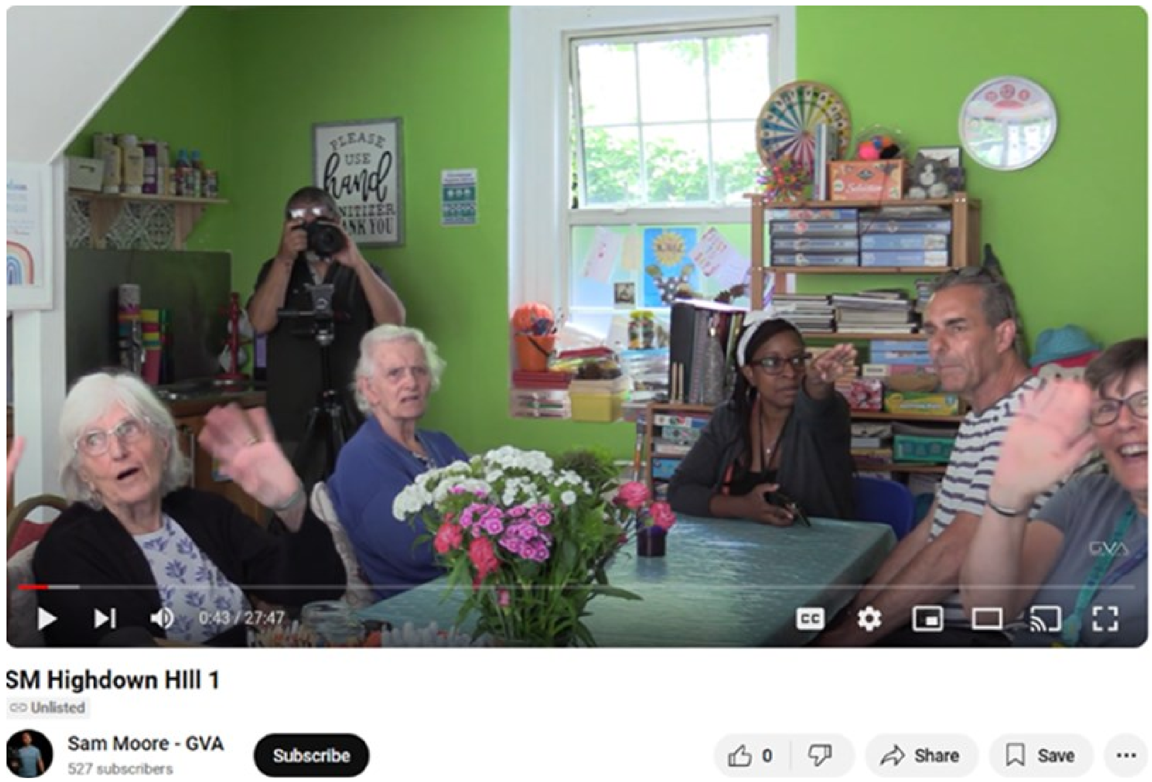


Figure 65 Participants wave back at pilot

The visualisation and interaction with the pilot, also address the disembodied nature of drone visuals, emphasizes that it is a live experience, that is happening in the here and now. This temporal frame is more accessible to people living with dementia and therefore makes the experience more accessible.

The next shot is an Ascent Shot, widening the view of the landscape from the pilot to Devil's Dyke Valley. This has a reveal effect which, combined with the Moving Aerial Pan that follows and the Moving Aerial Tilt after that, "unboxing" the landscape, presenting it in its "[T]he iconic views" (Heritage Expert Nine). The emotional responses from participants align with this auratic showcase of the Devil's Dyke.

This perspective elicits enjoyment and aesthetic appreciation from participants, evident in their comments and body language "That's nice, isn't it?" (Lovejoy Centre Participant One, 2023a 01:53). Nevertheless, the participants' comments may be prompted by the comments of the staff

“Oh look at all that. All that space. Wow. Oh, isn’t it beautiful?” (Lovejoy Centre Staff Participant Two, 2023a, 01:53).

The next shot is a Moving Aerial Pan, which the pilot uses as part of showing the context. This gives participants the opportunity to have an overview of what they are seeing, which is part of a process of orientation and familiarisation with the landscape. The pilot notes elements of landscape within the Long Shot range such as the sea, and Zooms in.

The next shot analysed is prompted by Lovejoy Centre Participant Three’s question “What’s the dot on the- the sheep are up- “ illustrated by Figure 66, (2023a, 03:26). This is part of the search for detail and is an example of Serafinelli and O’Hagan’s (2022, p. 229) drone visual characteristic of “Defamiliarizing the familiar”.

Figure 66 Participants responding to the experience with interest and curiosity

For the rest of the video and livestreaming experience of the Devil's Dyke, the camera perspectives alternate between two key types:

i. Extreme Long and Long Shots (as in Figure 67) that are functional and used to provide context (Moving Aerial Pan, Moving Aerial Tilt and Survey Shots). They are part of a cartographic approach to landscape and help the group to orient themselves within the space. For example: After zooming in on the seafront in Brighton through a Moving Aerial Pan, at the request from staff, I ask the group "Do you want him (the pilot) to go a bit further up, a bit left, show all around again?" Lovejoy Centre Participant Three answers "Oh, yeah, yes please" (2023a, 18:02).

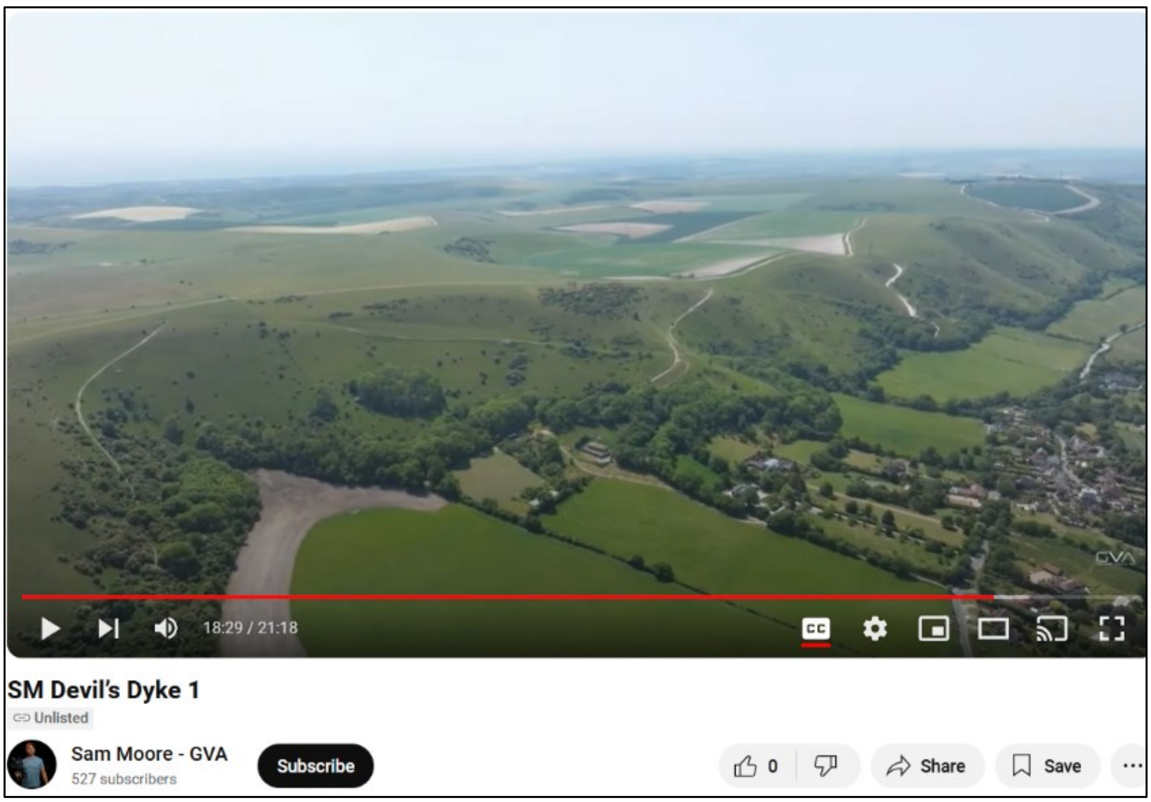


Figure 67 Long Shots provide contexts

ii. Zoom shots into Medium or Medium Close-up frames that bring the viewer to a closer distance to the landscape. These focus on the element individuals spontaneously elicit in their conversations during the viewing experience or directly ask questions about. For instance: when I ask the group if there is anywhere, they would like to see closer up, (Lovejoy Centre, 2023a, 10:21). Lovejoy Centre Participant One comments “Nice little village down there” (2023a, 10:23). As a result, the pilot changes the perspective from a Moving Aerial Pan to a Zoom into a Medium Close-up, Figure 68 (Lovejoy Centre, 2023a, 10:49).

Figure 68 Moving Aerial Pan offers a large-scale view of the landscape following a detailed shot

The inclusion of the consumer/producer in the drone video enables me to explore how the experiences of participants living with dementia and centre staff participants relate to the themes identified by heritage and drone experts (discussed in Chapter Five and in section 6.1).

Based on the analysis on the verbal and body language responses from participants and staff, during the livestreaming sessions, there are five key themes that emerge in the data. These relate to the opportunities of drone technology to engage underserved audiences identified in my heritage and drone experts' data, in Chapter Five. They are outlined below, including examples from participants' and staff's comments:

"A broader perspective that shows context" expressed through language articulating scale, position, or orientation elicited in Lovejoy Centre Participant One's comments such as "It is quite big, really." (2023b, 19:47); "That was a big shot" (2023a, 18:04); and that of Lovejoy Centre Staff Participant One: "Just shows how distinctive it is from a distance, doesn't it? (2023b, 17:05)

"Access and Accessibility" as an effect from accessing a place remotely and being able to engage with an activity the group enjoy and in which they are able to participate in, in an equitable way: "It is nice that you can see it" (Lovejoy Centre Participant One, 2023a, 04:57); Lovejoy Centre Staff Participant One asked: "Would it be too far to walk if you did that walk yourself?" Lovejoy Centre Participant Three answered; "It would for me."

"A greater range of experiences" expressed through comments and body language associated with a positive experience, such as "It is great that you can see the beach" (Lovejoy Centre Participant Three, 2023a, 07:09); "Neat." (Lovejoy Centre Participant One, 2023b, 22:24); "It's really good" (Lovejoy Centre Staff Participant One, 2023b, 06:56).

"A storytelling tool" emerging through the abstraction into relatable imagined experiences in the landscape: "Can you imagine walking around there, with your sandwich and your knapsack on your back?" (Lovejoy Centre Staff Participant Two, 2023a, 01:34).

Comparing the livestream experience with the drone content viewing activity in Session Three of the co-research, participants' responses in this session show participants experience the opportunities of drone technology as above, in both. Lovejoy Centre Participant Nine highlights access: " You can't see it down there, can you? You have to be up there to see it." Lovejoy Centre Participant Three perceives drone views as providing "A greater range of experiences" articulated through positive feelings, “It just feels good to be walking there” and mental simulation effects. The key difference between the two viewing events is that when passively viewing heritage drone content, participants’ behaviour in relation to the viewing experience, is not proactive in that they do not make any comments on the content and remain quiet until after the session, when they are asked about their feelings about what they saw, whether they liked it and why. In the livestreaming experiences, it is observed participants not only comment on what they are seeing, they also engage in conversations about the content, relating it to their own geographies.

The experience of one participant is of particular interest because its description may articulate a key aspect in the negotiations of power in the production of drone content and its effects in connecting with heritage. Lovejoy Centre Participant Two, who was a drone pilot in the past, compares the aerial views they watched in the passive viewing session with a “sword fight”, when asked what they thought about what they had viewed. They further described the views as “Colossal” agreeing, when prompted by the staff, they are amazing and beautiful but stating they do not like it too much.

This idea of vastness of landscape is also expressed by Lovejoy Centre Staff Participants: “It’s just massive” (Lovejoy Centre Staff Participant Two) and in heritage expert data as Heritage Expert Nine talks about the expanse of the landscape drone technology enables you to view. The effect of aerial views of defamiliarizing familiar sights (Serafinelli and O’Hagan, 2022) results in a feeling of awe that is overwhelming. Lovejoy Centre Staff Participant Two’s reflections corroborate this in reference to the Livestreaming events:*” Seeing something from a distance, you can’t really make out what it is and then being able to zoom in (…) when you got closer you could see the gardens and the swimming pools. It’s like coming to land on an aeroplane, isn’t it, when you’ve been on holiday”* (2023b, 09:01*).* The aspect of a return home journey is important, as it emphasizes the elements of belonging and placemaking that are inherent to the exploration of detail in aerial views. These aspects are part of projecting self on to the landscape and becoming visible within it.

Consistent with the other drone content videos generated by underserved communities, there is, in both livestreams, a trend to alternate between wider, longer shots and more detailed focus on the landscape. The former help the viewer/producer orientate themselves, fulfilling a cartographic purpose. The latter are followed by an interest in focusing on details that are relatable, sometimes autobiographic. These emerge in references to home and community "It is a nice little village, maybe" (Lovejoy Centre Participant One, 2023a, 11:09); "I have got some lavender in my garden", Lovejoy Centre Participant Three's comment when we zoomed in on a lavender field in Highdown Hill (Lovejoy Centre, 2023b, 18:05).

As previously discussed, this exploration of detail is considered here as a way of regaining control over the effect of defamiliarization resulting from aerial views. Detail is a strategy to manage the feelings of awe resulting from aerial views. These feelings are expressed in both Session Three of the co-research, when participants watched heritage drone content I selected and in Session Five, when Drone Livestreaming events took place.

De Cruz (2024) argues awe helps us to successfully engage with aspects of our environment. What is observed in this project is that whilst in the passive viewing of drone content, participants do not have agency to address the overwhelm from the views they are consuming, because they are engaging with ready-made content, this is not the case in the livestream. Here, the ability to control the drone views, helps negotiate the effects of reverence. This is because participants can mitigate it through refamiliarizing themselves with the views, through exploring and zooming in on the subject of their interest, making the experience accessible to themselves and therefore reclaiming power. I consider that awe helps us being effective, if we are able to use our agency to make sense of the environment whilst, admiring it.